THE MIDPOINT RIEMANN–VON MANGOLDT SYNTHETIC ZETA SYSTEM: CANONICAL PRODUCTS, EXPONENTIAL SUMS, PRIME COUNTS, MERTENS TRANSFORMS, AND THE TRANSFORMATION TO ζ_{Z1}

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Abstract. We study the zeta-like system generated by the midpoint inverse of the smooth Riemann–von Mangoldt counting law. Its nontrivial zeros are $\frac{1}{2} \pm i\tau_k$, where τ_k is the midpoint ordinate of the clock. We prove normal convergence of the canonical product, the functional symmetry, the exact zero divisor, a uniform logarithmic bound for the exponential sums of the τ_k, compact-uniform existence of the explicit-formula prime-power count with an explicit branch of Ei, exact finite Möbius inversion to a synthetic prime count π_{RVM}, and the unconditional square-root estimate $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$; with the prime number theorem this gives $\mathrm{li}/\pi_{\mathrm{RVM}} \to 1$, $\pi/\mathrm{li} \to 1$ and $\pi/\pi_{\mathrm{RVM}} \to 1$. The transition theorem $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ is proved in full; it gives a constant vertical limit of $1/\zeta_{\mathrm{RVM}}$ on every line to the right of $\frac{1}{2}$, excludes absolutely convergent Dirichlet expansions of $\zeta_{\mathrm{RVM}}^{\pm 1}$, and proves the unsmoothed synthetic Mertens bound $M_{\mathrm{RVM}}^{\mathrm{crit}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$, $L = 1/\varpi_{\mathrm{RVM}}$. A contour-free form of the same decomposition shows that ζ_{RVM} is bounded above and below on every half-plane $\Re s \geq \sigma_0 > 1$ and tends to ϖ_{RVM} as $\Re s \to +\infty$.

The core of the method is a transformation of ζ_{RVM} depending on a parameter $\lambda \in [0,1]$: the ordinates move along $\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k$, where γ_k are the ordinates of the zeros of the Riemann zeta function, every member ζ_λ has all its nontrivial zeros on $\Re s = \frac{1}{2}$ by construction, $\zeta_1 = \zeta_{\mathrm{RVM}}$, and $\zeta_0 = \zeta_{Z1}$. It is impossible for ζ_{Z1}, or for any ζ_λ, to have a zero off the critical line other than the trivial zeros. All differences between the moving zetas lie in the numbers $\gamma_k - \tau_k$. The final step of the method is the identity $\zeta_{Z1} = \zeta$. This identity is the single and complete test: while $\zeta_{Z1} = \zeta$ there is nothing outside $\Re s = \frac{1}{2}$ to speak of, and $\zeta_{Z1} \neq \zeta$ is the only possible obstruction.

For completeness we reprove the classical implication RH $\Rightarrow M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ in the endpoint notation: if $\zeta_{Z1} = \zeta$, then $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$, with a self-contained proof (Borel–Carathéodory, three circles, truncated Perron). By Hamburger's theorem the identity $\zeta_{Z1} = \zeta$ is equivalent to the condition (D) that ζ_{Z1} is an absolutely convergent Dirichlet series in $\Re s > 1$. We prove three further forms of (D): the value of the endpoint at infinity is $\varpi_{Z1} = \lim_{\sigma\to+\infty}\zeta_{Z1}(\sigma) = \varpi_{\mathrm{RVM}}\lim_{K\to\infty}\prod_{k\leq K}(\frac{1}{4}+\tau_k^2)/(\frac{1}{4}+\gamma_k^2) \geq 1$, and (D) holds exactly when $\varpi_{Z1} = 1$, that is, when the ordered product $\lim_{K\to\infty}\prod_{k\leq K}(\frac{1}{4}+\gamma_k^2)/(\frac{1}{4}+\tau_k^2) = \varpi_{\mathrm{RVM}}$; the prime-power counts of ζ_{RVM} and ζ_{Z1} are the Mellin transforms of their logarithms and satisfy a distributional explicit formula, which proves the Mellin–Stieltjes identification for the integer weights; and (D) holds exactly when the prime-power count of ζ_{Z1} is constant between consecutive integers and its jumps satisfy $\sum_{n\leq X}|\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$, an identity on each cell between the displacements $\gamma_k - \tau_k$ and the clock. Along the family, $\log\varpi_\lambda$ has derivative $-\Upsilon(\lambda)$, the generator at infinity, and (D) holds exactly when $\int_{0^+}^1 \Upsilon\, d\lambda = \log(1/\varpi_{\mathrm{RVM}})$. Finally we atomize every member of the family by smooth cells: the cell masses have mean square $X^{1+\varepsilon}$ on every dyadic block, uniformly in λ, so every atomized function $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde{h}_\lambda}$, the endpoint $\lambda = 0$ included, is an absolutely convergent Dirichlet series in $\Re s > 1$; the joining factor $\tilde{h}_\lambda$ is written cell by cell and is holomorphic in $\Re s > \frac{1}{2}$, and (D) is reduced to a single statement at $\lambda = 0$, the vanishing of every cell defect, which is the remaining step of the programme. We prove further that no member ζ_λ with $0 < \lambda \leq 1$ is a Dirichlet series, that $\lambda \mapsto \log\zeta_\lambda(\sigma)$ is strictly convex for every real $\sigma > 1$, and that $\zeta_{Z1}(\sigma) \geq \zeta(\sigma)$ for every real $\sigma > 1$, with equality at one point exactly when $\zeta_{Z1} = \zeta$. Hamburger's theorem is extended to the pair $(\zeta_\lambda, \tilde{\zeta}_{\mathbb{Z},\lambda})$: $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta$ holds as soon as the joining factor is symmetric under $s \mapsto 1-s$, equivalently when the atomized coefficients satisfy the theta relation of ζ, equivalently when the vertical mean

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square of $\tilde\zeta_{\mathbb{Z},\lambda}/\zeta$ on a line $\Re s=\sigma>1$, which is always at least 1, equals 1; at the clock end the symmetry fails, and $\tilde\zeta_{\mathbb{Z},\lambda}=\zeta$ holds for at most finitely many λ. (D) holds exactly when ζ_{Z1} is the Mellin–Stieltjes transform of a positive measure: the functional equation then gives $\varpi_{Z1}\le 1$ by an extremal summation argument, and in general $\varpi_\lambda-1$ is one explicit number computed from the summation formula of ζ_λ. At the clock end the absolutely continuous part of $d\pi_{\mathrm{RVM}}$ has density $\frac{1}{\log y}+2\cos(2\pi y)-2^{-1/2}y^{-1/4}\cos(4\pi\sqrt{y}+\frac{\pi}{4})+O(y^{-1/3})$, while $d\pi_{\mathrm{RVM}}$ also has the explicit atoms $\mu(j)J_{\mathrm{RVM}}(2)\delta_{2^j}/j$. Finally the constructions are carried out for every list of ordinates within a bounded distance of the clock: the lists whose continuous and atomized zeta functions coincide form a set with at most one element, which is non-empty exactly when (D) holds, so that the final step is an existence statement in a compact space of lists. In version 18 we retain these equivalences but remove the earlier continuation claim: the interpolation tests show that the exact-atomic linear path has no established interior solution, while the additive interpolation is only a level-set reformulation of $\zeta/\zeta_{\mathrm{RVM}}$. We add a corrected clock-only injectivity problem, a rigorously stated a priori bounded-displacement proposition for any exact rung solution satisfying the stated local-density hypothesis, and documented bounded finite-section diagnostics. None of these proves (D).

What is proved and what is not. Every statement above is proved in this paper without any hypothesis. At the clock end we further prove a multiscale theorem for the linearized rung system: up to height T the ordinate block has a uniform lower frame bound on u-windows of length $(1+\varkappa_{\mathrm{win}})\log(T/2\pi)$, no T-independent lower bound is possible on a fixed window, the infinite clock linearization has dense range on every finite window, and smooth transport tails beyond height T are $O_{A,\mathrm{f}}(T^{1-A}\log T)$. These statements use only the midpoint clock and no individual displacement $\eta_k=\tau_k-\gamma_k$. They rule out a fixed-window implicit-function argument. The later interpolation tests show that no multiscale continuation theorem is presently established; the remaining endpoint problem is not replaced by a path of exact intermediate objects.

The identity $\zeta_{Z1}=\zeta$, equivalently (D), equivalently each of the criteria above, is not proved; it is the open step of the method.

Version 19 adds a section on route B for finite sections of the rung system: compactness of near-solutions in a fixed box; with the von Mangoldt masses prescribed, the Laplace transform of the residual is identified as $\frac{1}{2}(P'_{\boldsymbol\kappa}/P_{\boldsymbol\kappa}-\xi'/\xi)$ and every exact solution at bounded distance from the clock is the zeta list, so the finite-section statement B′ is equivalent to the Riemann hypothesis (an equivalence classical in substance); a certificate theorem for staged sections (Newton–Banach with a frozen right inverse); Gonek's symmetrised Euler products as explicit lists in a tube around the clock; an a priori counting bound for quasi-solutions; a Kadec–Avdonin criterion for (A_2) with blocks of variable length, and the unconditional failure of (A_2) for the unfolded zeta ordinates on every horizontal line; and floating-point Banach certificates of exact square sections up to height 487, with the certified basin shrinking like T^{-2}.

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Notation

Every symbol below has a single meaning throughout the paper. The last column links to the defining equation; hovering over the link shows the formula. Symbols introduced inside a single proof (indices j, k_0, m, auxiliary variables v, y, thresholds t_0, t_1, t_2, radii r_1, r_2, r_3) are local to that proof and do not appear elsewhere.

Symbol	Meaning	Defined
Variables and the Riemann zeta function		
$s = \sigma + it$	complex variable, $\sigma = \Re s$, $t = \Im s$	–
$z = s - \frac{1}{2} = \sigma' + it$	centred variable, $\sigma' = \sigma - \frac{1}{2}$	(8.2)
$q(s) = s(s-1)$	symmetric quadratic, $q(1-s) = q(s)$	(3.1)

Symbol	Meaning	Defined
ζ, ρ	Riemann zeta function; ρ runs over its nontrivial zeros with multiplicity	§2
γ_k	kth positive ordinate of the nontrivial zeros of ζ, with multiplicity, nondecreasing	(2.1)
$N_\zeta(T)$, $N_\zeta^*(T)$	zero count up to height T; its symmetric version	(2.1), (2.10)
$S(T)$	remainder in the Riemann–von Mangoldt formula	(2.9)
$G(s)$	$s(s-1)\Gamma(s/2)\pi^{-s/2}$	(3.3)
$P_\zeta = 2\xi$	completed product $G\zeta$	(9.1)
μ, $M(x)$	Möbius function; Mertens function $\sum_{n\le x}\mu(n)$	(1.1)
$\pi(x)$, li	prime-counting function; $\mathrm{li}(x) := \mathrm{PV}\int_0^x dt/\log t$	–
$\chi(s)$, ϑ	functional-equation factor; Riemann–Siegel theta function	(8.1), (8.29)
The midpoint clock		
$\mathcal{N}_{\mathrm{RVM}}(T)$	midpoint Riemann–von Mangoldt clock	(2.3)
τ_k	midpoint ordinates: the unique solutions in $[2\pi,\infty)$ of $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$	(2.4)
W_0	principal branch of Lambert's W-function	(2.5)
$\mathcal{T}$	inverse function of the clock	(4.4)
ℓ_k	$\log(\tau_k/2\pi)$	text after (4.4)
$n_{\mathrm{RVM}}(H)$	number of $\tau_k \le H$	(4.12)
η_k, C_η	$\eta_k = \tau_k - \gamma_k$; a bound $\lvert\eta_k\rvert \le C_\eta$	(2.11), (2.12)
The synthetic zeta function and its sums		
$P_{\mathrm{RVM}}^{\mathrm{crit}}$, ξ_{RVM}, ζ_{RVM}	canonical product, completed and synthetic zeta	(3.2), (3.4)
ρ_k^{RVM}	synthetic zero $\frac{1}{2} + i\tau_k$	(3.6)
$S_{\mathrm{RVM}}(N;u)$, $C_{\mathrm{RVM}}^{\mathrm{pts}}$, $C_{\mathrm{RVM}}^{\mathrm{path}}$	exponential sum; point and path centroids	(4.1), (4.2)
u_1, u_2	a frequency interval $[u_1,u_2] \subset (0,\infty)$	(4.3)
$M_H(x)$, $M_H^{\mathrm{RVM}}(x)$	finite-height phase averages (not Mertens functions)	(4.14), (4.13)
Ei	exponential integral on $\Im w \neq 0$	(5.1)
J_{RVM}, J_{RVM}^*, π_{RVM}	explicit-formula prime-power count; truncation; synthetic prime count	(5.2), (5.6), (5.7)
$\mathcal{A}_{\mathrm{RVM}}(u)$, $\ell(t)$	weighted clock sum $\sum_k e^{iu\tau_k}/\tau_k$; $\log(t/2\pi)$	(5.10)
$\mathfrak{s}_1$; $\mathfrak{e}$, $\mathfrak{e}_1$	primitive of the midpoint sawtooth; $N_\zeta - \mathcal{N}_{\mathrm{RVM}}$ and its primitive	(8.36), (12.1)
ϖ_{Z1}	value of ζ_{Z1} at $+\infty$	(12.4)
J_X^{M}, J_X, C_X	Mellin and explicit-formula prime-power counts of ζ_X, $X \in \{\mathrm{RVM}, Z1\}$; their difference	(12.14), (12.16)
b_n	jumps of J_{Z1}^{M} at the integers	Thm. 12.7
ϖ_λ; $\Upsilon(\lambda)$	value of ζ_λ at $+\infty$; flow at infinity	(13.1), (13.3)
φ_n, $\tilde{c}_n(\lambda)$, $\tilde{a}_N(\lambda)$	smooth cells; atomized masses and coefficients	(13.7), (13.8)
$\tilde{\zeta}_{\mathbb{Z},\lambda}$, $\tilde{h}_\lambda$	atomized family; joining factor	(13.7), (13.13)
$\mathfrak{a}_s$, $\mathfrak{h}_n(\lambda;s)$, $\mathfrak{L}$	atomization kernel; cell defects; $\int_1^x (1-y^{-1})dy/\log y$	(13.10)
V_n, W_n	ordinate kernels of the cell n	(13.19)
$\tilde{\mu}_\lambda$, $\tilde{M}_\lambda$	atomized Möbius coefficients; summatory function	(13.32)
B_{Z1}	Mellin form of the endpoint Mertens function	(13.34)
c_n^{RVM}, $\zeta_{\mathbb{Z},\mathrm{RVM}}$	integer weights; weighted Euler product	(6.1), (6.2)

Symbol	Meaning	Defined
$J_{\mathbb{Z},\mathrm{RVM}}$, $J^{*,\mathbb{Z}}_{\mathrm{RVM}}$, h_{RVM}	integer-step counts; Mellin correction	(6.3), (6.4), (6.20)
θ	unit-cell step, $0 < \theta \le 1$	(6.5)
p_j^{RVM}	real RVM prime levels $\pi^{-1}_{\mathrm{RVM}}(j)$ on monotone branches	(1.6)
$\mathcal{M}(n)$, ς_n, μ_h, M_h	target samples of $M^{\mathrm{crit}}_{\mathrm{RVM}}$; signs; ternary walk and its sum (companion object; h is a label)	(1.9), (1.10)
$a_N^{\mathbb{Z}}$	Dirichlet coefficients of $\zeta_{\mathbb{Z},\mathrm{RVM}}$	(6.14)
C_J	Lipschitz constant of J_{RVM}	(6.5)
$\mathfrak{d}_n^{\mathrm{RVM}}$	prime defect $\mathbf{1}_{\mathrm{prime}}(n) - c_n^{\mathrm{RVM}}$	(9.11)
Mertens transforms and the transition theorem		
$F_{\mathrm{RVM}} = \widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}$	$1/\zeta_{\mathrm{RVM}}$	(7.1)
c, U	Perron abscissa; truncation height	(8.45)
$M^{\mathrm{crit}}_{\mathrm{RVM}}$, $M^{\mathrm{crit}}_{\mathrm{RVM},G}$	symmetric Perron inverse; Gaussian-smoothed inverse	(8.45), (7.5)
τ_*, $w(s)$	transition point $-iz = t - i\sigma'$; transition variable $e^{-2\pi i \mathcal{N}_{\mathrm{RVM}}(\tau_*)}$	(8.2)
B; σ_1', σ_2'	upper bound for σ'; a range $[\sigma_1', \sigma_2'] \subset (0, \frac{1}{2})$	(8.4), (8.5)
$\mathcal{E}_0, \mathcal{E}$	error factors of the transition theorem	(8.4), (8.5)
ϖ_{RVM}, L	transition constant 0.9619415807 (Richardson value at the reported precision); $L = 1/\varpi_{\mathrm{RVM}}$	(8.28), (8.33)
$f, \Lambda, \Lambda_{\mathrm{sm}}, \Lambda_{\mathrm{saw}}$	logarithm of the product and its smooth and sawtooth parts (used in the transition and related clock estimates)	(8.6), (8.9)
$\psi, \psi_r, \Phi_r, \nu_1, \dots, \nu_4$	sawtooth, Abel means, ray remainder, real constants (used in the transition and related clock estimates)	(8.8), (8.20)
α_c	$\min(c - \frac{1}{2}, 1)$	(8.35)
Comparison with ζ		
$\mathcal{Q}_{\mathrm{RVM}}$, $\mathcal{Q}^{\vee}_{\mathrm{RVM}}$	quotient $\zeta/\zeta_{\mathrm{RVM}}$; its inverse Mellin transform	(9.6), (9.8)
The λ-family and the endpoint ζ_{Z1}		
λ	family parameter; $\lambda = 1$: ζ_{RVM}, $\lambda = 0$: ζ_{Z1}	(10.1)
$\gamma_k(\lambda)$	moving ordinates $(1-\lambda)\gamma_k + \lambda\tau_k$	(10.1)
P_λ, ζ_λ, F_λ	family product, family zeta, $1/\zeta_\lambda$	(10.2), (10.3)
$A_{k,\lambda}$	$\frac{1}{4} + \gamma_k(\lambda)^2$	(10.10)
$\mathcal{V}_\lambda$	infinitesimal generator $\partial_\lambda \log P_\lambda$	(10.10)
Π_{tr}	total multiplier $P_{Z1}/P^{\mathrm{crit}}_{\mathrm{RVM}}$	(10.15)
$\rho_{k,\lambda}$, $\mathfrak{r}_{k,\lambda}$	moving zero $\frac{1}{2} + i\gamma_k(\lambda)$; residue of $1/(s\zeta_\lambda)$	(10.19)
P_{Z1}, ζ_{Z1}, F_{Z1}	endpoint product, endpoint zeta, $1/\zeta_{Z1}$	(11.1), (11.34)
$\mathscr{R}_{Z1}$	ratio ζ_{Z1}/ζ	(11.7)
$\mathcal{K}$	multiplier $\zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$	(11.12)
(D), a_n	Dirichlet condition for ζ_{Z1} and its coefficients	(11.8)
C_H	constant in Hamburger's theorem	(11.10)
μ_{Z1}, M_{Z1}	coefficients of F_{Z1}; summatory function	(11.37)
$M_{\lambda,U}$, $M_{Z1,U}$	finite-height Perron integrals	(11.35)
δ, ε, $\omega(\delta)$	margins $\sigma \ge \frac{1}{2} + \delta$, exponent $\varepsilon > 0$, three-circle exponent	(11.16)
Criteria in the half-plane, the rung system and the models		
$\mathsf{D}_n(\lambda)$, $B^{\vee}_\lambda$	$\tilde{c}_n(\lambda) - \tilde{c}_n(0)$; dual function of the atomized family	(14.57), (14.58)
d, Y, $\mathsf{e}_1, \mathsf{e}_2$	difference of two counts; auxiliary entire function; exponents	(14.61), (14.63), (14.67)

Symbol	Meaning	Defined
J_f, J_g, b_n, ϖ_f	counts and coefficients of a general pair (f,g); value of f at $+\infty$	(14.65), Thm. 14.36
(B1), (B2)	the two hypotheses of the bridge theorem	Thm. 14.36
v_λ	phase defect of the family on the critical line	(14.69)
o_n, L_n, j	one-ordinate difference of cell masses; $-\log(1-n^{-s})$; one-ordinate joining factor	(14.72), (14.75)
f, c_f, k, k_f	test function in $u=\log x$; its cosine transform; the part of G and the pole	(14.77)
R_f, S_f, $\mathsf{m}_j(\mathsf{f})$	rung functional; transport; moments sampled on the clock	(14.78), (14.79), (14.82)
$\mathfrak{t}_0$, I_m	zero of the clock, $\mathcal{N}_{\mathrm{RVM}}(\mathfrak{t}_0)=0$; Poisson integrals	(14.83)
a_n, $\dot\kappa_k$, $\dot{\mathsf{a}}_n$	free masses of the rung system; a direction of variation	(15.5), (15.9)
J, x_0, H_1, y, H, Z	objects of the proof of Theorem 15.6	(15.10)–(15.12)
$\langle\cdot,\cdot\rangle_\sigma$, $\mathsf{K}_{\boldsymbol\kappa}$, r_m, u_m, $\mathsf{Q}_{\boldsymbol\kappa}$	pairing of Dirichlet series; $\tilde\zeta_{\mathbb{Z},\boldsymbol\kappa}/\zeta$; gradient data of V_σ	(15.13), (15.14)
z, $\mathfrak{k}$, $\mathfrak{C}_{\boldsymbol\kappa}$, N	variable, half-type and canonical product of the models; $e^{i\mathfrak{k}\mathsf{z}}\mathfrak{C}_{\boldsymbol\kappa}$	(16.1)
$\mathscr{F}$, $\mathscr{F}^+$, $\mathfrak{l}$, $\mathfrak{w}_\mathfrak{l}$, $\mathfrak{b}_\mathfrak{l}$, $\mathfrak{c}_\mathfrak{l}$	frequencies of a discrete category; a frequency; masses, exponents, coefficients	(16.3)–(16.5)
$\mathfrak{p},\mathfrak{q},\mathfrak{v}$, U, X, $\mathfrak{D}$, $\mathfrak{S}$	data, unitary matrix, diagonal factor, determinant and sine function of the two-edge model	(16.6)–(16.8)
F, $\mathsf{F}_{\boldsymbol\kappa}$, F_λ, $\mathfrak{y}_k$, $\mathfrak{x}_k$	finite Euler product, its list functions and family; its ordinates and its clock	(16.10)–(16.13)
$\mathscr{L}$, $\mathfrak{P}$	ℓ^2 over the primes; $\sum_p p^{-s}$	(16.18), (16.19)
$\theta_{\mathrm{RVM}}(s)$, τ_0	bounded half-plane phase of the clock; fixed transition threshold $\tau_0=4\pi$ (not a clock ordinate)	(8.37), §8.1
$\mathfrak{m}_n$, $\mathfrak{u}_\lambda$, N_λ, $\mathfrak{e}^{(\lambda)}$, C_λ	cell baseline; first-cell term; moving count and count defect; Mellin/explicit-count constant	(13.20), (13.13), (13.2)
A_λ, $\mathsf{W}(\lambda)$, T_λ, $k_n(\lambda)$, V_σ	representing function; one-number defect; theta defect; quotient coefficients; vertical defect	(14.23), (14.6), (14.7)
Θ_λ, $\mathfrak{G}_\lambda$, $\mathcal{J}_T$, Δ_T, $K(T)$, L_T	auxiliary criteria and multiscale clock-Jacobian quantities	§14
E_n, Ξ_n, ε_r	cell-error and defect-kernel quantities; local regularization parameter where displayed	§14
$\boldsymbol\kappa$, $N_{\boldsymbol\kappa}$, τ_0^+	generic ordinate list, its count, and lower list threshold	(15.1)
$\mathcal{G}_\varepsilon$, $\mathsf{P}_{\varepsilon,M}$, $\mathsf{R}_{\varepsilon,M}$	additive interpolation and finite completed Dirichlet polynomials	(E.1), (E.6)
$\nu_5,\dots,\nu_{12}$	real constants, each local to the statement where it occurs	–
Route B for finite sections (Section 17)		
$\mathbb{K}(C,\mathbb{M})$, $\mathbb{M}_*$, a_n^ζ	box of C-admissible lists with masses $\le\mathbb{M}$; $\log 7/(2\sqrt{7})$; prescribed masses $\Lambda(n)/(2\sqrt{n})$	§17.3
$\mathfrak{R}(\boldsymbol\kappa,\mathsf{a})$, c_φ, $\mathscr{T}_m$, s_m	residual distribution of the rung system; cosine transform of a test; nested test sets; section residual	(17.5), (17.6)
$\mathcal{L}$, $\mathscr{K}$, $\mathscr{S}_{\boldsymbol\kappa}$, $\mathscr{A}_\zeta$, $\mathscr{E}_{\boldsymbol\kappa}$, $\mathfrak{H}_{\boldsymbol\kappa}$, $\mathfrak{g}$, $\mathsf{c}_\mathscr{K}$	Laplace transform; Archimedean, spectral, arithmetic and total residual distributions; $\sum_k \kappa_k^{-2}\cos(\kappa_k u)$; auxiliary function and constant	(17.7)–(17.12)

Symbol	Meaning	Defined
$\mathbb{F}_m$, $\mathbb{J}_m$, $\mathbb{Q}_m$, $\mathbb{L}_m$, $\mathbb{H}_m$, h_m, l_m, k_m^*, t_m, $\mathbb{T}_m$	residual map, Jacobian, right inverse and its norm, Lipschitz constant, start residual and contraction index of stage m; numbers of unknowns and tests; contraction map	(17.15)–(A$_m$)
q_m	graded contraction factors	Prop. 17.23
P_{X}, ζ_{X}, Λ_{X}, p_{X}, m_{X}, n_{X}, Θ_{X}, $\boldsymbol{\kappa}^{(\mathrm{X})}$	Gonek's truncated Euler product and symmetrised function with parameter X; its prime phase and two constants; the Gonek phase; the Gonek list	(17.20)–(17.23)
$\widehat{\mathsf{R}}$, $\mathbb{E}(T)$, υ, $\mathfrak{B}(C)$, $\mathbb{B}, \mathbb{B}_0, \dots, \mathbb{B}_3$, $\mathrm{c}_1, \dots, \mathrm{c}_6$, c_ℓ	Laplace residual of a section; its L^1 size on $\Re s = 2$ and growth ratio; constants of the a priori bound	(17.27)–(17.38)
$\mathcal{Y}$, $\mathcal{D}_{\mathsf{a}}$, $\mathbb{W}$, $\mathbb{S}$, $\mathbb{J}$, $\mathfrak{f}$, $\mathfrak{F}_T$, $\mathfrak{R}_T$, $\mathbb{G}_T$, n_T, $\mathbb{D}(T)$	Dirichlet factor and polynomial; $\log\lvert\zeta_{\boldsymbol{\kappa}}\rvert$; half-strip; compact set; log-factor; Backlund function; rectangle, path, zero count and bound of the counting argument	§17.4
$\mathscr{X}$, x_k, $\Psi_{\mathscr{X}}$, $\mathbb{P}_{\mathrm{y}}$, $\mathbb{Y}_{\mathrm{y}}$, $\mathfrak{G}_{\mathscr{X}}$, $\mathcal{F}_{\mathscr{X}}$, $\mathcal{F}_0$, $\mathfrak{z} = \mathrm{x} + i\mathrm{y}$	unfolded list and its points; discrepancy; Poisson and conjugate operators; partial-fraction series; generating function; $\cos \pi\mathfrak{z}$; unfolded complex variable	(17.41)–(17.46)
A, B, $\mathrm{d}_\star$, $\mathbb{U}$, $\mathcal{W}$, w, d_k, $\mathfrak{Z}_m$, $\mathfrak{I}_m$, $\mathfrak{n}_m$, D_m, $\mathbb{O}$, $\mathbb{V}$, $\mathbb{U}_{\mathrm{loc}}$	mean and oscillating parts of the discrepancy and their bounds; sector function; weight; displacements, index blocks, intervals, lengths, oscillations and the constants of the variable-length theorem	§17.5
$\hbar$, $\hbar_T$, $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}$, $\mathbb{P}_t^{\circ}$, c_{JN}, $\mathfrak{I}_T$, c_F	line height and its t-scale; the two kernels, the John–Nirenberg constant, the interval and Fujii's constant in the obstruction theorem	§17.6.3
w_m, T_m, u_m^*, $\mathbb{J}'$, $\mathbb{A}^{(3)}$, $\mathsf{l}(\mathrm{r})$	test width, height and window of stage m; second- and third-derivative matrices; Banach index of the computed certificates	§17.7

Collision check. Global symbols in the table have one meaning throughout the paper. Local proof variables that formerly collided with global notation have been renamed in version 18: in particular c_0 is the de la Vall'ee Poussin constant, $\varkappa_j$ and $\mathcal{P}_j$ are local van-der-Corput parameters, ω_0 and ω_{test} are test weights, d_ρ is a horizontal zero displacement, T_{ver} is the verified-zero height, $\mathcal{H}^{\pm}$ are transition half-planes, and $\Psi_m^{\pm}$ are local stationary phases; R_0 is reserved for the clock half-plane threshold and R_{end} for the endpoint clock-decomposition threshold. The symbol c is reserved for a Perron abscissa, ε for an exponent/margin, ε for the additive-family parameter, θ for the unit-cell step, and ϑ for the Riemann–Siegel function. The fixed number $\tau_0 = 4\pi$ is a transition threshold and is not a member of the sequence (τ_k). Dotted variables $\dot{\kappa}_k, \dot{\mathsf{a}}_n$ denote a direction of variation.

1. Introduction

1.1. **Background.** The theory of the Riemann zeta function joins four structures that are usually introduced in different languages. On the spectral side there are the nontrivial zeros and their Riemann–von Mangoldt counting law; on the analytic side the completed function ξ, its functional equation and its Hadamard product; on the prime side the prime-power measure and the explicit formula; and on the inverse side the reciprocal Dirichlet series $1/\zeta$ and the Mertens function

$$M(x) := \sum_{n \le x} \mu(n). \tag{1.1}$$

Riemann's memoir made the first three structures inseparable, and the work of von Mangoldt, Hadamard and Hamburger supplied, respectively, the rigorous zero count and explicit formula, the entire-product factorization, and a converse characterization of ζ from its Dirichlet-series and functional-equation data [47, 52, 32, 33, 51, 28]. This paper asks what remains of that architecture

when the zero staircase is replaced, at the outset, by the exact midpoint inverse of its smooth Riemann–von Mangoldt clock, and how the resulting object is carried back to ζ.

1.2. **The midpoint RVM system.** The point of departure is the smooth counting function (2.3),

$$\mathcal{N}_{\mathrm{RVM}}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.2}$$

and the real ordinates τ_k determined by the midpoint condition $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$. The half-integer convention is the midpoint convention for a staircase and leads to the exact Lambert inversion (2.5),

$$\tau_k = \frac{2\pi(k-11/8)}{W_0\big((k-11/8)/e\big)}. \tag{1.3}$$

Lambert-W approximations to individual zero ordinates and smooth inversions of the Riemann–von Mangoldt law occur in the literature, including explicit equations for the nth zero [29, 20, 27]. Here the midpoint inverse is treated as the exact spectrum of a deterministic system and is propagated, without the fluctuating term $S(T)$, through canonical products, explicit prime counting and Mertens inversion.

The first object is the canonical product and the synthetic zeta function

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(s) = \prod_{k\geq 1}\left(1 + \frac{s(s-1)}{\frac{1}{4}+\tau_k^2}\right), \qquad \zeta_{\mathrm{RVM}}(s) = \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)}, \qquad G(s) = s(s-1)\Gamma(s/2)\pi^{-s/2}. \tag{1.4}$$

Because $\tau_k \asymp k/\log k$, the factors form a normally convergent product in the symmetric variable $s(s-1)$; $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and invariant under $s \mapsto 1-s$, and ζ_{RVM} has a residue-one pole at 1, the trivial zeros, and the simple nontrivial zeros $\rho_k^{\mathrm{RVM}} = \frac{1}{2} \pm i\tau_k$ (Theorem 3.1). The real part $\frac{1}{2}$ is part of the definition.

Generalized prime systems. In Beurling's theory the generalized primes are arbitrary reals $1 < p_1 \leq p_2 \leq \cdots \to \infty$; their multiplicative semigroup defines generalized integers and a zeta function by a Dirichlet series, an Euler product or a Mellin–Stieltjes transform, separating prime distribution from the additive structure of the integers [15, 5, 24]. Beurling systems with prescribed analytic behaviour, critical zeros and large oscillation show that zeta-like zero data alone do not determine the arithmetic realization [25, 17]. The RVM system runs in the opposite direction: it starts from a prescribed zero clock, constructs an entire completed product first, obtains a continuous prime-power count from its explicit formula, and only afterwards asks for an atomic realization on the integers.

Continuous and discrete prime systems. The passage from a continuous prime system to a discrete one is classical in the theory of Beurling generalized primes [15, 25, 18, 26]: the atoms are placed at suitably chosen real points, and the discretization changes the zeta function by a factor holomorphic and zero-free in $\Re s > \frac{1}{2}$ (compare Theorem 13.12(iii)). Measures with a continuous part and non-integer multiplicities also appear in the theory of complex dimensions [38, Ch. 3]. Grosswald and Schnitzer [30] replaced each prime p_n by a real q_n with $p_n \leq q_n \leq p_{n+1}$ and proved that the modified Euler product is ζ times a factor holomorphic and zero-free in $\Re s > 0$, so that it has the same zeros as ζ there; Corollary 6.5 and Theorem 13.12(iv) are of the same type for ζ_{RVM}. Here the masses of $d\pi_\lambda$ are placed at the integers, with real exponents $\tilde{c}_n(\lambda)$; this yields an ordinary Dirichlet series for every λ, absolutely convergent in $\Re s > 1$ (Corollary 13.11), and makes Hamburger's theorem [33, 49] available at the endpoint. Hamburger's theorem and its extensions [16, 34] need absolute convergence in $\Re s > 1$, and Knopp [36] showed that a half-plane of absolute convergence further to the right does not suffice. We are not aware of an earlier use of this combination.

Prime coordinate. From the RVM zeros we define the prime-power count J_{RVM} by (5.2), with the branch (5.1) of Ei. Finite Möbius inversion in the exponent variable, which relates Riemann's

prime-power count to $\pi(x)$ [47, 51, 28], is used here as the *definition* of the synthetic prime count (5.7),

$$\pi_{\mathrm{RVM}}(x) = \sum_{1\le m\le \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}), \qquad J^*_{\mathrm{RVM}}(x) = \sum_{1\le m\le \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(x^{1/m}). \tag{1.5}$$

The function π_{RVM} is continuous away from the points 2^j with $\mu(j)\neq 0$, where it has the jump $\mu(j)J_{\mathrm{RVM}}(2)/j$; in particular it is not a staircase on the integers. It supports two notions of RVM prime: the continuous measure $d\pi_{\mathrm{RVM}}$, and, on every interval where π_{RVM} is continuous and strictly increasing, the real level crossings

$$p_j^{\mathrm{RVM}} := \pi_{\mathrm{RVM}}^{-1}(j), \qquad \pi_{\mathrm{RVM}}(p_j^{\mathrm{RVM}}) = j, \tag{1.6}$$

which need not be integers. The synthetic count has square-root accuracy without any hypothesis on ζ: $J_{\mathrm{RVM}}(x)-\mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x)-\mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 5.5). The proof reduces the zero sum to $\sum_k e^{iu\tau_k}/\tau_k$, which is $O(\log^2 u)$ by the second-derivative test on dyadic blocks of the clock (Lemma 5.4). Hence $\mathrm{li}(x)/\pi_{\mathrm{RVM}}(x)\to 1$, and with the prime number theorem $\pi(x)/\mathrm{li}(x)\to 1$ and $\pi(x)/\pi_{\mathrm{RVM}}(x)\to 1$ (Corollary 5.6, Figure 12). Sampling on unit cells gives the masses $c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(n)-\pi_{\mathrm{RVM}}(n-1)$ and the weighted Euler object $\zeta_{\mathbb{Z},\mathrm{RVM}}$ of (6.2). The masses are not 0 or 1 and need not be positive, but they are bounded: J_{RVM} is Lipschitz on $[2,\infty)$ (Theorem 6.1), so $c_n^{\mathrm{RVM}} = O(1)$ and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s>1$ (Corollaries 6.2 and 6.3). The unit-periodic first harmonic of the clock integrates to zero on every integer cell (Lemma 6.6); under the cell hypothesis one obtains $c_n^{\mathrm{RVM}}\sim 1/\log n$ (Proposition 6.8).

Mertens coordinate. The Mertens coordinate is the symmetric inverse Mellin object of $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$. A smooth/sawtooth Stieltjes decomposition of the clock product with Abel regularization of the sawtooth gives on every band to the right of $\frac12$

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\big(1+w(s)\big)\big(1+O(1/|t|)\big), \qquad w(s)=\exp\{-2\pi i\,\mathcal{N}_{\mathrm{RVM}}(-i(s-\tfrac12))\}, \tag{1.7}$$

and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+\chi)(1+O(1/|t|))$ on closed substrips of $\frac12<\Re s<1$ (Theorem 8.1). Hence $F_{\mathrm{RVM}}(\sigma+it)\to L = 1/\varpi_{\mathrm{RVM}}$ for every fixed $\sigma>\frac12$; neither ζ_{RVM} nor F_{RVM} is an absolutely convergent Dirichlet series in any half-plane (Theorem 8.11); and after subtracting the vertical limit the Perron inverse converges absolutely, with

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon}) \tag{1.8}$$

(Theorem 8.12). The RVM Mertens function therefore exists as a continuous spectral inverse although no coefficient sequence for F_{RVM} exists. Since the primitive of the midpoint sawtooth is bounded, the same decomposition without contours gives $C^{-1}\le|\zeta_{\mathrm{RVM}}|\le C$ on every half-plane $\Re s\ge\sigma_0>1$ and $\zeta_{\mathrm{RVM}}(s)\to\varpi_{\mathrm{RVM}}$ as $\Re s\to+\infty$ (Theorem 8.10).

Ternary discretization. An integer-valued representative of $M^{\mathrm{crit}}_{\mathrm{RVM}}$ is obtained by sampling the target $\mathcal{M}(n) := M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ and choosing signs on the squarefree support:

$$\mu_h(n) = \mu(n)^2\varsigma_n, \qquad \varsigma_n\in\{-1,+1\}, \qquad M_h(n) := \sum_{m\le n}\mu_h(m), \tag{1.9}$$

$$\varsigma_n = \begin{cases} +1, & M_h(n-1) < \mathcal{M}(n),\\ -1, & M_h(n-1)\ge \mathcal{M}(n),\end{cases} \qquad (\mu(n)^2=1). \tag{1.10}$$

This is a greedy sigma–delta feedback quantization; feedback quantization and noise shaping are standard themes of sigma–delta theory [21, 31, 6]. Here the target is a zero-generated Mertens inverse and the admissible times are the squarefree integers. The walk μ_h is a comparison object: it is not claimed to be multiplicative or equal to μ. The two discretizations are distinct: the cell masses c_n^{RVM} come from π_{RVM}, the signs ς_n track $M^{\mathrm{crit}}_{\mathrm{RVM}}$.

Comparison with ζ. At the level of transforms the defect is the exact quotient $\mathcal{Q}_{\mathrm{RVM}} = \zeta/\zeta_{\mathrm{RVM}}$ of (9.6), with $F_{\mathrm{RVM}} = \mathcal{Q}_{\mathrm{RVM}}\cdot(1/\zeta)$. Since $|\zeta_{\mathrm{RVM}}|$ is bounded above and below on $\frac12+\varepsilon\le\sigma\le c'$,

$|t| \geq t_0$ (Corollary 9.3), on each bounded band $\frac{1}{2}+\varepsilon \leq \sigma \leq c'$, every bound for ζ or $1/\zeta$ is equivalent to the corresponding bound for $\mathcal{Q}_{\mathrm{RVM}}$ by Corollary 9.3; for $\sigma > c'$ one uses Theorem 8.10. For M itself the bound $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ is equivalent to the Riemann hypothesis [51, §14.25], and the disproof of the Mertens conjecture shows that the ε cannot simply be dropped [43].

1.3. **The core of the method.** The transfer from ζ_{RVM} to ζ is a transformation of ζ_{RVM} depending on a parameter $\lambda \in [0,1]$. The ordinates move along

$$\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k, \qquad \zeta_\lambda(s) = \frac{1}{G(s)} \prod_{k\geq 1} \left(1 + \frac{s(s-1)}{\frac{1}{4}+\gamma_k(\lambda)^2}\right), \tag{1.11}$$

where γ_k are the ordinates of the nontrivial zeros of ζ; thus $\zeta_1 = \zeta_{\mathrm{RVM}}$ and $\zeta_0 = \zeta_{Z1}$ (Section 10).

(i) Every member ζ_λ has its nontrivial zeros exactly at $\frac{1}{2} \pm i\gamma_k(\lambda)$. The real part $\frac{1}{2}$ is part of the definition of the family, like the gamma factor, and never changes along it: it is impossible for ζ_{Z1}, or for any ζ_λ, to have a zero off the critical line other than the trivial zeros. For $0 < \lambda \leq 1$ all zeros are simple.

(ii) All differences between the moving zetas lie in the numbers $\gamma_k - \tau_k$, which are bounded (Lemma 2.2); the quotient of two members is an explicit product in γ_k and $\lambda(\tau_k - \gamma_k)$ (Remark 10.2).

(iii) The final step is the identity $\zeta_{Z1} = \zeta$. It is the single and complete test: while $\zeta_{Z1} = \zeta$ there is nothing outside $\Re s = \frac{1}{2}$ to speak of, since the identity itself places every nontrivial zero of ζ at $\frac{1}{2} \pm i\gamma_k$, and $\zeta_{Z1} \neq \zeta$ is the only possible obstruction.

The endpoint has the functional equation, the pole, the residue and the order of ζ (Proposition 11.1). Hamburger's theorem belongs to the family of converse theorems that recover Dirichlet series from analytic continuation, functional equation and growth [33, 45]; applied to ζ_{Z1} it gives (Theorem 11.2)

$$\zeta_{Z1} = \zeta \quad \Longleftrightarrow \quad \text{(D): } \zeta_{Z1} \text{ is an absolutely convergent Dirichlet series in } \Re s > 1. \tag{1.12}$$

With the integer-weighted zeta $\zeta_{\mathbb{Z},\mathrm{RVM}}$, which is a Dirichlet series, (D) is equivalent to an absolutely convergent Dirichlet expansion of the explicit multiplier $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}} = \Pi_{\mathrm{tr}} e^{-h_{\mathrm{RVM}}}$ (Theorem 11.3); since $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is also an absolutely convergent Dirichlet series in $\Re s > 1$, the equivalence holds on the whole half-plane. Section 12 gives two sharper forms of (D). First, the clock decomposition applied to the ordinates γ_k gives the value of the endpoint at infinity,

$$\varpi_{Z1} = \lim_{\sigma\to+\infty} \zeta_{Z1}(\sigma) = \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\leq K} \frac{\frac{1}{4}+\tau_k^2}{\frac{1}{4}+\gamma_k^2} \;\geq 1, \tag{1.13}$$

and $\zeta_{Z1} = \zeta$ holds if and only if $\varpi_{Z1} = 1$ (Theorems 12.1 and 12.2, Corollary 12.3): the final step is the single ordered-limit identity $\lim_{K\to\infty} \prod_{k\leq K} (\frac{1}{4}+\gamma_k^2)/(\frac{1}{4}+\tau_k^2) = \varpi_{\mathrm{RVM}}$ between the displacements and the transition constant. Second, the prime-power counts $J^{\mathrm{M}}_{\mathrm{RVM}}$ and J^{M}_{Z1} are the Mellin transforms of $\log \zeta_{\mathrm{RVM}}$ and $\log \zeta_{Z1}$ and satisfy a distributional explicit formula (Theorems 12.5 and 12.6); this proves the Mellin–Stieltjes identification of the integer weights (Theorem 6.4), and (D) holds if and only if J^{M}_{Z1} is constant between consecutive integers and its jumps satisfy the growth hypothesis of Theorem 12.7. Third, along the family $\frac{d}{d\lambda}\log\varpi_\lambda = -\Upsilon(\lambda)$, the generator at infinity, and (D) is the identity $\int_{0^+}^{1} \Upsilon(\lambda)\, d\lambda = \log L$ (Proposition 13.2). Section 13 atomizes every member: $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde{h}_\lambda}$ keeps the Euler product and the Dirichlet series while ζ_λ keeps the functional equation; the joining factor $\tilde{h}_\lambda$ is written cell by cell and is holomorphic in $\Re s > \frac{1}{2}$ (Theorems 13.6 and 13.12); the cell masses have mean square $X^{1+\varepsilon}$ on $(X, 2X]$ uniformly in λ, so every $\tilde{\zeta}_{\mathbb{Z},\lambda}$, the endpoint included, is an absolutely convergent Dirichlet series in $\Re s > 1$ (Theorem 13.10, Corollary 13.11); and (D) is equivalent to the vanishing of every cell defect at $\lambda = 0$ (Theorem 13.13). At the clock endpoint the joining factor $\tilde{h}_1$ is not constant, and the endpoint Mertens function has a Mellin form B_{Z1} with $B_{Z1} = M$ exactly when (D) holds (Propositions 13.15 and 13.16). The remaining

steps form the programme of Subsection 13.8. In RVM language, the final step asks whether the zero-generated analytic object ζ_{Z1} belongs to the discrete Dirichlet category; if it does, Hamburger uniqueness makes it ζ, and the prime and Mertens coordinates become those of the integers. Step 3 is Theorem 11.8:

$$\zeta_{Z1} = \zeta \quad \Longrightarrow \quad M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (\varepsilon > 0), \tag{1.14}$$

with a self-contained proof by the Borel–Carathéodory lemma, Hadamard's three-circle theorem and the truncated Perron formula. Theorem 11.9 rewrites the same endpoint consequence through the synthetic factorization. Its estimate for M is a corollary of Theorem 11.8; the synthetic Mertens bound is not an independent input to the classical implication. The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), is not proved in this paper; it is the open step of the method (Remark 11.12).

The relations among the principal objects are

$$\{\tau_k\} \longrightarrow \zeta_{\mathrm{RVM}} \longrightarrow J_{\mathrm{RVM}} \longleftrightarrow \pi_{\mathrm{RVM}} \longrightarrow (d\pi_{\mathrm{RVM}},\, p_j^{\mathrm{RVM}},\, c_n^{\mathrm{RVM}}), \tag{1.15}$$

$$F_{\mathrm{RVM}} = \zeta_{\mathrm{RVM}}^{-1} \longrightarrow M_{\mathrm{RVM}}^{\mathrm{crit}} \longrightarrow (\mathcal{M}, \mu_h, M_h), \tag{1.16}$$

$$\zeta_{\mathrm{RVM}} = \zeta_1 \xrightarrow{\lambda \downarrow 0} \zeta_0 = \zeta_{Z1} \overset{?}{=} \zeta \implies M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{1.17}$$

The first chain is the prime coordinate, the second the Mertens coordinate, the third the λ-family; they share the midpoint spectrum and solve different realization problems.

1.4. **Originality, scope and status.** The ingredients are standard or belong to neighbouring theories: the Riemann–von Mangoldt law, Lambert inversion, Hadamard products, explicit formulas, Möbius and Perron inversion, Beurling generalized primes, sigma–delta feedback, and Hamburger's converse theorem. The contribution is their assembly into one midpoint RVM space with four coordinated levels: (i) an exact midpoint zero clock and its canonical zeta-like product; (ii) a continuous prime-power and prime coordinate, with real prime levels and non-integral cell masses; (iii) a continuous synthetic Mertens inverse with its ternary feedback discretization; (iv) the λ-family from the midpoint spectrum to the ordinates of ζ, ending in the single identity $\zeta_{Z1} = \zeta$ and the Mertens bound that follows from it. To the author's knowledge the cited literature contains the individual mechanisms but not this chain. Standard material is cited at the point of use; statements without citation are original.

Statement	Status
Midpoint clock, Lambert inversion, bounded displacement	theorem
Canonical product $P_{\mathrm{RVM}}^{\mathrm{crit}}$ and ζ_{RVM}	theorem
$S_{\mathrm{RVM}}(N;u) \ll 1 + \log N$; centroid and M_H^{RVM} decay	theorem
Compact-uniform existence of J_{RVM}; finite inversion	theorem
Real prime levels p_j^{RVM}	defined on monotone branches of π_{RVM}
Square-root estimate for $J_{\mathrm{RVM}}, \pi_{\mathrm{RVM}}$	theorem
$\mathrm{li}\,/\pi_{\mathrm{RVM}} \to 1$; $\pi/\pi_{\mathrm{RVM}} \to 1$	theorem (with the prime number theorem)
J_{RVM} Lipschitz, $c_n^{\mathrm{RVM}} = O(1)$	theorem
$\zeta_{\mathbb{Z},\mathrm{RVM}}$ absolutely convergent Dirichlet series in $\Re s > 1$	theorem
Mellin–Stieltjes identification; $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$	theorem
$\lvert\zeta\rvert \asymp \lvert\mathcal{Q}_{\mathrm{RVM}}\rvert$ for $\sigma \geq \frac{1}{2} + \varepsilon$, $\lvert t\rvert \geq t_0$	theorem
Transition theorem $\varpi_{\mathrm{RVM}}(1+w)$, $\varpi_{\mathrm{RVM}}(1+\chi)$	theorem
$C^{-1} \leq \lvert\zeta_{\mathrm{RVM}}\rvert \leq C$ on $\Re s \geq \sigma_0 > 1$; $\zeta_{\mathrm{RVM}} \to \varpi_{\mathrm{RVM}}$ at $+\infty$	theorem
Unsmoothed $M_{\mathrm{RVM}}^{\mathrm{crit}} = L + O_\varepsilon(x^{1/2+\varepsilon})$	theorem
Dirichlet expansion of $\zeta_{\mathrm{RVM}}^{\pm 1}$	impossible (theorem)
Ternary walk μ_h, M_h	definition; comparison object
Exact Hadamard product of 2ξ	theorem

Statement	Status
λ-family, generator, multiplier, simplicity for $\lambda > 0$	theorem
Analytic class of ζ_{Z1}	theorem
$\zeta_{Z1} = \zeta \Leftrightarrow$ (D) $\Leftrightarrow$ RH	theorem
(D) $\Leftrightarrow$ Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$	theorem
$\varpi_{Z1} = \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K} \frac{1/4+\tau_k^2}{1/4+\gamma_k^2}$ (ordered product); (D) $\Leftrightarrow \varpi_{Z1} = 1$	theorem
Mellin prime-power counts; distributional explicit formula	theorem
(D) $\Leftrightarrow J_{Z1}^{\mathrm{M}}$ constant between integers with the jump-growth bound of Theorem 12.7	theorem
$\frac{d}{d\lambda} \log \varpi_\lambda = -\Upsilon$; (D) $\Leftrightarrow \int_{0^+}^{1} \Upsilon = \log L$	theorem
Atomized family; joining factor holomorphic in $\Re s > \frac{1}{2}$	theorem
Atomized family: Dirichlet series in $\Re s > 1$ for every λ	theorem
(D) $\Leftrightarrow$ vanishing cell defects at $\lambda = 0$	theorem
ζ_λ not a Dirichlet series in $\Re s > 1$ for $0 < \lambda \le 1$	theorem
$\lambda \mapsto \log \zeta_\lambda(\sigma)$ strictly convex for real $\sigma > 1$	theorem
Hamburger's theorem for the pair $(\zeta_\lambda, \tilde{\zeta}_{\mathbb{Z},\lambda})$ with symmetric joining factor	theorem
(D) $\Leftrightarrow \zeta_{Z1}$ has a positive representing measure; then $\varpi_{Z1} \le 1$	theorem
Density of $d\pi_{\mathrm{RVM}}$; cell masses and defects at $\lambda = 1$	theorem
Height kernel of the cell masses; Gaussian defect of the atomization	theorem
$\zeta_{Z1}(\sigma) \ge \zeta(\sigma)$ for real $\sigma > 1$; (D) $\Leftrightarrow$ equality at one σ	theorem
Hamburger's theorem for the pair in theta form and in vertical mean-square form	theorem
Symmetry of the joining factor fails at $\lambda = 1$; $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta$ for finitely many λ at most	theorem
Summation formula for every ζ_λ; $\varpi_\lambda - 1$ as one number $\mathsf{W}(\lambda)$	theorem
(D) $\Leftrightarrow \varpi_\lambda < 1$ on the open path $0 < \lambda \le 1$	theorem
Space $\mathscr{O}$ of ordinate lists: rigidity of the coincidence; (D) $\Leftrightarrow$ existence of a coincidence list	theorem
(D) $\Leftrightarrow \tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n; every point of E is a witness of (D)	theorem
Symmetric joining factors: order one is needed; the symmetry criterion has no slack at $\lambda = 0$	theorem
Ladder: $\tilde{h}_\lambda(\sigma) = O(\sigma N^{-\sigma}) \Leftrightarrow$ no mass of dJ_λ^{M} outside the integers below N	theorem
Bridge theorem in phase form; (D) $\Leftrightarrow$ (B1), (B2) for $\tilde{h}_0$; impossible for $\lambda > 0$	theorem
Phase of ζ_λ on the critical line; cell masses as sums of a one-ordinate function	theorem
Transport $\mathsf{S}_{\mathsf{f}}(\lambda)$: entire of exponential type; moment series on the clock; defect of the clock by Poisson summation	theorem
Multiscale clock Jacobian: lower frame bound on $U \asymp \log T$, no fixed-window bound, dense range, smooth tails	theorem; quantitative nonlinear continuation open

Statement	Status
(D) $\Leftrightarrow$ solvability of the rung system with free masses; the solution is nondegenerate	theorem
Gradient of V_σ on $\mathscr{O}$	theorem
Acceptance criterion (window and frequencies); two-edge model; finite Euler products with the final step proved	theorem
No object of finite rank in the window of ζ; the three abscissas of the diagonal operator over the primes	theorem
Phase defect along the path (rotation of the one-ordinate terms)	heuristic
Endpoint-only interpolation audit; clock-only injectivity problem; a priori rung displacement bound	Appendix E: theorem/proposition/open problem as labelled
Route B: compactness of near-solutions in a box; selection under uniqueness in the box	theorem (Section 17)
Laplace transform of the residual with prescribed masses; uniqueness; RH $\Leftrightarrow$ B$'(C)$	theorem; classical in substance
Certificate theorem for staged sections; graded extension; Gonek lists in a tube	theorem; the certificate hypotheses (A$_m$), (B) and (G1), (G2) are open and imply RH
Counting bound $\lvert N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}\rvert \le (\mathrm{c}_1 + \mathrm{c}_2 v)\log(T+8) + \mathrm{c}_3\mathbb{B} + \mathrm{c}_4$ for quasi-solutions	theorem
Block Kadec–Avdonin criterion; variable-length Avdonin theorem	theorem; novelty not checked
No (A_2) for the unfolded zeta ordinates on any horizontal line	theorem, unconditional
Banach certificates of exact square sections up to $T = 487$; basin $\sim T^{-2}$	computation (floating point)
$\zeta_{Z1} = \zeta \Rightarrow M(x) = O_\varepsilon(x^{1/2+\varepsilon})$	theorem (Step 3)
$\zeta_{Z1} = \zeta$	open

What is proved and what is not. Three groups of statements are proved in full. *(1) The synthetic system.* The clock and its Lambert inversion, the canonical product and ζ_{RVM}, the logarithmic bound for the exponential sums, the prime coordinate J_{RVM}, π_{RVM} with its square-root estimate and its density (Theorem 14.15), the transition theorem, the impossibility of a Dirichlet expansion of $\zeta_{\mathrm{RVM}}^{\pm 1}$ and the synthetic Mertens bound. *(2) The family and its endpoint.* Every ζ_λ has the functional equation, order one and all its nontrivial zeros on $\Re s = \frac{1}{2}$; $\zeta_{Z1} = \zeta$ is equivalent to (D), to $\varpi_{Z1} = 1$ and to the constancy of J^{M}_{Z1} between consecutive integers together with the jump-growth bound of Theorem 12.7; $\varpi_{Z1} \ge 1$ and $\zeta_{Z1}(\sigma) \ge \zeta(\sigma)$ for real $\sigma > 1$; $\log \zeta_\lambda(\sigma)$ is strictly convex in λ; and $\zeta_{Z1} = \zeta$ implies $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ (Step 3). *(3) The atomized family and the criteria.* Every $\tilde{\zeta}_{\mathbb{Z},\lambda}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, zero-free in $\Re s > \frac{1}{2}$, joined to ζ_λ by an explicit factor; no ζ_λ with $\lambda > 0$ is a Dirichlet series; Hamburger's theorem holds for the pair in three forms (Theorems 14.3, 14.4); the positivity criterion (Theorem 14.7); and the rigidity of the coincidence in the space of ordinate lists (Theorem 15.1). *(4) Criteria written where the joining factor is defined, and models.* The reduction of (D) to the cell masses (Corollary 14.26); the ladder and the half-line form of Hamburger's theorem for a pair (Theorems 14.33, 14.34); the bridge theorem in phase form (Theorem 14.36, Corollary 14.37); the transport from the clock and the exact defect of the clock (Theorem 14.42, Proposition 14.43); the existence form with free masses and the nondegeneracy of the solution (Theorems 15.5, 15.6); the acceptance criterion and the models in which the final step is a theorem (Theorems 16.1, 16.3, Proposition 16.2).

Not proved: the identity $\zeta_{Z1} = \zeta$. Each of the following statements is proved to be equivalent to it, and none of them is proved: (D); $\varpi_{Z1} \le 1$; $\zeta_{Z1}(\sigma) \le \zeta(\sigma)$ for one $\sigma > 1$; $\varpi_\lambda < 1$ for all

$\lambda \in (0,1]$; J_{Z1}^{M} non-decreasing; $\mathsf{W}(0) \geq 0$ (14.23); $\tilde{h}_0$ constant; the theta relation $T_0 \equiv 0$ (14.6); $\mathsf{V}_\sigma(0) = 0$ (14.7); the existence of a list in $\mathscr{O}$ whose continuous and atomized zeta functions coincide (Corollary 15.3); $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every $n \geq 2$ (Corollary 14.26); $\tilde{h}_0(\sigma) = O_N(\sigma N^{-\sigma})$ for every N (14.66); the conditions (B1) and (B2) for $\tilde{h}_0$ (Corollary 14.37); $\mathsf{S}_{\mathsf{f}}(0) = \mathsf{R}_{\mathsf{f}}(1)$ for every f whose support avoids the points $\log n$, together with the cell jump-growth condition (Proposition 14.41); the solvability of the rung system (15.5) in $\mathscr{O}$ (Theorem 15.5). The symmetry of the joining factor $e^{\tilde{h}_0}$ with finite order is a sufficient condition (Theorem 14.3), and it is also necessary (Remark 14.30).

1.5. **Organization.** Section 2 treats the clock, its Lambert inversion and the bounded displacement; Section 3 the canonical product; Section 4 the exponential sums; Sections 5 and 6 the prime coordinate and the integer weights; Sections 7 and 8 the Mertens transform and the transition theorem; Section 9 the exact Hadamard product and the comparison with ζ; Section 10 the λ-family; Section 11 the endpoint, Hamburger's theorem and Step 3; Section 12 the value of the endpoint at infinity and the Euler products of ζ_{RVM} and ζ_{Z1}; Section 13 the atomized family, Hamburger's hypotheses at the two endpoints and the programme for the final step; Section 14 the open path, convexity, Hamburger's theorem for the pair, the positivity criterion, the density of the clock, the Gaussian defect, the reduction to the cell masses, the half-line form, the bridge theorem and the transport from the clock; Section 15 the space of ordinate lists, the rigidity of the coincidence, the existence form of the final step, the rung system with free masses and the gradient of the vertical defect; Section 16 the acceptance criterion and the models in which the final step is a theorem; Section 17 route B for finite sections, the prescribed-mass uniqueness, the staged certificates, the Gonek lists, the a priori counting bound, the block Kadec–Avdonin criterion and its failure for the zeta list, with the certificate computations; Section 18 the numerical illustration; Section 19 the conclusions and open problems; Appendix E records the endpoint-only interpolation audit, the autonomous clock-rigidity question, the a priori rung bound, and the documented bounded finite-section diagnostics.

2. The midpoint Riemann–von Mangoldt clock

Let ρ run over the nontrivial zeros of ζ, counted with multiplicity. No nontrivial zero is real, so the positive ordinates can be listed with multiplicity as $0 < \gamma_1 \leq \gamma_2 \leq \cdots$. Put

$$N_\zeta(T) := \#\{\rho : \ 0 < \Im\rho \leq T\} = \#\{k : \gamma_k \leq T\}. \tag{2.1}$$

The Riemann–von Mangoldt formula [51, Thms. 9.3–9.4], [52] states that for $T \geq 2$, $T \neq \gamma_k$,

$$N_\zeta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + S(T) + O\Big(\frac{1}{T}\Big), \qquad S(T) := \frac{1}{\pi}\arg\zeta\big(\tfrac{1}{2}+iT\big) = O(\log T), \tag{2.2}$$

where $\arg\zeta(\frac{1}{2}+iT)$ is obtained by continuous variation of the argument along the segments from 2 to $2+iT$ and from $2+iT$ to $\frac{1}{2}+iT$, starting from $\arg\zeta(2) = 0$. The $O(1/T)$ term comes from Stirling's formula for Γ and is explicit; $S(T)$ is the fluctuating part, with a jump $+m$ at an ordinate of multiplicity m. The *midpoint RVM clock* is the smooth part of (2.2):

$$\mathcal{N}_{\mathrm{RVM}}(T) := \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}. \tag{2.3}$$

It is strictly increasing on $[2\pi, \infty)$. The midpoint ordinates τ_k are the unique solutions in $[2\pi, \infty)$ of

$$\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}, \qquad k \geq 1, \tag{2.4}$$

that is, the abscissae on the increasing branch where the smooth count crosses the midpoints $k - \frac{1}{2}$ of the steps of a unit staircase. For $k = 1$ the same level has a second positive solution in $(0, 2\pi)$ on the W_{-1} branch; it is not part of the midpoint clock used here. Thus $\mathcal{N}_{\mathrm{RVM}}$ replaces the staircase N_ζ by its smooth part, and τ_k is the point where the kth step would be crossed half-way (Figure 3).

The Lambert W-function is the multivalued inverse of $v \mapsto ve^v$: $W(z)e^{W(z)} = z$. On $[-1/e, \infty)$ the equation $ve^v = z$ has exactly one solution $v \geq -1$; this defines the principal branch $W_0 : [-1/e, \infty) \to [-1, \infty)$, which is real, increasing, with $W_0(0) = 0$ and $W_0(z) \sim \log z - \log\log z$ as $z \to \infty$ [27, §4.13], [20].

Proposition 2.1 (Lambert inversion)**.** *For every $k \geq 1$,*

$$\tau_k = \frac{2\pi(k - 11/8)}{W_0\big((k - 11/8)/e\big)}, \tag{2.5}$$

The quotient is positive also for $k = 1$.

Proof. Put $y = \tau/(2\pi)$. Equation (2.4) reads

$$y(\log y - 1) = k - \frac{11}{8}. \tag{2.6}$$

With $v = \log y - 1$ one has $y = e^{v+1}$ and $ve^v = (k - 11/8)/e$, so $v = W_0((k - 11/8)/e)$ and $y = (k - 11/8)/W_0((k - 11/8)/e)$. For $k = 1$, $(k - 11/8)/e = -3/(8e) \in (-1/e, 0)$, where W_0 is real and negative; numerator and denominator are both negative. □

The clock derivative and the resulting growth are

$$\mathcal{N}'_{\mathrm{RVM}}(T) = \frac{1}{2\pi} \log \frac{T}{2\pi}, \tag{2.7}$$

$$\tau_k \asymp \frac{k}{\log k} \qquad (k \geq 2). \tag{2.8}$$

In terms of the clock, (2.2) reads

$$N_\zeta(T) = \mathcal{N}_{\mathrm{RVM}}(T) + S(T) + O(T^{-1}), \qquad S(T) = O(\log T) \qquad (T \geq 2). \tag{2.9}$$

Let

$$N^*_\zeta(T) := \tfrac{1}{2}\big(N_\zeta(T + 0) + N_\zeta(T - 0)\big) \tag{2.10}$$

be the symmetric count, and define the displacements

$$\eta_k := \tau_k - \gamma_k. \tag{2.11}$$

Lemma 2.2 (Bounded displacement)**.** *There is a constant C_η such that*

$$|\eta_k| \leq C_\eta \qquad (k \geq 1). \tag{2.12}$$

Proof. Let $\gamma = \gamma_k$ and let $m = m(\gamma)$ be the number of indices j with $\gamma_j = \gamma$; these indices form a block $j_0 + 1, \ldots, j_0 + m$ containing k. Then $N_\zeta(\gamma - 0) = j_0$ and $N_\zeta(\gamma + 0) = j_0 + m$, so $N^*_\zeta(\gamma) = j_0 + m/2$ and

$$\big|N^*_\zeta(\gamma_k) - (k - \tfrac{1}{2})\big| \leq \frac{m(\gamma_k)}{2}. \tag{2.13}$$

By (2.9), $m(\gamma) \leq N_\zeta(\gamma + 1) - N_\zeta(\gamma - 1) = O(\log\gamma)$, and $N^*_\zeta(\gamma) = \mathcal{N}_{\mathrm{RVM}}(\gamma) + O(\log\gamma)$. With (2.4),

$$\big|\mathcal{N}_{\mathrm{RVM}}(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)\big| = O(\log\gamma_k). \tag{2.14}$$

Both γ_k and τ_k tend to infinity and (2.14) gives $\log\gamma_k \asymp \log\tau_k \asymp \log k$. By the mean-value theorem and (2.7), $\mathcal{N}'_{\mathrm{RVM}}$ is $\gg \log k$ between γ_k and τ_k for large k, so $|\gamma_k - \tau_k| = O(1)$. Finitely many initial indices are absorbed into C_η. □

3. The pure critical RVM zeta function

Put

$$q(s) := s(s-1), \tag{3.1}$$

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(s) := \prod_{k\geq 1}\left(1+\frac{q(s)}{\frac{1}{4}+\tau_k^2}\right), \tag{3.2}$$

$$G(s) := s(s-1)\Gamma(s/2)\pi^{-s/2}, \tag{3.3}$$

and

$$\xi_{\mathrm{RVM}}(s) := \frac{1}{2}P^{\mathrm{crit}}_{\mathrm{RVM}}(s), \qquad \zeta_{\mathrm{RVM}}(s) := \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)}. \tag{3.4}$$

Theorem 3.1 (Canonical characterization of ζ_{RVM})**.** *The product* (3.2) *converges absolutely and locally uniformly in $s\in\mathbb{C}$. It defines an entire function with*

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(1-s) = P^{\mathrm{crit}}_{\mathrm{RVM}}(s), \qquad P^{\mathrm{crit}}_{\mathrm{RVM}}(\bar{s}) = \overline{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}, \qquad P^{\mathrm{crit}}_{\mathrm{RVM}}(0) = P^{\mathrm{crit}}_{\mathrm{RVM}}(1) = 1. \tag{3.5}$$

Its zeros are exactly the simple zeros

$$\rho_k^{\mathrm{RVM}} = \frac{1}{2}+i\tau_k, \qquad \overline{\rho_k^{\mathrm{RVM}}} = \frac{1}{2}-i\tau_k, \qquad k\geq 1. \tag{3.6}$$

Consequently ζ_{RVM} has a simple pole at $s=1$ with residue 1, the trivial zeros $s=-2,-4,-6,\dots$, and nontrivial zeros exactly at (3.6)*; moreover $\xi_{\mathrm{RVM}}(s)=\xi_{\mathrm{RVM}}(1-s)$.*

Proof. By (2.8), $\sum_k \tau_k^{-2}<\infty$. On a compact set K,

$$\sup_{s\in K}\left|\frac{q(s)}{\frac{1}{4}+\tau_k^2}\right| \ll_K \tau_k^{-2}, \tag{3.7}$$

so the Weierstrass criterion gives normal convergence. Each factor depends on s only through $q(s)$ and equals 1 at $s=0,1$, which gives (3.5); real coefficients give the conjugation symmetry. The kth factor vanishes exactly when

$$\left(s-\frac{1}{2}\right)^2+\tau_k^2=0. \tag{3.8}$$

The clock is strictly increasing on $[2\pi,\infty)$ and $\tau_1>2\pi$, so the τ_k are distinct; the derivative $2s-1=\pm 2i\tau_k\neq 0$ of the vanishing factor and the non-vanishing of the other factors make the zeros simple. At $s=1$, $P^{\mathrm{crit}}_{\mathrm{RVM}}(1)=1$ and $G(s)=(s-1)(1+O(s-1))$, so the residue is 1. At $s=-2n$ ($n\geq 1$) the pole of $\Gamma(s/2)$ gives a zero of $1/G$, while $P^{\mathrm{crit}}_{\mathrm{RVM}}(-2n)\neq 0$ because $q(-2n)>0$. At $s=0$, $s\Gamma(s/2)\to 2$, so $G(0)\neq 0$ and $s=0$ is not a zero. □

4. RVM exponential sums, centroids, and the height transform

For $u\in\mathbb{R}$ and $N\geq 1$ define

$$S_{\mathrm{RVM}}(N;u) := \sum_{k\leq N} e^{iu\tau_k}, \qquad C^{\mathrm{pts}}_{\mathrm{RVM}}(N;u) := \frac{S_{\mathrm{RVM}}(N;u)}{N}, \tag{4.1}$$

and

$$C^{\mathrm{path}}_{\mathrm{RVM}}(N;u) := \frac{1}{N}\sum_{n\leq N} S_{\mathrm{RVM}}(n;u) = \sum_{k\leq N}\frac{N-k+1}{N}e^{iu\tau_k}. \tag{4.2}$$

Theorem 4.1 (Uniform logarithmic exponential-sum bound)**.** *Let $0<u_1\leq u_2<\infty$. Then*

$$\sup_{u\in[u_1,u_2]}|S_{\mathrm{RVM}}(N;u)| \ll_{u_1,u_2} 1+\log N. \tag{4.3}$$

Proof. Let $\mathcal{T} : [-\frac{1}{8}, \infty) \to [2\pi, \infty)$ be the inverse of the clock, so $\tau_k = \mathcal{T}(k - \frac{1}{2})$, and put $\ell_k = \log(\tau_k/2\pi) > 0$. Differentiating $\mathcal{N}_{\mathrm{RVM}}(\mathcal{T}(y)) = y$,

$$\mathcal{T}'(y) = \frac{2\pi}{\log(\mathcal{T}(y)/2\pi)}, \qquad \mathcal{T}''(y) = -\frac{4\pi^2}{\mathcal{T}(y)\log^3(\mathcal{T}(y)/2\pi)}. \tag{4.4}$$

Since $\mathcal{T}'$ decreases and $|\mathcal{T}''|$ decreases,

$$\frac{2\pi}{\ell_{k+1}} \le \tau_{k+1} - \tau_k \le \frac{2\pi}{\ell_k}, \tag{4.5}$$

$$|\tau_{k+2} - 2\tau_{k+1} + \tau_k| \le \frac{4\pi^2}{\tau_k \ell_k^3}. \tag{4.6}$$

From (2.4), $\tau_k \ell_k = 2\pi(k - \frac{11}{8}) + \tau_k$, hence

$$\tau_k \ell_k \asymp k \qquad (k \ge 2). \tag{4.7}$$

Fix $u \in [u_1, u_2]$ and put $\Delta_k = u(\tau_{k+1} - \tau_k)$, $E_k = e^{iu\tau_k}$. Choose $k_0 = k_0(u_2)$ with $\ell_{k_0} \ge 2u_2$; then $0 < \Delta_k \le \pi$ for $k \ge k_0$. With $\mathfrak{b}_k^{\mathrm{sbp}} := (e^{i\Delta_k} - 1)^{-1}$ one has $E_{k+1} = e^{i\Delta_k} E_k$, hence

$$E_k = \mathfrak{b}_k^{\mathrm{sbp}}(E_{k+1} - E_k). \tag{4.8}$$

Summing (4.8) over $k_0 \le k \le N$ and shifting the index in the first part gives the exact identity

$$\sum_{k=k_0}^{N} E_k = \mathfrak{b}_N^{\mathrm{sbp}} E_{N+1} - \mathfrak{b}_{k_0}^{\mathrm{sbp}} E_{k_0} + \sum_{k=k_0+1}^{N} (\mathfrak{b}_{k-1}^{\mathrm{sbp}} - \mathfrak{b}_k^{\mathrm{sbp}}) E_k. \tag{4.9}$$

For $0 < x \le \pi$, $|e^{ix} - 1| = 2\sin(x/2) \ge 2x/\pi$. Using (4.5) and (4.6),

$$|\mathfrak{b}_k^{\mathrm{sbp}}| \le \frac{\pi}{2\Delta_k} \le \frac{\ell_{k+1}}{4u_1}, \quad |\mathfrak{b}_{k-1}^{\mathrm{sbp}} - \mathfrak{b}_k^{\mathrm{sbp}}| \le |\Delta_k - \Delta_{k-1}|\,|\mathfrak{b}_{k-1}^{\mathrm{sbp}}||\mathfrak{b}_k^{\mathrm{sbp}}| \le \frac{\pi^2 u_2}{4u_1^2} \frac{\ell_{k+1}^2}{\tau_{k-1}\ell_{k-1}^3} \ll_{u_1,u_2} \frac{1}{\tau_k \ell_k} \ll \frac{1}{k}, \tag{4.10}$$

the last step by (4.7). Inserting (4.10) into (4.9) gives $|\sum_{k_0}^{N} E_k| \ll \ell_{N+1} + \ell_{k_0+1} + \sum_{k \le N} k^{-1} \ll 1 + \log N$. The indices $k < k_0$ contribute at most k_0. All constants depend only on u_1, u_2. □

Corollary 4.2 (Point centroid)**.** *Uniformly for $u \in [u_1, u_2] \subset (0, \infty)$,*

$$C_{\mathrm{RVM}}^{\mathrm{pts}}(N; u) = O_{u_1,u_2}\Big(\frac{1 + \log N}{N}\Big) \longrightarrow 0. \tag{4.11}$$

Theorem 4.1 alone does not imply convergence of the path centroid (4.2)*.*

Proof. The first assertion is (4.3) divided by N. For the second, the identity $C_{\mathrm{RVM}}^{\mathrm{path}} = N^{-1}\sum_{n \le N} S_{\mathrm{RVM}}(n; u)$ with $S_{\mathrm{RVM}}(n; u) = O(\log n)$ gives only $O(\log N)$. □

Define the midpoint count

$$n_{\mathrm{RVM}}(H) := \#\{k : \tau_k \le H\}. \tag{4.12}$$

For $x > 0$ and $n_{\mathrm{RVM}}(H) \ge 1$ define

$$M_H^{\mathrm{RVM}}(x) := \frac{1}{n_{\mathrm{RVM}}(H)} \sum_{\tau_k \le H} x^{i\tau_k}, \tag{4.13}$$

and for the ordinates of ζ

$$M_H(x) := \frac{1}{N_\zeta(H)} \sum_{\gamma_k \le H} x^{i\gamma_k}. \tag{4.14}$$

Corollary 4.3 (Synthetic height-transform decay)**.** *For every fixed $x > 1$, with $N = n_{\mathrm{RVM}}(H)$,*

$$M_H^{\mathrm{RVM}}(x) = O_x\Big(\frac{1 + \log N}{N}\Big) \longrightarrow 0 \qquad (H \to \infty). \tag{4.15}$$

Proof. Take $u = \log x > 0$ in Theorem 4.1; (4.13) is the point centroid at $N = n_{\mathrm{RVM}}(H)$. □

Remark 4.4. Neither (4.13) nor (4.14) is the Mertens function; both are normalized phase averages.

5. The RVM explicit formula, J_{RVM}, and π_{RVM}

For $w = a + ib$ with $b > 0$ define

$$\mathrm{Ei}(w) := \int_{-\infty}^{a} \frac{e^{v+ib}}{v+ib}\, dv, \qquad \mathrm{Ei}(\bar{w}) := \overline{\mathrm{Ei}(w)}. \tag{5.1}$$

The path is the horizontal line $\Im = b$, which avoids 0; this is the branch of $\mathrm{Ei}(\rho \log x)$ used in Riemann's explicit formula [28]. For $x > 1$ put

$$J_{\mathrm{RVM}}(x) := \mathrm{li}(x) - 2\Re \sum_{k\geq 1} \mathrm{Ei}\Big(\big(\frac{1}{2} + i\tau_k\big)\log x\Big) - \log 2 + \int_x^{\infty} \frac{dy}{y(y^2-1)\log y}, \tag{5.2}$$

the zero sum being taken in the order of k.

Lemma 5.1 (Asymptotics of Ei off the real axis)**.** *For $w = a + ib$, $b > 0$,*

$$\mathrm{Ei}(w) = \frac{e^w}{w} + \varrho(w), \qquad |\varrho(w)| \leq \frac{e^a}{b^2}. \tag{5.3}$$

Proof. Integrate (5.1) by parts: $\int_{-\infty}^{a} e^{v+ib}(v+ib)^{-1}dv = e^w/w + \int_{-\infty}^{a} e^{v+ib}(v+ib)^{-2}dv$, the boundary term at $-\infty$ being 0. The last integral is bounded by $\int_{-\infty}^{a} e^v b^{-2} dv = e^a/b^2$. □

Theorem 5.2 (Compact-uniform existence of the RVM zero sum)**.** *The series $\sum_k \Re\,\mathrm{Ei}((\frac{1}{2} + i\tau_k)u)$ converges uniformly for u in every compact interval $[u_1, u_2] \subset (0, \infty)$. Hence J_{RVM} is a continuous function on $(1, \infty)$.*

Proof. Apply Lemma 5.1 with $w = (\frac{1}{2} + i\tau_k)u$, $a = u/2$, $b = \tau_k u$. Since $\Re(e^{iu\tau_k}/(\frac{1}{2} + i\tau_k)) = (\frac{1}{2}\cos u\tau_k + \tau_k \sin u\tau_k)/(\frac{1}{4} + \tau_k^2)$,

$$\Re\,\mathrm{Ei}\big((\tfrac{1}{2} + i\tau_k)u\big) = \frac{e^{u/2}}{u}\,\frac{\sin(u\tau_k)}{\tau_k} + O_{u_1,u_2}(\tau_k^{-2}), \tag{5.4}$$

uniformly for $u \in [u_1, u_2]$. The error series converges absolutely and uniformly. For the main series, Abel summation with $S_{\mathrm{RVM}}(K; u) = O_{u_1,u_2}(1 + \log K)$ (Theorem 4.1) and

$$\frac{1}{\tau_K} - \frac{1}{\tau_{K+1}} \leq \frac{\tau_{K+1} - \tau_K}{\tau_K^2} \leq \frac{2\pi}{\ell_K \tau_K^2} \ll \frac{\log K}{K^2} \tag{5.5}$$

gives a uniformly convergent majorant $\sum_K (\log K)^2 K^{-2}$, while the boundary term $S_{\mathrm{RVM}}(K; u)/\tau_K \to 0$ uniformly. Hence $\sum_k e^{iu\tau_k}/\tau_k$ converges uniformly, and so does its imaginary part. □

Define the support-truncated version

$$J^*_{\mathrm{RVM}}(x) := \begin{cases} J_{\mathrm{RVM}}(x), & x \geq 2, \\ 0, & 0 < x < 2, \end{cases} \tag{5.6}$$

set $\pi_{\mathrm{RVM}}(x) = 0$ for $0 < x < 2$, and for $x \geq 2$

$$\pi_{\mathrm{RVM}}(x) := \sum_{1 \leq m \leq \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}). \tag{5.7}$$

Proposition 5.3 (Exact finite prime-power inversion)**.** *For every $x \geq 2$,*

$$J^*_{\mathrm{RVM}}(x) = \sum_{1 \leq m \leq \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(x^{1/m}). \tag{5.8}$$

Proof. All sums are finite because $x^{1/m} < 2$ for $m > \log_2 x$. Substituting (5.7) into the right side of (5.8), the coefficient of $J^*_{\mathrm{RVM}}(x^{1/r})$ is

$$\frac{1}{r}\sum_{d\mid r}\mu(d), \tag{5.9}$$

which is 1 for $r = 1$ and 0 otherwise. □

5.1. **The square-root estimate.** For $u > 0$ put

$$\mathcal{A}_{\mathrm{RVM}}(u) := \sum_{k\ge 1}\frac{e^{iu\tau_k}}{\tau_k}, \tag{5.10}$$

summed in the order of k, and write $\ell(t) = \log(t/2\pi)$, so that $\mathcal{N}'_{\mathrm{RVM}}(t) = \ell(t)/2\pi$ and $\ell(\tau_k) = \ell_k$.

Lemma 5.4 (The weighted clock sum)**.** *For every $u \ge \log 2$ the series* (5.10) *converges, and*

$$|\mathcal{A}_{\mathrm{RVM}}(u)| \le \frac{\log^2(2+2u)}{4\pi} + \frac{\log(2+2u)}{2\pi} + C_{\mathcal{A}}, \tag{5.11}$$

with an absolute constant $C_{\mathcal{A}}$.

Proof. Put $Y_j = 2^j \cdot 4\pi(1+u)$ for $j \ge 0$.

Head $\tau_k \le Y_0$. Let $n_{\mathrm{RVM}}(t) = \#\{k : \tau_k \le t\}$. By (2.4), $\tau_k \le t$ if and only if $k \le \mathcal{N}_{\mathrm{RVM}}(t) + \frac12$, so $n_{\mathrm{RVM}}(t) \le \mathcal{N}_{\mathrm{RVM}}(t) + \frac12 = \frac{t}{2\pi}(\ell(t) - 1) + \frac{11}{8} \le \frac{t\,\ell(t)}{2\pi}$ for $t \ge \tau_1 > \frac{11\pi}{4}$. Stieltjes integration gives

$$\sum_{\tau_k\le Y_0}\frac{1}{\tau_k} = \frac{n_{\mathrm{RVM}}(Y_0)}{Y_0} + \int_{\tau_1}^{Y_0}\frac{n_{\mathrm{RVM}}(t)}{t^2}\,dt \le \frac{\ell(Y_0)}{2\pi} + \int_{\tau_1}^{Y_0}\frac{\ell(t)}{2\pi t}\,dt \le \frac{\ell(Y_0)}{2\pi} + \frac{\ell(Y_0)^2}{4\pi}, \tag{5.12}$$

and $\ell(Y_0) = \log(2+2u)$.

Blocks. Let $\mathcal{B}_j = \{k : Y_j < \tau_k \le Y_{j+1}\}$, let $\mathcal{T}$ be the inverse clock of (4.4), and put $\phi(y) = u\,\mathcal{T}(y - \frac12)/2\pi$, so that $e^{2\pi i\phi(k)} = e^{iu\tau_k}$. By (4.4),

$$\phi''(y) = -\frac{2\pi u}{\mathcal{T}\,\ell(\mathcal{T})^3}, \qquad \mathcal{T} = \mathcal{T}(y - \tfrac12). \tag{5.13}$$

On $\mathcal{B}_j$ put $\varkappa_j = 2\pi u/(Y_{j+1}\ell(Y_{j+1})^3)$. Since $Y_{j+1} = 2Y_j$ and $\ell(Y_{j+1}) \le 2\ell(Y_j)$ (because $\ell(Y_j) \ge \ell(Y_0) \ge \log 2$), we have $\varkappa_j \le -\phi'' \le 16\varkappa_j$ there. The number of indices in $\mathcal{B}_j$ is at most $\mathcal{N}_{\mathrm{RVM}}(Y_{j+1}) - \mathcal{N}_{\mathrm{RVM}}(Y_j) + 1 \le Y_j\ell(Y_{j+1})/2\pi + 1$. The second-derivative test for exponential sums [51, Thm. 5.9] gives, for every run $a < k \le b$ of consecutive indices of $\mathcal{B}_j$,

$$\Big|\sum_{a<k\le b} e^{iu\tau_k}\Big| \le A_{\mathrm{vdC}}\Big(16\Big(\frac{Y_j\ell(Y_{j+1})}{2\pi} + 1\Big)\varkappa_j^{1/2} + \varkappa_j^{-1/2}\Big) =: \mathcal{P}_j, \tag{5.14}$$

with an absolute constant A_{vdC}; a run of length less than 1 has at most one term, and $1 \le \varkappa_j^{-1/2}$ because $\varkappa_j \le u/(4(1+u)\log^3 2) < 1$. The weights $1/\tau_k$ are positive and decreasing, and $1/\tau_k < 1/Y_j$ on $\mathcal{B}_j$, so Abel's inequality bounds the contribution of any run in $\mathcal{B}_j$ by $\mathcal{P}_j/Y_j$:

$$\frac{\mathcal{P}_j}{Y_j} \le A_{\mathrm{vdC}}\left(\frac{16}{2\pi}\Big(\frac{\pi u}{Y_j\ell(Y_{j+1})}\Big)^{1/2} + \frac{16}{Y_j} + \Big(\frac{\ell(Y_{j+1})^3}{\pi u Y_j}\Big)^{1/2}\right). \tag{5.15}$$

With $Y_j = 2^j \cdot 4\pi(1+u)$ and $\ell(Y_{j+1}) = \log(2^{j+2}(1+u)) \le j + 2 + \log(1+u)$, each of the three terms is at most an absolute constant times $(j+2)^{3/2}2^{-j/2}$, uniformly in $u \ge \log 2$, because $\sup_{u\ge\log 2}(1+\log(1+u))^3/(u(1+u)) < \infty$. Hence $\sum_{j\ge 0}\mathcal{P}_j/Y_j \le C_{\mathcal{A}}$ with $C_{\mathcal{A}}$ absolute.

Convergence and bound. If $\tau_K \in \mathcal{B}_J$, the terms with $k > K$ form a run of $\mathcal{B}_J$ followed by the full blocks $\mathcal{B}_j$, $j > J$, so $|\sum_{k>K'}^{K''} e^{iu\tau_k}/\tau_k| \le \sum_{j\ge J}\mathcal{P}_j/Y_j \to 0$ for $K'' > K' \ge K$, $K \to \infty$. The series converges, and (5.12) together with $\sum_j \mathcal{P}_j/Y_j \le C_{\mathcal{A}}$ gives (5.11). □

Theorem 5.5 (Square-root estimate for J_{RVM} and π_{RVM})**.** *For $x \geq 8$,*

$$J_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{5.16}$$

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{5.17}$$

with absolute implied constants. No hypothesis on ζ is used.

Proof. Let $u = \log x \geq \log 8$ and $\rho_k^{\mathrm{RVM}} = \frac{1}{2} + i\tau_k$. Lemma 5.1 with $w = \rho_k^{\mathrm{RVM}} u$, $a = u/2$, $b = \tau_k u$ gives

$$\mathrm{Ei}(\rho_k^{\mathrm{RVM}} u) = \frac{e^{\rho_k^{\mathrm{RVM}} u}}{\rho_k^{\mathrm{RVM}} u} + \varrho_k, \qquad |\varrho_k| \leq \frac{e^{u/2}}{\tau_k^2 u^2}. \tag{5.18}$$

Moreover $\frac{1}{\rho_k^{\mathrm{RVM}}} = -\frac{i}{\tau_k} + r_k$ with $r_k = \frac{i/2}{\tau_k \rho_k^{\mathrm{RVM}}}$, so $|r_k| \leq \frac{1}{2\tau_k^2}$. Since $\sum_k \tau_k^{-2} < \infty$, Lemma 5.4 shows that the series in (5.2) converges in the order of k and

$$\sum_{k\geq 1} \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u) = \frac{e^{u/2}}{u}\Big(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + \sum_k r_k e^{iu\tau_k}\Big) + \sum_k \varrho_k = \frac{e^{u/2}}{u}(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + O(1)). \tag{5.19}$$

For $u \geq 2$ one has $\log(2+2u) \leq \log(2u) + \log 2 \leq 3\log u$, so (5.11) gives $\mathcal{A}_{\mathrm{RVM}}(u) = O(\log^2 u)$. The remaining terms $-\log 2 + \int_x^\infty dy/(y(y^2-1)\log y)$ of (5.2) are bounded for $x \geq 2$. With $e^{u/2} = \sqrt{x}$ and $\log u = \log\log x$ this proves (5.16).

For (5.17), (5.7) gives

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = (J_{\mathrm{RVM}}(x) - \mathrm{li}(x)) + \sum_{2\leq m\leq \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}). \tag{5.20}$$

By (5.16), $\mathrm{li}(y) \ll y/\log y$, and the continuity of J_{RVM} on $[2,8]$ (Theorem 5.2), $|J^*_{\mathrm{RVM}}(y)| \leq C\,y/\log y$ for all $y \geq 2$. The term $m = 2$ is therefore $O(\sqrt{x}/\log x)$, and the terms $3 \leq m \leq \log_2 x$ add up to at most $C\,x^{1/3}\log_2 x/\log x = O(x^{1/3})$. Both are absorbed in (5.17). □

Corollary 5.6 (The three limits)**.** *As $x \to \infty$,*

$$\frac{\mathrm{li}(x)}{\pi_{\mathrm{RVM}}(x)} \to 1, \qquad \frac{\pi(x)}{\mathrm{li}(x)} \to 1, \qquad \frac{\pi(x)}{\pi_{\mathrm{RVM}}(x)} \to 1. \tag{5.21}$$

More precisely, for some absolute $c_0 > 0$,

$$\pi(x) - \pi_{\mathrm{RVM}}(x) = O\Big(x\, e^{-c_0\sqrt{\log x}}\Big). \tag{5.22}$$

Proof. Since $\mathrm{li}(x) \sim x/\log x$, (5.17) gives $\pi_{\mathrm{RVM}}(x) = \mathrm{li}(x)(1 + O(x^{-1/2}(\log\log x)^2))$; in particular $\pi_{\mathrm{RVM}}(x) > 0$ for large x, and the first limit follows. The second is the prime number theorem [32, 23], in the form $\pi(x) = \mathrm{li}(x) + O(x\,e^{-c_0\sqrt{\log x}})$ [22, Ch. 18]. Subtracting (5.17) gives (5.22), and dividing by $\pi_{\mathrm{RVM}}(x) \sim x/\log x$ gives the third limit. □

The first limit and (5.17) concern only the synthetic system; the second and third compare it with the primes and use the prime number theorem. Figures 10 and 12 show them numerically.

6. Integer weights and the continuous-to-atomic problem

Define

$$c_n^{\mathrm{RVM}} := \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1), \qquad n \geq 2, \tag{6.1}$$

and, initially for $\Re s > 1$,

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) := \prod_{n\geq 2} (1 - n^{-s})^{-c_n^{\mathrm{RVM}}}. \tag{6.2}$$

No positivity or 0/1 property is built into (6.1). Define the integer-step prime-power count and the right-continuous extension

$$J_{\mathbb{Z},\mathrm{RVM}}(x) := \sum_{1\leq m\leq \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(\lfloor x^{1/m} \rfloor), \tag{6.3}$$

$$J^{*,\mathbb{Z}}_{\mathrm{RVM}}(x) := J^{*}_{\mathrm{RVM}}(\lfloor x \rfloor). \tag{6.4}$$

Theorem 6.1 (Lipschitz control of J_{RVM})**.** *There is an absolute constant C_J such that*

$$|J_{\mathrm{RVM}}(x+\theta) - J_{\mathrm{RVM}}(x)| \leq C_J\, \theta \qquad (x \geq 2,\ 0 < \theta \leq 1). \tag{6.5}$$

Proof. Put $u_1 = \log x$, $u_2 = \log(x+\theta)$, so $0 < u_2 - u_1 = \log(1+\theta/x) \leq \theta/x$ and $u_2 - u_1 \leq \log 2$. The terms li and $\int_x^\infty dy/(y(y^2-1)\log y)$ of (5.2) have derivatives bounded by $1/\log 2$ and $3/\log 2$ on $[2,\infty)$, so they change by $O(\theta)$. It remains to bound $\sum_k D_k$, where, by (5.1) and $\frac{d}{dv}\,\mathrm{Ei}(\rho v) = e^{\rho v}/v$,

$$D_k := \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_2) - \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_1) = \int_{u_1}^{u_2} \frac{e^{v/2} e^{iv\tau_k}}{v}\, dv. \tag{6.6}$$

The series $\sum_k D_k$ converges (Theorem 5.2 and its proof). By Abel's theorem for the increasing exponents τ_k, $\sum_k D_k = \lim_{\alpha\downarrow 0} \sum_k e^{-\alpha\tau_k} D_k$, and for $\varepsilon > 0$ Fubini gives

$$\sum_k e^{-\alpha\tau_k} D_k = \int_{u_1}^{u_2} \frac{e^{v/2}}{v}\, S_\alpha(v)\, dv, \qquad S_\alpha(v) := \sum_{k\geq 1} g(\tau_k), \quad g(\tau) := e^{(iv-\alpha)\tau}. \tag{6.7}$$

Clock decomposition. With $\tau_0 = 4\pi$ and the sawtooth count (8.8) (proved later in Section 8), Stieltjes integration exactly as in the forward reference (8.9) ($g(\infty) = 0$, $g' \in L^1$) gives

$$S_\alpha(v) = \int_{\tau_0}^\infty g\, d\mathcal{N}_{\mathrm{RVM}} + g(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0) + \int_{\tau_0}^\infty \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}) g'(\tau)\, d\tau. \tag{6.8}$$

The middle term is at most $\frac{1}{2}$. For the first, two integrations by parts with $\mathcal{N}'_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi}\log\frac{\tau}{2\pi}$, $\mathcal{N}''_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi\tau}$ and $v \geq \log 2$ give $|\int_{\tau_0}^\infty g\, d\mathcal{N}_{\mathrm{RVM}}| \leq C/v$, uniformly in $\varepsilon \in (0,1]$.

Harmonics. The partial sums of $\psi(y+\frac{1}{2}) = -\sum_{m\geq 1}(-1)^m \sin(2\pi m y)/(\pi m)$ are uniformly bounded and converge off a null set, and $g' \in L^1$ for $\varepsilon > 0$, so termwise integration is allowed:

$$\int_{\tau_0}^\infty \psi(\mathcal{N}_{\mathrm{RVM}} + \tfrac{1}{2}) g'\, d\tau = -\sum_{m\geq 1} \frac{(-1)^m}{2\pi i\, m}(K_m^+ - K_m^-), \qquad K_m^\pm := \int_{\tau_0}^\infty (iv-\alpha) e^{-\varepsilon\tau} e^{i\Phi_m^\pm(\tau)}\, d\tau, \tag{6.9}$$

with $\Psi_m^\pm(\tau) = v\tau \pm 2\pi m \mathcal{N}_{\mathrm{RVM}}(\tau)$. The weight $e^{-\varepsilon\tau}$ is positive and decreasing, so by the second mean-value theorem each integral is at most $2(v+1)\sup_\xi |\int_a^\xi e^{i\Phi} d\tau|$ over the relevant interval $[a,b]$. We use the first- and second-derivative tests [51, Lemmas 4.2 and 4.4]: $|\int_a^b e^{i\Phi}| \leq 4/\mu$ if Φ' is monotone with $|\Phi'| \geq \mu$, and $\leq 8/\sqrt{\lambda}$ if $|\Phi''| \geq \lambda$.

For K_m^+: $(\Phi_m^+)' = v + m\log\frac{\tau}{2\pi} \geq v + m\log 2$ is increasing, so $|K_m^+| \leq 8(v+1)/(v+m\log 2)$.

For K_m^-: $(\Phi_m^-)' = v - m\log\frac{\tau}{2\pi}$ is decreasing and vanishes at $T_m := 2\pi e^{v/m} > 2\pi$, and $(\Phi_m^-)'' = -m/\tau$. On $[\tau_0, T_m/2]$ (if nonempty) $(\Phi_m^-)' \geq m\log 2$; on $[2T_m,\infty)$, $(\Phi_m^-)' \leq -m\log 2$; on

$[\max(\tau_0, T_m/2), 2T_m]$, $|(\Phi_m^-)''| \geq m/(2T_m)$. Hence

$$|K_m^-| \leq 2(v+1)\Big(\frac{8}{m\log 2} + 8\sqrt{\frac{2T_m}{m}}\Big) = 2(v+1)\Big(\frac{8}{m\log 2} + 16\sqrt{\pi}\,\frac{e^{v/(2m)}}{\sqrt{m}}\Big). \tag{6.10}$$

Summing with the weights $1/(2\pi m)$ and using $\sum_m m^{-3/2}e^{v/(2m)} \leq e^{v/2} + \zeta(\frac{3}{2})e^{v/4}$,

$$|S_\alpha(v)| \leq C\,(v+1)\,e^{v/2} \qquad (v \geq \log 2,\ 0 < \alpha \leq 1), \tag{6.11}$$

with an absolute constant C.

Conclusion. Inserting (6.11) into (6.7), for $v \in [u_1, u_2]$ we have $e^{v/2} \leq \sqrt{2x}$ and $(v+1)/v \leq 1 + 1/\log 2$, so

$$\Big|\sum_k e^{-\alpha\tau_k} D_k\Big| \leq C'(u_2 - u_1)\,x \leq C'\theta, \tag{6.12}$$

uniformly in ε. Letting $\alpha \downarrow 0$ and taking $2\Re$ proves (6.5). □

The $m = 1$ harmonic, with stationary point $T_1 = 2\pi x$, produces the unit-periodic ripple of J_{RVM} (Figure 10); the harmonics $m \geq 2$ have stationary points $2\pi x^{1/m}$ and contribute $O(x^{-1/2+1/(2m)}m^{-3/2})$ to a unit increment.

Corollary 6.2 (Bounded cell masses)**.**

$$\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y \rfloor) = O(1), \qquad c_n^{\mathrm{RVM}} = O(1). \tag{6.13}$$

Proof. By (5.7), $c_n^{\mathrm{RVM}} = \sum_{m \leq \log_2 n} \frac{\mu(m)}{m}\big(J^*_{\mathrm{RVM}}(n^{1/m}) - J^*_{\mathrm{RVM}}((n-1)^{1/m})\big)$. The term $m = 1$ is $O(1)$ by (6.5). For $m \geq 2$ and $(n-1)^{1/m} \geq 2$, (6.5) with $\theta = n^{1/m} - (n-1)^{1/m} \leq n^{1/m-1}$ bounds the term by $C_J n^{1/m-1} \leq C_J n^{-1/2}$, and there are at most $\log_2 n$ such terms. The jump of J^*_{RVM} at 2 enters only for $n = 2^m$, once, with size $|J_{\mathrm{RVM}}(2)|/m$. The same argument applies to $\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y \rfloor)$. □

Corollary 6.3 (The integer-weighted zeta is a Dirichlet series)**.** *The product* (6.2) *converges absolutely for* $\Re s > 1$ *and*

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{N \geq 1} \frac{a_N^{\mathbb{Z}}}{N^s}, \qquad a_1^{\mathbb{Z}} = 1, \qquad \sum_{N \geq 1} \frac{|a_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma > 1). \tag{6.14}$$

The inverse is also an absolutely convergent Dirichlet series on the whole half-plane:

$$\frac{1}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)} = \sum_{N \geq 1} \frac{\bar{a}_N^{\mathbb{Z}}}{N^s}, \qquad \bar{a}_1^{\mathbb{Z}} = 1, \qquad \sum_{N \geq 1} \frac{|\bar{a}_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma > 1). \tag{6.15}$$

Proof. By Corollary 6.2,

$$\log \zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{n \geq 2} c_n^{\mathrm{RVM}} \sum_{m \geq 1} \frac{n^{-ms}}{m} = \sum_{N \geq 2} \frac{b_N}{N^s}, \qquad b_N := \sum_{n^m = N} \frac{c_n^{\mathrm{RVM}}}{m}, \tag{6.16}$$

with $\sum_N |b_N| N^{-\sigma} \leq C \sum_{n \geq 2} \sum_m n^{-m\sigma}/m < \infty$ for $\sigma > 1$. The exponential of an absolutely convergent Dirichlet series without constant term is an absolutely convergent Dirichlet series with $a_1^{\mathbb{Z}} = 1$, and $\sum_N |a_N^{\mathbb{Z}}| N^{-\sigma} \leq \exp\big(\sum_N |b_N| N^{-\sigma}\big)$. The same applies to $1/\zeta_{\mathbb{Z},\mathrm{RVM}} = \exp\big(-\sum_N b_N N^{-s}\big)$, which gives (6.15). □

Numerically the cell masses lie in $[-0.38, 1.04]$ for $n \leq 1000$ (Experiment 18.6).

Theorem 6.4 (Mellin–Stieltjes identification)**.** *There is a function* $J^{\mathrm{M}}_{\mathrm{RVM}}$ *on* $(1, \infty)$ *with* $J^{\mathrm{M}}_{\mathrm{RVM}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ *for every* $c > 1$ *such that*

$$\log \zeta_{\mathrm{RVM}}(s) = s \int_1^\infty J^{\mathrm{M}}_{\mathrm{RVM}}(x)\, x^{-s-1}\, dx \qquad (\Re s > 1), \tag{6.17}$$

and $J^{\mathrm{M}}_{\mathrm{RVM}} - J_{\mathrm{RVM}}$ *is constant on* $(1,\infty)$*. In particular* $J^{\mathrm{M}}_{\mathrm{RVM}} = J^{*}_{\mathrm{RVM}} + C_{\mathrm{RVM}}$ *on* $[2,\infty)$ *for a constant* C_{RVM}*.*

The proof is given in Section 12 (Theorems 12.5 and 12.6); it uses the half-plane bound of Theorem 8.10.

Corollary 6.5 (Weighted factorization)**.**

$$J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{*}_{\mathrm{RVM}}(x) = O(1+\log\log x), \tag{6.18}$$

and

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \zeta_{\mathrm{RVM}}(s)\, e^{h_{\mathrm{RVM}}(s)}, \tag{6.19}$$

where

$$h_{\mathrm{RVM}}(s) := s\int_1^{\infty} (J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{\mathrm{M}}_{\mathrm{RVM}}(x)) x^{-s-1}\, dx \tag{6.20}$$

is holomorphic for $\Re s > 0$*. The two functions have the same zeros in* $\Re s > 0$*; compare the modified Euler products of Grosswald and Schnitzer* [30]*.*

Proof. By (6.13) replacing $x^{1/m}$ by its integer part costs $O(1/m)$ in the mth summand; summing m^{-1} up to $\log_2 x$ gives (6.18). The product (6.2) is initially defined for $\Re s > 1$. For non-integral exponents we use the principal logarithmic branch, equivalently $-\log(1-n^{-s}) = \sum_{m\ge 1} n^{-ms}/m$, which is unambiguous because $|n^{-s}| < 1$ there. It gives $\log \zeta_{\mathbb{Z},\mathrm{RVM}}(s) = s\int_1^{\infty} J_{\mathbb{Z},\mathrm{RVM}}(x) x^{-s-1} dx$ for $\Re s > 1$, because $\sum_{n\le y} c^{\mathrm{RVM}}_n = \pi_{\mathrm{RVM}}(\lfloor y \rfloor)$. Subtracting (6.17) gives (6.19) on $\Re s > 1$. On $[2,\infty)$, $J_{\mathbb{Z},\mathrm{RVM}} - J^{\mathrm{M}}_{\mathrm{RVM}} = (J_{\mathbb{Z},\mathrm{RVM}} - J^{*}_{\mathrm{RVM}}) - C_{\mathrm{RVM}} = O(1+\log\log x)$ by (6.18); on $(1,2)$, $J_{\mathbb{Z},\mathrm{RVM}} = 0$ and $J^{\mathrm{M}}_{\mathrm{RVM}}$ is square-integrable. Hence (6.20) converges locally normally for $\Re s > 0$, and (6.19) defines the holomorphic continuation of $\zeta_{\mathbb{Z},\mathrm{RVM}}$ from $\Re s > 1$ to $\Re s > 0$; exponentiation introduces no zeros. □

Lemma 6.6 (Exact first-harmonic cell cancellation)**.** *For every integer* n*,*

$$\int_{n-1}^{n} 2\cos(2\pi x)\, dx = 0. \tag{6.21}$$

Hypothesis 6.7 (Uniform cell remainder away from the exceptional powers of 2)**.** For $n \ge 3$ with $n \notin \{2^j : \mu(j) \neq 0\}$,

$$c^{\mathrm{RVM}}_n - \int_{n-1}^{n} \left(\frac{1}{\log x} + 2\cos(2\pi x)\right) dx = o\!\left(\frac{1}{\log n}\right). \tag{6.22}$$

Proposition 6.8 (Conditional eventual positivity)**.** *Hypothesis 6.7 implies*

$$c^{\mathrm{RVM}}_n \sim \frac{1}{\log n}, \tag{6.23}$$

hence $0 < c^{\mathrm{RVM}}_n < 1$ *for all sufficiently large* n *in that set.*

Proof. The main-density integral is $1/\log n + O(1/(n\log^2 n))$; the first harmonic vanishes by Lemma 6.6. □

7. The critical RVM Mertens transform

Define

$$F_{\mathrm{RVM}}(s) = \widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}(s) := \frac{1}{\zeta_{\mathrm{RVM}}(s)} = \frac{G(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{7.1}$$

Formally the associated summatory object is the inverse Mellin transform

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = \frac{1}{2\pi i}\int_{(c)} F_{\mathrm{RVM}}(s)\frac{x^s}{s}\, ds; \tag{7.2}$$

its existence as a symmetric limit, for every $c > \frac{1}{2}$, is proved in Theorem 8.12.

Proposition 7.1 (Synthetic pole abscissa)**.** *F_{RVM} is holomorphic in $\Re s > \frac{1}{2}$. Its poles are the simple poles*

$$\frac{1}{2} \pm i\tau_k \quad (k \geq 1) \qquad \text{and} \qquad s = -2, -4, -6, \dots \tag{7.3}$$

and $s = 1$ is a simple zero of F_{RVM}.

Proof. This is Theorem 3.1 applied to $F_{\mathrm{RVM}} = G/P^{\mathrm{crit}}_{\mathrm{RVM}}$: the zeros of $P^{\mathrm{crit}}_{\mathrm{RVM}}$ are (3.6), and the poles of G are those of $\Gamma(s/2)$ except $s = 0$, where $s\Gamma(s/2) \to 2$. □

Proposition 7.2 (Gaussian square-root bound under vertical growth)**.** *Fix $c > 1$ and $0 < \varepsilon < c - \frac{1}{2}$. Assume, uniformly for $\frac{1}{2} + \varepsilon \leq \sigma \leq c$,*

$$F_{\mathrm{RVM}}(\sigma + it) = \exp(o(t^2)) \qquad (|t| \to \infty). \tag{7.4}$$

Define

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) := \frac{1}{2\pi i} \int_{(c)} e^{s^2} F_{\mathrm{RVM}}(s) \frac{x^s}{s}\, ds. \tag{7.5}$$

Then

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{7.6}$$

Proof. On $s = \sigma + it$, $|e^{s^2}| = e^{\sigma^2 - t^2}$, so (7.4) gives absolute convergence on vertical lines and makes the horizontal integrals tend to 0. The strip $\frac{1}{2} + \varepsilon \leq \Re s \leq c$ contains no pole of F_{RVM} (Proposition 7.1) and does not contain $s = 0$, so Cauchy's theorem moves the line to $\Re s = \frac{1}{2} + \varepsilon$, where $|x^s| = x^{1/2+\varepsilon}$. □

Theorem 8.12 below removes both the assumption (7.4) and the smoothing.

8. The transition theorem and the unsmoothed synthetic Mertens bound

Let

$$\chi(s) := 2^s \pi^{s-1} \sin\Big(\frac{\pi s}{2}\Big) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \tag{8.1}$$

the factor of the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ [51, §2.1]. Write

$$s = \tfrac{1}{2} + z, \quad z = \sigma' + it, \qquad \tau_* := -iz = t - i\sigma', \qquad w(s) := \exp\{-2\pi i\, \mathcal{N}_{\mathrm{RVM}}(\tau_*)\}. \tag{8.2}$$

Stirling's formula gives, uniformly on compact σ-sets,

$$|\chi(\sigma + it)| = \Big(\frac{|t|}{2\pi}\Big)^{1/2-\sigma} (1 + O(|t|^{-1})). \tag{8.3}$$

Theorem 8.1 (Transition evaluation of the midpoint product)**.** *There is a constant $\varpi_{\mathrm{RVM}} > 0$ with the following properties.*

(a) *For every $B > 0$ there are $t_0(B)$ and C_B such that, for $0 < \sigma' \leq B$ and $t \geq t_0(B)$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1 + w(s))(1 + \mathcal{E}_0(s)), \qquad |\mathcal{E}_0(s)| \leq \frac{C_B}{t}. \tag{8.4}$$

(b) *For every $0 < \sigma_1' < \sigma_2' < \frac{1}{2}$ there are t_0 and C such that, for $\sigma_1' \leq \sigma' \leq \sigma_2'$ and $t \geq t_0$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1 + \chi(s))(1 + \mathcal{E}(s)), \qquad |\mathcal{E}(s)| \leq \frac{C}{t}. \tag{8.5}$$

For $t < 0$ the same statements hold by $\zeta_{\mathrm{RVM}}(\bar{s}) = \overline{\zeta_{\mathrm{RVM}}(s)}$.

The phase identity $w = \chi(s)(1+O(1/t))$ alone does not give (8.4); the product has to be evaluated across the transition region $\tau_k \asymp t$. Log denotes the principal logarithm and $\operatorname{Log}\Gamma$ the branch of $\log\Gamma$ on $\Re v > 0$ that is real on $(0, \infty)$. Nothing about ζ is used.

8.1. **Proof of Theorem 8.1.** Throughout, $t > 0$, $0 < \sigma' \le B$, $\tau_0 := 4\pi$, $t^2 \ge 2(\tau_0^2 + B^2)$, and implied constants depend only on B (and on σ_1', σ_2' in part (b)).

Step 0: reduction to a sum over the clock. Since $q(s) = z^2 - \frac{1}{4}$, each factor of (3.2) equals $(\tau_k^2 + z^2)/(\tau_k^2 + \frac{1}{4})$, so

$$\frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2})} = \prod_{k\ge1}\Big(1 + \frac{z^2}{\tau_k^2}\Big) = \exp\Lambda(z), \qquad \Lambda(z) := \sum_{k\ge1} f(\tau_k), \quad f(\tau) := \mathrm{Log}\Big(1 + \frac{z^2}{\tau^2}\Big). \tag{8.6}$$

For $\tau > 0$, $\Im(1 + z^2/\tau^2) = 2\sigma' t/\tau^2 > 0$, so f is smooth on $(0,\infty)$, $f(\tau) = O(|z|^2\tau^{-2})$, the series converges absolutely, and

$$f'(\tau) = -\frac{2z^2}{\tau(\tau^2 + z^2)} = -\frac{2}{\tau} + \frac{1}{\tau - iz} + \frac{1}{\tau + iz}. \tag{8.7}$$

Step 1: the midpoint sawtooth. The clock increases on $[2\pi,\infty)$, $\mathcal{N}_{\mathrm{RVM}}(\tau_0) = 2\log\frac{2}{e} + \frac{7}{8} \in (0, \frac{1}{2})$, and $\tau_1 > \tau_0$. With $\psi(y) := y - \lfloor y\rfloor - \frac{1}{2}$, the count (4.12) satisfies, for $\tau \ge \tau_0$,

$$n_{\mathrm{RVM}}(\tau) = \lfloor \mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}\rfloor = \mathcal{N}_{\mathrm{RVM}}(\tau) - \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}), \qquad n_{\mathrm{RVM}}(\tau_0) = 0. \tag{8.8}$$

Stieltjes integration of f against (8.8), followed by integration by parts in the ψ-part ($f(\infty) = 0$, $|\psi| \le \frac{1}{2}$, $\psi(\mathcal{N}_{\mathrm{RVM}}(\tau_0) + \frac{1}{2}) = \mathcal{N}_{\mathrm{RVM}}(\tau_0)$), gives

$$\begin{aligned} &\Lambda(z) = \Lambda_{\mathrm{sm}}(z) + \Lambda_{\mathrm{saw}}(z), \qquad \Lambda_{\mathrm{sm}}(z) := \int_{\tau_0}^{\infty} f\, d\mathcal{N}_{\mathrm{RVM}} + f(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0), \\ &\Lambda_{\mathrm{saw}}(z) := \int_{\tau_0}^{\infty} \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}) f'(\tau)\, d\tau. \end{aligned} \tag{8.9}$$

Lemma 8.2 (Smooth part)**.** *With* $\log G(s) := \mathrm{Log}(z^2 - \frac{1}{4}) + \mathrm{Log}\,\Gamma(\frac{s}{2}) - \frac{s}{2}\log\pi$,

$$\Lambda_{\mathrm{sm}}(z) - \log G(s) = \nu_3 + O(t^{-1}), \qquad \nu_3 \in \mathbb{R} \text{ independent of } z. \tag{8.10}$$

Proof. A second integration by parts ($f\mathcal{N}_{\mathrm{RVM}} \to 0$ at ∞) cancels the boundary term:

$$\Lambda_{\mathrm{sm}}(z) = -\int_{\tau_0}^{\infty} \mathcal{N}_{\mathrm{RVM}} f'\, d\tau = \int_{\tau_0}^{\infty} \mathcal{N}_{\mathrm{RVM}}(\tau)\Big(\frac{2}{\tau} - \frac{2\tau}{\tau^2 + z^2}\Big) d\tau. \tag{8.11}$$

The constant $\frac{7}{8}$ of the clock contributes $\frac{7}{8}\mathrm{Log}(1 + z^2/\tau_0^2)$, because $\mathrm{Log}\,\frac{\tau^2}{\tau^2+z^2}$ is an antiderivative of the bracket on $(0,\infty)$ tending to 0 at ∞. As $0 < \arg z < \frac{\pi}{2}$, $\arg(1 + \tau_0^2/z^2) = O(t^{-2})$ and

$$\tfrac{7}{8}\mathrm{Log}\Big(1 + \frac{z^2}{\tau_0^2}\Big) = \tfrac{7}{4}\mathrm{Log}\, z - \tfrac{7}{4}\log\tau_0 + O(t^{-2}). \tag{8.12}$$

The term $\frac{\tau}{2\pi}\log\frac{\tau}{2\pi e}$ contributes $\frac{1}{\pi}\int_{\tau_0}^{\infty}\log\frac{\tau}{2\pi e}\frac{z^2}{\tau^2+z^2}\,d\tau$. Put

$$\mathcal{I}(z) := \frac{1}{\pi}\int_0^{\infty} \log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2 + z^2}\, d\tau. \tag{8.13}$$

The integrand is holomorphic in z on $\Re z > 0$ (its singularities are at $z = \pm i\tau$), $O(|z|^2\tau^{-2}\log\tau)$ at ∞ and $O(|\log\tau|)$ at 0, locally uniformly; so $\mathcal{I}$ is holomorphic on $\Re z > 0$. For real $z = x > 0$ the substitution $\tau = xv$ and $\int_0^\infty \frac{dv}{1+v^2} = \frac{\pi}{2}$, $\int_0^\infty \frac{\log v}{1+v^2}\,dv = 0$ give $\mathcal{I}(x) = \frac{x}{2}\log\frac{x}{2\pi e}$; by the identity theorem $\mathcal{I}(z) = \frac{z}{2}\mathrm{Log}\,\frac{z}{2\pi e}$ on $\Re z > 0$. On $[0,\tau_0]$, $|\tau^2 + z^2| \ge t^2/2$, so $\frac{z^2}{\tau^2+z^2} = 1 + O(t^{-2})$ there. Hence

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\mathrm{Log}\,\frac{z}{2\pi e} + \frac{7}{4}\mathrm{Log}\, z + \nu_1 + O(t^{-2}), \qquad \nu_1 = -\tfrac{7}{4}\log\tau_0 - \tfrac{1}{\pi}\int_0^{\tau_0}\log\tfrac{\tau}{2\pi e}\,d\tau. \tag{8.14}$$

Stirling's formula $\operatorname{Log}\Gamma(v) = (v-\frac12)\operatorname{Log} v - v + \frac12\log 2\pi + O(|v|^{-1})$ [27, §5.11] with $v = \frac{s}{2} = \frac{z+1/2}{2}$ and $\operatorname{Log} v = \operatorname{Log} z - \log 2 + \frac{1}{2z} + O(t^{-2})$ gives

$$\operatorname{Log}\Gamma(\tfrac{s}{2}) = \tfrac{z}{2}\operatorname{Log}\tfrac{z}{2} - \tfrac14\operatorname{Log} z + \tfrac14\log 2 - \tfrac{z}{2} + \tfrac12\log 2\pi + O(t^{-1}), \tag{8.15}$$

and with $\operatorname{Log}(z^2-\frac14) = 2\operatorname{Log} z + O(t^{-2})$, $-\frac{s}{2}\log\pi = -\frac{z}{2}\log\pi - \frac14\log\pi$,

$$\log G(s) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac74\operatorname{Log} z + \nu_2 + O(t^{-1}), \qquad \nu_2 = \tfrac14\log 2 + \tfrac12\log 2\pi - \tfrac14\log\pi. \tag{8.16}$$

Subtracting gives (8.10) with $\nu_3 = \nu_1 - \nu_2$. □

Lemma 8.3 (The clock as a holomorphic phase)**.** *$\mathcal{N}_{\mathrm{RVM}}(\tau) = \frac{\tau}{2\pi}(\operatorname{Log}\frac{\tau}{2\pi} - 1) + \frac78$ is holomorphic on $\Re\tau > 0$, $\mathcal{N}_{\mathrm{RVM}}(\bar\tau) = \overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$, and for $\Re\tau \ge \tau_0$,*

$$2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \ge (\log 2)\,\Im\tau \quad (\Im\tau \ge 0), \qquad 2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \le (\log 2)\,\Im\tau \quad (\Im\tau \le 0). \tag{8.17}$$

Proof. For $\tau = X + iY$, $X \ge \tau_0$, $Y > 0$, put $v = Y/X$. Then

$$2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau) = Y\log\frac{|\tau|}{2\pi e} + X\arg\tau = Y\Big[\log\frac{X}{2\pi e} + \tfrac12\log(1+v^2) + \frac{\arctan v}{v}\Big] \ge Y\big[\log 2 + \varphi(v)\big], \tag{8.18}$$

using $X \ge 4\pi$, where $\varphi(v) = \frac12\log(1+v^2) + \frac{\arctan v}{v} - 1$. Since $\varphi(0^+) = 0$ and $\varphi'(v) = (v - \arctan v)/v^2 \ge 0$, $\varphi \ge 0$. The case $Y < 0$ follows by conjugation. □

In particular $|w(s)| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} \le 2^{-\sigma'} < 1$, and on the vertical rays $|e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0 \pm iy)}| \le 2^{-my}$.

Lemma 8.4 (Abel-regularized harmonic sum)**.**

$$\Lambda_{\mathrm{saw}}(z) = \operatorname{Log}\big(1 + w(s)\big) + \nu_4 + O(t^{-2}), \qquad \nu_4 \in \mathbb{R} \text{ independent of } z. \tag{8.19}$$

Proof. Regularization. For $0 < r < 1$ let

$$\psi_r(y) := -\sum_{m\ge1} r^m \frac{\sin 2\pi m y}{\pi m} = \frac{1}{\pi}\Im\operatorname{Log}\big(1 - re^{2\pi i y}\big). \tag{8.20}$$

Since $\Re(1 - re^{2\pi iy}) > 0$, $|\psi_r| \le \frac12$, and $\psi_r(y) \to \psi(y)$ as $r \uparrow 1$ for $y \notin \mathbb{Z}$ (Abel summation of the convergent Fourier series of ψ). The set $\{\tau : \mathcal{N}_{\mathrm{RVM}}(\tau) + \frac12 \in \mathbb{Z}\}$ is discrete and $f' \in L^1(\tau_0, \infty)$, so by dominated convergence

$$\Lambda_{\mathrm{saw}}(z) = \lim_{r\uparrow1}\Lambda_r(z), \qquad \Lambda_r(z) := \int_{\tau_0}^{\infty} \psi_r\big(\mathcal{N}_{\mathrm{RVM}} + \tfrac12\big) f'\,d\tau. \tag{8.21}$$

Termwise integration. For fixed $r < 1$ the series for $\psi_r(\mathcal{N}_{\mathrm{RVM}} + \frac12)$, with $\sin 2\pi m(y+\frac12) = (-1)^m \sin 2\pi m y$, is dominated by $\sum_m r^m/m$; by Fubini,

$$\Lambda_r(z) = -\sum_{m\ge1} \frac{(-r)^m}{2\pi i\, m}\big(I_m^+ - I_m^-\big), \qquad I_m^{\pm} := \int_{\tau_0}^{\infty} e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau)} f'(\tau)\,d\tau. \tag{8.22}$$

Contours. Let $\mathcal{H}^{\pm} = \{\Re\tau \ge \tau_0,\ \pm\Im\tau \ge 0\}$. On $\mathcal{H}^{\pm}$, $|e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}}| \le 1$ by Lemma 8.3, and $|f'(\tau)| \le 4|z|^2|\tau|^{-3}$ for $|\tau| \ge 2|z|$, so the quarter-circle arcs of radius R contribute $O(|z|^2R^{-2}) \to 0$. The poles of f' are 0, $iz = -t + i\sigma' \notin U^{\pm}$, and $\tau_* = -iz = t - i\sigma' \in U^-$, the last with residue 1 (from $\frac{1}{\tau+iz}$). Define the ray integrals

$$R_m^+ := i\int_0^{\infty} e^{2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0+iy)} f'(\tau_0+iy)\,dy, \qquad R_m^- := -i\int_0^{\infty} e^{-2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0-iy)} f'(\tau_0-iy)\,dy. \tag{8.23}$$

On U^+ Cauchy's theorem gives $I_m^+ = R_m^+$. On U^- the positively oriented boundary of the quarter disc runs down the ray, along the arc, and back along $[\tau_0, \tau_0 + R]$ from right to left; letting $R \to \infty$,

$$R_m^- - I_m^- = 2\pi i\, e^{-2\pi i m \mathcal{N}_{\mathrm{RVM}}(\tau_*)} = 2\pi i\, w^m. \tag{8.24}$$

Therefore, using $|rw| < 1$,

$$\Lambda_r(z) = -\sum_{m\ge 1} \frac{(-rw)^m}{m} + \Phi_r(z) = \mathrm{Log}(1+rw) + \Phi_r(z), \qquad \Phi_r(z) := -\sum_{m\ge 1} \frac{(-r)^m}{2\pi i\, m}(R_m^+ - R_m^-). \tag{8.25}$$

The rays. Split $f'(\tau) = -\frac{2}{\tau} + g_z(\tau)$, $g_z(\tau) = \frac{2\tau}{\tau^2+z^2}$, and correspondingly $R_m^\pm = R_m^{\pm,(0)} + R_m^{\pm,(z)}$, $\Phi_r = \Phi_r^{(0)} + \Phi_r^{(z)}$. The ray integrands carry the factor 2^{-my}. Since $|2/\tau| \le 2/\tau_0$ on the rays, $|R_m^{\pm,(0)}| \le 2/(\tau_0 m \log 2)$, so the series for $\Phi_r^{(0)}$ is dominated by $\sum_m C/m^2$ uniformly in r, and $\Phi_r^{(0)} \to \nu_4 := \Phi_1^{(0)}$, a constant depending only on τ_0. It is real: $\mathcal{N}_{\mathrm{RVM}}(\bar\tau) = \overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$ gives $R_m^{-,(0)} = \overline{R_m^{+,(0)}}$, so each term equals $-(-r)^m \Im R_m^{+,(0)}/(\pi m)$. For $\tau = \tau_0 \pm iy$, $\Re(\tau^2+z^2) = \tau_0^2 + \sigma'^2 - y^2 - t^2 \le -\frac12(y^2+t^2)$ because $t^2 \ge 2(\tau_0^2 + B^2)$; hence $|g_z(\tau)| \le 4(\tau_0+y)t^{-2}$ and

$$|R_m^{\pm,(z)}| \le \frac{4}{t^2}\int_0^\infty 2^{-my}(\tau_0+y)\,dy = \frac{4}{t^2}\Big(\frac{\tau_0}{m\log 2} + \frac{1}{(m\log 2)^2}\Big). \tag{8.26}$$

So $|\Phi_r^{(z)}(z)| \le C\,t^{-2}\sum_m m^{-2}$, uniformly in $0 < r < 1$ and $0 < \sigma' \le B$, and $\Phi_r^{(z)}$ converges as $r \uparrow 1$ by dominated convergence of the series. Since $\mathrm{Log}(1+rw) \to \mathrm{Log}(1+w)$, letting $r \uparrow 1$ in (8.25) and (8.21) gives (8.19). □

Step 2: assembly. By (8.6), (8.9) and Lemmas 8.2 and 8.4,

$$\begin{aligned}\zeta_{\mathrm{RVM}}(s) &= \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\exp\big(\Lambda_{\mathrm{sm}}(z) - \log G(s) + \Lambda_{\mathrm{saw}}(z)\big)\\ &= P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{\nu_3+\nu_4}(1+w)\big(1+O(t^{-1})\big),\end{aligned} \tag{8.27}$$

uniformly for $0 < \sigma' \le B$. This is (8.4) with

$$\varpi_{\mathrm{RVM}} := P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{\nu_3+\nu_4} > 0, \tag{8.28}$$

which depends only on the clock; its numerical value $0.9619415807\ldots$ plays no role in the proof.

Step 3: from the clock phase to χ. Let

$$\vartheta(v) := \frac{1}{2i}\Big(\mathrm{Log}\,\Gamma\big(\tfrac14 + \tfrac{iv}{2}\big) - \mathrm{Log}\,\Gamma\big(\tfrac14 - \tfrac{iv}{2}\big)\Big) - \frac{v}{2}\log\pi, \tag{8.29}$$

holomorphic on $|\Im v| < \frac12$. The identity $\chi(\frac12 + iv) = e^{-2i\vartheta(v)}$ holds for real v, hence on the strip, and $s = \frac12 + i\tau_*$ with $|\Im\tau_*| = \sigma' < \frac12$. Stirling's formula in the strip gives $\vartheta(v) = \frac{v}{2}\,\mathrm{Log}\,\frac{v}{2\pi} - \frac{v}{2} - \frac{\pi}{8} + O(|v|^{-1})$, i.e.

$$\mathcal{N}_{\mathrm{RVM}}(v) = \frac{\vartheta(v)}{\pi} + 1 + O(|v|^{-1}). \tag{8.30}$$

Therefore

$$w = e^{-2i\vartheta(\tau_*) - 2\pi i + O(1/t)} = \chi(s)\big(1 + O(t^{-1})\big). \tag{8.31}$$

For $\sigma' \ge \sigma_1'$, (8.3) gives $|\chi(s)| \le \frac12$ when $t \ge t_0(\sigma_1')$, so $1 + w = (1+\chi)(1 + (w-\chi)/(1+\chi)) = (1+\chi)(1+O(t^{-1}))$, which with (8.4) gives (8.5). This completes the proof of Theorem 8.1. □

Lemma 8.5 (Extension to the critical line)**.** *The function $\mathcal{E}_0 = \zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1$ extends holomorphically to $\{|\sigma'| < \frac12,\ t > t_0\}$, and $|\mathcal{E}_0| \le C/t$ for $0 \le \sigma' \le \frac14$.*

Proof. By Lemma 8.3, $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) \neq 0$ for $\sigma' \neq 0$, so $1+w$ vanishes in the strip only on $\sigma' = 0$, exactly at $t = \tau_k$; these zeros are simple because $\frac{d}{dt}w = -i\log(\tau_*/2\pi)\, w \neq 0$. The zeros of ζ_{RVM} in the strip are the same simple points. Hence $\mathcal{E}_0$ has only removable singularities, and the bound extends to $\sigma' = 0$ by continuity. □

Corollary 8.6 (Derivative at the zeros)**.** *For k with $\tau_k > t_0$,*

$$\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}}) = \varpi_{\mathrm{RVM}} \log\frac{\tau_k}{2\pi}\Big(1 + O\Big(\frac{1}{\tau_k}\Big)\Big). \tag{8.32}$$

Proof. By Lemma 8.5, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ holds holomorphically near $\sigma' = 0$. At $s = \rho_k^{\mathrm{RVM}}$, $1 + w = 0$, so $\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}}) = \varpi_{\mathrm{RVM}} w'(\rho_k^{\mathrm{RVM}})(1 + \mathcal{E}_0(\rho_k^{\mathrm{RVM}}))$. From (8.2), $w'(s) = -2\pi i\,\mathcal{N}'_{\mathrm{RVM}}(\tau_*) \cdot (-i)\, w(s) = -\log(\tau_*/2\pi)\, w(s)$, and at the zero $w = -1$, $\tau_* = \tau_k$. Finally $|\mathcal{E}_0| \leq C/\tau_k$. □

Remark 8.7 (Structure of the proof). The delicate points are handled as follows: the smooth part is computed with principal branches by the identity theorem (Lemma 8.2); the clock estimate is quantified (Lemma 8.3); the sawtooth series is replaced by its Abel means, termwise integration is justified for $r < 1$, and $r \uparrow 1$ is taken after the contour argument uniformly in r and z (Lemma 8.4); the orientation of the lower contour and the sign of the residue are written out in (8.24); the z-dependence of the rays is isolated in g_z and is $O(t^{-2})$, the rest is the explicit real constant ν_4; Stirling is used uniformly for $0 < \sigma' \leq B$, and the only place where $\sigma' < \frac{1}{2}$ enters is Step 3.

Corollary 8.8 (Vertical constant limit)**.** *For every fixed $\sigma > \frac{1}{2}$,*

$$\zeta_{\mathrm{RVM}}(\sigma + it) \longrightarrow \varpi_{\mathrm{RVM}}, \qquad F_{\mathrm{RVM}}(\sigma + it) \longrightarrow L := \frac{1}{\varpi_{\mathrm{RVM}}} \qquad (|t| \to \infty). \tag{8.33}$$

More precisely, with $\sigma' = \sigma - \frac{1}{2}$,

$$|w(\sigma + it)| \leq (|t|/2\pi)^{-\sigma'}(1 + O_B(|t|^{-1})) \qquad (0 < \sigma' \leq B), \tag{8.34}$$

and, with $\alpha_c := \min(c - \frac{1}{2}, 1)$ for $c > \frac{1}{2}$,

$$F_{\mathrm{RVM}}(c + it) - L = O_c(|t|^{-\alpha_c}). \tag{8.35}$$

Proof. Expanding the clock at $\tau_* = t - i\sigma'$ gives $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) = -\frac{\sigma'}{2\pi}\log\frac{t}{2\pi} + O_\sigma(t^{-1})$, hence $|w| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} = O_\sigma(t^{-\sigma'})$. Inverting (8.4) for large t gives $F_{\mathrm{RVM}} = L(1 + O(|w|) + O(t^{-1}))$. □

8.2. **Bounds on the whole half-plane.** The constants of Theorem 8.1 depend on the width B of the band. The following estimate holds on the whole half-plane and does not use contours. Put

$$\mathfrak{s}_1(\tau) := \int_{\tau_0}^{\tau} \psi(\mathcal{N}_{\mathrm{RVM}}(v) + \tfrac{1}{2})\, dv \qquad (\tau \geq \tau_0). \tag{8.36}$$

Lemma 8.9 (Bounded primitive of the midpoint sawtooth)**.** $|\mathfrak{s}_1(\tau)| \leq \dfrac{\pi}{4\log 2}$ *for all $\tau \geq \tau_0$.*

Proof. Substitute $y = \mathcal{N}_{\mathrm{RVM}}(v)$, $dv = \mathcal{T}'(y)\, dy$ with $\mathcal{T}'(y) = 2\pi/\log(\mathcal{T}(y)/2\pi)$ from (4.4). The weight $\mathcal{T}'$ is positive and decreasing, with $\mathcal{T}' \leq 2\pi/\log 2$ on $y \geq \mathcal{N}_{\mathrm{RVM}}(\tau_0)$ because $\log(\tau_0/2\pi) = \log 2$. The primitive of $y \mapsto \psi(y + \frac{1}{2})$ is periodic with values in $[-\frac{1}{8}, 0]$, so every integral of $\psi(y + \frac{1}{2})$ over an interval has modulus at most $\frac{1}{8}$. The second mean-value theorem gives $|\mathfrak{s}_1(\tau)| \leq \frac{2\pi}{\log 2} \cdot \frac{1}{8}$. □

Theorem 8.10 (Two-sided bound and limit at $+\infty$)**.** *There is an absolute constant R_0 such that, for $s = \frac{1}{2} + z$, $z = \sigma' + it$, $\sigma' > \frac{1}{2}$ and $|z| \geq R_0$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\, e^{\theta_{\mathrm{RVM}}(s)}, \qquad |\theta_{\mathrm{RVM}}(s)| \leq \frac{\pi^2}{2\log 2} \cdot \frac{1}{\sigma'} + \frac{C}{|z|}. \tag{8.37}$$

Consequently, for every $\sigma_0 > 1$ there is $C(\sigma_0)$ with

$$C(\sigma_0)^{-1} \leq |\zeta_{\mathrm{RVM}}(s)| \leq C(\sigma_0) \qquad (\Re s \geq \sigma_0), \tag{8.38}$$

and $\zeta_{\mathrm{RVM}}(s) \to \varpi_{\mathrm{RVM}}$ *uniformly in* t *as* $\Re s \to +\infty$. *Moreover*

$$\nu_4 = -2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau. \tag{8.39}$$

Proof. Smooth part. For $\Re z > 0$ the computation in the proof of Lemma 8.2 is exact:

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac{7}{8}\operatorname{Log}\Big(1+\frac{z^2}{\tau_0^2}\Big) - \frac{1}{\pi}\int_0^{\tau_0} \log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2+z^2}\, d\tau. \tag{8.40}$$

For $|z| \ge 2\tau_0$ one has $|\tau^2+z^2| \ge \frac{3}{4}|z|^2$ on $[0,\tau_0]$, and Stirling's formula is uniform in $|\arg(s/2)| < \frac{\pi}{2}$; the expansions (8.14) and (8.16) therefore hold with $O(|z|^{-1})$ in place of $O(t^{-1})$, uniformly on $\Re z > \frac{1}{2}$, and $\Lambda_{\mathrm{sm}}(z) - \log G(s) = \nu_3 + O(|z|^{-1})$.

Sawtooth part. Integrating (8.9) by parts with (8.36) ($\mathfrak{s}_1(\tau_0) = 0$, $f'(\tau) = O(|z|^2\tau^{-3})$),

$$\Lambda_{\mathrm{saw}}(z) = -\int_{\tau_0}^{\infty} \mathfrak{s}_1 f''\, d\tau, \qquad f''(\tau) = \frac{2}{\tau^2} - \frac{1}{(\tau - iz)^2} - \frac{1}{(\tau+iz)^2}. \tag{8.41}$$

Since $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$ and $\int_{\mathbb{R}} \frac{d\tau}{(\tau-a)^2+\sigma'^2} = \frac{\pi}{\sigma'}$, Lemma 8.9 gives

$$\Big|\Lambda_{\mathrm{saw}}(z) + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau\Big| \le \frac{\pi}{4\log 2}\cdot\frac{2\pi}{\sigma'}. \tag{8.42}$$

Assembly. By (8.27), $\log\zeta_{\mathrm{RVM}}(s) = \log\varpi_\infty + \theta_{\mathrm{RVM}}(s)$ with $\varpi_\infty := P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2})\exp(\nu_3 - 2\int_{\tau_0}^{\infty}\mathfrak{s}_1\tau^{-2}d\tau) > 0$ and θ_{RVM} bounded as in (8.37). For fixed σ' and $t \to \infty$, Corollary 8.8 gives $|\log\varpi_{\mathrm{RVM}} - \log\varpi_\infty| \le \pi^2/(2\sigma'\log 2)$; a priori the logarithms can differ by $2\pi i k$. For sufficiently large σ', the bound in (8.37) is $< 2\pi$, forcing $k = 0$; letting $\sigma' \to \infty$ then gives $\varpi_\infty = \varpi_{\mathrm{RVM}}$, which is (8.39) by (8.28). On the compact set $\{\Re s \ge \sigma_0,\ |z| \le R_0\}$ the function ζ_{RVM} is continuous and has no zeros or poles, which gives (8.38). □

Theorem 8.10 extends the two-sided bound (9.9) of Corollary 9.3 from bands to the whole half-plane $\Re s \ge \sigma_0 > 1$, and identifies ϖ_{RVM} also as the limit of $\zeta_{\mathrm{RVM}}(\sigma)$ as $\sigma \to +\infty$.

Theorem 8.11 (No absolutely convergent Dirichlet expansion)**.** *Neither* ζ_{RVM} *nor* $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$ *is represented by a Dirichlet series* $\sum_n a_n n^{-s}$ *converging absolutely in some half-plane* $\Re s > \sigma_0$.

Proof. Suppose $F_{\mathrm{RVM}}(s) = \sum_{n\ge1} a_n n^{-s}$ absolutely for $\Re s > \sigma_0$. Fix $\sigma > \max(\sigma_0, \frac{1}{2})$ and put $\tilde a_1 = a_1 - L$, $\tilde a_n = a_n$ $(n \ge 2)$, so $F_{\mathrm{RVM}}(\sigma+it) - L = \sum_n \tilde a_n n^{-\sigma-it}$ with $\sum_n |\tilde a_n| n^{-\sigma} < \infty$. Then

$$\frac{1}{2T}\int_{-T}^{T}\big|F_{\mathrm{RVM}}(\sigma+it) - L\big|^2 dt = \sum_{m,n} \tilde a_m \overline{\tilde a_n}(mn)^{-\sigma}\,\frac{1}{2T}\int_{-T}^{T}\Big(\frac{n}{m}\Big)^{it} dt, \tag{8.43}$$

the interchange being justified by absolute convergence of the double series, uniformly in t. The inner average is 1 for $m = n$ and is bounded by $\min(1, 1/(T|\log(n/m)|)) \to 0$ for $m \ne n$; each term is bounded by $|\tilde a_m \tilde a_n|(mn)^{-\sigma}$, which is summable, so dominated convergence gives

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\big|F_{\mathrm{RVM}}(\sigma+it) - L\big|^2 dt = \sum_{n\ge1}|\tilde a_n|^2 n^{-2\sigma}. \tag{8.44}$$

By (8.33) the integrand tends to 0 and is bounded on the line (continuous, with a finite limit), so the left side of (8.44) is 0. Hence every $\tilde a_n = 0$, $F_{\mathrm{RVM}} \equiv L$ in the half-plane and, by analytic continuation, everywhere, which contradicts $F_{\mathrm{RVM}}(1) = 0$. The same argument applied to $\zeta_{\mathrm{RVM}} - \varpi_{\mathrm{RVM}}$ forces an absolutely convergent Dirichlet series for ζ_{RVM} to be the constant ϖ_{RVM}, which contradicts its pole at $s = 1$. □

Theorem 8.12 (Unsmoothed synthetic Mertens bound)**.** *For every $x > 1$ and every $c > \frac{1}{2}$ the symmetric Perron integral*

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) := \lim_{U\to\infty} \frac{1}{2\pi i} \int_{c-iU}^{c+iU} F_{\mathrm{RVM}}(s) \frac{x^s}{s}\, ds \tag{8.45}$$

exists, is independent of $c > \frac{1}{2}$, and equals the absolutely convergent expression

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + \frac{1}{2\pi i} \int_{(c)} (F_{\mathrm{RVM}}(s) - L) \frac{x^s}{s}\, ds. \tag{8.46}$$

For every $\varepsilon \in (0, \frac{1}{2})$,

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon}). \tag{8.47}$$

Proof. Fix $c > \frac{1}{2}$. By (8.35) and continuity of F_{RVM} on $\Re s = c$ (no poles there by Proposition 7.1), the integral in (8.46) converges absolutely. The Perron kernel gives, for $x > 1$ and $c > 0$ [22, Ch. 17],

$$\lim_{U\to\infty} \frac{1}{2\pi i} \int_{c-iU}^{c+iU} \frac{x^s}{s}\, ds = 1. \tag{8.48}$$

Splitting $F_{\mathrm{RVM}} = L + (F_{\mathrm{RVM}} - L)$ proves existence of (8.45) and (8.46). If $\frac{1}{2} < c < c'$, the integrand of (8.45) is holomorphic on the closed strip $c \le \Re s \le c'$ (Proposition 7.1); by Theorem 8.1(a) with $B = c' - \frac{1}{2}$, F_{RVM} is bounded on the horizontal segments $\Im s = \pm U$, $U \ge t_0(B)$, so these contribute $O(x^{c'}U^{-1}) \to 0$, and Cauchy's theorem gives independence of c. Take $c = \frac{1}{2} + \varepsilon$ in (8.46):

$$|M^{\mathrm{crit}}_{\mathrm{RVM}}(x) - L| \le \frac{x^{1/2+\varepsilon}}{2\pi} \int_{-\infty}^{\infty} \frac{|F_{\mathrm{RVM}}(\frac{1}{2} + \varepsilon + it) - L|}{|\frac{1}{2} + \varepsilon + it|}\, dt, \tag{8.49}$$

and the integral is finite by (8.35), the integrand being $O(|t|^{-1-\alpha_c})$. □

Remark 8.13 (Scope). Theorem 8.12 concerns $1/\zeta_{\mathrm{RVM}}$. The transfer to M goes through the endpoint of the λ-family and is Theorem 11.8.

9. The exact Hadamard product and the comparison with ζ

Put

$$P_\zeta(s) := G(s)\zeta(s) = 2\xi(s), \qquad \xi(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{9.1}$$

Theorem 9.1 (Exact Hadamard product)**.** *Let ρ run over the nontrivial zeros of ζ with $\Im\rho > 0$, with multiplicity, ordered by $\Im\rho$. The product*

$$P_\zeta(s) = \prod_{\Im\rho>0} \left(1 + \frac{q(s)}{\rho(1-\rho)}\right) \tag{9.2}$$

converges absolutely and locally uniformly in $s \in \mathbb{C}$, and the identity holds with no exponential or constant factor.

Proof. ξ is entire of order one with $\xi(0) = \frac{1}{2}$, so Hadamard's factorization [32], [51, §2.12], [22, Ch. 12] gives

$$2\xi(s) = e^{Bs} \prod_\rho \Big(1 - \frac{s}{\rho}\Big) e^{s/\rho}, \tag{9.3}$$

the product running over all nontrivial zeros and converging absolutely because $\sum_\rho |\rho|^{-2} < \infty$. The functional equation $\xi(s) = \xi(1-s)$ makes $\rho \mapsto 1-\rho$ an involution of the zero multiset; no nontrivial zero is real, since $(1-2^{1-\sigma})\zeta(\sigma) = \sum_{n\ge1}(-1)^{n-1}n^{-\sigma} > 0$ for $0 < \sigma < 1$; so this involution exchanges $\Im\rho > 0$ and $\Im\rho < 0$. Grouping ρ with $1-\rho$, which absolute convergence permits,

$$\Big(1 - \frac{s}{\rho}\Big)\Big(1 - \frac{s}{1-\rho}\Big) e^{s/\rho + s/(1-\rho)} = \Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big) e^{s/(\rho(1-\rho))}. \tag{9.4}$$

Since $|\rho(1-\rho)| \geq (\Im\rho)^2$ and $\sum_{\Im\rho>0}(\Im\rho)^{-2} < \infty$, the numbers $1/(\rho(1-\rho))$ are absolutely summable, so the exponentials collect into a single factor:

$$2\xi(s) = e^{B's}\,\Pi_\zeta(s), \qquad \Pi_\zeta(s) := \prod_{\Im\rho>0}\Big(1+\frac{q(s)}{\rho(1-\rho)}\Big), \qquad B' = B + \sum_{\Im\rho>0}\frac{1}{\rho(1-\rho)}. \tag{9.5}$$

Π_ζ depends on s only through $q(s)$, hence $\Pi_\zeta(1-s) = \Pi_\zeta(s)$. From $\xi(1-s) = \xi(s)$ we get $e^{B's} = e^{B'(1-s)}$ on the open set where $\Pi_\zeta \neq 0$, so $B' = 0$. □

Define the exact quotient

$$\mathcal{Q}_{\mathrm{RVM}}(s) := \frac{\zeta(s)}{\zeta_{\mathrm{RVM}}(s)} = \frac{P_\zeta(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{9.6}$$

Proposition 9.2 (Exact transform relation)**.** *For $\Re s > 1$, and by meromorphic continuation,*

$$F_{\mathrm{RVM}}(s) = \mathcal{Q}_{\mathrm{RVM}}(s)\,\frac{1}{\zeta(s)}. \tag{9.7}$$

If the inverse Mellin transform $\mathcal{Q}^\vee_{\mathrm{RVM}}$ of $\mathcal{Q}_{\mathrm{RVM}}$ and the inverse defining $M^{\mathrm{crit}}_{\mathrm{RVM}}$ exist in a common distribution space with Mellin convolution, then

$$M^{\mathrm{crit}}_{\mathrm{RVM}} = \mathcal{Q}^\vee_{\mathrm{RVM}} *_{\mathrm{Mel}} M. \tag{9.8}$$

(9.7) is the identity $1/\zeta_{\mathrm{RVM}} = (\zeta/\zeta_{\mathrm{RVM}})(1/\zeta)$. Equation (9.8) is only notation for the Mellin-convolution theorem when both inverse transforms belong to a specified common convolution space; no such space is constructed here.

Corollary 9.3 (All growth of ζ to the right of $\frac12$ lies in $\mathcal{Q}_{\mathrm{RVM}}$)**.** *For every $c' > \frac12$ and $0 < \varepsilon < c' - \frac12$ there are $t_0 = t_0(\varepsilon, c')$ and $0 < c_-(\varepsilon, c') \leq c_+(\varepsilon, c')$ such that*

$$c_- \leq |\zeta_{\mathrm{RVM}}(\sigma+it)| \leq c_+ \qquad (\tfrac12+\varepsilon \leq \sigma \leq c',\ |t| \geq t_0), \tag{9.9}$$

and therefore, in the same region,

$$c_-|\mathcal{Q}_{\mathrm{RVM}}(s)| \leq |\zeta(s)| \leq c_+|\mathcal{Q}_{\mathrm{RVM}}(s)|, \qquad \frac{1}{c_+|\mathcal{Q}_{\mathrm{RVM}}(s)|} \leq \Big|\frac{1}{\zeta(s)}\Big| \leq \frac{1}{c_-|\mathcal{Q}_{\mathrm{RVM}}(s)|}. \tag{9.10}$$

Consequently:

(i) *the Lindelöf hypothesis holds if and only if $\mathcal{Q}_{\mathrm{RVM}}(\sigma+it) = O_{\sigma,\varepsilon}(|t|^\varepsilon)$ for every $\sigma > \frac12$;*
(ii) *every bound for ζ or $1/\zeta$ on $\frac12+\varepsilon \leq \sigma \leq c'$, $|t| \geq t_0$, is equivalent, up to the constants $c_\pm$, to the same bound for $\mathcal{Q}_{\mathrm{RVM}}$ or $1/\mathcal{Q}_{\mathrm{RVM}}$.*

Proof. By Theorem 8.1(a) with $B = c' - \frac12$, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|\mathcal{E}_0| \leq C_B/|t|$ and, by the uniform form (8.34), $|w| \leq C_B(|t|/2\pi)^{-\varepsilon}$; for $|t| \geq t_0$ this gives $\frac12 \leq |1+w| \leq 2$ and $\frac12 \leq |1+\mathcal{E}_0| \leq 2$, hence (9.9) with $c_- = \varpi_{\mathrm{RVM}}/4$, $c_+ = 4\varpi_{\mathrm{RVM}}$. Then (9.10) follows from $\zeta = \mathcal{Q}_{\mathrm{RVM}}\zeta_{\mathrm{RVM}}$. For (i): the Lindelöf hypothesis is equivalent to $\zeta(\sigma+it) = O(|t|^\varepsilon)$ for every $\sigma > \frac12$ [51, Ch. XIII], and (9.10) transfers this to $\mathcal{Q}_{\mathrm{RVM}}$. (ii) is (9.10). □

On the prime side define

$$\mathfrak{d}^{\mathrm{RVM}}_n := \mathbf{1}_{\{n \text{ prime}\}} - c^{\mathrm{RVM}}_n. \tag{9.11}$$

Since $\pi_{\mathrm{RVM}}(1) = 0$, for every integer $N \geq 2$

$$\sum_{2\leq n\leq N}\mathfrak{d}^{\mathrm{RVM}}_n = \pi(N) - \pi_{\mathrm{RVM}}(N). \tag{9.12}$$

10. The λ-family from ζ_{RVM} to ζ_{Z1}

For $\lambda \in [0,1]$ define the moving ordinates, the family product and the family zeta function

$$\gamma_k(\lambda) := (1-\lambda)\gamma_k + \lambda\tau_k = \gamma_k + \lambda\eta_k, \tag{10.1}$$

$$P_\lambda(s) := \prod_{k\geq 1}\left(1 + \frac{q(s)}{\frac{1}{4}+\gamma_k(\lambda)^2}\right), \tag{10.2}$$

$$\zeta_\lambda(s) := \frac{P_\lambda(s)}{G(s)}, \qquad F_\lambda(s) := \frac{1}{\zeta_\lambda(s)}, \tag{10.3}$$

and the endpoint

$$P_{Z1} := P_0, \qquad \zeta_{Z1} := \zeta_0, \qquad \zeta_1 = \zeta_{\mathrm{RVM}}. \tag{10.4}$$

Theorem 10.1 (The λ-family)**.** *The product* (10.2) *converges absolutely and locally uniformly jointly on* $[0,1]\times\mathbb{C}$*. For every* λ*,* P_λ *is entire of order exactly one,* $P_\lambda(1-s) = P_\lambda(s)$*,* $P_\lambda(\bar{s}) = \overline{P_\lambda(s)}$*,* $P_\lambda(0) = P_\lambda(1) = 1$*, and its zeros, with multiplicity, are exactly*

$$\frac{1}{2} \pm i\gamma_k(\lambda), \qquad k \geq 1. \tag{10.5}$$

For $0 < \lambda \leq 1$ *the ordinates* $\gamma_k(\lambda)$ *are strictly increasing in* k *and all zeros of* P_λ *are simple; at* $\lambda = 0$ *the multiplicities are those of the ordinate list* (γ_k)*. For every* $j \geq 0$*,* $P_\lambda^{(j)} \to P_{\lambda_0}^{(j)}$ *locally uniformly as* $\lambda \to \lambda_0$*. In particular no member* ζ_λ *of the family,* $\zeta_{Z1} = \zeta_0$ *included, can have a zero off the critical line other than* $-2, -4, \ldots$*.*

Proof. By Lemma 2.2, $|\gamma_k(\lambda) - \gamma_k| \leq C_\eta$, and $\gamma_k \asymp k/\log k$; so $\gamma_k(\lambda) \geq \frac{1}{2}\gamma_k$ for $k \geq k_0$, uniformly in λ. On a compact $K \subset \mathbb{C}$,

$$\sum_{k\geq 1}\sup_{s\in K,\ 0\leq\lambda\leq 1}\left|\frac{q(s)}{\frac{1}{4}+\gamma_k(\lambda)^2}\right| < \infty, \tag{10.6}$$

which gives joint normal convergence and, by Cauchy's formula, convergence of all s-derivatives. For $|s| \leq R_{\mathrm{fam}}$, uniformly in λ,

$$\log\max_{|s|\leq R_{\mathrm{fam}}}|P_\lambda(s)| \leq \sum_{k\geq 1}\log\Bigl(1 + \frac{R_{\mathrm{fam}}^2 + R_{\mathrm{fam}}}{\frac{1}{4}+\gamma_k(\lambda)^2}\Bigr) = O(R_{\mathrm{fam}}\log R_{\mathrm{fam}}), \tag{10.7}$$

so the order is at most one; the zero count is $\asymp T\log T$, so the exponent of convergence is one and the order is exactly one. Symmetries, normalization and divisor follow factor by factor as in Theorem 3.1. For $0 < \lambda \leq 1$,

$$\gamma_{k+1}(\lambda) - \gamma_k(\lambda) = (1-\lambda)(\gamma_{k+1}-\gamma_k) + \lambda(\tau_{k+1}-\tau_k) \geq \lambda(\tau_{k+1}-\tau_k) > 0, \tag{10.8}$$

so distinct factors vanish at distinct points and each zero is simple; at $\lambda = 0$ equal ordinates give equal factors. □

Remark 10.2 (All differences lie in $\gamma_k - \tau_k$)*.* For $\lambda, \lambda' \in [0,1]$ and s off the divisors,

$$\frac{\zeta_\lambda(s)}{\zeta_{\lambda'}(s)} = \prod_{k\geq 1}\frac{\bigl(\frac{1}{4}+(\gamma_k+\lambda'\eta_k)^2\bigr)\bigl(\frac{1}{4}+(\gamma_k+\lambda\eta_k)^2+q(s)\bigr)}{\bigl(\frac{1}{4}+(\gamma_k+\lambda\eta_k)^2\bigr)\bigl(\frac{1}{4}+(\gamma_k+\lambda'\eta_k)^2+q(s)\bigr)}. \tag{10.9}$$

The gamma factor cancels; the members differ only through $\eta_k = \tau_k - \gamma_k$, and all of them coincide if every η_k vanishes.

Theorem 10.3 (Infinitesimal generator)**.** *Put* $A_{k,\lambda} := \frac{1}{4} + \gamma_k(\lambda)^2$*. Off the divisor of* P_λ*,*

$$\mathcal{V}_\lambda(s) := \partial_\lambda \log P_\lambda(s) = -2q(s)\sum_{k\geq 1}\frac{\gamma_k(\lambda)\,\eta_k}{A_{k,\lambda}\bigl(A_{k,\lambda}+q(s)\bigr)}, \tag{10.10}$$

the series converging locally uniformly on compact sets separated from the divisor. Since G does not depend on λ,

$$\partial_\lambda \log \zeta_\lambda = \mathcal{V}_\lambda, \qquad \partial_\lambda \log F_\lambda = -\mathcal{V}_\lambda. \tag{10.11}$$

Proof. Since $\partial_\lambda A_{k,\lambda} = 2\gamma_k(\lambda)\eta_k$,

$$\partial_\lambda \log\Big(1 + \frac{q}{A_{k,\lambda}}\Big) = -\frac{2q\,\gamma_k(\lambda)\eta_k}{A_{k,\lambda}(A_{k,\lambda}+q)}. \tag{10.12}$$

On compact sets separated from the divisor the summand is $O_K(|\eta_k|\gamma_k^{-3})$, and $\sum_k \gamma_k^{-3} < \infty$, so termwise differentiation is justified. □

Proposition 10.4 (Integrated multiplier)**.** *On every component of $\mathbb{C} \setminus \bigcup_{\lambda'\in[0,1]}\{\frac12 \pm i\gamma_k(\lambda') : k \ge 1\}$,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s)\exp\Big(\int_0^\lambda \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big), \tag{10.13}$$

$$F_\lambda(s) = F_{Z1}(s)\exp\Big(-\int_0^\lambda \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big), \tag{10.14}$$

and the total multiplier is

$$\Pi_{\mathrm{tr}}(s) := \exp\Big(-\int_0^1 \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big) = \frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}, \qquad \zeta_{Z1} = \zeta_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}. \tag{10.15}$$

The identities extend meromorphically to $\mathbb{C}$ through the product form.

Proof. (10.13) and (10.14) are the fundamental theorem of calculus applied to (10.11). For (10.15), integrate each mode with $r = \gamma_k(\lambda')$, $dr = \eta_k\,d\lambda'$:

$$\int_0^1 \partial_{\lambda'} \log\Big(1 + \frac{q}{A_{k,\lambda'}}\Big)d\lambda' \equiv \log\frac{1 + q/(\frac14 + \tau_k^2)}{1 + q/(\frac14 + \gamma_k^2)} \pmod{2\pi i}. \tag{10.16}$$

Normal convergence permits summation, and exponentiation removes the branch ambiguity. □

Corollary 10.5 (Expansion at the endpoint)**.** *Let $K \subset \mathbb{C}$ be compact, disjoint from the zeros of P_{Z1} and from $s = 1$ (equivalently one may state the expansion for the product ratio P_λ/P_{Z1} without this exclusion). Uniformly for $s \in K$, as $\lambda \downarrow 0$,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s)\big(1 + \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big), \tag{10.17}$$

$$F_\lambda(s) = F_{Z1}(s)\big(1 - \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big). \tag{10.18}$$

Proof. By Theorem 10.1 the zeros of P_λ converge to those of P_{Z1}, so K avoids the moving divisor for small λ. The series for $\mathcal{V}_\lambda$ and its λ-derivative converge normally on K (the differentiated summands are $O_K(\gamma_k^{-3})$), so $\mathcal{V}_\lambda = \mathcal{V}_0 + O_K(\lambda)$, and (10.13) gives $\log(\zeta_\lambda/\zeta_{Z1}) = \lambda\mathcal{V}_0 + O_K(\lambda^2)$. □

Proposition 10.6 (Residues along the family)**.** *Suppose γ_k is a simple ordinate (it occurs once in the list). Put $\rho_{k,\lambda} = \frac12 + i\gamma_k(\lambda)$ and, for $0 < \lambda \le 1$,*

$$\mathfrak{r}_{k,\lambda} := \operatorname*{Res}_{s=\rho_{k,\lambda}} \frac{1}{s\,\zeta_\lambda(s)} = \frac{G(\rho_{k,\lambda})}{\rho_{k,\lambda}P'_\lambda(\rho_{k,\lambda})}. \tag{10.19}$$

Then

$$\mathfrak{r}_{k,\lambda} \longrightarrow \frac{G(\frac12 + i\gamma_k)}{(\frac12 + i\gamma_k)P'_{Z1}(\frac12 + i\gamma_k)} \qquad (\lambda \downarrow 0). \tag{10.20}$$

Proof. Theorem 10.1 gives convergence of the zero locations and local uniform convergence of P'_λ; the limit denominator is nonzero because the endpoint zero is simple, and G is continuous. □

Remark 10.7 (Multiplicity). For $0 < \lambda \le 1$ every zero of ζ_λ is simple. At $\lambda = 0$ a repeated ordinate of ζ appears in ζ_{Z1} with the same multiplicity, and the residues of Proposition 10.6 are replaced by the full principal part. Multiplicity plays no role in Step 3: Theorem 11.8 uses only that ζ does not vanish in $\Re s > \frac{1}{2}$.

11. The endpoint ζ_{Z1}, Hamburger's theorem, and Step 3

11.1. The analytic class of the endpoint.

Proposition 11.1 (Analytic class of ζ_{Z1})**.** *The endpoint*

$$P_{Z1}(s) = \prod_{k\ge 1} \left(1 + \frac{q(s)}{\frac{1}{4} + \gamma_k^2}\right), \qquad \zeta_{Z1}(s) = \frac{P_{Z1}(s)}{G(s)}, \tag{11.1}$$

has the following properties.

(i) *P_{Z1} is entire of order exactly one, $P_{Z1}(1-s) = P_{Z1}(s)$, $P_{Z1}(0) = P_{Z1}(1) = 1$;*

(ii) *the nontrivial zeros of ζ_{Z1} are exactly*

$$\frac{1}{2} \pm i\gamma_k, \qquad k \ge 1, \tag{11.2}$$

with the multiplicities of the ordinate list; its other zeros are $s = -2, -4, \dots$. It is impossible for ζ_{Z1} to vanish at any point off the critical line other than $-2, -4, \dots$: every factor of P_{Z1} vanishes only at $\frac{1}{2} \pm i\gamma_k$;

(iii) *ζ_{Z1} has a simple pole at $s = 1$ with residue 1;*

(iv) *ζ_{Z1} satisfies the functional equation of ζ:*

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta_{Z1}(1-s); \tag{11.3}$$

(v) *$(s-1)\zeta_{Z1}(s)$ is entire of order exactly one.*

Proof. (i) and the divisor in (ii) are the case $\lambda = 0$ of Theorem 10.1; the trivial zeros and the absence of other zeros come from $1/G$ as in Theorem 3.1. For (iii), $P_{Z1}(1) = 1$ and

$$G(s) = (s-1) + O\big((s-1)^2\big) \qquad (s \to 1). \tag{11.4}$$

For (iv),

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s) = \frac{P_{Z1}(s)}{q(s)}, \tag{11.5}$$

and both P_{Z1} and q are invariant under $s \mapsto 1-s$. For (v),

$$(s-1)\zeta_{Z1}(s) = P_{Z1}(s)\,\frac{\pi^{s/2}}{s\,\Gamma(s/2)}, \tag{11.6}$$

where $s\Gamma(s/2) \to 2$ at $s = 0$ and $1/\Gamma$ is entire of order one; the product has order at most one, and the zero count $\asymp T\log T$ gives order at least one. □

Define the ratio

$$\mathscr{R}_{Z1}(s) := \frac{\zeta_{Z1}(s)}{\zeta(s)} = \frac{P_{Z1}(s)}{P_\zeta(s)}, \tag{11.7}$$

a meromorphic function; $\mathscr{R}_{Z1} \equiv 1$ is the identity $\zeta_{Z1} = \zeta$.

11.2. **Hamburger's theorem at the endpoint.**

Theorem 11.2 (Hamburger closure)**.** *Consider the condition*

$$\text{(D)} \qquad \zeta_{Z1}(s) = \sum_{n\geq 1} \frac{a_n}{n^s} \qquad \text{absolutely for } \Re s > 1. \tag{11.8}$$

Then

$$\text{(D)} \iff \mathscr{R}_{Z1} \equiv 1 \iff \zeta_{Z1} = \zeta \iff \text{RH}. \tag{11.9}$$

Proof. Assume (D). Then $\zeta_{Z1}(1-s) = \sum_n a_n n^{-(1-s)}$ converges absolutely for $\Re s < 0$, $(s-1)\zeta_{Z1}$ is entire of finite order (Proposition 11.1(v)), and (11.3) holds. These are the hypotheses of Hamburger's theorem [33], [51, §2.13], which gives

$$\zeta_{Z1} = C_H\, \zeta \tag{11.10}$$

for a constant C_H. Both functions have residue 1 at $s = 1$, so $C_H = 1$. If $\zeta_{Z1} = \zeta$, the nontrivial zeros of ζ are those of ζ_{Z1}, namely (11.2), which is RH. If RH holds, every ρ with $\Im\rho > 0$ is $\frac{1}{2} + i\gamma_k$, so $\rho(1-\rho) = \frac{1}{4} + \gamma_k^2$, and Theorem 9.1 gives

$$P_\zeta(s) = \prod_{k\geq 1} \Big(1 + \frac{q(s)}{\frac{1}{4} + \gamma_k^2}\Big) = P_{Z1}(s) \tag{11.11}$$

exactly, hence $\zeta = \zeta_{Z1}$. Finally $\zeta = \sum_n n^{-s}$ converges absolutely for $\Re s > 1$, so $\zeta_{Z1} = \zeta$ gives (D). □

11.3. **Condition (D) and the integer-weighted zeta.** By Corollary 6.3, $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ with first coefficient 1, unconditionally. Define the multiplier

$$\mathcal{K}(s) := \frac{\zeta_{Z1}(s)}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)}. \tag{11.12}$$

Theorem 11.3 (Reduction of (D) to the multiplier)**.** (i) *If $\mathcal{K}$ has an expansion $\mathcal{K}(s) = \sum_{n\geq 1} k_n n^{-s}$ that converges absolutely in $\Re s > 1$, then* (11.8) *holds.*

(ii) *If* (11.8) *holds, then $\mathcal{K}(s) = \sum_{n\geq 1} k_n n^{-s}$ absolutely on the whole half-plane $\Re s > 1$.*

Thus (D) holds if and only if $\mathcal{K}$ is an absolutely convergent Dirichlet series in $\Re s > 1$. Hence, by Theorem 11.2, an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s > 1$ gives $\zeta_{Z1} = \zeta$, RH, and, by Theorem 11.8, $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$.

Proof. (i) $\zeta_{Z1} = \zeta_{\mathbb{Z},\mathrm{RVM}}\,\mathcal{K}$ is the Dirichlet convolution of two absolutely convergent series in $\Re s > 1$, and $\sum_N |(a^{\mathbb{Z}} * k)_N| N^{-\sigma} \leq \sum_N |a^{\mathbb{Z}}_N| N^{-\sigma} \sum_n |k_n| n^{-\sigma} < \infty$. (ii) By (6.15), $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, and so is its product $\mathcal{K}$ with (11.8). □

By (10.15) and Corollary 6.5,

$$\mathcal{K}(s) = \Pi_{\mathrm{tr}}(s)\, e^{-h_{\mathrm{RVM}}(s)} = \frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}\, e^{-h_{\mathrm{RVM}}(s)}, \tag{11.13}$$

an explicit function of the displacements $\gamma_k - \tau_k$ and of the cell masses c_n^{RVM}. Condition (D) is thereby located in a single object built from the two constructions of this paper: the λ-family and the integer weights.

Remark 11.4 (Continuous-to-atomic meaning of (D))*.* The prime-power count J^{M}_{Z1} of ζ_{Z1} (Theorem 12.5) is an L^2_{loc} count built from the zeros, and (D) makes its distributional derivative the atomic measure $\sum_{p,m} m^{-1}\delta_{p^m}$. The final step of the method is therefore the passage of ζ_{Z1} into the discrete Dirichlet category. Theorem 12.7 states it exactly: (D) holds if and only if J^{M}_{Z1} is constant between consecutive integers and its jumps obey $\sum_{n\leq X} |\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$, and Corollary 12.3 reduces it to the single identity (12.12) between the displacements $\gamma_k - \tau_k$ and ϖ_{RVM}.

11.4. **Step 3: the Mertens bound from $\zeta_{Z1} = \zeta$.**

Lemma 11.5 (Vertical growth of ζ)**.** *For $\frac{1}{4} \le \sigma \le 4$ and $|t| \ge 2$,*

$$|\zeta(\sigma + it)| \le 15|t|. \tag{11.14}$$

Proof. For $\sigma > 0$ [51, (2.1.5)],

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{y\} y^{-s-1}\, dy. \tag{11.15}$$

For $|t| \ge 2$ and $\sigma \le 4$ one has $|s| \le 3|t|$, $|s/(s-1)| \le |s|/|t| \le 3$, and the integral is at most $1/\sigma \le 4$; hence $|\zeta(s)| \le 3 + 12|t| \le 15|t|$. □

Lemma 11.6 (Logarithm of ζ off the critical line)**.** *Assume $\zeta(s) \ne 0$ for $\Re s > \frac{1}{2}$. Let $0 < \delta \le \frac{1}{4}$ and*

$$\omega(\delta) := \frac{\log(3-2\delta)}{\log(3-\delta)} < 1. \tag{11.16}$$

Then, with the branch of $\log\zeta$ equal to $\sum_{n\ge2}\Lambda(n)n^{-s}/\log n$ on $\Re s > 1$,

$$|\log\zeta(\sigma+it)| \le \Big(\frac{48}{\delta}\log|t|\Big)^{\omega(\delta)} \qquad (\sigma \ge \tfrac{1}{2}+\delta,\ |t| \ge 4). \tag{11.17}$$

Consequently, for every $\varepsilon > 0$ there is $t_2 = t_2(\delta,\varepsilon)$ with

$$\Big|\frac{1}{\zeta(\sigma+it)}\Big| \le |t|^{\varepsilon} \qquad (\sigma \ge \tfrac{1}{2}+\delta,\ |t| \ge t_2). \tag{11.18}$$

Proof. Fix t with $|t| \ge 4$ and put $s_0 = 2 + it$, $R = \frac{3}{2} - \frac{\delta}{4}$, $r_3 = \frac{3}{2} - \frac{\delta}{2}$. The closed disc $|s - s_0| \le R$ lies in $\frac{1}{2} + \frac{\delta}{4} \le \Re s \le \frac{7}{2}$, $|\Im s - t| \le \frac{3}{2}$; it avoids $s = 1$ and, by hypothesis, every zero of ζ. Hence $\log\zeta$ has a holomorphic branch on a neighbourhood of this disc, chosen to agree with the Dirichlet-series branch on the connected overlap with $\Re s > 1$. Such branches are defined component by component on zero-free regions; no global branch on $\Re s > \frac{1}{2}$ is asserted. On the disc $|\Im s| \ge |t|/2 \ge 2$ and $|\Im s| \le 2|t|$, so Lemma 11.5 gives $\Re\log\zeta = \log|\zeta| \le \log(30|t|)$ there, while $|\log\zeta(s_0)| \le \log\zeta(2)$. The Borel–Carathéodory lemma [50, §5.5], applied with radii $r_3 < R$,

$$\max_{|s-s_0|\le r_3} |\log\zeta(s)| \le \frac{2r_3}{R-r_3}\max_{|s-s_0|=R}\Re\log\zeta(s) + \frac{R+r_3}{R-r_3}|\log\zeta(s_0)|, \tag{11.19}$$

together with $R - r_3 = \delta/4$, $2r_3 \le 3$, $R + r_3 \le 3$ and $\log 30 + \log\zeta(2) < 4 \le 3\log|t|$, gives

$$\max_{|s-s_0|\le r_3} |\log\zeta(s)| \le \frac{12}{\delta}\big(\log(30|t|) + \log\zeta(2)\big) \le \frac{48}{\delta}\log|t|. \tag{11.20}$$

Now let $\frac{1}{2} + \delta \le \sigma < \frac{3}{2}$ and set $r_1 = \frac{1}{2}$, $r_2 = 2 - \sigma \in (\frac{1}{2}, \frac{3}{2} - \delta]$. On $|s - s_0| = r_1$ one has $\Re s \ge \frac{3}{2}$, so $|\log\zeta(s)| \le \log\zeta(\frac{3}{2}) < 1$. Hadamard's three-circle theorem [50, §5.3] gives, with $M_j = \max_{|s-s_0|=r_j}|\log\zeta|$,

$$M_2 \le M_1^{1-\omega} M_3^{\omega}, \qquad \omega = \frac{\log(r_2/r_1)}{\log(r_3/r_1)} \le \frac{\log(3-2\delta)}{\log(3-\delta)} = \omega(\delta). \tag{11.21}$$

Since $M_1 < 1$ and $M_3 \le \frac{48}{\delta}\log|t|$ with $\frac{48}{\delta}\log|t| > 1$, (11.21) gives (11.17) at $s = \sigma + it$, which lies on the circle of radius r_2. For $\sigma \ge \frac{3}{2}$, $|\log\zeta| \le \log\zeta(\frac{3}{2}) < 1$. Finally $|1/\zeta| = \exp(-\Re\log\zeta) \le \exp|\log\zeta|$, and $(\frac{48}{\delta}\log|t|)^{\omega(\delta)} \le \varepsilon\log|t|$ for $|t| \ge t_2(\delta,\varepsilon)$ because $\omega(\delta) < 1$. □

Lemma 11.7 (Perron kernel)**.** *For $y > 0$, $y \ne 1$, $c > 0$, $T > 0$ [22, Ch. 17],*

$$\left|\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{y^s}{s}\,ds - \mathbf{1}_{\{y>1\}}\right| \le y^c \min\Big(1, \frac{1}{T|\log y|}\Big). \tag{11.22}$$

Theorem 11.8 (Step 3)**.** *If $\zeta_{Z1} = \zeta$, then for every $\varepsilon > 0$*

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (x \geq 1). \tag{11.23}$$

Proof. Non-vanishing. By Proposition 11.1(ii), ζ_{Z1} has no zeros in $\Re s > \frac{1}{2}$; since $\zeta = \zeta_{Z1}$, neither has ζ, and Lemma 11.6 applies.

Truncation. Let $x = N + \frac{1}{2}$ with $N \geq 3$ an integer, $c = 1 + 1/\log x$ and $T = x$. Applying (11.22) with $y = x/n$ to each term of $\sum_n \mu(n)n^{-s} = 1/\zeta(s)$, which converges absolutely on $\Re s = c$, gives

$$M(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\,ds + O\Big(\sum_{n\geq 1}\Big(\frac{x}{n}\Big)^c \min\Big(1, \frac{1}{T|\log(x/n)|}\Big)\Big). \tag{11.24}$$

For $n \leq x/2$ or $n \geq 2x$, $|\log(x/n)| \geq \log 2$, and these terms contribute at most $x^c\zeta(c)/(T\log 2) \leq e\,x(1+\log x)/(T\log 2)$. For $x/2 < n < 2x$, $(x/n)^c \leq 4$ and $|\log(x/n)| \geq |n-x|/(2x)$; as $|n-x| \in \{j - \frac{1}{2} : j \geq 1\}$, each value taken at most twice, these terms contribute at most $8\sum_{1\leq j\leq 2x}\min(1, 4x/(T(2j-1))) \ll 1 + (x/T)\log x$. Hence

$$M(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\,ds + O(\log x). \tag{11.25}$$

Contour shift. Fix $0 < \delta \leq \frac{1}{4}$. The integrand is holomorphic on the closed rectangle with vertices $c \pm iT$, $\frac{1}{2} + \delta \pm iT$: ζ has no zeros there, $s = 1$ is a zero of $1/\zeta$, and $s = 0$ lies outside. By Cauchy's theorem

$$\int_{c-iT}^{c+iT} = \int_{\frac{1}{2}+\delta-iT}^{\frac{1}{2}+\delta+iT} + \int_{\frac{1}{2}+\delta+iT}^{c+iT} - \int_{\frac{1}{2}+\delta-iT}^{c-iT}. \tag{11.26}$$

For $x \geq t_2(\delta,\varepsilon)$, (11.18) on $|t| = T = x$ bounds the horizontal integrals by

$$\frac{1}{2\pi}\int_{\frac{1}{2}+\delta}^{c} \frac{x^\sigma T^\varepsilon}{T}\,d\sigma \leq x^c T^{\varepsilon-1} = e\,x^\varepsilon. \tag{11.27}$$

On the left side, $1/\zeta$ is continuous on the compact segment $\{\frac{1}{2} + \delta + it : |t| \leq t_2\}$, bounded by a constant $C(\delta,\varepsilon)$, and (11.18) holds for $|t| \geq t_2$; hence

$$\frac{1}{2\pi}\int_{-T}^{T} \frac{x^{1/2+\delta}}{|\frac{1}{2}+\delta+it|\,|\zeta(\frac{1}{2}+\delta+it)|}\,dt \leq x^{1/2+\delta}\Big(C'(\delta,\varepsilon) + \int_{t_2}^{T} t^{\varepsilon-1}dt\Big) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}. \tag{11.28}$$

Conclusion. (11.25)–(11.28) give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$ for $x = N + \frac{1}{2}$; taking $\delta = \varepsilon = \varepsilon_0/2$ (with $\varepsilon_0 \leq \frac{1}{2}$) gives $O_{\varepsilon_0}(x^{1/2+\varepsilon_0})$. For arbitrary $x \geq 3$, $M(x) = M(\lfloor x\rfloor + \frac{1}{2})$ and $\lfloor x\rfloor + \frac{1}{2} \leq 2x$; for $1 \leq x < 3$, $|M(x)| \leq 2$. □

Theorem 11.9 (Transfer of the synthetic Mertens bound to M)**.** *Assume $\zeta_{Z1} = \zeta$ and let $0 < \delta \leq \frac{1}{4}$, $\varepsilon > 0$. Then:*

(i) *on $\Re s > \frac{1}{2}$,*

$$\frac{1}{\zeta(s)} = F_{\mathrm{RVM}}(s)\,\Pi_{\mathrm{tr}}(s)^{-1}, \qquad \Pi_{\mathrm{tr}} = \frac{P_{Z1}}{P_{\mathrm{RVM}}^{\mathrm{crit}}} = \frac{\zeta}{\zeta_{\mathrm{RVM}}}, \tag{11.29}$$

where Π_{tr} is holomorphic and zero-free in $\Re s > \frac{1}{2}$ and $\Pi_{\mathrm{tr}}(1) = 1$;

(ii) *there is $t_3 = t_3(\delta,\varepsilon)$ such that, for $\frac{1}{2} + \delta \leq \sigma \leq 2$ and $|t| \geq t_3$,*

$$|F_{\mathrm{RVM}}(\sigma+it)| \leq 4L, \qquad |\Pi_{\mathrm{tr}}(\sigma+it)^{\pm 1}| \leq |t|^\varepsilon; \tag{11.30}$$

(iii) *the following classical consequence of RH is recovered at the endpoint:*

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{11.31}$$

Proof. (i) By (10.15), $\zeta_{Z1} = \zeta_{\mathrm{RVM}}\Pi_{\mathrm{tr}}$; with $\zeta_{Z1} = \zeta$ this is (11.29). In $\Re s > \frac{1}{2}$, $\zeta = \zeta_{Z1}$ has no zeros (Proposition 11.1(ii)) and ζ_{RVM} has none (Theorem 3.1); both have a simple pole of residue 1 at $s = 1$, so Π_{tr} is holomorphic there with $\Pi_{\mathrm{tr}}(1) = 1$, and it is zero-free in $\Re s > \frac{1}{2}$.

(ii) By Theorem 8.1(a) with $B = \frac{3}{2}$, for $\delta \le \sigma' \le \frac{3}{2}$ and $|t| \ge t_0$ we have $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|w| \le C|t|^{-\delta} \le \frac{1}{2}$ and $|\mathcal{E}_0| \le \frac{1}{2}$ for $|t|$ large; hence $|F_{\mathrm{RVM}}| \le 4L$ and

$$\log \zeta_{\mathrm{RVM}} = \log \varpi_{\mathrm{RVM}} + \mathrm{Log}(1+w) + \mathrm{Log}(1+\mathcal{E}_0) = O(1). \tag{11.32}$$

By Lemma 11.6, which applies because $\zeta = \zeta_{Z1}$ has no zeros in $\Re s > \frac{1}{2}$, $|\log \zeta(\sigma+it)| \le (\frac{48}{\delta}\log|t|)^{\omega(\delta)}$ with $\omega(\delta) < 1$. Therefore

$$|\log \Pi_{\mathrm{tr}}(\sigma+it)| = |\log\zeta - \log\zeta_{\mathrm{RVM}}| \le \Big(\frac{48}{\delta}\log|t|\Big)^{\omega(\delta)} + O(1) \le \varepsilon \log|t| \tag{11.33}$$

for $|t| \ge t_3(\delta,\varepsilon)$, which gives (11.30).

(iii) The required estimate for $1/\zeta$ already follows from the zero-free hypothesis $\zeta = \zeta_{Z1}$ through Lemma 11.6 and Theorem 11.8. The factorization (11.29) is therefore a reformulation of the same estimate, not an independent transfer from the synthetic Mertens bound. By (i)–(ii), Π_{tr}^{-1} is holomorphic in $\Re s > \frac{1}{2}$ and grows at most like $|t|^{\varepsilon}$. Hence $|1/\zeta(\sigma+it)| \le 4L|t|^{\varepsilon}$ for $\frac{1}{2}+\delta \le \sigma \le 2$, $|t| \ge t_3$, and the truncated Perron formula, contour shift and estimates (11.25)–(11.28) of Theorem 11.8, applied with this bound, give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$. Choosing $\delta = \varepsilon = \varepsilon_0/2$ gives (11.31). □

Remark 11.10 (The transfer in Mellin form). Formally, (11.29) reads $M = (\Pi_{\mathrm{tr}}^{-1})^{\vee} *_{\mathrm{Mel}} M_{\mathrm{RVM}}^{\mathrm{crit}}$: the Mertens function is the synthetic Mertens function convolved with the inverse Mellin transform of the inverse transport multiplier. The synthetic factor F_{RVM} contributes the exponent $\frac{1}{2}$ and is bounded (Theorem 8.1); the multiplier contributes at most $|t|^{\varepsilon}$. All the information of the λ-family enters through $\Pi_{\mathrm{tr}} = P_{Z1}/P_{\mathrm{RVM}}^{\mathrm{crit}}$, that is, through the displacements $\gamma_k - \tau_k$.

Corollary 11.11 (Step 3 through (D))**.** *Condition* (11.8) *implies* $M(x) = O_{\varepsilon}(x^{1/2+\varepsilon})$ *for every* $\varepsilon > 0$.

Proof. Theorem 11.2 and Theorem 11.8. □

Remark 11.12 (Status of Step 3). The implication $\zeta_{Z1} = \zeta \Rightarrow M(x) = O_{\varepsilon}(x^{1/2+\varepsilon})$ is the classical consequence RH $\Rightarrow M(x) = O_{\varepsilon}(x^{1/2+\varepsilon})$ (see [51, §14.25]) rewritten and reproved in the endpoint notation. It requires nothing beyond the identity: no separate statement about the zeros of ζ enters, because $\zeta = \zeta_{Z1}$ already carries the divisor (11.2). The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), equivalently an absolutely convergent Dirichlet expansion of the multiplier $\mathcal{K}$ of (11.12) in $\Re s > 1$ (Theorem 11.3), is not proved in this paper. It is the open step of the method.

11.5. **Reciprocal endpoint and finite-height Perron integrals.** Put

$$F_{Z1}(s) := \frac{1}{\zeta_{Z1}(s)} = F_{\mathrm{RVM}}(s)\,\Pi_{\mathrm{tr}}(s)^{-1}, \tag{11.34}$$

the second identity by (10.15). For $c > 1$ and finite U define

$$M_{\lambda,U}(x) := \frac{1}{2\pi i}\int_{c-iU}^{c+iU} F_{\lambda}(s)\frac{x^s}{s}\,ds, \qquad M_{Z1,U} := M_{0,U}. \tag{11.35}$$

Proposition 11.13 (Finite-height Perron integrals along the family)**.** *For fixed* $c > 1$, $U < \infty$ *and* $x > 0$,

$$M_{\lambda,U}(x) \longrightarrow M_{Z1,U}(x) \qquad (\lambda \downarrow 0). \tag{11.36}$$

Proof. The segment $\{c+it : |t| \le U\}$ is compact and disjoint from every divisor (10.5), and $P_{\lambda} \to P_{Z1}$ uniformly on it, so $F_{\lambda} \to F_{Z1}$ uniformly there. □

Theorem 11.14 (Reciprocal-coefficient criterion)**.** *Assume that in* $\Re s > 1$

$$F_{Z1}(s) = \sum_{n\geq 1} \frac{\mu_{Z1}(n)}{n^s} \tag{11.37}$$

absolutely, and let $M_{Z1}(x) = \sum_{n\leq x} \mu_{Z1}(n)$*, so that*

$$\mu_{Z1}(n) = M_{Z1}(n) - M_{Z1}(n-1). \tag{11.38}$$

Then the following are equivalent: (i) $M_{Z1}(n) = M(n)$ *for every* $n \geq 1$*; (ii)* $\mu_{Z1} = \mu$*; (iii)* $F_{Z1} = 1/\zeta$ *in* $\Re s > 1$*; (iv)* $\zeta_{Z1} = \zeta$*; (v) RH.*

Proof. (i)⇔(ii) by (11.38); (ii)⇔(iii) by uniqueness of absolutely convergent Dirichlet coefficients; (iii)⇔(iv) by analytic continuation, both functions being nonzero in $\Re s > 1$; (iv)⇔(v) by Theorem 11.2. □

Remark 11.15 (A route that does not close Step 3). The following approach does not work. One may try to expand both factors of $F_{Z1} = F_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}^{-1}$ in (11.34) as absolutely convergent Dirichlet series and identify their Dirichlet convolution with μ. By Theorem 8.11 the expansion of F_{RVM} does not exist in any half-plane, so there are no coefficients to convolve. The endpoint criterion concerns ζ_{Z1} itself, as in Theorem 11.2.

12. The endpoint at infinity and the Euler products

This section applies the clock decomposition of Theorem 8.10 to the ordinates γ_k of the endpoint, and constructs the prime-power counts of ζ_{RVM} and ζ_{Z1} as Mellin transforms of their logarithms. No hypothesis is used.

12.1. **The value of the endpoint at infinity.** With the zero count N_ζ of (2.2) put

$$\mathfrak{e}(\tau) := N_\zeta(\tau) - \mathcal{N}_{\mathrm{RVM}}(\tau), \qquad \mathfrak{e}_1(\tau) := \int_{\tau_0}^{\tau} \mathfrak{e}(v)\,dv \qquad (\tau \geq \tau_0 = 4\pi). \tag{12.1}$$

By (2.2), $\mathfrak{e}(\tau) = S(\tau) + O(1/\tau)$, and Littlewood's bound $\int_0^T S(t)\,dt = O(\log T)$ [51, Thm. 9.9] gives an absolute constant $C_{\mathfrak{e}}$ with

$$|\mathfrak{e}_1(\tau)| \leq C_{\mathfrak{e}} \log \tau \qquad (\tau \geq \tau_0). \tag{12.2}$$

For the clock the corresponding primitive is $-\mathfrak{s}_1$, bounded by Lemma 8.9.

Theorem 12.1 (Clock decomposition of the endpoint)**.** *Let* $s = \frac{1}{2} + z$*,* $z = \sigma' + it$*,* $\sigma' > 0$*. There is an absolute constant* $R_{\mathrm{end}} > 0$ *such that*

$$\log \zeta_{Z1}(s) = \log \varpi_{Z1} - \int_{\tau_0}^{\infty} \mathfrak{e}_1(\tau)\Big(\frac{1}{(\tau - iz)^2} + \frac{1}{(\tau + iz)^2}\Big)d\tau + O\Big(\frac{1}{|z|}\Big) \qquad (|z| \geq R_{\mathrm{end}}), \tag{12.3}$$

where

$$\varpi_{Z1} := P_{Z1}(\tfrac{1}{2}) \exp\Big(\nu_3 + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau)}{\tau^2}\,d\tau\Big) > 0. \tag{12.4}$$

Consequently:

(i) *for* $\sigma' > 0$ *and* $|z| \geq R_{\mathrm{end}}$*,*

$$\big|\log \zeta_{Z1}(s) - \log \varpi_{Z1}\big| \leq \frac{2(\pi+2)C_{\mathfrak{e}}}{\sigma'}\Big(1 + \log(2|t| + 2\sigma' + \tau_0)\Big) + \frac{C}{|z|}; \tag{12.5}$$

(ii) $\zeta_{Z1}(\sigma) \to \varpi_{Z1}$ *as* $\sigma \to +\infty$*;*

(iii) *the value at infinity is the ordered product*

$$\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_{k\geq 1} \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k^2} := \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\leq K} \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k^2}. \tag{12.6}$$

Proof. As in (8.6), $\log P_{Z1}(s) - \log P_{Z1}(\frac12) = \sum_k f(\gamma_k)$ with $f(\tau) = \operatorname{Log}(1 + z^2/\tau^2)$. Since $\gamma_1 > \tau_0$, $N_\zeta(\tau_0) = 0$ and $\mathfrak{e}(\tau_0) = -\mathcal{N}_{\mathrm{RVM}}(\tau_0)$. Stieltjes integration against $N_\zeta = \mathcal{N}_{\mathrm{RVM}} + \mathfrak{e}$ and two integrations by parts ($f\mathfrak{e} \to 0$ and $\mathfrak{e}_1 f' \to 0$ at ∞, $\mathfrak{e}_1(\tau_0) = 0$) give

$$\sum_k f(\gamma_k) = \Lambda_{\mathrm{sm}}(z) + \int_{\tau_0}^{\infty} \mathfrak{e}_1 f''\, d\tau, \tag{12.7}$$

with the same Λ_{sm} as in (8.9). For $\sigma' > \frac12$ the proof of Theorem 8.10 gives $\Lambda_{\mathrm{sm}} - \log G = \nu_3 + O(|z|^{-1})$; for $0 < \sigma' \le \frac12$ the same identity follows from Lemma 8.2 by analytic continuation in the zero-free component. Thus, inserting f'' from (8.41) gives (12.3).

For (i), $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$. For real a, splitting at $X = 2|a| + 2\sigma' + \tau_0$ and using $(\tau - a)^2 \ge \tau^2/4$ for $\tau \ge X$,

$$\int_{\tau_0}^{\infty} \frac{\log \tau}{(\tau - a)^2 + \sigma'^2}\, d\tau \le \frac{\pi}{\sigma'} \log X + 4\,\frac{1 + \log X}{X} \le \frac{\pi + 2}{\sigma'}(1 + \log X). \tag{12.8}$$

With (12.2) this gives (12.5). For (ii) take $t = 0$: the right side of (12.5) tends to 0 as $\sigma' \to \infty$.

For (iii), Theorem 8.10 gives $\varpi_{\mathrm{RVM}} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)\exp(\nu_3 - 2\int \mathfrak{s}_1 \tau^{-2} d\tau)$, so

$$\frac{\varpi_{Z1}}{\varpi_{\mathrm{RVM}}} = \frac{P_{Z1}(\frac12)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)} \exp\Big(2\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau) + \mathfrak{s}_1(\tau)}{\tau^2}\, d\tau\Big). \tag{12.9}$$

Since $\mathfrak{e} + \psi(\mathcal{N}_{\mathrm{RVM}} + \frac12) = N_\zeta - n_{\mathrm{RVM}}$ and $(\mathfrak{e}_1 + \mathfrak{s}_1)(X)/X \to 0$,

$$\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1 + \mathfrak{s}_1}{\tau^2}\, d\tau = \lim_{X\to\infty} \int_{\tau_0}^{X} \frac{N_\zeta(\tau) - n_{\mathrm{RVM}}(\tau)}{\tau}\, d\tau = \lim_{K\to\infty} \sum_{k \le K} \log \frac{\tau_k}{\gamma_k}. \tag{12.10}$$

In the last step, $\int_{\tau_0}^{X} (N_\zeta - n_{\mathrm{RVM}})\tau^{-1} d\tau = \sum_{\gamma_k \le X} \log \frac{X}{\gamma_k} - \sum_{\tau_k \le X} \log \frac{X}{\tau_k}$; the indices counted by only one of the two sums are $O(\log X)$ in number and, by Lemma 2.2, have ordinates in $[X - C_\eta, X]$, so they contribute $O(\log X/X)$. Finally $P_{Z1}(\frac12)/P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12) = \prod_k \frac{\gamma_k^2}{\frac14 + \gamma_k^2} \cdot \frac{\frac14 + \tau_k^2}{\tau_k^2}$, an absolutely convergent product, and multiplying by $\prod_{k \le K} \tau_k^2/\gamma_k^2$ gives (12.6). □

The ordinates enter ϖ_{Z1} only through the displacements $\gamma_k - \tau_k$, in agreement with Remark 10.2.

Theorem 12.2 (The value at infinity decides the endpoint identity)**.** *$\varpi_{Z1} \ge 1$, and $\varpi_{Z1} = 1$ if and only if $\zeta_{Z1} = \zeta$.*

Proof. Write $\zeta_{Z1} = \zeta\,\mathscr{R}_{Z1}$ with $\mathscr{R}_{Z1} = P_{Z1}/P_\zeta$ as in (11.7), and compare the products factor by factor, with the zeros $\rho_k = \beta_k + i\gamma_k$ listed with the ordinates (Theorem 9.1). Factors with $\rho_k(1 - \rho_k) = \frac14 + \gamma_k^2$ cancel. The remaining factors of P_ζ come in pairs $\{\rho, 1 - \bar\rho\}$ with the same ordinate γ; with $d_\rho = \Re\rho - \frac12$ and $W_\rho = (s - \frac12)^2 + \gamma^2$, the corresponding factor of $\mathscr{R}_{Z1}$ is

$$\frac{W_\rho^2}{(W_\rho - d_\rho^2)^2 + 4d_\rho^2\gamma^2} \cdot c_\gamma, \qquad c_\gamma = \frac{|\rho(1-\rho)|^2}{(\frac14 + \gamma^2)^2} = 1 + \frac{2d_\rho^2(\gamma^2 - \frac14) + d_\rho^4}{(\frac14 + \gamma^2)^2} > 1. \tag{12.11}$$

For real $\sigma > 1$, $W_\rho \ge \gamma^2 \ge \gamma_1^2$, and the logarithm of the first factor is at most $14d_\rho^2/W_\rho \le 14d_\rho^2/\gamma^2$ in modulus; since $|d_\rho| \le \frac12$ and $\sum \gamma^{-2} < \infty$, dominated convergence and $\zeta(\sigma) \to 1$ give $\varpi_{Z1} = \lim_{\sigma\to\infty} \zeta_{Z1}(\sigma) = \prod c_\gamma \ge 1$, with equality exactly when there are no such pairs, that is, when $\mathscr{R}_{Z1} \equiv 1$. □

Corollary 12.3 (The endpoint identity as an identity of the displacements)**.** *The following are equivalent: (D); $\zeta_{Z1} = \zeta$; $\varpi_{Z1} = 1$;*

$$\lim_{K\to\infty} \prod_{k \le K} \frac{\frac14 + \gamma_k^2}{\frac14 + \tau_k^2} = \varpi_{\mathrm{RVM}} \tag{12.12}$$

(ordered product); and $\lim_{\sigma\to+\infty} \Pi_{\mathrm{tr}}(\sigma) = L = 1/\varpi_{\mathrm{RVM}}$.

Proof. Theorems 11.2, 12.1 and 12.2, and $\Pi_{\mathrm{tr}} = \zeta_{Z1}/\zeta_{\mathrm{RVM}} \to \varpi_{Z1}/\varpi_{\mathrm{RVM}}$ by Theorems 8.10 and 12.1(ii). □

Proposition 12.4 (Boundedness of the endpoint)**.** *Let $\sigma_0 > 1$. Then $|\zeta_{Z1}|$ is bounded above and below on $\Re s \ge \sigma_0$ if and only if*

$$\sup_{\sigma' \ge \sigma_0 - \frac{1}{2},\ t \ge 0} \Big| \Re \int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau) + \mathfrak{s}_1(\tau)}{(\tau - t + i\sigma')^2}\, d\tau \Big| < \infty. \tag{12.13}$$

Proof. $\tau + iz = \tau - t + i\sigma'$. For $t \ge 0$ the term with $(\tau - iz)^{-2}$ in (12.3) is bounded by (12.2), since $|\tau - iz| \ge \tau + t$; the separate integral $\int_{\tau_0}^{\infty} \mathfrak{s}_1(\tau)(\tau + iz)^{-2} d\tau$ is bounded directly by Lemma 8.9 and $\int_{\tau_0}^{\infty} |\tau + iz|^{-2} d\tau \ll 1/\sigma'$. By (12.3), $\log|\zeta_{Z1}|$ is bounded on $\Re s \ge \sigma_0$, $t \ge 0$, exactly when (12.13) holds, and $\zeta_{Z1}(\bar{s}) = \overline{\zeta_{Z1}(s)}$ covers $t < 0$. □

The function in (12.13) is, after one integration by parts, the conjugate Poisson integral at distance σ' of the count difference $N_\zeta - n_{\mathrm{RVM}}$; it depends only on the displacements $\gamma_k - \tau_k$.

12.2. The Euler products of ζ_{RVM} and ζ_{Z1}.

Theorem 12.5 (Prime-power counts as Mellin transforms)**.** *Let $X \in \{\mathrm{RVM}, Z1\}$. There is a unique function J_X^{M} on $(1,\infty)$ with $J_X^{\mathrm{M}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$ and*

$$\log \zeta_X(s) = s \int_1^{\infty} J_X^{\mathrm{M}}(x)\, x^{-s-1}\, dx \qquad (\Re s > 1), \tag{12.14}$$

the branch of $\log\zeta_X$ being real on $(1,\infty)$. We use the notation $\int x^{-s}\, dJ_X^{\mathrm{M}} := s\int_1^{\infty} J_X^{\mathrm{M}}(x)x^{-s-1}dx$ distributionally; no bounded-variation representative is asserted. With this convention $\zeta_X(s) = \exp\big(\int x^{-s}\, dJ_X^{\mathrm{M}}(x)\big)$.

Proof. ζ_X has no zeros or poles in $\Re s > 1$, so $\log\zeta_X$ is holomorphic there. For $c > 1$, Theorem 8.10 ($X = \mathrm{RVM}$) and Theorem 12.1 ($X = Z1$), together with continuity on the compact part, give $|\log\zeta_X(s)| \le C_c(1 + \log(|t| + 2))$ on $\Re s \ge c$. Hence $F_\sigma(t) := \log\zeta_X(\sigma + it)/(\sigma + it)$ lies in $L^2(\mathbb{R})$ with $\|F_\sigma\|_2 \le B_c$ for all $\sigma \ge c$. In the variable $u = \log x$, define $J_X^{\mathrm{M}}(e^u)e^{-\sigma u}$ as the inverse Fourier–Plancherel transform of F_σ; Cauchy's theorem and the decay of F on horizontal segments show that J_X^{M} does not depend on σ. Plancherel gives $\int_{\mathbb{R}} |J_X^{\mathrm{M}}(e^u)|^2 e^{-2\sigma u} du = \|F_\sigma\|_2^2/2\pi \le B_c^2/2\pi$ for all $\sigma \ge c$; letting $\sigma \to \infty$ forces $J_X^{\mathrm{M}}(e^u) = 0$ for almost every $u < 0$. Plancherel again gives (12.14), and uniqueness follows from the injectivity of the Fourier transform. □

Theorem 12.6 (Distributional explicit formula)**.** *Let $X \in \{\mathrm{RVM}, Z1\}$, with ordinates $\kappa_k = \tau_k$ for $X = \mathrm{RVM}$ and $\kappa_k = \gamma_k$ for $X = Z1$. In $\mathcal{D}'(1,\infty)$,*

$$(\log x)\, \frac{dJ_X^{\mathrm{M}}}{dx} = 1 - \frac{2}{\sqrt{x}}\, \Re \sum_{k \ge 1} x^{i\kappa_k} - \frac{1}{x(x^2 - 1)}, \tag{12.15}$$

the series converging in $\mathcal{D}'(1,\infty)$. The explicit-formula count

$$J_X(x) := \mathrm{li}(x) - 2\Re \sum_{k \ge 1} \mathrm{Ei}\Big(\big(\tfrac{1}{2} + i\kappa_k\big) \log x \Big) - \log 2 + \int_x^{\infty} \frac{dy}{y(y^2 - 1)\log y} \tag{12.16}$$

converges in $\mathcal{D}'(1,\infty)$ (J_{RVM} is the function (5.2)), and $J_X^{\mathrm{M}} - J_X$ is a constant C_X on $(1,\infty)$.

Proof. With $\rho_k = \frac{1}{2} + i\kappa_k$, the factorization $\zeta_X = P_X/G$, $P_X'/P_X = \sum_k (\frac{1}{s - \rho_k} + \frac{1}{s - \bar{\rho}_k})$, and the digamma expansion $\frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2}) + \frac{1}{s} = -\frac{\gamma_E}{2} + \sum_{n \ge 1}(\frac{1}{2n} - \frac{1}{s + 2n})$ give, for $\Re s > 1$,

$$-\frac{\zeta_X'}{\zeta_X}(s) = \frac{1}{s - 1} - \sum_{k \ge 1}\Big(\frac{1}{s - \rho_k} + \frac{1}{s - \bar{\rho}_k}\Big) + \sum_{n \ge 1}\Big(\frac{1}{2n} - \frac{1}{s + 2n}\Big) - \frac{\gamma_E}{2} - \frac{1}{2}\log\pi. \tag{12.17}$$

Each term is a Mellin–Stieltjes transform $\int_{[1,\infty)} x^{-s}d\mu$: $dx \mapsto \frac{1}{s-1}$, $x^{\rho-1}dx \mapsto \frac{1}{s-\rho}$, $x^{-2n-1}dx \mapsto \frac{1}{s+2n}$, $\delta_1 \mapsto 1$. After multiplication by x^{-c} and the change $u = \log x$, the Fourier transforms of the partial sums over k and over n converge locally uniformly with polynomial bounds, so the series converge in $\mathcal{S}'$, and on $(1,\infty)$ the limit measure is the right side of (12.15) times dx. Differentiating (12.14) in s shows that $-\zeta_X'/\zeta_X$ is the distributional Mellin transform of $(\log x)\,dJ_X^{\mathrm{M}}$; injectivity gives (12.15). The series over k converges in $\mathcal{D}'(1,\infty)$ because $\int \varphi(x)x^{-1/2+i\kappa}dx$ is rapidly decreasing in κ for $\varphi \in C_c^\infty(1,\infty)$ and $\kappa_k \asymp k/\log k$. By Lemma 5.1, $\mathrm{Ei}(\rho_k u) = e^{\rho_k u}/(\rho_k u) + \varrho_k$ with $\rho_k = \frac{1}{2} + i\kappa_k$ and $\sum_k |\varrho_k|$ locally uniformly convergent, so (12.16) converges in $\mathcal{D}'(1,\infty)$ for the same reason; differentiating termwise with $\frac{d}{dx}\mathrm{Ei}(\rho\log x) = x^{\rho-1}/\log x$, $(\log x)J_X'$ equals the right side of (12.15). A distribution on an interval with zero derivative is constant. □

Proof of Theorem 6.4. Take $X = \mathrm{RVM}$ in Theorems 12.5 and 12.6. By Theorem 5.2, J_{RVM} is continuous on $(1,\infty)$, so $J^{\mathrm{M}}_{\mathrm{RVM}} = J_{\mathrm{RVM}} + C_{\mathrm{RVM}}$ pointwise, and $J^{*}_{\mathrm{RVM}} = J_{\mathrm{RVM}}$ on $[2,\infty)$. □

Theorem 12.7 (Cell form of the endpoint identity)**.** *The following are equivalent.*

(i) *(D), that is, $\zeta_{Z1} = \zeta$.*

(ii) *J^{M}_{Z1} is constant on every interval $(n, n+1)$, $n \geq 1$, and its jumps $\mathsf{b}_n := v_n - v_{n-1}$, where v_n is its value on $(n, n+1)$ and $v_0 := 0$, satisfy $\sum_{n\leq X} |\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$ for every $\varepsilon > 0$.*

(iii) *For every $n \geq 1$, in $\mathcal{D}'(n, n+1)$,*

$$2\,\Re \sum_{k\geq 1} \big(x^{i\gamma_k} - x^{i\tau_k}\big) = \sqrt{x}\,\log x\; J'_{\mathrm{RVM}}(x), \tag{12.18}$$

and the jumps of J^{M}_{Z1} satisfy the bound in (ii).

In this case J^{M}_{Z1} is Riemann's prime-power count $J(x) = \sum_{p^m \leq x} 1/m$.

Proof. (ii)⇒(i): $|v_n| \leq \sum_{m\leq n} |\mathsf{b}_m| \ll n^{1+\varepsilon}$, so for $\Re s > 1$ summation by parts in (12.14) gives $\log \zeta_{Z1}(s) = \sum_n v_n(n^{-s} - (n+1)^{-s}) = \sum_{n\geq 1} \mathsf{b}_n n^{-s}$, absolutely convergent by partial summation. The exponential of an absolutely convergent Dirichlet series is one, so ζ_{Z1} satisfies (11.8).

(i)⇒(ii): by Theorem 11.2, $\zeta_{Z1} = \zeta$, and $\log\zeta(s) = s\int_1^\infty J(x)x^{-s-1}dx$ for $\Re s > 1$; by uniqueness in Theorem 12.5, $J^{\mathrm{M}}_{Z1} = J$, which is constant between consecutive integers, with jumps $1/m$ at p^m and $\sum_{n\leq X} |\mathsf{b}_n| = J(X) \ll X$.

(ii)⇔(iii): Theorem 12.6 for $X = Z1$ and $X = \mathrm{RVM}$, with $dJ^{\mathrm{M}}_{\mathrm{RVM}}/dx = J'_{\mathrm{RVM}}$, gives in $\mathcal{D}'(1,\infty)$

$$(\log x)\Big(\frac{dJ^{\mathrm{M}}_{Z1}}{dx} - J'_{\mathrm{RVM}}(x)\Big) = -\frac{2}{\sqrt{x}}\,\Re\sum_{k\geq 1}\big(x^{i\gamma_k} - x^{i\tau_k}\big), \tag{12.19}$$

so $dJ^{\mathrm{M}}_{Z1}/dx = 0$ on $(n, n+1)$ is (12.18). □

Remark 12.8 (The final step in cell form). Theorem 12.7 is the exact form of the continuous-to-atomic passage of Remark 11.4. Both sides of (12.18) are built from the midpoint construction: the right side from the clock alone (J'_{RVM} is bounded on $[2,\infty)$ by Theorem 6.1, equal to $1/\log x$ plus the unit-periodic first harmonic and the harmonics with stationary points $2\pi x^{1/m}$, Section 6), the left side from the displacements $\gamma_k - \tau_k$. The final step of the method is the identity (12.18) on every cell, equivalently (12.12). Numerically, the explicit formula (12.16) with the γ_k is constant between consecutive integers up to the truncation of the list (Experiment 18.6, Figure 10), and the partial products of (12.12) approach ϖ_{RVM} (Experiment 18.3).

13. The atomized family and the programme for the final step

The continuous family ζ_λ keeps, for every λ, the functional equation and the zeros on the critical line. Its atomization $\tilde{\zeta}_{\mathbb{Z},\lambda}$, built from the cell masses of the prime coordinate, keeps the Euler product and the Dirichlet series. The two are joined by a factor $e^{\tilde{h}_\lambda}$, and both families have a limit at

$\lambda = 0$. Hamburger's theorem needs the functional equation and the Dirichlet series on one and the same function; at $\lambda = 0$ this is the vanishing of the joining factor. This section constructs the atomized family for every λ, proves its analytic properties, writes the joining factor cell by cell, and lists in Subsection 13.8 the steps that remain, in the order in which they will be followed. The passage from a continuous prime system to a discrete one is classical for Beurling generalized primes [15, 25, 18, 26]; the atomization used here places the masses at the integers, which produces ordinary Dirichlet series.

Throughout, $\eta_k = \tau_k - \gamma_k$, $\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k$, $\rho_{k,\lambda} = \frac{1}{2} + i\gamma_k(\lambda)$, and λ' denotes an integration variable in $[0,1]$.

13.1. **The family at infinity.** Put $N_\lambda(\tau) = \#\{k : \gamma_k(\lambda) \le \tau\}$, $\mathfrak{e}^{(\lambda)} = N_\lambda - \mathcal{N}_{\mathrm{RVM}}$ and $\mathfrak{e}_1^{(\lambda)}(\tau) = \int_{\tau_0}^{\tau} \mathfrak{e}^{(\lambda)}$, so that $\mathfrak{e}_1^{(0)} = \mathfrak{e}_1$ and $\mathfrak{e}_1^{(1)} = -\mathfrak{s}_1$.

Proposition 13.1 (Clock decomposition along the family)**.** *There is an absolute constant C with $|\mathfrak{e}_1^{(\lambda)}(\tau)| \le C \log \tau$ for $\tau \ge \tau_0$ and all $\lambda \in [0,1]$. Consequently Theorem 12.1 holds for every ζ_λ with $\mathfrak{e}_1$ replaced by $\mathfrak{e}_1^{(\lambda)}$, uniformly in λ; in particular $\varpi_\lambda := \lim_{\sigma\to+\infty} \zeta_\lambda(\sigma)$ exists and*

$$\varpi_\lambda = \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K} \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k(\lambda)^2}, \qquad \varpi_1 = \varpi_{\mathrm{RVM}}, \quad \varpi_0 = \varpi_{Z1}. \tag{13.1}$$

Proof. $\gamma_k(\lambda)$ lies between γ_k and τ_k, so $\gamma_1(\lambda) > \tau_0$. For $X \ge \tau_0$,

$$\int_{\tau_0}^{X} (N_\lambda - n_{\mathrm{RVM}}) = \sum_k \big((X - \gamma_k(\lambda))^+ - (X - \tau_k)^+\big) = (1-\lambda) \sum_{\gamma_k(\lambda), \tau_k \le X} \eta_k + O(\log X), \tag{13.2}$$

the indices counted by only one of the two sums being $O(\log X)$ in number with ordinates in $[X - C_\eta, X]$ (Lemma 2.2). The case $\lambda = 0$ of (13.2) identifies the last sum with $\mathfrak{e}_1 + \mathfrak{s}_1 + O(\log X) = O(\log X)$, by (12.2) and Lemma 8.9. Since $\mathfrak{e}_1^{(\lambda)} = \int (N_\lambda - n_{\mathrm{RVM}}) - \mathfrak{s}_1$, the bound follows, and the proof of Theorem 12.1 applies verbatim. □

Define the *flow at infinity*

$$\Upsilon(\lambda) := \sum_{k\ge1} \frac{2\gamma_k(\lambda)\,\eta_k}{\frac{1}{4} + \gamma_k(\lambda)^2} \qquad \text{(summed in the order of } k\text{)}. \tag{13.3}$$

Proposition 13.2 (The value at infinity along the family)**.** *The series* (13.3) *converges uniformly on $[0,1]$, $\lambda \mapsto \log \varpi_\lambda$ is C^1 on $[0,1]$, and*

$$\frac{d}{d\lambda} \log \varpi_\lambda = -\Upsilon(\lambda) = \lim_{\sigma\to+\infty} \mathcal{V}_\lambda(\sigma), \qquad \Upsilon'(\lambda) = \sum_{k\ge1} \frac{2\eta_k^2\big(\frac{1}{4} - \gamma_k(\lambda)^2\big)}{\big(\frac{1}{4} + \gamma_k(\lambda)^2\big)^2} < 0, \tag{13.4}$$

with $\mathcal{V}_\lambda$ the generator of Theorem 10.3. Moreover

$$\int_{0^+}^{1} \Upsilon(\lambda)\, d\lambda = \sum_{k\ge1} \log \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k^2} = \log L + \log \varpi_{Z1}, \tag{13.5}$$

so that (D) holds if and only if $\int_{0^+}^1 \Upsilon(\lambda)\, d\lambda = \log L$, and $\Upsilon(1) < \log L + \log \varpi_{Z1} < \Upsilon(0)$.

Proof. Put $g(u) = 2u/(\frac{1}{4} + u^2)$. The series $\Upsilon(1) = \sum_k g(\tau_k)\eta_k$ converges by summation by parts against $N_\zeta - n_{\mathrm{RVM}}$, exactly as in (12.10). Since $|g(\gamma_k(\lambda)) - g(\tau_k)| \le C|\eta_k|\gamma_k^{-2}$ and $\sum_k \eta_k^2\gamma_k^{-2} < \infty$, the series $\Upsilon(\lambda) - \Upsilon(1)$ converges absolutely and uniformly; the same holds for the termwise derivative, which is the series in (13.4) (each term is negative because $\gamma_k(\lambda) > \frac{1}{2}$). Differentiating (13.1) termwise gives $\frac{d}{d\lambda} \log \varpi_\lambda = -\Upsilon(\lambda)$, and the limit $\sigma \to +\infty$ in the generator (10.10) is justified by Abel summation of the ordered ordinate sum: its partial sums are controlled by the count-difference integral, while the multiplier $q/(A+q)$ is bounded and monotone in A. This gives the same ordered

series. Integrating from 0 to 1 and using $\varpi_1 = \varpi_{\mathrm{RVM}}$, $\varpi_0 = \varpi_{Z1}$ gives (13.5); the last statement follows from Theorem 12.2 and the monotonicity of Υ. □

Proposition 13.3 (The value at infinity and the verified zeros)**.** *If every zero $\beta+i\gamma$ with $0 < \gamma \le T_{\mathrm{ver}}$ has $\beta = \frac{1}{2}$, then*

$$0 \le \log \varpi_{Z1} \le 0.26 \sum_{\gamma > T_{\mathrm{ver}}} \frac{1}{\gamma^2}. \tag{13.6}$$

With $T_{\mathrm{ver}} = 3 \cdot 10^{12}$ [46], the right side is $\approx \frac{0.26}{2\pi T_{\mathrm{ver}}}(\log \frac{T_{\mathrm{ver}}}{2\pi} + 1) < 4 \cdot 10^{-13}$.

Proof. By the proof of Theorem 12.2, $\log \varpi_{Z1} = \sum \log c_\gamma$ over the pairs $\{\rho, 1-\bar\rho\}$ with $\rho(1-\rho) \ne \frac{1}{4} + \gamma^2$, and all of them have $\gamma > T_{\mathrm{ver}}$. For $|d_\rho| \le \frac{1}{2}$ and $\gamma \ge T_{\mathrm{ver}}$, (12.11) gives $\log c_\gamma \le (2d_\rho^2\gamma^2 + d_\rho^4)\gamma^{-4} \le 0.51\,\gamma^{-2}$; a pair consists of two zeros, which gives (13.6). The numerical value follows from $N_\zeta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(\log T)$. □

13.2. **Atomization by smooth cells.** Fix $\varphi_n \in C_c^\infty(n-1, n+1)$, $n \ge 2$, with $\sum_{n\ge2}\varphi_n = 1$ on $[2,\infty)$ and $\varphi_n(m) = \delta_{nm}$ for integers m. For $\lambda \in [0,1]$ let J_λ be the explicit-formula count (12.16) with the ordinates $\gamma_k(\lambda)$ (a distribution on $(1,\infty)$ by Theorem 12.6, which applies to every ζ_λ by Theorem 12.1), $J_\lambda^* = J_\lambda \mathbf{1}_{[2,\infty)}$, and $\pi_\lambda = \sum_{j\ge1} \frac{\mu(j)}{j} J_\lambda^*(x^{1/j})$.

Definition 13.4 (Atomized family)**.** The cell masses, the atomized zeta function and its coefficients are

$$\tilde c_n(\lambda) := \langle d\pi_\lambda, \varphi_n\rangle, \qquad \tilde\zeta_{\mathbb{Z},\lambda}(s) := \prod_{n\ge2}\bigl(1-n^{-s}\bigr)^{-\tilde c_n(\lambda)} = \sum_{N\ge1}\frac{\tilde a_N(\lambda)}{N^s}, \tag{13.7}$$

$$\tilde a_N(\lambda) = \sum_{\prod_n n^{r_n}=N}\ \prod_{n:\, r_n\ge1}\binom{\tilde c_n(\lambda)+r_n-1}{r_n}, \qquad \tilde a_1(\lambda) = 1. \tag{13.8}$$

Unlike the discretizations of Beurling systems, where the atoms are placed at chosen real points [25, 18, 26], the masses of $d\pi_\lambda$ are placed at the integers with real exponents; measures with continuous part and non-integer multiplicities of this kind appear also in [38, Ch. 3]. Thus $\tilde a_p(\lambda) = \tilde c_p(\lambda)$ for primes p, $\tilde a_4 = \tilde c_4 + \frac{1}{2}\tilde c_2(\tilde c_2+1)$, $\tilde a_6 = \tilde c_6 + \tilde c_2\tilde c_3$. The pairing with a fixed smooth partition defines $\tilde c_n(\lambda)$ for every λ without point values of π_λ, and $\lambda \mapsto \tilde c_n(\lambda)$ is continuous. If $d\pi_0 = \sum_p \delta_p$, then $\tilde c_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ and $\tilde a_N(0) = 1$ for all N. Along the family, in $\mathcal{D}'(1,\infty)$,

$$\frac{\partial J_\lambda}{\partial\lambda}(x) = 2\sum_{k\ge1}\eta_k\,\Im\frac{x^{\rho_{k,\lambda}}}{\rho_{k,\lambda}}, \qquad \frac{\partial}{\partial\lambda}\Bigl((\log x)\frac{dJ_\lambda}{dx}\Bigr) = \frac{2\log x}{\sqrt{x}}\sum_{k\ge1}\eta_k\sin\bigl(\gamma_k(\lambda)\log x\bigr). \tag{13.9}$$

13.3. **The joining factor cell by cell.** Define the *atomization kernel* and the *cell defects*

$$\mathfrak{a}_s(x) := \sum_{n\ge2}\varphi_n(x)\,n^{-s} - x^{-s}, \qquad \mathfrak{h}_n(\lambda; s) := \bigl\langle d\pi_\lambda,\ \varphi_n(x)\bigl(n^{-s}-x^{-s}\bigr)\bigr\rangle. \tag{13.10}$$

$\mathfrak{a}_s$ vanishes at every integer ≥ 2, and $\mathfrak{h}_n(\lambda; s)$ measures how far the mass of the cell n is from being concentrated at n.

The explicit-formula count J_λ need not vanish on $(1,2)$, whereas π_λ is built from $J_\lambda^* = J_\lambda\mathbf{1}_{[2,\infty)}$. The first cell $(1,2)$ enters the joining factor through

$$\mathfrak{u}_\lambda(s) := C_\lambda\,2^{-s} + s\int_1^2 J_\lambda^{\mathrm{M}}(x)\,x^{-s-1}\,dx, \qquad C_\lambda := J_\lambda^{\mathrm{M}} - J_\lambda, \tag{13.11}$$

where J_λ^{M} is the Mellin count of ζ_λ (Theorem 12.5, which applies to every ζ_λ by Proposition 13.1) and C_λ is the constant of Theorem 12.6.

Proposition 13.5 (The first cell)**.** *Let $\lambda \in [0,1]$.*

(i) $\mathfrak{u}_\lambda$ *is entire, and for* $\Re s > 1$

$$\log \zeta_\lambda(s) = \mathfrak{u}_\lambda(s) + \langle dJ_\lambda^*, x^{-s}\rangle, \qquad \langle dJ_\lambda^*, x^{-s}\rangle := s\int_2^\infty J_\lambda(x)\, x^{-s-1}\, dx. \tag{13.12}$$

(ii) $\mathfrak{u}_\lambda(\sigma + it) = o(|t|)$ *as* $|t| \to \infty$*, uniformly for* σ *in compact sets.*
(iii) $\mathfrak{u}_\lambda$ *is constant if and only if* $J_\lambda = 0$ *on* $(1,2)$*; in that case* $\mathfrak{u}_\lambda \equiv C_\lambda$*.*

For Riemann's count $J(x) = \sum_{p^m \le x} 1/m$*, which vanishes on* $(1,2)$*, the corresponding term is* 0*.*

Proof. (i) $J_\lambda^{\mathrm{M}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for $c > 1$, so $J_\lambda^{\mathrm{M}} \in L^1(1,2)$ and the integral in (13.11) is entire in s. By (12.14), $\log\zeta_\lambda(s) = s\int_1^2 J_\lambda^{\mathrm{M}} x^{-s-1}dx + s\int_2^\infty (J_\lambda + C_\lambda)x^{-s-1}dx$, and $s\int_2^\infty C_\lambda x^{-s-1}dx = C_\lambda 2^{-s}$. The second integral is the pairing of x^{-s} with the distributional derivative of J_λ^*, which vanishes on $(0,2)$. (ii) $C_\lambda 2^{-s}$ is bounded on vertical strips, and $\int_1^2 J_\lambda^{\mathrm{M}}(x)x^{-s-1}dx = \int_0^{\log 2} J_\lambda^{\mathrm{M}}(e^u)e^{-\sigma u}e^{-itu}du \to 0$ as $|t| \to \infty$ by the Riemann–Lebesgue lemma, uniformly for σ in compact sets. (iii) For a constant c one has $c = c\,2^{-s} + s\int_1^2 c\,x^{-s-1}dx$, so $\mathfrak{u}_\lambda(s) - c = s\int_1^\infty H(x)x^{-s-1}dx$ with $H = (J_\lambda^{\mathrm{M}} - c)\mathbf{1}_{(1,2)} + (C_\lambda - c)\mathbf{1}_{[2,\infty)}$. By the injectivity used in Theorem 12.5, $\mathfrak{u}_\lambda \equiv c$ if and only if $H = 0$ almost everywhere, that is, $C_\lambda = c$ and $J_\lambda^{\mathrm{M}} = C_\lambda$ on $(1,2)$, which is $J_\lambda = 0$ on $(1,2)$. □

Theorem 13.6 (The joining factor)**.** *Let* $\tilde h_\lambda := \log\tilde\zeta_{\mathbb{Z},\lambda} - \log\zeta_\lambda$ *on the half-plane where* $\tilde\zeta_{\mathbb{Z},\lambda}$ *converges absolutely (Theorem 13.12). Then*

$$\tilde\zeta_{\mathbb{Z},\lambda} = \zeta_\lambda\, e^{\tilde h_\lambda}, \qquad \tilde h_\lambda(s) = -\mathfrak{u}_\lambda(s) + \sum_{m\ge1}\frac{1}{m}\sum_{n\ge2}\mathfrak{h}_n(\lambda; ms), \tag{13.13}$$

and, explicitly,

$$\tilde h_\lambda(s) = -\mathfrak{u}_\lambda(s) + \sum_{m\ge1}\frac{1}{m}\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty \mathfrak{a}_{ms}(y^j)\,\frac{1}{\log y}\Big(1 - \frac{2}{\sqrt{y}}\Re\sum_{k\ge1} y^{i\gamma_k(\lambda)} - \frac{1}{y(y^2-1)}\Big)dy, \tag{13.14}$$

the series over k *converging absolutely after integration. Its derivative along the family is*

$$\partial_\lambda\tilde h_\lambda(s) = -\partial_\lambda\mathfrak{u}_\lambda(s) + \sum_{m\ge1}\frac{1}{m}\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty \mathfrak{a}_{ms}(y^j)\,\frac{2}{\sqrt{y}}\sum_{k\ge1}\eta_k\sin\big(\gamma_k(\lambda)\log y\big)\,dy. \tag{13.15}$$

Here $\partial_\lambda\mathfrak{u}_\lambda(s) = \mathcal{V}_\lambda(s) - \partial_\lambda\langle dJ_\lambda^*, x^{-s}\rangle$ *by* (13.12) *and* (10.11)*. Moreover* $\mathfrak{u}_\lambda(\sigma) \to \log\varpi_\lambda$ *as* $\sigma \to +\infty$*, and* $\mathfrak{u}_\lambda \equiv \log\varpi_\lambda$ *if* J_λ *vanishes on* $(1,2)$*.*

Proof. By the Euler product, $\log\tilde\zeta_{\mathbb{Z},\lambda}(s) = \sum_{m\ge1}\frac{1}{m}\sum_n \tilde c_n(\lambda)n^{-ms}$. By Proposition 13.5, $\log\zeta_\lambda(s) = \mathfrak{u}_\lambda(s) + \langle dJ_\lambda^*, x^{-s}\rangle$; finite inversion (Proposition 5.3, whose proof applies to J_λ^*) gives $dJ_\lambda^* = \sum_m \frac{1}{m}\,d[\pi_\lambda(x^{1/m})]$, and $\langle d[\pi_\lambda(x^{1/m})], x^{-s}\rangle = \langle d\pi_\lambda, y^{-ms}\rangle$. Inserting $1 = \sum_n \varphi_n$ on $[2,\infty)$ gives $\langle d\pi_\lambda, y^{-ms}\rangle = \sum_n(\tilde c_n(\lambda)n^{-ms} - \mathfrak{h}_n(\lambda; ms))$, hence (13.13). Letting $\sigma \to +\infty$, $\log\tilde\zeta_{\mathbb{Z},\lambda} \to 0$ (because $\tilde a_1 = 1$), $\log\zeta_\lambda \to \log\varpi_\lambda$ (Proposition 13.1) and the double series tends to 0 by the bounds of Theorem 13.12; so $\mathfrak{u}_\lambda(\sigma) \to \log\varpi_\lambda$, and a constant $\mathfrak{u}_\lambda$ (Proposition 13.5(iii)) equals $\log\varpi_\lambda$. Atoms of $d\pi_\lambda$ at integers contribute nothing because the test functions vanish there. Formula (13.14) is (13.13) with $d\pi_\lambda$ written by (12.15) and the substitution $x = y^j$; the integrated terms in k are $O_{m,j,s}(\gamma_k(\lambda)^{-2})$ because $\mathfrak{a}_{ms}(2) = 0$. The remaining double summation is absolutely controlled by $\sum_{m\ge1} m^{-1}\sum_{j\ge1} j^{-1}(mj|s|)^2 2^{-jm\sigma} < \infty$, so the exchanges of the m- and j-sums with the integrated ordinate sum are justified. Differentiating in λ with (13.9) gives (13.15); $\mathfrak{u}_\lambda(s)$ is differentiable in λ because J_λ^{M} is the Fourier–Plancherel transform of $\log\zeta_\lambda(\sigma+it)/(\sigma+it)$, whose λ-derivative $\mathcal{V}_\lambda(\sigma+it)/(\sigma+it)$ is bounded and continuous in $L^2(\mathbb{R})$ locally uniformly in λ by Lemma 13.9 and dominated convergence on the ordinate sum. Hence $\lambda \mapsto J_\lambda^{\mathrm{M}}$ is C^1 in the corresponding weighted L^2 space; the constant $C_\lambda = J_\lambda^{\mathrm{M}} - J_\lambda$ is obtained by pairing with one fixed test function and is C^1 as well. In particular $\|\partial_\lambda J_\lambda\|_{L^2(2,3)}$ is locally uniformly bounded. □

Proposition 13.7 (Explicit cell defects)**.** *For every $n \geq 2$, $\lambda \in [0,1]$ and $\Re s > 1$,*

$$\mathfrak{h}_n(\lambda; s) = \sum_{j\geq 1} \frac{\mu(j)}{j} \int_2^\infty \varphi_n(y^j)(n^{-s} - y^{-js}) \, \frac{1}{\log y}\Big(1 - \frac{2}{\sqrt{y}} \Re \sum_{k\geq 1} y^{i\gamma_k(\lambda)} - \frac{1}{y(y^2-1)}\Big) dy, \quad (13.16)$$

$$\partial_\lambda \mathfrak{h}_n(\lambda; s) = \sum_{j\geq 1} \frac{\mu(j)}{j} \int_2^\infty \varphi_n(y^j)(n^{-s} - y^{-js}) \, \frac{2}{\sqrt{y}} \sum_{k\geq 1} \eta_k \sin\big(\gamma_k(\lambda) \log y\big) \, dy, \quad (13.17)$$

the sums over j being finite and the sums over k converging absolutely after integration, because the test functions are smooth with compact support. At $\lambda = 1$ the cell defect $\mathfrak{h}_n(1;\cdot)$ is the dispersion of the cell n for ζ_{RVM}, computed from the clock alone; the absolutely continuous part of $d\pi_{\mathrm{RVM}}$ has bounded density there by Theorem 6.1, while the atoms at 2^j are treated separately, so $\mathfrak{h}_n(1;\cdot) \equiv 0$ holds only if this density vanishes on the support of φ_n.

Proof. Insert (12.15) for $d\pi_\lambda = \sum_j \frac{\mu(j)}{j} \, d[J_\lambda^*(x^{1/j})]$ into the definition (13.10) and substitute $x = y^j$; the atoms of $d\pi_\lambda$ at integers, for instance the jump of J_λ^* at 2, contribute nothing because $n^{-s} - x^{-s}$ vanishes at $x = n$ and φ_n vanishes at the other integers. Differentiation in λ uses (13.9); the factor $\log y$ cancels. For the last statement, $\mathfrak{h}_n(1;\cdot) \equiv 0$ forces $\varphi_n \, d\pi_{\mathrm{RVM}} = \tilde{c}_n(1)\delta_n$ by the injectivity used in Proposition 13.8, and, away from the explicit atoms at 2^j, an absolutely continuous measure equals a multiple of δ_n only if both vanish; at the powers of 2 the boundary atom is handled by (13.21). □

In the form of the path, the final step requires that in every cell the flow of the displacements cancel exactly the dispersion of the clock, (13.30).

Proposition 13.8 (The limit $\lambda \to 0$ and the meaning of a vanishing cell)**.** *For every $n \geq 2$ and $\Re s > 1$, $\mathfrak{h}_n(\lambda; s) \to \mathfrak{h}_n(0; s)$ and $\tilde{c}_n(\lambda) \to \tilde{c}_n(0)$ as $\lambda \to 0^+$, and*

$$\mathfrak{h}_n(0; s) = \mathfrak{h}_n(1; s) - \int_{0^+}^1 \partial_\lambda \mathfrak{h}_n(\lambda; s) \, d\lambda. \quad (13.18)$$

Moreover $\mathfrak{h}_n(0; s) = 0$ for all $\Re s > 1$ if and only if $\varphi_n \, d\pi_{Z1} = \tilde{c}_n(0) \, \delta_n$.

Proof. The test function $\varphi_n(x)(n^{-s} - x^{-s})$ is smooth with compact support, and the integrated terms over k are dominated uniformly in λ because $|\gamma_k(\lambda) - \gamma_k| \leq C_\eta$; dominated convergence gives the limits for cells away from the powers of 2. For $n = 2^j$, the boundary term (13.21) also converges: by Plancherel, $\log \zeta_\lambda(\sigma + it)/(\sigma + it) \to \log \zeta_{Z1}(\sigma + it)/(\sigma + it)$ in $L^2(dt)$ on a fixed line $\sigma > 1$, hence $J_\lambda^{\mathrm{M}} \to J_{Z1}^{\mathrm{M}}$ in the corresponding weighted L^2 space and therefore $J_\lambda \to J_{Z1}$ in $L^2(2,3)$. This proves all stated limits and (13.18). Since $\mathfrak{h}_n(0; s) = \tilde{c}_n(0) n^{-s} - \langle \varphi_n \, d\pi_{Z1}, x^{-s} \rangle$, its vanishing for all s says that the compactly supported distribution $\varphi_n \, d\pi_{Z1}$ has the Mellin transform of $\tilde{c}_n(0)\delta_n$; the Mellin transform is injective on such distributions. □

13.4. **Mean square of the cell masses.** From now on the cells are translates of one function: $\varphi_n(x) = \varphi(x - n)$, where $\varphi = \beta_0 / \sum_{m\in\mathbb{Z}} \beta_0(\cdot - m)$ for a fixed bump $\beta_0 \in C_c^\infty(-1, 1)$ positive on $[-\frac{1}{2}, \frac{1}{2}]$. Then $\sum_{n\geq 2} \varphi_n = 1$ on $[2, \infty)$, $\varphi_n(m) = \delta_{nm}$, and $\|\varphi_n^{(A)}\|_\infty = \|\varphi^{(A)}\|_\infty =: C_A$ for every $A \geq 0$, uniformly in n. For $n \geq 2$ put

$$V_n(\gamma) := \sum_{j\geq 1} \frac{\mu(j)}{j} \int_2^\infty \varphi_n(y^j) \, y^{-1/2} e^{i\gamma \log y} \, dy, \qquad W_n(\gamma) := \sum_{j\geq 1} \frac{\mu(j)}{j} \int_2^\infty \varphi_n(y^j) \, \frac{y^{-1/2} e^{i\gamma \log y}}{\log y} \, dy, \quad (13.19)$$

so that $W_n' = iV_n$; both are rapidly decreasing in γ when n is not a power of 2 (for $n = 2^j$ the term j carries the cut of J_λ^* at $y = 2$ and is treated in the proof of Theorem 13.10). For n not a power of 2, the density of $d\pi_\lambda$ used in (13.16) and the derivative (13.9) give, with $\eta_k = \tau_k - \gamma_k$:

$$\tilde{c}_n(\lambda) = \mathfrak{m}_n - 2\,\Re \sum_{k\geq 1} W_n\big(\gamma_k(\lambda)\big), \qquad \partial_\lambda \tilde{c}_n(\lambda) = 2 \sum_{k\geq 1} \eta_k \, \Im V_n\big(\gamma_k(\lambda)\big), \quad (13.20)$$

where $\mathfrak{m}_n$ collects the terms that do not involve the ordinates and is $O(1)$. If $n = 2^j$, the jth Möbius-inversion term crosses the cut at $y = 2$ and contributes instead the additional boundary term

$$-\frac{\mu(j)}{j}\int_2^3 J_\lambda(y)\,\frac{d}{dy}\varphi_n(y^j)\,dy, \tag{13.21}$$

whose λ-derivative is obtained by replacing J_λ by $\partial_\lambda J_\lambda$.

Lemma 13.9 (Mean value over the moving ordinates)**.** *There is C such that for all $\lambda \in [0,1]$, $T \geq 2$, $u_0 \in \mathbb{R}$ and complex a_k with $|a_k| \leq 1$,*

$$\int_{u_0-1}^{u_0+1}\Big|\sum_{\gamma_k(\lambda)\leq T} a_k\, e^{i\gamma_k(\lambda)u}\Big|^2 du \leq C\,T\log^2 T. \tag{13.22}$$

Proof. Let $\omega_0(u) = C_0(\sin(\frac{u-u_0}{2})/\frac{u-u_0}{2})^2$ with C_0 such that $\omega_0 \geq 1$ on $|u-u_0| \leq 1$; the Fourier transform of ω_0 is bounded and supported in $[-1,1]$. Expanding the square,

$$\begin{aligned}\int \omega_0(u)\Big|\sum_k a_k e^{i\gamma_k(\lambda)u}\Big|^2 du &= \sum_{k,k'} a_k\bar{a}_{k'}\, e^{i(\gamma_k(\lambda)-\gamma_{k'}(\lambda))u_0}\,\widehat{\omega_0}\big(\gamma_k(\lambda)-\gamma_{k'}(\lambda)\big)\\ &\ll \#\{(k,k') : \gamma_k(\lambda)\leq T,\ |\gamma_k(\lambda)-\gamma_{k'}(\lambda)|\leq 1\}.\end{aligned} \tag{13.23}$$

Since $|\gamma_k(\lambda)-\tau_k| \leq C_\eta$, every interval $[t,t+1]$ contains $O(\log(t+2))$ of the $\gamma_k(\lambda)$, uniformly in λ, because it contains that many τ_k after widening by C_η. The number of pairs is therefore $O(T\log^2 T)$. □

Theorem 13.10 (Mean square of the cell masses along the family)**.** *For every $\varepsilon > 0$ there is C_ε such that for all $X \geq 2$ and all $\lambda \in [0,1]$*

$$\sum_{X<n\leq 2X}|\tilde{c}_n(\lambda)|^2 \leq C_\varepsilon X^{1+\varepsilon}, \qquad \sum_{X<n\leq 2X}|\partial_\lambda\tilde{c}_n(\lambda)|^2 \leq C_\varepsilon X^{1+\varepsilon}. \tag{13.24}$$

In particular the same bound already includes $\lambda = 0$: $\sum_{X<n\leq 2X}|\tilde{c}_n(0)|^2 \leq C_\varepsilon X^{1+\varepsilon}$.

Proof. Fix ε, X, λ and put $T = X^{1+\varepsilon}$. For n not a power of 2, (13.20) gives $\tilde{c}_n(\lambda) - \mathfrak{m}_n = -2\Re\sum_j \frac{\mu(j)}{j}A_{n,j}$, with

$$A_{n,j} = \sum_k \int_2^\infty \varphi_n(y^j)\,\frac{y^{-1/2+i\gamma_k(\lambda)}}{\log y}\,dy, \qquad 1 \leq j \leq J := \Big\lfloor\frac{\log(2X+1)}{\log 2}\Big\rfloor. \tag{13.25}$$

Large ordinates. In $u = \log y$ the function $\varphi_n(e^{ju})e^{u/2}/u$ has support of length $O((jn)^{-1})$, size $O(n^{1/(2j)})$ and r-th derivative $O_r((jn)^r n^{1/(2j)})$: by Faà di Bruno's formula the r-th derivative of $\varphi_n(e^{ju})$ is a sum of terms $\varphi_n^{(m)}(e^{ju})(je^{ju})^m j^{r-m}$, $m \leq r$, and $\|\varphi_n^{(m)}\|_\infty = C_m$ does not depend on n; the derivatives of $e^{u/2}/u$ are $O_r(e^{u/2})$ because $u \geq \log 2$. Integrating by parts A times, each term with $\gamma_k(\lambda) > T$ is $O_A\big((jn/\gamma_k(\lambda))^A n^{1/(2j)-1}\big)$, and their sum over k is $O(X^{-1})$ once $A > (3+\varepsilon)/\varepsilon$. *The cell $n = 2^j$.* Each block contains exactly one $n = 2^j$; for it the term j carries the cut of J_λ^* at $y = 2$ and is treated directly. It equals $-\frac{\mu(j)}{j}\int_2^3 J_\lambda(y)\,\frac{d}{dy}\varphi_n(y^j)\,dy$; the weight has L^2 norm $O((jn)^{1/2})$, and $\|J_\lambda\|_{L^2(2,3)}$ is bounded uniformly in λ (Theorem 12.5 with the uniform bound of Proposition 13.1). This term is therefore $O((n/j)^{1/2})$ and contributes $O(X)$ to the sum of squares; the same holds for $\partial_\lambda\tilde{c}_n$, with $\partial_\lambda J_\lambda$ in place of J_λ. *Small ordinates.* Let $\mathcal{S}_{n,j,\lambda}(y) = y^{-1/2}(\log y)^{-1}\sum_{\gamma_k(\lambda)\leq T} y^{i\gamma_k(\lambda)}$ and let $I_{n,j}$ be the support of $y \mapsto \varphi_n(y^j)$, of length $O(X^{1/j-1})$. By Cauchy–Schwarz in the cell,

$$\Big|\int \varphi_n(y^j)\mathcal{S}_{n,j,\lambda}(y)\,dy\Big|^2 \leq \|\varphi\|_\infty^2\,|I_{n,j}|\int_{I_{n,j}}|\mathcal{S}_{n,j,\lambda}|^2 dy. \tag{13.26}$$

For fixed j the intervals $I_{n,j}$, $X < n \le 2X$, cover each point at most twice and lie in $[(X-1)^{1/j}, (2X+1)^{1/j}]$, an interval of length at most 2 in $u = \log y$. With $dy/y = du$ and Lemma 13.9 ($a_k = 1$),

$$\sum_{X<n\le 2X} \Big| \int \varphi_n(y^j) S \, dy \Big|^2 \ll X^{1/j-1} \Big(\frac{j}{\log X}\Big)^2 \int_{u_0-1}^{u_0+1} \Big| \sum_{\gamma_k(\lambda)\le T} e^{i\gamma_k(\lambda)u} \Big|^2 du \ll j^2 X^{1/j+2\varepsilon}. \tag{13.27}$$

Assembly. By Minkowski's inequality over j, the ℓ^2 norm over $X < n \le 2X$ of $\tilde{c}_n(\lambda) - \mathfrak{m}_n$ is $\ll \sum_{j\le J} (X^{1/(2j)+\varepsilon} + X^{-1/2}) \ll X^{1/2+\varepsilon}$, and $\sum_{X<n\le 2X} \mathfrak{m}_n^2 \ll X$. This is the first bound with 3ε in place of ε. Away from powers of 2, the argument for $\partial_\lambda \tilde{c}_n$ is the same, by (13.20), with coefficients $a_k = \eta_k / C_\eta$ in Lemma 13.9 and without the factor $(\log y)^{-1}$. □

Corollary 13.11 (Abscissa 1 for every member)**.** *For every $\lambda \in [0,1]$ and $\sigma > 1$,*

$$\sum_{n\ge 2} |\tilde{c}_n(\lambda)| \, n^{-\sigma} < \infty; \tag{13.28}$$

hence $\tilde{\zeta}_{\mathbb{Z},\lambda}$ and $1/\tilde{\zeta}_{\mathbb{Z},\lambda}$ are absolutely convergent Dirichlet series in $\Re s > 1$, for every λ, including the endpoint $\lambda = 0$. Moreover, uniformly in λ, $\#\{n \le X : |\tilde{c}_n(\lambda)| > n^\delta\} \ll_\varepsilon X^{1-2\delta+\varepsilon}$ for every $\delta > 0$.

Proof. By Cauchy–Schwarz on dyadic blocks, $\sum_{X<n\le 2X} |\tilde{c}_n(\lambda)| \le (X \cdot C_\varepsilon X^{1+\varepsilon})^{1/2}$, and summing over $X = 2^i$ gives (13.28) for $\sigma > 1 + \varepsilon/2$, with ε arbitrary. Then $\log \tilde{\zeta}_{\mathbb{Z},\lambda} = \sum_m \frac{1}{m} \sum_n \tilde{c}_n(\lambda) n^{-ms}$ converges absolutely in $\Re s > 1$, and the exponentials of $\pm$ this series are absolutely convergent Dirichlet series there. The last statement is Chebyshev's inequality on dyadic blocks. □

The abscissa $\frac{3}{2}$ obtained below from the primitive $\pi_\lambda - \mathfrak{L}$ (Theorem 13.12(ii)) is not the true one: a bound on the size of $\pi_\lambda - \mathfrak{L}$ does not see the cancellation of the density of $d\pi_\lambda$ in mean square, which Theorem 13.10 captures. Absolute convergence in $\Re s > 1$ is exactly what Hamburger's theorem requires, and Knopp's obstruction (Subsection 13.7) does not arise.

13.5. **Analytic continuation of the atomized family.** Let $\mathfrak{L}(x) = \int_1^x (1 - y^{-1})\, dy/\log y$; then $\log \frac{s}{s-1} = \int_1^\infty x^{-s}\, d\mathfrak{L}(x)$ for $\Re s > 1$ (both sides have derivative $\frac{1}{s} - \frac{1}{s-1}$ and tend to 0 as $s \to +\infty$), and $\mathfrak{L}(x) = \mathrm{li}(x) - \log\log x - \gamma_E$.

Theorem 13.12 (Continuation of the atomized family)**.** *Let $\lambda \in [0,1]$.*

(i) *For every $c > \frac{1}{2}$, $(\pi_\lambda - \mathfrak{L})\, x^{-c} \in L^2\big((2,\infty), dx/x\big)$.*

(ii) *$|\tilde{c}_n(\lambda)| \le C(1 + I_n^{1/2})$ with I_n as below, and $\sum_n |\tilde{c}_n(\lambda)| n^{-\sigma} < \infty$ for $\sigma > \frac{3}{2}$; by Corollary 13.11 this holds for $\sigma > 1$, and $\tilde{\zeta}_{\mathbb{Z},\lambda}$ is an absolutely convergent Dirichlet series in $\Re s > 1$.*

(iii) *$\tilde{h}_\lambda$ extends holomorphically to $\Re s > \frac{1}{2}$. Hence $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde{h}_\lambda}$ is meromorphic in $\Re s > \frac{1}{2}$, with a single pole at $s = 1$ and no zeros.*

(iv) *For $\lambda = 1$, $\tilde{h}_1$ extends holomorphically to $\Re s > 0$, and the zeros of $\tilde{\zeta}_{\mathbb{Z},1}$ in $\Re s > 0$ are exactly $\frac{1}{2} \pm i\tau_k$.*

Proof. (i) By Proposition 13.1, for $c > \frac{1}{2}$ there is C_c with $|\log \zeta_\lambda(s) - \log \frac{s}{s-1}| \le C_c(1 + \log(|t| + 2))$ on $\Re s \ge c$; the function $\log \zeta_\lambda - \log \frac{s}{s-1}$ is holomorphic on $\Re s > \frac{1}{2}$, where ζ_λ has no zeros, and its pole at 1 is removed. It equals $s \int_1^\infty (J_\lambda^{\mathrm{M}} - \mathfrak{L}) x^{-s-1} dx$ for $\Re s > 1$. The argument of Theorem 12.5, applied on the lines $\Re s = c > \frac{1}{2}$, gives $(J_\lambda^{\mathrm{M}} - \mathfrak{L}) x^{-c} \in L^2$. The constant $C_\lambda = J_\lambda^{\mathrm{M}} - J_\lambda$, the truncation at 2, the terms $j \ge 2$ of the Möbius inversion (of size $O(\sqrt{x})$ in mean) and $\mathfrak{L}(x) - \mathfrak{L}(x^{1/j})$ preserve the property for $c > \frac{1}{2}$.

(ii) Put $E = \pi_\lambda - \mathfrak{L}$ and $I_n = \int_{n-1}^{n+1} |E|^2 dx$. Since $\int \varphi_n' = 0$, $\tilde{c}_n(\lambda) = \int \varphi_n \, d\mathfrak{L} - \int E \varphi_n' \, dx$, so $|\tilde{c}_n(\lambda)| \le C(1 + I_n^{1/2})$. By (i), $\sum_n I_n n^{-2c-1} < \infty$, and Cauchy–Schwarz gives $\sum_n I_n^{1/2} n^{-\sigma} < \infty$ for $\sigma > c + 1$.

(iii) With $\psi_n = \varphi_n(x)(n^{-s} - x^{-s})$ one has $|\psi_n'| \le C|s|n^{-\sigma-1}$ and ψ_n vanishes at the ends of the cell, so

$$|\mathfrak{h}_n(\lambda; s)| \le C|s|\, n^{-\sigma-1}(1 + I_n^{1/2}), \qquad \sum_n I_n^{1/2} n^{-\sigma-1} < \infty \quad (\sigma > c). \tag{13.29}$$

Choosing $c \in (\frac{1}{2}, \sigma)$, the series (13.13) converges locally uniformly on $\Re s > \frac{1}{2}$; the terms $m \ge 2$ are evaluated at $\Re(ms) \ge 2\sigma$. Since $\mathfrak{u}_\lambda$ is entire (Proposition 13.5), $\tilde{h}_\lambda$ is holomorphic on $\Re s > \frac{1}{2}$, and ζ_λ has no zeros there.

(iv) By Corollary 6.2, π_{RVM} oscillates by $O(1)$ on every unit cell; integrating by parts, $|\mathfrak{h}_n(1; s)| \le C|s|n^{-\sigma-1}$, and the series converges for $\sigma > 0$. The factor $e^{\tilde{h}_1}$ has no zeros, so the zeros of $\tilde{\zeta}_{\mathbb{Z},1}$ in $\Re s > 0$ are those of ζ_{RVM}. □

Theorem 13.13 (Closing criterion; via Hamburger's theorem [33])**.** *The following are equivalent.*

(i) *(D), that is,* $\zeta_{Z1} = \zeta$.
(ii) $\tilde{h}_0 \equiv 0$.
(iii) $J_{Z1} = 0$ *on* $(1, 2)$*, and for every* $n \ge 2$ *and every* $\Re s > 1$*,*

$$\int_{0^+}^{1} \partial_\lambda \mathfrak{h}_n(\lambda; s)\, d\lambda = \mathfrak{h}_n(1; s). \tag{13.30}$$

In this case $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$*,* $\tilde{a}_N(0) = 1$ *for all* N*, and* $\varpi_{Z1} = 1$.

Proof. (ii)⇒(i): $\tilde{h}_0 \equiv 0$ gives $\zeta_{Z1} = \tilde{\zeta}_{\mathbb{Z},0} = \exp(\sum_m \frac{1}{m} \sum_n \tilde{c}_n(0) n^{-ms})$, an absolutely convergent Dirichlet series in $\Re s > 1$ by Corollary 13.11. (iii)⇒(ii): by (13.18) every $\mathfrak{h}_n(0; \cdot)$ vanishes, and $\mathfrak{u}_0 \equiv \log \varpi_{Z1}$ (Theorem 13.6), so $\tilde{h}_0 \equiv -\log \varpi_{Z1}$ is constant; then $\zeta_{Z1} = \varpi_{Z1}\tilde{\zeta}_{\mathbb{Z},0}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Corollary 13.11), Theorem 11.2 gives $\zeta_{Z1} = \zeta$, and comparing the first coefficients, $\varpi_{Z1} = 1$. (i)⇒(iii): if $\zeta_{Z1} = \zeta$, Theorem 12.7 gives $J_{Z1}^{\mathrm{M}} = J$, and $J_{Z1} = J$ by von Mangoldt's explicit formula [28], so J_{Z1} vanishes on $(1, 2)$, $d\pi_{Z1} = \sum_p \delta_p$, $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, every $\mathfrak{h}_n(0; \cdot)$ vanishes, and (13.18) gives (13.30). □

Remark 13.14 (The pointwise size of the cell masses). Theorem 13.13 needs no pointwise bound on $\tilde{c}_n(0)$: its proof uses the cell masses only through (13.28). The bound $\tilde{c}_n(0) = O(n^\varepsilon)$ for every n is not proved directly; Corollary 13.11 restricts the exceptions to a set of $n \le X$ of size $O(X^{1-2\delta+\varepsilon})$ for the threshold n^δ, uniformly in λ, and under the closure every cell mass is bounded, $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$. For n not a power of 2, (13.20) gives the trivial pointwise bound $\tilde{c}_n(\lambda) = O(n^{1/2+\varepsilon})$; the mean square (13.24) shows that the typical cell mass is $O(n^\varepsilon)$.

13.6. **The two endpoints of the atomized family.**

Proposition 13.15 (Atomization at the clock endpoint)**.** *At* $\lambda = 1$*:*

(i) $\tilde{c}_n(1) = O(1)$*, and* $\tilde{\zeta}_{\mathbb{Z},1}$ *is an absolutely convergent Dirichlet series in* $\Re s > 1$*;*
(ii) $\tilde{h}_1(\sigma) \to -\log \varpi_{\mathrm{RVM}} = \log L \ne 0$ *as* $\sigma \to +\infty$*;*
(iii) $\tilde{h}_1$ *is not constant: for every* $\sigma > 1$*,* $\zeta_{\mathrm{RVM}}(\sigma + it) \to \varpi_{\mathrm{RVM}}$ *as* $|t| \to \infty$*, whereas* $\tilde{\zeta}_{\mathbb{Z},1}(\sigma + it)$ *has no limit as* $|t| \to \infty$.

Proof. (i) By Corollary 6.2, π_{RVM} oscillates by $O(1)$ on every unit cell, and $\tilde{c}_n(1) = -\int \pi_{\mathrm{RVM}} \varphi_n'\, dx$ with $\int \varphi_n' = 0$. (ii) is (13.13) with $\sigma \to +\infty$, since $\varpi_{\mathrm{RVM}} < 1$. (iii) The first limit is (8.33). If $\tilde{\zeta}_{\mathbb{Z},1}(\sigma + it)$ had a limit ℓ as $|t| \to \infty$, the mean-square argument (8.43)–(8.44) applied to $\tilde{\zeta}_{\mathbb{Z},1} - \ell$ on the line $\Re s = \sigma$ would give $\tilde{a}_N(1) = 0$ for $N \ge 2$, so $\tilde{\zeta}_{\mathbb{Z},1} \equiv 1$, which contradicts its pole at $s = 1$ (Theorem 13.12(iii)). If $\tilde{h}_1$ were constant, ζ_{RVM} would be a constant multiple of an absolutely convergent Dirichlet series, which Theorem 8.11 excludes. □

The two families thus keep complementary structures and are joined by a zero-free factor:

$$\underbrace{\zeta_\lambda}_{\substack{\text{functional equation}\\ \text{zeros } \frac12\pm i\gamma_k(\lambda)}} \cdot e^{\tilde h_\lambda} \;=\; \underbrace{\tilde\zeta_{\mathbb{Z},\lambda}}_{\substack{\text{Dirichlet series}\\ \text{Euler product}}}, \qquad \tilde h_1 \text{ not constant}, \qquad (D) \iff \tilde h_0 \equiv 0. \tag{13.31}$$

At $\lambda = 1$ the joining factor carries the whole difference between the clock and an Euler product; Theorem 13.13 states that this difference disappears at $\lambda = 0$.

The reciprocal of the atomized family is again an Euler product:

$$\frac{1}{\tilde\zeta_{\mathbb{Z},\lambda}(s)} = \prod_{n\ge2}\bigl(1-n^{-s}\bigr)^{\tilde c_n(\lambda)} = \sum_{N\ge1}\frac{\tilde\mu_\lambda(N)}{N^s}, \qquad \tilde\mu_\lambda(N) = \sum_{\prod_n n^{r_n}=N}\ \prod_{n:\,r_n\ge1}(-1)^{r_n}\binom{\tilde c_n(\lambda)}{r_n}, \tag{13.32}$$

absolutely convergent in $\Re s > 1$ for every λ (Corollary 13.11), because it is the exponential of the absolutely convergent series $-\sum_m \frac1m \sum_n \tilde c_n(\lambda) n^{-ms}$. Thus $\tilde\mu_\lambda(p) = -\tilde c_p(\lambda)$, and the atomized Mertens functions

$$\tilde M_\lambda(x) := \sum_{N\le x}\tilde\mu_\lambda(N), \qquad \frac{1}{\tilde\zeta_{\mathbb{Z},\lambda}} = e^{-\tilde h_\lambda}F_\lambda, \tag{13.33}$$

exist for every $\lambda \in [0,1]$. Under the conditions of Theorem 13.13, $\tilde c_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, so $\tilde\mu_0 = \mu$ and $\tilde M_0 = M$.

Proposition 13.16 (Mellin form of the endpoint Mertens function)**.** *There are $c_1 \ge 2$ and a unique measurable B_{Z1} on $(0,\infty)$, vanishing on $(0,1)$, with $x^{-c}B_{Z1} \in L^2\bigl((1,\infty),dx/x\bigr)$ and*

$$F_{Z1}(s) = s\int_1^\infty B_{Z1}(x)\,x^{-s-1}\,dx = \exp\Bigl(-\int_{[1,\infty)} x^{-s}\,dJ^{\mathrm M}_{Z1}(x)\Bigr) \qquad (\Re s > c_1), \tag{13.34}$$

for every $c > c_1$. Moreover (D) holds if and only if $B_{Z1} = M$ almost everywhere; and if F_{Z1} has an absolutely convergent Dirichlet expansion (11.37)*, then $B_{Z1} = M_{Z1}$.*

Proof. Put $A = 2(\pi+2)C_{\mathfrak e}$. By (12.5), for $\sigma' \ge \max(4A_{\mathrm{end}}, R_{\mathrm{end}})$, $|F_{Z1}(s)| \le C(|t|+\sigma)^{1/4}$; take $c_1 = \max(2, \frac12+4A, \frac12+R_{\mathrm{end}})$. Then $\int_{\mathbb R}|F_{Z1}(\sigma+it)|^2|\sigma+it|^{-2}dt \le C'\sigma^{-1/2}$ for $\sigma > c_1$, and the argument of Theorem 12.5 gives B_{Z1}, its vanishing on $(0,1)$, its independence of c and its uniqueness; the second equality in (13.34) is Theorem 12.5. If $\zeta_{Z1} = \zeta$, then $F_{Z1}(s) = 1/\zeta(s) = s\int_1^\infty M(x)x^{-s-1}dx$ and uniqueness gives $B_{Z1} = M$. Conversely, $B_{Z1} = M$ gives $F_{Z1} = 1/\zeta$ on $\Re s > c_1$, hence $\zeta_{Z1} = \zeta$ by analytic continuation. Under (11.37), partial summation gives $F_{Z1}(s) = s\int_1^\infty M_{Z1}(x)x^{-s-1}dx$, and uniqueness gives $B_{Z1} = M_{Z1}$. □

13.7. **Hamburger's hypotheses at the two endpoints.** Hamburger's theorem [33], in the form of [51, §2.13] (see also [49, 16]), has four hypotheses: (H0) $f = G/P$ with G entire of finite order and P a polynomial; (H1) $f(s) = \sum_n a_n n^{-s}$ absolutely for $\sigma > 1$, with arbitrary complex a_n; (H2) $f(s)\Gamma(\frac s2)\pi^{-s/2} = g(1-s)\Gamma(\frac{1-s}{2})\pi^{-(1-s)/2}$; (H3) $g(1-s) = \sum_n b_n n^{s-1}$ absolutely for $\sigma < -\alpha < 0$. It contains no hypothesis on the location of zeros, and its conclusion is $f = C\zeta$. Absolute convergence for $\sigma > 1$ in (H1) cannot be weakened to some half-plane: Knopp constructed infinitely many linearly independent solutions of (H0), (H2), (H3) that are Dirichlet series absolutely convergent only in some half-plane [36, 44]. Within the Selberg class, the functions of degree 1 are classified in [34]. The table records the status at the two endpoints and is updated as the programme advances.

	ζ_{Z1} (continuous limit)	$\tilde\zeta_{\mathbb{Z},0}$ (atomized limit)
H0	proved: $(s-1)\zeta_{Z1}$ entire of order 1 (Proposition 11.1)	open: meromorphic in $\Re s > \frac12$ only (Theorem 13.12)
H1	open: this is (D)	proved in $\Re s > 1$ (Corollary 13.11)
H2	proved, with $g = f$ (Proposition 11.1)	open
H3	follows from H1	open
zeros	no hypothesis	no hypothesis

The two columns meet when $\tilde{h}_0 \equiv 0$: then $\zeta_{Z1} = \tilde{\zeta}_{\mathbb{Z},0}$ has H0–H3, and Hamburger's theorem gives $\zeta_{Z1} = \zeta$ (Theorem 13.13). It suffices that the joining factor be symmetric (Theorem 14.3), or that ζ_{Z1} have a positive representing measure (Theorem 14.7).

The parallel between the two families, topic by topic of the paper, is the following.

Topic of the paper	continuous family ζ_λ	atomized family $\tilde{\zeta}_{\mathbb{Z},\lambda}$
clock, τ_k, bounded displacement	same	same
definition	Hadamard product over $\frac{1}{2} \pm i\gamma_k(\lambda)$	Euler product over $n \geq 2$ with masses $\tilde{c}_n(\lambda) = \langle d\pi_\lambda, \varphi_n \rangle$
zeros	exactly $\frac{1}{2} \pm i\gamma_k(\lambda)$; impossible elsewhere (Theorem 10.1)	none in $\Re s > \frac{1}{2}$ for every λ, since $\tilde{h}_\lambda$ is holomorphic there (Theorem 13.12(iii)); for $\lambda = 1$, exactly $\frac{1}{2} \pm i\tau_k$ in $\Re s > 0$ (Theorem 13.12(iv))
functional equation	proved for every λ	open; the factor $e^{\tilde{h}_\lambda}$ carries no functional equation
Dirichlet series	not for $\lambda = 1$ in any half-plane (Theorem 8.11); not in $\Re s > 1$ for $0 < \lambda < 1$ (Proposition 14.1); at $\lambda = 0$ this is (D)	proved in $\Re s > 1$ for every λ, including $\lambda = 0$ (Corollary 13.11)
first coefficient $(\sigma \to +\infty)$	$\varpi_\lambda = \varpi_{\mathrm{RVM}} \prod_k \frac{1/4+\tau_k^2}{1/4+\gamma_k(\lambda)^2}$ (13.1)	$\tilde{a}_1(\lambda) = 1$ for every λ
vertical behaviour	$\lambda = 1$: $\zeta_{\mathrm{RVM}}(\sigma + it) \to \varpi_{\mathrm{RVM}}$ as $\lvert t\rvert \to \infty$ (8.33)	almost periodic on every line of absolute convergence; $\lambda = 1$: no limit as $\lvert t\rvert \to \infty$ (Proposition 13.15)
half-plane bounds	$\lambda = 1$: $C^{-1} \leq \lvert\zeta_{\mathrm{RVM}}\rvert \leq C$ on $\Re s \geq \sigma_0 > 1$ (Theorem 8.10); every λ: $\lvert \log \zeta_\lambda - \log \frac{s}{s-1} \rvert \ll_c 1 + \log(\lvert t\rvert + 2)$ on $\Re s \geq c > \frac{1}{2}$	$\lvert\tilde{\zeta}_{\mathbb{Z},\lambda}\rvert \leq \sum_N \lvert\tilde{a}_N(\lambda)\rvert N^{-\sigma_0}$ and $\lvert\tilde{\zeta}_{\mathbb{Z},\lambda}^{-1}\rvert \leq \sum_N \lvert\tilde{\mu}_\lambda(N)\rvert N^{-\sigma_0}$ on $\Re s \geq \sigma_0 > 1$, for every λ
prime coordinate	counts J_λ, π_λ (distributions, (12.16))	masses $\tilde{c}_n(\lambda)$ at the integers; $\tilde{c}_n(1) = O(1)$; mean square $X^{1+\varepsilon}$ on $(X, 2X]$ for every λ (Theorem 13.10)
reciprocal	$F_\lambda = 1/\zeta_\lambda$; F_{Z1} is the Mellin transform of B_{Z1} (Proposition 13.16)	Dirichlet series with coefficients $\tilde{\mu}_\lambda(N)$ (13.32)
Mertens	$M_{\mathrm{RVM}}^{\mathrm{crit}} = L + O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 8.12); $B_{Z1} = M \iff (D)$	$\tilde{M}_\lambda = \sum_{N \leq x} \tilde{\mu}_\lambda(N)$ exists for every λ (13.33); the bound $x^{1/2+\varepsilon}$ is open (it needs control of $e^{-\tilde{h}_\lambda}$)
flow in λ	generator $\mathcal{V}_\lambda$ on the Hadamard product; $\Upsilon(\lambda)$ at $+\infty$ (Proposition 13.2)	$\log \tilde{\zeta}_{\mathbb{Z},\lambda} = \sum_n \tilde{c}_n(\lambda)\big(-\log(1 - n^{-s})\big)$, each mass differentiable in λ by (13.9); joining factor (13.15)
limit $\lambda \to 0^+$	$\zeta_\lambda \to \zeta_{Z1}$	$\tilde{c}_n(\lambda) \to \tilde{c}_n(0)$, $\mathfrak{h}_n(\lambda; \cdot) \to \mathfrak{h}_n(0; \cdot)$ (Proposition 13.8)
endpoint: Hamburger's hypotheses	H0 and H2 proved; H1 is (D), open	H1 proved in $\Re s > 1$ (Corollary 13.11); H0 and H2 open
relation between the endpoints	$\tilde{\zeta}_{\mathbb{Z},0} = \zeta_{Z1}\, e^{\tilde{h}_0}$	same
what closes the route	ζ_{Z1} satisfies H1	$\tilde{\zeta}_{\mathbb{Z},0}$ satisfies H0 and H2
common form of the closure	$\tilde{h}_0 \equiv 0$	$\tilde{h}_0 \equiv 0$

Topic of the paper	continuous family ζ_λ	atomized family $\tilde\zeta_{\mathbb{Z},\lambda}$
equivalent forms of the closure	ζ_{Z1} has a positive representing measure (Theorem 14.7)	$e^{\tilde h_0}$ of finite order and symmetric under $s\mapsto 1-s$ (Theorem 14.3)
consequence of the closure	$\zeta_{Z1}=\zeta$; Step 3 gives $M(x)=O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 11.8)	$\tilde a_N(0)=1$, $\tilde c_n(0)=\mathbf{1}_{\mathrm{prime}}(n)$, $\tilde\mu_0=\mu$, $\tilde M_0=M$; $\tilde\zeta_{\mathbb{Z},0}=\zeta=\zeta_{Z1}$

How the two routes complement each other. Along λ each family keeps exactly the half of Hamburger's hypotheses that the other one lacks:

$$\underbrace{\zeta_\lambda}_{\substack{\text{keeps the functional equation}\\ \text{and the zeros on } \Re s=\frac12}} \quad\longleftrightarrow\quad \underbrace{\tilde\zeta_{\mathbb{Z},\lambda}}_{\substack{\text{keeps the Dirichlet series}\\ \text{and the Euler product}}}, \qquad \tilde\zeta_{\mathbb{Z},\lambda}=\zeta_\lambda\, e^{\tilde h_\lambda}. \tag{13.35}$$

The closure is the same in both columns, the vanishing of the joining factor at the endpoint: by (13.13) and Theorem 13.13,

$$\tilde h_0\equiv 0 \iff J_{Z1}=0 \text{ on } (1,2),\quad \mathfrak{h}_n(0;s)=0 \text{ for all } n\ge 2,\ \Re s>1,\ \text{and } \varpi_{Z1}=1. \tag{13.36}$$

No further condition is needed in the atomized column: the cell masses have mean square $X^{1+\varepsilon}$ uniformly in λ (Theorem 13.10), so $\tilde\zeta_{\mathbb{Z},0}$ satisfies H1 (Corollary 13.11).

13.8. **Programme.** The steps are numbered in the order in which they are followed. The status column is updated with each step.

Step	Statement	Status
A1	Atomization by smooth cells, coefficients $\tilde a_N(\lambda)$	proved (Definition 13.4)
A2	Joining factor and explicit cell defects, (13.13)–(13.17)	proved (Thm. 13.6, Prop. 13.7)
A3	L^2 square-root estimate for every λ; Dirichlet series in $\Re s>\frac32$	proved (Theorem 13.12); superseded by A6
A4	$\tilde h_\lambda$ holomorphic in $\Re s>\frac12$; in $\Re s>0$ for $\lambda=1$	proved (Theorem 13.12)
A5	Limits $\lambda\to 0^+$ of $\tilde c_n$, $\mathfrak{h}_n$, ϖ_λ; flow at infinity Υ	proved (Propositions 13.2, 13.8)
A5′	Endpoints: $\tilde h_1$ not constant; atomized Möbius coefficients $\tilde\mu_\lambda$; Mellin form B_{Z1}, with $B_{Z1}=M_{Z1}$ only under (11.37)	proved (Propositions 13.15, 13.16)
A6	Mean square of the cell masses uniformly in λ; Dirichlet series in $\Re s>1$ for every λ, including $\lambda=0$	proved (Theorem 13.10, Corollary 13.11)
A7	Cell identity (13.30) for every $n\ge 2$, and $J_{Z1}=0$ on the first cell $(1,2)$	open; reduced to one number for each cell in A20
A8	Equivalent scalar form: $\int_{0^+}^1 \Upsilon\, d\lambda=\log L$, i.e. (12.12)	open; follows from A7
A9	$\zeta_{Z1}=\zeta$ by Hamburger; $M(x)=O_\varepsilon(x^{1/2+\varepsilon})$ by Step 3	proved as implication from A7 (Thms. 13.13, 11.8)
A10	Continuation of $\tilde h_\lambda$ to $0<\Re s\le\frac12$ for $\lambda<1$	open; not needed for A9; at $\lambda=0$ it follows from (B1), (B2) (A23)
A11	ζ_λ is not a Dirichlet series and $\tilde h_\lambda$ is not constant for $0<\lambda\le 1$; convexity of $\log\zeta_\lambda(\sigma)$ in λ	proved (Propositions 14.1, 14.2)
A12	Hamburger's theorem for the pair: A9 follows if $e^{\tilde h_0}$ is of finite order and symmetric under $s\mapsto 1-s$	proved as implication (Theorem 14.3); at $\lambda=0$ the hypothesis is equivalent to (D) (Remark 14.30)
A13	Positivity: A9 follows if ζ_{Z1} has a positive representing measure; then $\varpi_{Z1}\le 1$	proved as implication (Theorem 14.7, Corollary 14.8)
A14	Clock end: density plus explicit atoms of $d\pi_{\mathrm{RVM}}$, cell masses $\frac{1}{\log n}+\mathbf{1}_{2^j}(n)\frac{\mu(j)}{j}J_{\mathrm{RVM}}(2)+O(n^{-1/4})$, defects $O(\lvert s\rvert n^{-\sigma-4/3})$	proved (Theorem 14.15, Corollary 14.16)

Step	Statement	Status
A15	Height kernel of the cell masses; flow as count difference; Gaussian defect and the density condition $\tilde{h}_0(1) = 0$	proved (Propositions 14.17, 14.18, 14.22)
A16	Hamburger's theorem for the pair without continuation: A9 follows if $T_0 \equiv 0$, or if $\mathsf{V}_\sigma(0) = 0$; symmetry fails at $\lambda = 1$; exceptional set finite	proved (Theorem 14.4, Propositions 14.5, 14.25)
A17	$\zeta_{Z1}(\sigma) \geq \zeta(\sigma)$ for $\sigma > 1$; A9 follows from the reverse inequality at one σ, or from $\varpi_\lambda \leq 1$ along $\lambda \to 0^+$, or from $\mathsf{W}(0) \geq 0$	proved (Theorem 14.11, Propositions 14.10, 14.12)
A18	Space $\mathscr{O}$ of ordinate lists: at most one coincidence list; A9 is equivalent to its existence	proved (Theorem 15.1, Corollary 15.3)
A19	Existence of a list in $\mathscr{O}$ with $\tilde{h}_{\boldsymbol{\kappa}}$ constant, equivalently $\min_{\mathscr{O}} \mathsf{V}_\sigma = 0$	open; equivalent to A7
A20	A9 holds if and only if $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n; every $\lambda \in \mathsf{E}$ is a witness; the identity (14.59) cell by cell	proved (Corollary 14.26, Propositions 14.27, 14.29); the identity (14.59) itself is open
A21	Symmetric joining factors: order one is needed, and A12 has no slack at $\lambda = 0$	proved (Proposition 14.31, Remarks 14.30, 14.32)
A22	Half-line form: A9 follows if $\tilde{h}_0(\sigma) = O_N(\sigma N^{-\sigma})$ for every N; the Nth rung is the constancy of J^{M}_{Z1} on $(N-1, N)$	proved (Theorems 14.33, 14.34, Corollary 14.35)
A23	Bridge theorem: A9 follows if $\Im\tilde{h}_0$ vanishes on $\Re s = \frac{1}{2}$ and $\tilde{h}_0$ is temperate; the two conditions hold for no $\lambda > 0$	proved (Theorem 14.36, Corollary 14.37); (B1), (B2) at $\lambda = 0$ open
A24	Transport from the clock: S_f entire of type $C_\eta \log N$, moment series on the clock, defect of the clock by Poisson summation	proved (Proposition 14.41, Theorem 14.42, Proposition 14.43)
A24b	Multiscale Jacobian at the clock: stable finite ordinate block on windows $U \asymp \log T$, fixed-window obstruction, dense range, and rapidly summable smooth tails	proved (Theorem 14.44); quantitative multiscale continuation to a solution open
A25	Rung system with free masses: A9 holds if and only if it has a solution in $\mathscr{O}$; the solution is unique and nondegenerate	proved (Theorems 15.5, 15.6); existence open
A26	Coordinate gradient of V_σ on $\mathscr{O}$; A9 would follow from a boundary-aware coordinate/KKT criterion forcing every minimizer to have zero defect	proved (Proposition 15.8); the forcing statement is open
A27	Acceptance criterion and models: the final step is a theorem for the finite Euler products; no object of finite rank in the window of ζ	proved (Theorems 16.1, 16.3, Propositions 16.2, 16.4)

Remark 13.17 (Input for A7). Steps A1–A6 use only two properties of an ordinate list: bounded displacement from the clock and a primitive of the count difference of size $O(\log \tau)$. The list (τ_k) has both properties, and for it the corresponding function ζ_{RVM} is not a Dirichlet series (Theorem 8.11). Step A7 must therefore use a property of the ordinates γ_k that the clock ordinates do not have; the natural candidate is the arithmetic of ζ carried by the γ_k, for instance Landau's formula $\sum_{0<\gamma\leq T} x^\rho = -\frac{T}{2\pi}\Lambda(x) + O(\log T)$, in which the prime powers appear as resonances of the ordinates. The multiscale analysis below separates a different issue: the linearized clock has enough directions, but only on windows whose length grows like $\log T$; thus the obstruction is not fixed-height injectivity but quantitative nonlinear continuation across scales.

14. Criteria for the final step and the clock end of the family

This section collects the consequences of Corollary 13.11 for the final step. Along the open path $0 < \lambda \leq 1$ no member ζ_λ is a Dirichlet series (Proposition 14.1); the family is convex in λ on the real axis (Proposition 14.2); Hamburger's theorem applies to the pair $(\zeta_\lambda, \tilde{\zeta}_{\mathbb{Z},\lambda})$ as soon as

the joining factor is symmetric (Theorem 14.3); positivity of the measure representing ζ_{Z1} gives $\varpi_{Z1} \le 1$ and closes the route (Theorem 14.7). At the clock end the density of $d\pi_{\mathrm{RVM}}$ is computed explicitly (Theorem 14.15), and the flow of the cell masses is written in the heights of the ordinates (Propositions 14.17 and 14.18). Subsections 14.9–14.13 reduce the final step to the cell masses (Corollary 14.26), state two criteria whose hypotheses are written where the joining factor is defined, the half-line form (Theorem 14.34) and the bridge theorem (Theorem 14.36), and compute the transport from the clock on each cell (Theorem 14.42).

Throughout this section the function χ of Subsection 13.4 is even, so that φ is even, and

$$\hat\varphi(\xi) := \int_{\mathbb{R}} \varphi(t) e^{-2\pi i \xi t}\, dt, \qquad \mathsf{M}_r := \int_{\mathbb{R}} t^r \varphi(t)\, dt, \qquad \mathsf{C}_r := \int_{\mathbb{R}} t^r \varphi(t) \cos(2\pi t)\, dt. \tag{14.1}$$

$\hat\varphi$ is real and even, and $\mathsf{M}_1 = \mathsf{C}_1 = 0$. Since $\sum_{m\in\mathbb{Z}} \varphi(t-m) = 1$,

$$\hat\varphi(q) = \int_0^1 \sum_{m\in\mathbb{Z}} \varphi(t-m)\, e^{-2\pi i q t}\, dt = \delta_{q0} \qquad (q \in \mathbb{Z}), \tag{14.2}$$

so $\mathsf{M}_0 = 1$ and $\mathsf{C}_0 = \hat\varphi(1) = 0$.

14.1. **The open path.**

Proposition 14.1 (No member of the open path is a Dirichlet series)**.** *Let $0 < \lambda \le 1$. Then ζ_λ is not a constant multiple of a Dirichlet series $\sum_n a_n n^{-s}$ converging absolutely in $\Re s > 1$. In particular the joining factor $\tilde h_\lambda$ is not constant.*

Proof. Suppose $\zeta_\lambda(s) = \sum_n a_n n^{-s}$ absolutely for $\Re s > 1$. By Theorem 10.1, exactly as in Proposition 11.1, $(s-1)\zeta_\lambda(s)$ is entire of order one and ζ_λ satisfies (11.3); these are the hypotheses of Hamburger's theorem, so $\zeta_\lambda = C\zeta$ with $C \neq 0$. The nontrivial zeros of ζ with positive ordinate are then $\frac12 + i\gamma_k(\lambda)$, $k \ge 1$, with multiplicity. The ordinates of ζ, listed in non-decreasing order, are the γ_k, and the $\gamma_k(\lambda)$ are increasing; hence $\gamma_k(\lambda) = \gamma_k$, that is $\lambda\eta_k = 0$, for every k. Since $\lambda > 0$, every η_k vanishes, all members of the family coincide (Remark 10.2), and $\zeta_{\mathrm{RVM}} = \zeta_\lambda$ is an absolutely convergent Dirichlet series, against Theorem 8.11. If $\tilde h_\lambda$ were a constant c, then $\zeta_\lambda = e^{-c}\tilde\zeta_{\mathbb{Z},\lambda}$ would be such a series by Corollary 13.11. □

The two columns of (13.35) exist for every λ, and for every $\lambda \in (0,1]$ they are different functions. The joining factor converges, $\tilde h_\lambda \to \tilde h_0$ as $\lambda \to 0^+$, through non-constant functions; (D) is the statement that the limit is constant. Any property of the displacement sequence that would imply (D) must distinguish the full sequence (η_k) from every scaled sequence $(c\,\eta_k)$ with $0 \le c < 1$: the latter correspond to members $\lambda > 0$ of the family, for which Proposition 14.1 excludes closure. Thus this observation is a necessary discrimination test for a proposed criterion, not an assertion that the actual displacement sequence already satisfies such a criterion.

14.2. **Convexity along the family.**

Proposition 14.2 (Convexity on the real axis)**.** *For every real $\sigma > 1$ the function $\lambda \mapsto \log\zeta_\lambda(\sigma)$ is strictly convex on $[0,1]$:*

$$\partial_\lambda^2 \log\zeta_\lambda(\sigma) = \sum_{k\ge1} \eta_k^2 \Big(\mathcal{C}(\sigma - \tfrac12, \gamma_k(\lambda)) - \mathcal{C}(\tfrac12, \gamma_k(\lambda)) \Big) > 0, \qquad \mathcal{C}(a,\gamma) := \frac{2(a^2-\gamma^2)}{(a^2+\gamma^2)^2}. \tag{14.3}$$

Consequently, with the generator $\mathcal{V}_\lambda$ of Theorem 10.3,

$$\log\zeta_{\mathrm{RVM}}(\sigma) - \mathcal{V}_1(\sigma) < \log\zeta_{Z1}(\sigma) < \log\zeta_{\mathrm{RVM}}(\sigma) - \mathcal{V}_0(\sigma) \qquad (\sigma > 1). \tag{14.4}$$

Proof. With $a = \sigma - \frac12 > \frac12$ one has $q(\sigma) = a^2 - \frac14$, so by (10.2) $\log\zeta_\lambda(\sigma) = -\log G(\sigma) + \sum_k \big(\log(a^2 + \gamma_k(\lambda)^2) - \log(\frac14 + \gamma_k(\lambda)^2)\big)$. Since $\partial_\gamma^2 \log(a^2+\gamma^2) = \mathcal{C}(a,\gamma)$ and $\partial_\lambda \gamma_k(\lambda) = \eta_k$, termwise differentiation gives the series in (14.3); its terms are $O(\eta_k^2 \gamma_k^{-2})$, so it converges uniformly on $[0,1]$. Fix $\gamma > \frac12$. Then

$\mathcal{C}(\frac12,\gamma)<0$; $\partial_a\mathcal{C}(a,\gamma)=4a(3\gamma^2-a^2)(a^2+\gamma^2)^{-3}>0$ for $0<a<\sqrt{3}\,\gamma$; and $\mathcal{C}(a,\gamma)>0$ for $a>\gamma$. Hence $\mathcal{C}(a,\gamma)>\mathcal{C}(\frac12,\gamma)$ for every $a>\frac12$. All ordinates satisfy $\gamma_k(\lambda)>\tau_0>\frac12$ and $\eta_1=\tau_1-\gamma_1\neq0$ ($\tau_1=14.52\ldots$ by (2.5), $\gamma_1=14.13\ldots$), so the sum is positive. A strictly convex function F on $[0,1]$ satisfies $F(1)-F'(1)<F(0)<F(1)-F'(0)$; with $F(\lambda)=\log\zeta_\lambda(\sigma)$ and $F'=\mathcal{V}_\lambda(\sigma)$ (10.11) this is (14.4). □

As $\sigma\to+\infty$, (14.3) becomes $-\Upsilon'(\lambda)>0$ and (14.4) becomes the last statement of Proposition 13.2. The sign in (14.3) comes from the test function $x^{-\sigma}$, whose transform is not oscillatory in the ordinate; for the cell functions φ_n the corresponding second derivative $\partial_\lambda^2\tilde c_n(\lambda)=-2\Re\sum_k\eta_k^2W_n''(\gamma_k(\lambda))$ has terms of both signs.

14.3. **Hamburger's theorem for the pair.**

Theorem 14.3 (Hamburger's theorem for the pair $(\zeta_\lambda,\tilde\zeta_{\mathbb{Z},\lambda})$)**.** *Let $\lambda\in[0,1]$. Suppose that the joining factor $e^{\tilde h_\lambda}$ extends to a meromorphic function on $\mathbb{C}$ of finite order with finitely many poles, and that*

$$e^{\tilde h_\lambda(s)}=e^{\tilde h_\lambda(1-s)}. \tag{14.5}$$

Then $\tilde\zeta_{\mathbb{Z},\lambda}=\zeta$. For $\lambda=0$, moreover, $\zeta_{Z1}=\zeta$ and $\tilde h_0\equiv0$.

Proof. Put $f=\tilde\zeta_{\mathbb{Z},\lambda}=\zeta_\lambda e^{\tilde h_\lambda}$. By Theorem 10.1, $(s-1)\zeta_\lambda$ is entire of order one, so $f=\mathsf{G}/\mathsf{P}$ with G entire of finite order and P a polynomial: this is (H0). By (11.3) for ζ_λ and (14.5), $f(s)\Gamma(\frac{s}{2})\pi^{-s/2}=f(1-s)\Gamma(\frac{1-s}{2})\pi^{-(1-s)/2}$: this is (H2) with $g=f$. (H1) is Corollary 13.11, and it gives (H3) for $g=f$. Hamburger's theorem gives $f=C\zeta$, and $C=1$ because $\tilde a_1(\lambda)=1$. By Theorem 13.12(iii) and the functional equation, the nontrivial zeros of $f=\zeta$ lie on $\Re s=\frac12$; they are the points $\frac12\pm i\gamma_k$. For $\lambda=0$, Theorem 11.2 then gives $\zeta_{Z1}=\zeta=\tilde\zeta_{\mathbb{Z},0}$, and $e^{\tilde h_0}=1$. □

Theorem 14.3 uses the functional equation of ζ_λ and the Dirichlet series of $\tilde\zeta_{\mathbb{Z},\lambda}$ as they stand; the only hypothesis is on the factor that joins them, and it asks for symmetry instead of constancy. At $\lambda=1$ the two hypotheses do not hold together: the conclusion $\tilde\zeta_{\mathbb{Z},1}=\zeta$ would place the zeros of ζ at $\frac12\pm i\tau_k$ (Theorem 13.12(iv)), so $\zeta=\zeta_{\mathrm{RVM}}$ by Theorem 9.1, against Theorem 8.11.

Theorem 14.4 (Real-variable and vertical forms of Hamburger's theorem for the pair)**.** *For $\lambda\in[0,1]$, $y>0$ and $\sigma>1$ put*

$$T_\lambda(y):=\sum_{N\ge1}\tilde a_N(\lambda)\,e^{-\pi N^2y}-y^{-1/2}\sum_{N\ge1}\tilde a_N(\lambda)\,e^{-\pi N^2/y}-\tfrac12(y^{-1/2}-1), \tag{14.6}$$

$$k_n(\lambda):=\sum_{d\mid n}\tilde a_d(\lambda)\,\mu(n/d),\qquad \mathsf{V}_\sigma(\lambda):=\sum_{n\ge2}|k_n(\lambda)|^2\,n^{-2\sigma}. \tag{14.7}$$

(i) *$\tilde\zeta_{\mathbb{Z},\lambda}/\zeta=\sum_{n\ge1}k_n(\lambda)n^{-s}$ converges absolutely in $\Re s>1$, $k_1(\lambda)=1$, and*

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\left|\frac{\tilde\zeta_{\mathbb{Z},\lambda}(\sigma+it)}{\zeta(\sigma+it)}\right|^2dt=1+\mathsf{V}_\sigma(\lambda)\ \ge1. \tag{14.8}$$

(ii) *For each λ the following are equivalent:* (a) $\tilde\zeta_{\mathbb{Z},\lambda}=\zeta$; (b) $T_\lambda(y)=0$ *for every $y>0$;* (c) $\mathsf{V}_\sigma(\lambda)=0$ *for some, equivalently every, $\sigma>1$. Each of them implies (D).*

(iii) *$\lambda\mapsto T_\lambda(y)$ and $\lambda\mapsto\mathsf{V}_\sigma(\lambda)$ are continuous on $[0,1]$, and*

$$\text{(D)}\iff T_0\equiv0\iff\mathsf{V}_\sigma(0)=0\iff\lim_{\lambda\to0^+}T_\lambda(y)=0\ \ (y>0)\iff\lim_{\lambda\to0^+}\mathsf{V}_\sigma(\lambda)=0. \tag{14.9}$$

At the clock end $T_1\not\equiv0$ and $\mathsf{V}_\sigma(1)>0$.

Proof. (i) $1/\zeta = \sum_n \mu(n) n^{-s}$ and $\tilde\zeta_{\mathbb{Z},\lambda}$ (Corollary 13.11) converge absolutely in $\Re s > 1$, and so does their product, with $k_1 = \tilde a_1 = 1$. The mean square of an absolutely convergent Dirichlet series on a vertical line is $\sum_n |k_n|^2 n^{-2\sigma}$, as in (8.43)–(8.44). (ii) (a)⇔(c) because $\mathsf{V}_\sigma(\lambda) = 0$ means $k_n(\lambda) = 0$ for $n \geq 2$. (a)⇒(b) is the theta relation $\sum_{n\geq 1} e^{-\pi n^2 y} = y^{-1/2} \sum_{n\geq 1} e^{-\pi n^2/y} + \frac{1}{2}(y^{-1/2} - 1)$. (b)⇒(a): the relation $T_\lambda \equiv 0$ is the identity reached in the first half of the proof of Hamburger's theorem in [51, §2.13], with $b_n = a_n = \tilde a_n(\lambda)$; the second half of that proof uses only this identity and the absolute convergence of $\sum_n a_n n^{-\sigma'}$ for $\sigma' > 1$, and gives $a_k = a_{k+1}$ for every k, hence $\tilde a_N(\lambda) = \tilde a_1(\lambda) = 1$. If (a) holds, then by Theorem 13.12(iii) and the functional equation the nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$; they are the points $\frac{1}{2} \pm i\gamma_k$, and Theorem 11.2 gives (D). (iii) By Theorem 13.10, for every $\sigma' > 1$ the sums $\sum_n |\tilde c_n(\lambda)| n^{-\sigma'}$ are bounded uniformly in λ; hence so are $\sum_N |\tilde a_N(\lambda)| N^{-\sigma'} \leq \exp\bigl(\sum_m \frac{1}{m} \sum_n |\tilde c_n(\lambda)| n^{-m\sigma'}\bigr)$ and $\sum_n |k_n(\lambda)| n^{-\sigma'}$. Each $\tilde a_N(\lambda)$ and each $k_n(\lambda)$ is a polynomial in finitely many cell masses, which are continuous in λ. The tails $\sum_{N>N_0}$ of (14.6) and of (14.7) (with $1 < \sigma' < \sigma$) are small uniformly in λ, which gives the continuity. Under (D), $\tilde a_N(0) = 1$ (Theorem 13.13), so $T_0 \equiv 0$ and $\mathsf{V}_\sigma(0) = 0$; the converse is (ii) at $\lambda = 0$, and the two limits are the values at $\lambda = 0$ by continuity. Finally $\tilde\zeta_{\mathbb{Z},1} \neq \zeta$ by the remark after Theorem 14.3. □

Theorem 14.4 needs no continuation of the joining factor: (14.6) is a real function of y built from the atomized coefficients alone, and (14.8) lives in $\Re s > 1$, where

$$\frac{\tilde\zeta_{\mathbb{Z},\lambda}}{\zeta} = e^{\tilde h_\lambda} \frac{\zeta_\lambda}{\zeta} \tag{14.10}$$

is a quotient of three explicit functions. Its vertical mean is 1 and its vertical mean square is at least 1 for every λ; (D) is the statement that at $\lambda = 0$ the mean square does not exceed 1.

Proposition 14.5 (At the clock end the symmetry fails)**.** *The joining factor $e^{\tilde h_1}$ has no meromorphic continuation to $\mathbb{C}$ that satisfies* (14.5)*.*

Proof. $e^{\tilde h_1}$ is holomorphic and zero-free in $\Re s > 0$ (Theorem 13.12(iv)). A meromorphic continuation with (14.5) is holomorphic and zero-free also in $\Re s < 1$, hence entire and zero-free, so it equals e^g with g entire, $g = \tilde h_1$ on $\Re s > 0$, and $g(s) - g(1-s)$ constant. For $\Re s \geq \frac{1}{4}$, the bound $|\mathfrak{h}_n(1; ms)| \leq C\, m|s|\, n^{-m\sigma-1}$ of Theorem 13.12(iv) gives $\sum_m \frac{1}{m} \sum_n |\mathfrak{h}_n(1; ms)| \leq C|s| \sum_{n\geq 2} n^{-1}(n^{1/4} - 1)^{-1}$, and $|\mathfrak{u}_1(s)| \leq C(1 + |s|)$ by (13.11); so $|g(s)| \leq C(1 + |s|)$ on $\Re s \geq \frac{1}{4}$, and by the symmetry also on $\Re s \leq \frac{3}{4}$. An entire function of linear growth is a polynomial of degree at most 1, and $g(s) - g(1-s)$ constant then forces g to be constant. This contradicts Proposition 13.15(iii). □

At $\lambda = 1$ it is therefore the symmetry (14.5) itself that fails in Theorem 14.3, whatever the order of the continuation.

14.4. **Positivity and the value at infinity.** For an even Schwartz function g on $\mathbb{R}$ put $\hat{\mathsf{g}}(\xi) = \int_{\mathbb{R}} \mathsf{g}(u) e^{-2\pi i \xi u} du$ and $\mathsf{g}^\sharp(s) = \int_0^\infty \mathsf{g}(u) u^{s-1} du$.

Lemma 14.6 (Summation formula for a representing measure)**.** *Let $\lambda \in [0, 1]$ and let A be a positive measure on $[1, \infty)$ with*

$$\zeta_\lambda(s) = \int_{[1,\infty)} u^{-s}\, d\mathsf{A}(u) \qquad (\Re s > 1). \tag{14.11}$$

Then for every even Schwartz function g

$$\mathsf{g}(0) + 2\int_{[1,\infty)} \mathsf{g}\, d\mathsf{A} = \hat{\mathsf{g}}(0) + 2\int_{[1,\infty)} \hat{\mathsf{g}}\, d\mathsf{A}. \tag{14.12}$$

Proof. $\mathsf{g}^\sharp$ is meromorphic with simple poles at $s = 0, -2, -4, \ldots$, residue $\mathsf{g}(0)$ at $s = 0$, and decays faster than any power of $|t|$ on vertical strips; $\mathsf{g}^\sharp(1) = \frac{1}{2}\hat{\mathsf{g}}(0)$; and

$$\pi^{-s/2}\Gamma(\tfrac{s}{2})\, \hat{\mathsf{g}}^\sharp(1-s) = \pi^{-(1-s)/2}\Gamma(\tfrac{1-s}{2})\, \mathsf{g}^\sharp(s) \tag{14.13}$$

for $0 < \Re s < 1$, and then by meromorphic continuation: the Fourier transform of the even homogeneous distribution $|\xi|^{-s}$ is a multiple of $|u|^{s-1}$, and the multiple is fixed by $\mathsf{g}(u) = e^{-\pi u^2}$, for which $\hat{\mathsf{g}} = \mathsf{g}$ and $\mathsf{g}^\sharp(s) = \frac{1}{2}\pi^{-s/2}\Gamma(\frac{s}{2})$. Fix $1 < c < 2$. By Mellin inversion and Fubini, $\int \mathsf{g}\, d\mathsf{A} = \frac{1}{2\pi i}\int_{(c)} \zeta_\lambda(s)\mathsf{g}^\sharp(s)\, ds$, since $\int u^{-c} d\mathsf{A} < \infty$. On $\Re s \ge c$, $|\zeta_\lambda(s)| \le \zeta_\lambda(c)$; with the functional equation (11.3), valid for every ζ_λ by Theorem 10.1, and the finite order of $(s-1)\zeta_\lambda$, the Phragmén–Lindelöf principle bounds ζ_λ by a power of $|t|$ on $1-c \le \Re s \le c$, $|t| \ge 1$. Moving the line to $\Re s = 1-c$ crosses $s = 1$, where ζ_λ has residue 1, and $s = 0$, where $\zeta_\lambda(0) = P_\lambda(0)/G(0) = -\frac{1}{2}$:

$$\int \mathsf{g}\, d\mathsf{A} = \tfrac{1}{2}\hat{\mathsf{g}}(0) - \tfrac{1}{2}\mathsf{g}(0) + \frac{1}{2\pi i}\int_{(1-c)} \zeta_\lambda(s)\mathsf{g}^\sharp(s)\, ds. \tag{14.14}$$

By (11.3) and (14.13), $\zeta_\lambda(s)\mathsf{g}^\sharp(s) = \zeta_\lambda(1-s)\hat{\mathsf{g}}^\sharp(1-s)$, and the last integral is $\frac{1}{2\pi i}\int_{(c)} \zeta_\lambda(s)\hat{\mathsf{g}}^\sharp(s)\, ds = \int \hat{\mathsf{g}}\, d\mathsf{A}$. □

Theorem 14.7 (Positivity criterion)**.** *Let $\lambda \in [0,1]$. If ζ_λ has a representation* (14.11) *with a positive measure* A*, then*

$$\varpi_\lambda = \mathsf{A}(\{1\}) \le 1. \tag{14.15}$$

In particular, if ζ_{Z1} has such a representation, then $\varpi_{Z1} = 1$ and $\zeta_{Z1} = \zeta$.

Proof. By monotone convergence $\zeta_\lambda(\sigma) = \int u^{-\sigma} d\mathsf{A} \to \mathsf{A}(\{1\})$ as $\sigma \to +\infty$, and the limit is ϖ_λ (Proposition 13.1). Let $\mathsf{T}(\xi) = (1-|\xi|)_+$ and $\mathsf{B} = \mathsf{T} * \mathsf{T}$, so that $\mathsf{B}(\xi) = \frac{2}{3} - \xi^2 + \frac{1}{2}|\xi|^3$ for $|\xi| \le 1$, $\mathsf{B}(\xi) = \frac{1}{6}(2-|\xi|)^3$ for $1 \le |\xi| \le 2$, $\mathsf{B} = 0$ elsewhere, and $\mathsf{B}''(\xi) = \mathsf{T}(\xi+1) - 2\mathsf{T}(\xi) + \mathsf{T}(\xi-1)$. Put

$$\mathsf{g}(u) := \Big(\frac{\sin \pi u}{\pi u}\Big)^4 (1-u^2), \qquad \hat{\mathsf{g}}(\xi) = \mathsf{B}(\xi) + \frac{1}{4\pi^2}\mathsf{B}''(\xi). \tag{14.16}$$

Then $\mathsf{g}(0) = 1$; $\mathsf{g}(u) \le 0$ for $|u| \ge 1$; $\hat{\mathsf{g}}(0) = \frac{2}{3} - \frac{1}{2\pi^2}$ and $\hat{\mathsf{g}}(1) = \frac{1}{6} + \frac{1}{4\pi^2}$; and $\hat{\mathsf{g}} \ge 0$ on $\mathbb{R}$: for $0 \le \xi \le 1$, $\hat{\mathsf{g}}(\xi) = \frac{2}{3} - \xi^2 + \frac{1}{2}\xi^3 + \frac{3\xi - 2}{4\pi^2}$ is positive at $\xi = 0$ and $\xi = 1$ and its derivative $-2\xi + \frac{3}{2}\xi^2 + \frac{3}{4\pi^2}$ changes sign once, from positive to negative, and for $1 \le \xi \le 2$, $\hat{\mathsf{g}}(\xi) = \frac{1}{6}(2-\xi)^3 + \frac{2-\xi}{4\pi^2} \ge 0$. For $\epsilon > 0$ let $\mathsf{g}_\epsilon(u) = \mathsf{g}(u)e^{-\epsilon\pi u^2}$, an even Schwartz function with $\mathsf{g}_\epsilon(0) = 1$, $\mathsf{g}_\epsilon \le 0$ on $|u| \ge 1$, and $\hat{\mathsf{g}}_\epsilon = \hat{\mathsf{g}} * (\epsilon^{-1/2}e^{-\pi\xi^2/\epsilon}) \ge 0$. By Lemma 14.6, since $\mathsf{A} \ge 0$ is carried by $[1,\infty)$ and has mass ϖ_λ at $u = 1$,

$$1 \;\ge\; \mathsf{g}_\epsilon(0) + 2\int \mathsf{g}_\epsilon\, d\mathsf{A} = \hat{\mathsf{g}}_\epsilon(0) + 2\int \hat{\mathsf{g}}_\epsilon\, d\mathsf{A} \;\ge\; \hat{\mathsf{g}}_\epsilon(0) + 2\varpi_\lambda\, \hat{\mathsf{g}}_\epsilon(1). \tag{14.17}$$

Letting $\epsilon \to 0$, $\hat{\mathsf{g}}_\epsilon \to \hat{\mathsf{g}}$ pointwise because $\hat{\mathsf{g}}$ is continuous, and

$$\varpi_\lambda \le \frac{1 - \hat{\mathsf{g}}(0)}{2\hat{\mathsf{g}}(1)} = \frac{\frac{1}{3} + \frac{1}{2\pi^2}}{\frac{1}{3} + \frac{1}{2\pi^2}} = 1. \tag{14.18}$$

The last statement follows from Theorem 12.2. □

Corollary 14.8 (Positive form of the final step)**.** *The following are equivalent:* (i) *(D);* (ii) *ζ_{Z1} is the Mellin–Stieltjes transform* (14.11) *of a positive measure on $[1,\infty)$;* (iii) *J^{M}_{Z1}, extended by* 0 *to $(0,1]$, is non-decreasing on $(0,\infty)$.*

Proof. (i)⇒(iii): then $J^{\mathrm{M}}_{Z1} = J$ (Theorem 12.7). (iii)⇒(ii): $J^{\mathrm{M}}_{Z1}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$, so a non-decreasing J^{M}_{Z1} is $O_c(x^c)$ and dJ^{M}_{Z1} is a positive measure on $[1,\infty)$ with $\int x^{-\sigma} dJ^{\mathrm{M}}_{Z1} < \infty$ for $\sigma > 1$. By Theorem 12.5, $\zeta_{Z1}(s) = \exp\int x^{-s} dJ^{\mathrm{M}}_{Z1} = \int u^{-s} d\mathsf{A}(u)$, where $\mathsf{A} = \sum_{r\ge 0} \frac{1}{r!}(dJ^{\mathrm{M}}_{Z1})^{*r}$ (multiplicative convolution powers) is a positive measure on $[1,\infty)$. (ii)⇒(i) is Theorem 14.7. □

In Theorem 14.7 the Dirichlet series is not used: (H1) is replaced by the positivity of the representing measure, the functional equation gives $\varpi_{Z1} \le 1$ through (14.17), and the Hadamard product gives $\varpi_{Z1} \ge 1$ (Theorem 12.2). The function (14.16) is extremal for the integers: for $\mathsf{A} = \sum_{n\ge 1} \delta_n$ both inequalities in (14.17) are equalities in the limit, since $\mathsf{g}(n) = 0$ for $n \ge 1$ and

$\hat{\mathsf{g}}(n) = 0$ for $n \geq 2$. At the clock end $\varpi_{\mathrm{RVM}} < 1$, and the density of dJ_{RVM} takes negative values in every cell $(n, n+1)$ with n large (Theorem 14.15).

Lemma 14.9 (Mellin form and summation formula for every member)**.** *There is $c_1 \geq 2$ such that for every $\lambda \in [0,1]$ there is a unique measurable function A_λ on $(0,\infty)$, vanishing on $(0,1)$, with $x^{-c}\mathsf{A}_\lambda \in L^2((1,\infty), dx/x)$ for every $c > c_1$ and*

$$\zeta_\lambda(s) = s\int_1^\infty \mathsf{A}_\lambda(x)\, x^{-s-1}\, dx \qquad (\Re s > c_1). \tag{14.19}$$

If ζ_λ has a representation (14.11)*, then $\mathsf{A}_\lambda(x) = \mathsf{A}([1,x])$ almost everywhere. With the pairing $\langle \mathsf{A}_\lambda, \mathsf{g}\rangle := -\int_1^\infty \mathsf{A}_\lambda(x)\, \mathsf{g}'(x)\, dx$, every even Schwartz function g satisfies*

$$\mathsf{g}(0) + 2\langle \mathsf{A}_\lambda, \mathsf{g}\rangle = \hat{\mathsf{g}}(0) + 2\langle \mathsf{A}_\lambda, \hat{\mathsf{g}}\rangle. \tag{14.20}$$

Equivalently, the even tempered distribution $\delta_0 + d\mathsf{A}_\lambda(|u|)$ on $\mathbb{R}$ is its own Fourier transform.

Proof. By Proposition 13.1 the bound (12.5) holds for every ζ_λ with constants independent of λ, so, as in the proof of Proposition 13.16, $|\zeta_\lambda(s)| \leq C(|t|+\sigma)^{1/4}$ for $\Re s > c_1$, and the argument of Theorem 12.5 applied to $\zeta_\lambda(s)/s$ gives A_λ, its vanishing on $(0,1)$ and its uniqueness. For a positive measure A as in (14.11), partial integration gives (14.19) with $\mathsf{A}([1,x])$, and uniqueness applies.

Let $c > c_1$. In the variable $u = \log x$, the function $\mathsf{A}_\lambda(e^u)e^{-cu}$ has Fourier transform $\zeta_\lambda(c+it)/(c+it)$, and $-\mathsf{g}'(e^u)e^{(c+1)u}$ has Mellin transform $\int_0^\infty(-\mathsf{g}'(x))x^s dx = s\,\mathsf{g}^\sharp(s)$ on $\Re s = c$; Parseval's formula gives

$$\langle \mathsf{A}_\lambda, \mathsf{g}\rangle = \frac{1}{2\pi i}\int_{(c)} \zeta_\lambda(s)\, \mathsf{g}^\sharp(s)\, ds. \tag{14.21}$$

ζ_λ is bounded by a power of $|t|$ on every vertical strip (finite order, the functional equation and the bound on $\Re s \geq c$), and $\mathsf{g}^\sharp$ decays faster than any power. Move the line to $\Re s = 1-c$. The poles of $\mathsf{g}^\sharp$ at $s = -2, -4, \ldots$ are cancelled by the trivial zeros of ζ_λ; the remaining poles are $s = 1$ and $s = 0$, exactly as in (14.14). On the new line (11.3) and (14.13) give $\zeta_\lambda(s)\mathsf{g}^\sharp(s) = \zeta_\lambda(1-s)\hat{\mathsf{g}}^\sharp(1-s)$, and (14.21) for $\hat{\mathsf{g}}$ gives (14.20). □

Proposition 14.10 (The value at infinity as one number)**.** *Let g_ϵ be as in the proof of Theorem 14.7, $\mathsf{w}_\epsilon := \hat{\mathsf{g}}_\epsilon - \mathsf{g}_\epsilon$, which is non-negative on $|u| \geq 1$, and $\mathsf{A}'_\lambda(x) := \mathsf{A}_\lambda(x) - \varpi_\lambda$ for $x \geq 1$. For every $\lambda \in [0,1]$ and $\epsilon > 0$,*

$$\varpi_\lambda\, \mathsf{w}_\epsilon(1) = \tfrac{1}{2}(1 - \hat{\mathsf{g}}_\epsilon(0)) - \langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle, \qquad \langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle := -\int_1^\infty \mathsf{A}'_\lambda(x)\, \mathsf{w}'_\epsilon(x)\, dx, \tag{14.22}$$

the limit $\mathsf{W}(\lambda) := \lim_{\epsilon\to 0}\langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle$ exists, and

$$\varpi_\lambda - 1 = -\frac{\mathsf{W}(\lambda)}{\hat{\mathsf{g}}(1)}, \qquad \hat{\mathsf{g}}(1) = \frac{1}{6} + \frac{1}{4\pi^2}. \tag{14.23}$$

Consequently $\mathsf{W}(0) \leq 0$, $\mathsf{W}(1) > 0$, and (D) holds if and only if $\mathsf{W}(0) \geq 0$. If A_λ is non-decreasing, then $\langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle = \int_{(1,\infty)} \mathsf{w}_\epsilon\, d\mathsf{A}_\lambda \geq 0$, which is Theorem 14.7.

Proof. By (14.20) for g_ϵ, with $\mathsf{g}_\epsilon(0) = 1$, $2\langle \mathsf{A}_\lambda, \mathsf{w}_\epsilon\rangle = 1 - \hat{\mathsf{g}}_\epsilon(0)$. Since $\mathsf{A}_\lambda = \varpi_\lambda + \mathsf{A}'_\lambda$ on $[1,\infty)$ and $-\int_1^\infty \mathsf{w}'_\epsilon = \mathsf{w}_\epsilon(1)$, $\langle \mathsf{A}_\lambda, \mathsf{w}_\epsilon\rangle = \varpi_\lambda \mathsf{w}_\epsilon(1) + \langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle$; this is (14.22). As $\epsilon \to 0$, $\mathsf{w}_\epsilon(1) = \hat{\mathsf{g}}_\epsilon(1) \to \hat{\mathsf{g}}(1)$, because $\mathsf{g}_\epsilon(1) = 0$, and $\frac{1}{2}(1 - \hat{\mathsf{g}}_\epsilon(0)) \to \frac{1}{2}(1 - \hat{\mathsf{g}}(0)) = \hat{\mathsf{g}}(1)$ by (14.18); hence the limit exists and (14.23) holds. The sign statements follow from $\varpi_{Z1} \geq 1$ (Theorem 12.2) and $\varpi_{\mathrm{RVM}} < 1$. □

Theorem 14.11 (The endpoint dominates ζ on the real axis)**.** *For every real $\sigma > 1$,*

$$\zeta_{Z1}(\sigma) \;\geq\; \zeta(\sigma), \tag{14.24}$$

and equality for one $\sigma > 1$ holds if and only if $\zeta_{Z1} = \zeta$. In particular

$$\prod_{k\geq 1}\Big(1 + \frac{2}{\frac{1}{4} + \gamma_k^2}\Big) \;\geq\; \frac{\pi}{3}, \qquad \textit{with equality if and only if } (\mathrm{D}). \tag{14.25}$$

Proof. By the proof of Theorem 12.2, $\zeta_{Z1}(\sigma)/\zeta(\sigma) = \mathscr{R}_{Z1}(\sigma)$ is the convergent product of the factors (12.11), with $W = (\sigma - \frac{1}{2})^2 + \gamma^2$. At $\sigma = 1$, $W = \frac{1}{4} + \gamma^2$ and the factor equals 1 by the definition of c_γ. The logarithmic derivative in W of $W_\rho^2/((W_\rho - d_\rho^2)^2 + 4d_\rho^2\gamma^2)$ is

$$\frac{2}{W} - \frac{2(W - d_\rho^2)}{(W - d_\rho^2)^2 + 4d_\rho^2\gamma^2} = \frac{2d_\rho^2\,(4\gamma^2 + d_\rho^2 - W)}{W\big((W - d_\rho^2)^2 + 4d_\rho^2\gamma^2\big)}, \tag{14.26}$$

positive for $W < 4\gamma^2 + d_\rho^2$ and negative afterwards, and the factor tends to $c_\gamma > 1$ as $W \to \infty$. A function that equals 1 at $\sigma = 1$, increases and then decreases to a limit greater than 1 is greater than 1 for every $\sigma > 1$. Hence $\mathscr{R}_{Z1}(\sigma) \geq 1$, with equality only if the product is empty, that is, $\mathscr{R}_{Z1} \equiv 1$. For $\sigma = 2$: $q(2) = 2$, $G(2) = 2/\pi$ and $P_\zeta(2) = G(2)\zeta(2) = \pi/3$, so (14.24) reads $P_{Z1}(2) \geq \pi/3$. □

Proposition 14.12 (The final step on the open path)**.** *The following are equivalent:* (i) *(D);* (ii) $\varpi_\lambda < 1$ *for every* $\lambda \in (0, 1]$*;* (iii) $\varpi_{\lambda_m} \leq 1$ *for some sequence* $\lambda_m \to 0^+$*;* (iv) $\mathsf{W}(\lambda) > 0$ *for every* $\lambda \in (0, 1]$.

Proof. $f(\lambda) = \log \varpi_\lambda$ is continuous and strictly convex on $[0, 1]$ (Proposition 13.2: $f' = -\Upsilon$, $\Upsilon' < 0$), $f(1) = \log \varpi_{\mathrm{RVM}} < 0$ and $f(0) \geq 0$ (Theorem 12.2). If $f(0) = 0$, convexity gives $f(\lambda) < (1 - \lambda)f(0) + \lambda f(1) < 0$ for $0 < \lambda < 1$. If $f(\lambda_m) \leq 0$ along $\lambda_m \to 0^+$, then $f(0) \leq 0$ by continuity, so $f(0) = 0$, which is (D) by Theorem 12.2. (ii)⇔(iv) is (14.23). □

A statement that holds as an identity only at the endpoint cannot be reached as a limit of the same identity along the path, where it is false (Proposition 14.1); an inequality can. Proposition 14.12 is of this form: the inequality $\varpi_\lambda \leq 1$ on the open path passes to the limit, and the opposite inequality $\varpi_{Z1} \geq 1$ holds at the endpoint. By Theorem 14.11 the same scheme is available at every real $\sigma > 1$: (D) holds if and only if $\zeta_{Z1}(\sigma) \leq \zeta(\sigma)$ for one $\sigma > 1$.

Remark 14.13 (The hypothesis added to the functional equation). The functional equation, the order and the zeros on the critical line do not characterize ζ: every ζ_λ has them. A converse theorem for ζ_{Z1} needs one further hypothesis on the function A_{Z1} of Lemma 14.9, and each of the following suffices and is equivalent to (D): A_{Z1} is a step function with jumps at the integers (Hamburger's theorem); A_{Z1} is non-decreasing (Theorem 14.7); $\mathsf{W}(0) \geq 0$ (Proposition 14.10), which is the weakest of the three, being equivalent to $\varpi_{Z1} \leq 1$. The inequality (14.17) is the one-dimensional case of the linear programming bound of Cohn and Elkies [19]. By Lemma 14.9 the measure $\delta_0 + d\mathsf{A}_\lambda(|u|)$ is self-dual for every λ; measures on $\mathbb{R}$ whose support and spectrum are both uniformly discrete are classified in [39].

14.5. **The density of the clock.** Let $\Psi(t) := t\log t - t$ for $t > 0$, and for $M \geq 1$, $0 < t_0 \leq \frac{1}{2}$ let

$$\mathcal{U}(M, t_0) := \int_{t_0}^{\infty} e^{-iM\Psi(t)}\,dt := \lim_{T\to\infty} \int_{t_0}^{T} e^{-iM\Psi(t)}\,dt. \tag{14.27}$$

Lemma 14.14 (Stationary phase for Ψ)**.** *The limit* (14.27) *exists, and there is an absolute constant* $C_{\mathcal{U}}$ *with*

$$\Big|\mathcal{U}(M, t_0) - \sqrt{\frac{2\pi}{M}}\,e^{i(M - \pi/4)}\Big| \leq \frac{C_{\mathcal{U}}}{M} \qquad (M \geq 1,\ 0 < t_0 \leq \tfrac{1}{2}). \tag{14.28}$$

Proof. $\Psi'(t) = \log t$ is increasing and unbounded, so the limit exists by the first-derivative test. Ψ is convex with minimum $\Psi(1) = -1$. Put $\mathsf{w} := \operatorname{sgn}(t - 1)\sqrt{2(\Psi(t) + 1)}$. Near $t = 1$, $\Psi(t) + 1 = \frac{1}{2}(t - 1)^2(1 + O(t - 1))$ with an analytic factor, so $t \mapsto \mathsf{w}$ is analytic with derivative 1 at $t = 1$; elsewhere $\mathsf{w}\,d\mathsf{w} = \log t\,dt$. Hence $t \mapsto \mathsf{w}$ is an increasing analytic bijection of $(0, \infty)$ onto $(-\sqrt{2}, \infty)$, with

$$\mathsf{p}(\mathsf{w}) := \frac{dt}{d\mathsf{w}} = \frac{\mathsf{w}}{\log t}, \qquad \mathsf{p}(0) = 1, \qquad \mathsf{q}(\mathsf{w}) := \frac{\mathsf{p}(\mathsf{w}) - 1}{\mathsf{w}} = \frac{1}{\log t} - \frac{1}{\mathsf{w}}. \tag{14.29}$$

q is analytic on $(-\sqrt{2},\infty)$, bounded, tends to 0 as $\mathsf{w}\to\infty$, and $\mathsf{q}'=-\mathsf{w}/(t\log^3 t)+\mathsf{w}^{-2}$. Away from $\mathsf{w}=0$ one has $\int|\mathsf{w}|\,|t\log^3 t|^{-1}d\mathsf{w}=\int dt/(t\log^2 t)$, which converges at $t=0$ and at $t=\infty$, and $\int\mathsf{w}^{-2}d\mathsf{w}$ converges at ∞; so $\int_{-\sqrt{2}}^{\infty}|\mathsf{q}'|\,d\mathsf{w}<\infty$. With $\mathsf{w}_0=\mathsf{w}(t_0)\le\mathsf{w}(\frac12)=-\sqrt{1-\log 2}<-\frac12$ and $e^{-iM\Psi}=e^{iM}e^{-iM\mathsf{w}^2/2}$,

$$e^{-iM}\,\mathcal{U}(M,t_0)=\int_{\mathsf{w}_0}^{\infty}e^{-iM\mathsf{w}^2/2}\,d\mathsf{w}+\int_{\mathsf{w}_0}^{\infty}e^{-iM\mathsf{w}^2/2}\,\mathsf{w}\,\mathsf{q}(\mathsf{w})\,d\mathsf{w}. \tag{14.30}$$

The first integral is $\int_{\mathbb{R}}e^{-iM\mathsf{w}^2/2}d\mathsf{w}=\sqrt{2\pi/M}\,e^{-i\pi/4}$ minus $\int_{-\infty}^{\mathsf{w}_0}$, and the latter is at most $4/(M|\mathsf{w}_0|)\le 8/M$ by the first-derivative test [51, Lemma 4.2]. Since $\mathsf{w}\,e^{-iM\mathsf{w}^2/2}=\frac{i}{M}\frac{d}{d\mathsf{w}}e^{-iM\mathsf{w}^2/2}$, integration by parts bounds the second integral by $M^{-1}(\sup|\mathsf{q}|+\int|\mathsf{q}'|)$. □

For $m\ge1$ and $y>1$ put

$$\mathcal{R}_m(y):=-2(-1)^m\,m^{-3/2}\,y^{\frac{1}{2m}-\frac12}\cos\Big(2\pi m\,y^{1/m}-\frac{(7m+1)\pi}{4}\Big), \tag{14.31}$$

so that $\mathcal{R}_1(y)=2\cos(2\pi y)$ and $\mathcal{R}_2(y)=-2^{-1/2}y^{-1/4}\cos(4\pi\sqrt{y}+\frac{\pi}{4})$.

Theorem 14.15 (The density of the clock)**.** *J_{RVM} is continuously differentiable on $(2,\infty)$, and for $y\ge16$*

$$J'_{\mathrm{RVM}}(y)=\frac{1}{\log y}+2\cos(2\pi y)-\frac{y^{-1/4}}{\sqrt{2}}\cos\Big(4\pi\sqrt{y}+\frac{\pi}{4}\Big)+O(y^{-1/3}). \tag{14.32}$$

On $[2,\infty)$ the measure $d\pi_{\mathrm{RVM}}$ is the sum of the atoms $\frac{\mu(j)}{j}J_{\mathrm{RVM}}(2)\,\delta_{2^j}$, $j\ge1$, and of a density that is continuous off the points 2^j and equals $J'_{\mathrm{RVM}}(y)+O(y^{-1/2})$.

Proof. An exact series. We use the notation of the proof of Theorem 6.1: $v=\log y$, $S_\alpha(v)=\sum_k e^{(iv-\alpha)\tau_k}$, the decomposition (6.8) and the harmonics $K_m^{\pm}$ of (6.9), with phases $\Psi_m^{\pm}(\tau)=v\tau\pm2\pi m\mathcal{N}_{\mathrm{RVM}}(\tau)$. Put

$$\mathcal{H}_m^{\pm}(v):=\int_{\tau_0}^{\infty}e^{i\Phi_m^{\pm}(\tau)}\,d\tau, \tag{14.33}$$

improper integrals that converge, locally uniformly in $v\ge\log2$, by the first-derivative test, because $(\Psi_m^{\pm})'$ is monotone and unbounded. As $\alpha\downarrow0$, $K_m^{\pm}\to iv\,\mathcal{H}_m^{\pm}(v)$, and the first two terms of (6.8) tend to a continuous function $S_{\mathrm{sm}}(v)$ with $|S_{\mathrm{sm}}(v)|\le C/v+\frac12$. The bounds for $K_m^{\pm}$ in that proof are uniform in the Abel parameter and, for K_m^{-}, include the factor displayed in (6.10); only the finite range $m\le v/\log2$ can contain the stationary point. After this finite range is separated, the remaining bounds are summable against $1/(2\pi m)$, so dominated convergence in (6.7) and (6.9)

$$\sum_k D_k=\int_{u_1}^{u_2}\frac{e^{v/2}}{v}\,S_0(v)\,dv,\qquad S_0(v):=S_{\mathrm{sm}}(v)+\sum_{m\ge1}\frac{(-1)^m\,v}{2\pi m}\big(\mathcal{H}_m^{-}(v)-\mathcal{H}_m^{+}(v)\big), \tag{14.34}$$

the series converging uniformly on compact sets. Thus S_0 is continuous, J_{RVM} is continuously differentiable on $(2,\infty)$, and by (5.2)

$$J'_{\mathrm{RVM}}(y)=\frac{1}{\log y}\Big(1-\frac{1}{y(y^2-1)}-\frac{2}{\sqrt{y}}\,\Re S_0(\log y)\Big). \tag{14.35}$$

The terms without a stationary point. $(\Phi_m^{+})'\ge v+m\log2$, so $|\mathcal{H}_m^{+}|\le4/(v+m\log2)$ and $\sum_m\frac{v}{2\pi m}|\mathcal{H}_m^{+}|\ll1+\log v$. For $m>v/\log2$, $(\Phi_m^{-})'\le v-m\log2<0$ on $[\tau_0,\infty)$; the first-derivative test, and the second-derivative test on $[\tau_0,2\tau_0]$ with $|(\Phi_m^{-})''|=m/\tau$, give $|\mathcal{H}_m^{-}|\ll\min\big((m\log2-v)^{-1},m^{-1/2}\big)$, and $\sum_{m>v/\log2}\frac{v}{2\pi m}|\mathcal{H}_m^{-}|\ll1+\log v$.

The terms $3\le m\le v/\log2$. By the argument for (6.10), $|\mathcal{H}_m^{-}|\le\frac{8}{m\log2}+16\sqrt{\pi}\,e^{v/(2m)}m^{-1/2}$, so these terms contribute to S_0 at most $C\,v\sum_{m\ge3}\big(m^{-2}+m^{-3/2}e^{v/(2m)}\big)\ll v\,e^{v/6}$.

The terms $m = 1, 2$. The stationary point of Φ_m^- is $\mathtt{t}_m := 2\pi e^{v/m} = 2\pi y^{1/m}$. With $\tau = \mathtt{t}_m t$ and (2.3), $\Phi_m^-(\mathtt{t}_m t) = -m\mathtt{t}_m\Psi(t) - \frac{7\pi m}{4}$, hence

$$\mathcal{H}_m^-(v) = \mathtt{t}_m\, e^{-7\pi i m/4}\,\mathcal{U}(m\mathtt{t}_m, \tau_0/\mathtt{t}_m), \qquad \tau_0/\mathtt{t}_m = 2e^{-v/m} \le \tfrac{1}{2} \quad (y \ge 4^m). \tag{14.36}$$

By Lemma 14.14 with $M = m\mathtt{t}_m$,

$$\frac{(-1)^m v}{2\pi m}\,\mathcal{H}_m^-(v) = (-1)^m\,\frac{v}{m^{3/2}}\,y^{\frac{1}{2m}}\exp\Big(i\Big(2\pi m\, y^{1/m} - \frac{(7m+1)\pi}{4}\Big)\Big) + O\Big(\frac{v}{m^2}\Big). \tag{14.37}$$

Inserting the four estimates into (14.35) gives $J'_{\mathrm{RVM}}(y) = \frac{1}{\log y} + \mathcal{R}_1(y) + \mathcal{R}_2(y) + O(y^{-1/3})$, which is (14.32).

The prime coordinate. By (5.7), on each interval $(2^r, 2^{r+1})$ the function π_{RVM} is the finite sum of the continuously differentiable functions $\frac{\mu(j)}{j}J_{\mathrm{RVM}}(x^{1/j})$, $j \le r$; the terms $j \ge 2$ have derivative $\frac{\mu(j)}{j^2}x^{1/j-1}J'_{\mathrm{RVM}}(x^{1/j})$, and $|J'_{\mathrm{RVM}}| \le C_J$ by (6.5), so their sum is $O(x^{-1/2})$. At $x = 2^j$ the term j jumps by $\frac{\mu(j)}{j}J_{\mathrm{RVM}}(2)$. □

Corollary 14.16 (Cell masses and cell defects at the clock end)**.** *For $n \ge 17$, with $\mathbf{1}_{2^j}(n) = 1$ if $n = 2^j$ and 0 otherwise,*

$$c_n^{\mathrm{RVM}} = \int_{n-1}^{n}\Big(\frac{1}{\log x} + 2\cos(2\pi x)\Big)dx + \mathbf{1}_{2^j}(n)\,\frac{\mu(j)}{j}\,J_{\mathrm{RVM}}(2) + O\big(n^{-1/4}\big), \tag{14.38}$$

$$\tilde{c}_n(1) = \frac{1}{\log n} + \mathbf{1}_{2^j}(n)\,\frac{\mu(j)}{j}\,J_{\mathrm{RVM}}(2) + O\big(n^{-1/4}\big), \tag{14.39}$$

and for $0 < \Re s \le \sigma_1$

$$|\mathfrak{h}_n(1;s)| \le C_{\sigma_1}\big(|s|\,n^{-\sigma-4/3} + |s|(1+|s|)\,n^{-\sigma-2}\big). \tag{14.40}$$

In particular Hypothesis 6.7 holds along the integers n that are not of the form 2^j with $\mu(j) \ne 0$, and $c_n^{\mathrm{RVM}} \sim 1/\log n$, $0 < c_n^{\mathrm{RVM}} < 1$ for all large such n; at $n = 2^j$, $c_n^{\mathrm{RVM}} = \big(1 + \mu(j)J_{\mathrm{RVM}}(2)\log 2\big)/\log n + O(n^{-1/4})$.

Proof. (14.38): $c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1)$ is the integral of the density over $(n-1, n)$ plus the atom at n, and $\int_{n-1}^{n}(|\mathcal{R}_2| + O(x^{-1/3}))dx = O(n^{-1/4})$. (14.39): $\varphi_n(2^j) = \mathbf{1}_{2^j}(n)$; $\int \varphi(t)\,dt/\log(n+t) = \frac{1}{\log n} + O(\frac{1}{n\log^2 n})$ because $\mathsf{M}_0 = 1$; and $\int \varphi(t)\,2\cos(2\pi(n+t))\,dt = 2\mathsf{C}_0 = 0$ by (14.2). (14.40): the test function $\varphi_n(x)(n^{-s} - x^{-s})$ vanishes at the atoms, and $n^{-s} - (n+t)^{-s} = s\,n^{-s-1}t + O_{\sigma_1}(|s|(1+|s|)n^{-\sigma-2})$ for $|t| \le 1$. The first moment of the density is $\int t\varphi(t)\pi'_{\mathrm{RVM}}(n+t)\,dt = \frac{\mathsf{M}_1}{\log n} + 2\mathsf{C}_1 + O(n^{-1/3}) = O(n^{-1/3})$, because $\mathsf{M}_1 = \mathsf{C}_1 = 0$, $\mathcal{R}_2(n+t) = \mathcal{R}_2(n) + O(n^{-3/4})$ and the density is bounded. The last statements follow from (14.38) and Lemma 6.6. □

By (14.32), $\frac{d\pi_{\mathrm{RVM}}}{dy} = (\frac{1}{\log y} - 1) + (1 + 2\cos(2\pi y)) + O(y^{-1/4})$: the clock carries the first harmonic of the comb $\sum_n \delta_n = 1 + 2\sum_{q\ge1}\cos(2\pi q y)$ of all integers, with weight 1, on a smooth background. This harmonic comes from the ordinates τ_k near the height $2\pi y$, the stationary point of Φ_1^-; by (14.2) it does not enter the cell masses (14.39), which are $1/\log n$ to leading order, and it enters the cell defects only through the moments C_r, $r \ge 2$.

14.6. **The cell masses in the heights of the ordinates.**

Proposition 14.17 (Height kernel of the cell masses)**.** *For every $\varepsilon > 0$, $\lambda \in [0,1]$ and $n \ge 3$ that is not a power of 2,*

$$\tilde{c}_n(\lambda) = \mathfrak{m}_n - \frac{2}{\sqrt{n}\,\log n}\sum_{k\ge1}\hat{\varphi}\Big(\frac{\gamma_k(\lambda)}{2\pi n}\Big)\cos(\gamma_k(\lambda)\log n) + E_n(\lambda), \tag{14.41}$$

where $\sum_{X<n\le 2X}|E_n(\lambda)|^2 \le C_\varepsilon X^{1/2+\varepsilon}$ uniformly in λ (sum over the n that are not powers of 2). The kernel $\hat{\varphi}(\gamma/2\pi n)$ vanishes at the heights $\gamma = 2\pi q n$, $q = 1, 2, \dots$

Proof. The term $j=1$ of W_n in (13.19) is $\int\varphi(t)(n+t)^{-1/2+i\gamma}\,dt/\log(n+t)$. For $0<\gamma\le n^{1+\varepsilon}$, $(1+t/n)^{-1/2+i\gamma}=e^{i\gamma t/n}\big(1+O((1+\gamma/n)/n)\big)$ and $\frac{1}{\log(n+t)}=\frac{1}{\log n}(1+O(\frac{1}{n}))$ on $|t|\le 1$, so this term equals $\frac{n^{-1/2+i\gamma}}{\log n}\hat\varphi(\frac{\gamma}{2\pi n})+O\big(n^{-3/2}(1+\gamma/n)\big)$, because $\int\varphi(t)e^{i\gamma t/n}dt=\hat\varphi(\gamma/2\pi n)$ for even φ. Summed over the $O(n^{1+\varepsilon}\log n)$ ordinates $\gamma_k(\lambda)\le n^{1+\varepsilon}$ the error is $O(n^{-1/2+3\varepsilon})$. For $\gamma>n^{1+\varepsilon}$ both the term and $\hat\varphi(\gamma/2\pi n)$ are $O_A((n/\gamma)^A)$, as in the proof of Theorem 13.10. The terms $j\ge 2$ have sum of squares $O(X^{1/2+3\varepsilon})$ over $X<n\le 2X$ by (13.27). The zeros of the kernel are (14.2). □

Since $\gamma_k(\lambda)\log n=\tau_k\log n-(1-\lambda)\eta_k\log n$, the passage along the family rotates each clock phase $n^{i\tau_k}$ by the angle $-(1-\lambda)\eta_k\log n$: up to the change of $\hat\varphi$ over a distance $O(1/n)$, the sum in (14.41) is $\Re\,\Xi_n\big((1-\lambda)\log n\big)$ with

$$\Xi_n(\Omega):=\sum_{k\ge1}\hat\varphi\Big(\frac{\tau_k}{2\pi n}\Big)n^{i\tau_k}e^{-i\Omega\eta_k}\qquad(0\le\Omega\le\log n).\tag{14.42}$$

At $\Omega=0$ this is the clock sum, whose real part is $O(n^{1/4}\log n)$ by (14.39) and Theorem 14.15, because the resonant heights $\tau_k\approx 2\pi n$ are zeros of the kernel; the endpoint of the family is $\Omega=\log n$.

Proposition 14.18 (The flow of a cell mass and the count difference)**.** *For every $n\ge3$ that is not a power of 2,*

$$\tilde c_n(0)-\tilde c_n(1)=-2\int_{0^+}^{1}\sum_{k\ge1}\eta_k\,\Im V_n\big(\gamma_k(\lambda)\big)\,d\lambda=-2\,\Im\int_{\tau_0}^{\infty}V_n(t)\big(N_\zeta(t)-n_{\mathrm{RVM}}(t)\big)\,dt.\tag{14.43}$$

Proof. The first equality is (13.20) and the continuity of $\tilde c_n$ at $\lambda=0$. By (13.20), $\tilde c_n(0)-\tilde c_n(1)=2\Re\sum_k\big(W_n(\tau_k)-W_n(\gamma_k)\big)=2\Re\int_{\tau_0}^{\infty}W_n\,d(n_{\mathrm{RVM}}-N_\zeta)$. Both counts vanish at τ_0, their difference is $O(\log t)$ by Lemma 2.2, and W_n is rapidly decreasing; integration by parts with $W_n'=iV_n$ gives $2\Re\big(i\int V_n(N_\zeta-n_{\mathrm{RVM}})\,dt\big)$. □

Proposition 14.19 (Height kernel of the cell defects)**.** *For every $\varepsilon>0$, $\lambda\in[0,1]$, $\Re s>0$ and $n\ge3$ that is not a power of 2,*

$$\mathfrak{h}_n(\lambda;s)=-\frac{s\,n^{-s-3/2}}{\pi\log n}\sum_{k\ge1}\hat\varphi'\Big(\frac{\gamma_k(\lambda)}{2\pi n}\Big)\sin\big(\gamma_k(\lambda)\log n\big)+O_{s,\varepsilon}\big(n^{-\sigma-3/4+\varepsilon}\big).\tag{14.44}$$

Proof. For $|t|\le1$, $n^{-s}-(n+t)^{-s}=s\,n^{-s-1}t+\psi_2(t)$, where ψ_2 and all its derivatives in t are $O_s(n^{-\sigma-2})$. *The term $j=1$.* The part of the density that does not involve the ordinates contributes $O_s(n^{-\sigma-2})$, because $\int t\varphi(t)\,dt/\log(n+t)=O(1/(n\log^2 n))$ ($\mathsf{M}_1=0$). The pairing of $\varphi\psi_2$ with the ordinate sum is $O_s(n^{-\sigma-2})$ times $O(n^{1/2+\varepsilon})$, as in Remark 13.14. For the real function $t\varphi(t)$, the computation in the proof of Proposition 14.17, with $\int t\varphi(t)e^{2\pi i\xi t}dt=-\frac{i}{2\pi}\hat\varphi'(\xi)$ for even φ and $-2\Re(-\frac{i}{2\pi}n^{i\gamma})=-\frac{1}{\pi}\sin(\gamma\log n)$, gives the sum in (14.44) with an error $O_s(n^{-\sigma-3/2+3\varepsilon})$. *The terms $j\ge2$.* The test function $\varphi_n(y^j)(n^{-s}-y^{-js})$ is $O(|s|n^{-\sigma-1})$ on a support of length $O(n^{1/j-1})$; with $O(n^{1+\varepsilon}\log n)$ relevant ordinates and $y^{-1/2}=O(n^{-1/(2j)})$, these terms are $O_s(n^{-\sigma-1+1/(2j)+2\varepsilon})$, at most $O_s(n^{-\sigma-3/4+2\varepsilon})$. □

The kernel $\hat\varphi$ of the masses vanishes at the heights $2\pi qn$ and the kernel $\hat\varphi'$ of the defects in general does not: the masses and the defects of a cell read complementary heights of the ordinates. At $\lambda=1$ the resonant heights $\tau_k\approx2\pi n$ contribute to $\sum_k n^{i\tau_k}$ the real quantity $-\sqrt{n}\log n\,e^{2\pi in}$ (proof of Theorem 14.15), which the sine in (14.44) does not see: the first harmonic of the clock is centred at the integers.

Remark 14.20 (The scale of the displacements)*.* By Proposition 14.1, a property of the sequence (η_k) that yields (D) is false for every sequence $(c\,\eta_k)$ with $0\le c<1$. Properties that are invariant under $\eta_k\mapsto c\,\eta_k$ with $c>0$ (the signs of the η_k, the non-vanishing of the η_k, their boundedness) hold along the whole open path and do not suffice; the property fixes the size of the displacements.

In the form (14.42) it is a condition on the angles $\eta_k \log n$ modulo 2π, jointly in k and n. The construction contains one exact relation of this kind: for a simple ordinate, (2.4) and (2.13) give $\mathcal{N}_{\mathrm{RVM}}(\tau_k) - \mathcal{N}_{\mathrm{RVM}}(\gamma_k) = N^*_\zeta(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\gamma_k)$, that is,

$$\eta_k \, \mathcal{N}'_{\mathrm{RVM}}(\xi_k) = N^*_\zeta(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\gamma_k) \qquad \text{for some } \xi_k \text{ between } \gamma_k \text{ and } \tau_k. \tag{14.45}$$

14.7. **The Gaussian defect of the atomization.**

Lemma 14.21 (Vertical growth of the joining factor)**.** *For every $\lambda \in [0,1]$ and $\frac12 < \sigma_0 < \sigma_1$, $\tilde h_\lambda(\sigma + it) = o(|t|)$ as $|t| \to \infty$, uniformly for $\sigma_0 \le \sigma \le \sigma_1$.*

Proof. By (13.13) and Proposition 13.5(ii) it suffices to treat the double series. Fix $c \in (\frac12, \sigma_0)$. By (13.29), $\sum_{m\ge1} \frac1m \sum_{n>N} |\mathfrak{h}_n(\lambda; ms)| \le C|s| \sum_{n>N} n^{-\sigma_0-1}(1+I_n^{1/2})$ and $\sum_{m>N} \frac1m \sum_{n\ge2} |\mathfrak{h}_n(\lambda; ms)| \le C|s| \sum_{n\ge2} n^{-N\sigma_0-1}(1 + I_n^{1/2})$; both are at most $\epsilon_N|s|$ with $\epsilon_N \to 0$ as $N \to \infty$. For the finitely many pairs $m, n \le N$, with $E = \pi_\lambda - \mathfrak{L}$ as in the proof of Theorem 13.12,

$$\begin{aligned} \mathfrak{h}_n(\lambda; ms) = {} & \int \varphi_n(x)\big(n^{-ms} - x^{-ms}\big) d\mathfrak{L}(x) - \int E(x)\varphi_n'(x)\big(n^{-ms} - x^{-ms}\big)dx \\ & - ms \int E(x)\varphi_n(x)\, x^{-ms-1} dx. \end{aligned} \tag{14.46}$$

The first two terms are bounded in t, and the last integral tends to 0 as $|t| \to \infty$ by the Riemann–Lebesgue lemma, because $E \in L^1(n-1, n+1)$. Hence $\limsup_{|t|\to\infty} |t|^{-1} \sup_\sigma |\tilde h_\lambda(\sigma + it)| \le 2\epsilon_N$ for every N. □

For $y > 0$ and $c > 1$ put

$$\Theta_\lambda(y) := \frac{1}{4\pi i} \int_{(c)} \zeta_\lambda(s)\, \Gamma\!\left(\tfrac{s}{2}\right) \pi^{-s/2}\, y^{-s/2}\, ds, \qquad \mathfrak{G}_\lambda(y) := \sum_{N\ge1} \tilde a_N(\lambda)\, e^{-\pi N^2 y} - \Theta_\lambda(y). \tag{14.47}$$

For ζ itself $\Theta(y) = \sum_{n\ge1} e^{-\pi n^2 y}$, and if ζ_λ has a representation (14.11), then $\Theta_\lambda(y) = \int e^{-\pi u^2 y}\, d\mathsf{A}(u)$. $\mathfrak{G}_\lambda$ is the change produced by the atomization in the Gaussian count of the integers of the system.

Proposition 14.22 (Gaussian defect)**.** *Let $\lambda \in [0,1]$.*

(i) *$\Theta_\lambda(y) = y^{-1/2}\,\Theta_\lambda(1/y) + \frac12(y^{-1/2} - 1)$ for $y > 0$, and $\Theta_\lambda(y) = O_A(y^{-A})$ as $y \to \infty$ for every A.*

(ii) *The coefficients $\tilde a_N(\lambda)$ satisfy the theta relation of ζ,*

$$\sum_{N\ge1} \tilde a_N(\lambda)\, e^{-\pi N^2 y} = y^{-1/2} \sum_{N\ge1} \tilde a_N(\lambda)\, e^{-\pi N^2/y} + \tfrac12(y^{-1/2} - 1) \qquad (y > 0), \tag{14.48}$$

if and only if

$$\mathfrak{G}_\lambda(y) = y^{-1/2}\, \mathfrak{G}_\lambda(1/y) \qquad (y > 0). \tag{14.49}$$

Under the hypotheses of Theorem 14.3, (14.49) *holds.*

(iii) *For every $\delta > 0$, as $y \to 0^+$,*

$$\mathfrak{G}_\lambda(y) = \tfrac12(e^{\tilde h_\lambda(1)} - 1)\, y^{-1/2} + O_\delta(y^{-1/4-\delta}), \tag{14.50}$$

and $e^{\tilde h_\lambda(1)} = \mathrm{Res}_{s=1}\, \tilde\zeta_{\mathbb{Z},\lambda}(s)$.

(iv) *If* (14.49) *holds, then $\tilde h_\lambda(1) = 0$.*

Proof. (i) ζ_λ is bounded by a power of $|t|$ on vertical strips (proof of Lemma 14.6; on $\Re s \ge c$ use Proposition 13.1), and $\Gamma(\frac{s}{2})$ decays exponentially. Moving the line to $\Re s = 1 - c$ crosses $s = 1$, with residue $\Gamma(\frac12)\pi^{-1/2}y^{-1/2} = y^{-1/2}$, and $s = 0$, with residue $2\zeta_\lambda(0) = -1$; on the new line (11.3) and $y^{-s/2} = y^{-1/2}(1/y)^{-(1-s)/2}$ give $y^{-1/2}\Theta_\lambda(1/y)$. The decay follows by moving the line to $\Re s = 2A$. (ii) By $e^{-x} = \frac{1}{2\pi i}\int_{(c/2)} \Gamma(w) x^{-w} dw$ and Corollary 13.11, $\sum_N \tilde a_N(\lambda) e^{-\pi N^2 y} =$

$\frac{1}{4\pi i}\int_{(c)}\tilde\zeta_{\mathbb{Z},\lambda}(s)\Gamma(\frac{s}{2})\pi^{-s/2}y^{-s/2}ds$; subtracting (i) from the theta relation gives the equivalence. Under the hypotheses of Theorem 14.3, $\tilde\zeta_{\mathbb{Z},\lambda}=\zeta$. (iii) By (ii), $2\mathfrak{G}_\lambda(y)=\frac{1}{2\pi i}\int_{(c)}\zeta_\lambda(s)(e^{\tilde h_\lambda(s)}-1)\Gamma(\frac{s}{2})\pi^{-s/2}y^{-s/2}ds$. By Lemma 14.21, $|e^{\tilde h_\lambda(s)}|\le e^{o(|t|)}$ on $\frac12+2\delta\le\Re s\le c$, so the line can be moved to $\Re s=\frac12+2\delta$. The only pole crossed is $s=1$, where ζ_λ has residue 1 and $\tilde h_\lambda$ is holomorphic; the residue is $(e^{\tilde h_\lambda(1)}-1)y^{-1/2}$, and the remaining integral is $O_\delta(y^{-1/4-\delta})$. (iv) $\sum_N\tilde a_N(\lambda)e^{-\pi N^2y}=O(e^{-\pi y})$ as $y\to\infty$, so by (i) $\mathfrak{G}_\lambda(y)=O_A(y^{-A})$ there; then (14.49) gives $\mathfrak{G}_\lambda(y)=O_A(y^A)$ as $y\to0^+$ for every A, and (14.50) forces $e^{\tilde h_\lambda(1)}=1$; $\tilde h_\lambda$ is real on the real axis. □

Corollary 14.23 (The density of the atomized integers)**.** $\sum_{N\le x}\tilde a_N(\lambda)$ *has mean density* $e^{\tilde h_\lambda(1)}$ *in the sense of* (14.50)*, and (D) implies*

$$\tilde h_0(1)=0,\qquad \textit{that is,}\qquad \mathfrak{u}_0(1)=\sum_{m\ge1}\frac1m\sum_{n\ge2}\mathfrak{h}_n(0;m). \tag{14.51}$$

Proof. Under (D), $\tilde h_0\equiv0$ (Theorem 13.13); the second form is (13.13) at $s=1$. □

The identity $\tilde h_0(\sigma)=0$ for real σ has two distinguished members: $\sigma=+\infty$ is $\varpi_{Z1}=1$, the first coefficient (Corollary 12.3), and $\sigma=1$ is (14.51), the density of the integers.

By Proposition 14.22(i), the function (14.6) is $T_\lambda(y)=\mathfrak{G}_\lambda(y)-y^{-1/2}\mathfrak{G}_\lambda(1/y)$, so (14.49) is $T_\lambda\equiv0$. At the clock end the expansion (14.50) holds with a smaller remainder: for every $\delta>0$,

$$\mathfrak{G}_1(y)=\tfrac12(e^{\tilde h_1(1)}-1)\,y^{-1/2}+O_\delta(y^{-\delta})\qquad(y\to0^+), \tag{14.52}$$

because $\tilde h_1$ is holomorphic in $\Re s>0$ with $\tilde h_1(\sigma+it)=o(|t|)$ uniformly on $\sigma_0\le\sigma\le\sigma_1$ for every $\sigma_0>0$ (the proof of Lemma 14.21 with the bound $|\mathfrak{h}_n(1;s)|\le C|s|n^{-\sigma-1}$ of Theorem 13.12(iv)), and the line of integration can be moved to $\Re s=2\delta$.

14.8. **Entire dependence on λ and the exceptional set.**

Proposition 14.24 (The cell masses as entire functions of λ)**.** *Let $n\ge3$ not be a power of* 2.

(i) *The real-axis function $\lambda\mapsto\tilde c_n(\lambda)$ has the entire extension*

$$\widetilde C_n(\lambda):=\mathfrak{m}_n-\sum_k\Big(W_n(\gamma_k+\lambda\eta_k)+\overline{W_n(\gamma_k+\bar\lambda\eta_k)}\Big),$$

which is of exponential type at most $C_\eta\log(n+1)$ and agrees with $\tilde c_n(\lambda)$ for real λ.

(ii) *For every $r\ge0$, $\varepsilon>0$ and $\lambda\in[0,1]$,*

$$\partial_\lambda^r\tilde c_n(\lambda)=-2\,\Re\sum_{k\ge1}\eta_k^r\,W_n^{(r)}(\gamma_k(\lambda)),\qquad |\partial_\lambda^r\tilde c_n(\lambda)|\le C_{r,\varepsilon}\big(C_\eta\log(n+1)\big)^r\,n^{1/2+\varepsilon}. \tag{14.53}$$

(iii) $|\tilde c_n(\lambda)-\tilde c_n(\lambda')|\le C_\varepsilon|\lambda-\lambda'|\,n^{1/2+\varepsilon}$ *on $[0,1]$. Hence, for sets of integers n that are not powers of* 2, $\tilde c_n(0)=O_\varepsilon(n^\varepsilon)$ *if and only if $\tilde c_n(\lambda)=O_\varepsilon(n^\varepsilon)$ uniformly for $0\le\lambda\le n^{-1/2}$, and $\tilde c_n(\lambda)=O_\varepsilon(n^\varepsilon)$ uniformly for $1-n^{-1/2}\le\lambda\le1$.*

Proof. For complex γ, $|y^{i\gamma}|\le e^{|\Im\gamma|\log y}$ with $\log y\le\log(n+1)$ on the support of $\varphi_n(y^j)$, and repeated integration by parts in (13.19) gives $|W_n^{(r)}(\gamma)|\le C_{A,r,n}\,e^{|\Im\gamma|\log(n+1)}(1+|\Re\gamma|)^{-A}$ for every A. With $\gamma=\gamma_k+\lambda\eta_k$, $|\Im\gamma|\le C_\eta|\Im\lambda|$, the two reflected series defining $\widetilde C_n$ converge locally uniformly in $\lambda\in\mathbb{C}$, so $\widetilde C_n$ is entire; on the real axis it equals $\tilde c_n$. Termwise differentiation on the real axis gives the first formula in (14.53). For real λ, $W_n^{(r)}$ carries the factor $(i\log y)^r$ with $\log y\le\log(n+1)$, the ordinates $\gamma_k(\lambda)>n^{1+\varepsilon}$ contribute $O(n^{-1})$, and each of the $O(n^{1+\varepsilon}\log n)$ remaining ordinates contributes $O\big((\log(n+1))^r n^{-1+1/(2j)}\big)$ to the term j, as in Remark 13.14; this is the bound in (14.53). (iii) is the case $r=1$ and Corollary 14.16. □

The bounds (14.53) do not by themselves control $\tilde{c}_n(0)$ from $\tilde{c}_n(1)$: the function $n^{1/2}\sin\big((1-\lambda)C_\eta \log n\big)$ is entire of the same type, satisfies the same bounds, vanishes at $\lambda = 1$ and has size $n^{1/2}$ at $\lambda = 0$ for infinitely many n. The Taylor expansion at the clock end, $\tilde{c}_n(0) = \sum_{r\ge 0} \frac{(-1)^r}{r!}\partial_\lambda^r \tilde{c}_n(1)$, has the terms $-2\Re\sum_k \eta_k^r W_n^{(r)}(\tau_k)$, clock sums weighted by the powers of the displacements.

Proposition 14.25 (The exceptional set)**.** *The set*

$$\mathsf{E} := \big\{\lambda \in [0,1] :\ \tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta\big\} \tag{14.54}$$

is finite, $1 \notin \mathsf{E}$*, and (D) holds if and only if* $0 \in \mathsf{E}$*. For every* $\lambda \notin \mathsf{E}$ *the hypotheses of Theorem 14.3 do not hold together,* $T_\lambda \not\equiv 0$ *and* $\mathsf{V}_\sigma(\lambda) > 0$*.*

Proof. If $\lambda \in \mathsf{E}$, then $\sum_n \tilde{c}_n(\lambda)\sum_m \frac{n^{-ms}}{m} = \log\zeta(s) = \sum_p \sum_m \frac{p^{-ms}}{m}$ for $\Re s > 1$. Comparing the coefficients of N^{-s} in increasing order of N determines the exponents uniquely, so $\tilde{c}_n(\lambda) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n. Choose a composite n, not a power of 2, so large that $\tilde{c}_n(1) > 0$ by (14.39). Then $\tilde{c}_n$ is an entire function of λ (Proposition 14.24) that does not vanish identically, and it has finitely many zeros in $[0,1]$; they contain E, and 1 is not one of them. $0 \in \mathsf{E}$ implies (D) by Theorem 14.4(ii), and (D) implies $\tilde{\zeta}_{\mathbb{Z},0} = \zeta_{Z1} = \zeta$ by Theorem 13.13. The last statement is Theorems 14.3 and 14.4(ii). □

Proposition 14.1 excludes that ζ_λ and $\tilde{\zeta}_{\mathbb{Z},\lambda}$ coincide for $\lambda > 0$; Proposition 14.25 excludes, outside a finite set, that $\tilde{\zeta}_{\mathbb{Z},\lambda}$ coincides with ζ. For $0 < \lambda < 1$ outside E it is not decided which of the two hypotheses of Theorem 14.3 fails; at $\lambda = 1$ it is the symmetry (Proposition 14.5).

14.9. **The final step in the cell masses.**

Corollary 14.26 (The final step is one number per cell)**.** *The following are equivalent:*

(i) *(D);*
(ii) $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ *for every* $n \ge 2$*;*
(iii) *there is a list* $\boldsymbol{\kappa} \in \mathscr{O}$ *(Section 15) with* $\tilde{c}_n(\boldsymbol{\kappa}) = \mathbf{1}_{\mathrm{prime}}(n)$ *for every* $n \ge 2$*.*

Proof. (i)⇒(ii) is Theorem 13.13, and (ii)⇒(iii) with the list (γ_k). If (iii) holds, then

$$\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}(s) = \prod_{n\ge 2}\big(1-n^{-s}\big)^{-\tilde{c}_n(\boldsymbol{\kappa})} = \prod_p \big(1-p^{-s}\big)^{-1} = \zeta(s) \qquad (\Re s > 1), \tag{14.55}$$

and Corollary 15.3, (iii)⇒(i), gives (D). □

Step A7 of the programme asks for the vanishing of the functions $\mathfrak{h}_n(0;\cdot)$, one function of s for each cell, and for $J_{Z1} = 0$ on the first cell. Corollary 14.26 asks for one real number for each cell; the position of the mass inside the cell and the first cell then follow from Hamburger's theorem.

Proposition 14.27 (Every point of the exceptional set is a witness)**.** *For the set* E *of* (14.54)*,*

$$\mathsf{E} \neq \varnothing \iff \text{(D)}, \qquad \text{so that either } \mathsf{E} = \varnothing \ \text{ or } \ 0 \in \mathsf{E}. \tag{14.56}$$

If (D) holds, then

$$\mathsf{E}\setminus\{0\} = \big\{\lambda \in (0,1] :\ \mathsf{D}_n(\lambda) = 0 \text{ for every } n \ge 2\big\}, \qquad \mathsf{D}_n(\lambda) := \tilde{c}_n(\lambda) - \tilde{c}_n(0), \tag{14.57}$$

and for $n \ge 3$ *not a power of* 2*, the real-axis function* $\mathsf{D}_n(\lambda) = -2\Re\sum_{k\ge 1}\big(W_n(\gamma_k+\lambda\eta_k) - W_n(\gamma_k)\big)$ *has the entire extension obtained by replacing each real part by the corresponding holomorphic plus reflected-conjugate pair; this extension is of exponential type at most* $C_\eta \log(n+1)$ *that vanishes at* $\lambda = 0$*.*

Proof. If $\lambda_0 \in \mathsf{E}$, the list $(\gamma_k(\lambda_0))$ belongs to $\mathscr{O}$ and its atomized zeta function is ζ; Corollary 15.3, (iii)⇒(i), gives (D), and then $0 \in \mathsf{E}$ by Proposition 14.25. Under (D), $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, and $\lambda \in \mathsf{E}$ means $\tilde{c}_n(\lambda) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n (proof of Proposition 14.25). The last statement is (13.20) and Proposition 14.24. □

Thus a proof that $\mathsf{E}\cap(0,1]=\varnothing$ does not approach the final step: if (D) fails, E is empty, and if (D) holds, $\mathsf{E}\cap(0,1]=\varnothing$ is the uniqueness of the witness on the path, which is the statement that the entire functions D_n have no common zero in $(0,1]$. This uniqueness is not proved; Proposition 14.25 gives that E is finite and $1\notin\mathsf{E}$.

Remark 14.28 (The dual function of the atomized family). For $\Re s<\frac12$ put $B_\lambda^\vee(s):=\zeta_\lambda(s)\,e^{\tilde h_\lambda(1-s)}$. By the functional equation of ζ_λ,

$$\pi^{-s/2}\Gamma\big(\tfrac{s}{2}\big)B_\lambda^\vee(s)=\pi^{-(1-s)/2}\Gamma\big(\tfrac{1-s}{2}\big)\,\tilde\zeta_{\mathbb{Z},\lambda}(1-s)\qquad(\Re s<\tfrac12),\tag{14.58}$$

so $B_\lambda^\vee$ is the function attached to $\tilde\zeta_{\mathbb{Z},\lambda}$ by the functional equation. To ask that $B_\lambda^\vee$ continue to $\Re s>1$ as an absolutely convergent Dirichlet series is to ask for the hypotheses of Theorem 14.3; no monotonicity in λ is known, and a parameter such as $\inf\{\lambda>0: B_\lambda^\vee$ is not a Dirichlet series$\}$ is not shown to be a transition point.

Proposition 14.29 (The identity that remains, cell by cell)**.** *Let $n\ge17$ not be a power of 2. Then $\tilde c_n(0)=\mathbf{1}_{\mathrm{prime}}(n)$ if and only if*

$$\mathbf{1}_{\mathrm{prime}}(n)-\tilde c_n(1)=-2\,\Im\int_{\tau_0}^{\infty}V_n(t)\big(N_\zeta(t)-n_{\mathrm{RVM}}(t)\big)\,dt,\qquad \tilde c_n(1)=\frac{1}{\log n}+O\big(n^{-1/4}\big).\tag{14.59}$$

Proof. This is (14.43) and (14.39). □

The left side of (14.59) is explicit. The right side is the pairing of the count difference with the kernel V_n, whose term $j=1$ is $n^{-1/2+it}\hat\varphi(t/2\pi n)$ up to $O(n^{-3/2}(1+t/n))$ (proof of Proposition 14.19). For $n=2^j$ the same identity holds with the cut term of Theorem 13.10 added. No evaluation of the right side of (14.59) is given in this paper.

14.10. **Symmetric joining factors.**

Remark 14.30 (The symmetry criterion has no slack at $\lambda=0$). At $\lambda=0$ the hypothesis of Theorem 14.3 is equivalent to its conclusion. If (D) holds, then $\tilde h_0\equiv0$ (Theorem 13.13), and $e^{\tilde h_0}=1$ is symmetric and of finite order; hence

$$e^{\tilde h_0}\text{ symmetric of finite order}\iff\tilde h_0\equiv0\iff\text{(D)},\tag{14.60}$$

and in the same way $T_0\equiv0$ and $\mathsf{V}_\sigma(0)=0$ are equivalent forms of (D), not weaker conditions. Moreover (14.5) presupposes a continuation of $\tilde h_0$ across the line $\Re s=\frac12$ (step A10 of the programme): $\tilde h_0$ is holomorphic in $\Re s>\frac12$, and $s\mapsto1-s$ maps this half-plane onto $\Re s<\frac12$. Theorems 14.34 and 14.36 below are criteria whose hypotheses are written on the half-line $\sigma\to+\infty$ and on the closed half-plane $\Re s\ge\frac12$.

Proposition 14.31 (Symmetric joining factors in general)**.** *Let f be meromorphic in $\mathbb{C}$ with $(s-1)f(s)$ entire of order at most one, and let f satisfy the functional equation* (11.3)*. Let $g=f\,e^h$, where e^h is entire, without zeros, of finite order, and $e^{h(s)}=e^{h(1-s)}$. If $g(s)=\sum_{n\ge1}a_nn^{-s}$ converges absolutely in $\Re s>1$, then $g=C_H\zeta$ and e^h is constant. If moreover $\lim_{\sigma\to+\infty}g(\sigma)=\lim_{\sigma\to+\infty}f(\sigma)\neq0$, then $e^h\equiv1$ and $f=g=C_H\zeta$.*

Proof. g satisfies (H0)–(H3) of Subsection 13.7 with the second function equal to g, so $g=C_H\zeta$ by Hamburger's theorem. If $C_H=0$ then $f\,e^h=0$, which is excluded when $g\not\equiv0$; assume $C_H\neq0$. Then $e^h=C_H(s-1)\zeta(s)/\big((s-1)f(s)\big)$ is an entire quotient of two entire functions of order at most one, hence of order at most one by Hadamard's factorization, and it has no zeros: h is a polynomial of degree at most one. The symmetry gives $h(s)-h(1-s)\in2\pi i\mathbb{Z}$, which forces the degree to be zero. The last statement compares the limits at $+\infty$. □

Remark 14.32 (Order one cannot be dropped). For real $\nu_5 \neq 0$ put $f(s) = \zeta(s)\exp\bigl(-\nu_5(s-\frac{1}{2})^2\bigr)$ and $e^{h(s)} = \exp\bigl(\nu_5(s-\frac{1}{2})^2\bigr)$. Then f is meromorphic of order two with the functional equation, the pole and the zeros of ζ; e^h is entire, without zeros, of finite order and symmetric; $g = f\,e^h = \zeta$ is an absolutely convergent Dirichlet series in $\Re s > 1$; and e^h is not constant. In Theorem 14.3 the conclusion $\tilde{h}_0 \equiv 0$ uses that $(s-1)\zeta_{Z1}$ has order one (Proposition 11.1). The location of the zeros of f plays no role in Proposition 14.31.

14.11. **The half-line form of Hamburger's theorem for a pair.** The next two theorems are stated for a general pair: a function with the functional equation and a Dirichlet series, joined on the real half-line only.

Theorem 14.33 (The ladder). *Let* $\mathsf{d}\colon [1,\infty) \to \mathbb{R}$ *be measurable with* $\int_1^\infty |\mathsf{d}(x)|\,x^{-\mathsf{e}_1-1}\,dx < \infty$ *for some* $\mathsf{e}_1 \geq 0$*, and put*

$$h(\sigma) := \sigma \int_1^\infty \mathsf{d}(x)\,x^{-\sigma-1}\,dx \qquad (\sigma > \mathsf{e}_1). \tag{14.61}$$

For every integer $N \geq 2$*,*

$$h(\sigma) = O\bigl(\sigma N^{-\sigma}\bigr) \quad (\sigma \to +\infty) \iff \mathsf{d}(x) = 0 \text{ for almost every } x \in [1,N). \tag{14.62}$$

Proof. If $\mathsf{d} = 0$ almost everywhere on $[1,N)$, then $|h(\sigma)| \leq \sigma N^{-(\sigma-\mathsf{e}_1)} \int_N^\infty |\mathsf{d}|x^{-\mathsf{e}_1-1}dx$ for $\sigma > \mathsf{e}_1$. Conversely, put $d(u) := \mathsf{d}(e^u)$, so that $d\,e^{-\mathsf{e}_1 u} \in L^1(0,\infty)$, and

$$\mathsf{Y}(s) := \int_0^{\log N} d(u)\,e^{s(\log N - u)}\,du. \tag{14.63}$$

Y is entire of exponential type at most $\log N$, bounded on $\Re s \leq 0$ by $\int_0^{\log N}|d|$, and $\mathsf{Y}(\sigma) \to 0$ as $\sigma \to -\infty$. For real $\sigma > \mathsf{e}_1$,

$$\mathsf{Y}(\sigma) = \frac{N^\sigma h(\sigma)}{\sigma} - \int_{\log N}^\infty d(u)\,e^{-\sigma(u-\log N)}\,du, \tag{14.64}$$

where the first term is bounded by hypothesis and the second by $N^{\mathsf{e}_1}\int |d|e^{-\mathsf{e}_1 u}du$; so Y is bounded on the real axis. The Phragmén–Lindelöf theorem in the two quadrants $\{\Re s > 0,\ \pm\Im s > 0\}$ (angle $\pi/2$, order one, bounded on both sides) shows that Y is bounded in $\mathbb{C}$, hence constant, hence 0. By the uniqueness of the Laplace transform on $(0,\log N)$, $d = 0$ almost everywhere there. □

No functional equation, no continuation of h and no uniformity in t enter Theorem 14.33: the hypothesis lives on $\sigma \to +\infty$.

Theorem 14.34 (Hamburger's theorem for a pair, half-line form). *Let* f *satisfy (H0) and the functional equation* (11.3)*, and suppose that for* $\Re s > 1$

$$\log f(s) = s\int_1^\infty J_f(x)\,x^{-s-1}\,dx, \qquad \int_1^\infty |J_f(x)|\,x^{-\sigma-1}\,dx < \infty \quad (\sigma > 1), \tag{14.65}$$

with J_f *real and measurable. Let* $g(s) = \sum_{n\geq 1} a_n n^{-s}$ *converge absolutely in* $\Re s > 1$*, with* $a_1 = 1$*, and let* $J_g(x) := \sum_{2\leq n\leq x} b_n$*, where* $\log g(s) = \sum_{n\geq 2} b_n n^{-s}$ *for large* σ*. Put* $h(\sigma) := \log g(\sigma) - \log f(\sigma)$ *for large real* σ*. The following are equivalent:*

(i) $h(\sigma) = O_N\bigl(\sigma N^{-\sigma}\bigr)$ *as* $\sigma \to +\infty$*, for every integer* $N \geq 2$*;*
(ii) $f = g$*;*
(iii) $f = g = \zeta$*.*

In this case J_f *is almost everywhere the step function* $\sum_{p^m \leq x} \frac{1}{m}$*.*

Proof. For large σ, $\log g(\sigma) = \sigma\int_1^\infty J_g(x)x^{-\sigma-1}dx$ by partial summation, so h has the form (14.61) with $\mathsf{d} = J_g - J_f$. (i)⇒(ii): Theorem 14.33 gives $\mathsf{d} = 0$ almost everywhere, so $\log f = \log g$ for large σ and $f = g$ by analytic continuation. (ii)⇒(iii): g then satisfies (H0)–(H3), so $g = C_H\zeta$, and $a_1 = 1$ gives $C_H = 1$. (iii)⇒(i) because $h \equiv 0$. □

Corollary 14.35 (The ladder for the pair $(\zeta_\lambda, \tilde\zeta_{\mathbb{Z},\lambda})$)**.** *Let $\lambda \in [0,1]$ and $N \geq 2$. Then $\tilde h_\lambda(\sigma) = O(\sigma N^{-\sigma})$ as $\sigma \to +\infty$ if and only if the Mellin count J_λ^{M} of ζ_λ coincides almost everywhere on $[1,N)$ with the step function $\sum_{n^m \leq x} \tilde c_n(\lambda)/m$; in particular J_λ^{M} then vanishes on $(1,2)$ and is constant between consecutive integers below N. Consequently*

$$\text{(D)} \iff \tilde h_0(\sigma) = O_N(\sigma N^{-\sigma}) \text{ for every } N \geq 2, \tag{14.66}$$

and at the clock end the first rung already fails: $\tilde h_1(\sigma) \to \log L \neq 0$.

Proof. Theorem 14.33 with $\mathsf{d} = J_g - J_\lambda^{\mathrm{M}}$, where J_g is the count of $\log \tilde\zeta_{\mathbb{Z},\lambda} = \sum_m \frac{1}{m} \sum_n \tilde c_n(\lambda) n^{-ms}$; the integrability is Theorem 12.5, whose proof applies to every ζ_λ, and Corollary 13.11. (14.66) is Theorem 14.34 with Corollary 13.11 and Theorem 12.7, and the last statement is Proposition 13.15. □

The joining factor therefore measures the mass of dJ_λ^{M} outside the integers, and its rate of decay on the real axis is the first integer below which there is such mass. The Nth rung at $\lambda = 0$ is the statement that J_{Z1}^{M} is constant on $(N-1, N)$; it is (D) restricted to one cell.

14.12. **The bridge theorem: the phase of the joining factor on the critical line.**

Theorem 14.36 (Bridge theorem, phase form)**.** *Let f be meromorphic in $\mathbb{C}$, real on the real axis, with (H0), the functional equation* (11.3)*, and $\lim_{\sigma\to+\infty} f(\sigma) = \varpi_f > 0$. Let $g(s) = \sum_{n\geq 1} a_n n^{-s}$ converge absolutely in $\Re s > 1$, with real a_n and $a_1 = 1$. Suppose that $g = f\, e^h$ in $\Re s > \frac12$ with h holomorphic there and real on $(1,\infty)$, and that*

(B1) *$\Im h$ extends continuously to $\Re s \geq \frac12$ and vanishes on the line $\Re s = \frac12$;*
(B2) *there are $\mathsf{e}_1, \mathsf{e}_2 \geq 0$ with*

$$|h(s)| \leq C\,(1+|s|)^{\mathsf{e}_1}\Big(1 + (\sigma - \tfrac12)^{-\mathsf{e}_2}\Big) \qquad (\Re s > \tfrac12). \tag{14.67}$$

Then $h \equiv -\log \varpi_f$, $g = \zeta$ and $f = \varpi_f\, \zeta$.

Proof. Reflection. $v := \Im h$ is harmonic in $\Re s > \frac12$, continuous on the closed half-plane and 0 on its boundary. By the reflection principle [1, Ch. 4, §6.5] the function equal to $v(s)$ for $\Re s \geq \frac12$ and to $-v(1-\bar s)$ for $\Re s < \frac12$ is harmonic in $\mathbb{C}$. It is the imaginary part of an entire function, which can be chosen equal to h in $\Re s > \frac12$; in $\Re s < \frac12$ it differs from $\overline{h(1-\bar s)}$ by a real constant, and this constant is 0 because the real part of an entire function is continuous across the line. Thus h is entire and $h(1-\bar s) = \overline{h(s)}$. Since h is real on $(1,\infty)$, $h(\bar s) = \overline{h(s)}$, and therefore $h(1-s) = h(s)$.

Growth. By (14.67) and the symmetry, $|h(s)| \leq C(1+|s|)^{\mathsf{e}_1}(1+|\sigma-\frac12|^{-\mathsf{e}_2})$ for $\Re s \neq \frac12$. The function $\log|h|$ is subharmonic, so for every s

$$\log|h(s)| \leq \frac{1}{2\pi}\int_0^{2\pi} \log\big|h\big(s+e^{i\phi}\big)\big|\, d\phi \leq \log C + \mathsf{e}_1 \log(2+|s|) + \log 2 + \frac{\mathsf{e}_2}{2\pi}\int_0^{2\pi} \log^+ \frac{1}{|\sigma - \frac12 + \cos\phi|}\, d\phi. \tag{14.68}$$

For real y, $\frac{1}{2\pi}\int_0^{2\pi} \log|y+\cos\phi|\, d\phi \geq -\log 2$, so the last integral is at most $2\pi\big(\log 2 + \log(2+|s|)\big)$. Hence $|h(s)| \leq C'(2+|s|)^{\mathsf{e}_1+\mathsf{e}_2}$ in $\mathbb{C}$, and h is a polynomial. On the real axis $h(\sigma) = \log g(\sigma) - \log f(\sigma) \to -\log\varpi_f$ as $\sigma \to +\infty$, so the polynomial is the constant $-\log\varpi_f$.

Hamburger. $g = f/\varpi_f$ satisfies (H0), (H1), and (H2)–(H3) with the second function equal to g; so $g = C_H\zeta$, and $a_1 = 1$ gives $C_H = 1$. □

The symmetry of the joining factor, its continuation to $\Re s \leq \frac12$ and its finite order are not hypotheses of Theorem 14.36: they follow from (B1) and (B2), which are written on the closed half-plane $\Re s \geq \frac12$ where the joining factor of the family is defined. Theorem 14.3 is the special case in which the continuation is assumed.

Corollary 14.37 (The bridge theorem for the family)**.** (i) *For $0 < \lambda \le 1$ the joining factor $\tilde h_\lambda$ does not satisfy both* (B1) *and* (B2)*.*
(ii) *(D) holds if and only if $\tilde h_0$ satisfies* (B1) *and* (B2)*.*

Proof. Theorem 14.36 applies with $f = \zeta_\lambda$ (Theorem 10.1, Proposition 13.1), $g = \tilde\zeta_{\mathbb{Z},\lambda}$ (Corollary 13.11) and $h = \tilde h_\lambda$ (Theorem 13.12(iii)). If (B1) and (B2) hold, then $\zeta_\lambda = \varpi_\lambda\zeta$ is an absolutely convergent Dirichlet series in $\Re s > 1$, which is excluded for $0 < \lambda \le 1$ by Proposition 14.1; for $\lambda = 0$ it is (D). Conversely (D) gives $\tilde h_0 \equiv 0$ (Theorem 13.13). □

Define the *phase defect* of the family, when the limit exists,

$$\mathsf{v}_\lambda(t) := \lim_{\sigma\to\frac12^+} \Im\tilde h_\lambda(\sigma+it) = \lim_{\sigma\to\frac12^+}\Big(\Im\log\tilde\zeta_{\mathbb{Z},\lambda}(\sigma+it) - \Im\log\zeta_\lambda(\sigma+it)\Big). \tag{14.69}$$

Proposition 14.38 (The phase of the continuous side)**.** *Let $\lambda \in [0,1]$ and $t > 0$ not an ordinate $\gamma_k(\lambda)$. With the branch of $\log\zeta_\lambda$ that is real on $(1,\infty)$, continued in $\{\Re s > \frac12,\ \Im s > 0\}$,*

$$\lim_{\sigma\to\frac12^+} \Im\log\zeta_\lambda(\sigma+it) = -\vartheta(t) + \pi\big(N_\lambda(t) - 1\big), \qquad N_\lambda(t) = \#\{k : \gamma_k(\lambda) \le t\}, \tag{14.70}$$

and in the sense of distributions in $t > 0$

$$\lim_{\sigma\to\frac12^+} \Im\mathcal{V}_\lambda(\sigma+it) = -\pi\sum_{k\ge1}\eta_k\,\delta\big(t-\gamma_k(\lambda)\big). \tag{14.71}$$

Consequently $\mathsf{v}_\lambda(t) = \lim_{\sigma\to\frac12^+}\Im\log\tilde\zeta_{\mathbb{Z},\lambda}(\sigma+it) + \vartheta(t) - \pi\big(N_\lambda(t)-1\big)$.

Proof. $\zeta_\lambda = P_\lambda/G$. On $s = \frac12 + it$, $q(s) = -(\frac14 + t^2)$ and $1 + q(s)/A_{k,\lambda} = (\gamma_k(\lambda)^2 - t^2)/A_{k,\lambda}$ are real, so $P_\lambda(\frac12 + it)$ is real and changes sign at each $\gamma_k(\lambda)$; approaching from $\sigma > \frac12$ the argument of the factor k increases by π as t passes $\gamma_k(\lambda)$, since $\Im(\gamma_k(\lambda)^2 + (\sigma - \frac12 + it)^2) = 2t(\sigma - \frac12) > 0$. The argument of $G(\frac12 + it) = -(\frac14 + t^2)\Gamma(\frac14 + \frac{it}{2})\pi^{-1/4-it/2}$ is $\pi + \vartheta(t)$ by (8.29), with the branch fixed by $\zeta_\lambda(\sigma) < 0$ for $\frac12 < \sigma < 1$. This gives (14.70). For (14.71) it is preferable not to interchange a boundary limit and ∂_λ. Put $z = \sigma' + it$ and $\gamma = \gamma_k(\lambda)$. The kth term of (10.11) is

$$2\gamma\eta_k\left(\frac{1}{\gamma^2+z^2} - \frac{1}{\frac14+\gamma^2}\right).$$

The second summand is real, while

$$\frac{2\gamma}{\gamma^2+z^2} = -i\left(\frac{1}{z-i\gamma} - \frac{1}{z+i\gamma}\right).$$

The Poisson-kernel boundary limit on $t > 0$ is therefore $-\pi\eta_k\delta(t-\gamma)$; the pole at $-\gamma$ lies outside the positive half-line. Summing against a compactly supported test function is justified by the bounded displacement and the same mean-value/summation-by-parts estimates used for the generator (equivalently by Cauchy–Schwarz with $\sum_k \eta_k^2\gamma_k^{-2} < \infty$ after one integration by parts). This proves (14.71) directly in distributions. □

By Proposition 14.38, $\mathsf{v}_\lambda \equiv 0$ asks two things of the discrete function on the critical line: jumps of π of its argument exactly at the ordinates $\gamma_k(\lambda)$, and the phase $-\vartheta(t)$ between them; equivalently, $e^{i\vartheta(t)}\tilde\zeta_{\mathbb{Z},\lambda}(\frac12 + it)$ is real. At the clock end $\tilde h_1$ is holomorphic in $\Re s > 0$ (Theorem 13.12(iv)), so $\mathsf{v}_1(t) = \Im\tilde h_1(\frac12 + it)$ is a real-analytic function; it does not vanish identically, by Corollary 14.37(i) and the bound $|\tilde h_1(s)| \le C(1 + |s|)$ on $\Re s \ge \frac14$ of the proof of Proposition 14.5.

Proposition 14.39 (The cell masses are sums of a one-ordinate function)**.** *For $n \geq 2$ there is a real function $\mathsf{o}_n(\kappa, \kappa')$ of two real variables, depending only on n and the cell φ_n, such that for all lists $\boldsymbol{\kappa}, \boldsymbol{\kappa}' \in \mathscr{O}$*

$$\tilde{c}_n(\boldsymbol{\kappa}) - \tilde{c}_n(\boldsymbol{\kappa}') = \sum_{k\geq 1} \mathsf{o}_n(\kappa_k, \kappa_k'), \qquad \mathsf{o}_n(\kappa, \kappa') = -2\,\Re\big(W_n(\kappa) - W_n(\kappa')\big) \quad \text{if } n \text{ is not a power of } 2, \tag{14.72}$$

and, with $\mathsf{L}_n(s) := -\log(1 - n^{-s})$, for $\Re s > 1$

$$\log \tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}(s) - \log \tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}'}(s) = \sum_{n\geq 2} \Big(\sum_{k\geq 1} \mathsf{o}_n(\kappa_k, \kappa_k') \Big) \mathsf{L}_n(s), \tag{14.73}$$

$$\log \zeta_{\boldsymbol{\kappa}}(s) - \log \zeta_{\boldsymbol{\kappa}'}(s) = \sum_{k\geq 1} \log \frac{1 + q(s)/(\frac{1}{4} + \kappa_k^2)}{1 + q(s)/(\frac{1}{4} + {\kappa_k'}^2)}. \tag{14.74}$$

Proof. $\tilde{c}_n(\boldsymbol{\kappa}) = \langle d\pi_{\boldsymbol{\kappa}}, \varphi_n \rangle$, where $d\pi_{\boldsymbol{\kappa}}$ is obtained from $J_{\boldsymbol{\kappa}}$ by truncation at 2 and finite Möbius inversion, both linear, and by (12.16) the difference $J_{\boldsymbol{\kappa}} - J_{\boldsymbol{\kappa}'} = -2\Re \sum_k \big(\mathrm{Ei}((\frac{1}{2} + i\kappa_k)\log x) - \mathrm{Ei}((\frac{1}{2} + i\kappa_k')\log x)\big)$ is a sum over k converging in $\mathcal{D}'(1,\infty)$. This gives (14.72), with the value (13.20) off the powers of 2. (14.73) is the Euler product with Corollary 13.11, and (14.74) is (15.2). □

By Proposition 14.39, the difference of two joining factors is, term by term in n and in k, a sum over the ordinates of one universal function of one ordinate, the joining factor of a single zero factor:

$$\mathsf{j}(\kappa, \kappa'; s) := \sum_{n\geq 2} \mathsf{o}_n(\kappa, \kappa')\, \mathsf{L}_n(s) - \log \frac{1 + q(s)/(\frac{1}{4} + \kappa^2)}{1 + q(s)/(\frac{1}{4} + \kappa'^2)}, \qquad \tilde{h}_{\boldsymbol{\kappa}} - \tilde{h}_{\boldsymbol{\kappa}'} = \sum_{k\geq 1} \mathsf{j}(\kappa_k, \kappa_k'; s) \ \text{ formally.} \tag{14.75}$$

With $\boldsymbol{\kappa}' = (\tau_k)$ this writes the phase defect as $\mathsf{v}_{\boldsymbol{\kappa}} = \mathsf{v}_1 + \sum_k \Im \mathsf{j}(\kappa_k, \tau_k; \frac{1}{2} + it)$, the defect of the clock plus a transport term.

Heuristic 14.40 (What happens to the phase defect along the path)**.** The following is a computation by stationary phase without remainder terms; it is not proved. For n much larger than κ the cell resolves the oscillation, $\Re W_n(\kappa) \approx n^{-1/2} \cos(\kappa \log n)/\log n$, and for n much smaller than $\kappa/2\pi$ the kernel $W_n(\kappa)$ is negligible because φ is smooth (Proposition 14.17). The transition is at $n \approx \kappa/2\pi$, where the phase is $\kappa \log \frac{\kappa}{2\pi} - \kappa = 2\pi(\mathcal{N}_{\mathrm{RVM}}(\kappa) - \frac{7}{8})$. This suggests

$$\Im \mathsf{j}(\kappa, \cdot\,; \tfrac{1}{2} + it) \approx \text{amplitude} \times \cos\big(2\pi \mathcal{N}_{\mathrm{RVM}}(\kappa) + \text{phase}(t)\big), \qquad \big|\mathsf{j}(\kappa, \cdot\,; \sigma + it)\big| \approx \kappa^{-1/2-\sigma}, \tag{14.76}$$

so that the sum over the ordinates converges absolutely exactly for $\sigma > \frac{1}{2}$, in agreement with Theorem 13.12(iii), and is at the edge of convergence on the critical line. On the clock, $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$ and the phases are aligned, which is why $\tilde{h}_1$ crosses the line. Along the path each term turns by the angle $(1 - \lambda)\eta_k \ell_k$, the rotation of (14.42). At $\lambda = 0$, (B1) is the statement that the rotated sum, with the phases $e^{2\pi i \mathcal{N}_{\mathrm{RVM}}(\gamma_k)}$, converges on the critical line to minus the defect v_1 of the clock.

14.13. **Transport from the clock on the rungs.** For a real function $\mathsf{f} \in C_c^\infty(0,\infty)$ put

$$\mathsf{c}_{\mathsf{f}}(t) := \int_0^\infty \mathsf{f}(u) \cos(tu)\, du, \qquad \mathsf{k}(u) := \frac{1}{2}\Big(e^{u/2} - \frac{e^{-u/2}}{e^{2u} - 1}\Big), \qquad \mathsf{k}_{\mathsf{f}} := \int_0^\infty \mathsf{f}(u)\, \mathsf{k}(u)\, du. \tag{14.77}$$

Proposition 14.41 (The rung functional)**.** *For every list $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 15),*

$$\mathsf{R}_{\mathsf{f}}(\boldsymbol{\kappa}) := \frac{1}{2} \int_1^\infty \mathsf{f}(\log x)\, x^{-1/2} \log x\, dJ_{\boldsymbol{\kappa}}^{\mathrm{M}}(x) = \mathsf{k}_{\mathsf{f}} - \sum_{k\geq 1} \mathsf{c}_{\mathsf{f}}(\kappa_k), \tag{14.78}$$

the series converging absolutely. Along the family, with $\mathsf{R}_{\mathsf{f}}(\lambda) := \mathsf{R}_{\mathsf{f}}\big((\gamma_k(\lambda))\big)$,

$$\mathsf{R}_{\mathsf{f}}(\lambda) = \mathsf{R}_{\mathsf{f}}(1) - \mathsf{S}_{\mathsf{f}}(\lambda), \qquad \mathsf{S}_{\mathsf{f}}(\lambda) := \sum_{k\geq 1} \Big(\mathsf{c}_{\mathsf{f}}\big(\gamma_k(\lambda)\big) - \mathsf{c}_{\mathsf{f}}(\tau_k) \Big), \qquad \mathsf{S}_{\mathsf{f}}(1) = 0. \tag{14.79}$$

If the support of f *contains no point* $\log n$, $n \in \mathbb{N}$, *then*

$$\text{(D)} \implies \mathsf{S}_{\mathsf{f}}(0) = \mathsf{R}_{\mathsf{f}}(1), \tag{14.80}$$

and (D) holds if and only if $\mathsf{S}_{\mathsf{f}}(0) = \mathsf{R}_{\mathsf{f}}(1)$ *for every such* f *and the resulting integer jumps satisfy the growth hypothesis of Theorem 12.7.*

Proof. Pair (12.15), which holds for every list of $\mathscr{O}$ with the same proof, with $\frac{1}{2}\mathsf{f}(\log x)x^{-1/2}$ and put $u = \log x$: the term 1 gives $\frac{1}{2}\int \mathsf{f}(u)e^{u/2}du$, the term $x^{-1}(x^2-1)^{-1}$ gives $\frac{1}{2}\int \mathsf{f}(u)e^{-u/2}(e^{2u}-1)^{-1}du$, and the sum gives $\sum_k \mathsf{c}_{\mathsf{f}}(\kappa_k)$; c_{f} is rapidly decreasing. (14.79) is the difference of two cases of (14.78). By Theorem 12.7, the vanishing $\mathsf{R}_{\mathsf{f}}(0) = 0$ for every f whose support avoids the points $\log n$ is exactly the constancy part of the cell criterion; (D) follows when the resulting integer jumps also satisfy $\sum_{n\le X}|\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$. □

The number $\mathsf{R}_{\mathsf{f}}(1)$ belongs to the clock; $\mathsf{S}_{\mathsf{f}}(0)$ is the transport from the clock to the ordinates of ζ. (D) is the statement that the transport cancels the defect of the clock on every rung.

Theorem 14.42 (The transport is band-limited in λ)**.** *Let* $\mathsf{f} \in C_c^\infty(0, \log N)$. *The function* $\lambda \mapsto \mathsf{S}_{\mathsf{f}}(\lambda)$ *extends to an entire function with*

$$|\mathsf{S}_{\mathsf{f}}(\lambda)| \le C_{\mathsf{f}}(1 + |\lambda|\log(2+|\lambda|))\, e^{C_\eta \log N\,|\Im\lambda|} \qquad (\lambda \in \mathbb{C}), \tag{14.81}$$

and for every $j \ge 1$

$$\partial_\lambda^j \mathsf{S}_{\mathsf{f}}(1) = \mathsf{m}_j(\mathsf{f}) := \sum_{k\ge1} \eta_k^j\, \mathsf{c}_{\mathsf{f}}^{(j)}(\tau_k), \qquad \mathsf{S}_{\mathsf{f}}(0) = \sum_{j\ge1} \frac{(-1)^j}{j!}\, \mathsf{m}_j(\mathsf{f}). \tag{14.82}$$

If the support of f *lies in* $[u_1, \log N]$ *with* $u_1 > 0$, *then* $|\mathsf{m}_j(\mathsf{f})| \le C_{\mathsf{f}}(1+j)^2(C_\eta \log N)^j$.

Proof. By the Paley–Wiener theorem, $|\mathsf{c}_{\mathsf{f}}^{(j)}(t)| \le C_{A,j}\, e^{\log N|\Im t|}(1+|t|)^{-A}$ for complex t and every A. With $\gamma_k(\lambda) = \gamma_k + \lambda\eta_k$, $|\Im\gamma_k(\lambda)| \le C_\eta|\Im\lambda|$ and $|\gamma_k(\lambda)| \ge \gamma_k - C_\eta|\lambda|$. The ordinates with $\gamma_k \le 2C_\eta|\lambda|$ are $O(|\lambda|\log(2+|\lambda|))$ in number and each contributes at most $C\, e^{C_\eta \log N|\Im\lambda|}$; for the others $|\mathsf{c}_{\mathsf{f}}(\gamma_k(\lambda))| \le C\, e^{C_\eta \log N|\Im\lambda|}(\gamma_k/2)^{-2}$. This gives (14.81), the termwise differentiation and the convergence of the Taylor series at $\lambda = 1$ evaluated at $\lambda = 0$. Finally $\mathsf{c}_{\mathsf{f}}^{(j)}(t) = \int \mathsf{f}(u)u^j\cos(tu + \frac{j\pi}{2})du$ and two integrations by parts give $(1+t^2)|\mathsf{c}_{\mathsf{f}}^{(j)}(t)| \le C_{\mathsf{f}}(1+j)^2(\log N)^j$. □

The scaled lists $(\tau_k - c'\eta_k)$ are the members $\lambda = 1 - c'$ of the family, so the scale test of Remark 14.20 is the behaviour of the entire function S_{f} on the real axis. In (14.82) the nodes τ_k and the weights $\mathsf{c}_{\mathsf{f}}^{(j)}(\tau_k)$ belong to the clock, and the powers η_k^j do not; on the first rung, $N = 2$, the term j is at most $C_{\mathsf{f}}(1+j)^2(C_\eta\log 2)^j/j!$.

Proposition 14.43 (The defect of the clock on a rung)**.** *Let* $\mathsf{t}_0 \in (2\pi, \tau_0)$ *be the point with* $\mathcal{N}_{\mathrm{RVM}}(\mathsf{t}_0) = 0$, *and let* $\mathsf{f} \in C_c^\infty(0,\infty)$. *Then*

$$\sum_{k\ge1} \mathsf{c}_{\mathsf{f}}(\tau_k) = \mathsf{I}_0 + 2\sum_{m\ge1}(-1)^m \mathsf{I}_m, \qquad \mathsf{I}_m := \int_{\mathsf{t}_0}^\infty \mathsf{c}_{\mathsf{f}}(t)\cos(2\pi m\mathcal{N}_{\mathrm{RVM}}(t))\mathcal{N}'_{\mathrm{RVM}}(t)\,dt = O_{\mathsf{f}}(m^{-2}), \tag{14.83}$$

$$\mathsf{I}_0 = -\int_0^\infty \frac{\mathsf{f}(u)}{2\pi u}\Big(\sin(u\mathsf{t}_0)\log\frac{\mathsf{t}_0}{2\pi} + \frac{\pi}{2} - \int_0^{u\mathsf{t}_0}\frac{\sin y}{y}\,dy\Big)du, \tag{14.84}$$

and therefore $\mathsf{R}_{\mathsf{f}}(1) = (\mathsf{k}_{\mathsf{f}} - \mathsf{I}_0) - 2\sum_{m\ge1}(-1)^m\mathsf{I}_m$.

Proof. The function $g(v) := \mathsf{c}_{\mathsf{f}}(\mathcal{T}(v))$, $v \ge 0$, with $\mathcal{T}$ the inverse of the clock on $[\mathsf{t}_0,\infty)$, is smooth and rapidly decreasing with its derivatives, and $\tau_k = \mathcal{T}(k - \frac{1}{2})$. Poisson's summation formula for the even extension of g at the half-integers gives $2\sum_{k\ge1} g(k-\frac{1}{2}) = \sum_{m\in\mathbb{Z}}(-1)^m 2\int_0^\infty g(v)\cos(2\pi m v)\,dv$, the Fourier coefficients being $O(m^{-2})$ after two integrations by parts; the substitution $v = \mathcal{N}_{\mathrm{RVM}}(t)$ gives

(14.83). For $m=0$, $\mathcal{N}'_{\mathrm{RVM}}(t)=\frac{1}{2\pi}\log\frac{t}{2\pi}$; integrating by parts with the primitive $\int \mathsf{f}(u)\sin(tu)u^{-1}du$ of c_{f} and exchanging the integrals gives (14.84). □

In l_m the phases $2\pi m\mathcal{N}_{\mathrm{RVM}}(t)\mp tu$ have the derivative $m\log\frac{t}{2\pi}\mp u$, which vanishes at $t=2\pi e^{u/m}$: the defect of the clock at the point $x=e^u$ is produced at the heights $2\pi x^{1/m}$, which is the origin of the terms $2\cos(2\pi y)$ and $y^{-1/4}\cos(4\pi\sqrt{y}+\frac{\pi}{4})$ of (14.32). On the rung $(n,n+1)$ the height for $m=1$ lies in $(2\pi n,2\pi(n+1))$. On the first rung it lies in $(2\pi,4\pi)$, below τ_1 and on both sides of $\mathfrak{t}_0$: the defect of the clock on $(1,2)$ is an effect of the start of the clock. Experiments 18.10 and 18.11 evaluate (14.79)–(14.83) for one function f.

14.14. **The multiscale Jacobian of the midpoint clock.** The rung system of Subsection 15.2 asks for a list and free masses simultaneously. Before attempting nonlinear continuation from the clock, one must understand the linearization with respect to the ordinates. The point is genuinely multiscale: the gaps of the midpoint clock tend to zero, so no fixed u-window can carry a uniform inverse, while a window of length comparable with $\log T$ is exactly large enough to resolve the ordinates up to height T.

For $T>\tau_2$ put

$$K(T):=\max\{k:\tau_k\le T\},\qquad \Delta_T:=\min_{1\le k<K(T)}(\tau_{k+1}-\tau_k),$$

and define the finite ordinate block of the linearized rung map by

$$(\mathcal{J}_T\mathbf{v})(u):=-u\sum_{\tau_k\le T}v_k\sin(\tau_k u). \tag{14.85}$$

No genuine zero ordinate and no individual displacement $\eta_k=\tau_k-\gamma_k$ enters this definition.

Theorem 14.44 (Multiscale Jacobian of the midpoint clock)**.** *The following statements hold.*

(i) *For every $T>\tau_2$,*

$$\Delta_T\ge\frac{2\pi}{\log(T/2\pi)}. \tag{14.86}$$

(ii) *Fix $a>0$ and $\varkappa_{\mathrm{win}}>0$, and put*

$$L_T:=(1+\varkappa_{\mathrm{win}})\log(T/2\pi),\qquad I_T=[a,a+L_T].$$

There is $c_{a,\varkappa_{\mathrm{win}}}>0$, independent of T, such that for all finitely supported $\mathbf{v}=(v_k)$ on $\{k:\tau_k\le T\}$,

$$\|\mathcal{J}_T\mathbf{v}\|^2_{L^2(I_T)}\ge c_{a,\varkappa_{\mathrm{win}}}L_T\sum_{\tau_k\le T}|v_k|^2. \tag{14.87}$$

(iii) *No analogue of* (14.87) *with a positive T-independent lower constant can hold on a fixed interval $(0,U)$. Indeed, for the unit vector $\mathbf{v}^{(k)}=(e_k-e_{k+1})/\sqrt{2}$,*

$$\|\mathcal{J}\mathbf{v}^{(k)}\|_{L^2(0,U)}\le\frac{U^{5/2}}{\sqrt{10}}(\tau_{k+1}-\tau_k)\longrightarrow 0. \tag{14.88}$$

(iv) *For every finite $U>0$,*

$$\overline{\mathrm{span}}\{u\sin(\tau_k u):k\ge1\}=L^2(0,U).$$

Thus the infinite linearized clock has dense range on every finite window, despite the absence of a bounded inverse on a fixed window.

(v) *Let $\mathsf{f}\in C_c^\infty(0,U)$ and $\mathsf{c}_{\mathsf{f}}(t)=\int_0^\infty\mathsf{f}(u)\cos(tu)\,du$. For every list $\boldsymbol{\kappa}$ satisfying $|\kappa_k-\tau_k|\le C_\eta$ and every $A>1$,*

$$\sum_{\tau_k>T}|\mathsf{c}_{\mathsf{f}}(\kappa_k)-\mathsf{c}_{\mathsf{f}}(\tau_k)|\ll_{A,\mathsf{f},C_\eta}T^{1-A}\log T. \tag{14.89}$$

Proof. The midpoint equation is $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$ with

$$\mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}.$$

For $\tau_{k+1} \le T$, the mean-value theorem gives

$$1 = \mathcal{N}'_{\mathrm{RVM}}(\xi_k)(\tau_{k+1} - \tau_k), \qquad \tau_k < \xi_k < \tau_{k+1} \le T,$$

which proves (14.86).

For (ii), apply Ingham's inequality for non-harmonic exponentials [14] to the separated frequency set $\{\pm\tau_k : \tau_k \le T\}$. By (14.86),

$$L_T \Delta_T \ge 2\pi(1 + \varkappa_{\mathrm{win}}),$$

so after scaling the frequencies by Δ_T the interval lies a fixed positive distance beyond Ingham's critical length. The lower Ingham constant is therefore bounded below by a positive constant depending only on $\varkappa_{\mathrm{win}}$; rescaling gives a bound of size L_T for the sine sum. Since $u \ge a$ on I_T, multiplication by u gives (14.87).

For (iii), $|\sin(\tau_{k+1}u) - \sin(\tau_k u)| \le (\tau_{k+1} - \tau_k)u$, whence

$$\|\mathcal{J}\mathbf{v}^{(k)}\|^2_{L^2(0,U)} \le \frac{(\tau_{k+1} - \tau_k)^2}{2} \int_0^U u^4 \, du = \frac{(\tau_{k+1} - \tau_k)^2 U^5}{10}.$$

The same mean-value formula used above shows $\tau_{k+1} - \tau_k \to 0$.

For (iv), suppose $F \in L^2(0, U)$ is orthogonal to every $u\sin(\tau_k u)$. Put $H(u) = uF(u)$ and extend H oddly to $(-U, U)$. Its Fourier transform

$$\widehat{H}(z) = \int_{-U}^{U} H(u) e^{-izu} \, du$$

is entire of exponential type at most U and vanishes at every $\pm\tau_k$. A nonzero entire function of finite exponential type has $O(r)$ zeros in $|z| \le r$ by Jensen's formula, whereas the midpoint clock has $2\mathcal{N}_{\mathrm{RVM}}(r) + O(1) \asymp r \log r$ points $\pm\tau_k$ in that disk. Hence $\widehat{H} \equiv 0$, so $H = 0$ and therefore $F = 0$ almost everywhere.

For (v), the Paley–Wiener decay of the cosine transform gives $|\mathsf{c}'_{\mathsf{f}}(t)| \ll_{A,\mathsf{f}} (1 + |t|)^{-A}$ for every A. The mean-value theorem and $|\kappa_k - \tau_k| \le C_\eta$ give

$$|\mathsf{c}_{\mathsf{f}}(\kappa_k) - \mathsf{c}_{\mathsf{f}}(\tau_k)| \ll_{A,\mathsf{f},C_\eta} \tau_k^{-A}$$

for large k. Partial summation with $\mathcal{N}_{\mathrm{RVM}}(t) = O(t \log t)$ yields $\sum_{\tau_k > T} \tau_k^{-A} = O(T^{1-A} \log T)$. □

The theorem identifies the correct scale but does not prove existence of a solution of the rung system. In particular, a fixed-window Banach-space implicit-function argument cannot work because of (14.88). A quantitative continuation must grow the resolved window together with the spectral height.

One precise residual formulation is the following. Fix $r > \frac{1}{2}$, so point masses belong to $H^{-r}(0, U)$, and put

$$\mathsf{D}_{\mathrm{RVM}}(u) := \mathsf{k}(u) - \sum_{k \ge 1} \cos(\tau_k u)$$

in $\mathcal{D}'(0, \infty)$. For $T, U > 0$ define

$$\varepsilon_r(T, U) := \inf_{h,b} \left\| \mathsf{D}_{\mathrm{RVM}} + u \sum_{\tau_k \le T} h_k \sin(\tau_k u) - \sum_{\log n \le U} b_n \delta(u - \log n) \right\|_{H^{-r}(0,U)}. \tag{14.90}$$

Theorem 14.44 supplies the finite-height separation, dense-range and tail ingredients. What is not proved is a bound for (14.90), together with control of the minimizing h strong enough to dominate the quadratic remainder in

$$\cos((\tau_k + h_k)u) = \cos(\tau_k u) - h_k u \sin(\tau_k u) + O(h_k^2 u^2),$$

uniformly along a nested sequence with $U \asymp \log T$. Such an estimate would turn the finite-section continuation observed numerically into an existence proof for the infinite rung system. By Theorem 15.5, any resulting solution would then be the zeta ordinates with the von Mangoldt masses. The individual displacements η_k are not data in this formulation.

15. The space of ordinate lists and the existence form of the final step

The constructions of Sections 10–14 use two properties of the ordinates (Remark 13.17). This section applies them to every list with these two properties. The final step then becomes an existence statement in a compact space, and the list of ζ is not part of its formulation.

15.1. **Lists, their zeta functions and their atomizations.** Let $\mathscr{O}$ be the set of sequences $\boldsymbol{\kappa} = (\kappa_k)_{k\ge1}$ of real numbers with

$$\tau_0^+ \le \kappa_1 \le \kappa_2 \le \cdots, \qquad |\kappa_k - \tau_k| \le C_\eta, \qquad \Big|\int_{\tau_0}^{X} \big(N_{\boldsymbol{\kappa}}(\tau) - \mathcal{N}_{\mathrm{RVM}}(\tau)\big)d\tau\Big| \le C\log X \quad (X \ge \tau_0), \tag{15.1}$$

where τ_0^+ is a fixed number with $\tau_0 < \tau_0^+ < \min(\gamma_1, \tau_1)$, $N_{\boldsymbol{\kappa}}(\tau) = \#\{k : \kappa_k \le \tau\}$, C_η is the constant of Lemma 2.2 and C the constant of Proposition 13.1. The clock (τ_k), the ordinates (γ_k) of ζ and every list $(\gamma_k(\lambda))$, $0 \le \lambda \le 1$, belong to $\mathscr{O}$. For $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 15) define, exactly as for the family,

$$\zeta_{\boldsymbol{\kappa}}(s) := \frac{1}{G(s)} \prod_{k\ge1} \Big(1 + \frac{q(s)}{\frac14 + \kappa_k^2}\Big), \qquad \tilde c_n(\boldsymbol{\kappa}), \quad \tilde a_N(\boldsymbol{\kappa}), \quad k_N(\boldsymbol{\kappa}), \qquad \tilde\zeta_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta_{\boldsymbol{\kappa}}\, e^{\tilde h_{\boldsymbol{\kappa}}}, \qquad \mathsf{V}_\sigma(\boldsymbol{\kappa}), \tag{15.2}$$

with the cell masses of Definition 13.4, the joining factor of Theorem 13.6 and the function (14.7). Theorem 10.1 (order one, functional equation, zeros exactly $\frac12 \pm i\kappa_k$), Theorems 12.5 and 12.6, Proposition 14.39, Theorem 12.1 in its list form, Theorem 13.10, Corollary 13.11 and Theorem 13.12(i)–(iii) hold for every $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 15), with constants depending only on those in (15.1): their proofs use the ordinates only through (15.1).

Theorem 15.1 (Rigidity of the coincidence)**.** *The set*

$$\mathscr{C} := \{\boldsymbol{\kappa} \in \mathscr{O} : \ \tilde h_{\boldsymbol{\kappa}} \text{ is constant}\} \tag{15.3}$$

of the lists whose continuous and atomized zeta functions coincide up to a constant factor has at most one element. If $\boldsymbol{\kappa} \in \mathscr{C}$, then $\zeta_{\boldsymbol{\kappa}} = \tilde\zeta_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$, $\boldsymbol{\kappa}$ is the list of the ordinates of ζ, and (D) holds. Conversely, (D) implies $\mathscr{C} = \{(\gamma_k)\}$.

Proof. If $\tilde h_{\boldsymbol{\kappa}} \equiv c$, then $\zeta_{\boldsymbol{\kappa}} = e^{-c}\tilde\zeta_{\mathbb{Z},\boldsymbol{\kappa}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Corollary 13.11), $(s-1)\zeta_{\boldsymbol{\kappa}}$ is entire of order one and $\zeta_{\boldsymbol{\kappa}}$ satisfies (11.3) (Theorem 10.1). Hamburger's theorem gives $\zeta_{\boldsymbol{\kappa}} = C_H\zeta$, and $C_H = 1$ because both functions have residue 1 at $s = 1$. Hence the nontrivial zeros of ζ with positive ordinate are the points $\frac12 + i\kappa_k$, with multiplicity; this determines $\boldsymbol{\kappa}$ as the list (γ_k), and $\zeta_{Z1} = \zeta_{\boldsymbol{\kappa}} = \zeta$. Then $e^{-c}\tilde\zeta_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$ and the first coefficients give $c = 0$. Conversely, under (D), $\tilde h_0 \equiv 0$ (Theorem 13.13), so $(\gamma_k) \in \mathscr{C}$. □

Proposition 15.2 (Compactness and continuity)**.** *The space $\mathscr{O}$ and the functions* (15.2) *have the following properties.*

(i) *$\mathscr{O}$ is compact in the product topology of $\prod_k [\tau_k - C_\eta, \tau_k + C_\eta]$.*
(ii) *For every $n \ge 2$, $N \ge 1$ and $\sigma > 1$, the functions $\boldsymbol{\kappa} \mapsto \tilde c_n(\boldsymbol{\kappa})$, $\tilde a_N(\boldsymbol{\kappa})$, $k_N(\boldsymbol{\kappa})$ and $\mathsf{V}_\sigma(\boldsymbol{\kappa})$ are continuous on $\mathscr{O}$.*
(iii) *For every real $\sigma > 1$, $\boldsymbol{\kappa} \mapsto \log\zeta_{\boldsymbol{\kappa}}(\sigma) = -\log G(\sigma) + \sum_k \log\frac{(\sigma-\frac12)^2+\kappa_k^2}{\frac14+\kappa_k^2}$ is a sum of strictly convex functions of the single coordinates κ_k.*

Proof. (i) The product of the intervals is compact, and each condition in (15.1) is closed: for fixed X, $\int_{\tau_0}^{X} N_{\boldsymbol{\kappa}} = \sum_k (X-\kappa_k)^+$ depends continuously on the finitely many coordinates with $\tau_k \le X + C_\eta$. (ii) For n not a power of 2, $\tilde{c}_n(\boldsymbol{\kappa}) = \mathfrak{m}_n - 2\Re \sum_k W_n(\kappa_k)$ by (13.20), and the series converges uniformly on $\mathscr{O}$ because W_n is rapidly decreasing and $|\kappa_k - \tau_k| \le C_\eta$. For $n = 2^j$ the term j is $-\frac{\mu(j)}{j}\int_2^3 J_{\boldsymbol{\kappa}}(y)\frac{d}{dy}\varphi_n(y^j)\,dy$ (proof of Theorem 13.10), and $\boldsymbol{\kappa} \mapsto J_{\boldsymbol{\kappa}} \in L^2(2,3)$ is continuous: $J^{\mathrm{M}}_{\boldsymbol{\kappa}}$ is the Fourier–Plancherel transform of $\log \zeta_{\boldsymbol{\kappa}}(\sigma+it)/(\sigma+it)$, which is continuous in $\boldsymbol{\kappa}$ pointwise and dominated in $L^2(\mathbb{R})$ by the uniform bound of Proposition 13.1, and the constant $J^{\mathrm{M}}_{\boldsymbol{\kappa}} - J_{\boldsymbol{\kappa}}$ is the difference of two pairings with a fixed test function, both continuous in $\boldsymbol{\kappa}$. $\tilde{a}_N$ and k_N are polynomials in finitely many cell masses. The sums $\sum_n |k_n(\boldsymbol{\kappa})| n^{-\sigma'}$, $\sigma' > 1$, are bounded uniformly on $\mathscr{O}$ (Theorem 13.10), so the tails of V_σ are uniformly small, as in the proof of Theorem 14.4(iii). (iii) is the computation in the proof of Proposition 14.2: $\partial_\kappa^2 \log \frac{a^2+\kappa^2}{1/4+\kappa^2} = \mathcal{C}(a,\kappa) - \mathcal{C}(\frac{1}{2},\kappa) > 0$ for $a > \frac{1}{2}$, $\kappa > \frac{1}{2}$. □

Corollary 15.3 (Existence form of the final step)**.** *For every $\sigma > 1$ the minimum $\min_{\mathscr{O}} \mathsf{V}_\sigma$ is attained, and the following are equivalent:* (i) *(D);* (ii) *$\mathscr{C} \neq \varnothing$;* (iii) *there is $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 15) with $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$;* (iv) $\min_{\mathscr{O}} \mathsf{V}_\sigma = 0$.

Proof. A continuous function on a compact space attains its minimum (Proposition 15.2). (i)⇔(ii) is Theorem 15.1, and (ii)⇒(iii) is part of it. (iii)⇔(iv) because $\mathsf{V}_\sigma(\boldsymbol{\kappa}) = 0$ means $k_n(\boldsymbol{\kappa}) = 0$ for $n \ge 2$. (iii)⇒(i): $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}$ has no zeros in $\Re s > \frac{1}{2}$ (Theorem 13.12(iii)), so, with the functional equation, the nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$; they are the points $\frac{1}{2} \pm i\gamma_k$, and Theorem 11.2 gives (D). □

In this form the final step does not ask to identify ζ_{Z1} with ζ: it asks for the existence of one list in $\mathscr{O}$ whose continuous and atomized zeta functions coincide, or whose atomized zeta function is ζ; Theorem 15.1 then identifies the list. The path $\lambda \mapsto (\gamma_k(\lambda))$ is one segment in $\mathscr{O}$, from the clock to the list of ζ, and along it V_σ is the function of Theorem 14.4.

Remark 15.4 (What an existence proof uses). Compactness, continuity and convexity do not decide whether $\min_{\mathscr{O}} \mathsf{V}_\sigma$ is 0: they do not involve the form of the functional equation. The existence statement uses the functional equation with the factor $\Gamma(\frac{s}{2})\pi^{-s/2}$, for which a function with both an absolutely convergent Dirichlet series and the functional equation exists, namely ζ. The first variation of the defect is explicit: by (13.20), for n not a power of 2,

$$\frac{\partial \tilde{c}_n(\boldsymbol{\kappa})}{\partial \kappa_k} = 2\,\Im V_n(\kappa_k), \tag{15.4}$$

so the vanishing of all coordinate derivatives of V_σ at a list is described by the functions V_n of (13.19) alone. Since $\mathscr{O}$ has empty interior in the product topology and the displayed derivatives are coordinate derivatives rather than a proved Fréchet gradient on a fixed Banach space, boundary minimizers must be treated by one-sided/KKT conditions.

15.2. **The rung system with free masses.** With the function k of (14.77), consider, for a list $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 15) and real numbers a_n, $n \ge 2$, on which nothing is assumed, the system

$$\sum_{k\ge 1} \cos(\kappa_k u) + \sum_{n\ge 2} \mathsf{a}_n\, \delta(u - \log n) = \mathsf{k}(u) \qquad \text{in } \mathcal{D}'(0,\infty). \tag{15.5}$$

In the variable $x = e^u$ it states that the Mellin count of $\zeta_{\boldsymbol{\kappa}}$ has no mass outside the integers, the masses at the integers being unknown.

Theorem 15.5 (Existence form with free masses)**.** *(D) holds if and only if* (15.5) *has a solution $(\boldsymbol{\kappa}, (\mathsf{a}_n))$ with $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 15). The solution is then unique: $\boldsymbol{\kappa} = (\gamma_k)$, and*

$$\mathsf{a}_n = \frac{\log p}{2\sqrt{n}} \quad \text{if } n = p^m \text{ is a prime power}, \qquad \mathsf{a}_n = 0 \quad \text{otherwise.} \tag{15.6}$$

Proof. By (12.15) for the list $\boldsymbol{\kappa}$, with $u = \log x$, (15.5) is the statement

$$(\log x)\, dJ^{\mathrm{M}}_{\boldsymbol{\kappa}}(x) = \sum_{n\ge 2} 2\sqrt{n}\, \mathsf{a}_n\, \delta_n \qquad \text{in } \mathcal{D}'(1,\infty). \tag{15.7}$$

If (D) holds, $J^{\mathrm{M}}_{Z1} = \sum_{p^m \le x} \frac{1}{m}$ (Theorem 12.7), which is (15.7) for the list (γ_k) with the masses (15.6). Conversely let (15.7) hold. Then $J^{\mathrm{M}}_{\boldsymbol{\kappa}}$ is constant on $(1,2)$ and between consecutive integers; the truncation at 2 and the finite Möbius inversion keep the support in the integers, so every cell defect vanishes and, by (13.11) and the proof of Theorem 13.6,

$$\zeta_{\boldsymbol{\kappa}}(s) = \exp\bigl(\nu_6 + \nu_7\, 2^{-s}\bigr)\, \tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}(s) \qquad (\Re s > 1) \tag{15.8}$$

with real constants ν_6, ν_7. Both factors are absolutely convergent Dirichlet series in $\Re s > 1$ (Corollary 13.11 for every list of $\mathscr{O}$). Hence $\zeta_{\boldsymbol{\kappa}}$ satisfies (H0)–(H3), Hamburger's theorem gives $\zeta_{\boldsymbol{\kappa}} = C_H \zeta$, and $C_H = 1$ by the residues at $s = 1$. The zeros of $\zeta_{\boldsymbol{\kappa}}$ are $\frac{1}{2} \pm i\kappa_k$, so $\boldsymbol{\kappa} = (\gamma_k)$ and $\zeta_{Z1} = \zeta$. Then $(\log x)\, dJ^{\mathrm{M}} = \sum_{p^m} \log p\, \delta_{p^m}$, which gives (15.6). □

In Theorem 15.5 the primes are not part of the data: the data are the function k, which comes from the factor G and the pole, the space $\mathscr{O}$, which is defined by the clock, and the condition that the support lie in the points $\log n$. The masses of von Mangoldt are an output. Experiment 18.12 solves a finite section of (15.5) numerically, starting from the clock.

Theorem 15.6 (Nondegeneracy of the solution)**.** *Let $(\boldsymbol{\kappa}, (\mathsf{a}_n))$ be a solution of (15.5) with $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 15). Let $(\dot\kappa_k)$ be a bounded real sequence and $(\dot{\mathsf{a}}_n)_{n\ge 2}$ real numbers with $\dot{\mathsf{a}}_n = O_\varepsilon(n^{-1/2+\varepsilon})$ for every $\varepsilon > 0$, such that*

$$\sum_{k\ge 1} \dot\kappa_k\, u \sin(\kappa_k u) = \sum_{n\ge 2} \dot{\mathsf{a}}_n\, \delta(u - \log n) \qquad \text{in } \mathcal{D}'(0,\infty), \tag{15.9}$$

the derivative of (15.5) in the direction $\bigl((\dot\kappa_k), (\dot{\mathsf{a}}_n)\bigr)$. Then $\dot{\mathsf{a}}_n = 0$ for every n, and $\sum_{k:\,\kappa_k = t} \dot\kappa_k = 0$ for every value t of the list. If the ordinates are simple, $\dot\kappa_k = 0$ for every k.

Proof. Step 1. The series $\mathsf{J} := \sum_k \dot\kappa_k\, u \sin(\kappa_k u)$ converges in the space of tempered distributions on $\mathbb{R}$, because $\bigl|\int \phi(u) u \sin(\kappa u) du\bigr| \le C_\phi \kappa^{-3}$ for Schwartz functions ϕ and $\sum_k \kappa_k^{-3} < \infty$; J is even. By (15.9), J vanishes on $(0, \log 2)$, so its restriction to $(-\log 2, \log 2)$ is supported at 0 and is a finite sum of even derivatives of δ_0. Fix an even $\mathsf{x}_0 \in C_c^\infty(-\log 2, \log 2)$ equal to 1 near 0, and for $\Re z > 0$ put

$$\mathsf{H}_1(z) := \sum_{k\ge 1} \dot\kappa_k \int_0^\infty (1 - \mathsf{x}_0(u))\, u \sin(\kappa_k u)\, e^{-zu}\, du = \tfrac{1}{2}\bigl\langle \mathsf{J}, (1 - \mathsf{x}_0)\, e^{-z|u|} \bigr\rangle. \tag{15.10}$$

Approximating by test functions with compact support in $(0,\infty)$ and using (15.9), $\mathsf{H}_1(z) = \sum_{n\ge 2} \dot{\mathsf{a}}_n n^{-z}$ for $\Re z > \frac{1}{2}$, where the series converges absolutely.

Step 2. For $\Re z > 0$, $\int_0^\infty u \sin(\kappa u) e^{-zu} du = 2\kappa z/(z^2 + \kappa^2)^2$, so

$$\mathsf{H}_1(z) = \sum_{k\ge 1} \dot\kappa_k \frac{2\kappa_k z}{(z^2 + \kappa_k^2)^2} - \mathsf{y}(z), \qquad \mathsf{y}(z) := \sum_{k\ge 1} \dot\kappa_k \int_0^{\log 2} \mathsf{x}_0(u)\, u \sin(\kappa_k u)\, e^{-zu}\, du. \tag{15.11}$$

Choose the Taylor degree $A \ge \max(\operatorname{ord}_0 \mathsf{J}, 2)$ and write e^{-zu} as its Taylor polynomial of degree A at $u = 0$ plus a remainder vanishing to order A. The Fourier coefficients of each regularized term are $O(\kappa^{-3})$, uniformly on compact z-sets. The even extension of x_0 times the remainder is of class C^A and vanishes to order A at 0; for A larger than the order of J near 0 its pairing with J is 0. The terms of the Taylor polynomial contribute constants times powers of z. Hence y is a polynomial. As $z = \sigma \to +\infty$, $\mathsf{H}_1(\sigma) \to 0$ and the first sum in (15.11) is $O(\sigma^{-1} \log \sigma)$, so $\mathsf{y} \equiv 0$.

Step 3. Put $\mathsf{H}(z) := \sum_{n\ge 2} \frac{\dot{\mathsf{a}}_n}{\log n} n^{-z}$ for $\Re z > \frac12$, so that $\mathsf{H}' = -\mathsf{H}_1$. By Step 2 there is a real constant ν_8 with

$$\mathsf{H}(z) = \nu_8 + \sum_{k\ge 1} \dot\kappa_k \Big(\frac{\kappa_k}{z^2+\kappa_k^2} - \frac{1}{\kappa_k} \Big), \tag{15.12}$$

which continues H to a meromorphic function in $\mathbb{C}$, even in z, with at most simple poles at $z = \pm i\kappa_k$, of residue $\pm\frac{1}{2i}\sum_{k':\kappa_{k'}=\kappa_k} \dot\kappa_{k'}$.

Step 4. By Theorem 15.5, $\zeta_{\boldsymbol\kappa} = \zeta$, with zeros $\frac12 \pm i\kappa_k$ of the multiplicity of the list. Put $\mathsf{Z}(s) := \mathsf{H}(s-\frac12)\,\zeta(s)$. The poles of $\mathsf{H}(s-\frac12)$ are cancelled by zeros of ζ, and there are arbitrarily large radii r whose circles stay at distance at least $1/\log r$ from all $\pm i\kappa_k$. Indeed, bounded displacement from the clock gives $N_\kappa(2R+1) - N_\kappa(R-1) = (R/(2\pi) + o(R))\log R$; hence the union in $[R, 2R]$ of intervals of radius $1/\log R$ around these ordinates has length $(1/\pi + o(1))R < R$ for large R. On such circles, splitting the sum into dyadic distance shells in (15.12) gives $O(\log^2 r)$; so $(s-1)\mathsf{Z}(s)$ is entire of finite order: this is (H0). Z is the product of $\sum_n \dot{\mathsf{a}}_n \sqrt{n}(\log n)^{-1} n^{-s}$ and $\zeta(s)$, both absolutely convergent in $\Re s > 1$: this is (H1). Since H is even, $\mathsf{Z}(1-s) = \mathsf{H}(s-\frac12)\zeta(1-s)$, so Z satisfies the functional equation of ζ: this is (H2) and (H3) with the second function equal to Z. Hamburger's theorem gives $\mathsf{Z} = C_H\zeta$, that is, $\mathsf{H} \equiv C_H$. Since $\mathsf{H}(\sigma) \to 0$ as $\sigma \to +\infty$, $\mathsf{H} \equiv 0$: every $\dot{\mathsf{a}}_n$ and every residue vanishes. □

Remark 15.7 (Scope of Theorem 15.6). The theorem is conditional on the existence of a solution; it shows that a solution is isolated to first order, the Jacobian of (15.5) being injective there on the directions considered, except for the splitting of multiple ordinates. This is a necessary endpoint nondegeneracy condition for any continuation scheme; it is not, by itself, an implicit-function theorem, because no isomorphism between fixed Banach spaces is proved. Steps 1–3 do not use that $\boldsymbol\kappa$ is a solution: at any list of $\mathscr{O}$ the kernel of the linearization consists of the Dirichlet series H that are even meromorphic functions with simple poles at $\pm i\kappa_k$ of the form (15.12). Such series exist for lists in arithmetic progression, for instance $(2^z + 2^{-z} - 2\cos\nu_9)^{-1}$, which is even, a power series in 2^{-z} for $\Re z > 0$, with simple poles at $z = \pm i(\nu_9 + 2\pi m)/\log 2$, $m \in \mathbb{Z}$; these lists do not belong to $\mathscr{O}$. For perturbations $\dot{\mathsf{a}}_n$ of polynomial growth the same proof applies with Hamburger's theorem for a finite abscissa of absolute convergence [16]. The injectivity of the Jacobian at the clock and along a path, and with it the existence of a solution, are not proved.

15.3. **The gradient of the vertical defect.** For Dirichlet series $A = \sum_n a_n n^{-s}$, $B = \sum_n b_n n^{-s}$ and $\sigma > 1$ put $\langle A, B\rangle_\sigma := \sum_{n\ge1} a_n \bar b_n\, n^{-2\sigma}$, and for $\boldsymbol\kappa \in \mathscr{O}$ (Section 15)

$$\mathsf{K}_{\boldsymbol\kappa} := \frac{\tilde\zeta_{\mathbb{Z},\boldsymbol\kappa}}{\zeta} = \sum_{n\ge1} k_n(\boldsymbol\kappa)\, n^{-s}, \qquad \mathsf{V}_\sigma(\boldsymbol\kappa) = \langle \mathsf{K}_{\boldsymbol\kappa}, \mathsf{K}_{\boldsymbol\kappa}\rangle_\sigma - 1, \qquad \mathsf{r}_m(\boldsymbol\kappa) := \Re\langle \mathsf{K}_{\boldsymbol\kappa}\mathsf{L}_m, \mathsf{K}_{\boldsymbol\kappa}\rangle_\sigma, \tag{15.13}$$

with $\mathsf{L}_m(s) = -\log(1-m^{-s})$ as in Proposition 14.39. By Proposition 14.39 the derivative $\partial\tilde c_m(\boldsymbol\kappa)/\partial\kappa_k$ is a function of m and of the value κ_k alone; denote it by $\mathsf{u}_m(\kappa_k)$, so that $\mathsf{u}_m = 2\,\Im V_m$ when m is not a power of 2 (15.4).

Proposition 15.8 (Critical points of the vertical defect)**.** *Let $\sigma > 1$ and $\boldsymbol\kappa \in \mathscr{O}$ (Section 15). Then $|\mathsf{r}_m(\boldsymbol\kappa)| \le 2m^{-\sigma}(1 + \mathsf{V}_\sigma(\boldsymbol\kappa))$, $|\mathsf{u}_m(t)| \le Cm^{-1/2}$ off the powers of 2 and $|\mathsf{u}_m(t)| \le C$ on them, and*

$$\frac{\partial \mathsf{V}_\sigma}{\partial\kappa_k}(\boldsymbol\kappa) = 2\,\mathsf{Q}_{\boldsymbol\kappa}(\kappa_k), \qquad \mathsf{Q}_{\boldsymbol\kappa}(t) := \sum_{m\ge2} \mathsf{r}_m(\boldsymbol\kappa)\,\mathsf{u}_m(t), \tag{15.14}$$

the series converging absolutely and uniformly in t. If $\mathsf{V}_\sigma(\boldsymbol\kappa) = 0$, then $\mathsf{r}_m(\boldsymbol\kappa) = 0$ for every m and $\mathsf{Q}_{\boldsymbol\kappa} \equiv 0$.

Proof. Since $\log \mathsf{K}_{\boldsymbol\kappa} = \sum_m (\tilde c_m(\boldsymbol\kappa) - \mathbf{1}_{\mathrm{prime}}(m))\mathsf{L}_m$, $\partial\mathsf{K}_{\boldsymbol\kappa}/\partial\tilde c_m = \mathsf{K}_{\boldsymbol\kappa}\mathsf{L}_m$ and $\partial\mathsf{V}_\sigma/\partial\tilde c_m = 2\mathsf{r}_m$. Multiplication by m^{-rs} multiplies the norm $\langle\cdot,\cdot\rangle_\sigma^{1/2}$ by $m^{-r\sigma}$, so $\langle \mathsf{K}\mathsf{L}_m, \mathsf{K}\mathsf{L}_m\rangle_\sigma^{1/2} \le \langle \mathsf{K},\mathsf{K}\rangle_\sigma^{1/2} \sum_{r\ge1} m^{-r\sigma}/r \le$

$2m^{-\sigma}\langle \mathsf{K},\mathsf{K}\rangle_\sigma^{1/2}$, and the bound for r_m follows by the Cauchy–Schwarz inequality. In (13.19) the support of $\varphi_m(y^j)$ has length at most $\frac{2}{j}m^{1/j-1}$ and the integrand is at most $m^{-1/(2j)}$ in absolute value, so $|V_m(t)|\le\sum_j \frac{2}{j^2}m^{-1+1/(2j)}\le Cm^{-1/2}$; for $m=2^j$ the cut term is bounded by the total variation of $\varphi_m(y^j)$ times a bound for $\partial_\kappa J_{\boldsymbol\kappa}$ on $(2,3)$. The chain rule is justified first for finite Euler products and then by passage to the limit in the weighted ℓ^1 space of cell masses: the bound below is uniform in the truncation and gives uniform convergence of $\sum_n n^{-2\sigma}\partial_{\kappa_k}|k_n|^2$. It gives (15.14), with $\sum_m|\mathsf{r}_m||\mathsf{u}_m|\le C\sum_m m^{-\sigma-1/2}+C\sum_j 2^{-j\sigma}<\infty$. If $\mathsf{V}_\sigma=0$, then $\mathsf{K}=1$ and $\mathsf{r}_m=\Re\langle\mathsf{L}_m,1\rangle_\sigma=0$, because L_m has no constant term. □

A list $\boldsymbol\kappa\in\mathscr{O}$ (Section 15) at which all coordinate derivatives of V_σ vanish is therefore a list all of whose ordinates are zeros of the single real-analytic function $\mathsf{Q}_{\boldsymbol\kappa}$. The minimum of V_σ on $\mathscr{O}$ is attained (Corollary 15.3); if every minimizer satisfied the appropriate coordinate/KKT stationarity conditions and those conditions forced $\mathsf{V}_\sigma=0$, the minimum would be 0 and (D) would follow. This is not proved. A count of zeros does not exclude critical points with $\mathsf{V}_\sigma>0$: $\mathsf{Q}_{\boldsymbol\kappa}(t)$ is a sum of terms of frequency $\log m$ with $m\gtrsim t/2\pi$, so it oscillates at the scale $1/\log t$, which is the scale of the gaps between the ordinates.

16. Acceptance criteria and models in which the final step is a theorem

In the construction of this paper the continuous function ζ_λ is built from a list of ordinates and the discrete function $\tilde\zeta_{\mathbb{Z},\lambda}$ from the cells, and the two are joined by $\tilde h_\lambda$. This section states, without ζ, the general criterion under which a list is accepted by both sides, and gives three models: a non-periodic list accepted by a discrete category with two generators, where the acceptance is proved by one unitary matrix; a finite Euler product over the primes, to which the constructions of Sections 2–15 apply verbatim and for which the final step is a theorem; and the diagonal operator over all primes, which shows where the argument of the finite models stops. The complex variable of the first two subsections is denoted by z, and the half-plane $\Im\mathsf{z}>0$ plays the role of $\Re s>1$.

16.1. **The acceptance criterion: window and frequencies.** Let $\mathfrak{k}>0$ and let $\boldsymbol\kappa=(\kappa_k)_{k\ge1}$ be a nondecreasing list of positive numbers with $|\kappa_k-k\pi/\mathfrak{k}|$ bounded. The *continuous side* is the canonical product

$$\mathfrak{C}_{\boldsymbol\kappa}(\mathsf{z}):=\mathsf{z}\prod_{k\ge1}\Big(1-\frac{\mathsf{z}^2}{\kappa_k^2}\Big),\tag{16.1}$$

an odd entire function of order one with real zeros; the oddness plays the role of the functional equation. We assume that $\mathfrak{C}_{\boldsymbol\kappa}$ has exponential type $\mathfrak{k}$ and polynomial growth on the real axis, as $\sin(\mathfrak{k}\mathsf{z})$ has for the list $\kappa_k=k\pi/\mathfrak{k}$. The *discrete side* is given by a closed discrete set $\mathscr{F}\subset(0,\infty)$ of frequencies that is closed under addition; put $\mathscr{F}^+:=\mathscr{F}\cup\{0\}$. The discrete category consists of the generalized Dirichlet series $\sum_{\mathfrak{l}\in\mathscr{F}^+}\mathfrak{c}_{\mathfrak{l}}\,e^{i\mathfrak{l}\mathsf{z}}$, absolutely convergent in some half-plane $\Im\mathsf{z}>\nu_{10}$ (the number ν_{10} is not prescribed and may be enlarged), and of their Euler forms $\exp(\sum_{\mathfrak{l}\in\mathscr{F}}\mathfrak{b}_{\mathfrak{l}}\,e^{i\mathfrak{l}\mathsf{z}})$. For $\Im\mathsf{z}>0$,

$$\frac{\mathfrak{C}'_{\boldsymbol\kappa}}{\mathfrak{C}_{\boldsymbol\kappa}}(\mathsf{z})=\frac{1}{\mathsf{z}}+\sum_{k\ge1}\frac{2\mathsf{z}}{\mathsf{z}^2-\kappa_k^2}=-i\int_0^\infty e^{i\mathsf{z}u}\Big(1+2\sum_{k\ge1}\cos(\kappa_k u)\Big)du,\tag{16.2}$$

the bracket being a tempered distribution in u: it is the spectral density of the list.

Theorem 16.1 (Acceptance criterion)**.** *For a list $\boldsymbol\kappa$ as above the following are equivalent:*

(i) *the spectral density is atomic on the frequencies: there are real $\mathfrak{w}_{\mathfrak{l}}$ with $\sum_{\mathfrak{l}\in\mathscr{F}}|\mathfrak{w}_{\mathfrak{l}}|e^{-\nu_{10}\mathfrak{l}}<\infty$ for some admissible ν_{10} and*

$$1+2\sum_{k\ge1}\cos(\kappa_k u)=\sum_{\mathfrak{l}\in\mathscr{F}}\mathfrak{w}_{\mathfrak{l}}\,\delta(u-\mathfrak{l})\qquad\text{in }\mathcal{D}'(0,\infty);\tag{16.3}$$

(ii) *the continuous function has an Euler form: for $\Im\mathsf{z} > \nu_{10}$, with a constant $\mathfrak{c}_0 \neq 0$,*

$$e^{i\mathfrak{k}\mathsf{z}}\,\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z}) = \mathfrak{c}_0 \exp\Big(\sum_{\mathfrak{l}\in\mathscr{F}} \mathfrak{b}_\mathfrak{l}\, e^{i\mathfrak{l}\mathsf{z}}\Big), \qquad \mathfrak{b}_\mathfrak{l} = -\frac{\mathfrak{w}_\mathfrak{l}}{\mathfrak{l}}; \tag{16.4}$$

(iii) *the continuous function is a finite sum in the discrete category:*

$$e^{i\mathfrak{k}\mathsf{z}}\,\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z}) = \sum_{\mathfrak{l}\in\mathscr{F}^+\cap[0,2\mathfrak{k}]} \mathfrak{c}_\mathfrak{l}\, e^{i\mathfrak{l}\mathsf{z}}, \qquad \mathfrak{c}_0 \neq 0. \tag{16.5}$$

Proof. (i)⇔(ii): by (16.2), (i) is the identity $\mathfrak{C}'_{\boldsymbol{\kappa}}/\mathfrak{C}_{\boldsymbol{\kappa}} = -i\sum_\mathfrak{l} \mathfrak{w}_\mathfrak{l} e^{i\mathfrak{l}\mathsf{z}}$ in $\Im\mathsf{z} > \nu_{10}$ up to a polynomial in z that accounts for the point $u = 0$; since $\mathfrak{C}'_{\boldsymbol{\kappa}}/\mathfrak{C}_{\boldsymbol{\kappa}}(iy) = -i\big(\frac{1}{y} + \sum_k \frac{2y}{y^2+\kappa_k^2}\big) \to -i\mathfrak{k}$ as $y \to +\infty$, the polynomial is the constant $-i\mathfrak{k}$. Integrating, $\log \mathfrak{C}_{\boldsymbol{\kappa}} = \text{const} - i\mathfrak{k}\mathsf{z} - \sum_\mathfrak{l} \frac{\mathfrak{w}_\mathfrak{l}}{\mathfrak{l}} e^{i\mathfrak{l}\mathsf{z}}$, and conversely. (ii)⇒(iii): expanding the exponential, $\mathsf{N}(\mathsf{z}) := e^{i\mathfrak{k}\mathsf{z}}\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z}) = \sum_{\mathfrak{l}\in\mathscr{F}^+} \mathfrak{c}_\mathfrak{l} e^{i\mathfrak{l}\mathsf{z}}$, absolutely convergent in $\Im\mathsf{z} > \nu_{10}$. On the other hand N is entire of exponential type with polynomial growth on the real axis and is bounded by a polynomial in $\Im\mathsf{z} \geq 0$; by the Paley–Wiener–Schwartz theorem the Fourier transform of its restriction to a horizontal line is supported in $[0, 2\mathfrak{k}]$. On the line $\Im\mathsf{z} = \nu_{10} + 1$ this transform is $\sum_\mathfrak{l} \mathfrak{c}_\mathfrak{l} e^{-\mathfrak{l}(\nu_{10}+1)}\delta_\mathfrak{l}$, so $\mathfrak{c}_\mathfrak{l} = 0$ for $\mathfrak{l} > 2\mathfrak{k}$, and the sum is finite because $\mathscr{F}^+$ is discrete. (iii)⇒(ii): the logarithm of a finite sum with $\mathfrak{c}_0 \neq 0$ is, for sufficiently large $\Im\mathsf{z}$, an absolutely convergent series of exponentials whose frequencies are sums of elements of $\mathscr{F}$, hence elements of $\mathscr{F}$; this proves (ii) in some half-plane, possibly after increasing ν_{10}. □

In model identities of the form $2 + 2\sum_k \cos(\kappa_k u)$, the leading 2 records a double zero at the origin; the normalized criterion (16.3) starts with 1 because $\mathfrak{C}_{\boldsymbol{\kappa}}$ has a simple zero there.

The continuous side contributes a *window*: by its type and its symmetry, the frequencies of $e^{i\mathfrak{k}\mathsf{z}}\mathfrak{C}_{\boldsymbol{\kappa}}$ lie in $[0, 2\mathfrak{k}]$. The discrete side contributes the *admissible frequencies* $\mathscr{F}^+$. A list is accepted by both sides if and only if its function lives in the intersection, and it is then determined by the finitely many coefficients $\mathfrak{c}_\mathfrak{l}$, $\mathfrak{l} \in \mathscr{F}^+ \cap [0, \mathfrak{k})$, the others being fixed by the oddness. If $\mathscr{F} = 2\mathfrak{k}\mathbb{N}$, the intersection is $\{0, 2\mathfrak{k}\}$, the only function is a multiple of $\sin(\mathfrak{k}\mathsf{z})$, and the only accepted list is the clock $\kappa_k = k\pi/\mathfrak{k}$. Hamburger's theorem is the statement that for the window fixed by the factor $\Gamma(\frac{s}{2})\pi^{-s/2}$ and the frequencies $\log n$ the space of accepted functions has dimension one.

16.2. **A non-periodic accepted list: one unitary matrix.** Let $\mathfrak{p}, \mathfrak{q} > 0$ with $\mathfrak{p}/\mathfrak{q} \notin \mathbb{Q}$, $0 < \mathfrak{v} < 1$, $\mathfrak{k} = \mathfrak{p} + \mathfrak{q}$, and

$$\mathsf{U} := \begin{pmatrix} \mathfrak{v} & \sqrt{1-\mathfrak{v}^2} \\ \sqrt{1-\mathfrak{v}^2} & -\mathfrak{v} \end{pmatrix}, \qquad \mathsf{X}(\mathsf{z}) := \begin{pmatrix} e^{2i\mathfrak{p}\mathsf{z}} & 0 \\ 0 & e^{2i\mathfrak{q}\mathsf{z}} \end{pmatrix}, \tag{16.6}$$

$$\mathfrak{D}(\mathsf{z}) := \det(I - \mathsf{U}\mathsf{X}(\mathsf{z})) = 1 - \mathfrak{v}\, e^{2i\mathfrak{p}\mathsf{z}} + \mathfrak{v}\, e^{2i\mathfrak{q}\mathsf{z}} - e^{2i(\mathfrak{p}+\mathfrak{q})\mathsf{z}} = -2i\, e^{i(\mathfrak{p}+\mathfrak{q})\mathsf{z}}\, \mathfrak{S}(\mathsf{z}), \tag{16.7}$$

$$\mathfrak{S}(\mathsf{z}) := \sin\big((\mathfrak{p}+\mathfrak{q})\mathsf{z}\big) + \mathfrak{v}\sin\big((\mathfrak{p}-\mathfrak{q})\mathsf{z}\big). \tag{16.8}$$

Proposition 16.2 (The two-edge model)**.** (i) *All zeros of $\mathfrak{S}$ are real. Each interval between consecutive points $(m + \frac{1}{2})\pi/\mathfrak{k}$, $m \in \mathbb{Z}$, contains a zero, and the number of zeros in $[0, T]$, with multiplicity, is $\mathfrak{k}T/\pi + O(1)$; hence the positive zeros form a list $\boldsymbol{\kappa}$ with $|\kappa_k - k\pi/\mathfrak{k}|$ bounded. The set of zeros is not periodic.*

(ii) *$\mathfrak{S}(-\mathsf{z}) = -\mathfrak{S}(\mathsf{z})$, and $\mathfrak{D}(\mathsf{z}) = -e^{2i\mathfrak{k}\mathsf{z}}\,\overline{\mathfrak{D}(\bar{\mathsf{z}})}$.*

(iii) *For $\Im\mathsf{z} > 0$,*

$$\log\mathfrak{D}(\mathsf{z}) = -\sum_{r\geq 1} \frac{1}{r}\,\mathrm{tr}\big(\mathsf{U}\mathsf{X}(\mathsf{z})\big)^r = \sum_{\mathfrak{l}\in\mathscr{F}} \mathfrak{b}_\mathfrak{l}\, e^{i\mathfrak{l}\mathsf{z}}, \qquad \mathscr{F} = \{2(m\mathfrak{p} + m'\mathfrak{q}) : m, m' \geq 0\} \setminus \{0\}, \tag{16.9}$$

absolutely convergent, and the spectral density of the zeros of $\mathfrak{S}$, counted with multiplicity, is atomic on $\mathscr{F}$ in the sense of (16.3).

(iv) *$\mathfrak{D}(\mathsf{z}) = -2i\,\mathfrak{S}'(0)\, e^{i\mathfrak{k}\mathsf{z}}\,\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z})$ for the list $\boldsymbol{\kappa}$ of all positive zeros of $\mathfrak{S}$: the continuous and the discrete functions coincide up to the factor $e^{i\mathfrak{k}\mathsf{z}}$, and the list is accepted.*

Proof. (i) For $\Im \mathsf{z} > 0$ the matrix $\mathsf{UX}(\mathsf{z})$ has norm at most $\max(e^{-2\mathfrak{p}\Im \mathsf{z}}, e^{-2\mathfrak{q}\Im \mathsf{z}}) < 1$, because U is unitary; so $I - \mathsf{UX}(\mathsf{z})$ is invertible and $\mathfrak{D}(\mathsf{z}) \neq 0$. $\mathfrak{S}$ is odd, so it has no zeros in $\Im \mathsf{z} < 0$ either. At $\mathsf{z}_m = (m + \frac{1}{2})\pi/\mathfrak{k}$, $\sin(\mathfrak{k}\mathsf{z}_m) = (-1)^m$ and $|\mathfrak{v} \sin((\mathfrak{p} - \mathfrak{q})\mathsf{z}_m)| < 1$, so $\mathfrak{S}(\mathsf{z}_m)$ has the sign $(-1)^m$ and there is a zero in $(\mathsf{z}_{m-1}, \mathsf{z}_m)$. For real t the eigenvalues of the unitary matrix $\mathsf{UX}(t)$ move counterclockwise as t increases, because $\frac{d}{dt}\mathsf{X}(t) = 2i \operatorname{diag}(\mathfrak{p}, \mathfrak{q})\mathsf{X}(t)$ with $\mathfrak{p}, \mathfrak{q} > 0$, and $\det \mathsf{UX}(t) = -e^{2i\mathfrak{k}t}$; the real zeros of $\mathfrak{D}$, with multiplicity, are the passages of an eigenvalue through 1, so their number in $[0, T]$ is $\mathfrak{k}T/\pi + O(1)$. If the zero set had a period $y \neq 0$, then $\mathfrak{S}(\mathsf{z} + y)$ and $\mathfrak{S}(\mathsf{z})$, both of type $\mathfrak{k}$ and bounded on the real axis with the same zeros, would be proportional; comparing the coefficients of $e^{\pm i(\mathfrak{p}+\mathfrak{q})\mathsf{z}}$ and $e^{\pm i(\mathfrak{p}-\mathfrak{q})\mathsf{z}}$ gives $2\mathfrak{p}y \in \pi\mathbb{Z}$ and $2\mathfrak{q}y \in \pi\mathbb{Z}$, against $\mathfrak{p}/\mathfrak{q} \notin \mathbb{Q}$. (ii) is immediate; the second identity follows from $\det \mathsf{U} = -1$. (iii) The series is the logarithm of the determinant, convergent because the norm is less than 1; each trace is a combination of exponentials with frequencies in $\mathscr{F}$. The statement on the density is Theorem 16.1, (ii)⇒(i), applied to (iv). (iv) $\mathfrak{S}$ is an odd entire function of exponential type $\mathfrak{k}$, bounded on the real axis, with real zeros; by Hadamard's factorization and the oddness, $\mathfrak{S}(\mathsf{z}) = \mathfrak{S}'(0)\, \mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z})$. □

In this model nothing has to be joined: the continuous function and the discrete function are the same determinant. The Euler form comes from the expansion of $\det(I - \mathsf{UX})$ in closed paths, and the real zeros and the symmetry come from the unitarity of U, through the contraction in $\Im \mathsf{z} > 0$. The acceptance criterion at this level is the existence of the common object U. In finite rank this is general: the zeros of $\mathsf{P}(e^{i\mathfrak{l}_1\mathsf{z}}, \dots, e^{i\mathfrak{l}_r\mathsf{z}})$, for a polynomial P without zeros in the product of the open unit discs and in the product of their exteriors, are real and their spectral density is atomic [37], and conversely the lists with unit masses whose spectral density is atomic with locally finite support are zero sets of exponential polynomials with real zeros [42].

16.3. **A finite Euler product: the constructions of this paper with the final step proved.** Let

$$\mathsf{F}(s) := (1 - 2^{-s})(1 - 3^{-s}) = 1 - 2^{-s} - 3^{-s} + 6^{-s} = \sum_{n|6} \mu(n)\, n^{-s}. \tag{16.10}$$

It is a Dirichlet series over the integers with an Euler product over the primes 2 and 3, and $\mathsf{F}(-s) = 6^s\, \mathsf{F}(s)$, with axis of symmetry $\Re s = 0$. Its zeros are $s = 0$, double, and $s = \pm i\mathfrak{y}_k$, where

$$\{\mathfrak{y}_k\}_{k\geq 1} = \Big\{\frac{2\pi m}{\log 2}\Big\}_{m\geq 1} \cup \Big\{\frac{2\pi m}{\log 3}\Big\}_{m\geq 1} \quad \text{in increasing order,} \tag{16.11}$$

all simple, a non-periodic list, and, with $2\sinh(\frac{s\log p}{2}) = s\log p \prod_{m\geq 1}(1 + \frac{s^2\log^2 p}{4\pi^2 m^2})$,

$$\mathsf{F}(s) = 6^{-s/2}\,(\log 2 \log 3)\, s^2 \prod_{k\geq 1}\Big(1 + \frac{s^2}{\mathfrak{y}_k^2}\Big). \tag{16.12}$$

The constructions of this paper apply as follows. The count of the ordinates is $\lfloor \frac{T\log 2}{2\pi} \rfloor + \lfloor \frac{T\log 3}{2\pi} \rfloor = \frac{T\log 6}{2\pi} - 1 + O(1)$ with a remainder of absolute value at most 1; the midpoint clock is $\frac{\mathfrak{x}_k \log 6}{2\pi} - 1 = k - \frac{1}{2}$, that is, $\mathfrak{x}_k = (2k+1)\pi/\log 6$, and $\mathfrak{x}_k - \mathfrak{y}_k$ is bounded. For a list $\boldsymbol{\kappa}$ at bounded distance from the clock,

$$\mathsf{F}_{\boldsymbol{\kappa}}(s) := 6^{-s/2}\,(\log 2 \log 3)\, s^2 \prod_{k\geq 1}\Big(1 + \frac{s^2}{\kappa_k^2}\Big), \qquad \mathsf{F}_{\boldsymbol{\kappa}}(-s) = 6^s\, \mathsf{F}_{\boldsymbol{\kappa}}(s), \tag{16.13}$$

and the family is $\mathsf{F}_\lambda := \mathsf{F}_{\boldsymbol{\kappa}(\lambda)}$ with $\kappa_k(\lambda) = (1-\lambda)\mathfrak{y}_k + \lambda\mathfrak{x}_k$, so that $\mathsf{F}_0 = \mathsf{F}$ and, since $\prod_{k\geq 0}(1 + s^2/\mathfrak{x}_k^2) = \cosh(\frac{s\log 6}{2})$,

$$\mathsf{F}_1(s) = \frac{\log 2 \log 3}{2} \cdot \frac{s^2\,(1 + 6^{-s})}{1 + s^2 \log^2 6/\pi^2}. \tag{16.14}$$

Theorem 16.3 (The final step in the model)**.** (i) (Poisson, prime by prime.) *In* $\mathcal{D}'(\mathbb{R})$*,*

$$2+2\sum_{k\ge1}\cos(\mathfrak{y}_k u)=\sum_{p\in\{2,3\}}\sum_{m\in\mathbb{Z}}\cos\frac{2\pi m u}{\log p}=\sum_{p\in\{2,3\}}\log p\sum_{m\in\mathbb{Z}}\delta(u-m\log p). \tag{16.15}$$

The spectral density of the list $(\mathfrak{y}_k)$ *is atomic on the points* $m\log p$*: the continuous function built from the list is the Dirichlet series with Euler product,* $\mathsf{F}_0=\mathsf{F}$*.*

(ii) (Hamburger's theorem of the model.) *Let* f *be entire of finite order with* $f(-s)=6^s f(s)$ *and* $f(0)=0$*, and let* $f(s)=\sum_{n\ge1}a_n n^{-s}$ *converge absolutely in some half-plane* $\Re s>\nu_{12}$*. Then* $f=a_1\mathsf{F}$*.*

(iii) (The family.) F_λ *is an absolutely convergent Dirichlet series in a half-plane if and only if* $\lambda=0$*. In particular the clock function* F_1 *is not a Dirichlet series.*

Proof. (i) is the Poisson summation formula $\sum_{m\in\mathbb{Z}}e^{2\pi i m u/\log p}=\log p\sum_{m\in\mathbb{Z}}\delta(u-m\log p)$ for $p=2$ and $p=3$; the second statement is (16.12). (ii) f is bounded in $\Re s\ge\nu_{12}+1$, and $|f(s)|=6^{-\sigma}|f(-s)|\le C\,6^{-\sigma}$ in $\Re s\le-\nu_{12}-1$; by the Phragmén–Lindelöf theorem f is bounded in every vertical strip. For every real σ, moving the line of integration,

$$a_n=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}f(\sigma+it)\,n^{\sigma+it}\,dt,\qquad |a_n|\le C\,(6/n)^{-\sigma}\quad(\sigma\le-\nu_{12}-1), \tag{16.16}$$

and $\sigma\to-\infty$ gives $a_n=0$ for $n>6$. The functional equation reads $\sum_{n\le6}a_n n^s=\sum_{n\le6}a_n(6/n)^s$; by the uniqueness of the exponents, $a_n=0$ unless $n\mid6$, $a_6=a_1$ and $a_3=a_2$. Finally $f(0)=2a_1+2a_2=0$. (iii) F_λ is entire of order one, satisfies the functional equation and vanishes at 0. If it is a Dirichlet series, (ii) gives $\mathsf{F}_\lambda=a_1\mathsf{F}$, the lists of zeros coincide, and $\lambda(\mathfrak{x}_k-\mathfrak{y}_k)=0$ for every k, so $\lambda=0$. □

In the model every object of the programme exists and the analogue of (D) is proved. What makes $\lambda=0$ close is a property of the list that the clock and the lists with $\lambda\neq0$ do not have,

$$\{\mathfrak{y}_k\}=\frac{2\pi}{\log2}\,\mathbb{N}\;\cup\;\frac{2\pi}{\log3}\,\mathbb{N}, \tag{16.17}$$

a superposition of one clock for each prime; it passes the scale test of Remark 14.20 and it is the hypothesis of the Poisson formula (16.15). The same statements hold for $\prod_{p\le X}(1-p^{-s})=\sum_{n\mid X\#}\mu(n)n^{-s}$, with $X\#$ the product of the primes up to X, the functional equation with the factor $(-1)^{\pi(X)}(X\#)^s$, a zero of order $\pi(X)$ at the origin and the list $\bigcup_{p\le X}\frac{2\pi}{\log p}\mathbb{N}$. As $X\to\infty$ these functions tend to $1/\zeta$ in $\Re s>1$ only; their axis of symmetry is $\Re s=0$, not $\Re s=\frac12$, and their lists do not converge to the ordinates of ζ, which are not a superposition of one clock for each prime.

16.4. **The diagonal operator over all primes and the three abscissas.** On $\mathscr{L}:=\ell^2(\{p\text{ prime}\})$ let $\mathsf{X}(s):=\operatorname{diag}(p^{-s})$. The closed paths of this model are the primes themselves, without scattering between them, and

$$\det\bigl(I-\mathsf{X}(s)\bigr)=\prod_p(1-p^{-s})=\frac{1}{\zeta(s)}\qquad(\Re s>1). \tag{16.18}$$

$\mathsf{X}(s)$ has norm $2^{-\sigma}<1$ for $\sigma>0$, it is of trace class if and only if $\sigma>1$, and Hilbert–Schmidt if and only if $\sigma>\frac12$. With $\mathfrak{P}(s):=\sum_p p^{-s}$ initially for $\Re s>1$, the regularized determinant is

$$\det{}_2\bigl(I-\mathsf{X}(s)\bigr)=\prod_p(1-p^{-s})e^{p^{-s}}=\frac{e^{\mathfrak{P}(s)}}{\zeta(s)}=\exp\Bigl(-\sum_{m\ge2}\frac{1}{m}\,\mathfrak{P}(ms)\Bigr)\qquad(\Re s>\tfrac12), \tag{16.19}$$

holomorphic and without zeros in $\Re s>\frac12$ because $I-\mathsf{X}(s)$ is invertible there.

Proposition 16.4 (No common object of finite rank in the window of ζ)**.** *Let* $X\ge2$*. There is no function* $f\not\equiv0$ *that satisfies (H0) and the functional equation* (11.3) *and is an absolutely convergent Dirichlet series in* $\Re s>1$ *supported on the integers all of whose prime factors are at most* X*.*

Proof. By Hamburger's theorem $f = C_H\zeta$, and the coefficients of ζ do not vanish at the integers with a prime factor larger than X; so $C_H = 0$. □

In the models of Subsections 16.2 and 16.3 three boundaries coincide: the boundary of the domain where the matrix is a strict contraction, the boundary of the domain of convergence of the Euler product, and the axis of symmetry. The real zeros come from this coincidence. For (16.18) they are different:

$$\text{contraction: } \sigma > 0; \qquad \text{convergence of the determinant: } \sigma > 1; \qquad \text{axis of symmetry: } \sigma = \tfrac{1}{2}. \tag{16.20}$$

The contraction proves that ζ has no zeros in $\sigma > 1$ and that $\zeta\, e^{-\mathfrak{P}}$ is holomorphic without zeros in $\sigma > \frac{1}{2}$; neither statement reaches the axis, and in the strip $\frac{1}{2} < \sigma \le 1$ the regularized determinant moves the question to the continuation of $\mathfrak{P}$. In the language of this paper, $\log \tilde{\zeta}_{\mathbb{Z},\lambda} = \sum_n \tilde{c}_n(\lambda)\mathsf{L}_n$ is the determinant (16.18) with real multiplicities $\tilde{c}_n(\lambda)$ at all integers, and the strip $\frac{1}{2} < \sigma \le 1$ is where the mean square of the cell masses (Theorem 13.10) and the continuation of the joining factor (Theorem 13.12) are used. By Proposition 16.4 the common object of the window of ζ has no finite-rank approximants in the discrete category; a unitary object of infinite rank from which both ζ_{Z1} and $\tilde{\zeta}_{\mathbb{Z},0}$ descend, as U does in Proposition 16.2, is not constructed in this paper. A model without ζ with the same separation of the three boundaries is a graph with infinitely many edges of lengths $\mathfrak{l}_j$ such that $\sum_j e^{-\mathfrak{l}_j y}$ diverges for $y \le y_0$, with $y_0 > 0$.

17. Route B for finite sections: compactness, prescribed masses, certificates, and a Kadec–Avdonin criterion with blocks of variable length

This section incorporates, with the notation of this paper, the companion note *Route B for the midpoint RvM system* (version 4; the file `supplementary/lemas_B_v4/` of the repository). Its subsections are self-contained and use the preceding sections only through the results quoted by label. The numerical part (Section 17.7) is a computation and is not part of any proof.

17.1. **Introduction to the section.** This paper constructs a synthetic zeta system on the midpoint clock τ_k, defined by $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, and reduces the remaining step to condition (D). Route B approaches (D) without constructing the zeta ordinates. One solves finite sections of the rung system (17.4), which imposes simultaneously a real critical spectrum and atomic arithmetic on $\{\log n\}$. If the residuals of these sections tend to zero in a fixed box, compactness yields an exact solution. The uniqueness theorem 15.5 then identifies that solution as the list of zeta ordinates with the von Mangoldt masses. In this scheme the ordinates γ_k are the output of an inverse problem, not an input. Section 17.3 makes this precise and isolates the two statements that remain open: existence of sections with vanishing residual, and uniqueness in the box without the third condition of $\mathscr{O}$. Section 17.3.1 shows that prescribing the von Mangoldt masses removes the second: the Laplace transform of the residual is identified (Lemma 17.11), and uniqueness follows from Hadamard's theorem and the functional equation, so that route B′ is equivalent to the Riemann hypothesis. Section 17.3.2 gives the existence part a certificate form: two computable quantities per stage, a right inverse of the Jacobian and the residual of the new tests, imply an exact solution of the stage. Section 17.3.3 shows that Gonek's symmetrised Euler products give explicit lists in a tube around the clock. Section 17.7 reports the computations that measure these quantities.

Section 17.4 proves an a priori bound for quasi-solutions. Section 17.5 proves a Kadec–Avdonin criterion with blocks of variable length, which is the tool needed for a Fredholm/stability formulation of the linearised problem. Section 17.6 applies that criterion to lists near the clock and proves that, for the zeta list, the uniform version fails.

References of the form Theorem 15.5 and the other results of the preceding sections are used as stated there. Except in Section 17.6.3, where the zeta list is the object of study, nothing below uses the ordinates of ζ.

17.2. **Setting.** Throughout, $q(s) := s(s-1)$ and $G(s) := s(s-1)\pi^{-s/2}\Gamma(s/2)$. The smooth clock count is

$$\mathcal{N}_{\mathrm{RVM}}(t) := \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}, \qquad \mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} \qquad (t>0). \tag{17.1}$$

It is strictly increasing on $[2\pi,\infty)$, with $\mathcal{N}_{\mathrm{RVM}}(2\pi) = -\frac{1}{8}$, and the clock points are the solutions $\tau_k > 2\pi$ of $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, $k \ge 1$ ($\tau_1 = 14.5213\ldots$). We write $n_{\mathrm{RVM}}(t) := \#\{k \ge 1 : \tau_k \le t\}$. Since $\mathcal{N}_{\mathrm{RVM}}$ is increasing,

$$n_{\mathrm{RVM}}(t) = \Big\lfloor \mathcal{N}_{\mathrm{RVM}}(t) + \tfrac{1}{2} \Big\rfloor, \qquad |n_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(t)| \le \tfrac{1}{2} \qquad (t \ge 2\pi). \tag{17.2}$$

Definition 17.1 (Admissible lists)**.** For $C > 0$, a list $\boldsymbol{\kappa} = (\kappa_k)_{k\ge1}$ is *C-admissible* if it is real and nondecreasing, $\kappa_1 \ge 2\pi$, and $|\kappa_k - \tau_k| \le C$ for all k. Put $N_{\boldsymbol{\kappa}}(t) := \#\{k : \kappa_k \le t\}$ (with multiplicity) and

$$P_{\boldsymbol{\kappa}}(s) := \prod_{k\ge1}\Big(1 + \frac{q(s)}{\frac{1}{4}+\kappa_k^2}\Big), \qquad \zeta_{\boldsymbol{\kappa}} := \frac{P_{\boldsymbol{\kappa}}}{G}. \tag{17.3}$$

The clock list $\boldsymbol{\tau} = (\tau_k)$ gives $\zeta_{\boldsymbol{\tau}} = \zeta_{\mathrm{RVM}}$.

$P_{\boldsymbol{\kappa}}$ is entire and real on $\mathbb{R}$, and $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$. Its zeros are exactly $\frac{1}{2} \pm i\kappa_k$, with multiplicity, and it is positive on $[-1,2]$, because $q([-1,2]) = [-\frac{1}{4}, 2]$ and $\kappa_k^2 \ge 4\pi^2$. Writing $s = \frac{1}{2} + w$, one has $\frac{1}{4} + \kappa^2 + q(s) = \kappa^2 + w^2$.

The *rung system* 15.5 for a list $\boldsymbol{\kappa}$ and real masses $\mathsf{a} = (\mathsf{a}_n)_{n\ge2}$ is

$$\sum_{k\ge1}\cos(\kappa_k u) + \sum_{n\ge2}\mathsf{a}_n\,\delta(u - \log n) = \mathsf{k}(u) \qquad \text{in } \mathcal{D}'(0,\infty), \tag{17.4}$$

with the fixed distribution k of 14.77. Its residual is the distribution $\mathfrak{R}(\boldsymbol{\kappa},\mathsf{a})$ defined, for $\varphi \in C_c^\infty(0,\infty)$, by

$$\langle \mathfrak{R}(\boldsymbol{\kappa},\mathsf{a}), \varphi\rangle := \sum_{k\ge1}\mathsf{c}_\varphi(\kappa_k) + \sum_{n\ge2}\mathsf{a}_n\varphi(\log n) - \langle \mathsf{k}, \varphi\rangle, \qquad \mathsf{c}_\varphi(t) := \int_0^\infty \varphi(u)\cos(tu)\,du. \tag{17.5}$$

Two integrations by parts give $|\mathsf{c}_\varphi(t)| \le \|\varphi''\|_{L^1}\,t^{-2}$. Since $\kappa_k \ge \tau_k - C$ and $\tau_k \gg k/\log(k+1)$, the series in (17.5) converges absolutely. The mass sum is finite, because $\operatorname{supp}\varphi \subset [u_1, u^*]$ with $0 < u_1 < u^*$ leaves only $n \le e^{u^*}$.

17.3. **Route B: compactness, selection and the canonical box.** For $C, \mathbb{M} > 0$ let $\mathbb{K}(C,\mathbb{M})$ be the set of pairs $(\boldsymbol{\kappa},\mathsf{a})$ with $\boldsymbol{\kappa}$ C-admissible and $|\mathsf{a}_n| \le \mathbb{M}$ for all n. A *finite section* is an element of $\mathbb{K}(C,\mathbb{M})$ with $\kappa_k = \tau_k$ for $k > k^*$ and $\mathsf{a}_n = 0$ for $n > n^*$, for some k^*, n^*.

Fix finite sets $\mathscr{T}_1 \subset \mathscr{T}_2 \subset \cdots \subset C_c^\infty(0,\infty)$ whose union $\mathscr{T}_\infty$ is dense in $C_c^\infty(0,\infty)$. Such sets exist because $C_c^\infty(0,\infty)$ is separable for its usual LF topology: it is the countable strict inductive limit of the separable Fréchet spaces $C^\infty_{[1/j,j]}$, and a countable union of countable dense subsets of these is dense. For $(\boldsymbol{\kappa},\mathsf{a}) \in \mathbb{K}(C,\mathbb{M})$ put

$$\mathsf{s}_m(\boldsymbol{\kappa},\mathsf{a}) := \max_{\varphi\in\mathscr{T}_m}\big|\langle \mathfrak{R}(\boldsymbol{\kappa},\mathsf{a}),\varphi\rangle\big|. \tag{17.6}$$

Definition 17.2 (Statement B)**.** *Statement $B(C,\mathbb{M})$* says that there are finite sections $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathbb{K}(C,\mathbb{M})$ with $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$.

Lemma 17.3 (Canonical box of the exact solution)**.** *Let $\mathbb{M}_* := \max_{n\ge2}\Lambda(n)/(2\sqrt{n})$. Then $\mathbb{M}_* = \log 7/(2\sqrt{7}) = 0.367742\ldots < 1/e$. If condition (D) holds, the pair $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$, with $\mathsf{a}^\zeta_n := \Lambda(n)/(2\sqrt{n})$, solves* (17.4) *and lies in $\mathbb{K}(C_\eta, \mathbb{M}_*)$, where C_η is the constant of 2.2.*

Proof. $x \mapsto \log x/(2\sqrt{x})$ increases on $[1, e^2]$ and decreases on $[e^2, \infty)$, with $e^2 = 7.389\ldots$. Hence among primes its maximum is at $p = 7$ or $p = 11$; the values are $0.367742\ldots$ and $0.361540\ldots$. A prime power p^j with $j \geq 2$ gives $\log p/(2p^{j/2}) < \log p/(2\sqrt{p}) \leq \mathbb{M}_*$. The bound by $1/e$ is the continuous maximum, attained at $x = e^2$. Under (D), $\zeta_{Z1} = \zeta$, so all zeros of ζ are $\frac{1}{2} \pm i\gamma_k$. The rung system with list $\boldsymbol{\gamma}$ and masses a^ζ is then the distributional explicit formula 12.6. Finally, $|\gamma_k - \tau_k| \leq C_\eta$ by 2.2, and $\gamma_1 = 14.13\ldots \geq 2\pi$. □

Corollary 17.4 ((D) implies B)**.** *If (D) holds, then $B(C, \mathbb{M})$ holds for every $C \geq C_\eta$ and $\mathbb{M} \geq \mathbb{M}_*$.*

Proof. Let $(\boldsymbol{\kappa}^{(k^*,n^*)}, \mathsf{a}^{(k^*,n^*)})$ be the truncation of $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$. Its list is the nondecreasing rearrangement of the multiset $\{\gamma_1, \ldots, \gamma_K\} \cup \{\tau_{K+1}, \tau_{K+2}, \ldots\}$, and its masses are a_n^ζ for $n \leq n^*$ and 0 beyond. By Lemma 17.18 (Section 17.3.2) the rearranged list satisfies $|\kappa_k - \tau_k| \leq C_\eta$, coincides with (τ_k) beyond some index, and $\mathfrak{R}$ depends only on the multiset. So this is a finite section in $\mathbb{K}(C_\eta, \mathbb{M}_*)$. For fixed φ, $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(k^*,n^*)}, \mathsf{a}^{(k^*,n^*)}), \varphi\rangle - \langle \mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta), \varphi\rangle = \sum_{k>k^*}(\mathsf{c}_\varphi(\tau_k) - \mathsf{c}_\varphi(\gamma_k))$ as soon as $e^{u^*} \leq n^*$. This tends to 0 as $k^* \to \infty$. Since $\mathscr{T}_m$ is finite and $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$ by Lemma 17.3, k_m^*, N_m can be chosen with $\mathsf{s}_m \leq 1/m$. □

Proposition 17.5 (Compactness and passage to the limit)**.** *Let $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathbb{K}(C, \mathbb{M})$ with $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$. Then a subsequence converges coordinatewise to some $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathbb{K}(C, \mathbb{M})$, and every such limit satisfies $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ in $\mathcal{D}'(0, \infty)$.*

Proof. Each $\kappa_k^{(m)}$ lies in the compact interval $[\tau_k - C, \tau_k + C]$ and each $\mathsf{a}_n^{(m)}$ in $[-\mathbb{M}, \mathbb{M}]$. A diagonal extraction over the countably many coordinates gives a coordinatewise convergent subsequence. The limit is nondecreasing, satisfies $\kappa_1 \geq 2\pi$, $|\kappa_k - \tau_k| \leq C$ and $|\mathsf{a}_n| \leq \mathbb{M}$, and therefore lies in $\mathbb{K}(C, \mathbb{M})$.

Continuity of each pairing. Fix φ with support in $[u_1, u^*]$. The mass part of (17.5) involves only $n \leq e^{u^*}$ and converges. For the spectral part, $|\mathsf{c}_\varphi(\kappa_k^{(m)})| \leq \|\varphi''\|_1(\tau_k - C)^{-2}$ when $\tau_k > 2C$, and $|\mathsf{c}_\varphi| \leq \|\varphi\|_1$ for the finitely many other k. The majorant is summable, so dominated convergence for series gives $\sum_k \mathsf{c}_\varphi(\kappa_k^{(m)}) \to \sum_k \mathsf{c}_\varphi(\kappa_k)$. Hence $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle \to \langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle$ along the subsequence.

Vanishing. If $\varphi \in \mathscr{T}_\infty$, then $\varphi \in \mathscr{T}_m$ for all large m, so $|\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle| \leq \mathsf{s}_m \to 0$ and $\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle = 0$. The limit $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a})$ is a single distribution, hence continuous on $C_c^\infty(0, \infty)$. It vanishes on the dense set $\mathscr{T}_\infty$, so it vanishes identically. No uniform bound on the residuals is needed for this step. □

Hypothesis 17.6 (Uniqueness in the box)**.** $U(C, \mathbb{M})$: the only $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathbb{K}(C, \mathbb{M})$ with $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ is $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$.

Remark 17.7 (What is proved and what is open in U). 15.5 proves uniqueness for lists in $\mathscr{O}$. Besides the tube and the ordering, $\mathscr{O}$ (15.1) carries a third condition, $|\int(N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}})| \leq C \log \mathrm{X}$, and a coordinatewise limit of sections does not inherit it. Hypothesis 17.6 is therefore the statement that 15.5 holds for exact solutions in $\mathbb{K}(C, \mathbb{M})$ without the third condition. That condition is used to bound $\zeta_{\boldsymbol{\kappa}}$ on half-planes. For an exact solution, Lemma 17.30 below with $\widehat{\mathsf{R}} \equiv 0$ gives that bound directly on $\Re s \geq 2$. Whether this suffices for the rest of the proof of 15.5 has to be checked line by line against that proof. This is not done here, and Hypothesis 17.6 is not claimed.

Corollary 17.8 (Selection of the zeta list)**.** *Assume $B(C, \mathbb{M})$ and $U(C, \mathbb{M})$, and let $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)})$ be any sequence in $\mathbb{K}(C, \mathbb{M})$ with $\mathsf{s}_m \to 0$. Then the whole sequence converges coordinatewise to $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$. In particular $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$, that is, (D) holds.*

Proof. By Proposition 17.5, every subsequence has a further subsequence converging to a zero of $\mathfrak{R}$ in $\mathbb{K}(C, \mathbb{M})$. By $\mathrm{U}(C, \mathbb{M})$ that zero is $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$, so the whole sequence converges to it. Finally we show that $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$ implies (D). By Lemma 17.11 (Section 17.3.1), $\widehat{\mathsf{R}}$ of (17.27) is the Laplace transform of a distribution of finite order on $[0, \infty)$ whose restriction to $(0, \infty)$ is $\mathfrak{R}$. If $\mathfrak{R} = 0$, that

distribution is supported at $\{0\}$, so $\widehat{\mathsf{R}}$ is a polynomial $p(s)$. Since $\sum_n \mathsf{a}_n^\zeta n^{\frac12-s} = -\frac12\zeta'/\zeta(s)$, this reads $P_\gamma'/P_\gamma - (G\zeta)'/(G\zeta) = 2p$, so $P_\gamma = c\,G\zeta\,e^Q$ with $Q' = 2p$. Here $G\zeta = 2\xi$ and P_γ are entire of order 1, so $e^Q = P_\gamma/(2c\,\xi)$ is entire, zero-free and of order at most 1, and $\deg Q \le 1$. Both P_γ and ξ are invariant under $s \mapsto 1-s$, so $e^{Q(1-s)} = e^{Q(s)}$, which forces Q to be constant. Hence $\zeta_\gamma = c'\zeta$. Both functions have residue 1 at $s=1$ ($P_\gamma(1)=1$ and $G(s)\sim s-1$), so $c'=1$. This is (D), because $\zeta_\gamma = \zeta_{Z1}$. □

Remark 17.9 (Logical status of route B). Corollaries 17.4 and 17.8 give (D) ⇒ B$(C_\eta,\mathbb{M}_*)$ and [B$(C,\mathbb{M})$ and U$(C,\mathbb{M})$] ⇒ (D). Route B therefore consists of two independent pieces. The first is existence, B, which is not weaker than (D). The second is uniqueness without the third condition of $\mathscr{O}$, Hypothesis 17.6, which does not involve ζ. Compactness itself is not an issue in the box: Proposition 17.5 uses only the imposed bounds. In this scheme the ordinates γ_k never enter the construction; they appear only as the unique admissible zero of $\mathfrak{R}$. Lemma 17.3 does *not* show that near-solutions of sections lie in $\mathbb{K}(C_\eta,\mathbb{M}_*)$: it locates only the exact solution. Restricting the sections to that box is a choice, justified a posteriori by Corollary 17.4.

17.3.1. *Route B with prescribed von Mangoldt masses.* The masses $\mathsf{a}_n^\zeta = \Lambda(n)/(2\sqrt{n})$ are known and do not depend on (D) or on the ordinates of ζ. Prescribing them leaves the ordinates as the only unknowns, and the uniqueness question of Hypothesis 17.6 disappears. The tool is the identification of the Laplace transform of the residual. In version 3 of the companion note this was a convention; here it is a lemma.

Laplace transforms of distributions. Let $\mathscr{T} \in \mathcal{D}'(\mathbb{R})$ have support in $[0,\infty)$ and suppose that $e^{-au}\mathscr{T}$ is a tempered distribution for some $a \in \mathbb{R}$. For $\Re s > a + \frac12$ put $z := s - \frac12$ and

$$\mathcal{L}\mathscr{T}(s) := \langle \mathscr{T},\ \mathsf{j}(u)\,e^{-zu}\rangle, \tag{17.7}$$

where $\mathsf{j} \in C^\infty(\mathbb{R})$ equals 1 on $[-1,\infty)$ and 0 on $(-\infty,-2]$; the value does not depend on j, and $\mathcal{L}\mathscr{T}$ is holomorphic there [55, Ch. 8]. The shift by $\frac12$ in z is the normalisation of (17.27). Since j' vanishes on the support of $\mathscr{T}$, $\mathcal{L}(\partial_u\mathscr{T})(s) = z\,\mathcal{L}\mathscr{T}(s)$; also $\mathcal{L}\delta_0 = 1$, $\mathcal{L}\delta_0' = z$, and for a locally integrable g on $[0,\infty)$ with $|g(u)| \le ce^{au}$, $\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s) = \int_0^\infty e^{-zu}g(u)\,du$. We write $\mathbf{1}_+ := \mathbf{1}_{[0,\infty)}$ and $\operatorname{fp}\frac{\mathbf{1}_+}{u}$ for the distribution $\varphi \mapsto \int_0^1 \frac{\varphi(u)-\varphi(0)}{u}\,du + \int_1^\infty \frac{\varphi(u)}{u}\,du$, of order 1, supported in $[0,\infty)$, with restriction $1/u$ to $(0,\infty)$.

Lemma 17.10 (The Archimedean distribution)**.** *Let* $\mathfrak{g}(u) := \frac{e^{-u/2}}{e^{2u}-1} - \frac{1}{2u}$ *for* $u>0$*;* $\mathfrak{g}$ *is bounded,* $\mathfrak{g}(0^+) = -\frac34$*, and* $\mathfrak{g}(u) = O(e^{-u/2})$*. Put* $\mathrm{c}_{\mathscr{K}} := -\frac14\log(2\pi e^{\gamma_E})$*,* γ_E *Euler's constant, and*

$$\mathscr{K} := \tfrac12 e^{u/2}\mathbf{1}_+ - \tfrac12\,\mathfrak{g}\,\mathbf{1}_+ - \tfrac14\operatorname{fp}\frac{\mathbf{1}_+}{u} + \mathrm{c}_{\mathscr{K}}\,\delta_0. \tag{17.8}$$

Then $\mathscr{K}$ *is a distribution of order* 1 *supported in* $[0,\infty)$*, its restriction to* $(0,\infty)$ *is the function* k *of 14.77, and*

$$\mathcal{L}\mathscr{K}(s) = \tfrac12\,\frac{G'}{G}(s) \qquad (\Re s > 1). \tag{17.9}$$

Proof. On $(0,\infty)$, $\frac12 e^{u/2} - \frac12\mathfrak{g}(u) - \frac{1}{4u} = \frac12\big(e^{u/2} - \frac{e^{-u/2}}{e^{2u}-1}\big) = \mathsf{k}(u)$. The expansion $\frac{e^{-u/2}}{e^{2u}-1} = \frac{1}{2u} - \frac34 + O(u)$ gives the behaviour of $\mathfrak{g}$ at 0. With $z = s-\frac12$ and $\Re s > 1$: $\mathcal{L}(\frac12 e^{u/2}\mathbf{1}_+) = \frac{1}{2(z-\frac12)} = \frac{1}{2(s-1)}$. Gauss's integral $\frac{\Gamma'}{\Gamma}(x) = \int_0^\infty\big(\frac{e^{-t}}{t} - \frac{e^{-xt}}{1-e^{-t}}\big)dt$ $(\Re x > 0)$ for the logarithmic derivative of Γ [54, Ch. XII], with $t = 2u$ and $x = \frac{s}{2}$, reads

$$\tfrac12\frac{\Gamma'}{\Gamma}\big(\tfrac{s}{2}\big) = \int_0^\infty\Big(\frac{e^{-2u}}{2u} - \frac{e^{-su}}{1-e^{-2u}}\Big)du = \int_0^\infty\Big(\frac{e^{-2u}}{2u} - e^{-zu}\frac{e^{-u/2}}{e^{2u}-1}\Big)du - \frac{1}{s}, \tag{17.10}$$

because $\frac{e^{-su}}{1-e^{-2u}} = e^{-zu}\big(\frac{e^{-u/2}}{e^{2u}-1} + e^{-u/2}\big)$ and $\int_0^\infty e^{-zu}e^{-u/2}du = \frac1s$. Writing $\frac{e^{-u/2}}{e^{2u}-1} = \mathfrak{g}(u) + \frac{1}{2u}$ and using Frullani's integral $\int_0^\infty \frac{e^{-2u}-e^{-zu}}{u}\,du = \log\frac{z}{2}$ (both sides vanish at $z=2$ and have derivative

$\int_0^\infty e^{-zu}du = 1/z$ in z, for $\Re z > 0$), the integral in (17.10) equals $-\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s) + \frac{1}{2}\log\frac{z}{2}$. Hence

$$\tfrac{1}{2}\frac{G'}{G}(s) = \frac{1}{2s} + \frac{1}{2(s-1)} - \frac{1}{4}\log\pi + \frac{1}{4}\frac{\Gamma'}{\Gamma}(\tfrac{s}{2}) = \mathcal{L}(\tfrac{1}{2}e^{u/2}\mathbf{1}_+)(s) - \tfrac{1}{2}\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s) + \tfrac{1}{4}\log\frac{z}{2} - \tfrac{1}{4}\log\pi. \quad (17.11)$$

Finally $\mathcal{L}(\mathrm{fp}\,\frac{\mathbf{1}_+}{u})(s) = \int_0^1 \frac{e^{-zu}-1}{u}\,du + \int_1^\infty \frac{e^{-zu}}{u}\,du = -\gamma_E - \log z$: the left side has derivative $-\int_0^\infty e^{-zu}du = -1/z$ in z, and its value at $z=1$ is $\int_0^1 \frac{e^{-u}-1}{u}\,du + \int_1^\infty \frac{e^{-u}}{u}\,du = -\gamma_E$, the classical integral representation of Euler's constant [54, Ch. XII]. Substituting, $\frac{1}{4}\log\frac{z}{2} = -\frac{1}{4}\mathcal{L}(\mathrm{fp}\,\frac{\mathbf{1}_+}{u}) - \frac{1}{4}(\gamma_E + \log 2)$, and (17.11) becomes (17.9) with the constant $\mathrm{c}_{\mathscr{K}}$. (The identity was also checked numerically to 25 digits at four points s; file `laplace_check.py`.) □

Lemma 17.11 (Laplace transform of the residual)**.** *Let $\boldsymbol{\kappa}$ be C-admissible and put $\mathfrak{H}_{\boldsymbol{\kappa}}(u) := \sum_k \kappa_k^{-2}\cos(\kappa_k u)$, which converges absolutely and uniformly on $\mathbb{R}$ since $\mathfrak{H}_{\boldsymbol{\kappa}}(0) = \sum_k \kappa_k^{-2} < \infty$, and*

$$\mathscr{S}_{\boldsymbol{\kappa}} := -\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+) + \mathfrak{H}_{\boldsymbol{\kappa}}(0)\,\delta_0', \qquad \mathscr{A}_\zeta := \sum_{n\ge 2} \mathsf{a}_n^\zeta\,\delta_{\log n}, \qquad \mathscr{E}_{\boldsymbol{\kappa}} := \mathscr{S}_{\boldsymbol{\kappa}} + \mathscr{A}_\zeta - \mathscr{K}. \quad (17.12)$$

Then $\mathscr{E}_{\boldsymbol{\kappa}}$ is a distribution of order at most 2, supported in $[0,\infty)$, whose restriction to $(0,\infty)$ is $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta)$, and for $\Re s > 1$

$$\mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}}(s) = \widehat{\mathsf{R}}(s) = \tfrac{1}{2}\Big(\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}} - \frac{\xi'}{\xi}\Big)(s), \qquad \xi := \tfrac{1}{2}G\zeta, \quad (17.13)$$

where $\widehat{\mathsf{R}}$ is (17.27) *with the masses a^ζ for all n. The same holds for a finite section $(\boldsymbol{\kappa}, \mathsf{a})$ with $\mathscr{A}_\zeta$ replaced by $\sum_{n\le n^*} \mathsf{a}_n\delta_{\log n}$ and $-\frac{1}{2}\xi'/\xi$ by $-\frac{1}{2}G'/G + \sum_{n\le n^*}\mathsf{a}_n n^{\frac{1}{2}-s}$.*

Proof. Spectral part. $\kappa_k \ge \tau_k - C \gg k/\log(k+1)$, so $\mathfrak{H}_{\boldsymbol{\kappa}}(0) < \infty$ and $\mathfrak{H}_{\boldsymbol{\kappa}}$ is continuous and bounded by $\mathfrak{H}_{\boldsymbol{\kappa}}(0)$. In $\mathcal{D}'(0,\infty)$ a uniformly convergent series may be differentiated termwise, so $-\mathfrak{H}_{\boldsymbol{\kappa}}'' = \sum_k \cos(\kappa_k u)$ there; the term $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$ is supported at 0. Hence $\mathscr{S}_{\boldsymbol{\kappa}}|_{(0,\infty)} = \sum_k \cos(\kappa_k u)$, and $\mathscr{S}_{\boldsymbol{\kappa}}$ has order at most 2. For $\Re z > 0$, Fubini (the double integral $\sum_k \kappa_k^{-2}\int_0^\infty e^{-\Re z\, u}du$ is finite) gives $\mathcal{L}(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)(s) = \sum_k \kappa_k^{-2}\frac{z}{z^2+\kappa_k^2}$, so

$$\mathcal{L}\mathscr{S}_{\boldsymbol{\kappa}}(s) = -z^2\sum_k \frac{z}{\kappa_k^2(z^2+\kappa_k^2)} + z\sum_k \frac{1}{\kappa_k^2} = \sum_k \frac{z}{z^2+\kappa_k^2} = \tfrac{1}{2}\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s), \quad (17.14)$$

because $\frac{z}{\kappa^2} - \frac{z^3}{\kappa^2(z^2+\kappa^2)} = \frac{z}{z^2+\kappa^2}$, and $P_{\boldsymbol{\kappa}} = \prod_k \frac{z^2+\kappa_k^2}{\frac{1}{4}+\kappa_k^2}$ converges locally uniformly, so its logarithmic derivative is the termwise sum $\sum_k \frac{2z}{z^2+\kappa_k^2}$.

Arithmetic part. On $[0,u^*]$ only $n \le e^{u^*}$ contribute, so $\mathscr{A}_\zeta$ is a locally finite measure, of order 0, and $e^{-au}\mathscr{A}_\zeta$ is a finite measure for $a > \frac{1}{2}$. For $\Re s > 1$, $\mathcal{L}\mathscr{A}_\zeta(s) = \sum_n \frac{\Lambda(n)}{2\sqrt{n}}n^{-z} = \frac{1}{2}\sum_n \Lambda(n)n^{-s} = -\frac{1}{2}\frac{\zeta'}{\zeta}(s)$.

Assembly. By Lemma 17.10, $\mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}} = \frac{1}{2}P_{\boldsymbol{\kappa}}'/P_{\boldsymbol{\kappa}} - \frac{1}{2}\zeta'/\zeta - \frac{1}{2}G'/G$, and $(G\zeta)'/(G\zeta) = \xi'/\xi$ since $\xi = \frac{1}{2}G\zeta$. The restriction to $(0,\infty)$ is $\sum_k \cos(\kappa_k u) + \sum_n \mathsf{a}_n^\zeta\delta(u-\log n) - \mathsf{k}(u)$, which is (17.5). The finite-section case is identical with a finite sum. □

Remark 17.12 (The boundary term). Without the correction $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$ in (17.12), the transform of $-\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)$ is $\frac{1}{2}P_{\boldsymbol{\kappa}}'/P_{\boldsymbol{\kappa}} - (s-\frac{1}{2})\mathfrak{H}_{\boldsymbol{\kappa}}(0)$: the second derivative of the cut-off function produces $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0' = \mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$. Either normalisation gives the same restriction to $(0,\infty)$ and differs by a polynomial in s, which is harmless in the argument below. Both $\mathscr{K}$ and $\mathscr{S}_{\boldsymbol{\kappa}}$ carry singular terms at $u=0$; this is why the rung system is posed in $\mathcal{D}'(0,\infty)$ and not on $[0,\infty)$.

Proposition 17.13 (Uniqueness with prescribed masses)**.** *Let $\boldsymbol{\kappa}$ be C-admissible for some C, and suppose $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta) = 0$ in $\mathcal{D}'(0,\infty)$. Then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, and $\boldsymbol{\kappa} = \boldsymbol{\gamma}$.*

Proof. By Lemma 17.11, $\mathscr{E}_{\boldsymbol{\kappa}}$ is a distribution of finite order supported in $[0,\infty)$ whose restriction to $(0,\infty)$ vanishes, so it is supported at $\{0\}$. A distribution of finite order supported at a point is a finite linear combination of δ_0 and its derivatives [13, Thm. 2.3.4], so $\widehat{\mathsf{R}} = \mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}}$ is a polynomial p in $z = s - \frac{1}{2}$. By (17.13), $(P_{\boldsymbol{\kappa}} e^{-Q}/\xi)' = 0$ on $\Re s > 1$ with $Q' = 2p$, Q a polynomial. Hence $P_{\boldsymbol{\kappa}} = c\,\xi\, e^{Q}$ there with $c \neq 0$, and by analytic continuation on all of $\mathbb{C}$. In the variable $w = s - \frac{1}{2}$, $P_{\boldsymbol{\kappa}}$ is a constant multiple of $\prod_k (1 + w^2/\kappa_k^2)$, and $N_{\boldsymbol{\kappa}}(t) \ll t \log t$, so $P_{\boldsymbol{\kappa}}$ has order 1; ξ has order 1. Thus $e^{Q} = P_{\boldsymbol{\kappa}}/(c\xi)$ is entire, zero-free and of order at most 1, and $\deg Q \leq 1$ by Hadamard's theorem. Since $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$ and $\xi(1-s) = \xi(s)$, $e^{Q(1-s)} = e^{Q(s)}$; for $Q(s) = a + bs$ this forces $b(1-2s) \in 2\pi i\mathbb{Z}$ for all s, so $b = 0$. Therefore $P_{\boldsymbol{\kappa}} = c'\xi$. The zeros of $P_{\boldsymbol{\kappa}}$ are $\frac{1}{2} \pm i\kappa_k$, so every zero of ξ, that is, every nontrivial zero of ζ, has real part $\frac{1}{2}$, and the ordinates counted with multiplicity are the κ_k. Both lists are nondecreasing, so $\boldsymbol{\kappa} = \boldsymbol{\gamma}$. □

Definition 17.14 (Statement B′)**.** *Statement $B'(C)$* says that there are C-admissible lists $\boldsymbol{\kappa}^{(m)}$, each equal to the clock beyond some finite index, with $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{\zeta}) \to 0$.

Corollary 17.15 (Route B′)**.** (a) *If $B'(C)$ holds for some C, then every sequence as in Definition 17.14 converges coordinatewise to $\boldsymbol{\gamma}$, and all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$.*

(b) *If all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, then $B'(C)$ holds for every $C \geq C_\eta$.*

Hence the Riemann hypothesis is equivalent to $B'(C)$ for some $C < \infty$.

Proof. (a) $|\mathsf{a}^{\zeta}_n| \leq \mathbb{M}_*$ by the elementary computation in Lemma 17.3, so $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{\zeta}) \in \mathbb{K}(C, \mathbb{M}_*)$. By Proposition 17.5, every subsequence has a further subsequence converging coordinatewise to some $(\boldsymbol{\kappa}, \mathsf{a}^{\zeta})$ with $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^{\zeta}) = 0$ and $\boldsymbol{\kappa}$ C-admissible; the masses do not move. By Proposition 17.13, $\boldsymbol{\kappa} = \boldsymbol{\gamma}$ and the zeros lie on the line. Since every subsequence has a further subsequence with the same limit, the whole sequence converges. (b) Under this hypothesis the rung system with $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$ is the distributional explicit formula 12.6, and $|\gamma_k - \tau_k| \leq C_\eta$ by 2.2. The truncations in the proof of Corollary 17.4, with the masses kept equal to a^{ζ}, give the sections. □

Remark 17.16 (What prescribing the masses changes, and what it does not). Hypothesis 17.6 is no longer needed: uniqueness holds for every list at bounded distance from the clock, not only in $\mathscr{O}$, and it follows from Hadamard's theorem and the functional equation instead of 15.5. Route B′ has a single open piece, the existence statement B′, which by Corollary 17.15 is equivalent to the Riemann hypothesis. With prescribed masses the whole window $(0, u^*)$ constrains the ordinates; the part above $\log \Omega$ can no longer be absorbed by the masses, as it could in Heuristic 17.39.

The equivalence itself is classical in substance. The prime side of the rung system is the prime side of the explicit formula of Riemann and von Mangoldt in the form of Guinand and Weil [12, 53], and the uniqueness is Hadamard's factorisation. Arias de Reyna [2] proves that the Riemann hypothesis is equivalent to the temperedness of the measure $-\sum_n \frac{\Lambda(n)}{\sqrt{n}}(\delta_{\log n} + \delta_{-\log n}) + 2\cosh(x/2)\,dx$, the same prime side up to the factor 2 and the symmetrisation, and identifies its Fourier transform with $\sum_\gamma \delta_{\gamma/2\pi} - 2\vartheta'(2\pi t)\,dt$ under that hypothesis; by partial summation his criterion is von Koch's $\sum_{n \leq x} \Lambda(n) = x + O(x^{1/2}\log^2 x)$. Statement B′ is the finite-section form of the same equivalence: it asks for a real spectrum in a tube around the clock that reproduces the prime side against a dense family of tests. It is not a weaker intermediate statement, and the finite sections are not more constrained than the global statement; what they offer is a formulation in which each stage is a finite-dimensional problem with computable certificates (Section 17.3.2) and explicit candidates (Section 17.3.3). Dyson's suggestion that the zeros form a one-dimensional quasicrystal, and the classification of crystalline measures by Kurasov and Sarnak [37] and Olevskii and Ulanovskii [42], concern the same object; the zeta list is not in those classes because of its density, as Arias de Reyna points out.

17.3.2. *Staged construction: a certificate theorem and a graded extension.*

Definition 17.17 (Staged sections)**.** Fix $u_1^* < u_2^* < \cdots \to \infty$, finite test sets $\mathscr{T}_1 \subset \mathscr{T}_2 \subset \cdots$ supported in $(0, u_m^*)$ with dense union as in Section 17.3, $\mathrm{t}_m := |\mathscr{T}_m|$, and $k_m^* \to \infty$. Stage m starts from a list $\boldsymbol{\kappa}^{(m,0)}$ that equals the clock beyond some finite index, moves the coordinates $k \le k_m^*$ by $x \in \mathbb{R}^{k_m^*}$, and has masses a^{ζ}. Its residual map is

$$\mathbb{F}_m : \mathbb{R}^{k_m^*} \to \mathbb{R}^{\mathrm{t}_m}, \qquad \mathbb{F}_m(x)_\varphi := \big\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m,0)} + x, \mathsf{a}^{\zeta}), \varphi \big\rangle \quad (\varphi \in \mathscr{T}_m), \tag{17.15}$$

where $(\boldsymbol{\kappa} + x)_k = \kappa_k + x_k$ for $k \le k_m^*$ and κ_k otherwise, and its Jacobian at 0 is

$$(\mathbb{J}_m)_{\varphi,k} = \partial_{x_k} \mathbb{F}_m(0)_\varphi = -\int_0^\infty \varphi(u)\, u \sin(\kappa_k^{(m,0)} u)\, du \qquad (\varphi \in \mathscr{T}_m,\ k \le k_m^*). \tag{17.16}$$

We use the norms $\|x\|_\infty$ on $\mathbb{R}^{k_m^*}$ and $\mathbb{R}^{\mathrm{t}_m}$ and the induced operator norms $\|\cdot\|_{\infty\to\infty}$ (maximal absolute row sum). Put

$$\mathsf{h}_m := \|\mathbb{F}_m(0)\|_\infty, \qquad \mathbb{H}_m := k_m^* \max_{\varphi \in \mathscr{T}_m} \int_0^\infty |\varphi(u)|\, u^2\, du. \tag{17.17}$$

Lemma 17.18 (Monotone matching)**.** *Let $(\mathrm{a}_k)_{k\ge1}$ be real with $|\mathrm{a}_k - \tau_k| \le C$ for all k, and let $(\mathrm{a}_{(k)})$ be its nondecreasing rearrangement. Then $|\mathrm{a}_{(k)} - \tau_k| \le C$ for all k. If moreover $\mathrm{a}_k = \tau_k$ for $k > k^*$, then $\mathrm{a}_{(k)} = \tau_k$ for all $k > k^* + j$, where $j := \#\{k > k^* : \tau_k \le \tau_K + C\} < \infty$.*

Proof. At least k of the a_i (those with $i \le k$) are $\le \tau_k + C$, so $\mathrm{a}_{(k)} \le \tau_k + C$; at least k of them (those with $i \ge k$) are $\ge \tau_k - C$, hence at most $k-1$ are $< \tau_k - C$, so $\mathrm{a}_{(k)} \ge \tau_k - C$. For the second claim, the elements of the multiset not exceeding $\tau_K + C$ are the k^* moved ones and the j clock points τ_k, $k^* < k \le k^* + j$; the remaining elements are exactly τ_k, $k > k^* + j$, in increasing order. □

Lemma 17.19 (Lipschitz bound for the Jacobian)**.** *For all $x, x' \in \mathbb{R}^{k_m^*}$, $\|D\mathbb{F}_m(x) - D\mathbb{F}_m(x')\|_{\infty\to\infty} \le \mathbb{H}_m \|x - x'\|_\infty$, and*

$$\|\mathbb{F}_m(x) - \mathbb{F}_m(0) - \mathbb{J}_m x\|_\infty \le \tfrac{1}{2} \mathbb{H}_m \|x\|_\infty^2. \tag{17.18}$$

Proof. $\mathbb{F}_m(x)_\varphi = \sum_{k \le k_m^*} \mathsf{c}_\varphi(\kappa_k^{(m,0)} + x_k) + \text{const}$, so $D^2\mathbb{F}_m(x)_\varphi$ is diagonal with entries $\partial_{x_k}^2 \mathbb{F}_m(x)_\varphi = -\int \varphi(u) u^2 \cos((\kappa_k + x_k)u)\, du$, of modulus at most $\int |\varphi| u^2$. The row φ of $D\mathbb{F}_m(x) - D\mathbb{F}_m(x')$ has absolute sum at most $\sum_{k \le k_m^*} \int |\varphi| u^2\, |x_k - x'_k| \le \mathbb{H}_m \|x - x'\|_\infty$. The Taylor bound follows by integrating $D\mathbb{F}_m$ along the segment. □

Theorem 17.20 (Certificate for one stage)**.** *Suppose $\mathbb{J}_m$ has a right inverse $\mathbb{Q}_m : \mathbb{R}^{\mathrm{t}_m} \to \mathbb{R}^{k_m^*}$, $\mathbb{J}_m \mathbb{Q}_m = I_{\mathrm{t}_m}$, with $\|\mathbb{Q}_m\|_{\infty\to\infty} \le \mathbb{L}_m$, and*

$$\mathsf{l}_m := 2\mathbb{L}_m^2 \mathbb{H}_m \mathsf{h}_m < 1. \tag{A_m}$$

Then the map $\mathbb{T}_m(y) := y - \mathbb{F}_m(\mathbb{Q}_m y)$ is a contraction of the ball $\{\|y\|_\infty \le 2\mathsf{h}_m\} \subset \mathbb{R}^{\mathrm{t}_m}$ with constant l_m, and $x_* := \mathbb{Q}_m y_*$, y_* its fixed point, satisfies

$$\mathbb{F}_m(x_*) = 0, \qquad \|x_*\|_\infty \le 2\mathbb{L}_m \mathsf{h}_m. \tag{17.19}$$

Proof. $D\mathbb{T}_m(y) = I - D\mathbb{F}_m(\mathbb{Q}_m y)\mathbb{Q}_m = (\mathbb{J}_m - D\mathbb{F}_m(\mathbb{Q}_m y))\mathbb{Q}_m$ because $\mathbb{J}_m\mathbb{Q}_m = I$, so by Lemma 17.19, for $\|y\|_\infty \le 2\mathsf{h}_m$, $\|D\mathbb{T}_m(y)\|_{\infty\to\infty} \le \mathbb{H}_m \|\mathbb{Q}_m y\|_\infty \mathbb{L}_m \le 2\mathbb{H}_m\mathbb{L}_m^2\mathsf{h}_m = \mathsf{l}_m < 1$. Moreover $\mathbb{T}_m(y) = -\mathbb{F}_m(0) - \big(\mathbb{F}_m(\mathbb{Q}_m y) - \mathbb{F}_m(0) - \mathbb{J}_m\mathbb{Q}_m y\big)$, and by (17.18) the bracket has norm at most $\tfrac{1}{2}\mathbb{H}_m\mathbb{L}_m^2\|y\|_\infty^2 \le 2\mathbb{H}_m\mathbb{L}_m^2\mathsf{h}_m^2$. Hence $\|\mathbb{T}_m(y)\|_\infty \le \mathsf{h}_m(1 + \mathsf{l}_m) < 2\mathsf{h}_m$: the ball is mapped into itself. Banach's fixed point theorem gives a unique fixed point y_* in the ball, $y_* = y_* - \mathbb{F}_m(\mathbb{Q}_m y_*)$, that is, $\mathbb{F}_m(x_*) = 0$ with $\|x_*\|_\infty \le \mathbb{L}_m\|y_*\|_\infty \le 2\mathbb{L}_m\mathsf{h}_m$. The iteration $y_{q+1} = \mathbb{T}_m(y_q)$, $y_0 = 0$, converges to y_* geometrically with ratio l_m; in the variable x it is $x_{q+1} = x_q - \mathbb{Q}_m\mathbb{F}_m(x_q)$, a Newton iteration with the frozen right inverse $\mathbb{Q}_m$. □

Theorem 17.21 (Staged certificate implies route B′)**.** *Let $C < \tau_1 - 2\pi$. Start at $\boldsymbol{\kappa}^{(1,0)} := \boldsymbol{\tau}$ and, for $m \geq 1$, let $\boldsymbol{\kappa}^{(m)}$ be the nondecreasing rearrangement of $\boldsymbol{\kappa}^{(m,0)} + x_*^{(m)}$ given by Theorem 17.20, and $\boldsymbol{\kappa}^{(m+1,0)} := \boldsymbol{\kappa}^{(m)}$. Suppose* ($\mathrm{A}_m$) *holds for every m and*

$$\sup_{m\geq 1} \sum_{j\leq m} 2\mathbb{L}_j \mathsf{h}_j \leq C. \tag{B}$$

Then every $\boldsymbol{\kappa}^{(m)}$ is C-admissible and equals the clock beyond a finite index $\widetilde{k}_m^*$, $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta) = 0$ for every m, $B'(C)$ holds, and therefore (Corollary 17.15) $\boldsymbol{\kappa}^{(m)} \to \boldsymbol{\gamma}$ coordinatewise and the Riemann hypothesis holds. In (A_{m+1}) *and* (B) *the quantity h_{m+1} is $\max_{\varphi\in\mathscr{T}_{m+1}\setminus\mathscr{T}_m} |\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta), \varphi\rangle|$, the residual of the new tests only.*

Proof. Induction on m. Suppose $\boldsymbol{\kappa}^{(m,0)}$ is C_{m-1}-admissible with $C_{m-1} := \sum_{j<m} 2\mathbb{L}_j\mathsf{h}_j$ ($C_0 = 0$) and equals the clock beyond a finite index. By Theorem 17.20, $\boldsymbol{\kappa}^{(m,0)} + x_*^{(m)}$ satisfies $|(\cdot)_k - \tau_k| \leq C_{m-1} + 2\mathbb{L}_m\mathsf{h}_m = C_m$ for all k, and only finitely many coordinates differ from the clock. Lemma 17.18 shows that its rearrangement $\boldsymbol{\kappa}^{(m)}$ satisfies $|\kappa_k^{(m)} - \tau_k| \leq C_m \leq C$ and equals the clock beyond some $\widetilde{k}_m^*$; it is nondecreasing, and $\kappa_1^{(m)} \geq \tau_1 - C > 2\pi$. So $\boldsymbol{\kappa}^{(m)}$ is C-admissible. The residual $\mathfrak{R}$ depends only on the multiset of ordinates, so $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta), \varphi\rangle = \mathbb{F}_m(x_*^{(m)})_\varphi = 0$ for all $\varphi \in \mathscr{T}_m$, that is, $\mathsf{s}_m = 0$. Since $\mathscr{T}_m \subset \mathscr{T}_{m+1}$, the start of stage $m+1$ has zero residual on $\mathscr{T}_m$, which gives the stated form of h_{m+1}. $B'(C)$ holds with $\mathsf{s}_m = 0$, and Corollary 17.15(a) gives the rest. □

Remark 17.22 (What the certificate theorem does and does not do)*.* Theorem 17.21 replaces the extension lemma of version 3 of the companion note (there Hypothesis E) by two computable quantities per stage, $\mathsf{l}_m = 2\mathbb{L}_m^2\mathbb{H}_m\mathsf{h}_m$ and $2\mathbb{L}_m\mathsf{h}_m$. It requires no contraction factor q, no separate control of the perturbation of old tests, and no prior "entry into the basin": old tests are kept in $\mathscr{T}_{m+1}$ and are solved again exactly. The two conditions are not independent of the height: $\mathbb{H}_m \geq k_m^* \int |\varphi| u^2$ grows with the number of unknowns, and $\mathbb{L}_m$ is the ℓ^∞ norm of a right inverse of the Jacobian, whose growth is measured in Section 17.7. The conditions (A_m) for all m and (B) together imply the Riemann hypothesis, so they are not a weaker intermediate step; what changes is that each stage is a statement about one finite matrix and one finite residual vector, verifiable in principle by interval arithmetic. The restriction $C < \tau_1 - 2\pi = 8.24\ldots$ only ensures $\kappa_1 \geq 2\pi$ and can be removed by treating the first coordinate separately.

Proposition 17.23 (Graded extension)**.** *Let $\mathsf{s}_m \geq 0$ satisfy $\mathsf{s}_{m+1} \leq \mathrm{q}_m\mathsf{s}_m + \mathsf{h}_m$ with $0 \leq \mathrm{q}_m \leq 1$, $\prod_{m\geq 1} \mathrm{q}_m = 0$ and $\sum_m \mathsf{h}_m < \infty$. Then $\mathsf{s}_m \to 0$. If the lists $\boldsymbol{\kappa}^{(m)}$ producing $\mathsf{s}_m = \mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta)$ are C-admissible sections for a fixed C, then $B'(C)$ holds.*

Proof. Iterating, $\mathsf{s}_m \leq \mathsf{s}_1 \prod_{j<m} \mathrm{q}_j + \sum_{\ell<m} \mathsf{h}_\ell \prod_{\ell<j<m} \mathrm{q}_j$. Given $\epsilon > 0$ choose L with $\sum_{\ell\geq L} \mathsf{h}_\ell < \epsilon$. The terms with $\ell < L$ are at most $(\mathsf{s}_1 + \sum_{\ell<L} \mathsf{h}_\ell) \prod_{L\leq j<m} \mathrm{q}_j \cdot \max(1, \ldots)$; more precisely each product $\prod_{\ell<j<m} \mathrm{q}_j$ with $\ell < L$ is at most $\prod_{L\leq j<m} \mathrm{q}_j \to 0$ as $m \to \infty$, since all $\mathrm{q}_j \leq 1$ and $\prod_j q_j = 0$. The terms with $\ell \geq L$ are at most $\sum_{\ell\geq L} \mathsf{h}_\ell < \epsilon$. Hence $\limsup \mathsf{s}_m \leq \epsilon$. The second claim is Definition 17.14. □

Remark 17.24 (Tube budget, and the two geometries)*.* In Proposition 17.23 the tube $|\kappa_k^{(m)} - \tau_k| \leq C$, uniform in m, is a separate hypothesis: summability of the coordinate increments $\sum_m |\kappa_k^{(m+1)} - \kappa_k^{(m)}|$ gives convergence of each coordinate but not that the limit stays in a prescribed tube, so a construction must carry a displacement budget $\sum_{m\geq m(k)} |\kappa_k^{(m+1)} - \kappa_k^{(m)}| \leq C - |\kappa_k^{(m(k))} - \tau_k|$. For $B'(C)$ itself no convergence of the whole sequence is needed: compactness in the product of the intervals $[\tau_k - C, \tau_k + C]$ (Proposition 17.5) extracts a convergent subsequence, and uniqueness identifies the limit. A second point concerns norms. Theorem 17.20 is stated in ℓ^∞, where $\mathbb{H}_m$ has the simple form (17.17) because the Hessian of each c_φ is diagonal. A bound $\|\mathbb{J}_m^\dagger\|_{2\to 2} \leq 1/\sigma_{\min}(\mathbb{J}_m)$ is a Euclidean

statement; passing to ℓ^∞ costs factors $\sqrt{\mathrm{t}_m}$ and $\sqrt{K_m}$ unless the right inverse is controlled directly in $\ell^\infty_{\mathrm{t}_m} \to \ell^\infty_{K_m}$. The numerical constants of Section 17.7 are computed directly in ℓ^∞.

Heuristic 17.25 (Frame bound from exponential sums of the clock)**.** In the opposite regime $k^*_m \gg \mathrm{t}_m$ one may take $\mathbb{Q}_m := \mathbb{J}_m^T(\mathbb{J}_m\mathbb{J}_m^T)^{-1}$. If $\|\frac{1}{K_m}\mathbb{J}_m\mathbb{J}_m^T - D\|_{\infty\to\infty} \le \epsilon < d_-$ for a diagonal $D \ge d_- I$, and $|(\mathbb{J}_m)_{\varphi,k}| \le \mathrm{a}_J$, then $\mathbb{L}_m \le \mathrm{t}_m A/(k^*_m(d_- - \epsilon)) = O(\mathrm{t}_m/k^*_m)$ and $\mathbb{L}^2_m\mathbb{H}_m = O(\mathrm{t}^2_m B/k^*_m)$ with $\mathrm{B}_0 := \max_\varphi \int |\varphi| u^2$, so (A_m) becomes $k^*_m \gg \mathrm{t}^2_m B\mathsf{h}_m$: a wide block compensates a moderate residual. In the point model $\varphi \approx \delta_{u_j}$ the entries of $\frac{1}{k^*}\mathbb{J}\mathbb{J}^T$ are $u_j u_\ell \cdot \frac{1}{k}^* \sum_{k\in\mathfrak{B}} \sin(\tau_k u_j)\sin(\tau_k u_\ell)$, so the frame property reduces to cancellation in the exponential sums $\frac{1}{k}^* \sum_{k\in\mathfrak{B}} e^{i\mathrm{v}\tau_k}$ at the frequencies $\mathrm{v} = u_j \pm u_\ell \neq 0$. On a block of k^* consecutive clock points at height T the spacing is $\Delta_T \approx 2\pi/\log T$ with curvature $O((k^*)^2/(T\log^3 T))$, which suggests $|\frac{1}{k}^* \sum e^{i\mathrm{v}\tau_k}| \lesssim \frac{\log T}{K|v|} + \frac{|v|(k^*)^2}{T\log^3 T}$ and a window $\frac{\log T}{h} \ll k^* \ll (T\log^3 T/u^*)^{1/2}$ for tests on a grid of step Δu. None of this is proved here, and the point model omits a decisive feature of smooth tests: a test of width w in u has $|\partial_\kappa \mathsf{c}_\varphi(\kappa)| \lesssim \mathrm{w}^{-1}e^{-(\mathrm{w}\kappa)^2/2}$ for Gaussian φ, so ordinates with $\mathrm{w}\kappa \gg 1$ are invisible to it and the corresponding columns of $\mathbb{J}_m$ are exponentially small. The underdetermined right inverse then has $\mathbb{L}_m$ dominated by the invisible columns; in the computations of Section 17.7 this produced $\mathbb{L}_m$ of size 10^{11} within four stages, and the stable regime was the opposite one, $\mathrm{t}_m \ge k^*_m$ with the unknowns restricted to the visible ordinates. The quasi-orthogonality heuristic is therefore a statement about tests that are not in $C^\infty_c(0,\infty)$, and it is not pursued further here.

17.3.3. *Explicit lists in the tube: Gonek's symmetrised Euler products.* Gonek [11] introduced, for $\mathrm{X} \ge 2$, the truncated Euler product $P_{\mathrm{X}}(s) := \exp(\sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{n^s \log n})$ with $\Lambda_{\mathrm{X}}(n) := \Lambda(n)$ for $n \le \mathrm{X}$ and $\Lambda_{\mathrm{X}}(n) := \Lambda(n)\log(\mathrm{X}^2/n)/\log \mathrm{X}$ for $\mathrm{X} < n \le \mathrm{X}^2$, and the symmetrised function $\zeta_{\mathrm{X}}(s) := P_{\mathrm{X}}(s) + \chi(s)P_{\mathrm{X}}(1-s)$, where χ is the factor of the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$. On the critical line $\chi(\frac12 + it) = e^{-2i\vartheta(t)}$ and $P_{\mathrm{X}}(\frac12 - it) = \overline{P_{\mathrm{X}}(\frac12 + it)}$, so with

$$\mathrm{p}_{\mathrm{X}}(t) := \arg P_{\mathrm{X}}(\tfrac12 + it) = -\sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)\sin(t\log n)}{\sqrt{n}\log n}, \tag{17.20}$$

and the two constants

$$\mathrm{m}_{\mathrm{X}} := \sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{\sqrt{n}\log n}, \qquad \mathrm{n}_{\mathrm{X}} := \sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{\sqrt{n}}, \tag{17.21}$$

one has $\zeta_{\mathrm{X}}(\frac12 + it) = 2|P_{\mathrm{X}}(\frac12 + it)|\, e^{-i\vartheta(t)}\cos\big(\vartheta(t) + \mathrm{p}_{\mathrm{X}}(t)\big)$. Since P_{X} has no zeros, the zeros of ζ_{X} on the line are exactly the solutions of

$$\Theta_{\mathrm{X}}(t) := \vartheta(t) + \mathrm{p}_{\mathrm{X}}(t) \in \tfrac{\pi}{2} + \pi\mathbb{Z}. \tag{17.22}$$

Gonek proves [11, Thm. 6.2 of the arXiv version] that every zero of ζ_{X} in $0 \le \Re s \le 1$ with imaginary part at least an absolute constant C_0 lies on $\Re s = \frac12$, unconditionally, and that under the Riemann hypothesis the zeros of ζ_{X} approach those of ζ as $\mathrm{X} \to \infty$. The phase (17.22) is the clock phase plus a truncated prime sum: $\mathcal{N}_{\mathrm{RVM}}(t) = 1 + \vartheta(t)/\pi + \mathrm{r}_\vartheta(t)$ with $|\mathrm{r}_\vartheta(t)| \le \mathrm{c}_r/t$ for $t \ge 1$, by Stirling's formula [54, §12.33], so (17.22) reads $\mathcal{N}_{\mathrm{RVM}}(t) + \mathrm{p}_{\mathrm{X}}(t)/\pi \in \frac12 + \mathbb{Z} + \mathrm{r}_\vartheta(t)$, to be compared with $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac12$.

Lemma 17.26 (Gonek lists lie in a tube around the clock)**.** *Fix* $\mathrm{X} \ge 2$ *and put* $\mathrm{t}_{\mathrm{X}} := \max(2\pi e^{2\mathrm{n}_{\mathrm{X}}+2}, t_1)$*, where* t_1 *is such that* $|\vartheta'(t) - \frac12\log\frac{t}{2\pi}| \le \frac12$ *for* $t \ge t_1$*. Then* Θ_{X} *is strictly increasing on* $[\mathrm{t}_{\mathrm{X}}, \infty)$ *with* $\Theta'_{\mathrm{X}} \ge \frac12$*, the solutions of* (17.22) *in* $[\mathrm{t}_{\mathrm{X}}, \infty)$ *form a strictly increasing sequence* $\mathrm{t}^{(\mathrm{X})}_1 < \mathrm{t}^{(\mathrm{X})}_2 < \cdots$*, each a simple zero of* ζ_{X}*, and there is an integer* j_{X} *such that the list*

$$\kappa^{(\mathrm{X})}_k := \begin{cases} \tau_k, & k \le \mathrm{j}_{\mathrm{X}}, \\ \mathrm{t}^{(\mathrm{X})}_{k-\mathrm{j}_{\mathrm{X}}}, & k > \mathrm{j}_{\mathrm{X}}, \end{cases} \tag{17.23}$$

is nondecreasing and $\mathrm{c_X}$*-admissible for some* $\mathrm{c_X} < \infty$*, with*

$$|\kappa_k^{(\mathrm{X})} - \tau_k| \le \frac{2\mathrm{m_X} + o(1)}{\log(\tau_k/2\pi)} \qquad (k \to \infty). \tag{17.24}$$

Proof. $|\mathrm{p}'_{\mathrm{X}}(t)| = |\sum_{n\le \mathrm{X}^2} \Lambda_{\mathrm{X}}(n) n^{-1/2}\cos(t\log n)| \le \mathrm{n_X}$ and $|\mathrm{p_X}| \le \mathrm{m_X}$. For $t \ge \mathfrak{t}_{\mathrm{X}}$, $\vartheta'(t) \ge \frac12 \log\frac{t}{2\pi} - \frac12 \ge \mathrm{n_X} + \frac12$, so $\Theta'_{\mathrm{X}} \ge \frac12$. Hence Θ_{X} is a strictly increasing bijection of $[\mathfrak{t}_{\mathrm{X}}, \infty)$ onto $[\Theta_{\mathrm{X}}(\mathfrak{t}_{\mathrm{X}}), \infty)$, and the solutions of (17.22) are the preimages of the points $\frac{\pi}{2} + \pi j$, $j \ge j_0$, where j_0 is the least integer with $\frac{\pi}{2} + \pi j_0 \ge \Theta_{\mathrm{X}}(\mathfrak{t}_{\mathrm{X}})$; they are simple zeros because $\frac{d}{dt}\cos\Theta_{\mathrm{X}} = -\Theta'_{\mathrm{X}}\sin\Theta_{\mathrm{X}} \ne 0$ there. The i-th solution $\mathrm{t}_i^{(\mathrm{X})}$ satisfies $\Theta_{\mathrm{X}}(\mathrm{t}_i^{(\mathrm{X})}) = \frac{\pi}{2} + \pi(j_0 + i - 1)$; we set $\mathrm{j_X} := j_0$, so that $k = \mathrm{j_X} + i$ gives $\Theta_{\mathrm{X}}(\kappa_k^{(\mathrm{X})}) = \pi(k - \frac12)$. Since $\mathcal{N}_{\mathrm{RVM}} = 1 + \vartheta/\pi + \mathrm{r}_\vartheta$,

$$\mathcal{N}_{\mathrm{RVM}}(\kappa_k^{(\mathrm{X})}) = k + \tfrac12 - \frac{\mathrm{p_X}(\kappa_k^{(\mathrm{X})})}{\pi} + \mathrm{r}_\vartheta(\kappa_k^{(\mathrm{X})}), \tag{17.25}$$

so, after replacing $\mathrm{j_X}$ by $\mathrm{j_X} - 1$,

$$|\mathcal{N}_{\mathrm{RVM}}(\kappa_k^{(\mathrm{X})}) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)| \le \frac{\mathrm{m_X}}{\pi} + \frac{\mathrm{c}_r}{\kappa_k^{(\mathrm{X})}} \tag{17.26}$$

(the unit shift from the constant 1 is absorbed into the integer $\mathrm{j_X}$, fixed once and for all; in the computations of Section 17.7 the offset was determined by matching). The right-hand side is bounded, and $\mathcal{N}_{\mathrm{RVM}}(2t) - \mathcal{N}_{\mathrm{RVM}}(t) \to \infty$, so $\kappa_k^{(\mathrm{X})}/\tau_k \to 1$. By the mean value theorem, $|\kappa_k^{(\mathrm{X})} - \tau_k| \le (\mathrm{m_X}/\pi + o(1))/\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}_k)$ with s_k between $\kappa_k^{(\mathrm{X})}$ and τ_k, and $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}_k) = \frac{1}{2\pi}\log\frac{\tau_k}{2\pi}(1 + o(1))$. This is (17.24); it gives $|\kappa_k^{(\mathrm{X})} - \tau_k| \to 0$, so the displacements are bounded, the list is eventually nondecreasing, and finitely many initial indices (where the list is the clock) are harmless. If monotonicity of the concatenated list fails at the junction $k = \mathrm{j_X}$, one replaces the prefix τ_k, $k \le \mathrm{j_X}$, by the nondecreasing rearrangement of Lemma 17.18, which keeps the tube. □

Remark 17.27 (What the Gonek lists give and what remains). The lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ use the primes up to X^2 and the clock, and nothing about the zeros of ζ. Their prime side is built in: formally, $\sum_k \delta(t - \kappa_k^{(\mathrm{X})}) = \frac{\Theta'_{\mathrm{X}}(t)}{\pi}\big[1 + 2\sum_{j\ge1}(-1)^j\cos(2j\Theta_{\mathrm{X}}(t))\big]$ where $\Theta'_{\mathrm{X}} > 0$, and the $j = 0$ term carries $\mathrm{p}'_{\mathrm{X}}/\pi = -\frac{1}{\pi}\sum_{n\le \mathrm{X}^2}\Lambda_{\mathrm{X}}(n)n^{-1/2}\cos(t\log n)$, whose cosine transform is a sum of masses $\Lambda_{\mathrm{X}}(n)/(2\sqrt{n})$ at $u = \log n$, equal to a_n^{ζ} for $n \le \mathrm{X}$. Two statements would make them a solution of B′: (G1) a tube uniform in X, $\sup_{\mathrm{X}\ge \mathrm{X}_0}\sup_k|\kappa_k^{(\mathrm{X})} - \tau_k| \le C$, which by (17.24) concerns the size of $\mathrm{p_X}$ at its own crossing points and not its supremum ($\mathrm{m_X} \asymp \mathrm{X}/\log\mathrm{X}$ is useless for this); and (G2) $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(\mathrm{X})}, \mathsf{a}^{\zeta}), \varphi\rangle \to 0$ as $\mathrm{X} \to \infty$ for each test φ, whose content is the vanishing of the harmonics $j \ge 1$ against φ. Neither is proved here. The monotonicity threshold $\mathfrak{t}_{\mathrm{X}}$ of Lemma 17.26 grows like $e^{2\mathrm{n_X}}$, so the global construction cannot rely on monotonicity of Θ_{X} at fixed height as $\mathrm{X} \to \infty$; Gonek's theorem on the location of the zeros does not need it. For fixed X the tube constant in (17.24) is $2\mathrm{m_X}$: $\mathrm{m}_3 = 1.42$, $\mathrm{m}_8 = 3.02$, $\mathrm{m}_{20} = 5.98$. Numerically (Section 17.7) the lists with $3 \le \mathrm{X} \le 20$ lie within 0.19 to 0.035 mean spacings of the zeta ordinates for $k \le 400$, whereas the clock is within 0.66 spacings.

17.4. **An a priori bound for quasi-solutions.** Throughout this section $(\boldsymbol{\kappa}, \mathsf{a})$ is a finite section in $\mathbb{K}(C, \mathbb{M})$ with parameters k^*, n^*.

Definition 17.28 (Laplace residual)**.** For $\Re s > 1$ put

$$\widehat{\mathsf{R}}(s) := \tfrac12\,\frac{P'_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\kappa}}}(s) - \tfrac12\,\frac{G'}{G}(s) + \sum_{n=2}^{n^*} \mathsf{a}_n\, n^{\frac12 - s}. \tag{17.27}$$

For $T \geq 0$ put

$$\mathbb{E}(T) := \int_{-T}^{T} |\widehat{\mathsf{R}}(2+iv)|\, dv, \qquad \upsilon := \sup_{T\geq 2} \frac{\mathbb{E}(T)}{\log T} \in [0,\infty]. \tag{17.28}$$

Since $\int_0^\infty e^{-(s-\frac12)u}\cos(\kappa u)\,du = \frac{s-\frac12}{(s-\frac12)^2+\kappa^2} = \frac12 \frac{d}{ds}\log(1+\frac{q(s)}{\frac14+\kappa^2})$ and $\int_0^\infty e^{-(s-\frac12)u}\delta(u-\log n)\,du = n^{\frac12-s}$, $\widehat{\mathsf{R}}$ is formally the Laplace transform of the residual distribution of (17.4), with the normalisation of k fixed in 14.77; this is made precise in Lemma 17.11. We take (17.27) as the definition. Because the coefficients are real, $\widehat{\mathsf{R}}(\bar s) = \overline{\widehat{\mathsf{R}}(s)}$, so $\mathbb{E}(T) = 2\int_0^T |\widehat{\mathsf{R}}(2+iv)|\,dv$. We assume $\upsilon < \infty$ throughout this section; if $\upsilon = \infty$ every bound below is void. By Lemma 17.11, $\widehat{\mathsf{R}}$ is the Laplace transform of the residual distribution of the section.

Constants. Put $\mathrm{c}_\ell := 2\sum_{n\geq 2} n^{-3/2}/\log n < \infty$, and, for $s \in \mathbb{C}$ and real $x \geq 2\pi$, $\mathfrak{f}(x;s) := \log(1+\frac{q(s)}{\frac14+x^2})$, with the principal branch, which is defined whenever $|s-\frac12| < x$. Let $\mathbb{J} := \{2\} \cup \{\sigma + i : \ -1 \leq \sigma \leq 2\}$, a compact set on which $|s-\frac12| \leq \frac52 < 2\pi$, and

$$\mathfrak{B}(C) := \sum_{k\geq 1} \sup_{x\in[\max(\tau_k - C, 2\pi),\, \tau_k + C]} \sup_{s\in\mathbb{J}} |\mathfrak{f}(x;s) - \mathfrak{f}(\tau_k;s)|. \tag{17.29}$$

This is finite. On $\mathbb{J}$, $|q(s)| \leq 5$ and $|\partial_x f(x;s)| = |\frac{2xq(s)}{(\frac14+x^2)(x^2+(s-\frac12)^2)}| \leq \frac{20}{x^3}$ for $x \geq 4$. The k-th term is therefore at most $160C\tau_k^{-3}$ when $\tau_k \geq 2C$, and $\sum_k \tau_k^{-3} < \infty$. Further, $\mathrm{c}_\tau := \max_{-1\leq\sigma\leq 2} |\log|\zeta_{\mathrm{RVM}}(\sigma+i)||$ and $\mathrm{c}_{\mathrm{RVM}} := \sup_{\Re s\geq 2} |\log \zeta_{\mathrm{RVM}}(s)|$. Both are finite: the first because ζ_{RVM} has no zeros or poles on that segment, the second by 8.10.

Lemma 17.29 (Phragmén–Lindelöf, two forms)**.** (a) *Let $\mathbb{S} := \{\mathrm{l}_1 \leq \Re s \leq \mathrm{l}_2,\ \Im s \geq t_1\}$ have width $a := \mathrm{l}_2 - \mathrm{l}_1$. Let u be upper semicontinuous on $\mathbb{S}$ and subharmonic in its interior, with $u \leq \mathrm{M}_0$ on $\partial\mathbb{S}$ and $u(s) \leq \mathrm{A}_0 e^{c\Im s}$ for some $c < \pi/a$. Then $u \leq \mathrm{M}_0$ on $\mathbb{S}$.*
(b) *Let $\mathfrak{g}$ be holomorphic in $\Re s > 1$, continuous on $\Re s \geq 2$, with $|g| \leq 1$ on $\Re s = 2$ and $|\mathfrak{g}(s)| \leq \mathrm{A}_0(2+|s|)^{\mathrm{L}}$ on $\Re s \geq 2$. Then $|g| \leq 1$ on $\Re s \geq 2$.*

Proof. (a) Choose $c < c' < \pi/a$ and $m := \frac{\mathrm{l}_1+\mathrm{l}_2}{2}$, and put $\omega(s) := e^{c'\Im s}\cos(c'(\Re s - m))$. This is harmonic, being the real part of $e^{-ic'(s-m)}$. On $\mathbb{S}$, $\omega \geq \cos(c'a/2)\, e^{c'\Im s}$, and $\cos(c'a/2) > 0$. For $\mathrm{p} > 0$, $u - \mathrm{M}_0 - \mathrm{p}\omega$ is subharmonic, ≤ 0 on $\partial\mathbb{S}$, and tends to $-\infty$ as $\Im s \to \infty$. The maximum principle on $\mathbb{S} \cap \{\Im s \leq t_2\}$, with t_2 large, gives $u \leq \mathrm{M}_0 + \mathrm{p}\omega$. Let $\mathrm{p} \to 0$.
(b) For $\mathrm{p} > 0$, $\mathfrak{g}_\eta(s) := \mathfrak{g}(s)\exp(-\mathrm{p}(s-1)^{1/2})$, with the principal branch, satisfies $|\mathfrak{g}_\eta| \leq |g|$ on $\Re s \geq 2$, because $\Re(s-1)^{1/2} \geq \cos(\pi/4)|s-1|^{1/2} > 0$ there. Moreover $\mathfrak{g}_\eta(s) \to 0$ as $|s| \to \infty$ in $\Re s \geq 2$. The maximum principle on large half-discs gives $|\mathfrak{g}_\eta| \leq 1$. Let $\mathrm{p} \to 0$. □

Lemma 17.30 (The Dirichlet factor)**.** *Put $\mathcal{D}_{\mathsf{a}}(s) := \sum_{n=2}^{n^*} \frac{2\mathsf{a}_n}{\log n} n^{\frac12 - s}$ and $\mathcal{Y} := \zeta_{\boldsymbol{\kappa}} e^{-\mathcal{D}_{\mathsf{a}}}$. Then $\mathcal{Y}$ is holomorphic and zero-free in $\Re s > 1$, and $|\mathcal{D}_{\mathsf{a}}| \leq \mathbb{M}\mathrm{c}_\ell$ on $\Re s \geq 2$. For the branch of $\log\mathcal{Y}$ that is real on $(1,\infty)$,*

$$(\log\mathcal{Y})' = 2\,\widehat{\mathsf{R}}, \qquad |\log\mathcal{Y}(2)| \leq \mathbb{B}_0 := |\log\zeta_{\mathrm{RVM}}(2)| + \mathfrak{B}(C) + \mathbb{M}\mathrm{c}_\ell, \tag{17.30}$$

and, for all real t,

$$|\log\mathcal{Y}(2+it)| \leq \mathbb{B}_0 + \mathbb{E}(|t|) \leq \mathbb{B}_0 + \upsilon\log(2+|t|). \tag{17.31}$$

Proof. $P_{\boldsymbol{\kappa}}$ vanishes only on $\Re s = \frac12$, and G is holomorphic and zero-free in $\Re s > 1$. Hence $\zeta_{\boldsymbol{\kappa}}$ and $\mathcal{Y}$ are holomorphic and zero-free there. $|\mathcal{D}_{\mathsf{a}}(s)| \leq \sum_n 2Mn^{\frac12 - \Re s}/\log n \leq \mathbb{M}\mathrm{c}_\ell$ for $\Re s \geq 2$. Differentiating, $(\log\mathcal{Y})' = \frac{P'_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\kappa}}} - \frac{G'}{G} + 2\sum_n \mathsf{a}_n n^{\frac12 - s} = 2\widehat{\mathsf{R}}$. At $s = 2$, $\log\zeta_{\boldsymbol{\kappa}}(2) = \log\zeta_{\mathrm{RVM}}(2) + \sum_{k\leq k^*}(\mathfrak{f}(\kappa_k;2) - \mathfrak{f}(\tau_k;2))$, a finite sum of real numbers bounded by $\mathfrak{B}(C)$, and $|\mathcal{D}_{\mathsf{a}}(2)| \leq \mathbb{M}\mathrm{c}_\ell$. Finally, $\log\mathcal{Y}(2+it) - \log\mathcal{Y}(2) = 2i\int_0^t \widehat{\mathsf{R}}(2+iv)\,dv$ has modulus at most $\mathbb{E}(|t|)$. For $|t| \geq 2$, $\mathbb{E}(|t|) \leq \upsilon\log|t|$; for $|t| < 2$, $\mathbb{E}(|t|) \leq \mathbb{E}(2) \leq \upsilon\log 2$. □

Lemma 17.31 (Bound on the half-plane)**.** *With* $\mathbb{W} := \log|\zeta_{\boldsymbol{\kappa}}| = \log|P_{\boldsymbol{\kappa}}| - \log|G|$ *and* $\mathbb{B}_1 := \mathbb{B}_0 + \upsilon\log(2\sqrt{2})$, $\mathbb{B}_2 := \mathbb{B}_1 + \mathbb{M}\mathrm{c}_\ell$, *for* $\Re s \geq 2$,

$$|\log|\mathcal{Y}(s)|| \leq \mathbb{B}_1 + \upsilon\log|s-1|, \qquad |\mathbb{W}(s)| \leq \mathbb{B}_2 + \upsilon\log|s-1|. \tag{17.32}$$

Proof. On $\Re s = 2$, $|s-1| = \sqrt{1+t^2} \geq (2+|t|)/(2\sqrt{2})$, so (17.31) gives $|\mathcal{Y}(s)|^{\pm 1} \leq e^{B_1}|s-1|^{\upsilon}$. A priori growth: on $\Re s \geq 2$, $|\log|\zeta_{\mathrm{RVM}}|| \leq \mathrm{c}_{\mathrm{RVM}}$. The quotient $\zeta_{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}}$ is a finite product of k^* factors $\frac{(\kappa_k^2+w^2)(\frac{1}{4}+\tau_k^2)}{(\frac{1}{4}+\kappa_k^2)(\tau_k^2+w^2)}$ with $w = s - \frac{1}{2}$. Since $|\kappa^2+w^2| = |\kappa - \Im s + i\Re w|\,|\kappa + \Im s - i\Re w| \geq (\Re w)^2 \geq \frac{9}{4}$, each factor and its inverse are $O((2+|s|)^4)$. Hence $|\mathcal{Y}|^{\pm 1} \leq \mathrm{A}_0(2+|s|)^{4k^*}$. Apply Lemma 17.29(b) to $\mathfrak{g}_+ := \mathcal{Y}\,(s-1)^{-\upsilon}e^{-\mathbb{B}_1}$ and to $\mathfrak{g}_- := \mathcal{Y}^{-1}(s-1)^{-\upsilon}e^{-\mathbb{B}_1}$, with the principal branch of $(s-1)^{-\upsilon}$, holomorphic in $\Re s > 1$ with $|(s-1)^{-\upsilon}| = |s-1|^{-\upsilon}$; both are holomorphic in $\Re s > 1$ and of polynomial growth on $\Re s \geq 2$. Finally $\mathbb{W} = \log|\mathcal{Y}| + \Re\mathcal{D}_{\mathsf{a}}$. □

Lemma 17.32 (Bound on the half-strip)**.** *Let* $\mathbb{S} := \{-1 \leq \Re s \leq 2,\ \Im s \geq 1\}$ *and* $\mathbb{B}_3 := \max\bigl(\mathbb{B}_2 + \frac{3}{4}\log 2,\ \mathrm{c}_\tau + \mathfrak{B}(C)\bigr)$. *Then*

$$\mathbb{W}(s) \leq \mathbb{B}_3 + \bigl(\tfrac{3}{2}+\upsilon\bigr)\log|s+2| \qquad (s \in \mathbb{S}). \tag{17.33}$$

Proof. $\mathbb{W}$ is subharmonic in the interior of $\mathbb{S}$, since $\log|G|$ is harmonic there (G has no zeros or poles with $\Im s \geq 1$), and it is upper semicontinuous on $\mathbb{S}$. The majorant $\mathrm{h}_0(s) := (\frac{3}{2}+\upsilon)\log|s+2|$ is harmonic and nonnegative on $\mathbb{S}$. We check the three sides.

Right side. By (17.32), $\mathbb{W}(2+it) \leq \mathbb{B}_2 + \upsilon\log|1+it| \leq \mathbb{B}_2 + \mathrm{h}_0(2+it)$.

Left side. $P_{\boldsymbol{\kappa}}$ and G are real, so $\mathbb{W}(\bar{s}) = \mathbb{W}(s)$. Combined with $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$, this gives $\mathbb{W}(-1+it) = \mathbb{W}(2+it) + \log|G(2+it)/G(-1+it)|$. By $\Gamma(z+1) = z\Gamma(z)$ and the reflection formula, $|\Gamma(\frac{1}{2}+iy)|^2 = \pi/\cosh\pi y$ and $|\Gamma(1+iy)|^2 = \pi y/\sinh\pi y$, so

$$\Bigl|\frac{G(2+it)}{G(-1+it)}\Bigr| = \frac{|1+it|}{2}\,\pi^{-3/2}\Bigl(\frac{t}{2}\coth\frac{\pi t}{2}\Bigr)^{1/2} \leq (1+t)^{3/2} \qquad (t \geq 1). \tag{17.34}$$

For the inequality: $|1+it| \leq 1+t$, $\frac{t}{2}\coth\frac{\pi t}{2} \leq \frac{1+t}{2}\coth\frac{\pi}{2}$, and $\frac{1}{2}\pi^{-3/2}(\frac{1}{2}\coth\frac{\pi}{2})^{1/2} < 1$. Since $(1+t) \leq \sqrt{2}|1+it|$ and $|s+2| = |1+it|$ at $s = -1+it$, we get $\mathbb{W}(-1+it) \leq \mathbb{B}_2 + \frac{3}{4}\log 2 + \mathrm{h}_0(-1+it)$.

Bottom side. For $s = \sigma + i$, $\mathbb{W}(s) = \log|\zeta_{\mathrm{RVM}}(s)| + \sum_{k\leq k^*}\Re(\mathfrak{f}(\kappa_k;s) - \mathfrak{f}(\tau_k;s)) \leq \mathrm{c}_\tau + \mathfrak{B}(C) \leq \mathrm{c}_\tau + \mathfrak{B}(C) + \mathrm{h}_0(s)$.

Growth. $\log|P_{\boldsymbol{\kappa}}(s)| \leq \sum_k \log(1+|q(s)|/\kappa_k^2)$. Split at $\kappa_k \leq 2|s|$ and $\kappa_k > 2|s|$: the first part has $N_{\boldsymbol{\kappa}}(2|s|) = O(|s|\log|s|)$ terms, each at most $\log(1+|q(s)|/4\pi^2) = O(\log|s|)$; in the second, $\log(1+|q(s)|/\kappa_k^2) \leq |q(s)|\kappa_k^{-2}$, and $\sum_{\kappa_k > 2|s|}\kappa_k^{-2} = O(\log|s|/|s|)$ by partial summation with $N_{\boldsymbol{\kappa}}(t) = O(t\log t)$. Hence $\log|P_{\boldsymbol{\kappa}}(s)| = O(|s|\log^2|s|)$, and $-\log|G(s)| \leq \frac{\pi}{4}|\Im s| + O(\log|s|)$ on $\mathbb{S}$ by Stirling's formula [54, §12.33]. Hence $\mathbb{W} - \mathrm{h}_0 \leq \mathrm{A}_0 e^{c\Im s}$ for any $c > 0$. Lemma 17.29(a), with width 3 and $c < \pi/3$, applied to $\mathbb{W} - \mathrm{h}_0$ gives the claim. □

Lemma 17.33 (Backlund–Jensen on a horizontal segment)**.** *There are absolute constants* $T_0 \geq 3$ *and* c_5 *such that, for* $T \geq T_0$ *with* $T \notin \{\kappa_k\}$, *the variation of* $\arg P_{\boldsymbol{\kappa}}$ *along the segment from* $2+iT$ *to* $\frac{1}{2}+iT$ *is at most* $\pi(n_T+1)$, *where*

$$n_T \leq \frac{(\frac{5}{2}+\upsilon)\log(T+8) + \mathbb{E}(T) + \mathbb{B}_0 + \mathbb{B}_3 + \mathbb{M}\mathrm{c}_\ell + 2\mathrm{c}_5}{\log(4/3)}. \tag{17.35}$$

Proof. Choose $\mathrm{a}_T \in \mathbb{R}$ with $e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(2+iT) > 0$ and put $\mathfrak{F}_T(z) := \frac{1}{2}\bigl(e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(z+iT) + e^{-i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(z-iT)\bigr)$, which is entire. Since $P_{\boldsymbol{\kappa}}$ is real, $P_{\boldsymbol{\kappa}}(x-iT) = \overline{P_{\boldsymbol{\kappa}}(x+iT)}$ for real x, so $\mathfrak{F}_T(x) = \Re(e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(x+iT))$ and $\mathfrak{F}_T(2) = |P_{\boldsymbol{\kappa}}(2+iT)|$. If $\mathfrak{F}_T$ has n_T zeros on $[\frac{1}{2},2]$, the segment splits into at most n_T+1 pieces on which $\Re(e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}})$ has constant sign. On each piece arg varies by less than π, which gives the first claim [51, §9.4].

The segment $[\frac12,2]$ lies in the closed disc $|z-2|\le\frac32$. For $\frac32<r<2$, Jensen's formula on $|z-2|\le R=2$ gives $n_T\log\frac{R}{r}\le\max_{|z-2|=2}\log|\mathfrak{F}_T(z)|-\log|\mathfrak{F}_T(2)|$, because the zeros in the closed disc of radius r, which include the n_T zeros on the segment, contribute at least $n_T\log(R/r)$; letting $r\downarrow\frac32$ gives the inequality with $\log\frac43$. Moreover $|\mathfrak{F}_T(z)|\le\max_{|\zeta-2|\le2}|P_{\boldsymbol\kappa}(\zeta+iT)|$, because $|P_{\boldsymbol\kappa}(z-iT)|=|P_{\boldsymbol\kappa}(\bar z+iT)|$. Write $\log|P_{\boldsymbol\kappa}|=\mathbb{W}+\log|G|$. On the disc $\{|s-(2+iT)|\le2\}$, the part with $\Re s\le2$ lies in $\mathbb{S}$ and $|s+2|\le T+6$, so Lemma 17.32 gives $\mathbb{W}\le\mathbb{B}_3+(\frac32+\upsilon)\log(T+6)$. The part with $\Re s\ge2$ satisfies $\mathbb{W}\le\mathbb{B}_2+\upsilon\log(T+5)$ by Lemma 17.31, a smaller bound. At the centre, (17.31) gives $\mathbb{W}(2+iT)\ge-(\mathbb{B}_0+\mathbb{E}(T)+\mathbb{M}\mathrm{c}_\ell)$. For the oscillation of $\log|G|$ on the disc: by Stirling's formula for Γ'/Γ in $|\arg z|\le\frac34\pi$ [54, §12.33], $|G'/G(s)|=|\frac1s+\frac{1}{s-1}-\frac12\log\pi+\frac12\frac{\Gamma'}{\Gamma}(\frac s2)|\le\frac12\log(T+8)+\mathrm{c}_5$ on the disc for $T\ge T_0$. So the oscillation is at most $\log(T+8)+2\mathrm{c}_5$. Collecting terms gives (17.35). □

Lemma 17.34 (Argument principle)**.** *For $T>0$, $T\notin\{\kappa_k\}$, let $\mathbb{G}_T$ be the path $2\to2+iT\to\frac12+iT$. Then*

$$\pi N_{\boldsymbol\kappa}(T)=\Delta_{\mathbb{G}_T}\arg P_{\boldsymbol\kappa}=\pi+\vartheta(T)+\Delta_{\mathbb{G}_T}\arg\zeta_{\boldsymbol\kappa},\tag{17.36}$$

where $\vartheta(T)=\arg(\pi^{-iT/2}\Gamma(\frac14+\frac{iT}{2}))$ is the Riemann–Siegel theta function, defined by continuous variation from $\vartheta(0)=0$.

Proof. Take the rectangle $\mathfrak{R}_T$ with vertices $2\pm iT$, $-1\pm iT$. $P_{\boldsymbol\kappa}$ has no zeros on $\partial\mathfrak{R}_T$: its zeros lie on $\Re s=\frac12$, and $T\notin\{\kappa_k\}$, so $\frac12\pm iT$ are not zeros. Its zeros inside $\mathfrak{R}_T$ are $\frac12\pm i\kappa_k$ with $\kappa_k<T$, counted with multiplicity (coincident $\kappa_j=\kappa_k$ give a multiple zero), $2N_{\boldsymbol\kappa}(T)$ in total. Reflection $s\mapsto1-\bar s$ maps $\mathbb{G}_T$ to the path $-1\to-1+iT\to\frac12+iT$. Since $P_{\boldsymbol\kappa}(1-\bar s)=\overline{P_{\boldsymbol\kappa}(s)}$, the variation of $\arg P_{\boldsymbol\kappa}$ along the reversed image path $\frac12+iT\to-1+iT\to-1$ equals $\Delta_{\mathbb{G}_T}\arg P_{\boldsymbol\kappa}$. The two lower quarters are handled in the same way with $s\mapsto\bar s$. The four quarter paths make up $\partial\mathfrak{R}_T$, so $4\Delta_{\mathbb{G}_T}\arg P_{\boldsymbol\kappa}=2\pi\cdot2N_{\boldsymbol\kappa}(T)$.

For the second equality, write $P_{\boldsymbol\kappa}=G\zeta_{\boldsymbol\kappa}$. Along $\mathbb{G}_T$, $s(s-1)$ moves from 2 to $-(\frac14+T^2)$ inside the closed upper half-plane, so its argument varies by π. $\pi^{-s/2}\Gamma(s/2)$ is holomorphic and zero-free in $\Re s>0$, so its argument variation along $\mathbb{G}_T$ equals that along $[2,\frac12]\cup[\frac12,\frac12+iT]$, which is $0+\vartheta(T)$. □

Theorem 17.35 (Counting bound for quasi-solutions)**.** *There are absolute constants T_0 and c_5 such that, for every finite section in $\mathbb{K}(C,\mathbb{M})$ and every $T\ge T_0$,*

$$\bigl|N_{\boldsymbol\kappa}(T)-n_{\mathrm{RVM}}(T)\bigr|\le(\mathrm{c}_1+\mathrm{c}_2\upsilon)\log(T+8)+\mathrm{c}_3\mathbb{B}+\mathrm{c}_5,\tag{17.37}$$

with

$$\mathrm{c}_1=\frac{5}{2\log(4/3)}=8.690\ldots,\quad \mathrm{c}_2=\frac{2}{\log(4/3)}+\frac1\pi=7.270\ldots,\quad \mathrm{c}_3=\frac{1}{\log(4/3)}+\frac1\pi=3.794\ldots,\tag{17.38}$$

and $\mathbb{B}:=\mathbb{B}_0+\mathbb{B}_3+\mathbb{M}\mathrm{c}_\ell$. The constants $\mathrm{c}_1,\ldots,\mathrm{c}_5$ do not depend on k^*, n^*, C, $\mathbb{M}$ or υ; $\mathbb{B}$ depends on C (through $\mathfrak{B}(C)$), on $\mathbb{M}$, and linearly on υ (through $\mathbb{B}_2$), so the total coefficient of υ in (17.37) *is $\mathrm{c}_2\log(T+8)+\mathrm{c}_3\log(2\sqrt2)$. In Corollary 17.36 the parameters C, $\mathbb{M}$, υ are fixed before $k\to\infty$.*

Proof. By continuity from the right it suffices to take $T\notin\{\kappa_k\}$. By Lemma 17.34, $N_{\boldsymbol\kappa}(T)-1-\vartheta(T)/\pi=\pi^{-1}\Delta_{\mathbb{G}_T}\arg\zeta_{\boldsymbol\kappa}$.

Vertical piece. $\arg\zeta_{\boldsymbol\kappa}=\Im\log\mathcal{Y}+\Im\mathcal{D}_{\mathsf{a}}$, with $\Im\log\mathcal{Y}(2)=\Im\mathcal{D}_{\mathsf{a}}(2)=0$. Its variation is at most $\mathbb{E}(T)+\mathbb{M}\mathrm{c}_\ell$ by Lemma 17.30.

Horizontal piece. The variation is $\Delta\arg P_{\boldsymbol\kappa}-\Delta\arg G$. The first is at most $\pi(n_T+1)$ by Lemma 17.33. The second is $-\Im\int_{1/2}^{2}(G'/G)(\sigma+iT)\,d\sigma$. Here $|\Im(G'/G)|\le|\frac1s|+|\frac{1}{s-1}|+\frac12|\arg\frac s2|+O(1/T)\le1$ for $T\ge T_0$, by Stirling's formula for Γ'/Γ [54, §12.33]. So the second variation is at most $\mathrm{c}_6:=\frac32$.

Main term. By the Stirling expansion of ϑ [54, §12.33], $1+\vartheta(T)/\pi = \mathcal{N}_{\mathrm{RVM}}(T)+O(1/T)$, and this error is at most $1/T$ for $T \ge T_0$. With (17.2):

$$|N_{\boldsymbol{\kappa}}(T) - n_{\mathrm{RVM}}(T)| \le n_T + 1 + \frac{\mathbb{E}(T) + \mathbb{M}\mathrm{c}_\ell + \mathrm{c}_6}{\pi} + \frac{1}{T} + \frac{1}{2}. \tag{17.39}$$

Insert (17.35) and $\mathbb{E}(T) \le \upsilon \log(T+8)$. □

Corollary 17.36 (Displacement bound)**.** *Assume $T_0 \ge 2\pi e$. For every k with $\tau_k - C \ge T_0$,*

$$|\kappa_k - \tau_k| \le \frac{2\pi\big[(\mathrm{c}_1 + \mathrm{c}_2\upsilon)\log(\tau_k + C + 8) + \mathrm{c}_3\mathbb{B} + \mathrm{c}_5\big]}{\log((\tau_k - C)/2\pi)}. \tag{17.40}$$

In particular $\limsup_{k\to\infty} |\kappa_k - \tau_k| \le 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon)$ *uniformly over sections with given $C, \mathbb{M}, \upsilon$. For every* $\mathrm{p} > 0$ *there is* $k_0(C, \mathbb{M}, \upsilon, \mathrm{p})$ *such that* $|\kappa_k - \tau_k| \le 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$ *for* $k \ge k_0$.

Proof. Let $\mathbb{D}(T)$ be the right-hand side of (17.37); it is nondecreasing. Suppose $\kappa_k > \tau_k + \mathrm{q}$ with $0 < d \le C$, and let $T := \tau_k + \mathrm{q} - \mathrm{h}'$ with small $\mathrm{h}' > 0$. The list is nondecreasing, so $N_{\boldsymbol{\kappa}}(T) \le k-1$. By (17.2) and because $\mathcal{N}'_{\mathrm{RVM}}$ is increasing, $n_{\mathrm{RVM}}(T) \ge \mathcal{N}_{\mathrm{RVM}}(T) - \frac{1}{2} \ge k - 1 + (\mathrm{q}-\mathrm{h}')\mathcal{N}'_{\mathrm{RVM}}(\tau_k)$. Hence $(\mathrm{q}-\mathrm{h}')\frac{1}{2\pi}\log\frac{\tau_k}{2\pi} \le \mathbb{D}(\tau_k + C)$. Now suppose $\kappa_k < \tau_k - \mathrm{q}$, and let $T := \tau_k - \mathrm{q} + \mathrm{h}'$. Then $N_{\boldsymbol{\kappa}}(T) \ge k$ and $n_{\mathrm{RVM}}(T) \le \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \le k - (\mathrm{q}-\mathrm{h}')\mathcal{N}'_{\mathrm{RVM}}(\tau_k - C)$. Let $\mathrm{h}' \to 0$ in both cases. □

Remark 17.37 (What Corollary 17.36 does not give). The proof uses the imposed tube $|\kappa_k - \tau_k| \le C$ twice: through $\mathfrak{B}(C)$ in $\mathbb{B}$, and to replace $\mathbb{D}(\tau_k + d)$ by $\mathbb{D}(\tau_k + C)$. The conclusion is that, inside any finite tube, the coordinates of sufficiently high index lie in the smaller tube of radius $2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$. So the constraint is inactive there once $C > 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$. For the indices $k < k_0(C, \mathbb{M}, \upsilon, \mathrm{p})$ nothing is proved, and the corollary does not show that unconstrained near-minimisers lie in a universal tube. The proof uses finite sections: Lemma 17.31 needs the polynomial growth $\mathrm{A}_0(2+|s|)^{4k^*}$. Extending it to infinite lists requires a separate growth bound for $P_{\boldsymbol{\kappa}}/P_{\boldsymbol{\tau}}$ on $\Re s \ge 2$, which is not given here. The constant $2\pi\mathrm{c}_1 = 54.60\ldots$ is far from optimal.

Heuristic 17.38 (What a finite section controls)**.** The quantity in Theorem 17.35 is $\mathbb{E}(T)$, the L^1 norm on $\Re s = 2$ of the Laplace transform of the residual. It is not the Euclidean residual that a numerical section minimises over finitely many tests. Let the tests be $\mathrm{j}_0(u)\varphi(u)$ with a fixed window $\mathrm{j}_0 \in C_c^\infty(0,\infty)$, $\mathrm{j}_0 \equiv 1$ on $[u_0, u^*]$, and φ band-limited to $|\omega| \le \mathrm{w}^\sharp$. Then the kernel $\mathrm{j}_0(u)e^{-\frac{3}{2}u}e^{-ivu}$ is, approximately, in the span of the tests only for $|v| \lesssim \mathrm{w}^\sharp$. The residual on $(0, u_0)$ is then not controlled, and for a list at distance C from the clock it is expected to be of size $C\log(1/u)$ as $u \to 0$. Its contribution to $\mathbb{E}(T)$ is then $O(CTu_0\log(1/u_0))$. Since $u_0 \gtrsim 1/\mathrm{w}^\sharp$ in a band-limited section, Theorem 17.35 would apply with a given υ_0 only at heights $T \lesssim \upsilon_0 \mathrm{w}^\sharp/(C\log \mathrm{w}^\sharp)$. None of these size estimates is proved here. They indicate which norm a section must make small. Lemma 17.30 shows that this norm is the right one.

Heuristic 17.39 (Informative part of a section)**.** Where the atoms are denser than the resolution of the tests, $e^{-u} \ll \pi/\mathrm{w}^\sharp$, the masses alone can approximate a smooth target by a Riemann sum $\mathsf{a}_n \approx \mathfrak{g}(\log n)/n$ without moving any ordinate. That part of the window then carries no information on the ordinates. Counting unknowns against equations in the remaining window, $u^* \le \log \mathrm{w}^\sharp + O(1)$, gives an overdetermined system. This proves nothing about compatibility. It only explains why B is an existence statement and not a dimension count.

17.5. **A Kadec–Avdonin criterion with blocks of variable length.**

17.5.1. *Discrepancy and generating function.*

Definition 17.40 (Discrepancy, Poisson and conjugate operators)**.** Let $\mathscr{X} = (\mathrm{x}_k)_{k\in\mathbb{Z}}$ be a real nondecreasing sequence with

$$|\mathrm{x}_k - (k - \tfrac{1}{2})| \le \mathrm{c}_{\mathscr{X}} \log(2 + |k|) \qquad (k \in \mathbb{Z}). \tag{17.41}$$

Its *discrepancy* with respect to the half-integers is

$$\Psi_{\mathscr{X}}(\mathrm{x}) := \sum_{k\in\mathbb{Z}} \big(\mathbf{1}_{[\mathrm{x}_k,\infty)}(\mathrm{x}) - \mathbf{1}_{[k-\frac12,\infty)}(\mathrm{x})\big). \tag{17.42}$$

The k-th term is ±1 on the interval between x_k and $k-\frac12$ and 0 elsewhere. By (17.41), at most $O(\log(2+|\mathrm{x}|))$ terms are nonzero at x, so $|\Psi_{\mathscr{X}}(\mathrm{x})| \le \mathrm{c}'_{\mathscr{X}} \log(2+|\mathrm{x}|)$ and $\Psi_{\mathscr{X}} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$. For $\mathrm{g} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$ and $\mathrm{y}>0$ put

$$\mathbb{P}_{\mathrm{y}}\mathrm{g}(\mathrm{x}) := \frac{1}{\pi}\int \frac{\mathrm{y}\,\mathrm{g}(\mathrm{s})\,d\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}, \qquad \mathbb{Y}_{\mathrm{y}}\mathrm{g}(\mathrm{x}) := \frac{1}{\pi}\int \mathrm{g}(\mathrm{s})\Big[\frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2} + \frac{\mathrm{s}}{1+\mathrm{s}^2}\Big]d\mathrm{s}. \tag{17.43}$$

The last term in $\mathbb{Y}_{\mathrm{y}}$ only makes the integral converge. Changing it changes $\mathbb{Y}_{\mathrm{y}}\mathrm{g}$ by a constant, which is irrelevant for all statements below.

Lemma 17.41 (Generating function)**.** *Let $\mathscr{X} = (\mathrm{x}_k)_{k\in\mathbb{Z}}$ satisfy* (17.41)*, understood as a sequence indexed by $\mathbb{Z}$ in which a point of multiplicity m occurs m times. Let $\mathcal{F}_0(\mathfrak{z}) := \cos\pi\mathfrak{z}$, and fix $\mathfrak{z}_0 \in \mathbb{R}\setminus(\mathscr{X}\cup(\mathbb{Z}+\frac12))$. The series*

$$\mathfrak{G}_{\mathscr{X}}(\mathfrak{z}) := \sum_{k\in\mathbb{Z}}\Big(\frac{1}{\mathfrak{z}-\mathrm{x}_k} - \frac{1}{\mathfrak{z}-k+\frac12}\Big) \tag{17.44}$$

converges absolutely and locally uniformly off $\mathscr{X}\cup(\mathbb{Z}+\frac12)$, and

$$\mathcal{F}_{\mathscr{X}}(\mathfrak{z}) := \mathcal{F}_0(\mathfrak{z})\exp\int_{\mathfrak{z}_0}^{\mathfrak{z}} \mathfrak{G}_{\mathscr{X}}(\mathrm{z})\,d\mathrm{z} \tag{17.45}$$

is a real entire function whose zero set is $\mathscr{X}$, with multiplicities. Moreover:

(i) *for $\mathrm{y}>0$ there is a constant $\mathrm{c}''_{\mathscr{X}}$ such that*

$$\log|\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y})| = \log|\mathcal{F}_0(\mathrm{x}+i\mathrm{y})| + \pi\,\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x}) + \mathrm{c}''_{\mathscr{X}}; \tag{17.46}$$

(ii) *$\mathcal{F}_{\mathscr{X}}$ is of exponential type π, and its indicator diagram is the segment $[-i\pi, i\pi]$;*

(iii) *if $\mathscr{X}$ is symmetric, $\mathrm{x}_{1-k} = -\mathrm{x}_k$, and $\mathfrak{z}_0 = 0 \notin \mathscr{X}$, then $\mathcal{F}_{\mathscr{X}}$ is even.*

Proof. Each term of $\mathfrak{G}_{\mathscr{X}}$ equals $\frac{x_k-k+\frac12}{(\mathfrak{z}-\mathrm{x}_k)(\mathfrak{z}-k+\frac12)} = O(\log(2+|k|)/k^2)$ locally uniformly, so the series converges. $\mathfrak{G}_{\mathscr{X}}$ is meromorphic with simple poles, with residue $+m$ at a point of $\mathscr{X}$ of multiplicity m and -1 at each half-integer; coincident poles combine their residues. The residues are integers, so $\exp\int_{\mathfrak{z}_0}^{\mathfrak{z}}\mathfrak{G}_{\mathscr{X}}$ is single-valued. It has a pole of order one at each half-integer not in $\mathscr{X}$, cancelled by the zero of $\mathcal{F}_0$. Hence $\mathcal{F}_{\mathscr{X}}$ is entire with zero set $\mathscr{X}$. $\mathfrak{G}_{\mathscr{X}}(\bar{\mathfrak{z}}) = \overline{\mathfrak{G}_{\mathscr{X}}(\mathfrak{z})}$ and $\mathfrak{z}_0$ is real, so $\mathcal{F}_{\mathscr{X}}$ is real on $\mathbb{R}$. Under the symmetry in (iii), $\mathfrak{G}_{\mathscr{X}}$ is odd, so $\int_0^{\mathfrak{z}}\mathfrak{G}_{\mathscr{X}}$ is even, and $\mathcal{F}_{\mathscr{X}}$ is even.

(i) For $\Im\mathfrak{z}\neq0$ put $\mathbb{A}_{\mathscr{X}}(\mathfrak{z}) := \int\Psi_{\mathscr{X}}(\mathrm{s})\big[\frac{1}{\mathfrak{z}-\mathrm{s}}+\frac{\mathrm{s}}{1+\mathrm{s}^2}\big]d\mathrm{s}$. For $f(\mathrm{s}) := \frac{1}{\mathfrak{z}-\mathrm{s}}$, the identity $f(\mathrm{x}_k)-f(k-\frac12) = -\int f'(\mathrm{s})\big(\mathbf{1}_{[\mathrm{x}_k,\infty)} - \mathbf{1}_{[k-\frac12,\infty)}\big)(\mathrm{s})\,d\mathrm{s}$ summed over k (Fubini applies because $\sum_k\int|f'||\cdots| < \infty$ by (17.41)) gives $\mathfrak{G}_{\mathscr{X}}(\mathfrak{z}) = -\int\Psi_{\mathscr{X}}(\mathrm{s})(\mathfrak{z}-\mathrm{s})^{-2}d\mathrm{s} = \mathbb{A}'_{\mathscr{X}}(\mathfrak{z})$. So $\mathcal{F}_{\mathscr{X}} = \mathrm{c}_+\mathcal{F}_0 e^{\mathbb{A}_{\mathscr{X}}}$ on $\Im\mathfrak{z}>0$, with $\mathrm{c}_+\neq0$. Since $\Re\frac{1}{\mathfrak{z}-\mathrm{s}} = \frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}$, $\Re\mathbb{A}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y}) = \pi\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})$ and $\Im\mathbb{A}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y}) = -\pi\mathbb{P}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})$.

(ii) Split $\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})$ into $|\mathrm{s}-\mathrm{x}|\le1$, $1<|\mathrm{s}-\mathrm{x}|$ with $|\mathrm{s}|\le2|\mathfrak{z}|+2$, and $|\mathrm{s}|>2|\mathfrak{z}|+2$. Using $|\Psi_{\mathscr{X}}(\mathrm{s})|\le\mathrm{c}'_{\mathscr{X}}\log(2+|\mathrm{s}|)$, $\int_{|u|\le1}\frac{|u|\,du}{u^2+\mathrm{y}^2} = \log(1+\mathrm{y}^{-2})$, and, for $|\mathrm{s}|>2|\mathfrak{z}|+2$, $|\frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}+\frac{\mathrm{s}}{1+\mathrm{s}^2}| \le 8(1+|\mathfrak{z}|)/\mathrm{s}^2$, one gets

$$|\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})| \le \mathrm{c}'''_{\mathscr{X}}(1+\log(1+\mathrm{y}^{-2}))\log^2(4+|\mathfrak{z}|) \qquad (\mathfrak{z}=\mathrm{x}+i\mathrm{y},\ \mathrm{y}>0). \tag{17.47}$$

For $|\mathrm{y}|\ge1$, $\log|\mathcal{F}_{\mathscr{X}}(\mathfrak{z})| \le \pi|\mathrm{y}| + O(\log^2(4+|\mathfrak{z}|))$ by (17.46), (17.47) and $|\mathcal{F}_{\mathscr{X}}(\bar{\mathfrak{z}})| = |\mathcal{F}_{\mathscr{X}}(\mathfrak{z})|$. For $|\mathrm{y}_1|<1$, apply the sub-mean-value property of $\log|\mathcal{F}_{\mathscr{X}}|$ on the circle $|\mathfrak{z}-\mathfrak{z}_1| = \frac12$. The factor $\log(1+\mathrm{y}^{-2})$ is integrable along the circle uniformly in y_1, since it has at most two logarithmic singularities. This gives $\log|\mathcal{F}_{\mathscr{X}}(\mathfrak{z}_1)| \le \frac{3\pi}{2} + O(\log^2(4+|\mathfrak{z}_1|))$ for $|\mathrm{y}_1|<1$. Hence the type is at most π and the indicator satisfies $\mathrm{h}(\mathrm{u}) \le \pi|\sin\mathrm{u}|$. Since (17.46) is an equality and (17.47) is two-sided,

$\log|\mathcal{F}_{\mathscr{X}}(iy)| = \log|\cos(i\pi \mathrm{y})| + \pi\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(0) + \mathrm{c}''_{\mathscr{X}} = \pi|\mathrm{y}| + O(\log^2(4+|\mathrm{y}|))$ for $|\mathrm{y}| \ge 1$, so $\mathrm{h}(\pm\frac{\pi}{2}) = \pi$. The indicator diagram lies in $[-i\pi, i\pi]$, whose support function is $\pi|\sin \mathrm{u}|$, and it contains $\pm i\pi$, so it equals that segment. □

17.5.2. *The criterion.* A weight $\mathrm{w} > 0$ on $\mathbb{R}$ is in (A_2) if

$$[\mathrm{w}]_{A_2} := \sup_I \Big(\frac{1}{|I|}\int_I w\Big)\Big(\frac{1}{|I|}\int_I w^{-1}\Big) < \infty, \tag{17.48}$$

the supremum being over bounded intervals.

Theorem 17.42 (Block $\frac{1}{4}$-criterion)**.** *Let $\mathscr{X}$ satisfy* (17.41) *and suppose $\Psi_{\mathscr{X}} = \mathrm{A}+\mathrm{B}$ with measurable A, B such that:*

(i) $\|\mathrm{A}\|_\infty \le \mathrm{d}_\star < \frac{1}{4}$;
(ii) $\sup_x |\mathbb{Y}_1\mathrm{B}(\mathrm{x}) - \mathrm{c}_\mathrm{B}| \le \mathbb{U}$ *for some constant* c_B.

Then $\mathrm{w}(\mathrm{x}) := |\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i)|^2$ *is in* (A_2) *and*

$$[\mathrm{w}]_{A_2} \le \frac{\pi^2 \coth^2(\pi)\, e^{4\pi\mathbb{U}}}{\cos^2(2\pi \mathrm{d}_\star)}. \tag{17.49}$$

Proof. $\mathrm{B} = \Psi_{\mathscr{X}} - \mathrm{A} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$, so $\mathbb{Y}_1\Psi_{\mathscr{X}} = \mathbb{Y}_1\mathrm{A} + \mathbb{Y}_1\mathrm{B}$. By (17.46), $\mathrm{w} = e^{2\mathrm{c}''_{\mathscr{X}}}|\mathcal{F}_0(\mathrm{x}+i)|^2 e^{2\pi\mathbb{Y}_1\mathrm{A}} e^{2\pi\mathbb{Y}_1\mathrm{B}}$.

Put $\mathbb{A}_\mathrm{A}(\mathfrak{z}) := \int \mathrm{A}(\mathrm{s})\big[\frac{1}{\mathfrak{z}-\mathrm{s}} + \frac{\mathrm{s}}{1+\mathrm{s}^2}\big]d\mathrm{s}$. It is holomorphic on $\Im\mathfrak{z} > 0$, and $\mathbb{A}_\mathrm{A}(\mathfrak{o}+i)$ is holomorphic on $\Im\mathfrak{o} > -1$. Let $\mathcal{W}(\mathfrak{o}) := \exp(2\mathbb{A}_\mathrm{A}(\mathfrak{o}+i))$. For $\mathfrak{o} = \mathrm{x}_1 + i\mathrm{y}_1$ with $\mathrm{y}_1 \ge 0$,

$$|\mathcal{W}(\mathfrak{o})| = e^{2\pi\mathbb{Y}_{1+\mathrm{y}_1}\mathrm{A}(\mathrm{x}_1)}, \qquad \arg\mathcal{W}(\mathfrak{o}) = -2\pi\mathbb{P}_{1+\mathrm{y}_1}\mathrm{A}(\mathrm{x}_1) \in [-2\pi\mathrm{d}_\star, 2\pi\mathrm{d}_\star]. \tag{17.50}$$

So $\mathcal{W}$ maps the closed upper half-plane into the sector $|\arg| \le \mathrm{b}_\star := 2\pi\mathrm{d}_\star < \frac{\pi}{2}$, and $|\mathcal{W}(\mathrm{x})| = e^{2\pi\mathbb{Y}_1\mathrm{A}(\mathrm{x})}$ on $\mathbb{R}$.

$\Re\mathcal{W}$ is a positive harmonic function on $\Im\mathfrak{o} > 0$, continuous up to $\mathbb{R}$. By the Herglotz representation it equals $\mathrm{c}\,\mathrm{y}_1 + \int \mathbb{P}_\mathfrak{o}(t)\,d\mathrm{n}(t)$ with $\mathrm{c} \ge 0$, $\mathrm{n} \ge 0$, and $\mathrm{n} = \Re\mathcal{W}(t)\,dt$ by continuity. Here $\mathbb{P}_\mathfrak{o}(t) := \frac{1}{\pi}\frac{\mathrm{y}_1}{(t-\mathrm{x}_1)^2+\mathrm{y}_1^2}$. Hence $\Re\mathcal{W}(\mathfrak{o}) \ge \int \mathbb{P}_\mathfrak{o}(t)\Re\mathcal{W}(t)\,dt$. For an interval I with centre x_I, put $\mathfrak{o}_I := \mathrm{x}_I + i|I|/2$. Then $\mathbb{P}_{\mathfrak{o}_I}(t) \ge (\pi|I|)^{-1}$ for $t \in I$, and

$$\frac{1}{|I|}\int_I |\mathcal{W}| \le \frac{1}{\cos\mathrm{b}_\star}\frac{1}{|I|}\int_I \Re\mathcal{W} \le \frac{\pi}{\cos\mathrm{b}_\star}\Re\mathcal{W}(\mathfrak{o}_I) \le \frac{\pi}{\cos\mathrm{b}_\star}|\mathcal{W}(\mathfrak{o}_I)|. \tag{17.51}$$

$1/\mathcal{W}$ maps into the same sector, so the same holds for it. Multiplying, $[|\mathcal{W}|]_{A_2} \le \pi^2/\cos^2\mathrm{b}_\star$.

Finally $\mathrm{w} = \mathrm{g}\,|\mathcal{W}|$ with $\mathrm{g} := e^{2\mathrm{c}''_{\mathscr{X}}}|\mathcal{F}_0(\mathrm{x}+i)|^2 e^{2\pi\mathbb{Y}_1\mathrm{B}}$. Since $|\cos\pi(\mathrm{x}+i)|^2 = \cos^2(\pi\mathrm{x})\cosh^2\pi + \sin^2(\pi\mathrm{x})\sinh^2\pi \in [\sinh^2\pi, \cosh^2\pi]$ and $e^{2\pi\mathbb{Y}_1\mathrm{B}} \in [e^{2\pi(\mathrm{c}_\mathrm{B}-\mathbb{U})}, e^{2\pi(\mathrm{c}_\mathrm{B}+\mathbb{U})}]$, we have $\sup\mathrm{g}/\inf\mathrm{g} \le \coth^2(\pi)e^{4\pi\mathbb{U}}$. For any weight w_1, $[\mathrm{g}\mathrm{w}_1]_{A_2} \le (\sup\mathrm{g}/\inf\mathrm{g})[\mathrm{w}_1]_{A_2}$. □

Corollary 17.43 (Riesz bases)**.** *Under the hypotheses of Theorem 17.42:*

(a) *if $\mathscr{X}$ is uniformly discrete (in particular without repeated points), $\{e^{i\mathrm{x}_k\mathrm{v}}\}_{k\in\mathbb{Z}}$ is a Riesz basis of $L^2(-\pi,\pi)$;*
(b) *if $\mathscr{X}$ is a finite union of uniformly discrete sets, the generalised divided differences of $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ in the sense of* [4] *form a Riesz basis of $L^2(-\pi,\pi)$. Consequently, for every partition of $\mathbb{Z}$ into consecutive blocks $\mathfrak{B}_m$ that are unions of the groups of* [4]*, the subspaces spanned by $\{\mathrm{v}^j e^{i\mathrm{x}\mathrm{v}}\}$, with $\mathrm{x} = \mathrm{x}_k$ for $k \in \mathfrak{B}_m$ and j below the multiplicity of x, form a Riesz basis of subspaces.*

Proof. By Lemma 17.41, $\mathcal{F}_{\mathscr{X}}$ is entire of exponential type with indicator diagram of width 2π and zero set $\mathscr{X}$. By Theorem 17.42, $|\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i)|^2 \in (A_2)$, which is equivalent to the Helson–Szegő condition [10, Ch. VI]. Then (a) is Pavlov's criterion [4, Thm. 1], and the first statement of (b) is [4, Thm. 3]. Both are stated for $L^2(0,2\pi)$, and translation by π multiplies each exponential by a

unimodular constant. Grouping the vectors of a Riesz basis into disjoint groups gives a Riesz basis of subspaces. □

17.5.3. *Kadec, Avdonin, and variable block length.* Write $\mathrm{x}_k = k - \frac{1}{2} + \mathrm{d}_k$, $I_k := [k-1, k)$, $\mathrm{i}_k := \mathbf{1}_{[\mathrm{x}_k,\infty)} - \mathbf{1}_{[k-\frac{1}{2},\infty)}$, so that $\Psi_{\mathscr{X}} = \sum_k \mathrm{i}_k$ and $\int \mathrm{i}_k = -\mathrm{d}_k$.

Proposition 17.44 (Bounded conjugate for block-supported oscillation)**.** *Let $\{J_m\}$ be a partition of $\mathbb{R}$ into consecutive intervals of lengths $\mathfrak{n}_m \geq 1$. Let $\mathrm{B} = \sum_m B_m$ with $\operatorname{supp} \mathrm{B}_m \subset \mathfrak{J}_m$, $\int \mathrm{B}_m = 0$ and $|\mathrm{B}_m| \leq \mathrm{D}'_m$, $\sup_m \mathrm{D}'_m < \infty$. Put*

$$\mathbb{U}_{\mathrm{loc}} := \sup_m \sup_{\mathrm{x}\in\mathfrak{J}_{m-1}\cup\mathfrak{J}_m\cup\mathfrak{J}_{m+1}} \Big|\int_{J_m} \frac{(\mathrm{x}-\mathrm{s})\,\mathrm{B}_m(\mathrm{s})}{\pi((\mathrm{x}-\mathrm{s})^2+1)}\,d\mathrm{s}\Big|, \qquad \mathbb{O} := \sup_{m_0} \sum_{|m-m_0|\geq 2} \frac{N_m^2}{\operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)^2}. \tag{17.52}$$

If $\mathbb{U}_{\mathrm{loc}} < \infty$ and $\mathbb{O} < \infty$, then $\sup_x |\mathbb{Y}_1 \mathrm{B}(\mathrm{x}) - \mathrm{c}_{\mathrm{B}}| \leq 3\mathbb{U}_{\mathrm{loc}} + \frac{\mathbb{O}}{2\pi} \sup_m \mathrm{D}'_m$ for a constant c_{B}.

Proof. The term $\frac{1}{\pi}\int \frac{\mathrm{s}}{1+\mathrm{s}^2}\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}$ equals $\frac{1}{\pi}\int(\frac{\mathrm{s}}{1+\mathrm{s}^2} - \frac{\mathrm{m}_m}{1+\mathrm{m}_m^2})\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}$, where m_m is the centre of $\mathfrak{J}_m$. Since $|\frac{d}{d\mathrm{s}}\frac{\mathrm{s}}{1+\mathrm{s}^2}| \leq \frac{1}{1+\mathrm{s}^2}$, its modulus is at most $\mathrm{D}'_m N_m^2/(2\pi(1+\operatorname{dist}(0,\mathfrak{J}_m)^2))$. If m_0 is the index with $0 \in \mathfrak{J}_{m_0}$, then $\operatorname{dist}(0,\mathfrak{J}_m) \geq \operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)$, so these terms are summable by $\mathbb{O} < \infty$. Their sum is the constant c_{B}. Let $\mathrm{x} \in \mathfrak{J}_{m_0}$. The three blocks with $|m-m_0| \leq 1$ contribute at most $3\mathbb{U}_{\mathrm{loc}}$. For $|m-m_0| \geq 2$, $\int \mathrm{B}_m = 0$ gives $\big|\int_{J_m} \mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}\big| \leq \mathrm{D}'_m N_m \cdot \frac{N_m}{2}\sup_{\mathrm{s}\in\mathfrak{J}_m}|\mathrm{k}'(\mathrm{x}-\mathrm{s})|$ with $\mathrm{k}(\mathrm{v}) := \frac{\mathrm{v}}{\pi(\mathrm{v}^2+1)}$. Moreover $|\mathrm{k}'(\mathrm{v})| \leq \frac{1}{\pi\mathrm{v}^2}$ and $|\mathrm{x}-\mathrm{s}| \geq \operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)$. Sum over m. □

Corollary 17.45 (Kadec and Avdonin, with the constant $\frac{1}{4}$)**.** (a) [35] *If $|\mathrm{d}_k| \leq \mathrm{d}_\star < \frac{1}{4}$ for all k, then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*
(b) [3] *If $\mathscr{X}$ is uniformly discrete, $\sup_k |\mathrm{d}_k| \leq \mathrm{D}$, and for a fixed $\mathfrak{n}$ the means of d_k over consecutive blocks of $\mathfrak{n}$ indices are at most $\mathrm{d}_\star < \frac{1}{4}$ in modulus, then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*

Proof. Both statements follow from Theorem 17.46 below: in (a) take $\mathfrak{n}_m \equiv 1$, and in (b) take $\mathfrak{n}_m \equiv \mathfrak{n}$. In both cases the blocks at index distance $j \geq 2$ from $\mathfrak{J}_{m_0}$ are at distance $(j-1)\mathfrak{n}$, so $\mathbb{O} \leq 2\sum_{j\geq 2}\mathfrak{n}^2/((j-1)\mathfrak{n})^2 = \pi^2/3$ and $\mathbb{V} \leq 2\mathrm{D}\log(2+3\mathfrak{n})$; for $\mathfrak{n}=1$, $\mathrm{D}_m = 0$ and $\mathbb{V} = 0$. For (a), $\mathrm{d}_\star < \frac{1}{4}$ gives gaps $\geq 1-2L > \frac{1}{2}$, so $\mathscr{X}$ is uniformly discrete. □

Theorem 17.46 (Avdonin's theorem with variable block length)**.** *Let $\mathscr{X}$ be uniformly discrete with $\sup_k |\mathrm{d}_k| \leq \mathrm{D}$. Let $\{\mathfrak{B}_m\}$ be a partition of $\mathbb{Z}$ into consecutive blocks of $\mathfrak{n}_m$ indices, and $\mathfrak{J}_m := \bigcup_{k\in\mathfrak{B}_m} I_k$. Let $\bar{\mathrm{d}}_m$ be the mean of d_k over $\mathfrak{B}_m$, $\mathrm{D}_m := \max_{k\in\mathfrak{B}_m}|\mathrm{d}_k - \bar{\mathrm{d}}_m|$, and $\mathfrak{n}_m^* := \mathfrak{n}_{m-1} + \mathfrak{n}_m + \mathfrak{n}_{m+1}$. Assume*

(a) $\sup_m |\bar{\mathrm{d}}_m| \leq \mathrm{d}_\star < \frac{1}{4}$;
(b) $\mathbb{O} < \infty$, *with $\mathbb{O}$ as in* (17.52);
(c) $\mathbb{V} := \sup_m \mathrm{D}_m \log(2+\mathfrak{n}_m^*) < \infty$.

Then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$, and (17.49) *holds with*

$$\mathbb{U} = 16(1+\mathrm{D})^3 + \frac{6\mathbb{V}}{\pi} + \frac{V\mathbb{O}}{2\pi\log 2}. \tag{17.53}$$

Proof. Write $\Psi_{\mathscr{X}} = \mathrm{A} + \mathrm{B}_1 + \mathrm{B}_2$ with $\mathrm{A} := -\sum_m \bar{\mathrm{d}}_m \mathbf{1}_{J_m}$, $\mathrm{B}_1 := \sum_k b_k$ where $\mathrm{b}_k := \mathrm{i}_k + \mathrm{d}_k\mathbf{1}_{I_k}$, and $\mathrm{B}_2 := -\sum_m \sum_{k\in\mathfrak{B}_m}(\mathrm{d}_k - \bar{\mathrm{d}}_m)\mathbf{1}_{I_k}$. Then $\|\mathrm{A}\|_\infty \leq d$.

B_1. b_k has mean zero, $|\mathrm{b}_k| \leq 1+\mathrm{D}$, and support in an interval of length at most $2\mathrm{D}+1$ containing $\mathrm{m}_k := k - \frac{1}{2}$. With k as above, $\sup|\mathrm{k}| = \frac{1}{2\pi}$ and $|\mathrm{k}'(\mathrm{v})| \leq \frac{1}{\pi(1+\mathrm{v}^2)}$, so $|\int \mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{b}_k(\mathrm{s})\,d\mathrm{s}| = |\int(\mathrm{k}(\mathrm{x}-\mathrm{s}) - \mathrm{k}(\mathrm{x}-\mathrm{m}_k))\mathrm{b}_k(\mathrm{s})\,d\mathrm{s}| \leq (1+\mathrm{D})(2\mathrm{D}+1)\min(\frac{1}{\pi}, (2\mathrm{D}+1)\sup_{\mathrm{s}\in\operatorname{supp}\mathrm{b}_k}|\mathrm{k}'(\mathrm{x}-\mathrm{s})|)$. At most $8\mathrm{D}+5 \leq 8(1+\mathrm{D})$ indices have $|\mathrm{x}-\mathrm{m}_k| < 2(2\mathrm{D}+1)$, and they contribute at most $16(1+\mathrm{D})^3/\pi$. For the others $|\mathrm{x}-\mathrm{s}| \geq |\mathrm{x}-\mathrm{m}_k|/2$ on the support, so the term is at most $4(1+\mathrm{D})^3 \cdot \frac{4}{\pi}(4+(\mathrm{x}-\mathrm{m}_k)^2)^{-1}$. Since $\sum_k(4+(\mathrm{x}-\mathrm{m}_k)^2)^{-1} \leq \frac{1}{2} + \frac{\pi}{2}$, they contribute at most $\frac{16}{\pi}(\frac{1}{2}+\frac{\pi}{2})(1+\mathrm{D})^3$. The total is

below $16(1+\mathrm{D})^3$. The modifying term of $\mathbb{Y}_1$ contributes a convergent sum of constants, as in Proposition 17.44.

B_2. On $\mathfrak{J}_m$ it has mean zero and modulus at most D_m. For x in $\mathfrak{J}_{m-1}\cup\mathfrak{J}_m\cup\mathfrak{J}_{m+1}$, $|\int_{J_m}\mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{B}_2(\mathrm{s})\,d\mathrm{s}| \le \frac{\mathrm{D}_m}{\pi}\int_{|u|\le \mathfrak{n}_m^*}\frac{|u|\,du}{1+u^2} \le \frac{2}{\pi}\mathrm{D}_m\log(2+\mathfrak{n}_m^*) \le \frac{2\mathbb{V}}{\pi}$. Moreover $\sup_m \mathrm{D}_m \le \mathbb{V}/\log 2$. Proposition 17.44 gives $|\mathbb{Y}_1\mathrm{B}_2-\mathrm{c}| \le \frac{6\mathbb{V}}{\pi}+\frac{V\mathbb{O}}{2\pi\log 2}$.

Apply Theorem 17.42 and Corollary 17.43(a). □

Remark 17.47 (The price of variable length). For fixed $\mathfrak{n}$, conditions (b) and (c) hold automatically, and Theorem 17.46 is Avdonin's theorem. If $\mathfrak{n}_m\to\infty$, the averaging condition (a) is still only $|\bar{\mathrm{d}}_m| \le \mathrm{d}_\star < \frac14$. However, (c) then requires the oscillation inside the block to decay like $1/\log\mathfrak{n}_m$. In Theorem 17.42 the oscillation enters only through the bounded-conjugate condition (ii), which allows cancellation inside the blocks and is weaker than (c). Without some condition of this kind the conclusion fails for growing blocks: a block of length $\mathfrak{n}$ with oscillation of size D_m can make $|\mathbb{Y}_1\mathrm{B}|$ of size $\mathrm{D}_m\log\mathfrak{n}$. Lyubarskii and Seip derive Kadec's theorem from their Muckenhoupt characterisation ([40, Thm. 2] for $p=2$), and indicate that the same technique gives results of Avdonin type. I have not searched the literature for Theorem 17.46 or for the bounded-conjugate formulation of Theorem 17.42, so their novelty is not claimed. The proof of Theorem 17.42 is the sufficiency half of the Helson–Szegő argument, applied on the line $\Im\mathfrak{z}=1$.

17.6. **Application to lists near the RvM clock.**

17.6.1. *Unfolding.*

Definition 17.48 (Unfolded list)**.** Let $\mathfrak{t}_0 := 9.6769\ldots$ be the zero of $\mathcal{N}_{\mathrm{RVM}}$ in $(2\pi,\infty)$. For a C-admissible list $\boldsymbol{\kappa}$ with $\kappa_1>\mathfrak{t}_0$ (true for $\boldsymbol{\tau}$ and $\boldsymbol{\gamma}$), put $\mathrm{x}_k := \mathcal{N}_{\mathrm{RVM}}(\kappa_k)>0$ for $k\ge1$ and $\mathrm{x}_{1-k}:=-\mathrm{x}_k$. The resulting symmetric sequence $\mathscr{X}_{\boldsymbol{\kappa}}$ has discrepancy $\Psi_{\boldsymbol{\kappa}}:=\Psi_{\mathscr{X}_{\boldsymbol{\kappa}}}$, and $\mathcal{F}_{\boldsymbol{\kappa}}:=F_{\mathscr{X}_{\boldsymbol{\kappa}}}$ with $\mathfrak{z}_0=0$. For the clock, $\mathscr{X}_{\boldsymbol{\tau}}=\mathbb{Z}+\frac12$ and $\mathcal{F}_{\boldsymbol{\tau}}=\mathcal{F}_0$.

Lemma 17.49 (Unfolded discrepancy)**.** *For a C-admissible list with $\kappa_1>\mathfrak{t}_0$, $|\mathrm{x}_k-(k-\frac12)|\le\frac{C}{2\pi}\log\frac{\tau_k+C}{2\pi}$ for $k\ge1$, so* (17.41) *holds. Moreover, for $\mathrm{x}=\mathcal{N}_{\mathrm{RVM}}(t)$ with $\mathrm{x}>0$,*

$$\Psi_{\boldsymbol{\kappa}}(\mathrm{x}) = N_{\boldsymbol{\kappa}}(t)-n_{\mathrm{RVM}}(t), \qquad |N_{\boldsymbol{\kappa}}(t)-n_{\mathrm{RVM}}(t)| \le \frac{C}{\pi}\log\frac{t+C}{2\pi}+2, \tag{17.54}$$

and $\Psi_{\boldsymbol{\kappa}}(-\mathrm{x})=-\Psi_{\boldsymbol{\kappa}}(\mathrm{x})$ for almost every x.

Proof. By the mean value theorem and the monotonicity of $\mathcal{N}'_{\mathrm{RVM}}$, $|\mathcal{N}_{\mathrm{RVM}}(\kappa_k)-\mathcal{N}_{\mathrm{RVM}}(\tau_k)|\le C\mathcal{N}'_{\mathrm{RVM}}(\tau_k+C)$. For $\mathrm{x}>0$ the terms of (17.42) with $k\le0$ vanish, because both x_k and $k-\frac12$ are negative. For $k\ge1$, $\mathrm{x}_k\le\mathrm{x}$ if and only if $\kappa_k\le t$, by monotonicity of $\mathcal{N}_{\mathrm{RVM}}$. If $N_{\boldsymbol{\kappa}}(t)=j$, then $\tau_j\le t+C$ and $\tau_{j+1}>t-C$, so $n_{\mathrm{RVM}}(t-C)\le j\le n_{\mathrm{RVM}}(t+C)$. By (17.2), $n_{\mathrm{RVM}}(t+C)-n_{\mathrm{RVM}}(\max(t-C,2\pi))\le\mathcal{N}_{\mathrm{RVM}}(t+C)-\mathcal{N}_{\mathrm{RVM}}(\max(t-C,2\pi))+1$. This is at most $\frac{2C}{2\pi}\log\frac{t+C}{2\pi}+2$, the extra 1 covering $t-C<2\pi$, where $\mathcal{N}_{\mathrm{RVM}}\ge-\frac18$ and $n_{\mathrm{RVM}}=0$. Oddness follows from the symmetry of $\mathscr{X}_{\boldsymbol{\kappa}}$ and of $\mathbb{Z}+\frac12$. □

Proposition 17.50 (The condition of $\mathscr{O}$ gives the averaging hypothesis with blocks $\asymp\log^2 T$)**.** *Let $\boldsymbol{\kappa}$ be C-admissible with $\kappa_1>\mathfrak{t}_0$, and assume the third condition of $\mathscr{O}$ (15.1) in the form*

$$\Big|\int_{2\pi}^{R}(N_{\boldsymbol{\kappa}}(t)-\mathcal{N}_{\mathrm{RVM}}(t))dt\Big| \le C_2\log R \qquad (R\ge3). \tag{17.55}$$

Put $\mathrm{H}(T):=16C_2\log T$. There is $T_1=T_1(C,C_2)$ such that the following holds for the points $T_{m+1}:=T_m+\mathrm{H}(T_m)$ ($m\ge1$) and the blocks $\mathfrak{J}_m:=[\mathcal{N}_{\mathrm{RVM}}(T_m),\mathcal{N}_{\mathrm{RVM}}(T_{m+1}))$ with their mirror images $-\mathfrak{J}_m$. The step function $\mathrm{A}_{\boldsymbol{\kappa}}$, equal to the mean of $\Psi_{\boldsymbol{\kappa}}$ on each $\pm\mathfrak{J}_m$ and to 0 on $(-\mathcal{N}_{\mathrm{RVM}}(T_1),\mathcal{N}_{\mathrm{RVM}}(T_1))$, satisfies

$$\|\mathrm{A}_{\boldsymbol{\kappa}}\|_\infty \le \tfrac{3}{16}, \qquad |\mathfrak{J}_m| = \frac{8C_2}{\pi}\log^2 T_m\,(1+o(1)) \quad (m\to\infty). \tag{17.56}$$

The remainder $\mathrm{B}_{\boldsymbol{\kappa}} := \Psi_{\boldsymbol{\kappa}} - \mathrm{A}_{\boldsymbol{\kappa}}$ *has mean zero on every block. For* $\boldsymbol{\kappa} = \boldsymbol{\gamma}$, (17.55) *holds unconditionally for some* C_2.

Proof. Write $\mathfrak{J} = [\mathcal{N}_{\mathrm{RVM}}(T), \mathcal{N}_{\mathrm{RVM}}(T+\mathrm{H}))$. Then

$$\begin{aligned}\int_{\mathfrak{J}} \Psi_{\boldsymbol{\kappa}}\, d\mathrm{x} &= \int_{T}^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(t)\, \mathcal{N}'_{\mathrm{RVM}}(t)\, dt \\ &= \mathcal{N}'_{\mathrm{RVM}}(T) \int_{T}^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}) + \int_{T}^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(\mathcal{N}'_{\mathrm{RVM}}(t) - \mathcal{N}'_{\mathrm{RVM}}(T))\, dt.\end{aligned} \tag{17.57}$$

The last integral is at most $\mathrm{H} \cdot \frac{\mathrm{H}}{2\pi T}\big(\frac{C}{\pi} \log \frac{T+\mathrm{H}+C}{2\pi} + 1\big)$, by (17.54) and $\mathcal{N}''_{\mathrm{RVM}}(t) = \frac{1}{2\pi t}$.

Next, $\int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}) = \int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}}) + \int_T^{T+\mathrm{H}} (\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}})$. The first piece is at most $2C_2 \log(T+\mathrm{H})$ by (17.55). For the second, substitute $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t)$. By (17.2), $\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}} = \mathrm{saw}(\mathrm{x}) := \mathrm{x} - \lfloor \mathrm{x} + \frac{1}{2} \rfloor$, which is 1-periodic with mean zero and $|\mathrm{saw}| \le \frac{1}{2}$, and $dt = g(\mathrm{x})\, d\mathrm{x}$ with $g := 1/\mathcal{N}'_{\mathrm{RVM}}(t(\mathrm{x}))$ positive and decreasing. On a full period $[j - \frac{1}{2}, j + \frac{1}{2})$, $|\int \mathrm{saw}\, \mathfrak{g}_1| = |\int \mathrm{saw}\, (\mathfrak{g}_1 - \mathfrak{g}_1(j))| \le \frac{1}{4}(\mathfrak{g}_1(j - \frac{1}{2}) - \mathfrak{g}_1(j + \frac{1}{2}))$, which telescopes. The two partial periods contribute at most $\mathfrak{g}_1(\mathcal{N}_{\mathrm{RVM}}(T))$. Hence $|\int_T^{T+\mathrm{H}} (\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}})| \le \frac{5}{4\mathcal{N}'_{\mathrm{RVM}}(T)}$.

Since $|\mathfrak{J}| \ge \mathcal{N}'_{\mathrm{RVM}}(T)\mathrm{H}$,

$$\Big|\frac{1}{|\mathfrak{J}|} \int_{\mathfrak{J}} \Psi_{\boldsymbol{\kappa}}\Big| \le \frac{2C_2 \log(T+\mathrm{H})}{\mathrm{H}} + \frac{5}{4\mathcal{N}'_{\mathrm{RVM}}(T)\mathrm{H}} + \frac{\mathrm{H}}{2\pi T \mathcal{N}'_{\mathrm{RVM}}(T)} \Big(\frac{C}{\pi} \log \frac{T+\mathrm{H}+C}{2\pi} + 1\Big). \tag{17.58}$$

With $\mathrm{H} = 16C_2 \log T$, the first term tends to $\frac{1}{8}$ and the others tend to 0, so the bound is at most $\frac{3}{16}$ for $T \ge T_1(C, C_2)$. The length is $|\mathfrak{J}_m| = \mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})\mathrm{H}(T_m)$ with $\mathrm{s} \in [T_m, T_{m+1}]$. The mirror blocks follow by oddness. For $\boldsymbol{\gamma}$: $N_{\boldsymbol{\gamma}}(t) = 1 + \vartheta(t)/\pi + S(t)$ [51, §9.3], $\int_0^R S = O(\log R)$ [51, Thm. 9.9], and $1 + \vartheta/\pi - \mathcal{N}_{\mathrm{RVM}} = O(1/t)$. Finally, $\boldsymbol{\gamma}$ is C_η-admissible, unconditionally, by 2.2. In particular (17.41) holds for $\mathscr{X}_{\boldsymbol{\gamma}}$, by Lemma 17.49. □

Remark 17.51 (Correction of an earlier formulation). Proposition 17.50 is an implication. The averaging condition on blocks of length $\asymp \log^2 T$ does not imply (17.55): summing $|\int_{\mathfrak{J}_m} \Psi| \le \frac{3}{16}|\mathfrak{J}_m|$ over the blocks up to height R gives only $O(R \log R)$, not $O(\log R)$. The two conditions do not coincide.

Corollary 17.52 (Reduction to one condition)**.** *For every* $\boldsymbol{\kappa}$ *as in Proposition 17.50: if* $\mathbb{Y}_1 \mathrm{B}_{\boldsymbol{\kappa}}$ *is bounded up to a constant, then* $|\mathcal{F}_{\boldsymbol{\kappa}}(\mathrm{x} + i)|^2 \in (A_2)$*, with constant* (17.49) *for* $\mathrm{d}_\star = \frac{3}{16}$*. Corollary 17.43(a) then holds for* $\mathscr{X}_{\boldsymbol{\kappa}}$ *if it is uniformly discrete, and Corollary 17.43(b) if it is a finite union of uniformly discrete sets.*

Proof. Lemma 17.49, Proposition 17.50 and Theorem 17.42. □

17.6.2. *The weight is a quotient of synthetic zetas.*

Proposition 17.53 (Weight identity)**.** *Let* $\boldsymbol{\kappa}$ *be* C*-admissible and satisfy* (17.55)*. For* $0 < \hbar \le 1$ *and* $t \ge 2\pi$*,*

$$\log\Big|\frac{\zeta_{\boldsymbol{\kappa}}}{\zeta_{\mathrm{RVM}}}\Big(\tfrac{1}{2} + \hbar + it\Big)\Big| = \int_{2\pi}^{\infty} \frac{(t - \mathrm{s})\,(N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(\mathrm{s})}{(t-\mathrm{s})^2 + \hbar^2}\, d\mathrm{s} + \mathrm{c}_{\boldsymbol{\kappa}} + \mathrm{m}_{\boldsymbol{\kappa},\hbar}(t), \tag{17.59}$$

where the integral converges as an improper integral and $\mathrm{c}_{\boldsymbol{\kappa}}$ *is a constant. The remainder satisfies* $|\mathrm{m}_{\boldsymbol{\kappa},\hbar}(t)| \le \mathrm{c}(C, C_2) \log(2+t)/(1+t)$.

Proof. For $s = \frac{1}{2} + \hbar + it$, $|\frac{1}{4} + \kappa^2 + q(s)| = |\kappa^2 + (\hbar + it)^2| = |\kappa - t + i\hbar|\,|\kappa + t - i\hbar|$. Hence, with $\mathrm{l}_\hbar(u) := \frac{1}{2}\log(u^2 + \hbar^2)$,

$$\log\Big|\frac{P_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\tau}}}(s)\Big| = \sum_k [\mathrm{l}_\hbar(t - \kappa_k) - \mathrm{l}_\hbar(t - \tau_k)] + \sum_k [\mathrm{l}_\hbar(t + \kappa_k) - \mathrm{l}_\hbar(t + \tau_k)] - \sum_k \log \frac{\frac{1}{4} + \kappa_k^2}{\frac{1}{4} + \tau_k^2}. \tag{17.60}$$

Put $\mathfrak{g}_1 := N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}$ and $\mathfrak{G}_1(\mathrm{s}) := \int_{2\pi}^{\mathrm{s}} \mathfrak{g}_1$. By (17.55), by the sawtooth estimate in the proof of Proposition 17.50 on $[\tau_1, \infty)$, and by $|\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}}| \le \frac{1}{2}$ on $[2\pi, \tau_1]$, $|\mathfrak{G}_1(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$. For f smooth with $\mathrm{f}'(\mathrm{s}) = O(1/\mathrm{s})$ and $\mathrm{f}''(\mathrm{s}) = O(1/\mathrm{s}^2)$, the sums $\sum_{k\le k^*}[\mathrm{f}(\kappa_k) - \mathrm{f}(\tau_k)] = -\int \mathrm{f}'\mathfrak{g}_{1,k^*}$ converge as $k^* \to \infty$ to $-\int^{\infty} \mathrm{f}'\mathfrak{g}_1$, an improper integral. Integration by parts against $\mathfrak{G}_1$ shows both the convergence and that the tail contributions vanish. With $\mathrm{f} = \mathrm{l}_\hbar(t - \cdot)$ this gives the integral in (17.59). With $\mathrm{f} = \log(\frac{1}{4} + (\cdot)^2)$ it gives the constant $\mathrm{c}_{\boldsymbol{\kappa}}$. With $\mathrm{f} = \mathrm{l}_\hbar(t + \cdot)$ it gives $\mathrm{m}_{\boldsymbol{\kappa},\hbar}(t) = \int_{2\pi}^{\infty} \mathrm{k}_1'(\mathrm{s})\mathfrak{G}_1(\mathrm{s})\,d\mathrm{s}$, where $\mathrm{k}_1(\mathrm{s}) := \frac{t+\mathrm{s}}{(t+\mathrm{s})^2+\hbar^2}$ and $|\mathrm{k}_1'(\mathrm{s})| \le (t+\mathrm{s})^{-2}$. This yields the bound. □

Remark 17.54 (Local scale). With $\hbar = 1$ and $\hbar_T := \hbar/\mathcal{N}'_{\mathrm{RVM}}(T) = 2\pi/\log(T/2\pi)$ (the notation of the proof of Theorem 17.55), the substitution $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t)$ maps the kernel $\frac{(t-\mathrm{s})\,ds}{(t-\mathrm{s})^2+\hbar_T^2}$ to $\frac{(\mathrm{x}-\mathrm{x}')\,dx'}{(\mathrm{x}-\mathrm{x}')^2+1}$ to first order near $t = T$. Comparing (17.59) with (17.46) at $\mathrm{y} = 1$, the unfolded weight $|\mathcal{F}_{\boldsymbol{\kappa}}(\mathrm{x}+i)|^2$ near $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(T)$ is therefore, to first order, $|\zeta_{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}}|^2$ on the line $\Re s = \frac{1}{2} + \hbar_T$, up to the bounded factor $|\mathcal{F}_0(\mathrm{x}+i)|^2$. The identity (17.59) is exact. The identification with the unfolded weight is local, and its quantitative form for the Poisson part is Step 3 of the proof of Theorem 17.55.

17.6.3. *Obstruction for the zeta list.*

Theorem 17.55 (No (A_2) for the unfolded zeta ordinates)**.** *Let $\mathscr{X}_{\boldsymbol{\gamma}}$ be the unfolded list of the ordinates $\gamma_k > 0$ of the nontrivial zeros of ζ, counted with multiplicity, and let $\mathcal{F}$ be any entire function of exponential type whose zero set is $\mathscr{X}_{\boldsymbol{\gamma}}$, with multiplicities. Then $|\mathcal{F}(\mathrm{x}+i\hbar)|^2 \notin (A_2)$ for every real $\hbar \neq 0$. The proof does not assume the Riemann hypothesis.*

Proof. Write $\Psi := \Psi_{\boldsymbol{\gamma}}$ and $\mathcal{F}_{\boldsymbol{\gamma}}$ for the function of Definition 17.48. It is even and real by Lemma 17.41(iii), because $\mathrm{x}_1 = \mathcal{N}_{\mathrm{RVM}}(\gamma_1) = 0.45\ldots > 0$, so $0 \notin \mathscr{X}_{\boldsymbol{\gamma}}$.

Step 1: reduction to $\mathcal{F}_{\boldsymbol{\gamma}}$ and $\hbar > 0$. $\mathcal{F}$ and $\mathcal{F}_{\boldsymbol{\gamma}}$ are entire of order at most 1 (exponential type) with the same zeros, with multiplicity, so by Hadamard's factorisation theorem both are e^{linear} times the same canonical product, and $\mathcal{F}/\mathcal{F}_{\boldsymbol{\gamma}} = e^{\mathrm{a}_1 + \mathrm{b}_1\mathfrak{z}}$. Hence $|\mathcal{F}(\mathrm{x}+i\hbar)|^2 = \mathrm{c}\,e^{2\mathrm{b}_1'\mathrm{x}}\mathrm{w}_0(\mathrm{x})$ with $\mathrm{b}_1' = \Re\mathrm{b}_1$, a constant $\mathrm{c} > 0$, and $\mathrm{w}_0(\mathrm{x}) := |\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x}+i\hbar)|^2$, which is even in x. Moreover $|\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x}-i\hbar)| = |\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x}+i\hbar)|$. Suppose $e^{2\mathrm{b}_1'\mathrm{x}}\mathrm{w}_0 \in (A_2)$. Reflection preserves (A_2), so $e^{-2\mathrm{b}_1'\mathrm{x}}\mathrm{w}_0 \in (A_2)$ as well. By Step 2 below both logarithms are in BMO, so $4\mathrm{b}_1'\mathrm{x} \in \mathrm{BMO}$, which forces $\mathrm{b}_1' = 0$. So we may assume $\mathcal{F} = \mathcal{F}_{\boldsymbol{\gamma}}$ and $\hbar > 0$, and that $\mathrm{w}_0 \in (A_2)$.

Step 2: BMO. Let $\mathrm{f} := \log \mathrm{w}_0$ and f_I its mean on I. By Jensen's inequality, $\frac{1}{|I|}\int_I e^{\pm(f-\mathrm{f}_I)} \ge 1$. The product of the two averages is at most $[\mathrm{w}_0]_{A_2}$, so each is at most $[\mathrm{w}_0]_{A_2}$. Since $u^2 \le 2e^{|u|} \le 2(e^u + e^{-u})$,

$$\frac{1}{|I|}\int_I |f - \mathrm{f}_I|^2 \le 4[\mathrm{w}_0]_{A_2} \qquad \text{for every interval } I. \tag{17.61}$$

By (17.46) and the boundedness of $\log|\mathcal{F}_0(\mathrm{x}+i\hbar)|$, $\mathrm{g}_0 := \pi\mathbb{Y}_\hbar\Psi \in \mathrm{BMO}$. Now take $\mathbb{A}_\hbar(\mathfrak{z}) := \mathbb{A}_{\mathscr{X}}(\mathfrak{z} + i\hbar)$, in the notation of the proof of Lemma 17.41, which is holomorphic on $\Im\mathfrak{z} > -\hbar$. Its real part on $\mathbb{R}$ is g_0, and its imaginary part on $\mathbb{R}$ is $-\pi\mathbb{P}_\hbar\Psi$. By (17.47), $\Re\mathbb{A}_\hbar(\mathfrak{z}) = O(\log^2(4 + |\mathfrak{z}|))$ on $\Im\mathfrak{z} \ge 0$, and so is the Poisson integral $\mathbb{P}[\mathrm{g}_0]$. Hence $\Re\mathbb{A}_\hbar - \mathbb{P}[\mathrm{g}_0]$ is harmonic on $\Im\mathfrak{z} > 0$, continuous with zero boundary values, and $O(\log^2|\mathfrak{z}|)$. After odd reflection it is harmonic on $\mathbb{C}$ and $o(|\mathfrak{z}|)$, hence constant, hence zero. Therefore $\mathbb{A}_\hbar = \mathbb{P}[\mathrm{g}_0] + i\mathbb{Y}[\mathrm{g}_0] + ic$ on $\Im\mathfrak{z} > 0$, where $\mathbb{P}[\mathrm{g}_0] + i\mathbb{Y}[\mathrm{g}_0]$ is the holomorphic function with real part $\mathbb{P}[\mathrm{g}_0]$ and $\mathbb{Y}[\mathrm{g}_0](\mathrm{x}+i\mathrm{y}) = \mathbb{Y}_\mathrm{y}\mathrm{g}_0(\mathrm{x})$. Letting $\mathrm{y} \to 0$, $-\pi\mathbb{P}_\hbar\Psi = \tilde{\mathrm{g}}_0 + c$ almost everywhere on $\mathbb{R}$, where $\tilde{\mathrm{g}}_0$ is the conjugate function of g_0, the a.e. boundary value of $\mathbb{Y}_\mathrm{y}u$ [10, Ch. III]. Conjugation is bounded on BMO [10, Ch. VI], so $\mathbb{P}_\hbar\Psi \in \mathrm{BMO}$. By the John–Nirenberg inequality [10, Ch. VI], there is $\mathrm{c}_{\mathrm{JN}} < \infty$ with

$$\frac{1}{|I|}\int_I \left|\mathbb{P}_\hbar\Psi - (\mathbb{P}_\hbar\Psi)_I\right|^2 \le \mathrm{c}_{\mathrm{JN}} \qquad \text{for every interval } I. \tag{17.62}$$

Step 3: back to the ordinates. By [51, §9.3], $N_{\boldsymbol{\gamma}}(t) = 1 + \vartheta(t)/\pi + S(t)$, with $S(t) = O(\log t)$ [51, Thm. 9.4]. Lemma 17.49 and (17.2) then give, for $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t) > 0$, $\Psi(\mathrm{x}) = S(t) + \mathrm{r}(\mathrm{x})$ with

$|\mathrm{r}| \le \mathrm{c}_e$. Here c_e bounds $|1+\vartheta/\pi - \mathcal{N}_{\mathrm{RVM}}| + \frac{1}{2}$ on $t \ge 2\pi$. Let $\mathfrak{t}_0 := \mathcal{N}_{\mathrm{RVM}}^{-1}(0)$ and $T \to \infty$, with $t \in [T, 2T]$ and $\hbar_t := \hbar/\mathcal{N}'_{\mathrm{RVM}}(t)$, $\hbar_T := \hbar/\mathcal{N}'_{\mathrm{RVM}}(T)$.

(a) The part of $\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t))$ coming from $\mathrm{x}' < 0$ is $O(\log T/T)$, since $|\Psi(\mathrm{x}')| \le \mathrm{c}\log(2+|\mathrm{x}'|)$. The part coming from r is at most c_e. Substituting $\mathrm{x}' = \mathcal{N}_{\mathrm{RVM}}(\mathrm{s})$, the rest is $\int_{\mathfrak{t}_0}^\infty \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s})S(\mathrm{s})\,d\mathrm{s}$ with $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) := \mathbb{P}_\hbar(\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}))\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})$, where $\mathbb{P}_\hbar(\mathrm{v}) := \frac{\hbar}{\pi(\mathrm{v}^2+\hbar^2)}$.

(b) Put $\mathbb{P}_t^\circ(\mathrm{s}) := \mathbb{P}_{\hbar_t}(t-\mathrm{s}) = \mathbb{P}_\hbar(\mathcal{N}'_{\mathrm{RVM}}(t)(t-\mathrm{s}))\mathcal{N}'_{\mathrm{RVM}}(t)$. For $|\mathrm{s}-t| \le T^{1/3}$, Taylor's formula with $\mathcal{N}''_{\mathrm{RVM}} = \frac{1}{2\pi\mathrm{s}}$ gives $|\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) - \mathcal{N}'_{\mathrm{RVM}}(t)(t-\mathrm{s})| \le \mathrm{d}_T := T^{-1/3}/\pi$. Hence $|\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) - \mathbb{P}_t^\circ(\mathrm{s})| \le (5\mathrm{d}_T/\hbar)\mathbb{P}_t^\circ(\mathrm{s})$ for T large. With $|S| \le \mathrm{c}\log(3T)$ on that range, the core difference is $O(T^{-1/3}\log T)$. For $|\mathrm{s}-t| > T^{1/3}$ we use $\mathbb{P}_\hbar(\mathrm{v}) \le \frac{\hbar}{\pi\mathrm{v}^2}$, the convexity of $\mathcal{N}_{\mathrm{RVM}}$ and $|S(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$. If $\mathrm{s} > t + T^{1/3}$, then $\mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) - \mathcal{N}_{\mathrm{RVM}}(t) \ge \mathcal{N}'_{\mathrm{RVM}}(t)(\mathrm{s}-t) \ge \mathrm{s}-t$ for T large, so $\int_{t+T^{1/3}}^\infty \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s})|S(\mathrm{s})|\,d\mathrm{s} \le \frac{\mathrm{c}\hbar}{\pi}\int_{T^{1/3}}^\infty \frac{\mathcal{N}'_{\mathrm{RVM}}(t+\mathrm{v})\log(2+t+\mathrm{v})}{\mathrm{v}^2}\,d\mathrm{v} = O(T^{-1/3}\log^2 T)$, splitting the v-integral at $\mathrm{v} = T$. If $\mathfrak{t}_0 \le \mathrm{s} < t - T^{1/3}$, then $\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) \ge \mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})(t-\mathrm{s})$ and $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}) \ge \mathcal{N}'_{\mathrm{RVM}}(\mathfrak{t}_0) > 0$, so $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) \le \frac{\hbar}{\pi\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})(t-\mathrm{s})^2} \le \frac{\hbar}{\pi\mathcal{N}'_{\mathrm{RVM}}(\mathfrak{t}_0)(t-\mathrm{s})^2}$ and $\int_{\mathfrak{t}_0}^{t-T^{1/3}} \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}|S| \le \frac{\mathrm{c}\hbar\log(3T)}{\pi\mathcal{N}'_{\mathrm{RVM}}(\mathfrak{t}_0)}\int_{T^{1/3}}^\infty \mathrm{v}^{-2}\,d\mathrm{v} = O(T^{-1/3}\log T)$. The same bounds hold for $\mathbb{P}_t^\circ$ with $\hbar_t$ in place of $\hbar$ and $\mathcal{N}'_{\mathrm{RVM}}(t)$ in place of $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})$, the region $\mathrm{s} \gg T$ being handled by the split at $\mathrm{v} = T$. So that part tends to 0 as well.

(c) $\|\mathbb{P}_a - \mathbb{P}_b\|_{L^1} \le \frac{2}{\pi}|\log(a/b)|$, because $\int|\partial_c\mathbb{P}_c| = \frac{2}{\pi c}$. Since $\hbar_t/\hbar_T \in [1 - \frac{2\log 2}{\log(T/2\pi)}, 1]$, $|\int(\mathbb{P}_{\hbar_t} - \mathbb{P}_{\hbar_T})(t-\mathrm{s})S(\mathrm{s})\,d\mathrm{s}| = O(\log T/\log T) = O(1)$, again with the tails controlled by $|S(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$.

(d) $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} = O(\sqrt{T})$, with $S := 0$ on $(-\infty, \mathfrak{t}_0)$; this is Lemma 17.56 below.

Together, (a)–(d) give $\|\mathbb{P}_\hbar\Psi \circ \mathcal{N}_{\mathrm{RVM}} - S\|_{L^2[T,2T]} = O(\sqrt{T})$.

Step 4: contradiction. Let $\mathfrak{I}_T := [\mathcal{N}_{\mathrm{RVM}}(T), \mathcal{N}_{\mathrm{RVM}}(2T)]$ and $\bar{\mathrm{p}}_T := (\mathbb{P}_\hbar\Psi)_{\mathfrak{I}_T}$. By (17.62), $\int_T^{2T}|\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t)) - \bar{\mathrm{p}}_T|^2\mathcal{N}'_{\mathrm{RVM}}(t)\,dt \le \mathrm{c}_{\mathrm{JN}}|\mathfrak{I}_T| \le \mathrm{c}_{\mathrm{JN}}T\mathcal{N}'_{\mathrm{RVM}}(2T)$, so $\int_T^{2T}|\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t)) - \bar{\mathrm{p}}_T|^2\,dt \le 2\mathrm{c}_{\mathrm{JN}}T$. With Step 3, $\int_T^{2T}|S - \bar{\mathrm{p}}_T|^2 = O(T)$. But

$$\int_T^{2T}|S-\mathrm{c}|^2 \ge \int_T^{2T} S^2 - \frac{1}{T}\Big(\int_T^{2T} S\Big)^2 = \frac{1}{2\pi^2}T\log\log T\,(1+o(1)) \tag{17.63}$$

for every constant c, by Selberg's theorem $\int_0^T S^2 = \frac{T}{2\pi^2}\log\log T + O(T(\log\log T)^{1/2})$ [48], with the normalisation $S(t) = \pi^{-1}\arg\zeta(\frac{1}{2}+it)$ used here and in [7, p. 225], and $\int_T^{2T} S = O(\log T)$ [51, Thm. 9.9]. This is a contradiction. □

Lemma 17.56 (Smoothing S at scale $1/\log T$)**.** *Let $\hbar > 0$ be fixed, $\hbar_T := 2\pi\hbar/\log(T/2\pi)$, and $S := 0$ on $(-\infty, \mathfrak{t}_0)$. Then $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} = O_\hbar(\sqrt{T})$, unconditionally.*

Proof. Fujii's Main Theorem [7, Main Theorem], proved unconditionally by modifying Selberg's method, states for $q = 1$, $k = 1$, $T^{1/2+a} \le H \le T$ and $0 < u \le H - (H/\sqrt{T})^{1/8}$ that $\int_T^{T+H}(S(t+u) - S(t))^2 dt = c_1 H \cdot 2\log(2 + u\log T)\,(1 + O((\log(2+u\log T))^{-1/2}))$ with an absolute c_1. The relative error is $O(1)$ when $u\log T$ is bounded, so the statement carries no asymptotic information there, and we use it only as the upper bound it contains: for $0 < u \le 1$, $T \ge T_0$ and $H = T$,

$$\int_T^{2T}\big(S(t+u) - S(t)\big)^2 dt \le \mathrm{c}_F T\log(2+u\log T), \tag{17.64}$$

with an absolute constant c_F. The range $0 < u \le H - (H/\sqrt{T})^{1/8} = T - T^{1/16}$ of [7] contains $0 < u \le 1$, and the same bound holds for $-1 \le u < 0$: the substitution $t \mapsto t-u$ moves the integral to $[T-|u|, 2T-|u|] \subset [T/2, 2T]$, covered by (17.64) at the base points $T/2$ (with $H = T/2$) and T (with $H = T$). By Minkowski's inequality, $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} \le \int \mathbb{P}_{\hbar_T}(u)\|S(\cdot - u) - S\|_{L^2[T,2T]}\,du$. For

$0<|u|\le 1$ the norm is at most $(2\mathrm{c}_F T\log(2+|u|\log T))^{1/2}$. For $1<|u|\le T/2$, Selberg's theorem $\int_0^T S^2=\frac{1}{2\pi^2}T\log\log T+O(T(\log\log T)^{1/2})$ [48] bounds it by $\mathrm{c}(T\log\log T)^{1/2}$. For $|u|>T/2$, $|S(t)|\le \mathrm{c}\log(2+t)$ [51, Thm. 9.4] bounds it by $\mathrm{c}\sqrt{T}\log(3T+|u|)$. Substituting $u=\hbar_T v$ in the first range, where $\hbar_T\log T\to 2\pi\hbar$, and using $\int_{|u|>1}\mathbb{P}_{\hbar_T}=O(\hbar_T)$ and $\int_{|u|>T/2}\mathbb{P}_{\hbar_T}(u)\log(3T+|u|)\,du=O(\hbar_T\log T/T)$, all three contributions are $O(\sqrt{T})$. □

Remark 17.57 (A second proof). Lemma 17.56 also follows from Selberg's mean-square approximation of $S(t)$ by $S_{\mathrm{x}_0}(t):=-\frac{1}{\pi}\sum_{p\le \mathrm{x}}p^{-1/2}\sin(t\log p)$, $\mathrm{x}_0=T^\delta$ [48], and the Montgomery–Vaughan mean value theorem [41, Cor. 3]. Indeed $\mathbb{P}_{\hbar_T}*S_{\mathrm{x}_0}-S_{\mathrm{x}_0}=-\frac{1}{\pi}\sum_{p\le \mathrm{x}}p^{-1/2}(e^{-\hbar_T\log p}-1)\sin(t\log p)$ has mean square $\ll T\,\hbar_T^2\log^2\mathrm{x}\ll T$ on $[T,2T]$, and the remainder $S-S_{\mathrm{x}_0}$ is handled by Minkowski's inequality. Fujii's announcement [8] states the moment lemma of [7] in the form used in [9, Lemma 1].

Corollary 17.58 (Consequences for the zeta list)**.** (a) $\{e^{ix_k\xi}\}_{x_k\in\mathscr{X}_\gamma}$ *is not a Riesz basis of* $L^2(-\pi,\pi)$.
(b) *If* $\mathscr{X}_\gamma$ *is a finite union of uniformly discrete sets, its generalised divided differences are not a Riesz basis of* $L^2(-\pi,\pi)$.
(c) *For every decomposition* $\Psi_\gamma=A+B$ *with* $\|A\|_\infty<\frac14$, $\mathbb{Y}_1B$ *is unbounded. In particular this holds for the blocks of Proposition 17.50, where the averaging hypothesis is satisfied.*

Proof. (a) and (b): the criteria [4, Thm. 1, Thm. 3] are necessary and sufficient, and require a generating function satisfying Helson–Szegő, equivalently (A_2), on some line $\Im\mathfrak{z}=\hbar$. The line $\hbar=0$ is excluded, since $|\mathcal{F}(\mathrm{x})|^{-2}$ is not locally integrable at a real zero. All lines with $\hbar\neq 0$ are excluded by Theorem 17.55. (c): otherwise Theorem 17.42 would give $|\mathcal{F}_\gamma(\mathrm{x}+i)|^2\in(A_2)$. □

Heuristic 17.59 (Graduated version)**.** The proof of Theorem 17.55 uses intervals $\mathfrak{I}_T$ of length $\asymp T\log T$. On a window of L unfolded points at height T, the mean square oscillation of $\mathbb{P}_\hbar\Psi$ is governed by the variance of increments of S over t-intervals of length $L/\mathcal{N}'_{\mathrm{RVM}}(T)$. By Fujii's asymptotics this is expected to be of size $\frac{1}{\pi^2}\log L$. This suggests that local (A_2) constants on windows of L points grow with L but stay bounded for L bounded. That is the graduated stability that a Fredholm formulation would need. None of this is proved here. The computations of Section 17.7 measure the ℓ^∞ norm of the inverse of square finite sections directly and find growth like $T^{1.8}$ up to $T=487$.

Remark 17.60 (Bearing on route B). Theorem 17.55 rules out a uniform Riesz-basis (Fredholm) structure for the unfolded zeta list on the whole line, and hence a linear stability argument with constants independent of the height. It says nothing about statement B or B′. The compactness step of route B (Proposition 17.5) uses no stability; the open part of route B is existence (Definition 17.2) together with uniqueness in the box (Hypothesis 17.6), and the open part of route B′ is existence alone (Corollary 17.15).

17.7. **Numerical certificates for finite sections.** This section reports computations. They are not part of any proof; their role is to measure the quantities of Theorems 17.20 and 17.21 on actual sections and to locate where the staged construction fails. The programs and their outputs are listed in Appendix C and in `code/routeB/` of the repository. Floating point arithmetic was used throughout; no interval arithmetic was run.

17.7.1. *Set-up.* The tests are Gaussians $\varphi_{c,\mathrm{w}}(u):=e^{-(u-c)^2/2\mathrm{w}^2}$ with centre $c\ge 9\mathrm{w}$, so that their restriction to $u>0$ is, to working precision, a test supported in $(0,\infty)$; their cosine transform is $\mathsf{c}_{\varphi_{c,\mathrm{w}}}(t)=\mathrm{w}\sqrt{2\pi}\,e^{-(\mathrm{w}t)^2/2}\cos(ct)$. Stage m uses the width $\mathrm{w}_m:=0.08\cdot 0.85^{m-1}$, the height $T_m:=4/\mathrm{w}_m$, and the new tests φ_{c,w_m} with centres $c\in[9\mathrm{w}_m,\log(\max(T_m,7)/2\pi)+2]$ in steps of w_m; the test set $\mathscr{T}_m$ is the union of the levels $\le m$. The unknowns of stage m are the ordinates with $\tau_k\le 2.4/\mathrm{w}_m$ ("interior"); the factor $e^{-(\mathrm{w}_m\kappa)^2/2}$ makes the others invisible to the tests of level m. The masses are a_n^ζ for all $n\le e^{c+9\mathrm{w}}$, and k is 14.77. The pairings $\langle\mathsf{k},\varphi\rangle$ were computed by adaptive

quadrature. The zeta ordinates γ_k, $k \le 400$ ($\gamma_{400} = 679.74$), computed with `mpmath`, enter only in the diagnostic columns; no construction uses them.

Two checks fix the normalisation: $\langle \mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^{\zeta}), \varphi \rangle \approx 10^{-16}$ for all tests tried, and $\langle \mathfrak{R}(\boldsymbol{\tau}, \mathsf{a}^{\zeta}), \varphi \rangle$ between 10^{-3} and 10^{-1}.

17.7.2. *Initialisation: the clock is outside the basin, the Gonek lists are inside. Receding horizon from the clock.* Starting at $\boldsymbol{\kappa} = \boldsymbol{\tau}$, each stage solved the overdetermined least-squares problem for the interior ordinates with all tests of levels $\le m$, with bounds $|\kappa_k - \tau_k| \le 3$ and the non-interior ordinates frozen at their current values. The iteration stalls at the first Gram-type failure of the clock: the fit returns $\kappa_{34} = 109.75$ where $\gamma_{34} = 111.03$ ($\tau_{34} = 110.04$), and $\kappa_{41} = 125.50$ where $\gamma_{41} = 124.26$; from $m = 9$ on the residual on the lagged levels $\le m-4$ grows from $2 \cdot 10^{-5}$ to $4 \cdot 10^{-2}$ and the front of ordinates within 10^{-2} of $\boldsymbol{\gamma}$ stays at $k \approx 31$.

Receding horizon from a Gonek list. The same iteration started at $\boldsymbol{\kappa}^{(\mathrm{X})}$ with $\mathrm{X} = 8$ (Lemma 17.26; 983 roots up to $t = 1400$, computed from the Riemann–Siegel expansion of ϑ and the sum (17.20)) converged through $m = 16$, $T_{16} = 572$, 165 interior ordinates and 4985 tests: all interior ordinates within 10^{-2} of γ_k, 90 of them within 10^{-4}, the first twenty within $5 \cdot 10^{-6}$, residual on the lagged levels between 10^{-6} and $9 \cdot 10^{-6}$ at every stage, and $\max_k |\kappa_k - \tau_k| = 0.99$ throughout. The residual over all tests of level $\le m$ stays at $2 \cdot 10^{-3}$, which is the contribution of the frozen edge.

Why. The basin of the exact solution, measured by random perturbations of $\boldsymbol{\gamma}$ in the interior block, has radius 0.1 to 0.25 mean spacings. The clock lies at up to 0.66 spacings from $\boldsymbol{\gamma}$ for $k \le 400$, with 50 indices beyond 0.25; the lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ lie within 0.184 ($\mathrm{X} = 3$), 0.081 ($\mathrm{X} = 5$), 0.053 ($\mathrm{X} = 8$), 0.047 ($\mathrm{X} = 12$) and 0.035 ($\mathrm{X} = 20$) spacings for $k \le 160$, and within 0.103 spacings for $k \le 400$ at $\mathrm{X} = 8$ (Figure 1). The ℓ^2 conditioning of the interior block at $\boldsymbol{\gamma}$ is moderate and roughly uniform up to $T = 299$: condition numbers 17 to 75, smallest singular value 0.08 to 0.15. The coupling of the interior to the frozen edge, $\|\mathbb{J}_I^{+}\mathbb{J}_E\|_{\infty\to\infty}$, is between 0.25 and 0.47, and the shift it induces on the low ordinates decays from $2 \cdot 10^{-2}$ at $m = 2$ to 10^{-5} for $m \ge 10$.

17.7.3. *Certificates at the Gonek list: the direct bounds fail.* At $\boldsymbol{\kappa}^{(8)}$, with the interior unknowns of stage m and three choices of the right inverse, the quantities of Theorem 17.20 were computed for $m = 1, \dots, 14$ (T from 50 to 414).

Right inverse and norm	Certificate index	Range over $m = 1, \dots, 14$
square subsystem (tests chosen by pivoted QR), ℓ^{∞}	$2\mathbb{L}_m^2 \mathbb{H}_m \mathsf{h}_m$	$56 \to 1.2 \cdot 10^{7}$
pseudo-inverse of the full system, ℓ^2, $\beta_2 = 1/\sigma_{\min}$	$\beta_2 \mathrm{K}_2 \delta_2$ (Kantorovich)	$0.47 \to 233$
pseudo-inverse of the full system, ℓ^{∞}, diagonal Hessian	$2\mathrm{c}_{\mathrm{diag}} \mathsf{h}$	$1.3 \to 1.6 \cdot 10^{3}$

None of the certificates is satisfied, not even at $T = 50$. Yet one Gauss–Newton step from $\boldsymbol{\kappa}^{(8)}$ reduces $\max_k |\kappa_k - \gamma_k|$ by a factor between 4 and 15 at every height ($0.088 \to 0.016$ at $T = 572$), and $\sigma_{\min}$ in ℓ^2 stays between 1/17 and 1/5. The worst-case bounds lose a factor 10^3 to 10^4 through cancellation: $\mathbb{H}_m$ counts every unknown with the full weight $\int |\varphi| u^2$, while the actual second-order term of $\mathbb{F}_m$ along the Newton step is a sum of k_m^{*} oscillating terms.

17.7.4. *Certified exact finite sections.* The iterate of Section 17.7.2 was polished, stage by stage, by Newton's method on the square system (n tests selected by pivoted QR on $\mathbb{J}^T$, n interior unknowns) to residual 10^{-13}. At the polished point x_0 the certificate of Theorem 17.20 was evaluated in the sharper form that uses the separable structure of F_m: with Q the inverse of the square Jacobian, B

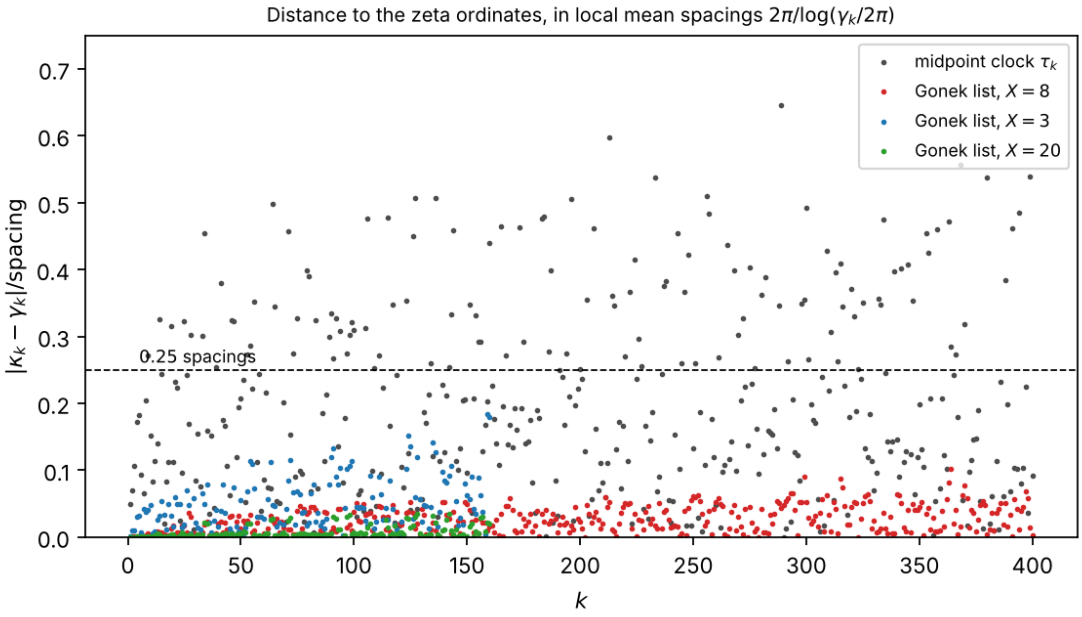


FIGURE 1. Distance to the zeta ordinates, in units of the local mean spacing $2\pi/\log(\gamma_k/2\pi)$, of the midpoint clock and of the Gonek lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ for $\mathrm{X} = 3, 8, 20$ ($k \le 160$ for $\mathrm{X} = 3, 20$; $k \le 400$ for $\mathrm{X} = 8$). The dashed line is the observed radius of the Newton basin. The ordinates γ_k are used here only to measure the distance.

the matrix of second derivatives $\partial^2_{\kappa_k} \mathsf{c}_\varphi$ at x_0, and $A_3(r)$ an entrywise bound for the third derivatives on $|x - x_0| \le \mathrm{r}$, the map $\mathbb{T}(x) = x - \mathbb{Q}\mathbb{F}(x)$ contracts the ball of radius r in ℓ^∞ if

$$\mathsf{l}(\mathrm{r}) := \|\mathbb{Q}\mathbb{J}'\|_{\infty\to\infty}\,\mathrm{r} + \tfrac{1}{2}\,\big\|\,|\mathbb{Q}|\mathbb{A}^{(3)}(\mathrm{r})\big\|_{\infty\to\infty}\,\mathrm{r}^2 < 1, \qquad \mathsf{h} \le (1 - \mathsf{l}(\mathrm{r}))\,\mathrm{r}, \tag{17.65}$$

in which case the ball contains an exact solution of the square system. The margin of a stage is $\max_{\mathrm{r}}(1 - \mathsf{l}(\mathrm{r}))\mathrm{r}$, the largest residual that (17.65) could certify at that point.

m	T_m	n	$\mathbb{L}_m = \|Q\|_{\infty\to\infty}$	margin	s_m on all tests $\le m$	$\max_k \|\kappa_k - \gamma_k\|$
1	50	4	41	$7.0 \cdot 10^{-3}$	$7.7 \cdot 10^{-8}$	$7 \cdot 10^{-7}$
4	81	9	50	$3.1 \cdot 10^{-3}$	$2.6 \cdot 10^{-7}$	$1 \cdot 10^{-6}$
8	156	26	150	$7.5 \cdot 10^{-4}$	$9.6 \cdot 10^{-7}$	$1 \cdot 10^{-5}$
12	299	68	460	$2.2 \cdot 10^{-4}$	$5.5 \cdot 10^{-6}$	$4 \cdot 10^{-5}$
15	487	133	1600	$6.7 \cdot 10^{-5}$	$3.3 \cdot 10^{-3}$	$4 \cdot 10^{-2}$
16	572	165	–	square Jacobian singular	–	–

Stages 1 to 15 are certified: for each, (17.65) holds with r equal to the polishing residual (10^{-15} to $5 \cdot 10^{-13}$), so an exact solution of the square section exists within r of x_0, inside the tube with $C = 1$. The margins exceed the residuals by 10^7 or more, which is far above the expected rounding error, so the same certificates are expected, but not verified, to survive interval arithmetic. The certified solutions are not unique: at $m = 15$ the solution is $4 \cdot 10^{-2}$ from $\boldsymbol{\gamma}$. At $m = 16$ the square Jacobian is singular to working precision and the stage is not certified. The scaling with height (Figure 2) is

$$\mathbb{L}_m \sim T^{1.8}, \qquad \|\mathbb{Q}\mathbb{J}'\|_{\infty\to\infty} \sim T^{2.1}, \qquad \text{margin} \sim T^{-2.1} \qquad (T \le 487). \tag{17.66}$$

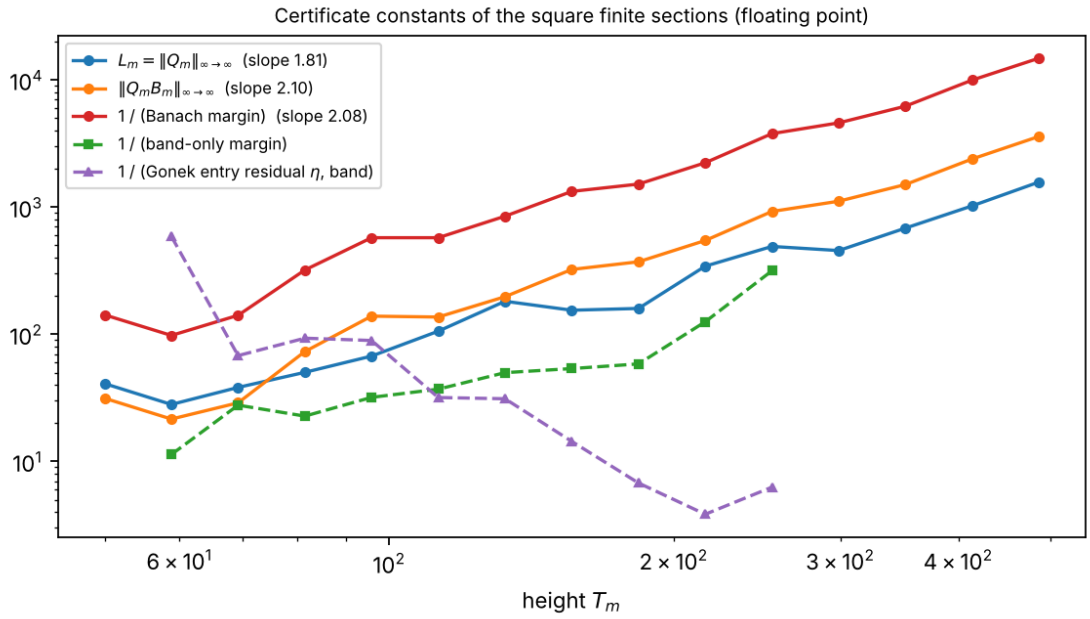


FIGURE 2. Constants of the Banach certificate (17.65) for the square sections of Section 17.7.4 (solid lines, with fitted slopes in $\log T$), and the band-only entry test of Section 17.7.5 (dashed): the reciprocal of the band margin and the reciprocal of the residual of the new band at its Gonek values. The entry test is certified where the triangles lie below the squares.

17.7.5. *The entry step.* The open point of the staged construction, in the form that Theorem 17.21 makes precise, is whether the residual h_{m+1} of the new tests at the start of stage $m+1$ is below the margin of that stage. This was tested as follows: start from the certified solution of stage $m-1$, put the new band of unknowns (those with $\tau_k \in (2.4/\mathrm{w}_{m-1}, 2.4/\mathrm{w}_m]$) and everything beyond at their Gonek values, and evaluate (17.65) (i) for the band alone, with the new tests of level m, and (ii) for the full square system of stage m, without polishing.

The band-only test is certified for $T \le 96$ and fails from $T = 113$ on; its margin falls from $9 \cdot 10^{-2}$ to $3 \cdot 10^{-3}$ while the Gonek residual of the band stays between 10^{-2} and $3 \cdot 10^{-1}$ (the Gonek error in the band is 0.02 to 0.09). The full-system test is certified only at $T = 59$. The margin decreases as a power of T by (17.66), whereas the error of the Gonek lists with fixed X does not decrease with T, and with $\mathrm{X} = T^{\mathrm{x}}$ it would decrease at most like $1/\log T$. Within the range computed, no worst-case certificate in ℓ^∞ closes the entry step, and the trend of the constants says that none will at larger T. This is a numerical statement, not a theorem.

17.7.6. *Reading of the computations.*

(i) The constructions that use only the clock and the primes (the Gonek lists, the square sections, the certificates) behave as predicted by the tube lemma and the certificate theorem, and the certified sections exist up to $T = 487$. These sections would exist equally if the Riemann hypothesis were false: a zero $\frac{1}{2} + \mathrm{b}_0 + i\gamma_0$ off the line contributes $2\cosh(\mathrm{b}_0 u)\cos(\gamma_0 u)$ to the prime side, which real ordinates near γ_0 can imitate against tests supported in $(0, u^*)$ as long as $\mathrm{b}_0 u^*$ is small. The information about the hypothesis sits entirely at heights $T \gtrsim (C \log \gamma_0)^{1/\mathrm{b}_0}$, which no finite computation reaches.

(ii) The quantities $\mathbb{L}_m$ and $\|QB\|$ grow polynomially in T. This is the quantitative face of the obstruction theorem of Section 17.6.3: no height-uniform linear stability, hence no

height-uniform Newton basin. A certificate for the infinite construction needs a basin that shrinks no faster than the initial error of a γ-free list decreases, and the computed trend is T^{-2} against $1/\log T$.

(iii) The observed contraction of Gauss–Newton (factor 4 to 15 per step at every height) is a mean-square phenomenon that the worst-case constants do not see. A proof of the entry step would have to bound $\|\mathbb{Q}\mathbb{J}'\operatorname{diag}(\mathrm{e})\,\mathrm{e}\|_\infty$ for the actual error vector e of the Gonek list, not for an arbitrary vector of the same norm; that error vector is γ-free, and its residual $\mathbb{F}_m(\boldsymbol{\kappa}^{(\mathrm{X})})$ is controlled through the explicit formula by the moments of $S - S_{\mathrm{X}}$, which are unconditional (Selberg, Fujii). Whether this closes is not known; any proof along this line would contain a proof of the Riemann hypothesis, by Corollary 17.15.

17.8. **Status of the statements.**

Statement	Status
Lemma 17.3, Corollary 17.4	proved; conditional on (D) as stated
Proposition 17.5, Lemma 17.18	proved, unconditional
Hypothesis 17.6	open; it extends 15.5; not needed with prescribed masses
Corollary 17.8	proved, assuming B and U
Lemmas 17.10, 17.11	proved, unconditional (the former Hypothesis (L))
Proposition 17.13, Corollary 17.15	proved, unconditional: RH $\Leftrightarrow$ B$'(C)$; classical in substance
Lemma 17.19, Theorems 17.20, 17.21, Proposition 17.23	proved; their hypotheses (A$_m$) for all m and (B) are open and imply RH
Heuristic 17.25	not proved; its point model is not a test of $C_c^\infty(0,\infty)$
Lemma 17.26	proved, unconditional; uses [11] only for the interpretation of the list
(G1), (G2) of Remark 17.27	open; together they imply B$'$ and RH
Lemmas 17.29–17.34, Theorem 17.35, Corollary 17.36	proved, unconditional; uses 8.10
Heuristics 17.38, 17.39, 17.59	not proved
Lemma 17.41, Theorem 17.42, Corollary 17.43	proved; Corollary 17.43 uses [4]
Corollary 17.45, Theorem 17.46, Proposition 17.44	proved; novelty not checked
Lemma 17.49, Proposition 17.50, Proposition 17.53, Corollary 17.52	proved, unconditional
Lemma 17.56, Theorem 17.55, Corollary 17.58	proved, unconditional; uses the upper bound contained in [7, Main Theorem], [48, 51, 10]
Section 17.7	computations in floating point; not part of any proof

18. Numerical illustration with 2 001 052 zeros

This section shows what is being constructed and checks that the theorems are consistent with computation. It uses the ordinates $\gamma_1, \dots, \gamma_N$ of the first $N = 2\,001\,052$ zeros of ζ (file `zeross.bin`, $\gamma_N = 1\,132\,490.6587$) and the midpoint ordinates $\tau_1, \dots, \tau_N$ computed from (2.5) (file `zLamb.bin`, $\tau_N = 1\,132\,490.7226$). The programs are listed in Appendix A and their complete console output in Appendix B; all figures have linear axes. Each item is an *experiment*: it reports a computation and does not replace a proof.

Products over the ordinates are evaluated as follows: the first 10^5 factors exactly, the factors $10^5 < k \le N$ through power sums of $(\frac14 + o_k^2)^{-1}$, and the factors $k > N$ by the midpoint rule for the clock, which gives the tail $\frac{1}{2\pi}\big(q\frac{L_U+1}{U} - \frac{q^2}{2}\frac{3L_U+1}{9U^3} + \frac{q^3}{3}\frac{5L_U+1}{25U^5}\big)$ with $\mathcal{N}_{\mathrm{RVM}}(U) = N$, $L_U = \log(U/2\pi)$.

18.1. **The clock and the zeros.**

Experiment 18.1 (Clock, midpoints and displacements)**.** *Object:* $\mathcal{N}_{\mathrm{RVM}}$, N_ζ, τ_k, $\eta_k = \tau_k - \gamma_k$. *Range:* $k \le N$. *Expected:* (2.4) to rounding, and $|\eta_k| \le C_\eta$ (Lemma 2.2). *Observed:* $\max_k |\mathcal{N}_{\mathrm{RVM}}(\tau_k) - (k - \frac{1}{2})| = 6.985 \cdot 10^{-10}$; $\max_k |\eta_k| = 0.9937$ (at $k = 34$), mean $|\eta_k| = 0.1512$, root mean square 0.1881, mean $\eta_k = 1.7 \cdot 10^{-7}$. By blocks of k, $\max|\eta_k|$ is $0.840, 0.994, 0.915, 0.834, 0.786, 0.763, 0.717$ on $[1, 10]$, $[11, 10^2]$, $[10^2, 10^3]$, $\dots$, $[10^6, N]$.

Figure 3 shows how the clock places the midpoint ordinates. The staircase N_ζ jumps by one at each γ_k; the smooth clock $\mathcal{N}_{\mathrm{RVM}}$ crosses the half-integer level $k - \frac{1}{2}$ at τ_k. Since $N_\zeta - \mathcal{N}_{\mathrm{RVM}} = S(T) + O(1/T)$ (Figure 4), the horizontal distance between the open square $(\gamma_k, k - \frac{1}{2})$ and the red point $(\tau_k, k - \frac{1}{2})$ is the remainder S converted into ordinate units by the slope $\mathcal{N}'_{\mathrm{RVM}} = \frac{1}{2\pi} \log \frac{T}{2\pi}$. The right panel repeats the picture at height $1.13 \cdot 10^6$, where the mean spacing is 0.519: every τ_k still lies next to its γ_k.

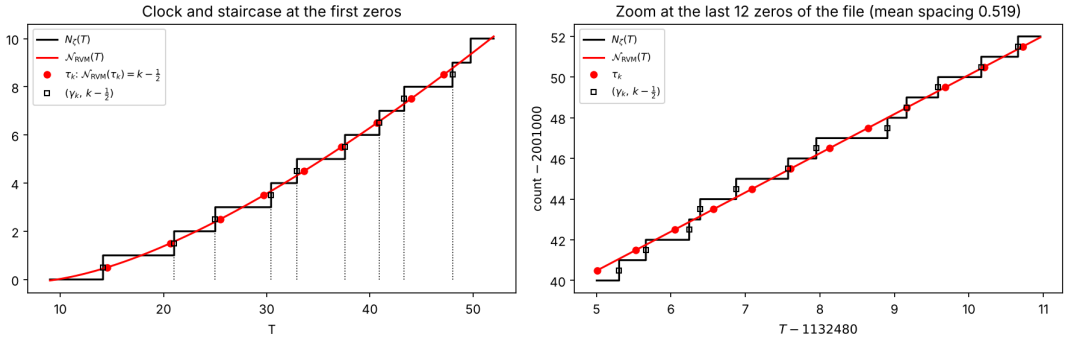


FIGURE 3. The midpoint clock $\mathcal{N}_{\mathrm{RVM}}$ (red), the zero staircase N_ζ (black), the midpoint ordinates τ_k (red points) and the points $(\gamma_k, k - \frac{1}{2})$ (squares). Left: the first nine zeros. Right: the last twelve zeros of the file.

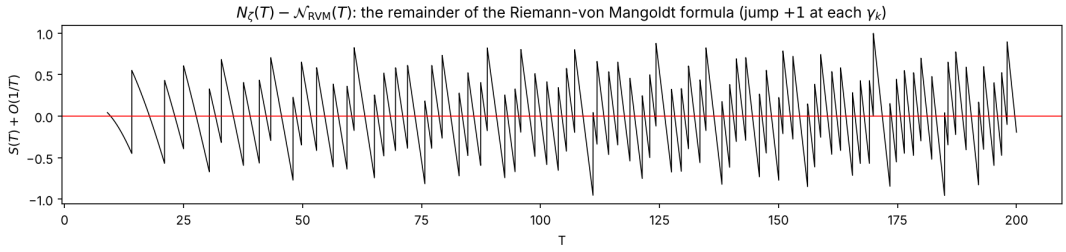


FIGURE 4. The remainder $N_\zeta(T) - \mathcal{N}_{\mathrm{RVM}}(T) = S(T) + O(1/T)$ of (2.2) for $T \le 200$.

18.2. **The λ-family and the endpoint.** Figure 5 shows the core of the method. The lower left panel draws the moving ordinates $\gamma_k(\lambda)$ in clock units, $\mathcal{N}_{\mathrm{RVM}}(\gamma_k(\lambda)) - (k - \frac{1}{2})$: at $\lambda = 1$ all lines meet at 0 (the τ_k), at $\lambda = 0$ they sit at the ordinates of ζ; for $\lambda > 0$ the ordinates $\gamma_k(\lambda)$ remain strictly increasing in k (Theorem 10.1). The lower right panel draws the real functions $Z_\lambda(t) = e^{i\vartheta(t)}\zeta_\lambda(\frac{1}{2} + it)$ for $\lambda = 0, 0.25, 0.5, 0.75, 1$ together with Hardy's function $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$: the zeros of every Z_λ are real, they move continuously with λ, and the curve for $\lambda = 0$ lies on top of Z.

Experiment 18.2 (The endpoint ζ_{Z1} against ζ)**.** *Object:* $\zeta_{Z1} = P_{Z1}/G$ computed from the N ordinates γ_k, and ζ computed by mpmath. *Range:* $\sigma \in \{2, 1.25, 0.75\}$, $1 \le t \le 2000$ (499 points per

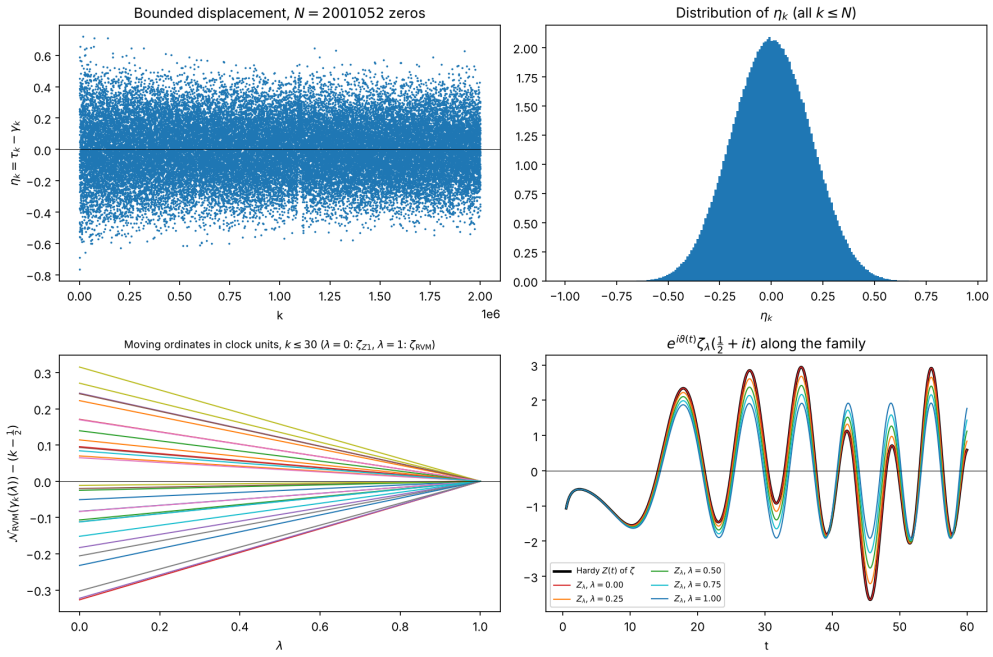


FIGURE 5. Top: η_k for every 50th $k \le N$ and its distribution. Bottom: moving ordinates in clock units, $k \le 30$, and the functions $Z_\lambda(t)$ on the critical line.

line), and the critical line $0.5 \le t \le 60$. *Expected:* if $\zeta_{Z1} = \zeta$, the ratio is 1 up to the truncation of the product. *Observed:* $\max|\zeta_{Z1}/\zeta - 1| = 8.7 \cdot 10^{-10}$, $1.1 \cdot 10^{-9}$, $2.6 \cdot 10^{-9}$ on the three lines, and $\max|Z_{Z1} - Z|/\max(1, |Z|) = 8.4 \cdot 10^{-10}$ on the critical line (Figure 6).

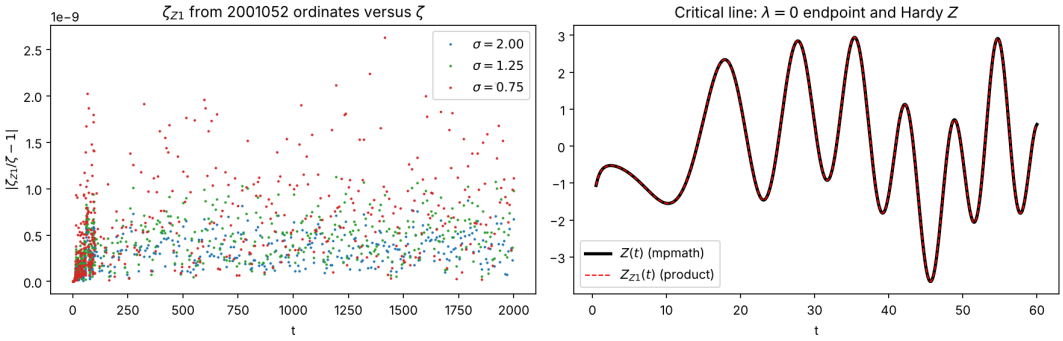


FIGURE 6. Left: $|\zeta_{Z1}/\zeta - 1|$ on three vertical lines. Right: Z_{Z1} and Hardy's Z on the critical line.

Experiment 18.2 shows numerical agreement at the displayed truncation precision. This does not test (D): Proposition 13.3 already forces $|\log \varpi_{Z1}| < 4 \cdot 10^{-13}$ from verified zeros, while the observed 10^{-9} scale is dominated by truncating the ordered clock tail.

Experiment 18.3 (The value at infinity: the product identity)**.** *Object:* the ordered partial products $P_K = \prod_{k \le K}(\frac{1}{4} + \gamma_k^2)/(\frac{1}{4} + \tau_k^2)$ of (12.12). *Range:* $K \le N$. *Expected:* by Corollary 12.3, $\zeta_{Z1} = \zeta$ is the statement $P_K \to \varpi_{\mathrm{RVM}}$. *Observed:* $P_K = 0.96256$, 0.96248, 0.96229, 0.961951, 0.961930, 0.9619418189, 0.9619415390 for $K = 10, 10^2, \ldots, 10^6, N$, against $\varpi_{\mathrm{RVM}} = 0.9619415807$ from Experiment 18.4; for $K \ge 10^6$, P_K stays in $[0.9619389, 0.9619440]$ with mean 0.9619416633

(Figure 7). Proposition 13.3 already bounds the endpoint discrepancy by $4 \cdot 10^{-13}$ under the verified-zero input, so the 10^{-9}–10^{-6} discrepancies in these finite products are truncation/tail errors and are not numerical evidence for (D). The flow at infinity (13.3) is $\Upsilon(\lambda) = 0.0436950$, 0.0412445, 0.0387987, 0.0363572, 0.0339198 for $\lambda = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$, decreasing as in (13.4) with slope close to $-\sum_k 2\eta_k^2/\gamma_k^2 = -0.0098211$, and $\int_0^1 \Upsilon\, d\lambda = \sum_{k\le N} \log \frac{1/4+\tau_k^2}{1/4+\gamma_k^2} = 0.0388016004$ against $\log L = 0.0388015571$.

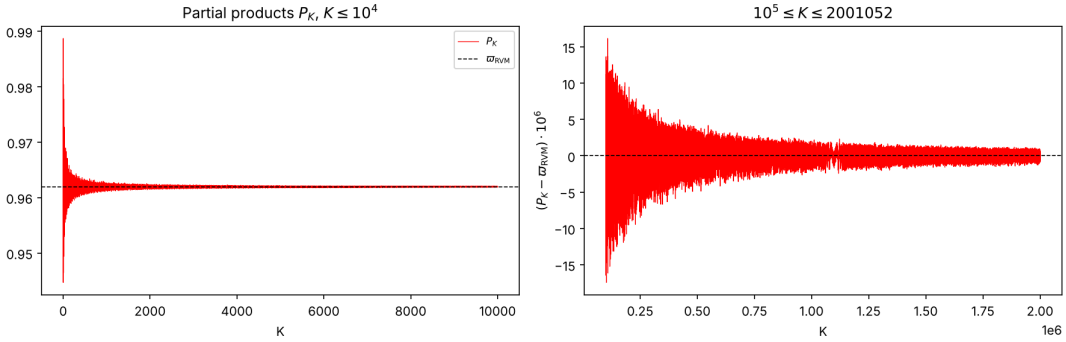


FIGURE 7. The partial products P_K of (12.12) against ϖ_{RVM} (dashed). Left: $K \le 10^4$. Right: $(P_K - \varpi_{\mathrm{RVM}}) \cdot 10^6$ for $10^5 \le K \le N$. The value of the endpoint at $+\infty$ is $\varpi_{Z1} = \varpi_{\mathrm{RVM}}/\lim_K P_K$ (Theorem 12.1).

18.3. The transition theorem and ζ_{RVM} against ζ.

Experiment 18.4 (Transition theorem). *Object:* ζ_{RVM}, $w(s)$, ϖ_{RVM}, L. *Range:* $\sigma' \in \{0.25, 0.5, 1.5\}$, $10 \le t \le 10^4$. *Expected:* Theorem 8.1 and Corollary 8.8: $t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1|$ bounded, $|w| = (t/2\pi)^{-\sigma'}(1 + O(1/t))$. *Observed:* Richardson extrapolation on $\Re s = 2$ gives $\varpi_{\mathrm{RVM}} = 0.9619415807$ and $L = 1.0395641690$; $\max_{t>100} t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1| = 0.0617, 0.0618, 0.0620$ for the three values of σ', and 0.0790 for $\varpi_{\mathrm{RVM}}(1+\chi)$ at $\sigma' = 0.25$; $|w|\,t^{\sigma'} = 1.5832$, 2.5066, 15.66–15.75, i.e. $(2\pi)^{\sigma'}$.

The bottom panels of Figure 8 compare $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on two vertical lines: ζ carries the Euler product and oscillates with the primes, whereas ζ_{RVM} approaches the constant ϖ_{RVM} on $\sigma = 2$ and $\varpi_{\mathrm{RVM}}|1+\chi|$ on $\sigma = 0.75$. This is the analytic content of Theorem 8.11.

18.4. Exponential sums.

Experiment 18.5 (Exponential sums over the two spectra). *Object:* $S(N; \log x) = \sum_{k\le N} x^{io_k}$ for $o_k = \tau_k$ and $o_k = \gamma_k$. *Range:* $N \le 2\,001\,052$, $x \in \{2, 3, 4, 6, 10\}$. *Expected:* $|S_{\mathrm{RVM}}| \ll 1 + \log N$ (Theorem 4.1); for γ_k, Landau's term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$. *Observed:* $\max_N |S_{\mathrm{RVM}}(N; \log x)| = 3.57, 3.98, 3.95, 4.53, 9.70$; at $N = 2\,001\,052$, $|\sum x^{i\gamma_k}| = 88\,301$, 114 268, 62 435, 0.26, 2.42, against Landau's 88 342, 114 324, 62 467, 0, 0.

18.5. Prime counts: π and π_{RVM}.

Experiment 18.6 (Prime counts and RVM primes). *Object:* J_{RVM}, π_{RVM}, c_n^{RVM}, p_j^{RVM} against J, π, the primes. *Range:* $2 \le x \le 100$ with all N ordinates of each list (3000 points); $2 \le x \le 1000$ with the first $K = 17\,061$ ordinates and a smooth cut (40 000 points). *Expected:* the explicit formula with the γ_k reproduces $\pi(x)$; π_{RVM} is continuous and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 5.5). *Observed:* away from the primes, $|\pi_\gamma - \pi| \le 0.40$ on $[2, 100]$; $\pi_{\mathrm{RVM}}(100) = 26.08$,

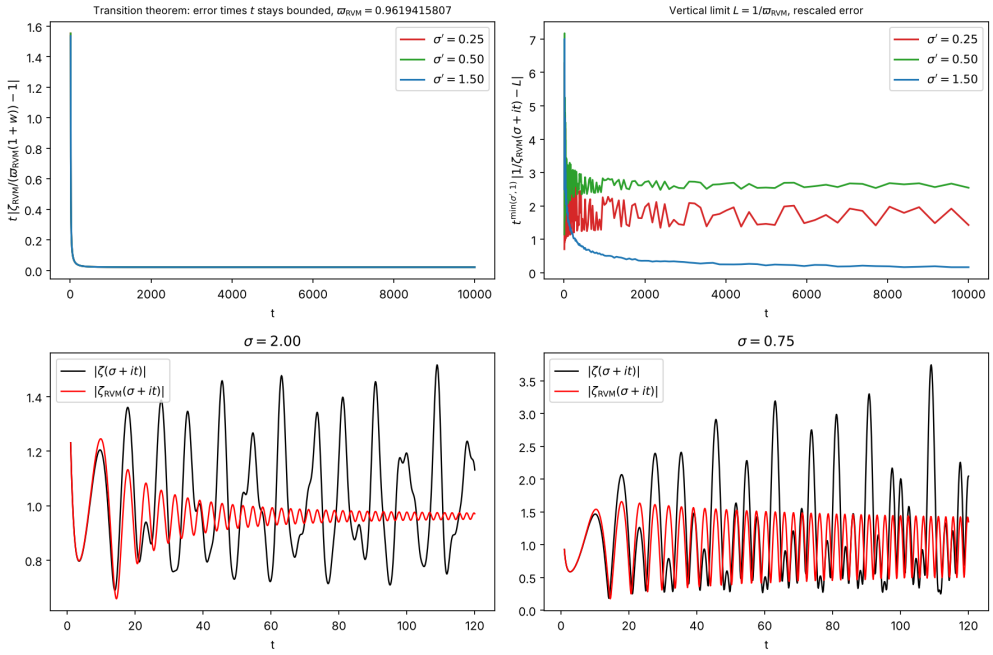


FIGURE 8. Top: the transition error multiplied by t, and the rescaled vertical limit $t^{\min(\sigma',1)}|1/\zeta_{\mathrm{RVM}} - L|$. Bottom: $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on $\sigma = 2$ and $\sigma = 0.75$.

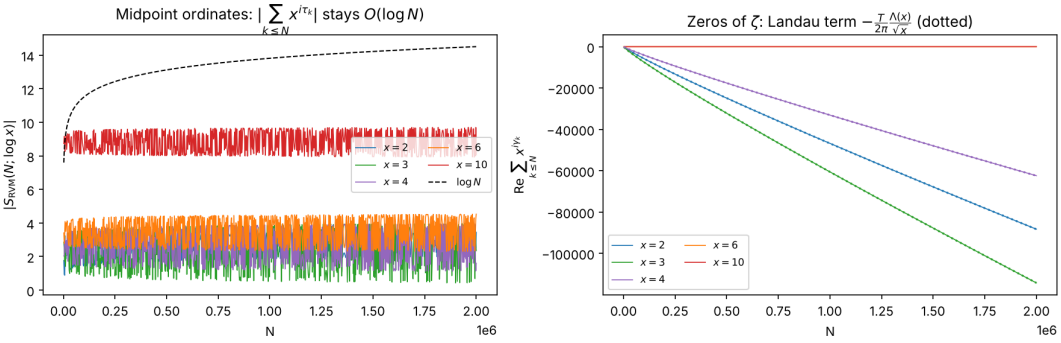


FIGURE 9. The midpoint spectrum carries no prime frequencies (left); the spectrum of ζ resonates at prime powers (right).

$\pi(100) = 25$, $\mathrm{li}(100) = 30.13$; $\pi_{\mathrm{RVM}}(1000) = 169.98$, $\pi(1000) = 168$, $\mathrm{li}(1000) = 177.61$. π_{RVM} is not monotone: it carries a ripple of unit scale, with minimum grid increment -0.25 on $[2, 100]$. The first crossings of the levels $j = 1, \ldots, 8$ are $p_j^{\mathrm{RVM}} = 2.21, 3.16, 5.19, 7.16, 11.14, 14.02, 17.26, 19.18$ against $p_j = 2, 3, 5, 7, 11, 13, 17, 19$.

Figure 10 shows π and π_{RVM} in the usual format. The RVM count follows the prime staircase much more closely than li: replacing each γ_k by τ_k moves every zero by less than one unit (Experiment 18.1), and the low ordinates, which shape $\pi(x)$ for small x, are moved least in relative terms. The steps are replaced by a continuous ripple, so the cell masses c_n^{RVM} of Figure 11 are real numbers, occasionally negative, and are not a prime indicator.

Experiment 18.7 (The three ratios and the square-root scale)**.** *Object:* $\mathrm{li}/\pi_{\mathrm{RVM}}$, π/li, π/π_{RVM}, and $(\pi_{\mathrm{RVM}} - \mathrm{li})/B$, $(\pi - \mathrm{li})/B$ with $B(x) = \sqrt{x}(\log\log x)^2/\log x$. *Range:* the grid of Experiment 18.6 on $[8, 1000]$ ($K = 17\,061$ ordinates). *Expected:* the three ratios tend to 1 (Corollary 5.6) and $(\pi_{\mathrm{RVM}} -$

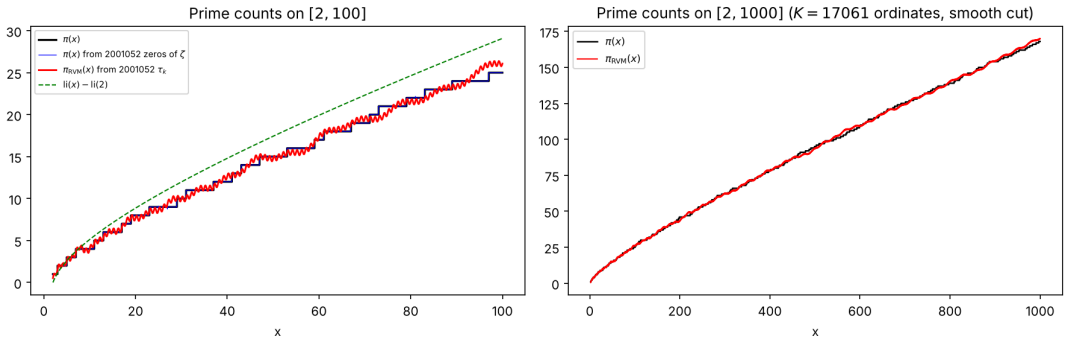


FIGURE 10. $\pi(x)$ (black staircase), the explicit formula with the zeros of ζ (blue), $\pi_{\mathrm{RVM}}(x)$ (red) and $\mathrm{li}(x) - \mathrm{li}(2)$ (green). The distance between the red and green curves is the quantity bounded in Theorem 5.5; the ratios are in Figure 12.

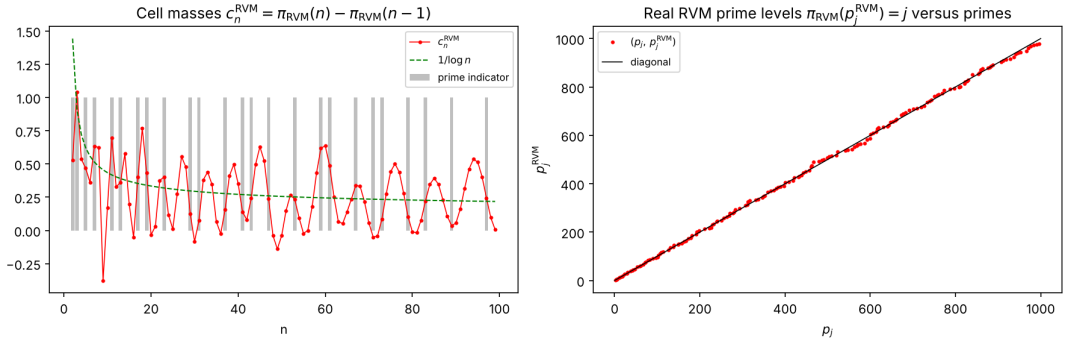


FIGURE 11. Left: cell masses c_n^{RVM}, the prime indicator and $1/\log n$. Right: real RVM prime levels against the primes up to 1000.

$\mathrm{li})/B$ stays bounded (Theorem 5.5). *Observed:* at $x = 100, 500, 1000$, $\mathrm{li}\,/\pi_{\mathrm{RVM}} = 1.155, 1.086, 1.045$, $\pi/\,\mathrm{li} = 0.830, 0.933, 0.946$, $\pi/\pi_{\mathrm{RVM}} = 0.959, 1.013, 0.988$; $\max|\pi_{\mathrm{RVM}} - \mathrm{li}\,|/B = 2.72$ on $[8, 1000]$, 1.35 on $[50, 1000]$ and 0.72 on $[500, 1000]$, while $\max|\pi - \mathrm{li}\,|/B = 2.45$ on $[8, 1000]$ (Figure 12).

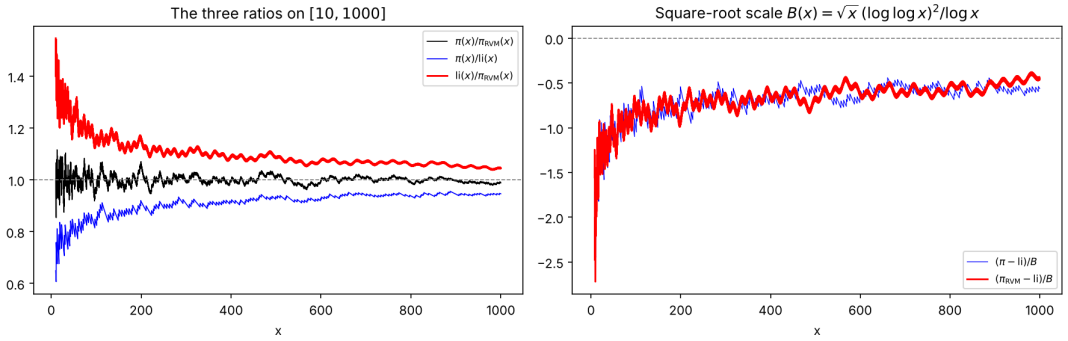


FIGURE 12. Left: the three ratios of Corollary 5.6. Right: $\pi_{\mathrm{RVM}} - \mathrm{li}$ and $\pi - \mathrm{li}$ in units of the scale $B(x)$ of Theorem 5.5. The ratio π/π_{RVM} (black) is the closest of the three to 1; the ratios with li approach 1 at the rate $\frac{1}{2}\,\mathrm{li}(\sqrt{x})/\,\mathrm{li}(x)$ of the bias $J - \pi$.

On $[2, 1000]$ the two normalized errors in Figure 12 have the same size and sign: both counts lie below li by about $\frac{1}{2}\,\mathrm{li}(\sqrt{x})$, the $m = 2$ term of the Möbius inversion (5.20). This is why π/π_{RVM} is much closer to 1 than either ratio with li.

18.6. **The Mertens functions and the ternary division.** The Mertens function is the sum of the ternary sequence $\mu(n) \in \{-1, 0, +1\}$. Its values split into two independent pieces of information:

(a) *the zero set* $\{n : \mu(n) = 0\}$, the non-squarefree integers, of density $1 - 6/\pi^2 = 0.392073$;

(b) *the signs* on the squarefree integers, $\mu(n) = (-1)^{\omega(n)}$, each sign of density $3/\pi^2 = 0.303964$.

The ternary walk μ_h of (1.9)–(1.10) keeps (a) exactly, $\mu_h(n) = 0 \Leftrightarrow \mu(n) = 0$, and replaces (b): at a squarefree n it steps up when $M_h(n-1)$ lies below the target $\mathcal{M}(n) = M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ and down otherwise. Thus

$$M_h(n) = M_h(n-1) + \mu(n)^2\,\varsigma_n, \qquad |M_h(n) - \mathcal{M}(n)| \text{ small}, \tag{18.1}$$

so that M_h is an integer walk with the zero set of μ that follows $M^{\mathrm{crit}}_{\mathrm{RVM}}$, in the same way as M is the integer walk with the zero set of μ whose signs are $(-1)^{\omega(n)}$.

n	$\mu(n)$	$\mu(n)^2$	$\mathcal{M}(n)$	$M_h(n-1)$	ς_n	$\mu_h(n)$	$M_h(n)$	$M(n)$
1	1	1	0.2925	0	+1	1	1	1
2	−1	1	0.4563	1	−1	−1	0	0
3	−1	1	−0.6523	0	−1	−1	−1	−1
4	0	0	−0.8720	−1	–	0	−1	−1
5	−1	1	−1.7255	−1	−1	−1	−2	−2
6	1	1	−1.3063	−2	+1	1	−1	−1
7	−1	1	−1.5931	−1	−1	−1	−2	−2
8	0	0	−2.4029	−2	–	0	−2	−2
9	0	0	−1.3292	−2	–	0	−2	−2
10	1	1	−1.3280	−2	+1	1	−1	−1
11	−1	1	−2.3272	−1	−1	−1	−2	−2
12	0	0	−2.2067	−2	–	0	−2	−2
13	−1	1	−1.8468	−2	+1	1	−1	−3
14	1	1	−1.9324	−1	−1	−1	−2	−2
15	1	1	−1.5435	−2	+1	1	−1	−1

TABLE 8. The ternary division for $n \le 15$ (the first 40 values are in Appendix B.6). The zero columns of μ and μ_h coincide; the signs of μ_h are chosen by comparing $M_h(n-1)$ with $\mathcal{M}(n)$. For $n \le 12$ the two walks agree; at $n = 13$ the target -1.85 lies above $M_h(12) = -2$, so $\mu_h(13) = +1$ while $\mu(13) = -1$.

Experiment 18.8 (Mertens functions and the ternary walk)**.** *Object:* M, $M^{\mathrm{crit}}_{\mathrm{RVM}}$, M_h; the explicit formulas. *Range:* $n \le 10^6$; explicit formulas on $[2, 2000]$ with all N ordinates of each list. *Expected:* the explicit formula with the zeros of ζ reproduces M; $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = O(x^{1/2+\varepsilon})$ (Theorem 8.12); M_h tracks $M^{\mathrm{crit}}_{\mathrm{RVM}}$ with the densities of μ. *Observed:*

- explicit formula for M with the γ_k: root mean square error 0.0005 on $[2, 2000]$ off the integers;
- $\zeta'_{\mathrm{RVM}}(\rho^{\mathrm{RVM}}_k)/\log(\tau_k/2\pi) = 0.96194$ at $k = 1000$, against $\varpi_{\mathrm{RVM}} = 0.96194$ (Corollary 8.6); the values $|\zeta'(\rho_k)|/\log(\gamma_k/2\pi)$ are spread over $(0, 3)$ (Figure 15);
- spectral sums $2\sum |a|^2$: 0.029034 for ζ, 0.027203 for the midpoint system;
- the integer target agrees with the full explicit formula for $M^{\mathrm{crit}}_{\mathrm{RVM}}$ to 0.019 at 100 integers $\le 10^5$;
- $\max |M_h - \mathcal{M}| = 2.62$ on $[1, 10^6]$ and 1.65 on $[10^4, 10^6]$;

- densities of $(-1, 0, +1)$ for μ: 0.303857, 0.392074, 0.304069;
- densities of $(-1, 0, +1)$ for μ_h: 0.304003, 0.392074, 0.303923;
- $\mu_h(n) = \mu(n)$ at 50.03% of the squarefree $n \le 10^6$, and $\sum_{n\le 10^6} \mu(n)\mu_h(n)/\#\{n \le 10^6 : \mu(n) \ne 0\} = 0.0007$: the sign sequences are uncorrelated although both walks have the same zero set and comparable size;
- $\max_{10\le n\le 10^6} |W(n)|/\sqrt{n} = 0.832, 0.754, 0.688$ for $W = M, M^{\mathrm{crit}}_{\mathrm{RVM}}, M_h$;
- energies $\frac{1}{N}\sum_{n\le N} W(n)^2$ at $N = 10^6$: 14 851, 12 538, 12 538.

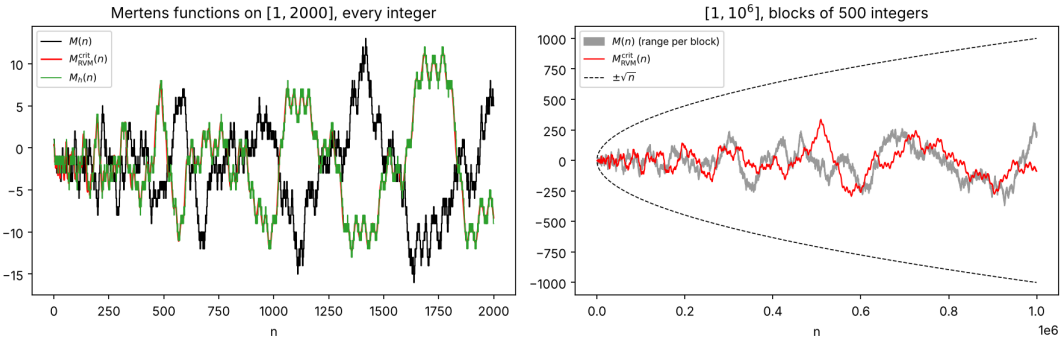


FIGURE 13. Left: M (black), $M^{\mathrm{crit}}_{\mathrm{RVM}}$ (red) and M_h (green) at every integer $n \le 2000$. Right: range of M and mean of $M^{\mathrm{crit}}_{\mathrm{RVM}}$ in blocks of 500 integers up to 10^6, with $\pm\sqrt{n}$.

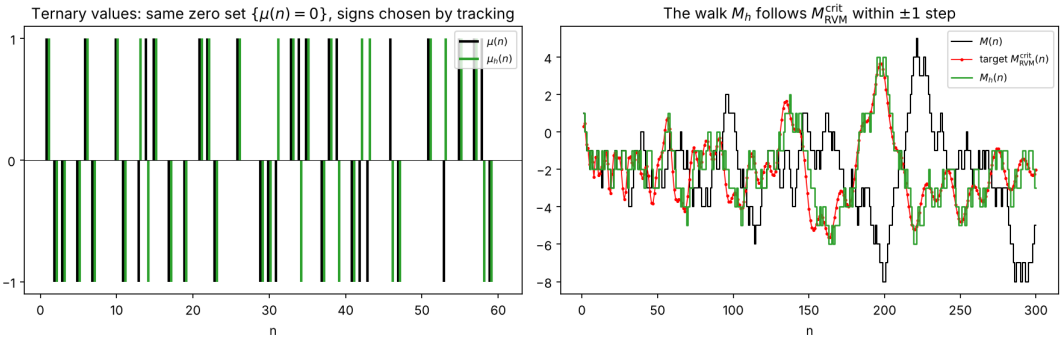


FIGURE 14. Left: $\mu(n)$ and $\mu_h(n)$ for $n \le 60$: same zeros, different signs. Right: M, the target $M^{\mathrm{crit}}_{\mathrm{RVM}}$ and M_h for $n \le 300$.

Figures 13–15 summarize the Mertens side. $M^{\mathrm{crit}}_{\mathrm{RVM}}$ has the size and the visual texture of M, it stays inside $\pm\sqrt{n}$ in agreement with the square-root-scale behaviour of Theorem 8.12, and its residues are deterministic; M_h is its ternary integer version with the zero set of μ. The passage from these synthetic objects to M itself is not numerical: it is the identity $\zeta_{Z1} = \zeta$ of Step 3, which Experiment 18.2 is consistent with.

Remark 18.9 (The opposite phase of M and $M^{\mathrm{crit}}_{\mathrm{RVM}}$ between 10^3 and $2 \cdot 10^3$). In the left panel of Figure 13 the curves of M and $M^{\mathrm{crit}}_{\mathrm{RVM}}$ are in opposite phase between $n \approx 10^3$ and $n \approx 2 \cdot 10^3$. Heuristically, the visible anticorrelation is largely explained by the first mode, $2\sqrt{x}\,\Re(x^{i\gamma_1}/(\rho_1\zeta'(\rho_1)))$ and $2\sqrt{x}\,\Re(x^{i\tau_1}/(\rho_1^{\mathrm{RVM}}\zeta'_{\mathrm{RVM}}(\rho_1^{\mathrm{RVM}})))$, two oscillations in $\log x$ with the frequencies γ_1 and τ_1. Their phase difference is $\eta_1 \log x$ with $\eta_1 = \tau_1 - \gamma_1 = 14.5213 - 14.1347 = 0.3866$, up to the constant difference of the arguments of the two residues, which is not computed here; $\eta_1 \log x$ equals π at

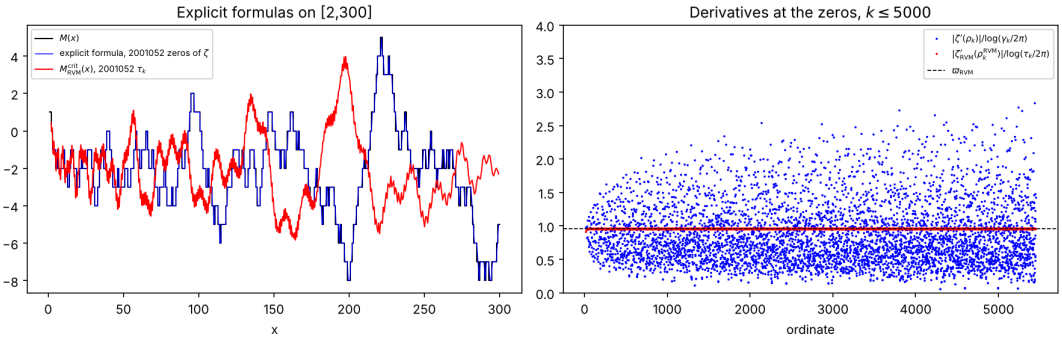


FIGURE 15. Left: $M(x)$, its explicit formula from the zeros of ζ and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x)$ from the τ_k on $[2,300]$. Right: normalized derivatives at the zeros: erratic for ζ, constant ϖ_{RVM} for ζ_{RVM}.

$x = e^{\pi/\eta_1} \approx 3.4\cdot 10^3$ and lies between 2.67 and 2.94 on $[10^3, 2\cdot 10^3]$. The beat has period $2\pi/\eta_1 \approx 16.3$ in $\log x$. This is the rotation (14.42) seen on the Mertens side: the passage from the clock to ζ turns the mode k by the angle $\eta_k \log x$. The higher modes turn at different rates, and at large scale neither a correlation nor an anticorrelation remains, in agreement with the value 0.0007 of Experiment 18.8. The effect depends on η_1 alone and changes sign with x; it does not distinguish $\lambda = 0$.

18.7. **The rung equations: transport, the defect of the clock, and the solution from the clock.** The four experiments of this subsection evaluate the objects of Subsections 14.13 and 15.2. In the first two, f is the function $\exp(-1/(1-y^2))$, $y = (u-0.36)/0.26$, on $[0.10, 0.62] \subset (0, \log 2)$, normalized to $\int \mathsf{f} = 1$; it tests the first rung. In the programs f, $\mathsf{k_f}$ and $\mathsf{c_f}$ are called `psi`, `W_psi` and `c_psi`.

Experiment 18.10 (Transport on the first rung)**.** *Object:* $\mathsf{R_f}(1)$, $\mathsf{S_f}(\lambda)$ and the moments $\mathsf{m}_j(\mathsf{f})$ of (14.78)–(14.82). *Range:* the first 20 000 ordinates of each list; $\max_{k\ge 1000} |\mathsf{c_f}(\tau_k)| = 2.5\cdot 10^{-10}$ and $\max_{k\ge 3000} |\mathsf{c_f}(\tau_k)| = 2.1\cdot 10^{-13}$. *Expected:* if $\zeta_{Z1} = \zeta$, then $\mathsf{S_f}(0) = \mathsf{R_f}(1)$ (Proposition 14.41); the series (14.82) converges to $\mathsf{S_f}(0)$ with terms bounded by $C(1+j)^2(C_\eta \log 2)^j/j!$ (Theorem 14.42). *Observed:*

- $\mathsf{k_f} = 0.133643530916$, $\sum_k \mathsf{c_f}(\tau_k) = 0.176106813076$, $\sum_k \mathsf{c_f}(\gamma_k) = 0.133643530929$;
- defect of the clock $\mathsf{R_f}(1) = -0.042463282161$; transport $\mathsf{S_f}(0) = -0.042463282147$; $\mathsf{R_f}(0) = \mathsf{R_f}(1) - \mathsf{S_f}(0) = -1.4\cdot 10^{-11}$;
- moment series (Table 9): the term $j=1$ is 95.1% of $\mathsf{R_f}(1)$, the partial sum with $j \le 2$ is 100.8%, and the partial sums with $j \le 3$, $j \le 5$, $j \le 8$ differ from $\mathsf{R_f}(1)$ by $3.7\cdot 10^{-6}$, $6.5\cdot 10^{-9}$, $1.0\cdot 10^{-11}$;
- along the family (Table 9) $\mathsf{R_f}(\lambda)$ is monotone and almost linear on $[-1,2]$, with slope about -0.044 and a single zero, at $\lambda = 0$;
- the partial sums $\sum_{k\le K_0}(\mathsf{c_f}(\gamma_k) - \mathsf{c_f}(\tau_k))$ are -0.02086, -0.04381, -0.04649, -0.04221, -0.04249, -0.0424634 for $K_0 = 1, 5, 10, 50, 100, 400$: the first ordinate gives half of the transport, and the first rung is decided by about fifty ordinates.

Experiment 18.11 (The defect of the clock by Poisson summation)**.** *Object:* the two parts of $\mathsf{R_f}(1)$ in Proposition 14.43. *Expected:* $\mathsf{R_f}(1) = (\mathsf{k_f} - \mathsf{l}_0) - 2\sum_{m\ge 1}(-1)^m \mathsf{l}_m$. *Observed:* $\mathsf{t}_0 = 9.676906787$; $\mathsf{l}_0 = 0.063533498702$ from (14.84); the mismatch of the smooth densities is $\mathsf{k_f} - \mathsf{l}_0 = +0.070110032214$, and the oscillating part, obtained as the difference $\mathsf{R_f}(1) - (\mathsf{k_f} - \mathsf{l}_0)$ and not by summing the l_m, is -0.112573314375. The stationary heights $2\pi e^u$, u in the support of f, fill $[6.944, 11.680]$; only those with $u > 0.4319$ lie above t_0.

j	$\mathsf{m}_j(\mathsf{f})$	$(-1)^j\mathsf{m}_j(\mathsf{f})/j!$	partial sum $-\ \mathsf{R}_\mathsf{f}(1)$	λ	$\mathsf{R}_\mathsf{f}(\lambda)$
1	$+4.0395\cdot10^{-2}$	$-4.0395\cdot10^{-2}$	$+2.07\cdot10^{-3}$	-1.00	$+0.045303554710$
2	$-4.8098\cdot10^{-3}$	$-2.4049\cdot10^{-3}$	$-3.37\cdot10^{-4}$	-0.50	$+0.022413444752$
3	$-2.0432\cdot10^{-3}$	$+3.4053\cdot10^{-4}$	$+3.69\cdot10^{-6}$	-0.25	$+0.011133149816$
4	$-5.5184\cdot10^{-5}$	$-2.2993\cdot10^{-6}$	$+1.39\cdot10^{-6}$	0.00	-0.000000000014
5	$+1.6734\cdot10^{-4}$	$-1.3945\cdot10^{-6}$	$-6.47\cdot10^{-9}$	$+0.25$	-0.010956709511
6	$+5.5721\cdot10^{-6}$	$+7.7390\cdot10^{-9}$	$+1.27\cdot10^{-9}$	$+0.50$	-0.021706821780
7	$+7.2608\cdot10^{-6}$	$-1.4406\cdot10^{-9}$	$-1.69\cdot10^{-10}$	$+1.00$	-0.042463282161
8	$+6.4039\cdot10^{-6}$	$+1.5883\cdot10^{-10}$	$-9.77\cdot10^{-12}$	$+2.00$	-0.080112190380

TABLE 9. Experiment 18.10. Left: the moments $\mathsf{m}_j(\mathsf{f}) = \sum_k \eta_k^j \mathsf{c}_\mathsf{f}^{(j)}(\tau_k)$ and the partial sums of the series (14.82) for $\mathsf{S}_\mathsf{f}(0)$, compared with the defect $\mathsf{R}_\mathsf{f}(1) = -0.042463282161$ of the clock. Right: the rung functional along the family.

Experiment 18.12 (From the clock to the ordinates and to the masses of the primes)**.** *Object:* a finite section of the system (15.5): the unknowns are the ordinates κ_k below a height and the masses a_n, $2 \le n \le 11$; the equations are the pairings of (15.5) with $\omega_{\mathrm{test}}(u)\cos(\omega u)$ and $\omega_{\mathrm{test}}(u)\sin(\omega u)$, where ω_{test} is a fixed smooth function supported in $(0, \log 13.5)$ and ω runs over a grid of step 0.1 in $[0, \Omega]$. The ordinates above $\Omega + 12$ are kept on the clock; only the first $KT = 6000$ clock ordinates are summed numerically. Although the support window $(0, \log 13.5)$ contains $\log 12$ and $\log 13$, the implementation drops mass columns whose bump weight is below 0.05, so those two masses are fixed to zero in this run. These are explicit truncation sources in the residual. *Method:* Levenberg–Marquardt iteration started at $\kappa_k = \tau_k$, the masses being eliminated linearly at each step, with continuation in height $\Omega = 18, 22, \dots, 50$. The file `zeross.bin` is read only after the iteration, for the comparison. *Expected:* by Theorem 15.5, if $\zeta_{Z1} = \zeta$ the system has the unique solution $\kappa_k = \gamma_k$, $\mathsf{a}_n = \log p/(2\sqrt{n})$ at prime powers and 0 elsewhere. *Observed:* the iteration converges at every stage (residual norms between $4\cdot10^{-4}$ and $5\cdot10^{-3}$, of the size of the contribution of the ordinates kept on the clock); the solution is in Table 10: the first ten ordinates agree with $\gamma_1, \dots, \gamma_{10}$ within $4.4\cdot10^{-4}$, the eleventh within $2.2\cdot10^{-3}$, and the masses agree with (15.6) within $8\cdot10^{-4}$, including the values 0 at $n = 6, 10$ and the prime-power values at $n = 4, 8, 9$. The ordinates $k = 12, 13, 14$, which lie above the band $\omega \le 50$, are not determined (errors 0.10, 0.77, 1.73). With masses up to $n \le 34$ and the band $\omega \le 60$ the section is not resolved and the iteration does not converge: resolving the ordinates up to a height Ω requires a strict margin $\log N > (1+\varkappa)\log(\Omega/2\pi)$ for some fixed $\varkappa > 0$ in a stable finite solve and resolving the masses up to N requires Ω of the order of πN, in agreement with the heights $2\pi x^{1/m}$ of Proposition 14.43. The experiment illustrates the existence form of the final step; it is a computation and not a proof of existence.

Experiment 18.13 (Multiscale Jacobian of the clock)**.** *Object:* the minimum midpoint-clock gap, its bound (14.86), the Ingham critical window $2\pi/\Delta_T$, and the adjacent-pair bound (14.88). *Input:* none of the genuine zeta ordinates; the program `s12_multiscale_jacobian.py` generates the τ_k directly from the Lambert formula up to $K = 2\,001\,052$. *Observed:*

K	$T = \tau_K$	Δ_T	$2\pi/\log(T/2\pi)$
10^2	235.987267	1.73462445	1.73286190
10^3	1419.517765	1.15930485	1.15921748
10^4	9877.629616	0.85368118	0.85367617
10^5	74920.891033	0.66939912	0.66939880
$5\cdot10^5$	319387.322632	0.57982852	0.57982848
$2\,001\,052$	1132490.722588	0.51918344	0.51918343

k	τ_k (start)	solution	γ_k	n	a_n (solution)	(15.6)
1	14.521346953	14.134660056	14.134725142	2	+0.245076213	0.245064536
2	20.655740356	21.022091631	21.022039639	3	+0.317145240	0.317142050
3	25.492675432	25.010827129	25.010857580	4	+0.173268937	0.173286795
4	29.739411632	30.424917668	30.424876126	5	+0.359884977	0.359881258
5	33.624531889	32.935072953	32.935061588	6	−0.000004422	0
6	37.257370087	37.586146548	37.586178159	7	+0.367785301	0.367742452
7	40.700551604	40.918919600	40.918719012	8	+0.122615622	0.122532268
8	43.994013953	43.327322117	43.327073281	9	+0.183143344	0.183102048
9	47.165128427	48.004716753	48.005150881	10	−0.000286111	0
10	50.233653391	49.773657726	49.773832478	11	+0.362276961	0.361496314
11	53.214422874	52.972535790	52.970321478			

TABLE 10. Experiment 18.12. Solution of a finite section of the rung system (15.5), started at the clock. Left: the ordinates. Right: the masses at the points $\log n$, which are not assumed, against $\log p/(2\sqrt{n})$ at prime powers $n = p^m$.

At the last height, $2\pi/\Delta_T = 12.102053$ while $\log(T/2\pi) = 12.10205288$. Thus the rigorous scale $U \asymp \log T$ is numerically almost exact for the midpoint clock. On every fixed window the adjacent-pair upper bound decreases with the clock gap, illustrating the obstruction in Theorem 14.44(iii). These calculations use no γ_k and no η_k.

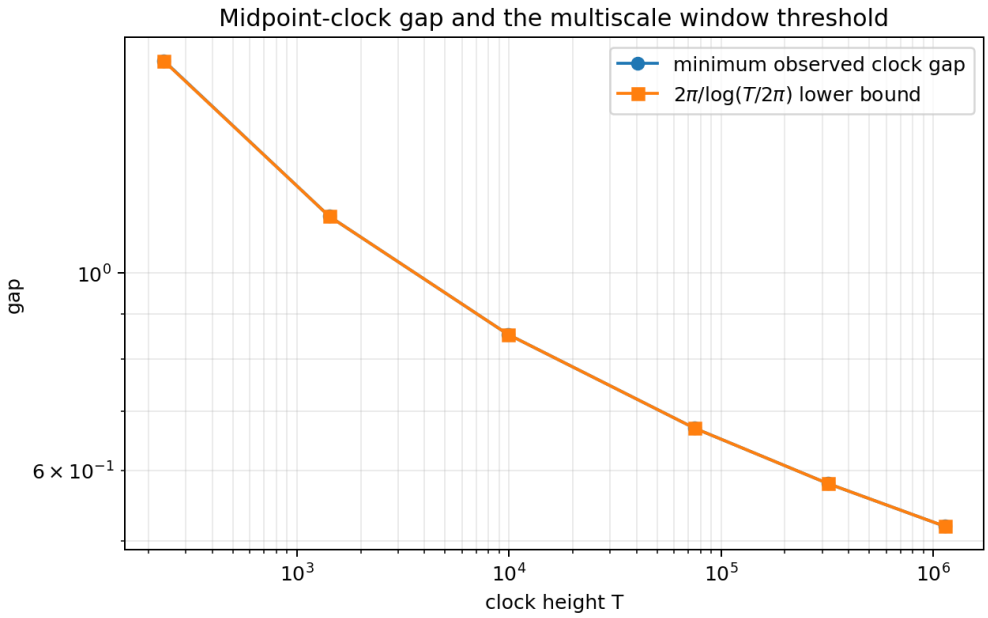


FIGURE 16. Minimum gap of the midpoint clock up to height T and the explicit lower bound $2\pi/\log(T/2\pi)$. Their ratio is already $1 + 2 \cdot 10^{-8}$ at $K = 2\,001\,052$.

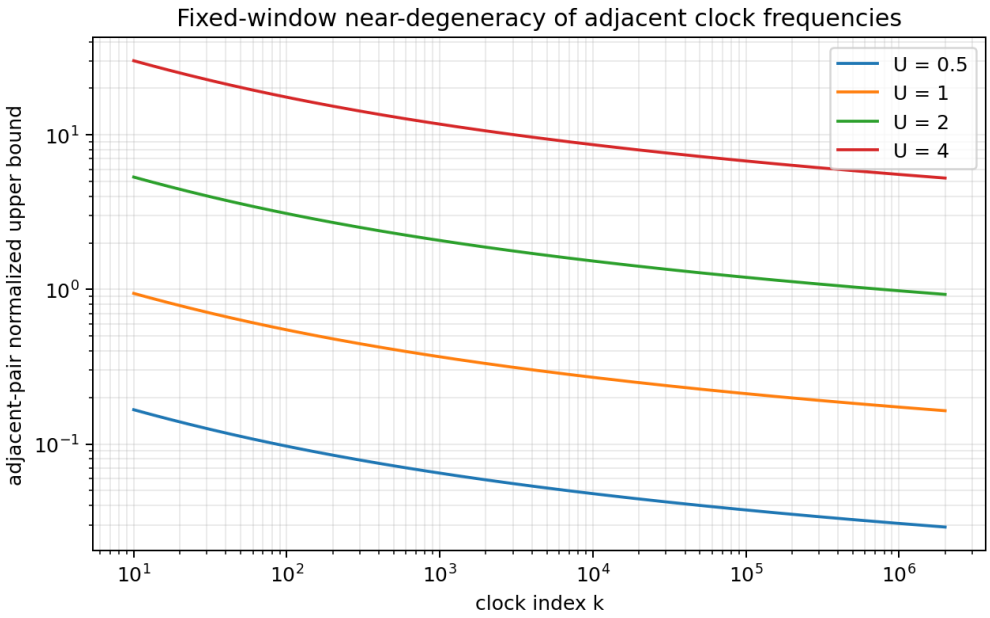


FIGURE 17. Upper bound in (14.88) for adjacent clock frequencies on four fixed u-windows. The decay is slow, of order $1/\log \tau_k$, but it is enough to exclude a height-uniform lower frame bound on any fixed window.

19. Conclusions and open problems

The following statements are proved (the results of Section 17 are summarised in its status table, Section 17.8):

(1) the midpoint ordinates τ_k are given by the Lambert formula (2.5) and stay within a bounded distance of the ordinates γ_k of ζ (Lemma 2.2);
(2) $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and symmetric, and ζ_{RVM} has exactly the simple zeros $\frac{1}{2} \pm i\tau_k$, the trivial zeros and a residue-one pole at 1 (Theorem 3.1);
(3) $S_{\mathrm{RVM}}(N;u) \ll 1 + \log N$ uniformly on compact positive frequency intervals, so the point centroid and $M_H^{\mathrm{RVM}}(x)$ tend to 0 (Section 4);
(4) J_{RVM} exists compact-uniformly and is related to π_{RVM} by exact finite Möbius inversion (Section 5);
(5) $J_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$, so $\mathrm{li}/\pi_{\mathrm{RVM}} \to 1$, and with the prime number theorem $\pi/\mathrm{li} \to 1$ and $\pi/\pi_{\mathrm{RVM}} \to 1$ (Theorem 5.5, Corollary 5.6);
(6) J_{RVM} is Lipschitz on $[2,\infty)$, $c_n^{\mathrm{RVM}} = O(1)$, and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Section 6);
(7) $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ on every band to the right of $\frac{1}{2}$ and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+\chi)(1+O(1/|t|))$ on closed substrips of $\frac{1}{2} < \Re s < 1$ (Theorem 8.1);
(8) $1/\zeta_{\mathrm{RVM}} \to L$ on every vertical line to the right of $\frac{1}{2}$, $\zeta_{\mathrm{RVM}}^{\pm 1}$ have no absolutely convergent Dirichlet expansion, and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ (Section 8);
(9) 2ξ equals its Hadamard product over $\rho(1-\rho)$ exactly (Theorem 9.1), and $|\zeta| \asymp |\mathcal{Q}_{\mathrm{RVM}}|$ on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$ (Corollary 9.3);
(10) the λ-family from ζ_{RVM} to ζ_{Z1} converges jointly, has simple zeros for $\lambda > 0$, an explicit generator and total multiplier, and its members differ only through $\gamma_k - \tau_k$ (Section 10);
(11) ζ_{Z1} has the functional equation, the pole, the residue and the order of ζ, and $\zeta_{Z1} = \zeta \Leftrightarrow$(D)$\Leftrightarrow$RH (Theorem 11.2);
(12) ζ_{RVM} is bounded above and below on every half-plane $\Re s \ge \sigma_0 > 1$ and tends to ϖ_{RVM} as $\Re s \to +\infty$ (Theorem 8.10); $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, so (D) is equivalent to an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s > 1$ (Theorem 11.3);
(13) no member of the λ-family, ζ_{Z1} included, can have a zero off the critical line other than the trivial zeros (Theorem 10.1);
(14) the endpoint has the value $\varpi_{Z1} = \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K}(\frac{1}{4}+\tau_k^2)/(\frac{1}{4}+\gamma_k^2) \ge 1$ at $+\infty$, and $\zeta_{Z1} = \zeta$ if and only if $\varpi_{Z1} = 1$ (Theorems 12.1, 12.2, Corollary 12.3);
(15) the prime-power counts of ζ_{RVM} and ζ_{Z1} are the Mellin transforms of their logarithms and satisfy a distributional explicit formula; this proves the Mellin–Stieltjes identification and the weighted factorization $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$, and, with the jump-growth hypothesis of Theorem 12.7(ii), (D) holds if and only if the prime-power count of ζ_{Z1} is constant between consecutive integers (Theorems 12.5, 12.6, 6.4, 12.7);
(16) along the family $\frac{d}{d\lambda}\log\varpi_\lambda = -\Upsilon(\lambda)$, and (D) is the identity $\int_{0^+}^1 \Upsilon\, d\lambda = \log L$; with the verified zeros, $0 \le \log\varpi_{Z1} < 4\cdot 10^{-13}$ (Propositions 13.2, 13.3);
(17) the atomized family $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde h_\lambda}$ consists of absolutely convergent Dirichlet series in $\Re s > 1$ for every λ, the endpoint included, because the cell masses have mean square $X^{1+\varepsilon}$ on dyadic blocks uniformly in λ; the joining factor is explicit cell by cell and holomorphic in $\Re s > \frac{1}{2}$ (in $\Re s > 0$ for $\lambda = 1$), and (D) is equivalent to the vanishing of every cell defect at $\lambda = 0$ and of J_{Z1} on the first cell $(1,2)$ (Proposition 13.5, Theorems 13.6, 13.10, 13.12, 13.13);
(18) at the clock endpoint the joining factor is not constant, $\tilde h_1(\sigma) \to \log L$; the atomized reciprocals have Möbius coefficients $\tilde\mu_\lambda$, and the endpoint Mertens function has a Mellin form B_{Z1} with $B_{Z1} = M$ if and only if (D) (Propositions 13.15, 13.16);

(19) Step 3: $\zeta_{Z1} = \zeta$ implies $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 11.8); equivalently, the synthetic bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ transfers to M through $1/\zeta = F_{\mathrm{RVM}}\Pi_{\mathrm{tr}}^{-1}$ (Theorem 11.9);

(20) for $0 < \lambda \le 1$ no ζ_λ is an absolutely convergent Dirichlet series in $\Re s > 1$ and no joining factor $\tilde h_\lambda$ is constant, and $\lambda \mapsto \log \zeta_\lambda(\sigma)$ is strictly convex for every real $\sigma > 1$ (Propositions 14.1, 14.2);

(21) if the joining factor $e^{\tilde h_\lambda}$ is meromorphic of finite order with finitely many poles and symmetric under $s \mapsto 1 - s$, Hamburger's theorem gives $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$, and for $\lambda = 0$ also $\zeta_{Z1} = \zeta$ (Theorem 14.3);

(22) if ζ_λ is the Mellin–Stieltjes transform of a positive measure, then $\varpi_\lambda \le 1$; (D) holds if and only if ζ_{Z1} has such a representation, if and only if J^{M}_{Z1} is non-decreasing (Theorem 14.7, Corollary 14.8);

(23) J_{RVM} is continuously differentiable on $(2,\infty)$ with $J'_{\mathrm{RVM}}(y) = \frac{1}{\log y} + 2\cos(2\pi y) - 2^{-1/2}y^{-1/4}\cos(4\pi\sqrt{y} + \frac{\pi}{4}) + O(y^{-1/3})$; consequently $\tilde c_n(1) = \frac{1}{\log n} + O(n^{-1/4})$ off the powers of 2, and Hypothesis 6.7 holds off the integers 2^j with $\mu(j) \ne 0$ (Theorem 14.15, Corollary 14.16);

(24) the cell masses are sums over the ordinates with the height kernel $\hat\varphi(\gamma/2\pi n)$, which vanishes at the heights $2\pi qn$; their flow is the pairing of V_n with the count difference $N_\zeta - n_{\mathrm{RVM}}$; and the Gaussian defect of the atomization is $\frac12(e^{\tilde h_\lambda(1)} - 1)y^{-1/2} + O(y^{-1/4-\delta})$ (Propositions 14.17, 14.18, 14.22);

(25) $\zeta_{Z1}(\sigma) \ge \zeta(\sigma)$ for every real $\sigma > 1$, with equality at one point if and only if $\zeta_{Z1} = \zeta$; in particular $\prod_k (1 + 2/(\frac14 + \gamma_k^2)) \ge \pi/3$ (Theorem 14.11);

(26) every ζ_λ is the Mellin transform of a function A_λ that satisfies the summation formula of ζ, and $\varpi_\lambda - 1 = -\mathsf{W}(\lambda)/\hat{\mathsf{g}}(1)$ for one explicit number $\mathsf{W}(\lambda)$; (D) is equivalent to $\mathsf{W}(0) \ge 0$ and to $\varpi_\lambda < 1$ for all $\lambda \in (0,1]$ (Lemma 14.9, Propositions 14.10, 14.12);

(27) Hamburger's theorem for the pair holds without continuation of the joining factor: $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$ if and only if the atomized coefficients satisfy the theta relation, if and only if the vertical mean square of $\tilde\zeta_{\mathbb{Z},\lambda}/\zeta$ on a line $\Re s = \sigma > 1$ is 1; both defects are continuous in λ (Theorem 14.4);

(28) at the clock end the joining factor has no symmetric meromorphic continuation, and $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$ for at most finitely many $\lambda \in [0,1]$, not for $\lambda = 1$ (Propositions 14.5, 14.25);

(29) the cell masses are entire functions of λ of exponential type $O(\log n)$, and the cell defects have the height kernel $\hat\varphi'(\gamma/2\pi n)$ (Propositions 14.24, 14.19);

(30) in the compact space $\mathscr{O}$ of ordinate lists at bounded distance from the clock, at most one list has coinciding continuous and atomized zeta functions; such a list exists if and only if $\zeta_{Z1} = \zeta$, and it is then the list of the ordinates of ζ (Theorem 15.1, Proposition 15.2, Corollary 15.3);

(31) (D) holds if and only if $\tilde c_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every $n \ge 2$, one real number for each cell; every λ with $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$ is a witness of (D), so the exceptional set is empty or contains 0; cell by cell the remaining identity is (14.59) (Corollary 14.26, Propositions 14.27, 14.29);

(32) for a pair joined by a symmetric factor Hamburger's theorem gives a constant factor when the continuous function has order one, and not in general; at $\lambda = 0$ the symmetry of the joining factor is equivalent to (D) (Proposition 14.31, Remarks 14.30, 14.32);

(33) the ladder: for every $N \ge 2$, $\tilde h_\lambda(\sigma) = O(\sigma N^{-\sigma})$ as $\sigma \to +\infty$ if and only if the Mellin count of ζ_λ has no mass outside the integers below N; (D) holds if and only if this holds at $\lambda = 0$ for every N, and at the clock end the first rung fails (Theorems 14.33, 14.34, Corollary 14.35);

(34) the bridge theorem: if the imaginary part of the joining factor of a pair extends continuously to $\Re s \ge \frac12$, vanishes on the critical line, and the factor satisfies the temperate bound (14.67), then the factor is constant, the discrete function is ζ and the continuous one is a multiple of ζ; for the family the two conditions hold for no $\lambda \in (0,1]$, and at $\lambda = 0$ they are equivalent to (D) (Theorem 14.36, Corollary 14.37);

(35) on the critical line $\Im \log \zeta_\lambda = -\vartheta(t) + \pi(N_\lambda(t) - 1)$, and the cell masses of two lists differ by a sum over the ordinates of one function of one ordinate (Propositions 14.38, 14.39);
(36) on each rung the passage from the clock to the ordinates of ζ is the transport $\mathsf{S}_\mathsf{f}(\lambda)$, an entire function of λ of exponential type at most $C_\eta \log N$ whose value at $\lambda = 0$ is the sum of the series of the moments $\sum_k \eta_k^j \mathsf{c}_\mathsf{f}^{(j)}(\tau_k)$; (D) holds if and only if the transport equals the defect of the clock on every rung, and this defect is given exactly by Poisson summation, with stationary heights $2\pi x^{1/m}$ (Proposition 14.41, Theorem 14.42, Proposition 14.43);
(37) (D) holds if and only if the rung system (15.5), with free masses at the points $\log n$, has a solution in $\mathscr{O}$; the solution is then the list of ζ with the masses of von Mangoldt, and its linearization is injective up to the splitting of multiple ordinates (Theorems 15.5, 15.6);
(38) at the midpoint clock, the finite ordinate Jacobian up to height T has a uniform lower frame bound on windows of length $(1 + \varkappa_{\mathrm{win}}) \log(T/2\pi)$, no uniform lower bound is possible on a fixed window, the infinite ordinate block has dense range on every finite window, and smooth bounded-displacement transport tails are $O_{A,\mathsf{f}}(T^{1-A} \log T)$ (Theorem 14.44);
(39) the gradient of the vertical defect on $\mathscr{O}$ is $\partial \mathsf{V}_\sigma / \partial \kappa_k = 2\mathsf{Q}_{\boldsymbol{\kappa}}(\kappa_k)$ with one real-analytic function $\mathsf{Q}_{\boldsymbol{\kappa}}$ (Proposition 15.8);
(40) a list is accepted by a continuous side and a discrete side if and only if its function lies in the intersection of the window of the first with the frequencies of the second; a non-periodic list is accepted through one unitary matrix; for the finite Euler products $\prod_{p \le X}(1 - p^{-s})$ the constructions of the paper apply and the endpoint $\lambda = 0$ is the only member of the family that is a Dirichlet series (Theorem 16.1, Proposition 16.2, Theorem 16.3);
(41) there is no function with the functional equation of ζ and an absolutely convergent Dirichlet series supported on the integers with bounded prime factors, and for the diagonal operator over the primes the contraction, the convergence of the determinant and the symmetry hold in $\sigma > 0$, in $\sigma > 1$ and on $\sigma = \frac{1}{2}$ respectively (Proposition 16.4, Subsection 16.4).

Not proved. The identity $\zeta_{Z1} = \zeta$ is not proved. The statements proved to be equivalent to it are listed in Subsection 1.4; in the form of Corollary 15.3 it is the existence of one list in $\mathscr{O}$ whose continuous and atomized zeta functions coincide, and in the form of Corollary 14.37 it is the vanishing on the critical line of the imaginary part of the joining factor $\tilde{h}_0$, with the bound (14.67). The theorems of Subsections 14.9–14.13 and of Sections 15 and 16 are formulations of this step and tools for it; none of them evaluates the pairing (14.59) of the count difference $N_\zeta - n_{\mathrm{RVM}}$, on which every one of them depends at $\lambda = 0$.

The computations of Section 18 with 2 001 052 zeros illustrate only the specific numerical statements explicitly tested there; they do not provide evidence for the unproved equivalent forms of (D).

Remark 19.1 (Open problems).
(i) Prove the identity $\zeta_{Z1} = \zeta$, equivalently condition (D), equivalently an absolutely convergent Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$, equivalently the product identity (12.12), equivalently the cell identity (12.18) with the jump bound of Theorem 12.7, equivalently step A7 of the programme in Subsection 13.8, equivalently the positivity of the measure representing ζ_{Z1} (Corollary 14.8), equivalently the existence of a coincidence list in $\mathscr{O}$ (Corollary 15.3, step A19). This is the final step of the method.
(ii) Hypothesis 6.7 is proved for the integers n that are not of the form 2^j with $\mu(j) \neq 0$ (Corollary 14.16); at $n = 2^j$ the cell mass is $(1 + \mu(j) J_{\mathrm{RVM}}(2) \log 2)/\log n + O(n^{-1/4})$. Determine the sign of $1 \pm J_{\mathrm{RVM}}(2) \log 2$.
(iii) Decide, for $0 < \lambda < 1$ outside the finite set E, which hypothesis of Theorem 14.3 fails, and which of (B1), (B2) fails in Theorem 14.36; decide whether the vanishing of all admissible coordinate derivatives (with the corresponding one-sided/KKT conditions at

active constraints) forces $\mathsf{V}_\sigma = 0$ (Proposition 15.8); and decide whether the atomized coefficients $\tilde{a}_N(0)$ control the sign of $\mathsf{W}(0)$ in Proposition 14.10.

(iv) Determine the monotone branches of π_{RVM}, on which the real prime levels p_j^{RVM} are defined, and explain the observed proximity $p_j^{\mathrm{RVM}} \approx p_j$ for small j (Experiment 18.6).

(v) Derive the bound (B2) for $\tilde{h}_\lambda$ from the bounds of the cell defects, and turn Heuristic 14.40 into a lemma with remainder; decide (B1) at $\lambda = 0$.

(vi) Evaluate from the construction the right side of (14.59), or the moments $\mathsf{m}_j(\mathsf{f})$ of (14.82); on the first rung the moments $j = 1, 2$ carry the transport up to $4 \cdot 10^{-6}$ (Experiment 18.10).

(vii) Theorem 14.44 gives the finite-height lower frame bound on the natural scale $U \asymp \log T$, the failure of fixed-window uniform invertibility, dense range on every fixed window, and rapidly summable smooth tails. The exact-atomic interpolation tested later has no established interior path. A remaining numerical existence programme is instead to control bounded endpoint finite sections on an exhausting nested family of tests and pass to a distributional limit; proving the required residual-to-zero and compactness statement would already be an existence theorem for the final rung system.

(viii) Under (D), decide whether $\mathsf{E} = \{0\}$, that is, whether the functions D_n of (14.57) have a common zero in $(0, 1]$.

(ix) Construct an object of infinite rank from which ζ_{Z1} and $\tilde{\zeta}_{\mathbb{Z},0}$ both descend, as the matrix U of Proposition 16.2 does for the two-edge model (Subsection 16.4).

Remark 19.2 (Statements derived in part or conjectural, not proved here). The following statements arose in the work on the final step; none of them is used in this paper.

(a) *Rates in λ.* Off the powers of 2 the formulas of Proposition 14.24 suggest

$$\tilde{h}_\lambda(s) - \tilde{h}_0(s) = O_s(\lambda), \qquad \bigl|\mathfrak{h}_n(\lambda; s) - \mathfrak{h}_n(0; s)\bigr| \ll_{s,\varepsilon} \lambda\, n^{-\sigma-\frac{1}{2}+\varepsilon}; \tag{19.1}$$

the cells $n = 2^j$ and the uniformity in n of the first-cell term are not written.

(b) *Flow form of the density condition.* $\tilde{h}_0(1) = 0$ if and only if $\int_{0^+}^1 \partial_\lambda \tilde{h}_\lambda(1)\, d\lambda = \tilde{h}_1(1)$; the differentiability at $s = 1$ uniformly in λ is not written.

(c) *Pointwise bound.* $|\tilde{c}_n(\lambda)| \le C_\varepsilon n^\varepsilon$ uniformly for λ in a neighbourhood of 0 is open; by Corollary 14.26 it is not needed for the final step.

(d) *Leading term of the clock defect.* The form $\mathfrak{h}_n(1; s) \approx \sqrt{2}\,\pi\,\mathsf{M}_2\, s\, n^{-s-7/4} \sin(4\pi\sqrt{n} + \frac{\pi}{4})$ is not proved; the bound (14.40) is.

(e) *A heuristic model of the atomized clock.* By (14.39), $\tilde{\zeta}_{\mathbb{Z},1}(s)$ is close to $\prod_{n\ge 2}(1 - n^{-s})^{-1/\log n}$, whose logarithm has the leading part $\sum_n n^{-s}/\log n = \int_s^\infty (\zeta(y) - 1)\, dy$; no error term is claimed.

(f) The comparison between the two columns of Subsection 13.7 and the use of the classification of [39] remain as discussed in the open-problem paragraph following Proposition 15.8 and are not proved.

Appendix A. Python programs

The programs below produce every number, table and figure of Section 18. They read the two data files `data/zeross.bin` (ordinates γ_k of the zeros of ζ) and `data/zLamb.bin` (midpoint ordinates τ_k), both little-endian float64 with 2 001 052 entries, and write their console output, CSV, MAT and binary files to `output/` and the figures to `figures/`. They require Python 3 with NumPy, SciPy, mpmath and Matplotlib, and are run from the directory `code/` in the order `s01`–`s10`; the programs `s08`–`s10` use NumPy only and read the same two data files.

A.1. `rvm_common.py`.

```
"""Common routines for the midpoint Riemann-von Mangoldt (RVM) zeta system.

Data files (little-endian float64, one ordinate per entry):
```

```
    data/zeross.bin  ordinates gamma_k of the nontrivial zeros of zeta (2,001,052 values)
    data/zLamb.bin   midpoint RVM ordinates tau_k, N_RVM(tau_k) = k - 1/2

Notation follows the paper: N_RVM is the clock, tau_k the midpoint ordinates,
gamma_k the ordinates of zeta, eta_k = tau_k - gamma_k, q(s) = s(s-1),
G(s) = s(s-1) Gamma(s/2) pi^(-s/2), zeta_lambda = P_lambda / G with ordinates
gamma_k(lambda) = (1-lambda) gamma_k + lambda tau_k.
"""
import os
import sys
import time
import numpy as np
from scipy import special

ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
DATA = os.path.join(ROOT, "data")
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)

TWO_PI = 2.0 * np.pi


class Tee:
    """Write console output both to the screen and to output/<name>_console.txt."""

    def __init__(self, name):
        self.f = open(os.path.join(OUT, name + "_console.txt"), "w")
        self.t0 = time.time()

    def __call__(self, fmt, *args):
        s = fmt % args if args else fmt
        sys.stdout.write(s + "\n")
        sys.stdout.flush()
        self.f.write(s + "\n")
        self.f.flush()

    def close(self):
        self("total run time %.1f s" % (time.time() - self.t0))
        self.f.close()


def read_bin(name):
    return np.fromfile(os.path.join(DATA, name), dtype="<f8")


def load_ordinates():
    gam = read_bin("zeross.bin")
    tau = read_bin("zLamb.bin")
    n = min(gam.size, tau.size)
    return gam[:n], tau[:n]


def clock(T):
    """N_RVM(T) = T/(2 pi) log(T/(2 pi e)) + 7/8 (real or complex T, principal log)."""
    return T / TWO_PI * (np.log(T / TWO_PI) - 1.0) + 0.875


def tail_coefficients(U):
    """Midpoint-rule tail of sum_{k>N} log(1 + q/tau_k^2), N = N_RVM(U):
    returns (c1, c2, c3) with tail(q) = c1 q - c2 q^2 + c3 q^3."""
    LU = np.log(U / TWO_PI)
    c1 = (LU + 1.0) / U / TWO_PI
    c2 = (3.0 * LU + 1.0) / (9.0 * U ** 3) / 2.0 / TWO_PI
```

```
    c3 = (5.0 * LU + 1.0) / (25.0 * U ** 5) / 3.0 / TWO_PI
    return c1, c2, c3


def clock_inverse(y):
    """Solve N_RVM(U) = y by Newton iteration."""
    U = TWO_PI * max(y, 2.0) / np.log(max(y, 3.0))
    for _ in range(80):
        U -= (clock(U) - y) / (np.log(U / TWO_PI) / TWO_PI)
    return U


class Product:
    """log P(s) = sum_k log(1 + q(s)/(1/4 + o_k^2)) for an ordinate list o_k.

    Exact summation for k <= K1, power sums for K1 < k <= N, and the clock tail
    for k > N.  Accurate for |q(s)| << o_{K1}^2."""

    def __init__(self, ords, K1=100000, nterms=6):
        self.o = np.asarray(ords, dtype=float)
        self.N = self.o.size
        self.K1 = min(K1, self.N)
        A = 0.25 + self.o[self.K1:] ** 2
        self.pw = [np.sum(A ** (-j)) for j in range(1, nterms + 1)]
        self.A1 = 0.25 + self.o[: self.K1] ** 2
        self.tail = tail_coefficients(clock_inverse(float(self.N)))

    def logP(self, s):
        s = np.atleast_1d(np.asarray(s, dtype=complex))
        q = s * (s - 1.0)
        out = np.empty(s.shape, dtype=complex)
        for i, qi in enumerate(q):
            v = np.sum(np.log1p(qi / self.A1))
            for j, p in enumerate(self.pw, start=1):
                v += (-1) ** (j + 1) * qi ** j / j * p
            c1, c2, c3 = self.tail
            v += c1 * qi - c2 * qi ** 2 + c3 * qi ** 3
            out[i] = v
        return out


def logG(s):
    s = np.asarray(s, dtype=complex)
    return np.log(s) + np.log(s - 1.0) + special.loggamma(s / 2.0) - s / 2.0 * np.log(np.pi)


def zeta_from_product(prod, s):
    s = np.atleast_1d(np.asarray(s, dtype=complex))
    return np.exp(prod.logP(s) - logG(s))


def chi(s):
    s = np.asarray(s, dtype=complex)
    return np.exp((s - 0.5) * np.log(np.pi) + special.loggamma((1 - s) / 2) - special.loggamma(s / 2))


def w_transition(s):
    """w(s) = exp(-2 pi i N_RVM(tau_*)), tau_* = -i (s - 1/2)."""
    s = np.asarray(s, dtype=complex)
    tstar = -1j * (s - 0.5)
    return np.exp(-2j * np.pi * clock(tstar))


def theta_rs(t):
    t = np.asarray(t, dtype=float)
```

```
    return t / 2 * np.log(t / TWO_PI) - t / 2 - np.pi / 8 + 1 / (48 * t) + 7 / (5760 * t ** 3)


def gamma_lambda(gam, tau, lam):
    return (1.0 - lam) * gam + lam * tau


def ei_upper(w):
    """Ei on Im w != 0 along the horizontal line (paper, eq. Ei-def): Ei(w) = -E1(-w)."""
    return -special.exp1(-np.asarray(w, dtype=complex))


def mobius_sieve(X):
    mu = np.ones(X + 1, dtype=np.int8)
    mu[0] = 0
    is_comp = np.zeros(X + 1, dtype=bool)
    for p in range(2, X + 1):
        if not is_comp[p]:
            is_comp[2 * p::p] = True
            mu[p::p] *= -1
            if p * p <= X:
                mu[p * p::p * p] = 0
    return mu


def primes_upto(X):
    s = np.ones(X + 1, dtype=bool)
    s[:2] = False
    for p in range(2, int(X ** 0.5) + 1):
        if s[p]:
            s[p * p::p] = False
    return np.nonzero(s)[0]


def write_csv(name, header, cols):
    A = np.column_stack(cols)
    np.savetxt(os.path.join(OUT, name), A, delimiter=",", header=",".join(header), comments="", fmt="%.12g")


def savefig(fig, name):
    fig.savefig(os.path.join(FIG, name), dpi=150, bbox_inches="tight")
```

A.2. **s01_clock_and_family.py.**

```
"""Experiment 1: the midpoint clock, the displacements eta_k = tau_k - gamma_k,
the lambda-family zeta_lambda, and the numerical comparison zeta_Z1 versus zeta."""
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, clock, Product, zeta_from_product, gamma_lambda,
                        write_csv, savefig, OUT)
import os

say = Tee("s01_clock_and_family")
gam, tau = load_ordinates()
N = gam.size
k = np.arange(1, N + 1)
say("zeros loaded: N = %d, gamma_1 = %.9f, gamma_N = %.6f, tau_1 = %.9f, tau_N = %.6f",
    N, gam[0], gam[-1], tau[0], tau[-1])

# ---- clock check and displacements
clock_err = np.max(np.abs(clock(tau) - (k - 0.5)))
```

```
eta = tau - gam
say("max |N_RVM(tau_k) - (k - 1/2)| = %.3e", clock_err)
say("eta_k = tau_k - gamma_k:  max |eta| = %.6f (k = %d),  mean |eta| = %.6f,  rms = %.6f,  mean eta = %.3e",
    np.max(np.abs(eta)), int(np.argmax(np.abs(eta))) + 1, np.mean(np.abs(eta)),
    np.sqrt(np.mean(eta ** 2)), np.mean(eta))
blocks = [(1, 10), (11, 100), (101, 1000), (1001, 10 ** 4), (10 ** 4 + 1, 10 ** 5), (10 ** 5 + 1, 10 ** 6),
     (10 ** 6 + 1, N)]
rows = []
say("%12s %12s %12s %12s", "k from", "k to", "max|eta|", "rms eta")
for a, b in blocks:
    e = eta[a - 1:b]
    rows.append([a, b, np.max(np.abs(e)), np.sqrt(np.mean(e ** 2))])
    say("%12d %12d %12.6f %12.6f", a, b, rows[-1][2], rows[-1][3])
rows = np.array(rows)
write_csv("s01_eta_blocks.csv", ["k_from", "k_to", "max_abs_eta", "rms_eta"], [rows[:, i] for i in range(4)])
say("first ordinates:")
say("%4s %16s %16s %12s", "k", "gamma_k", "tau_k", "eta_k")
for i in range(12):
    say("%4d %16.9f %16.9f %12.6f", i + 1, gam[i], tau[i], eta[i])

# ---- zeta_Z1 versus zeta on vertical lines
PZ = Product(gam)
PR = Product(tau)
tline = np.concatenate([np.linspace(1.0, 100.0, 199), np.linspace(101.0, 2000.0, 300)])
mp.mp.dps = 20
cmp_rows = []
for sigma in (2.0, 1.25, 0.75):
    s = sigma + 1j * tline
    zz1 = zeta_from_product(PZ, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    rel = np.abs(zz1 / ze - 1.0)
    cmp_rows.append([sigma, np.max(rel), np.median(rel)])
    say("sigma = %.2f, t in [1,2000] (%d points): max |zeta_Z1/zeta - 1| = %.3e, median = %.3e",
        sigma, tline.size, np.max(rel), np.median(rel))
    write_csv("s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaZ1", "rel_err"],
              [tline, np.abs(ze), np.abs(zz1), rel])
cmp_rows = np.array(cmp_rows)

# ---- Hardy-type functions Z_lambda(t) = exp(i theta(t)) zeta_lambda(1/2+it) on the critical line
def theta_exact(t):
    return np.array([float(mp.siegeltheta(x)) for x in t])

tc = np.linspace(0.5, 60.0, 1200)
th = theta_exact(tc)
lams = [0.0, 0.25, 0.5, 0.75, 1.0]
Zl = {}
for lam in lams:
    P = PZ if lam == 0.0 else (PR if lam == 1.0 else Product(gamma_lambda(gam, tau, lam)))
    Zl[lam] = np.real(np.exp(1j * th) * zeta_from_product(P, 0.5 + 1j * tc))
Zhardy = np.array([float(mp.siegelz(x)) for x in tc])
dZ = np.max(np.abs(Zl[0.0] - Zhardy) / np.maximum(1.0, np.abs(Zhardy)))
say("critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = %.3e", dZ)
write_csv("s01_Z_lambda_critical_line.csv", ["t", "Z_hardy"] + ["Z_lambda_%.2f" % l for l in lams],
          [tc, Zhardy] + [Zl[l] for l in lams])

# ---- figures
dec = np.arange(0, N, 50)
fig, ax = plt.subplots(2, 2, figsize=(12, 8))
ax[0, 0].plot(k[dec], eta[dec], ".", ms=1.5, color="tab:blue")
ax[0, 0].axhline(0, color="k", lw=0.5)
ax[0, 0].set_xlabel("k")
ax[0, 0].set_ylabel(r"$\eta_k=\tau_k-\gamma_k$")
ax[0, 0].set_title(r"Bounded displacement, $N=%d$ zeros" % N)
ax[0, 1].hist(eta, bins=200, density=True, color="tab:blue")
ax[0, 1].set_xlabel(r"$\eta_k$")
```

```
ax[0, 1].set_title(r"Distribution of $\eta_k$ (all $k\leq N$)")
lam = np.linspace(0, 1, 101)
for kk in range(30):
    gl = (1 - lam) * gam[kk] + lam * tau[kk]
    ax[1, 0].plot(lam, clock(gl) - (kk + 0.5), "-", lw=1)
ax[1, 0].axhline(0, color="k", lw=0.5)
ax[1, 0].set_xlabel(r"$\lambda$")
ax[1, 0].set_ylabel(r"$\mathcal{N}_{\rm RVM}(\gamma_k(\lambda))-(k-\frac{1}{2})$")
ax[1, 0].set_title(r"Moving ordinates in clock units, $k\leq30$ ($\lambda=0$: $\zeta_{Z1}$, $\lambda=1$: $\
    zeta_{\rm RVM}$)", fontsize=9)
ax[1, 1].plot(tc, Zhardy, "k-", lw=2.2, label=r"Hardy $Z(t)$ of $\zeta$")
cols = ["tab:red", "tab:orange", "tab:green", "tab:cyan", "tab:blue"]
for lam, c in zip(lams, cols):
    ax[1, 1].plot(tc, Zl[lam], "-", color=c, lw=1.0, label=r"$Z_\lambda$, $\lambda=%.2f$" % lam)
ax[1, 1].axhline(0, color="k", lw=0.5)
ax[1, 1].set_xlabel("t")
ax[1, 1].set_title(r"$e^{i\vartheta(t)}\zeta_\lambda(\frac{1}{2}+it)$ along the family")
ax[1, 1].legend(fontsize=7, ncol=2)
fig.tight_layout()
savefig(fig, "fig01_clock_family.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
for sigma, c in zip((2.0, 1.25, 0.75), ("tab:blue", "tab:green", "tab:red")):
    d = np.loadtxt(os.path.join(OUT, "s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma), delimiter=",", skiprows=1)
    ax[0].plot(d[:, 0], d[:, 3], ".", ms=2, color=c, label=r"$\sigma=%.2f$" % sigma)
ax[0].set_xlabel("t")
ax[0].set_ylabel(r"$|\zeta_{Z1}/\zeta-1|$")
ax[0].set_title(r"$\zeta_{Z1}$ from %d ordinates versus $\zeta$" % N)
ax[0].legend()
ax[1].plot(tc, Zhardy, "k-", lw=2.5, label=r"$Z(t)$ (mpmath)")
ax[1].plot(tc, Zl[0.0], "r--", lw=1.0, label=r"$Z_{Z1}(t)$ (product)")
ax[1].set_xlabel("t")
ax[1].legend()
ax[1].set_title(r"Critical line: $\lambda=0$ endpoint and Hardy $Z$")
fig.tight_layout()
savefig(fig, "fig02_zetaZ1_vs_zeta.png")

sio.savemat(os.path.join(OUT, "s01_results.mat"), {
    "N": N, "clock_err": clock_err, "eta_blocks": rows, "zetaZ1_cmp": cmp_rows,
    "t_critical": tc, "Z_hardy": Zhardy, "Z_lambda": np.array([Zl[l] for l in lams]), "lambdas": np.array(lams
    )})
say.close()

# ---- zoom: how the smooth clock places the midpoint ordinates next to the zeros
say2 = Tee("s01b_clock_zoom")
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
T = np.linspace(9.0, 52.0, 4000)
ax[0].step(T, np.searchsorted(gam, T, side="right"), where="post", color="k", lw=1.5, label=r"$N_\zeta(T)$")
ax[0].plot(T, clock(T), "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
for i in range(9):
    ax[0].plot([gam[i], gam[i]], [0, i + 1], "k:", lw=0.8)
ax[0].plot(tau[:9], np.arange(9) + 0.5, "ro", ms=5, label=r"$\tau_k$: $\mathcal{N}_{\rm RVM}(\tau_k)=k-\frac
    {1}{2}$")
ax[0].plot(gam[:9], np.arange(9) + 0.5, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$")
ax[0].set_xlabel("T")
ax[0].set_title("Clock and staircase at the first zeros")
ax[0].legend(loc="upper left", fontsize=8)
kz = np.arange(N - 12, N)
T3 = np.linspace(gam[kz[0]] - 0.3, gam[kz[-1]] + 0.3, 6000)
ax[1].step(T3 - 1132480, np.searchsorted(gam, T3, side="right") - 2001000, where="post", color="k", lw=1.5,
    label=r"$N_\zeta(T)$")
ax[1].plot(T3 - 1132480, clock(T3) - 2001000, "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
ax[1].plot(tau[kz] - 1132480, kz + 0.5 - 2001000, "ro", ms=5, label=r"$\tau_k$")
ax[1].plot(gam[kz] - 1132480, kz + 0.5 - 2001000, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$
    ")
```

```
ax[1].set_xlabel(r"$T-1132480$")
ax[1].set_ylabel(r"count $-\,2001000$")
ax[1].set_title(r"Zoom at the last 12 zeros of the file (mean spacing %.3f)" % (2 * np.pi / np.log(gam[-1] /
    (2 * np.pi))))
ax[1].legend(loc="upper left", fontsize=8)
fig.tight_layout()
savefig(fig, "fig00_clock_zoom.png")
fig, ax = plt.subplots(1, 1, figsize=(12, 3))
T2 = np.linspace(9.0, 200.0, 20000)
ax.plot(T2, np.searchsorted(gam, T2, side="right") - clock(T2), "k-", lw=0.8)
ax.axhline(0, color="r", lw=0.8)
ax.set_xlabel("T")
ax.set_ylabel(r"$S(T)+O(1/T)$")
ax.set_title(r"$N_\zeta(T)-\mathcal{N}_{\rm RVM}(T)$: the remainder of the Riemann-von Mangoldt formula (jump
    $+1$ at each $\gamma_k$)")
fig.tight_layout()
savefig(fig, "fig00b_remainder.png")
say2("zoom table near the top of the file:")
say2("%9s %18s %18s %10s %10s", "k", "gamma_k", "tau_k", "eta_k", "N*-N_RVM")
for i in kz:
    say2("%9d %18.6f %18.6f %10.6f %10.6f", i + 1, gam[i], tau[i], tau[i] - gam[i],
         np.searchsorted(gam, gam[i], side="right") - 0.5 - clock(gam[i]))
say2.close()
```

A.3. **s02_transition.py.**

```
"""Experiment 2: the transition theorem zeta_RVM = varpi (1 + w)(1 + O(1/t)),
the constant varpi_RVM, the vertical limit of 1/zeta_RVM, and zeta_RVM versus zeta."""
import os
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, zeta_from_product, chi, w_transition,
                        write_csv, savefig, OUT)

say = Tee("s02_transition")
gam, tau = load_ordinates()
PR = Product(tau)

# ---- varpi_RVM by Richardson extrapolation on Re s = 2
Ts = 1.0e4 * np.array([0.25, 0.5, 1.0])
v = np.array([zeta_from_product(PR, 2 + 1j * T)[0] / (1 + w_transition(2 + 1j * T)) for T in Ts])
A = np.column_stack([np.ones(3), 1 / Ts, 1 / Ts ** 2])
c = np.linalg.solve(A.astype(complex), v)
varpi = c[0].real
L = 1.0 / varpi
say("varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000) = %.10f %+.2e i", c[0].real, c[0].imag)
say("L = 1/varpi_RVM = %.10f", L)

# ---- relative error of the transition formula
tt = np.exp(np.linspace(np.log(10.0), np.log(1.0e4), 160))
rows = [tt]
hdr = ["t"]
fits = []
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    zr = zeta_from_product(PR, s)
    err_w = np.abs(zr / (varpi * (1 + w_transition(s))) - 1)
    rows.append(err_w)
    hdr.append("err_w_sigmap%.2f" % sp)
    sel = tt > 100
    slope = np.polyfit(np.log(tt[sel]), np.log(err_w[sel]), 1)[0]
```

```
    fits.append([sp, slope, np.max(err_w[sel] * tt[sel])])
    say("sigma' = %.2f: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = %.3f, max t*err = %.4f",
        sp, slope, np.max(err_w[sel] * tt[sel]))
    if sp < 0.5:
        err_c = np.abs(zr / (varpi * (1 + chi(s))) - 1)
        rows.append(err_c)
        hdr.append("err_chi_sigmap%.2f" % sp)
        say("sigma' = %.2f: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = %.4f",
            sp, np.max(err_c[sel] * tt[sel]))
    wmod = np.abs(w_transition(s))
    rows.append(wmod)
    hdr.append("abs_w_sigmap%.2f" % sp)
    Fv = 1 / zr
    rows.append(np.abs(Fv - L))
    hdr.append("abs_F_minus_L_sigmap%.2f" % sp)
    say("          |w| * t^sigma' in [%.4f, %.4f];  |1/zeta_RVM - L| at t = 1e4: %.3e",
        np.min(wmod * tt ** sp), np.max(wmod * tt ** sp), np.abs(Fv[-1] - L))
write_csv("s02_transition_errors.csv", hdr, rows)

# ---- zeta_RVM versus zeta on two vertical lines and on the critical line
mp.mp.dps = 20
tl = np.linspace(1.0, 120.0, 900)
cmp = {}
for sigma in (2.0, 0.75):
    s = sigma + 1j * tl
    zr = zeta_from_product(PR, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    cmp[sigma] = (zr, ze)
    write_csv("s02_zetaRVM_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaRVM", "
     abs_varpi_1_plus_w"],
              [tl, np.abs(ze), np.abs(zr), np.abs(varpi * (1 + w_transition(s)))])

fig, ax = plt.subplots(2, 2, figsize=(12, 8))
cols = {0.25: "tab:red", 0.5: "tab:green", 1.5: "tab:blue"}
j = 1
for sp in (0.25, 0.5, 1.5):
    ax[0, 0].plot(tt, tt * rows[j], "-", color=cols[sp], label=r"$\sigma'=%.2f$" % sp)
    j += 4 if sp < 0.5 else 3
ax[0, 0].set_xlabel("t")
ax[0, 0].set_ylabel(r"$t\,|\zeta_{\rm RVM}/(\varpi_{\rm RVM}(1+w))-1|$")
ax[0, 0].set_title(r"Transition theorem: error times $t$ stays bounded, $\varpi_{\rm RVM}=%.10f$" % varpi,
     fontsize=9)
ax[0, 0].legend()
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    ax[0, 1].plot(tt, np.abs(1 / zeta_from_product(PR, s) - L) * tt ** min(sp, 1.0), "-", color=cols[sp],
     label=r"$\sigma'=%.2f$" % sp)
ax[0, 1].set_xlabel("t")
ax[0, 1].set_ylabel(r"$t^{\min(\sigma',1)}\,|1/\zeta_{\rm RVM}(\sigma+it)-L|$")
ax[0, 1].set_title(r"Vertical limit $L=1/\varpi_{\rm RVM}$, rescaled error", fontsize=9)
ax[0, 1].legend()
for a, sigma in zip((ax[1, 0], ax[1, 1]), (2.0, 0.75)):
    zr, ze = cmp[sigma]
    a.plot(tl, np.abs(ze), "k-", lw=1.2, label=r"$|\zeta(\sigma+it)|$")
    a.plot(tl, np.abs(zr), "r-", lw=1.2, label=r"$|\zeta_{\rm RVM}(\sigma+it)|$")
    a.set_xlabel("t")
    a.set_title(r"$\sigma=%.2f$" % sigma)
    a.legend()
fig.tight_layout()
savefig(fig, "fig03_transition.png")

sio.savemat(os.path.join(OUT, "s02_results.mat"), {"varpi": varpi, "L": L, "richardson": c,
                                                   "slopes": np.array(fits), "t": tt})
say.close()
```

A.4. **s03_exponential_sums.py.**

```
"""Experiment 3: exponential sums S(N;u) = sum_{k<=N} exp(i u o_k) over the midpoint
ordinates tau_k and over the ordinates gamma_k of zeta, and the height transforms M_H(x)."""
import os
import numpy as np
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, write_csv, savefig, OUT

say = Tee("s03_exponential_sums")
gam, tau = load_ordinates()
N = gam.size
step = 2000
Nk = np.arange(step, N + 1, step)
xs = [2.0, 3.0, 4.0, 6.0, 10.0]
Lam = {2.0: np.log(2), 3.0: np.log(3), 4.0: np.log(2), 6.0: 0.0, 10.0: 0.0}
res = {}
say("%6s %14s %14s %14s %16s %14s" % ("x", "max|S_RVM|", "max|S_RVM|/log", "|S_zeta(N)|",
                                      "Landau -T/2pi*L/sqrt(x)", "|M_H^RVM(x)|"))
for x in xs:
    u = np.log(x)
    SR = np.cumsum(np.exp(1j * u * tau))[Nk - 1]
    SZ = np.cumsum(np.exp(1j * u * gam))[Nk - 1]
    res[x] = (SR, SZ)
    landau = -gam[-1] / (2 * np.pi) * Lam[x] / np.sqrt(x)
    say("%6.1f %14.4f %14.4f %14.2f %16.2f %14.3e" % (x, np.max(np.abs(SR)), np.max(np.abs(SR) / np.log(Nk)),
                                                      np.abs(SZ[-1]), landau, np.abs(SR[-1]) / N))
    write_csv("s03_expsum_x%g.csv" % x, ["N", "re_S_RVM", "im_S_RVM", "re_S_zeta", "im_S_zeta"],
              [Nk, SR.real, SR.imag, SZ.real, SZ.imag])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
cols = {2.0: "tab:blue", 3.0: "tab:green", 4.0: "tab:purple", 6.0: "tab:orange", 10.0: "tab:red"}
for x in xs:
    ax[0].plot(Nk, np.abs(res[x][0]), "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
ax[0].plot(Nk, np.log(Nk), "k--", lw=1, label=r"$\log N$")
ax[0].set_xlabel("N")
ax[0].set_ylabel(r"$|S_{\rm RVM}(N;\log x)|$")
ax[0].set_title(r"Midpoint ordinates: $|\sum_{k\leq N}x^{i\tau_k}|$ stays $O(\log N)$")
ax[0].legend(fontsize=8, ncol=2)
for x in xs:
    ax[1].plot(Nk, res[x][1].real, "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
T = np.interp(Nk, np.arange(1, N + 1), gam)
for x in (2.0, 3.0, 4.0):
    ax[1].plot(Nk, -T / (2 * np.pi) * Lam[x] / np.sqrt(x), ":", color=cols[x], lw=1.5)
ax[1].set_xlabel("N")
ax[1].set_ylabel(r"$\mathrm{Re}\sum_{k\leq N}x^{i\gamma_k}$")
ax[1].set_title(r"Zeros of $\zeta$: Landau term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$ (dotted)")
ax[1].legend(fontsize=8, ncol=2)
fig.tight_layout()
savefig(fig, "fig04_exponential_sums.png")
sio.savemat(os.path.join(OUT, "s03_results.mat"), {"N": Nk, "x": np.array(xs),
                                                   "S_RVM": np.array([res[x][0] for x in xs]),
                                                   "S_zeta": np.array([res[x][1] for x in xs])})
say.close()
```

A.5. **s04_primes.py.**

```
"""Experiment 4: the prime-power count J_RVM, the synthetic prime count pi_RVM, the cell
masses c_n^RVM and the real RVM prime levels p_j^RVM, compared with J, pi(x) and the primes.

J_o(x) = li(x) - 2 Re sum_k Ei((1/2 + i o_k) log x) - log 2 + int_x^inf dt/(t(t^2-1) log t)
for an ordinate list o_k (o = tau: J_RVM; o = gamma: Riemann's explicit formula for J),
pi_o(x) = sum_{m <= log2 x} mu(m)/m J_o(x^(1/m)).
The zero sums are evaluated on a uniform grid in u = log x by the recurrence
```

```
exp(i o_k (u + du)) = exp(i o_k u) exp(i o_k du); Ei is exact (scipy exp1) for k <= 300 and
uses Ei(w) = e^w (1/w + 1/w^2 + 2/w^3 + 6/w^4) for k > 300, where |w| > 400."""
import os
import numpy as np
import scipy.io as sio
from scipy import special, integrate
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, ei_upper, mobius_sieve, primes_upto, write_csv,
                        savefig, clock_inverse, OUT)

say = Tee("s04_primes")
gam, tau = load_ordinates()
KE = 300


def zero_sum_ei(ords, u, K=None, taper=None):
    """sum_k Ei((1/2 + i o_k) u) on a uniform grid u, k <= K, optional weights."""
    o = ords if K is None else ords[:K]
    wts = np.ones(o.size) if taper is None else taper
    rho = 0.5 + 1j * o
    out = np.zeros(u.size, dtype=complex)
    # exact part
    for i in range(0, KE):
        out += wts[i] * ei_upper(rho[i] * u)
    # asymptotic part by recurrence
    r = rho[KE:]
    b = [wts[KE:] / r ** m for m in (1, 2, 3, 4)]
    du = u[1] - u[0]
    stepf = np.exp(1j * o[KE:] * du)
    for j in range(u.size):
        if j % 200 == 0:
            E = np.exp(1j * o[KE:] * u[j])
        A = [E @ bm for bm in b]
        uj = u[j]
        out[j] += np.exp(uj / 2) * (A[0] / uj + A[1] / uj ** 2 + 2 * A[2] / uj ** 3 + 6 * A[3] / uj ** 4)
        E = E * stepf
    return out


def tail_integral(x):
    return integrate.quad(lambda t: 1.0 / (t * (t * t - 1.0) * np.log(t)), x, np.inf, limit=200)[0]


def J_from(ords, u, K=None, taper=None):
    x = np.exp(u)
    S = zero_sum_ei(ords, u, K, taper)
    ti = np.array([tail_integral(v) for v in x])
    return special.expi(u) - 2 * S.real - np.log(2.0) + ti


def pi_from_J(u, J):
    mu = mobius_sieve(64)
    out = np.zeros(u.size)
    for i, ui in enumerate(u):
        m = 1
        while ui / m >= np.log(2.0) - 1e-12:
            if mu[m] != 0:
                out[i] += mu[m] / m * np.interp(ui / m, u, J)
            m += 1
    return out


def exact_J_pi(x):
```

```
    P = primes_upto(int(x.max()) + 1)
    pi = np.searchsorted(P, x, side="right").astype(float)
    J = np.zeros(x.size)
    for m in range(1, 40):
        J += np.searchsorted(P, x ** (1.0 / m) + 1e-12, side="right") / m
    return J, pi


def run(xmax, npts, K, label):
    u = np.linspace(np.log(2.0), np.log(xmax), npts)
    if K is None:
        tg = tr = None
        Kg = Kr = None
    else:
        Kg = Kr = K
        v = np.arange(K) / K
        tg = tr = np.where(v < 0.6, 1.0, np.cos(np.pi / 2 * (v - 0.6) / 0.4) ** 2)
    JR = J_from(tau, u, Kr, tr)
    JG = J_from(gam, u, Kg, tg)
    PR = pi_from_J(u, JR)
    PG = pi_from_J(u, JG)
    x = np.exp(u)
    Je, Pe = exact_J_pi(x)
    write_csv("s04_prime_counts_%s.csv" % label, ["x", "J_exact", "J_from_zeta_zeros", "J_RVM",
                                                  "pi_exact", "pi_from_zeta_zeros", "pi_RVM", "li"],
              [x, Je, JG, JR, Pe, PG, PR, special.expi(u)])
    return x, Je, JG, JR, Pe, PG, PR


# x <= 100 with all zeros; x <= 1000 with a smooth cut at K zeros
x1, Je1, JG1, JR1, Pe1, PG1, PR1 = run(100.0, 3000, None, "x100_allzeros")
say("x <= 100, all %d zeros of each list, %d grid points:" % (gam.size, x1.size))
say("  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): %.4f",
    np.max(np.abs(PG1 - Pe1)[np.min(np.abs(x1[:, None] - primes_upto(101)[None, :]), axis=1) > 0.2]))
say("  pi_RVM(100) = %.4f, pi(100) = %d, li(100) = %.4f, J_RVM(100) = %.4f, J(100) = %.4f",
    PR1[-1], Pe1[-1], special.expi(np.log(100.0)), JR1[-1], Je1[-1])
dPR = np.diff(PR1)
say("  pi_RVM monotone on [2,100]: %s (min increment on the grid %.3e)", bool(np.all(dPR > 0)), dPR.min())

tauK = 2 * np.pi * 1000.0 * 2.5
K = int(np.searchsorted(tau, tauK))
x2, Je2, JG2, JR2, Pe2, PG2, PR2 = run(1000.0, 40000, K, "x1000_K%d" % K)
say("x <= 1000, smooth cut at K = %d zeros (tau_K = %.1f), %d grid points:" % (K, tau[K - 1], x2.size))
say("  pi_RVM(1000) = %.4f, pi(1000) = %d, li(1000) = %.4f", PR2[-1], Pe2[-1], special.expi(np.log(1000.0)))

# cell masses and real prime levels from the fine x<=1000 run
n = np.arange(2, 1001)
pin = np.interp(n, x2, PR2)
c = np.diff(np.concatenate([[0.0], pin]))
P = primes_upto(1000)
isp = np.isin(n, P).astype(float)
say("cell masses c_n = pi_RVM(n) - pi_RVM(n-1):")
say("%5s %10s %10s %8s" % ("n", "c_n", "1/log n", "prime"))
for i in range(0, 40):
    say("%5d %10.5f %10.5f %8d" % (n[i], c[i], 1 / np.log(n[i]), isp[i]))
say("sum_{n<=1000} c_n = %.4f,  pi(1000) = %d", c.sum(), len(P))
levels = np.arange(1, int(np.floor(PR2.max())) + 1)
pj = np.array([x2[np.argmax(PR2 >= j)] for j in levels])
say("real RVM prime levels p_j^RVM (first crossing of level j) versus p_j:")
say("%5s %12s %8s %10s" % ("j", "p_j^RVM", "p_j", "diff"))
for j in range(min(30, levels.size)):
    say("%5d %12.4f %8d %10.4f" % (levels[j], pj[j], P[j], pj[j] - P[j]))
write_csv("s04_cell_masses.csv", ["n", "c_n_RVM", "inv_log_n", "is_prime"], [n, c, 1 / np.log(n), isp])
m = min(levels.size, P.size)
write_csv("s04_rvm_prime_levels.csv", ["j", "p_j_RVM", "p_j"], [levels[:m], pj[:m], P[:m]])
```

```
# ---- figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].step(x1, Pe1, where="post", color="k", lw=1.6, label=r"$\pi(x)$")
ax[0].plot(x1, PG1, "b-", lw=0.6, label=r"$\pi(x)$ from %d zeros of $\zeta$" % gam.size)
ax[0].plot(x1, PR1, "r-", lw=1.4, label=r"$\pi_{\rm RVM}(x)$ from %d $\tau_k$" % tau.size)
ax[0].plot(x1, special.expi(np.log(x1)) - special.expi(np.log(2.0)), "g--", lw=1, label=r"$\mathrm{li}(x)-\
    mathrm{li}(2)$")
ax[0].set_xlabel("x")
ax[0].set_title(r"Prime counts on $[2,100]$")
ax[0].legend(fontsize=7)
ax[1].step(x2, Pe2, where="post", color="k", lw=1.2, label=r"$\pi(x)$")
ax[1].plot(x2, PR2, "r-", lw=1.0, label=r"$\pi_{\rm RVM}(x)$")
ax[1].set_xlabel("x")
ax[1].set_title(r"Prime counts on $[2,1000]$ ($K=%d$ ordinates, smooth cut)" % K)
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig05_prime_counts.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
nn = n[:98]
ax[0].bar(nn, isp[:98], width=0.8, color="0.75", label="prime indicator")
ax[0].plot(nn, c[:98], "r.-", lw=0.8, ms=4, label=r"$c_n^{\rm RVM}$")
ax[0].plot(nn, 1 / np.log(nn), "g--", lw=1, label=r"$1/\log n$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Cell masses $c_n^{\rm RVM}=\pi_{\rm RVM}(n)-\pi_{\rm RVM}(n-1)$")
ax[0].legend(fontsize=8)
ax[1].plot(P[:m], pj[:m], "r.", ms=4, label=r"$(p_j,\,p_j^{\rm RVM})$")
ax[1].plot([0, 1000], [0, 1000], "k-", lw=0.8, label="diagonal")
ax[1].set_xlabel(r"$p_j$")
ax[1].set_ylabel(r"$p_j^{\rm RVM}$")
ax[1].set_title(r"Real RVM prime levels $\pi_{\rm RVM}(p_j^{\rm RVM})=j$ versus primes")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig06_rvm_primes.png")

sio.savemat(os.path.join(OUT, "s04_results.mat"), {"x100": x1, "pi100": Pe1, "piZeta100": PG1, "piRVM100": PR1,

                                                   "x1000": x2, "pi1000": Pe2, "piRVM1000": PR2,
                                                   "c_n": c, "p_RVM": pj, "K": K})
say.close()
```

A.6. **s05_mertens.py.**

```
"""Experiment 5: the Mertens function M, the synthetic Mertens function M_RVM^crit and the
ternary walk M_h with values in {-1,0,+1} on the squarefree support.

Explicit formulas (simple zeros, residues a = 1/(rho zeta'(rho))):
  M_zeta(x) = -2 + 2 Re sum_k a_k x^(rho_k) + trivial terms        (zeros of zeta)
  M_RVM(x)  = -2 + 2 Re sum_k b_k x^(1/2 + i tau_k) + trivial terms (b_k = 1/(rho_k zeta_RVM'(rho_k)))
Ternary walk: mu_h(n) = mu(n)^2 * sign, sign = +1 if M_h(n-1) < target(n) else -1."""
import os
import math
import numpy as np
import mpmath as mp
import scipy.io as sio
from scipy.interpolate import PchipInterpolator
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, logG, clock_inverse, tail_coefficients,
                        theta_rs, mobius_sieve, write_csv, savefig, OUT)

say = Tee("s05_mertens")
gam, tau = load_ordinates()
```

```
Nz = gam.size
XMAX = 10 ** 6
K0 = 1000

# ---------------------------------------------------------------- RVM residues
c1, c2, c3 = tail_coefficients(clock_inverse(float(Nz)))
t2 = tau ** 2
zpR = np.zeros(K0, dtype=complex)
for k in range(K0):
    tk = tau[k]
    fac = (t2 - tk ** 2) / (t2 + 0.25)
    fac[k] = 1.0
    q = -0.25 - tk ** 2
    lmod = np.sum(np.log(np.abs(fac))) + c1 * q - c2 * q ** 2 + c3 * q ** 3
    rho = 0.5 + 1j * tk
    lP = np.log(2j * tk / (0.25 + tk ** 2)) + lmod
    zpR[k] = (-1) ** k * np.exp(lP - logG(rho))
ratio = zpR / np.log(tau[:K0] / (2 * np.pi))
PR = Product(tau)
Ts = 1.0e4 * np.array([0.25, 0.5, 1.0])
from rvm_common import zeta_from_product, w_transition
v = np.array([zeta_from_product(PR, 2 + 1j * T)[0] / (1 + w_transition(2 + 1j * T)) for T in Ts])
varpi = np.linalg.solve(np.column_stack([np.ones(3), 1 / Ts, 1 / Ts ** 2]).astype(complex), v)[0].real
sel = np.arange(K0 // 2, K0)
cfit = np.mean((ratio[sel] / varpi - 1) * tau[sel])
say("zeta_RVM'(rho_k) exact for k <= %d; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = %d: %.10f %+.2e i",
    K0, K0, ratio[-1].real, ratio[-1].imag)
say("varpi_RVM = %.10f; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = %.4f %+.4f i",
    varpi, cfit.real, cfit.imag)
zR = varpi * np.log(tau / (2 * np.pi)) * (1 + cfit / tau)
zR[:K0] = zpR
bR = 1.0 / ((0.5 + 1j * tau) * zR)

# ---------------------------------------------------------------- residues of zeta
def rs_Z(t):
    a = np.sqrt(t / (2 * np.pi))
    Nn = np.floor(a).astype(int)
    p = a - Nn
    nmax = Nn.max()
    n = np.arange(1, nmax + 1)
    ph = theta_rs(t)[:, None] - t[:, None] * np.log(n)[None, :]
    mask = n[None, :] <= Nn[:, None]
    Z = 2 * np.sum(np.cos(ph) * mask / np.sqrt(n)[None, :], axis=1)
    cc = np.cos(2 * np.pi * p)
    C0 = np.cos(2 * np.pi * (p ** 2 - p - 1.0 / 16)) / cc
    return Z + (-1.0) ** (Nn - 1) * (t / (2 * np.pi)) ** (-0.25) * C0


mp.mp.dps = 20
KM = 1000
zG = np.zeros(Nz, dtype=complex)
for k in range(KM + 50):
    zG[k] = complex(mp.zeta(mp.mpc(0.5, gam[k]), 1, 1))
h = 1e-5
for i0 in range(KM, Nz, 4000):
    i1 = min(Nz, i0 + 4000)
    t = gam[i0:i1]
    Zp = (rs_Z(t + h) - rs_Z(t - h)) / (2 * h)
    val = -1j * np.exp(-1j * theta_rs(t)) * Zp
    if i0 == KM:
        chk = np.max(np.abs(val[:50] / zG[KM:KM + 50] - 1))
        say("Riemann-Siegel zeta'(rho) versus mpmath at k = %d..%d: max relative difference %.2e",
            KM + 1, KM + 50, chk)
        zG[KM + 50:i1] = val[50:]
    else:
```

```
        zG[i0:i1] = val
aG = 1.0 / ((0.5 + 1j * gam) * zG)
S2G = 2 * np.sum(np.abs(aG) ** 2)
S2R = 2 * np.sum(np.abs(bR) ** 2)
say("median |zeta'(rho)| = %.4f;  spectral sums 2 sum|a|^2: zeta %.6f, RVM %.6f", np.median(np.abs(zG)), S2G,
    S2R)


def trivial_zeta(x, n=6):
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        v += (-1) ** (k - 1) * (2 * np.pi / x) ** (2 * k) / (math.factorial(2 * k) * k * float(mp.zeta(2 * k +
     1)))
    return v


def trivial_rvm(x, n=6):
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        s = -2.0 * k
        lP = np.real(PR.logP(s)[0])
        v -= x ** (-2 * k) * (2 * k + 1) * np.pi ** k * 2 * (-1) ** k / (math.factorial(k) * np.exp(lP))
    return v


def explicit_uniform(ords, wts, u):
    """2 Re sqrt(x) sum_k w_k exp(i o_k u) on a uniform u grid (recurrence)."""
    out = np.zeros(u.size)
    st = np.exp(1j * ords * (u[1] - u[0]))
    for j in range(u.size):
        if j % 200 == 0:
            E = np.exp(1j * ords * u[j])
        out[j] = 2 * np.exp(u[j] / 2) * np.real(E @ wts)
        E = E * st
    return out


# ---------------------------------------------------------------- exact M by sieve
mu = mobius_sieve(XMAX)
M = np.cumsum(mu.astype(np.int64))
sqf = mu != 0
say("M(10^6) = %d;  squarefree density Q(10^6)/10^6 = %.6f (6/pi^2 = %.6f)", M[XMAX], sqf[1:].mean(), 6 / np.
    pi ** 2)

# ---------------------------------------------------------------- explicit formulas on [2, 2000]
u = np.linspace(np.log(2.0), np.log(2000.0), 2500)
xg = np.exp(u)
Mz = -2 + explicit_uniform(gam, aG, u) + trivial_zeta(xg)
Mc = -2 + explicit_uniform(tau, bR, u) + trivial_rvm(xg)
Mfl = M[np.floor(xg).astype(int)]
off = np.abs(xg - np.round(xg)) > 0.05
say("explicit formula with %d zeros of zeta on [2,2000]: rms(M_zeta - M) = %.4f off integers (|x-n|>0.05)",
    Nz, np.sqrt(np.mean((Mz - Mfl)[off] ** 2)))
write_csv("s05_explicit_formulas.csv", ["x", "M_floor", "M_zeta_explicit", "M_RVM_explicit"], [xg, Mfl, Mz, Mc
    ])

# ---------------------------------------------------------------- integer target for M_RVM^crit
def target_grid(X, div=24):
    x = [1.0]
    while x[-1] < X:
        x.append(min(X, x[-1] + max(1.0, np.floor(np.sqrt(x[-1]) / div))))
    return np.array(x)


def tapered(x, ords, a, C1=2.0, C2=4.0):
```

```
    S = np.zeros(x.size, dtype=complex)
    lx = np.log(x)
    i0 = 0
    while i0 < x.size:
        i1 = min(x.size, i0 + 200)
        K = max(1, int(np.searchsorted(ords, C2 * 2 * np.pi * np.sqrt(x[i1 - 1]))))
        uu = ords[None, :K] / (2 * np.pi * np.sqrt(x[i0:i1, None]))
        w = np.where(uu <= C1, 1.0, np.where(uu >= C2, 0.0, np.cos(np.pi / 2 * (uu - C1) / (C2 - C1)) ** 2))
        E = np.exp(1j * lx[i0:i1, None] * ords[None, :K])
        S[i0:i1] = np.sqrt(x[i0:i1]) * np.sum(E * w * a[None, :K], axis=1)
        i0 = i1
    return S


xT = target_grid(float(XMAX))
TT = -2 + 2 * np.real(tapered(xT, tau, bR)) + trivial_rvm(xT)
nn = np.arange(1, XMAX + 1)
T = np.concatenate([[0.0], PchipInterpolator(xT, TT)(nn)])
nchk = np.arange(1000, 100001, 997)
full = -2 + 2 * np.real(np.array([np.sqrt(n) * np.sum(np.exp(1j * np.log(n) * tau) * bR) for n in nchk])) +
     trivial_rvm(nchk.astype(float))
say("target on %d grid points; at %d integers <= 1e5: max |M_RVM(n) (all zeros) - target(n)| = %.4f",
    xT.size, nchk.size, np.max(np.abs(full - T[nchk])))

# ------------------------------------------------------------------ ternary walk
mu_h = np.zeros(XMAX + 1, dtype=np.int8)
Mh = np.zeros(XMAX + 1)
sgn = np.zeros(XMAX + 1, dtype=np.int8)
cur = 0.0
for n in range(1, XMAX + 1):
    if sqf[n]:
        sgn[n] = 1 if cur < T[n] else -1
        cur += sgn[n]
        mu_h[n] = sgn[n]
    Mh[n] = cur
err = Mh[1:] - T[1:]
say("ternary walk: max |M_h - target| on [1,1e6] = %.4f, on [1e4,1e6] = %.4f", np.max(np.abs(err)), np.max(np.
     abs(err[9999:])))
dens = lambda a: [np.mean(a[1:] == -1), np.mean(a[1:] == 0), np.mean(a[1:] == 1)]
say("densities of (-1, 0, +1):  mu   %.6f %.6f %.6f", *dens(mu))
say("                           mu_h %.6f %.6f %.6f   (3/pi^2 = %.6f, 1 - 6/pi^2 = %.6f)", *dens(mu_h), 3 / np.
     pi ** 2, 1 - 6 / np.pi ** 2)
say("sign agreement mu_h = mu on squarefrees: %.6f;  sum mu mu_h / Q = %.6f",
    np.mean(mu_h[sqf] == mu[sqf]), np.sum(mu_h.astype(int) * mu.astype(int)) / sqf[1:].sum())
say("max |M(n)|/sqrt(n) on [10,1e6] = %.4f;  max |M_RVM(n)|/sqrt(n) = %.4f;  max |M_h(n)|/sqrt(n) = %.4f",
    np.max(np.abs(M[10:]) / np.sqrt(nn[9:])), np.max(np.abs(T[10:]) / np.sqrt(nn[9:])), np.max(np.abs(Mh[10:])
      / np.sqrt(nn[9:])))
say("\nNUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)")
say("%4s %4s %6s %10s %8s %6s %6s %6s %6s" % ("n", "mu", "mu^2", "target", "M_h(n-1)", "sign", "mu_h", "M_h",
     "M"))
rows = []
for n in range(1, 41):
    say("%4d %4d %6d %10.4f %8d %6s %6d %6d %6d" % (n, mu[n], abs(mu[n]), T[n], Mh[n - 1],
                                                    ("+1" if sgn[n] > 0 else ("-1" if sgn[n] < 0 else "-")),
                                                    mu_h[n], Mh[n], M[n]))
    rows.append([n, mu[n], abs(mu[n]), T[n], Mh[n - 1], sgn[n], mu_h[n], Mh[n], M[n]])
rows = np.array(rows, dtype=float)
write_csv("s05_ternary_example.csv", ["n", "mu", "mu2", "target", "Mh_prev", "sign", "mu_h", "Mh", "M"],
          [rows[:, i] for i in range(rows.shape[1])])

Ns = 10 ** np.arange(3, 7)
say("\nENERGY E_N = (1/N) sum_{n<=N} W(n)^2")
say("%10s %14s %14s %14s" % ("N", "E_N(M)", "E_N(M_RVM)", "E_N(M_h)"))
EN = []
for Nn in Ns:
```

```
    EN.append([Nn, np.mean(M[1:Nn + 1].astype(float) ** 2), np.mean(T[1:Nn + 1] ** 2), np.mean(Mh[1:Nn + 1] **
      2)])
    say("%10d %14.4f %14.4f %14.4f" % tuple(EN[-1]))
EN = np.array(EN)
write_csv("s05_energy.csv", ["N", "E_M", "E_MRVM", "E_Mh"], [EN[:, i] for i in range(4)])
nall = np.arange(1, XMAX + 1)
write_csv("s05_walks.csv", ["n", "M", "M_RVM_target", "M_h"], [nall, M[nall], T[nall], Mh[nall]])
mu_h[1:].astype(np.int8).tofile(os.path.join(OUT, "mu_h_int8_1e6.bin"))

# ---------------------------------------------------------------- figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 2001)
ax[0].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[0].plot(w, T[w], "r-", lw=1.2, label=r"$M_{\rm RVM}^{\rm crit}(n)$")
ax[0].step(w, Mh[w], where="post", color="tab:green", lw=0.8, label=r"$M_h(n)$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Mertens functions on $[1,2000]$, every integer")
ax[0].legend(fontsize=8)
nb = 2000
bw = XMAX // nb
xb = (np.arange(nb) + 0.5) * bw
Mb = M[1:].reshape(nb, bw)
Tb = T[1:].reshape(nb, bw)
ax[1].fill_between(xb, Mb.min(1), Mb.max(1), color="0.6", label=r"$M(n)$ (range per block)")
ax[1].plot(xb, Tb.mean(1), "r-", lw=1.0, label=r"$M_{\rm RVM}^{\rm crit}(n)$")
ax[1].plot(xb, np.sqrt(xb), "k--", lw=0.8, label=r"$\pm\sqrt{n}$")
ax[1].plot(xb, -np.sqrt(xb), "k--", lw=0.8)
ax[1].set_xlabel("n")
ax[1].set_title(r"$[1,10^6]$, blocks of %d integers" % bw)
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig07_mertens.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 61)
ax[0].vlines(w - 0.15, 0, mu[w], color="k", lw=2, label=r"$\mu(n)$")
ax[0].vlines(w + 0.15, 0, mu_h[w], color="tab:green", lw=2, label=r"$\mu_h(n)$")
ax[0].axhline(0, color="k", lw=0.5)
ax[0].set_xlabel("n")
ax[0].set_yticks([-1, 0, 1])
ax[0].set_title(r"Ternary values: same zero set $\{\mu(n)=0\}$, signs chosen by tracking")
ax[0].legend(fontsize=8)
w = np.arange(1, 301)
ax[1].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[1].plot(w, T[w], "r.-", lw=0.8, ms=3, label=r"target $M_{\rm RVM}^{\rm crit}(n)$")
ax[1].step(w, Mh[w], where="post", color="tab:green", lw=1.2, label=r"$M_h(n)$")
ax[1].set_xlabel("n")
ax[1].set_title(r"The walk $M_h$ follows $M_{\rm RVM}^{\rm crit}$ within $\pm1$ step")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig08_ternary_walk.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
sel = xg <= 300
ax[0].step(np.arange(1, 301), M[1:301], where="post", color="k", lw=1.0, label=r"$M(x)$")
ax[0].plot(xg[sel], Mz[sel], "b-", lw=0.7, label=r"explicit formula, %d zeros of $\zeta$" % Nz)
ax[0].plot(xg[sel], Mc[sel], "r-", lw=1.0, label=r"$M_{\rm RVM}^{\rm crit}(x)$, %d $\tau_k$" % Nz)
ax[0].set_xlabel("x")
ax[0].set_title("Explicit formulas on [2,300]")
ax[0].legend(fontsize=7)
kk = np.arange(5000)
ax[1].plot(gam[kk], np.abs(zG[kk]) / np.log(gam[kk] / (2 * np.pi)), "b.", ms=1.5, label=r"$|\zeta'(\rho_k)|/\
    log(\gamma_k/2\pi)$")
ax[1].plot(tau[kk], np.abs(zR[kk]) / np.log(tau[kk] / (2 * np.pi)), "r.", ms=1.5, label=r"$|\zeta_{\rm RVM}'(\
    rho^{\rm RVM}_k)|/\log(\tau_k/2\pi)$")
```

```
ax[1].axhline(varpi, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
ax[1].set_xlabel("ordinate")
ax[1].set_ylim(0, 4)
ax[1].set_title(r"Derivatives at the zeros, $k\leq5000$")
ax[1].legend(fontsize=7)
fig.tight_layout()
savefig(fig, "fig09_explicit_weights.png")

sio.savemat(os.path.join(OUT, "s05_results.mat"), {"varpi": varpi, "cfit": cfit, "S2G": S2G, "S2R": S2R,
                                                   "energy": EN, "ternary_example": rows,
                                                   "x_explicit": xg, "M_zeta_explicit": Mz, "M_RVM_explicit":
    Mc})
say.close()
```

A.7. **s06_prime_ratios.py.**

```
"""Experiment 6: the three ratios li/pi_RVM, pi/li, pi/pi_RVM and the square-root estimate
pi_RVM(x) - li(x) = O(sqrt(x) (log log x)^2 / log x), from the prime counts written by s04_primes.py.

Input: output/s04_prime_counts_x1000_K17061.csv (columns x, pi_exact, pi_RVM, li, ...).
B(x) = sqrt(x) (log log x)^2 / log x is the scale of the square-root estimate."""
import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, savefig, write_csv, OUT

say = Tee("s06_prime_ratios")
D = np.genfromtxt(os.path.join(OUT, "s04_prime_counts_x1000_K17061.csv"), delimiter=",", names=True)
x, pe, pr, li = D["x"], D["pi_exact"], D["pi_RVM"], D["li"]
say("grid: %d points on [%.0f, %.0f]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)"
    % (x.size, x[0], x[-1]))

sel = x >= 10
r1, r2, r3 = li[sel] / pr[sel], pe[sel] / li[sel], pe[sel] / pr[sel]
xs = x[sel]
say("")
say("ratios at selected x")
say("%8s %12s %12s %12s %12s %12s %12s" % ("x", "pi", "pi_RVM", "li", "li/pi_RVM", "pi/li", "pi/pi_RVM"))
for X in [10, 20, 50, 100, 200, 500, 1000]:
    i = int(np.argmin(np.abs(x - X)))
    say("%8.1f %12.4f %12.4f %12.4f %12.6f %12.6f %12.6f"
        % (x[i], pe[i], pr[i], li[i], li[i] / pr[i], pe[i] / li[i], pe[i] / pr[i]))

s8 = x >= 8
B = np.sqrt(x[s8]) * np.log(np.log(x[s8])) ** 2 / np.log(x[s8])
e_rvm = (pr[s8] - li[s8]) / B
e_pi = (pe[s8] - li[s8]) / B
say("")
say("square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]")
say("max |pi_RVM - li| / B = %.4f   (at x = %.2f)" % (np.max(np.abs(e_rvm)), x[s8][np.argmax(np.abs(e_rvm))]))
say("max |pi - li| / B     = %.4f   (at x = %.2f)" % (np.max(np.abs(e_pi)), x[s8][np.argmax(np.abs(e_pi))]))
for X in [100, 1000]:
    m = x[s8] >= X / 2
    say("max |pi_RVM - li| / B on [%d, 1000] = %.4f" % (X // 2, np.max(np.abs(e_rvm[m]))))

# CSV files: full grid
write_csv("s06_prime_ratios.csv", ["x", "li_over_pi_RVM", "pi_over_li", "pi_over_pi_RVM"],
          [xs, r1, r2, r3])
write_csv("s06_square_root_scale.csv", ["x", "pi_RVM_minus_li_over_B", "pi_minus_li_over_B"],
          [x[s8], e_rvm, e_pi])

# ---- figure (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
```

```
ax[0].plot(xs, r3, "k-", lw=0.8, label=r"$\pi(x)/\pi_{\rm RVM}(x)$")
ax[0].plot(xs, r2, "b-", lw=0.8, label=r"$\pi(x)/\mathrm{li}(x)$")
ax[0].plot(xs, r1, "r-", lw=1.4, label=r"$\mathrm{li}(x)/\pi_{\rm RVM}(x)$")
ax[0].axhline(1.0, color="0.5", lw=0.8, ls="--")
ax[0].set_xlabel("x")
ax[0].set_title(r"The three ratios on $[10,1000]$")
ax[0].legend(fontsize=8)
ax[1].plot(x[s8], e_pi, "b-", lw=0.6, label=r"$(\pi-\mathrm{li})/B$")
ax[1].plot(x[s8], e_rvm, "r-", lw=1.4, label=r"$(\pi_{\rm RVM}-\mathrm{li})/B$")
ax[1].axhline(0.0, color="0.5", lw=0.8, ls="--")
ax[1].set_xlabel("x")
ax[1].set_title(r"Square-root scale $B(x)=\sqrt{x}\,(\log\log x)^2/\log x$")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig10_prime_ratios.png")
say.close()
```

A.8. **s07_product_identity.py.**

```
"""Experiment 7: the value at infinity of the endpoint.

varpi_Z1 = lim_{sigma -> +inf} zeta_Z1(sigma) = varpi_RVM * prod_k (1/4 + tau_k^2) / (1/4 + gamma_k^2),
so zeta_Z1 = zeta holds exactly when the ordered partial products
P_K = prod_{k <= K} (1/4 + gamma_k^2) / (1/4 + tau_k^2)
converge to varpi_RVM.  The partial products are computed from data/zeross.bin (gamma_k) and
data/zLamb.bin (tau_k) and compared with varpi_RVM from s02_transition.py."""
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, savefig, write_csv

say = Tee("s07_product_identity")
gam, tau = load_ordinates()
VARPI = 0.9619415807          # Richardson value of varpi_RVM (s02_transition.py)
N = gam.size
logP = np.cumsum(np.log1p(0.25 / gam ** 2) + 2 * np.log(gam) - np.log1p(0.25 / tau ** 2) - 2 * np.log(tau))
P = np.exp(logP)
say("N = %d ordinates; varpi_RVM = %.10f" % (N, VARPI))
say("%10s %16s %14s" % ("K", "P_K", "P_K - varpi"))
for K in [10, 100, 1000, 10 ** 4, 10 ** 5, 10 ** 6, N]:
    say("%10d %16.10f %14.3e" % (K, P[K - 1], P[K - 1] - VARPI))
for K0 in [10 ** 4, 10 ** 5, 10 ** 6]:
    seg = P[K0 - 1:]
    say("K >= %7d: min P_K = %.10f, max P_K = %.10f, mean = %.10f" % (K0, seg.min(), seg.max(), seg.mean()))
# the flow along the family: Upsilon(lam) = sum_k 2 gamma_k(lam) eta_k / (1/4 + gamma_k(lam)^2),
# eta_k = tau_k - gamma_k, gamma_k(lam) = (1 - lam) gamma_k + lam tau_k; d/dlam log varpi_lam = -Upsilon(lam)
eta = tau - gam
say("")
say("flow along the lambda-family")
for lam in [0.0, 0.25, 0.5, 0.75, 1.0]:
    gl = (1 - lam) * gam + lam * tau
    say("Upsilon(%.2f) = %.10f" % (lam, np.sum(2 * gl * eta / (0.25 + gl ** 2))))
say("sum_k 2 eta_k^2 / gamma_k^2 = %.10f" % np.sum(2 * eta ** 2 / gam ** 2))
total = np.sum(np.log((0.25 + tau ** 2) / (0.25 + gam ** 2)))
say("integral_0^1 Upsilon = sum_k log((1/4+tau_k^2)/(1/4+gamma_k^2)) = %.10f" % total)
say("log L = log(1/varpi_RVM)                                  = %.10f" % np.log(1 / VARPI))
step = 50
idx = np.arange(step - 1, N, step)
write_csv("s07_partial_products.csv", ["K", "P_K"], [idx + 1, P[idx]])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
k1 = np.arange(1, 10 ** 4 + 1)
ax[0].plot(k1, P[:10 ** 4], "r-", lw=0.6, label=r"$P_K$")
ax[0].axhline(VARPI, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
```

```
ax[0].set_xlabel("K")
ax[0].set_title(r"Partial products $P_K$, $K\leq 10^4$")
ax[0].legend(fontsize=8)
sel = idx >= 10 ** 5 - 1
ax[1].plot(idx[sel] + 1, (P[idx[sel]] - VARPI) * 1e6, "r-", lw=0.6)
ax[1].axhline(0.0, color="k", lw=0.8, ls="--")
ax[1].set_xlabel("K")
ax[1].set_ylabel(r"$(P_K-\varpi_{\rm RVM})\cdot 10^{6}$")
ax[1].set_title(r"$10^5\leq K\leq %d$" % N)
fig.tight_layout()
savefig(fig, "fig11_product_identity.png")
say.close()
```

A.9. **s08_transport_moments.py.**

```
#!/usr/bin/env python3
"""s08_transport_moments.py
Transport tau -> gamma on the first rung (support of psi inside (0, log 2)).

Input (not recomputed): data/zeross.bin (gamma_k), data/zLamb.bin (tau_k), little-endian float64.

For a fixed normalized smooth bump psi supported in [A,B] subset (0, log 2):
  c_psi^{(j)}(t) = int psi(u) u^j cos(t u + j pi/2) du
  W_psi    = (1/2) int psi(u) ( e^{u/2} - e^{-u/2}/(e^{2u}-1) ) du       (lambda-independent part)
  R_psi(1) = W_psi - sum_k c_psi(tau_k)                                   (clock defect; clock side only)
  S_psi(lambda) = sum_k [ c_psi(gamma_k + lambda*eta_k) - c_psi(tau_k) ]  (transport), eta_k = tau_k - gamma_k
  m_j(psi) = sum_k eta_k^j c_psi^{(j)}(tau_k)                             (clock-sampled moments)
and the partial sums of  S_psi(0) = sum_{j>=1} (-1)^j m_j / j!.
Outputs: output/s08_transport_console.txt, s08_moments.csv, s08_transport_lambda.csv, s08_first_moment_terms.
    csv
"""
import os, math
import numpy as np

K = 20000
A, B = 0.10, 0.62            # support of psi, inside (0, log 2 = 0.6931)
NU = 4001                    # quadrature nodes
JMAX = 12
HERE = os.path.dirname(os.path.abspath(__file__))
OUT = os.path.join(HERE, '..', 'output')
DATA = os.path.join(HERE, '..', 'data')
os.makedirs(OUT, exist_ok=True)

u = np.linspace(A, B, NU)
v = (u - 0.5 * (A + B)) / (0.5 * (B - A))
psi = np.zeros_like(u)
ins = np.abs(v) < 1
psi[ins] = np.exp(-1.0 / (1.0 - v[ins] ** 2))
du = u[1] - u[0]
psi /= psi.sum() * du                      # int psi = 1

def c(t, j=0, chunk=2000):
    t = np.atleast_1d(np.asarray(t, dtype=float))
    w = psi * u ** j * du
    out = np.empty_like(t)
    for a in range(0, len_{\RVM}(t), chunk):
        out[a:a + chunk] = np.cos(np.outer(t[a:a + chunk], u) + j * np.pi / 2) @ w
    return out

def main():
    lines = []
    def P(s=''):
        print(s); lines.append(s)
    g_all = np.fromfile(os.path.join(DATA, 'zeross.bin'), dtype='<f8')
    t_all = np.fromfile(os.path.join(DATA, 'zLamb.bin'), dtype='<f8')
```

```
    g = g_all[:K]; tau = t_all[:K]; eta = tau - g
    W = 0.5 * (psi * (np.exp(u / 2) - np.exp(-u / 2) / (np.exp(2 * u) - 1))).sum() * du
    c_tau = c(tau); c_gam = c(g)
    R1 = W - c_tau.sum()
    S0 = (c_gam - c_tau).sum()
    P('s08: transport tau -> gamma on the first rung')
    P(f'entries in zeross.bin / zLamb.bin: {len(g_all)} / {len(t_all)};  used K = {K}')
    P(f'gamma_1 = {g[0]:.9f}, tau_1 = {tau[0]:.9f};  gamma_K = {g[-1]:.6f}, tau_K = {tau[-1]:.6f}')
    P(f'psi: normalized bump on [{A}, {B}] (log 2 = {math.log(2):.6f}), {NU} nodes')
    P(f'max|eta_k| = {np.abs(eta).max():.6f} at k = {np.abs(eta).argmax()+1};  rms eta = {np.sqrt((eta**2).
     mean()):.6f}')
    for k0 in (100, 300, 1000, 3000, 10000, K - 1):
        P(f'  max_(k>={k0:5d}) |c_psi(tau_k)| = {np.abs(c_tau[k0:]).max():.3e}')
    P()
    P(f'W_psi                          = {W:+.12f}')
    P(f'sum_k c_psi(tau_k)            = {c_tau.sum():+.12f}')
    P(f'sum_k c_psi(gamma_k)           = {c_gam.sum():+.12f}')
    P(f'R_psi(1) = W - sum c(tau)      = {R1:+.12f}   (clock defect)')
    P(f'S_psi(0) = sum[c(gam)-c(tau)]  = {S0:+.12f}   (transport)')
    P(f'R_psi(0) = R_psi(1) - S_psi(0) = {R1 - S0:+.3e}')
    P()
    P(' j        m_j(psi)        (-1)^j m_j/j!      partial sum      partial - R_psi(1)')
    rows = []; part = 0.0
    for j in range(1, JMAX + 1):
        mj = (eta ** j * c(tau, j)).sum()
        term = (-1) ** j * mj / math.factorial(j)
        part += term
        rows.append((j, mj, term, part, part - R1))
        P(f'{j:2d}  {mj:+.10e}  {term:+.10e}  {part:+.12f}  {part - R1:+.3e}')
    np.savetxt(os.path.join(OUT, 's08_moments.csv'), np.array(rows), delimiter=',',
               header='j,m_j,term,partial_sum,partial_minus_R1', comments='', fmt=['%d'] + ['%.15e'] * 4)
    # which k carry the first two moments
    t1 = -eta * c(tau, 1); t2 = 0.5 * eta ** 2 * c(tau, 2)
    kk = np.arange(1, K + 1)
    np.savetxt(os.path.join(OUT, 's08_first_moment_terms.csv'), np.c_[kk, tau, eta, c_tau, c_gam - c_tau, t1,
     t2][:400], delimiter=',',
               header='k,tau_k,eta_k,c_psi(tau_k),c_psi(gamma_k)-c_psi(tau_k),-eta_k*c1(tau_k),0.5*eta_k^2*c2(
     tau_k)', comments='',
               fmt=['%d'] + ['%.12e'] * 6)
    P()
    P(' cumulative over k (first K0 ordinates):  K0   sum[c(gam)-c(tau)]   W - sum c(tau)   -m_1 part   +m_2/2
      part')
    for K0 in (1, 2, 3, 5, 10, 20, 50, 100, 200, 400):
        P(f'   {K0:4d}   {(c_gam-c_tau)[:K0].sum():+.10f}   {W - c_tau[:K0].sum():+.10f}   {t1[:K0].sum():+.10
     f}   {t2[:K0].sum():+.10f}')
    P()
    P(' lambda     S_psi(lambda)     R_psi(lambda) = R_psi(1) - S_psi(lambda)')
    lam = np.linspace(-1.0, 2.0, 61); rows = []
    for L in lam:
        S = (c(g + L * eta) - c_tau).sum()
        rows.append((L, S, R1 - S))
    for L, S, R in rows[::5]:
        P(f'{L:+.2f}   {S:+.12f}   {R:+.12f}')
    np.savetxt(os.path.join(OUT, 's08_transport_lambda.csv'), np.array(rows), delimiter=',',
               header='lambda,S_psi,R_psi', comments='', fmt='%.15e')
    with open(os.path.join(OUT, 's08_transport_console.txt'), 'w') as f:
        f.write('\n'.join(lines) + '\n')

if __name__ == '__main__':
    main()
```

A.10. **s09_clock_defect_split.py.**

```
#!/usr/bin/env python3
"""s09_clock_defect_split.py
```

```
Splits the clock defect R_psi(1) of s08 (same psi) by Poisson summation on the clock:
  sum_{k>=1} c_psi(tau_k) = I_0 + 2 sum_{m>=1} (-1)^m I_m,
  I_m = int_{t_0}^infty c_psi(t) cos(2 pi m N_RVM(t)) N_RVM'(t) dt,   N_RVM(t_0) = 0,
  I_0 = int psi(u) K_0(u) du,  K_0(u) = -(1/(2 pi u)) [ sin(u t_0) log(t_0/(2 pi)) + pi/2 - Si(u t_0) ].
Reports W_psi, I_0, the smooth mismatch W_psi - I_0 and the oscillating part  -2 sum_{m>=1}(-1)^m I_m
(obtained as R_psi(1) - (W_psi - I_0), with sum_k c_psi(tau_k) read from data/zLamb.bin).
Output: output/s09_clock_defect_split_console.txt
"""
import os, math
import numpy as np
HERE = os.path.dirname(os.path.abspath(__file__))
OUT = os.path.join(HERE, '..', 'output'); DATA = os.path.join(HERE, '..', 'data')
A, B, NU, K = 0.10, 0.62, 4001, 20000
u = np.linspace(A, B, NU); v = (u - 0.5*(A+B)) / (0.5*(B-A))
psi = np.zeros_like(u); ins = np.abs(v) < 1; psi[ins] = np.exp(-1.0/(1.0 - v[ins]**2))
du = u[1]-u[0]; psi /= psi.sum()*du

def NR(t): return t/(2*np.pi)*np.log(t/(2*np.pi*np.e)) + 7.0/8.0
t0 = 9.7
for _ in range(60): t0 -= NR(t0) / (np.log(t0/(2*np.pi))/(2*np.pi))

def Si(x, n=20001):
    s = np.linspace(0.0, 1.0, n)[None, :] * np.atleast_1d(x)[:, None]
    f = np.ones_like(s); nz = s != 0; f[nz] = np.sin(s[nz]) / s[nz]
    h = np.atleast_1d(x) / (n - 1)
    return (f.sum(axis=1) - 0.5*(f[:, 0] + f[:, -1])) * h

K0 = -(1.0/(2*np.pi*u)) * (np.sin(u*t0)*np.log(t0/(2*np.pi)) + np.pi/2 - Si(u*t0))
I0 = (psi*K0).sum()*du
W = 0.5*(psi*(np.exp(u/2) - np.exp(-u/2)/(np.exp(2*u)-1))).sum()*du
tau = np.fromfile(os.path.join(DATA, 'zLamb.bin'), dtype='<f8')[:K]
w = psi*du; ctau = sum((np.cos(np.outer(tau[a:a+2000], u)) @ w).sum() for a in range(0, K, 2000))
R1 = W - ctau
L = [f's09: Poisson split of the clock defect (psi of s08 on [{A},{B}])',
     f't_0 (N_RVM(t_0)=0)                = {t0:.12f}   (2 pi = {2*np.pi:.6f}, tau_1 = {tau[0]:.6f})',
     f'W_psi                             = {W:+.12f}',
     f'I_0  (m = 0, smooth clock density) = {I0:+.12f}',
     f'sum_k c_psi(tau_k)                = {ctau:+.12f}',
     f'R_psi(1)                          = {R1:+.12f}',
     f'smooth mismatch  W_psi - I_0      = {W - I0:+.12f}',
     f'oscillating part -2 sum_(m>=1)(-1)^m I_m = {R1 - (W - I0):+.12f}',
     f'stationary heights 2 pi e^u, u in supp psi: [{2*np.pi*np.exp(A):.4f}, {2*np.pi*np.exp(B):.4f}];  part
     above t_0: u > {np.log(t0/(2*np.pi)):.6f}']
print('\n'.join(L)); open(os.path.join(OUT, 's09_clock_defect_split_console.txt'), 'w').write('\n'.join(L)+'\n
     ')
```

A.11. **s10_clock_to_zeros.py.**

```
#!/usr/bin/env python3
"""s10_clock_to_zeros.py
Solves, starting from the clock (eta = 0), the system of rung equations

   sum_k c_j(gamma_k) + sum_n a_n psi_j(log n) = W_j ,   j = 1..J,

for the unknown ordinates gamma_k (below a height) and the unknown masses a_n at the log-integers.
  psi_j(u) = w(u) cos(omega_j u), w(u) sin(omega_j u);  w = smooth bump on (0, U), U = log(N + 1/2)
  c_j(t)   = int psi_j(u) cos(t u) du
  W_j      = int psi_j(u) (1/2)( e^{u/2} - e^{-u/2}/(e^{2u}-1) ) du      (lambda-independent part)
Ordinates above the free range are kept on the clock (tau_k from data/zLamb.bin).
The masses enter linearly and are eliminated exactly at every step (ridge RIDGE).
Continuation in height: the band omega <= Omega is enlarged step by step (OMEGAS); at each stage the
ordinates with tau_k <= Omega + MARGIN are free.  Start: gamma_k = tau_k.  Method: Levenberg-Marquardt.
data/zeross.bin is used only AFTER the iteration, to compare.
Outputs: output/s10_clock_to_zeros_console.txt, s10_ordinates.csv, s10_masses.csv
"""
```

```
import os, math
import numpy as np
HERE = os.path.dirname(os.path.abspath(__file__))
OUT = os.path.join(HERE, '..', 'output'); DATA = os.path.join(HERE, '..', 'data')
N = 13; U = math.log(N + 0.5)
KT = 6000; DOM = 0.1; NU = 8001; ITER = 400
OMEGAS = [18, 22, 26, 30, 34, 38, 42, 46, 50]; MARGIN = 12.0
WMIN = 5e-2; STEP = 0.2; RIDGE = 1e-10

u = np.linspace(0.0, U, NU)[1:-1]; du = U / (NU - 1)
v = (u - U/2) / (U/2); w = np.exp(-1.0/(1.0 - v**2)); w /= w.sum()*du
Wfun = 0.5*(np.exp(u/2) - np.exp(-u/2)/(np.exp(2*u) - 1))
ns = np.array([n for n in range(2, N + 1) if np.interp(math.log(n), u, w)/w.max() >= WMIN])
tau = np.fromfile(os.path.join(DATA, 'zLamb.bin'), dtype='<f8')[:KT]
COS_TAU = np.zeros_like(u)                        # sum over all clock ordinates of cos(tau_k u)
for a in range(0, KT, 500): COS_TAU += np.cos(np.outer(u, tau[a:a+500])).sum(axis=1)

lines = []
def P(s=''):
    print(s); lines.append(s)
P('s10: from the clock to the ordinates through the rung equations (continuation in height)')
P(f'N = {N}, U = log({N}+1/2) = {U:.6f}; free masses n = {ns[0]}..{ns[-1]}; start gamma_k = tau_k')
g_all = tau.copy()
for OM in OMEGAS:
    om = np.arange(0.0, OM + 1e-9, DOM)
    Psi = np.vstack([w*np.cos(o*u) for o in om] + [w*np.sin(o*u) for o in om[1:]]) * du
    Wj = Psi @ Wfun
    Pn = np.vstack([np.interp(np.log(ns), u, Psi[j]/du) for j in range(Psi.shape[0])])
    Q = np.linalg.solve(Pn.T @ Pn + RIDGE*np.eye(len(ns)), Pn.T)
    K = int((tau <= OM + MARGIN).sum())
    g = g_all[:K].copy()
    fixed = Psi @ (COS_TAU - np.cos(np.outer(u, tau[:K])).sum(axis=1)) - Wj
    def r0(g): return Psi @ np.cos(np.outer(u, g)).sum(axis=1) + fixed
    def res(g):
        x = r0(g); a = -Q @ x; return x + Pn @ a, a
    r, a = res(g); n0 = np.linalg.norm(r); damp = 1e-3
    for it in range(1, ITER + 1):
        Jg = Psi @ (-u[:, None]*np.sin(np.outer(u, g))); Jg = Jg - Pn @ (Q @ Jg)
        A = Jg.T @ Jg; b = -Jg.T @ r
        while True:
            d = np.linalg.solve(A + damp*np.diag(np.diag(A) + 1e-14), b)
            d *= min(1.0, STEP/max(np.abs(d).max(), 1e-300))
            r2, a2 = res(g + d)
            if np.linalg.norm(r2) < np.linalg.norm(r) or damp > 1e12: break
            damp *= 4
        g = g + d; r, a = r2, a2; damp = max(damp/3, 1e-14)
        if np.abs(d).max() < 1e-11: break
    g_all[:K] = g
    P(f'band omega <= {OM:2d}: {K:2d} free ordinates, {Psi.shape[0]} tests, |residual| {n0:.3e} -> {np.linalg.
     norm(r):.3e} in {it} iterations')
P()
gam = np.fromfile(os.path.join(DATA, 'zeross.bin'), dtype='<f8')[:K]
P('  k      tau_k (start)    solved ordinate    gamma_k (zeross.bin)   solved - gamma_k    tau_k - gamma_k')
for k in range(K):
    P(f'{k+1:3d}   {tau[k]:14.9f}   {g_all[k]:14.9f}   {gam[k]:14.9f}   {g_all[k]-gam[k]:+.3e}   {tau[k]-gam[k
     ]:+.6f}')
P()
def Lam(n):
    for p in range(2, n + 1):
        if n % p == 0:
            m = n
            while m % p == 0: m //= p
            return math.log(p) if m == 1 else 0.0
    return 0.0
P('  n     solved mass a_n     Lambda(n)/(2 sqrt n)     difference     w(log n)/max w')
```

```
for i, n in enumerate(ns):
    ex = Lam(int(n))/(2*math.sqrt(n))
    P(f'{n:3d}   {a[i]:+.9f}   {ex:+.9f}   {a[i]-ex:+.3e}   {np.interp(math.log(n), u, w)/w.max():.3e}')
np.savetxt(os.path.join(OUT, 's10_ordinates.csv'), np.c_[np.arange(1, K+1), tau[:K], g_all[:K], gam],
     delimiter=',',
           header='k,tau_k,solved,gamma_k', comments='', fmt=['%d'] + ['%.12f']*3)
np.savetxt(os.path.join(OUT, 's10_masses.csv'), np.c_[ns, a, [Lam(int(n))/(2*math.sqrt(n)) for n in ns]],
     delimiter=',',
           header='n,solved_mass,Lambda(n)/(2sqrt(n))', comments='', fmt=['%d'] + ['%.12f']*2)
open(os.path.join(OUT, 's10_clock_to_zeros_console.txt'), 'w').write('\n'.join(lines) + '\n')
```

A.12. **`s11_rvm_z1_difference_analysis.py`.** This supplemental diagnostic reproduces the RVM–$Z1$ difference, shell and convergence calculations. It uses the genuine ordinates only for post-construction diagnostics and is not used in any proof.

```
#!/usr/bin/env python3
"""s11_rvm_z1_difference_analysis.py

Python reproduction of the focused MATLAB program RVM_Z1_DIFFERENCE_ANALYSIS.m.

The script reads the two separately distributed binary ordinate files and
computes, for K = 1e3, 1e4, 1e5, 5e5, Kmax on 2 <= x <= 1e5:
  * the Perron-based Mertens difference M_RVM,K^crit - M_Z1,K^crit;
  * the explicit-formula prime-counting difference pi_RVM,K - pi_Z1,K;
  * K-convergence errors, spectral-shell contributions, normalizations,
    running suprema, and band statistics.

The numerical conventions and default parameters match the MATLAB analysis
used for the figures discussed in the paper.  The raw binaries are not
included in the archive; place zeross.bin and zLamb.bin in data/ before use.
"""

import os
import math
import time
import numpy as np
from scipy.special import expi, loggamma
from scipy.integrate import quad
import matplotlib.pyplot as plt

HERE = os.path.dirname(os.path.abspath(__file__))
ROOT = os.path.abspath(os.path.join(HERE, ".."))
DATA = os.path.join(ROOT, "data")
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)

XMAX = 100000.0
NX = 500
CP = 0.75
TPERRON = 80.0
NTP = 401
KHEAD = 30000
RPROD = 5
SBLOCK = 40
KEI = 5000
REI = 5
ZERO_BLOCK = 100000
ARG_BLOCK = 16


def say(lines, text=""):
    print(text)
```

```
    lines.append(text)


def load_ordinates():
    tau_path = os.path.join(DATA, "zLamb.bin")
    gam_path = os.path.join(DATA, "zeross.bin")
    if not os.path.exists(tau_path) or not os.path.exists(gam_path):
        raise FileNotFoundError(
            "Place data/zLamb.bin and data/zeross.bin in the archive tree before running s11."
        )
    tau = np.fromfile(tau_path, dtype="<f8")
    gam = np.fromfile(gam_path, dtype="<f8")
    kmax = min(len(tau), len(gam))
    return tau[:kmax], gam[:kmax]


def zeta_snapshots(s, ords, klist):
    """Canonical-product snapshots using the same head+moment expansion as MATLAB."""
    s = np.asarray(s, dtype=np.complex128).ravel()
    ords = np.asarray(ords, dtype=float).ravel()
    klist = np.asarray(klist, dtype=np.int64).ravel()
    ns = len(s)
    nk = len(klist)
    kmax = int(klist.max())
    aord = 0.25 + ords[:kmax] ** 2

    moments = np.zeros((RPROD, nk), dtype=float)
    for ik, K in enumerate(klist):
        if K > KHEAD:
            tail = aord[KHEAD:K]
            for r in range(1, RPROD + 1):
                moments[r - 1, ik] = np.sum(tail ** (-r))

    zout = np.empty((ns, nk), dtype=np.complex128)
    for i0 in range(0, ns, SBLOCK):
        i1 = min(ns, i0 + SBLOCK)
        ss = s[i0:i1]
        q = ss * (ss - 1.0)
        logG = np.log(ss * (ss - 1.0)) + loggamma(ss / 2.0) - (ss / 2.0) * math.log(math.pi)

        for ik, K in enumerate(klist):
            kh = min(int(K), KHEAD)
            head = aord[:kh]
            logP = np.sum(np.log(1.0 + q[:, None] / head[None, :]), axis=1)
            if K > KHEAD:
                tail = np.zeros_like(q)
                qpow = np.ones_like(q)
                for r in range(1, RPROD + 1):
                    qpow *= q
                    tail += ((-1) ** (r + 1)) * qpow * moments[r - 1, ik] / r
                logP += tail
            zout[i0:i1, ik] = np.exp(logP - logG)
    return zout


def tail_integral(y):
    val, _ = quad(
        lambda q: 1.0 / (q * (q * q - 1.0) * math.log(q)),
        float(y), np.inf, epsrel=2e-9, epsabs=2e-11, limit=200,
    )
    return val


def j_snapshots(y, ords, klist, label, lines):
    y = np.asarray(y, dtype=float).ravel()
```

```
    ords = np.asarray(ords, dtype=float).ravel()
    klist = np.asarray(klist, dtype=np.int64).ravel()
    ny = len(y)
    nk = len(klist)
    kmax = int(klist.max())
    jout = np.zeros((ny, nk), dtype=float)
    nblocks = (ny + ARG_BLOCK - 1) // ARG_BLOCK

    for ib in range(nblocks):
        a0 = ib * ARG_BLOCK
        a1 = min(ny, a0 + ARG_BLOCK)
        yy = y[a0:a1]
        u = np.log(yy)
        B = len(u)
        sum_at_k = np.zeros((B, nk), dtype=np.complex128)

        kbase = min(KEI, kmax)
        rho = 0.5 + 1j * ords[:kbase]
        ehead = expi(rho[:, None] * u[None, :])
        cumulative = np.sum(ehead, axis=0)

        for ik, K in enumerate(klist):
            if K <= KEI:
                sum_at_k[:, ik] = np.sum(expi((0.5 + 1j * ords[:K])[:, None] * u[None, :]), axis=0)
            elif K == kbase:
                sum_at_k[:, ik] = cumulative

        kstart = kbase
        while kstart < kmax:
            higher = klist[klist > kstart]
            next_target = int(higher.min()) if len(higher) else kmax
            kend = min(kmax, kstart + ZERO_BLOCK, next_target)
            ok = ords[kstart:kend]
            rk = 0.5 + 1j * ok
            E = np.exp(1j * ok[:, None] * u[None, :])
            local = np.zeros(B, dtype=np.complex128)
            rpow = 1.0 / rk
            for qidx in range(REI):
                Sq = np.sum(rpow[:, None] * E, axis=0)
                coeff = math.factorial(qidx) * np.exp(u / 2.0) / (u ** (qidx + 1))
                local += coeff * Sq
                rpow /= rk
            cumulative += local
            hits = np.where(klist == kend)[0]
            for ik in hits:
                sum_at_k[:, ik] = cumulative
            kstart = kend

        lix = expi(u).real
        corr = np.array([tail_integral(v) for v in yy])
        jout[a0:a1, :] = lix[:, None] - 2.0 * sum_at_k.real - math.log(2.0) + corr[:, None]
        say(lines, f"  {label}: argument block {ib+1}/{nblocks}")
    return jout


def mobius_vector(N):
    mu = np.ones(N + 1, dtype=np.int8)
    is_prime = np.ones(N + 1, dtype=bool)
    is_prime[:2] = False
    for p in range(2, N + 1):
        if not is_prime[p]:
            continue
        mu[p::p] *= -1
        p2 = p * p
        if p2 <= N:
```

```
            mu[p2::p2] = 0
        if p * p <= N:
            is_prime[p * p::p] = False
    return mu


def save_lines_plot(x, data, labels, ylabel, title, filename, zero=True):
    fig, ax = plt.subplots(figsize=(7.4, 4.8))
    for j, lab in enumerate(labels):
        ax.plot(x, data[:, j], linewidth=1.0, label=lab)
    if zero:
        ax.axhline(0.0, linestyle="--", linewidth=0.8)
    ax.set_xlabel("x")
    ax.set_ylabel(ylabel)
    ax.set_title(title)
    ax.grid(True, alpha=0.25)
    ax.legend(fontsize=8)
    fig.tight_layout()
    fig.savefig(os.path.join(FIG, filename), dpi=180)
    plt.close(fig)


def main():
    t0 = time.time()
    lines = []
    tau, gam = load_ordinates()
    kmax = min(len(tau), len(gam))
    klist = np.unique(np.array([1000, 10000, 100000, 500000, kmax], dtype=np.int64))
    klist = klist[klist <= kmax]
    nk = len(klist)
    x = np.linspace(2.0, XMAX, NX)

    say(lines, "s11: Python reproduction of the MATLAB RVM-Z1 difference analysis")
    say(lines, f"Kmax = {kmax}")
    say(lines, "Klist = " + " ".join(str(int(v)) for v in klist))

    tp = np.linspace(0.0, TPERRON, NTP)
    sp = CP + 1j * tp
    say(lines, "Computing Perron-line zeta snapshots...")
    zRP = zeta_snapshots(sp, tau, klist)
    zGP = zeta_snapshots(sp, gam, klist)

    MR = np.zeros((NX, nk))
    MG = np.zeros((NX, nk))
    for ik in range(nk):
        baseR = (1.0 / zRP[:, ik]) / sp
        baseG = (1.0 / zGP[:, ik]) / sp
        for ix, xx in enumerate(x):
            ph = np.exp(1j * tp * math.log(xx))
            MR[ix, ik] = xx ** CP / math.pi * np.real(np.trapz(baseR * ph, tp))
            MG[ix, ik] = xx ** CP / math.pi * np.real(np.trapz(baseG * ph, tp))
    dM = MR - MG

    say(lines, "Building Mobius-inversion arguments...")
    mu = mobius_vector(int(math.floor(XMAX)))
    all_y = []
    all_coeff = []
    maps = []
    for xx in x:
        start = len(all_y)
        mmax = int(math.floor(math.log(xx) / math.log(2.0)))
        for m in range(1, mmax + 1):
            if mu[m] == 0:
                continue
            y = xx ** (1.0 / m)
```

```
            if y >= 2.0:
                all_y.append(y)
                all_coeff.append(float(mu[m]) / m)
        maps.append((start, len(all_y) - start))
    all_y = np.asarray(all_y)
    all_coeff = np.asarray(all_coeff)
    y_unique, inverse = np.unique(all_y, return_inverse=True)

    say(lines, "Computing J snapshots...")
    JRu = j_snapshots(y_unique, tau, klist, "RVM", lines)
    JGu = j_snapshots(y_unique, gam, klist, "Z1", lines)
    JRa = JRu[inverse]
    JGa = JGu[inverse]

    piR = np.zeros((NX, nk))
    piG = np.zeros((NX, nk))
    for ix, (start, count) in enumerate(maps):
        if count:
            sl = slice(start, start + count)
            piR[ix, :] = all_coeff[sl] @ JRa[sl, :]
            piG[ix, :] = all_coeff[sl] @ JGa[sl, :]
    dPi = piR - piG

    sqrtx = np.sqrt(x)
    logx = np.log(x)
    nPi = dPi / (sqrtx[:, None] / logx[:, None])
    nM_sqrt = dM / sqrtx[:, None]
    nM_sqrtlog = dM / (sqrtx[:, None] * logx[:, None])

    errM = dM - dM[:, [-1]]
    errPi = dPi - dPi[:, [-1]]
    shellM = np.column_stack([dM[:, 0], np.diff(dM, axis=1)])
    shellPi = np.column_stack([dPi[:, 0], np.diff(dPi, axis=1)])
    runSupPi = np.maximum.accumulate(np.abs(nPi), axis=0)
    runSupMsqrt = np.maximum.accumulate(np.abs(nM_sqrt), axis=0)
    runSupMsqrtlog = np.maximum.accumulate(np.abs(nM_sqrtlog), axis=0)

    rmsErrM = np.sqrt(np.mean(errM ** 2, axis=0))
    maxErrM = np.max(np.abs(errM), axis=0)
    rmsErrPi = np.sqrt(np.mean(errPi ** 2, axis=0))
    maxErrPi = np.max(np.abs(errPi), axis=0)
    shellRMSM = np.sqrt(np.mean(shellM ** 2, axis=0))
    shellMaxM = np.max(np.abs(shellM), axis=0)
    shellRMSPi = np.sqrt(np.mean(shellPi ** 2, axis=0))
    shellMaxPi = np.max(np.abs(shellPi), axis=0)

    summary = np.column_stack([
        klist, rmsErrM, maxErrM, rmsErrPi, maxErrPi,
        shellRMSM, shellMaxM, shellRMSPi, shellMaxPi,
        runSupPi[-1], runSupMsqrt[-1], runSupMsqrtlog[-1],
    ])
    np.savetxt(
        os.path.join(OUT, "s11_difference_summary.csv"), summary, delimiter=",",
        header=("K,rms_err_M,max_err_M,rms_err_pi,max_err_pi,rms_shell_M,max_shell_M,"
                "rms_shell_pi,max_shell_pi,final_run_sup_pi,final_run_sup_M_sqrt,"
                "final_run_sup_M_sqrtlog"), comments="", fmt="%.15e"
    )
    np.savez_compressed(
        os.path.join(OUT, "s11_difference_analysis.npz"),
        x=x, Klist=klist, MR=MR, MG=MG, dM=dM, piR=piR, piG=piG, dPi=dPi,
        nPi=nPi, nM_sqrt=nM_sqrt, nM_sqrtlog=nM_sqrtlog,
        errM=errM, errPi=errPi, shellM=shellM, shellPi=shellPi,
        runSupPi=runSupPi, runSupMsqrt=runSupMsqrt, runSupMsqrtlog=runSupMsqrtlog,
    )
```

```
    labels = [f"K={int(v)}" + (" (max)" if v == kmax else "") for v in klist]
    shell_labels = [f"1...{int(klist[0])}"] + [f"{int(klist[j-1])+1}...{int(klist[j])}" for j in range(1, nk)]
    save_lines_plot(x, dM, labels, "M_RVM,K^crit - M_Z1,K^crit",
                    "Mertens difference: RVM - Z1", "fig16_difference_mertens.png")
    save_lines_plot(x, dPi, labels, "pi_RVM,K - pi_Z1,K",
                    "Prime-counting difference: RVM - Z1", "fig17_difference_pi.png")
    save_lines_plot(x, shellM, shell_labels, "Mertens shell contribution",
                    "Spectral shell contributions to the Mertens difference", "fig18_shells_mertens.png")
    save_lines_plot(x, shellPi, shell_labels, "Prime-counting shell contribution",
                    "Spectral shell contributions to the prime-counting difference", "fig19_shells_pi.png")
    save_lines_plot(x, runSupMsqrt, labels, "max_{x<=X} |dM_K(x)| / sqrt(x)",
                    "Running normalized supremum: Mertens / sqrt(x)", "fig20_runsup_mertens.png", zero=False)
    save_lines_plot(x, runSupPi, labels, "max_{x<=X} |dpi_K(x)| / (sqrt(x)/log(x))",
                    "Running normalized supremum: prime-counting difference", "fig21_runsup_pi.png", zero=
     False)

    say(lines, "")
    say(lines, "K-convergence and shell summary")
    for j, K in enumerate(klist):
        say(lines, f"K={int(K):8d}  rmsErrM={rmsErrM[j]:.8g}  maxErrM={maxErrM[j]:.8g}  "
                   f"rmsErrPi={rmsErrPi[j]:.8g}  maxErrPi={maxErrPi[j]:.8g}")
        say(lines, f"            rmsShellM={shellRMSM[j]:.8g} maxShellM={shellMaxM[j]:.8g}  "
                   f"rmsShellPi={shellRMSPi[j]:.8g} maxShellPi={shellMaxPi[j]:.8g}")
    say(lines, f"Total elapsed time: {(time.time()-t0)/60.0:.2f} min")
    with open(os.path.join(OUT, "s11_difference_analysis_console.txt"), "w", encoding="utf-8") as f:
        f.write("\n".join(lines) + "\n")


if __name__ == "__main__":
    main()
```

A.13. **`s12_multiscale_jacobian.py`.** This program produces Experiment 18.13 and Figures 16 and its companion fixed-window diagnostic. It generates the clock from the Lambert formula and does not read the genuine ordinates.

```
#!/usr/bin/env python3
"""s12_multiscale_jacobian.py

Clock-only diagnostics for the multiscale Jacobian analysis.

No genuine zeta ordinates and no displacement data are used.  The midpoint
ordinates are generated directly from the Lambert-W formula

    tau_k = 2*pi*(k - 11/8) / W_0((k - 11/8)/e).

The program computes:
  1. the minimum clock gap up to several heights and the rigorous lower bound
     2*pi/log(T/(2*pi));
  2. the associated Ingham window threshold 2*pi/min_gap and the natural
     scale log(T/(2*pi));
  3. fixed-window adjacent-pair upper bounds showing that no uniform lower
     frame bound can hold on a fixed u-window as k -> infinity.

Outputs:
  output/s12_multiscale_summary.csv
  output/s12_fixed_window_pairs.csv
  output/s12_multiscale_jacobian_console.txt
  figures/fig14_multiscale_gap.png
  figures/fig15_fixed_window_degeneracy.png
"""

import os
import math
import numpy as np
```

```
from scipy.special import lambertw
import matplotlib.pyplot as plt

HERE = os.path.dirname(os.path.abspath(__file__))
ROOT = os.path.abspath(os.path.join(HERE, ".."))
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)

KMAX = 2_001_052
K_VALUES = np.array([100, 1_000, 10_000, 100_000, 500_000, KMAX], dtype=np.int64)
PAIR_K = np.unique(np.geomspace(10, KMAX - 1, 240).astype(np.int64))
FIXED_WINDOWS = (0.5, 1.0, 2.0, 4.0)


def midpoint_ordinates(k):
    """Return midpoint RVM ordinates for a NumPy array of positive indices."""
    k = np.asarray(k, dtype=np.float64)
    a = k - 11.0 / 8.0
    w = lambertw(a / math.e, k=0).real
    return 2.0 * math.pi * a / w


def main():
    lines = []

    def say(text=""):
        print(text)
        lines.append(text)

    say("s12: multiscale Jacobian diagnostics from the midpoint clock")
    say("No zeross.bin, zLamb.bin, or displacement sequence is used.")
    say(f"KMAX = {KMAX}")
    say()

    indices = np.arange(1, KMAX + 1, dtype=np.float64)
    tau = midpoint_ordinates(indices)
    gaps = np.diff(tau)
    prefix_min_gap = np.minimum.accumulate(gaps)

    rows = []
    say("       K              T        min gap      gap lower     min/lower      Ucrit       log(T/2pi)
     Ucrit/log")
    for K in K_VALUES:
        T = tau[K - 1]
        min_gap = prefix_min_gap[K - 2] if K >= 2 else np.nan
        gap_lower = 2.0 * math.pi / math.log(T / (2.0 * math.pi))
        ucrit = 2.0 * math.pi / min_gap
        unatural = math.log(T / (2.0 * math.pi))
        rows.append((K, T, min_gap, gap_lower, min_gap / gap_lower,
                     ucrit, unatural, ucrit / unatural))
        say(f"{K:9d} {T:14.6f} {min_gap:13.8f} {gap_lower:13.8f} "
            f"{min_gap/gap_lower:13.8f} {ucrit:11.6f} {unatural:14.8f} {ucrit/unatural:11.8f}")

    np.savetxt(
        os.path.join(OUT, "s12_multiscale_summary.csv"),
        np.asarray(rows, dtype=float),
        delimiter=",",
        header="K,T,min_gap,gap_lower_bound,min_gap_over_lower,Ucrit,log_T_over_2pi,Ucrit_over_log",
        comments="",
        fmt="%.15e",
    )

    # Fixed-window adjacent-pair upper bounds.
```

```
# For h_k = 1, h_{k+1} = -1, ||h||_2 = sqrt(2), and
# |sin(tau_{k+1}u)-sin(tau_ku)| <= gap_k * u, hence
# ||u[sin(tau_{k+1}u)-sin(tau_ku)]||_L2(0,U) / ||h||_2
# <= gap_k * U^(5/2) / sqrt(10).
pair_rows = []
for K in PAIR_K:
    gap = gaps[K - 1]
    row = [K, tau[K - 1], gap]
    for U in FIXED_WINDOWS:
        row.append(gap * U ** 2.5 / math.sqrt(10.0))
    pair_rows.append(row)
pair_rows = np.asarray(pair_rows, dtype=float)

header = "k,tau_k,gap_k," + ",".join(
    f"adjacent_pair_upper_U_{U:g}" for U in FIXED_WINDOWS
)
np.savetxt(
    os.path.join(OUT, "s12_fixed_window_pairs.csv"),
    pair_rows,
    delimiter=",",
    header=header,
    comments="",
    fmt="%.15e",
)

say()
say("Fixed-window adjacent-pair upper bounds at the largest sampled k:")
K = int(PAIR_K[-1])
gap = gaps[K - 1]
say(f"k = {K}, tau_k = {tau[K-1]:.6f}, gap = {gap:.9f}")
for U in FIXED_WINDOWS:
    bound = gap * U ** 2.5 / math.sqrt(10.0)
    say(f"  U = {U:3.1f}: normalized upper bound <= {bound:.9e}")

# Figure 14: minimum gap and rigorous lower bound.
fig, ax = plt.subplots(figsize=(7.2, 4.6))
arr = np.asarray(rows, dtype=float)
ax.loglog(arr[:, 1], arr[:, 2], marker="o", label="minimum observed clock gap")
ax.loglog(arr[:, 1], arr[:, 3], marker="s", label=r"$2\pi/\log(T/2\pi)$ lower bound")
ax.set_xlabel("clock height T")
ax.set_ylabel("gap")
ax.set_title("Midpoint-clock gap and the multiscale window threshold")
ax.grid(True, which="both", alpha=0.25)
ax.legend()
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig14_multiscale_gap.png"), dpi=180)
plt.close(fig)

# Figure 15: failure of a fixed-window lower bound.
fig, ax = plt.subplots(figsize=(7.2, 4.6))
for j, U in enumerate(FIXED_WINDOWS, start=3):
    ax.loglog(pair_rows[:, 0], pair_rows[:, j], label=f"U = {U:g}")
ax.set_xlabel("clock index k")
ax.set_ylabel("adjacent-pair normalized upper bound")
ax.set_title("Fixed-window near-degeneracy of adjacent clock frequencies")
ax.grid(True, which="both", alpha=0.25)
ax.legend()
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig15_fixed_window_degeneracy.png"), dpi=180)
plt.close(fig)

say()
say("Saved numerical tables and figures.")
say("The decreasing fixed-window bounds are a diagnostic of the rigorous fact")
say("that no T-independent lower frame bound can hold on a fixed u-window.")
```

```
    with open(os.path.join(OUT, "s12_multiscale_jacobian_console.txt"), "w", encoding="utf-8") as f:
        f.write("\n".join(lines) + "\n")


if __name__ == "__main__":
    main()
```

A.14. **`s13_nested_bounded_endpoint.py`.** This program produces the bounded and nested endpoint diagnostics of Appendix E. The solver uses the midpoint clock; genuine ordinates and von Mangoldt masses, when displayed, are read only after the solve as diagnostics.

```
"""Nested bounded finite-section experiment for the endpoint rung system.

This program is diagnostic only. It solves finite sections using the midpoint RVM
clock as input. Genuine zeta ordinates, if present, are loaded only after the solve
for diagnostics. The experiment enforces bounded ordinate displacement and bounded
free masses, and keeps every earlier test block active in later sections.
"""

from pathlib import Path
import math
import numpy as np
from scipy.optimize import least_squares

HERE = Path(__file__).resolve().parent
ROOT = HERE.parent
DATA = ROOT / "data"
OUT = ROOT / "output"
FIG = ROOT / "figures"
OUT.mkdir(exist_ok=True)
FIG.mkdir(exist_ok=True)

TAU_FILE = DATA / "zLamb.bin"
GAMMA_FILE = DATA / "zeross.bin"

SHIFT_BOUND = 1.2
MASS_BOUND = 0.5
U_MIN = 0.08
OMEGA_STEP = 1.0
MARGIN = 14.0
LEVELS = [(13, 50.0), (20, 70.0), (34, 110.0), (50, 150.0), (70, 210.0), (100, 300.0)]


def load_data():
    tau = np.fromfile(TAU_FILE, dtype="<f8")
    if tau.size == 0:
        raise RuntimeError("zLamb.bin is required in data/")
    gamma = None
    if GAMMA_FILE.exists():
        gamma = np.fromfile(GAMMA_FILE, dtype="<f8")
    return tau, gamma


def choose_grid_size(U, omega):
    """Keep the Nyquist frequency safely above the largest tested frequency."""
    target_nyquist = 2.5 * (omega + MARGIN)
    n = int(math.ceil(target_nyquist * (U - U_MIN) / math.pi)) + 1
    return max(701, n)


def build_block(tau, n_max, omega_max, active_k, mass_nodes):
    U = math.log(n_max + 0.5)
    n_grid = choose_grid_size(U, omega_max)
```

```
    grid = np.linspace(U_MIN, U, n_grid)
    u = grid[1:-1]
    du = (U - U_MIN) / (n_grid - 1)

    mid = 0.5 * (U_MIN + U)
    half = 0.5 * (U - U_MIN)
    v = (u - mid) / half
    raw = np.exp(-1.0 / (1.0 - v * v))
    norm = raw.sum() * du
    w = raw / norm

    omega = np.arange(0.0, omega_max + 1e-12, OMEGA_STEP)
    psi = np.vstack([
        np.cos(np.outer(omega, u)) * w[None, :] * du,
        np.sin(np.outer(omega[1:], u)) * w[None, :] * du,
    ])

    smooth = 0.5 * (np.exp(u / 2.0) - np.exp(-u / 2.0) / (np.exp(2.0 * u) - 1.0))
    target = psi @ smooth

    clock = np.zeros_like(u)
    for start in range(0, tau.size, 500):
        clock += np.cos(np.outer(u, tau[start:start + 500])).sum(axis=1)
    low_clock = psi @ np.cos(np.outer(u, tau[:active_k])).sum(axis=1)
    fixed = psi @ clock - low_clock - target

    logs = np.log(mass_nodes.astype(float))
    env = np.zeros_like(logs)
    mask = (logs > U_MIN) & (logs < U)
    if np.any(mask):
        vv = (logs[mask] - mid) / half
        env[mask] = np.exp(-1.0 / (1.0 - vv * vv)) / norm

    mass_matrix = np.vstack([
        np.cos(np.outer(omega, logs)) * env[None, :],
        np.sin(np.outer(omega[1:], logs)) * env[None, :],
    ])

    return {
        "n_max": n_max,
        "omega_max": omega_max,
        "u": u,
        "psi": psi,
        "fixed": fixed,
        "mass_matrix": mass_matrix,
        "n_grid": n_grid,
    }


def solve_nested_level(tau, gamma, level_index, previous=None):
    n_max, omega_max = LEVELS[level_index]
    mass_nodes = np.arange(2, n_max + 1, dtype=int)
    active_k = int(np.sum(tau <= omega_max + MARGIN))

    blocks = []
    for j in range(level_index + 1):
        nb, ob = LEVELS[j]
        blocks.append(build_block(tau, nb, ob, active_k, mass_nodes))

    x0 = np.r_[tau[:active_k], np.zeros(mass_nodes.size)]
    if previous is not None:
        k_copy = min(previous["active_k"], active_k)
        n_copy = min(previous["masses"].size, mass_nodes.size)
        x0[:k_copy] = previous["ordinates"][:k_copy]
        x0[active_k:active_k + n_copy] = previous["masses"][:n_copy]
```

```
    lower = np.r_[tau[:active_k] - SHIFT_BOUND, -MASS_BOUND * np.ones(mass_nodes.size)]
    upper = np.r_[tau[:active_k] + SHIFT_BOUND, MASS_BOUND * np.ones(mass_nodes.size)]
    x0 = np.clip(x0, lower + 1e-10, upper - 1e-10)

    def residual(z):
        ordinates = z[:active_k]
        masses = z[active_k:]
        pieces = []
        for block in blocks:
            spectral = block["psi"] @ np.cos(np.outer(block["u"], ordinates)).sum(axis=1)
            r = spectral + block["fixed"] + block["mass_matrix"] @ masses
            pieces.append(r / math.sqrt(r.size))
        return np.concatenate(pieces)

    def jacobian(z):
        ordinates = z[:active_k]
        pieces = []
        for block in blocks:
            jk = block["psi"] @ (-block["u"][:, None] * np.sin(np.outer(block["u"], ordinates)))
            j = np.hstack([jk, block["mass_matrix"]])
            pieces.append(j / math.sqrt(j.shape[0]))
        return np.vstack(pieces)

    sol = least_squares(
        residual,
        x0,
        jac=jacobian,
        method="trf",
        tr_solver="lsmr",
        bounds=(lower, upper),
        max_nfev=500,
        ftol=2e-10,
        xtol=2e-10,
        gtol=2e-10,
        x_scale="jac",
    )

    ordinates = sol.x[:active_k]
    masses = sol.x[active_k:]

    block_rows = []
    for block in blocks:
        spectral = block["psi"] @ np.cos(np.outer(block["u"], ordinates)).sum(axis=1)
        r = spectral + block["fixed"] + block["mass_matrix"] @ masses
        block_rows.append((block["n_max"], float(np.sqrt(np.mean(r * r))), float(np.max(np.abs(r))), block["
     n_grid"]))

    result = {
        "n_max": n_max,
        "omega_max": omega_max,
        "active_k": active_k,
        "mass_nodes": mass_nodes,
        "ordinates": ordinates,
        "masses": masses,
        "blocks": block_rows,
        "nfev": sol.nfev,
    }

    if gamma is not None and gamma.size >= min(10, active_k):
        q = min(10, active_k)
        result["rms_first10_diagnostic"] = float(np.sqrt(np.mean((ordinates[:q] - gamma[:q]) ** 2)))
    else:
        result["rms_first10_diagnostic"] = float("nan")
    return result
```

```
def main():
    tau, gamma = load_data()
    rows = []
    previous = None
    for level_index in range(len(LEVELS)):
        result = solve_nested_level(tau, gamma, level_index, previous)
        previous = result
        print(
            f"N={result['n_max']:3d} K={result['active_k']:3d} "
            f"max_shift={np.max(np.abs(result['ordinates'] - tau[:result['active_k']])):.6f} "
            f"max_mass={np.max(np.abs(result['masses'])):.6f} "
            f"diagnostic_rms10={result['rms_first10_diagnostic']:.3e}"
        )
        for block_n, rms, max_abs, n_grid in result["blocks"]:
            print(f"  block N={block_n:3d}: RMS={rms:.3e}, max={max_abs:.3e}, grid={n_grid}")
            rows.append((result["n_max"], block_n, rms, max_abs, n_grid))

    np.savetxt(
        OUT / "s13_nested_metric_summary.csv",
        np.asarray(rows, dtype=float),
        delimiter=",",
        header="section_N,block_N,rms,max_abs,n_grid",
        comments="",
    )


if __name__ == "__main__":
    main()
```

Appendix B. Console output of the programs

The complete console output of each program, as written to `output/*_console.txt`.

The post-v16 numerical programs that produce the multiscale and bounded-endpoint figures (files `fig14`–`fig17`) and the bounded/nested endpoint tables are also distributed with their machine-readable outputs in the repository. In particular, `s12_multiscale_jacobian.py` writes `output/s12_multiscale_jacobian_console.txt`, `output/s12_multiscale_summary.csv` and `output/s12_fixed_window_pairs.csv`; `s13_nested_bounded_endpoint.py` writes `output/s13_bounded_mass_summary.csv`, `output/s13_bounded_mass_ordinates.csv`, `output/s13_bounded_mass_masses.csv`, `output/s13_cross_validation.csv` and `output/s13_nested_metric_summary.csv`. Here "normalized residual" means the Euclidean residual divided by the square root of the number of test equations. The bump and permanent-block conventions are specified in the corresponding code and in Appendix E.

B.1. `s01_clock_and_family.`

```
zeros loaded: N = 2001052, gamma_1 = 14.134725142, gamma_N = 1132490.658714, tau_1 = 14.521346953, tau_N =
    1132490.722588
max |N_RVM(tau_k) - (k - 1/2)| = 6.985e-10
eta_k = tau_k - gamma_k:  max |eta| = 0.993711 (k = 34),  mean |eta| = 0.151209,  rms = 0.188066,  mean eta =
    1.731e-07
     k from         k to     max|eta|      rms eta
          1           10     0.840022     0.545768
         11          100     0.993711     0.453166
        101         1000     0.915450     0.345177
       1001        10000     0.833881     0.274177
      10001       100000     0.785835     0.226352
     100001      1000000     0.762816     0.192411
    1000001      2001052     0.717359     0.178900
first ordinates:
   k          gamma_k            tau_k        eta_k
```

```
   1     14.134725142     14.521346953     0.386622
   2     21.022039639     20.655740356    -0.366299
   3     25.010857580     25.492675432     0.481818
   4     30.424876126     29.739411632    -0.685464
   5     32.935061588     33.624531889     0.689470
   6     37.586178159     37.257370087    -0.328808
   7     40.918719012     40.700551604    -0.218167
   8     43.327073281     43.994013953     0.666941
   9     48.005150881     47.165128427    -0.840022
  10     49.773832478     50.233653391     0.459821
  11     52.970321478     53.214422874     0.244101
  12     56.446247697     56.118921948    -0.327326
sigma = 2.00, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 8.706e-10, median = 2.685e-10
sigma = 1.25, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 1.128e-09, median = 3.693e-10
sigma = 0.75, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 2.629e-09, median = 5.179e-10
critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = 8.370e-10
total run time 30.6 s
```

B.2. **s01b_clock_zoom.**

```
zoom table near the top of the file:
        k            gamma_k              tau_k      eta_k   N*-N_RVM
  2001041     1132485.303923     1132485.011569  -0.292355  -0.563104
  2001042     1132485.662638     1132485.530752  -0.131886  -0.254025
  2001043     1132486.244063     1132486.049936  -0.194127  -0.373908
  2001044     1132486.392256     1132486.569120   0.176863   0.340656
  2001045     1132486.874881     1132487.088303   0.213422   0.411072
  2001046     1132487.571868     1132487.607487   0.035619   0.068605
  2001047     1132487.950694     1132488.126670   0.175977   0.338949
  2001048     1132488.905328     1132488.645854  -0.259474  -0.499774
  2001049     1132489.162780     1132489.165037   0.002257   0.004348
  2001050     1132489.585215     1132489.684221   0.099006   0.190696
  2001051     1132490.165305     1132490.203404   0.038099   0.073383
  2001052     1132490.658714     1132490.722588   0.063873   0.123026
total run time 0.6 s
```

B.3. **s02_transition.**

```
varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000) = 0.9619415807 -1.28e-10 i
L = 1/varpi_RVM = 1.0395641690
sigma' = 0.25: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0617
sigma' = 0.25: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = 0.0790
          |w| * t^sigma' in [1.5832, 1.5832];  |1/zeta_RVM - L| at t = 1e4: 1.425e-01
sigma' = 0.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0618
          |w| * t^sigma' in [2.5061, 2.5066];  |1/zeta_RVM - L| at t = 1e4: 2.543e-02
sigma' = 1.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.182, max t*err = 0.0620
          |w| * t^sigma' in [15.6619, 15.7496];  |1/zeta_RVM - L| at t = 1e4: 1.598e-05
total run time 11.5 s
```

B.4. **s03_exponential_sums.**

```
     x     max|S_RVM| max|S_RVM|/log    |S_zeta(N)| Landau -T/2pi*L/sqrt(x)   |M_H^RVM(x)|
   2.0         3.5745         0.2695       88300.77        -88341.59      1.722e-06
   3.0         3.9750         0.3759      114267.92       -114324.31      1.175e-06
   4.0         3.9523         0.3732       62435.29        -62466.94      8.004e-07
   6.0         4.5264         0.4547           0.26            -0.00      2.262e-06
  10.0         9.6989         1.1495           2.42            -0.00      4.031e-06
total run time 1.7 s
```

B.5. **s04_primes.**

```
x <= 100, all 2001052 zeros of each list, 3000 grid points:
  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): 0.4035
  pi_RVM(100) = 26.0782, pi(100) = 25, li(100) = 30.1261, J_RVM(100) = 29.5889, J(100) = 28.5333
  pi_RVM monotone on [2,100]: False (min increment on the grid -2.528e-01)
x <= 1000, smooth cut at K = 17061 zeros (tau_K = 15707.6), 40000 grid points:
  pi_RVM(1000) = 169.9750, pi(1000) = 168, li(1000) = 177.6097
cell masses c_n = pi_RVM(n) - pi_RVM(n-1):
```

```
    n        c_n    1/log n    prime
    2    0.52944    1.44270        1
    3    1.04188    0.91024        1
    4    0.53593    0.72135        0
    5    0.46739    0.62133        1
    6    0.36089    0.55811        0
    7    0.63254    0.51390        1
    8    0.62393    0.48090        0
    9   -0.37669    0.45512        0
   10    0.17151    0.43429        0
   11    0.69779    0.41703        1
   12    0.32703    0.40243        0
   13    0.36227    0.38987        1
   14    0.57662    0.37892        0
   15    0.19671    0.36927        0
   16   -0.05284    0.36067        0
   17    0.39958    0.35296        1
   18    0.76865    0.34598        0
   19    0.43500    0.33962        1
   20   -0.03342    0.33381        0
   21    0.03009    0.32846        0
   22    0.37466    0.32352        0
   23    0.40277    0.31893        1
   24    0.11768    0.31466        0
   25    0.01006    0.31067        0
   26    0.27350    0.30693        0
   27    0.55508    0.30341        0
   28    0.47627    0.30010        0
   29    0.12577    0.29697        1
   30   -0.08367    0.29401        0
   31    0.07329    0.29121        1
   32    0.38040    0.28854        0
   33    0.43927    0.28600        0
   34    0.34619    0.28358        0
   35    0.06419    0.28127        0
   36   -0.02475    0.27906        0
   37    0.15448    0.27694        1
   38    0.40992    0.27491        0
   39    0.49628    0.27296        0
   40    0.35226    0.27109        0
   41    0.14030    0.26928        1
sum_{n<=1000} c_n = 169.9750,  pi(1000) = 168
real RVM prime levels p_j^RVM (first crossing of level j) versus p_j:
    j      p_j^RVM      p_j       diff
    1       2.2139        2     0.2139
    2       3.1550        3     0.1550
    3       5.1887        5     0.1887
    4       7.1593        7     0.1593
    5      11.1406       11     0.1406
    6      14.0208       13     1.0208
    7      17.2552       17     0.2552
    8      19.1781       19     0.1781
    9      26.0536       23     3.0536
   10      28.0447       29    -0.9553
   11      33.0707       31     2.0707
   12      38.0931       37     1.0931
   13      40.2282       41    -0.7718
   14      44.1861       43     1.1861
   15      46.1221       47    -0.8779
   16      57.1779       53     4.1779
   17      59.1201       59     0.1201
   18      61.0809       61     0.0809
   19      67.0384       67     0.0384
   20      74.0931       71     3.0931
   21      76.1235       73     3.1235
```

```
   22      83.0822       79     4.0822
   23      86.0513       83     3.0513
   24      92.0825       89     3.0825
   25      94.0781       97    -2.9219
   26      96.1319      101    -4.8681
   27     103.2219      103     0.2219
   28     112.1163      107     5.1163
   29     115.0277      109     6.0277
   30     118.1431      113     5.1431
total run time 107.7 s
```

B.6. **s05_mertens.**

```
zeta_RVM'(rho_k) exact for k <= 1000; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = 1000: 0.9619388429 -1.41e-05 i
varpi_RVM = 0.9619415807; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = -0.0052 -0.0208 i
Riemann-Siegel zeta'(rho) versus mpmath at k = 1001..1050: max relative difference 1.19e-05
median |zeta'(rho)| = 7.8899;  spectral sums 2 sum|a|^2: zeta 0.029034, RVM 0.027203
M(10^6) = 212;  squarefree density Q(10^6)/10^6 = 0.607926 (6/pi^2 = 0.607927)
explicit formula with 2001052 zeros of zeta on [2,2000]: rms(M_zeta - M) = 0.0005 off integers (|x-n|>0.05)
target on 49896 grid points; at 100 integers <= 1e5: max |M_RVM(n) (all zeros) - target(n)| = 0.0191
ternary walk: max |M_h - target| on [1,1e6] = 2.6155, on [1e4,1e6] = 1.6539
densities of (-1, 0, +1):  mu   0.303857 0.392074 0.304069
                          mu_h 0.304003 0.392074 0.303923   (3/pi^2 = 0.303964, 1 - 6/pi^2 = 0.392073)
sign agreement mu_h = mu on squarefrees: 0.500344;  sum mu mu_h / Q = 0.000688
max |M(n)|/sqrt(n) on [10,1e6] = 0.8321;  max |M_RVM(n)|/sqrt(n) = 0.7541;  max |M_h(n)|/sqrt(n) = 0.6882

NUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)
   n   mu   mu^2     target M_h(n-1)   sign   mu_h    M_h      M
   1    1      1     0.2925        0     +1      1      1      1
   2   -1      1     0.4563        1     -1     -1      0      0
   3   -1      1    -0.6523        0     -1     -1     -1     -1
   4    0      0    -0.8720       -1      -      0     -1     -1
   5   -1      1    -1.7255       -1     -1     -1     -2     -2
   6    1      1    -1.3063       -2     +1      1     -1     -1
   7   -1      1    -1.5931       -1     -1     -1     -2     -2
   8    0      0    -2.4029       -2      -      0     -2     -2
   9    0      0    -1.3292       -2      -      0     -2     -2
  10    1      1    -1.3280       -2     +1      1     -1     -1
  11   -1      1    -2.3272       -1     -1     -1     -2     -2
  12    0      0    -2.2067       -2      -      0     -2     -2
  13   -1      1    -1.8468       -2     +1      1     -1     -3
  14    1      1    -1.9324       -1     -1     -1     -2     -2
  15    1      1    -1.5435       -2     +1      1     -1     -1
  16    0      0    -1.0414       -1      -      0     -1     -1
  17   -1      1    -1.6970       -1     -1     -1     -2     -2
  18    0      0    -3.0095       -2      -      0     -2     -2
  19   -1      1    -3.2869       -2     -1     -1     -3     -3
  20    0      0    -2.2604       -3      -      0     -3     -3
  21    1      1    -1.2515       -3     +1      1     -2     -2
  22    1      1    -1.1120       -2     +1      1     -1     -1
  23   -1      1    -1.3607       -1     -1     -1     -2     -2
  24    0      0    -1.3669       -2      -      0     -2     -2
  25    0      0    -1.3810       -2      -      0     -2     -2
  26    1      1    -1.9805       -2     +1      1     -1     -1
  27    0      0    -3.0141       -1      -      0     -1     -1
  28    0      0    -3.6155       -1      -      0     -1     -1
  29   -1      1    -3.2066       -1     -1     -1     -2     -2
  30   -1      1    -2.1599       -2     -1     -1     -3     -3
  31   -1      1    -1.3649       -3     +1      1     -2     -4
  32    0      0    -1.2973       -2      -      0     -2     -4
  33    1      1    -1.6435       -2     +1      1     -1     -3
  34    1      1    -1.7783       -1     -1     -1     -2     -2
  35    1      1    -1.4711       -2     +1      1     -1     -1
  36    0      0    -1.0723       -1      -      0     -1     -1
  37   -1      1    -1.0774       -1     -1     -1     -2     -2
```

```
  38    1      1    -1.5897       -2     +1      1     -1     -1
  39    1      1    -2.2227       -1     -1     -1     -2      0
  40    0      0    -2.4881       -2      -      0     -2      0

ENERGY E_N = (1/N) sum_{n<=N} W(n)^2
         N         E_N(M)     E_N(M_RVM)       E_N(M_h)
      1000        14.6400        18.6280        18.5960
     10000       163.6747       195.6965       196.0683
    100000      1622.3189      1308.3041      1308.8186
   1000000     14851.0857     12537.9008     12538.1174
total run time 148.2 s
```

B.7. **s06_prime_ratios.**

```
grid: 40000 points on [2, 1000]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)

ratios at selected x
       x           pi       pi_RVM           li    li/pi_RVM        pi/li    pi/pi_RVM
    10.0       4.0000       3.9876       6.1657     1.546218     0.648747     1.003105
    20.0       8.0000       7.6636       9.9052     1.292496     0.807657     1.043893
    50.0      15.0000      14.7686      18.4695     1.250592     0.812151     1.015670
   100.0      25.0000      26.0813      30.1265     1.155098     0.829835     0.958541
   200.0      46.0000      43.3656      50.1910     1.157393     0.916499     1.060749
   500.0      95.0000      93.7489     101.7975     1.085854     0.933225     1.013346
  1000.0     168.0000     169.9750     177.6097     1.044916     0.945895     0.988381

square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]
max |pi_RVM - li| / B = 2.7213   (at x = 9.73)
max |pi - li| / B     = 2.4490   (at x = 11.00)
max |pi_RVM - li| / B on [50, 1000] = 1.3455
max |pi_RVM - li| / B on [500, 1000] = 0.7216
total run time 1.2 s
```

B.8. **s07_product_identity.**

```
N = 2001052 ordinates; varpi_RVM = 0.9619415807
         K              P_K    P_K - varpi
        10     0.9625640008      6.224e-04
       100     0.9624787401      5.372e-04
      1000     0.9622891707      3.476e-04
     10000     0.9619513787      9.798e-06
    100000     0.9619301865     -1.139e-05
   1000000     0.9619418189      2.382e-07
   2001052     0.9619415390     -4.166e-08
K >=   10000: min P_K = 0.9618023968, max P_K = 0.9620641927, mean = 0.9619418623
K >=  100000: min P_K = 0.9619210208, max P_K = 0.9619605076, mean = 0.9619417591
K >= 1000000: min P_K = 0.9619389488, max P_K = 0.9619439825, mean = 0.9619416633

flow along the lambda-family
Upsilon(0.00) = 0.0436949578
Upsilon(0.25) = 0.0412445375
Upsilon(0.50) = 0.0387987180
Upsilon(0.75) = 0.0363572218
Upsilon(1.00) = 0.0339197739
sum_k 2 eta_k^2 / gamma_k^2 = 0.0098210688
integral_0^1 Upsilon = sum_k log((1/4+tau_k^2)/(1/4+gamma_k^2)) = 0.0388016004
log L = log(1/varpi_RVM)                                       = 0.0388015571
total run time 1.5 s
```

B.9. **s08_transport_moments.**

```
s08: transport tau -> gamma on the first rung
entries in zeross.bin / zLamb.bin: 2001052 / 2001052;  used K = 20000
gamma_1 = 14.134725142, tau_1 = 14.521346953;  gamma_K = 18046.464296, tau_K = 18046.544665
psi: normalized bump on [0.1, 0.62] (log 2 = 0.693147), 4001 nodes
max|eta_k| = 0.993711 at k = 34;  rms eta = 0.266395
  max_(k>=  100) |c_psi(tau_k)| = 6.880e-05
```

```
  max_(k>=  300) |c_psi(tau_k)| = 7.263e-07
  max_(k>= 1000) |c_psi(tau_k)| = 2.527e-10
  max_(k>= 3000) |c_psi(tau_k)| = 2.074e-13
  max_(k>=10000) |c_psi(tau_k)| = 2.074e-13
  max_(k>=19999) |c_psi(tau_k)| = 1.616e-15

W_psi                          = +0.133643530916
sum_k c_psi(tau_k)             = +0.176106813076
sum_k c_psi(gamma_k)           = +0.133643530929
R_psi(1) = W - sum c(tau)      = -0.042463282161   (clock defect)
S_psi(0) = sum[c(gam)-c(tau)]  = -0.042463282147   (transport)
R_psi(0) = R_psi(1) - S_psi(0) = -1.376e-11

 j        m_j(psi)        (-1)^j m_j/j!      partial sum      partial - R_psi(1)
 1  +4.0395231784e-02  -4.0395231784e-02  -0.040395231784  +2.068e-03
 2  -4.8097910277e-03  -2.4048955139e-03  -0.042800127298  -3.368e-04
 3  -2.0431952768e-03  +3.4053254613e-04  -0.042459594751  +3.687e-06
 4  -5.5184329118e-05  -2.2993470466e-06  -0.042461894098  +1.388e-06
 5  +1.6734348340e-04  -1.3945290283e-06  -0.042463288628  -6.467e-09
 6  +5.5720624242e-06  +7.7389755891e-09  -0.042463280889  +1.272e-09
 7  +7.2607702259e-06  -1.4406290131e-09  -0.042463282329  -1.686e-10
 8  +6.4039357525e-06  +1.5882777164e-10  -0.042463282170  -9.773e-12
 9  -8.6516463634e-06  +2.3841618065e-11  -0.042463282147  +1.407e-11
10  -2.6791658297e-06  -7.3830628022e-13  -0.042463282147  +1.333e-11
11  +2.8798049874e-06  -7.2145186673e-14  -0.042463282147  +1.326e-11
12  +8.0540801176e-07  +1.6814307338e-15  -0.042463282147  +1.326e-11

 cumulative over k (first K0 ordinates):  K0   sum[c(gam)-c(tau)]   W - sum c(tau)   -m_1 part   +m_2/2 part
      1   -0.0208609181   +0.0155048033   -0.0168182977   -0.0039851021
      2   -0.0179065615   +0.0336283174   -0.0153170893   -0.0024894395
      3   -0.0230428015   -0.0613123463   -0.0185197872   -0.0044492860
      5   -0.0438112092   -0.0723587090   -0.0422769516   -0.0020012704
     10   -0.0464863677   -0.0403092255   -0.0440974925   -0.0029001855
     20   -0.0406060545   -0.0470689184   -0.0387914922   -0.0021299305
     50   -0.0422141654   -0.0432549118   -0.0401314609   -0.0024151466
    100   -0.0424897738   -0.0426540294   -0.0404216192   -0.0024058875
    200   -0.0424647471   -0.0424770847   -0.0403966024   -0.0024050814
    400   -0.0424633643   -0.0424636133   -0.0403953106   -0.0024049002

 lambda     S_psi(lambda)     R_psi(lambda) = R_psi(1) - S_psi(lambda)
-1.00   -0.087766836871   +0.045303554710
-0.75   -0.076275893121   +0.033812610960
-0.50   -0.064876726912   +0.022413444752
-0.25   -0.053596431977   +0.011133149816
+0.00   -0.042463282147   -0.000000000014
+0.25   -0.031506572650   -0.010956709511
+0.50   -0.020756460380   -0.021706821780
+0.75   -0.010243803437   -0.032219478724
+1.00   +0.000000000000   -0.042463282161
+1.25   +0.009943173538   -0.052406455698
+1.50   +0.019553725448   -0.062017007609
+1.75   +0.028799612928   -0.071262895088
+2.00   +0.037648908219   -0.080112190380
```

B.10. **s09_clock_defect_split.**

```
s09: Poisson split of the clock defect (psi of s08 on [0.1,0.62])
t_0 (N_RVM(t_0)=0)                = 9.676906787166   (2 pi = 6.283185, tau_1 = 14.521347)
W_psi                             = +0.133643530916
I_0  (m = 0, smooth clock density) = +0.063533498702
sum_k c_psi(tau_k)                = +0.176106813076
R_psi(1)                          = -0.042463282161
smooth mismatch  W_psi - I_0      = +0.070110032214
oscillating part -2 sum_(m>=1)(-1)^m I_m = -0.112573314375
stationary heights 2 pi e^u, u in supp psi: [6.9440, 11.6800];  part above t_0: u > 0.431865
```

B.11. `s10_clock_to_zeros.`

```
s10: from the clock to the ordinates through the rung equations (continuation in height)
N = 13, U = log(13+1/2) = 2.602690; free masses n = 2..11; start gamma_k = tau_k
band omega <= 18:  4 free ordinates, 361 tests, |residual| 8.136e-01 -> 3.892e-04 in 37 iterations
band omega <= 22:  5 free ordinates, 441 tests, |residual| 1.824e-02 -> 1.520e-03 in 20 iterations
band omega <= 26:  6 free ordinates, 521 tests, |residual| 1.985e-02 -> 8.002e-04 in 25 iterations
band omega <= 30:  7 free ordinates, 601 tests, |residual| 2.363e-02 -> 2.506e-03 in 21 iterations
band omega <= 34:  8 free ordinates, 681 tests, |residual| 4.635e-02 -> 2.827e-03 in 47 iterations
band omega <= 38:  9 free ordinates, 761 tests, |residual| 3.868e-02 -> 1.683e-03 in 11 iterations
band omega <= 42: 11 free ordinates, 841 tests, |residual| 1.881e-01 -> 1.595e-03 in 11 iterations
band omega <= 46: 12 free ordinates, 921 tests, |residual| 1.878e-02 -> 4.591e-03 in 12 iterations
band omega <= 50: 14 free ordinates, 1001 tests, |residual| 5.189e-02 -> 2.885e-03 in 22 iterations

  k      tau_k (start)    solved ordinate    gamma_k (zeross.bin)   solved - gamma_k    tau_k - gamma_k
  1     14.521346953     14.134660056     14.134725142   -6.509e-05   +0.386622
  2     20.655740356     21.022091631     21.022039639   +5.199e-05   -0.366299
  3     25.492675432     25.010827129     25.010857580   -3.045e-05   +0.481818
  4     29.739411632     30.424917668     30.424876126   +4.154e-05   -0.685464
  5     33.624531889     32.935072953     32.935061588   +1.137e-05   +0.689470
  6     37.257370087     37.586146548     37.586178159   -3.161e-05   -0.328808
  7     40.700551604     40.918919600     40.918719012   +2.006e-04   -0.218167
  8     43.994013953     43.327322117     43.327073281   +2.488e-04   +0.666941
  9     47.165128427     48.004716753     48.005150881   -4.341e-04   -0.840022
 10     50.233653391     49.773657726     49.773832478   -1.748e-04   +0.459821
 11     53.214422874     52.972535790     52.970321478   +2.214e-03   +0.244101
 12     56.118921948     56.548052093     56.446247697   +1.018e-01   -0.327326
 13     58.956266008     58.576523603     59.347044003   -7.705e-01   -0.390778
 14     61.733838843     62.561880181     60.831778525   +1.730e+00   +0.902060

  n     solved mass a_n     Lambda(n)/(2 sqrt n)     difference     w(log n)/max w
  2   +0.245076213   +0.245064536   +1.168e-05   7.562e-01
  3   +0.317145240   +0.317142050   +3.190e-06   9.754e-01
  4   +0.173268937   +0.173286795   -1.786e-05   9.957e-01
  5   +0.359884977   +0.359881258   +3.719e-06   9.423e-01
  6   -0.000004422   +0.000000000   -4.422e-06   8.474e-01
  7   +0.367785301   +0.367742452   +4.285e-05   7.225e-01
  8   +0.122615622   +0.122532268   +8.335e-05   5.733e-01
  9   +0.183143344   +0.183102048   +4.130e-05   4.062e-01
 10   -0.000286111   +0.000000000   -2.861e-04   2.344e-01
 11   +0.362276961   +0.361496314   +7.806e-04   8.642e-02
```

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Appendix C. Python programs of Section 17

The programs of `code/routeB/` produce the numbers of Section 17.7 and Figures 1 and 2. They use only the clock, the primes and the Gonek lists; `zeros.py` computes the zeta ordinates used in the diagnostic columns with `mpmath`. The exploratory scripts (`staged.py`, `lsq*.py`, `basin*.py`, `coupling.py`, `horizon.py`, `certG.py`, `cert2-6.py`) are distributed in the same directory with their logs and are described in `RESULTS.txt`.

C.1. `core.py`.

```
import numpy as np
from scipy.special import lambertw
from scipy.integrate import quad
SQ2PI=np.sqrt(2*np.pi)
def clock(K):
    k=np.arange(1,K+1); a=(k-11/8)/np.e
    return np.real(2*np.pi*(k-11/8)/lambertw(a))
def vonmangoldt(Nmax):
    L=np.zeros(Nmax+1)
    for p in range(2,Nmax+1):
        if all(p%q for q in range(2,int(p**0.5)+1)):
            q=p
            while q<=Nmax: L[q]=np.log(p); q*=p
    return L
LAM=vonmangoldt(200000)
def kfun(u): return np.cosh(u/2)-np.exp(-u/2)/(2*(1-np.exp(-2*u)))
def gauss(c,s,u): return np.exp(-(u-c)**2/(2*s*s))
def Fphi(c,s,kap):  # int_R gauss cos(kap u) du
    return s*SQ2PI*np.exp(-0.5*(s*kap)**2)*np.cos(c*kap)
def dFphi(c,s,kap):
    return s*SQ2PI*np.exp(-0.5*(s*kap)**2)*(-(s*s)*kap*np.cos(c*kap)-c*np.sin(c*kap))
_kcache={}
def kpair(c,s):
    key=(round(c,12),round(s,12))
    if key not in _kcache:
        a=max(c-9*s,1e-9); b=c+9*s
        _kcache[key]=quad(lambda u:kfun(u)*gauss(c,s,u),a,b,limit=400,epsabs=1e-13,epsrel=1e-12)[0]
    return _kcache[key]
def masspair(c,s):
    nmax=int(np.exp(c+9*s))+1
    n=np.arange(2,nmax+1); a=LAM[n]/(2*np.sqrt(n))
    return np.sum(a*gauss(c,s,np.log(n)))
def residual(tests,kap):
    """tests: list of (c,s); kap: full list (must extend far enough)"""
    out=[]
    for c,s in tests:
        out.append(np.sum(Fphi(c,s,kap))+masspair(c,s)-kpair(c,s))
    return np.array(out)
def jac(tests,kap,K):
    return np.array([dFphi(c,s,kap[:K]) for c,s in tests])
```

C.2. `zeros.py`.

```
import mpmath as mp, numpy as np, sys
mp.mp.dps=20
N=int(sys.argv[1])
g=[float(mp.zetazero(n).imag) for n in range(1,N+1)]
np.save('gamma.npy',np.array(g)); print(len(g), g[-1])
```

C.3. `gonek_fast.py`.

```
import numpy as np
from scipy.optimize import brentq
from core import LAM
def theta(t): t=np.asarray(t,float); return t/2*np.log(t/(2*np.pi))-t/2-np.pi/8+1/(48*t)+7/(5760*t**3)
def make_alpha(X):
```

```
    n=np.arange(2,int(X*X)+1); L=LAM[n]; w=np.where(n<=X,1.0,2-np.log(n)/np.log(X)); c=L*w/(np.sqrt(n)*np.log(
     n)); ln=np.log(n)
    return lambda t: -(np.sin(np.multiply.outer(np.atleast_1d(t),ln))@c)
def gonek_list(X,tmax,t0=10.0,h=0.005):
    al=make_alpha(X); ph=lambda t: theta(t)+al(t)
    ts=np.arange(t0,tmax,h); v=(ph(ts)-np.pi/2)/np.pi; j=np.floor(v); idx=np.where(np.diff(j)!=0)[0]; roots=[]
    for i in idx:
        lev=max(j[i],j[i+1]); f=lambda t:(ph(np.array([t]))[0]-np.pi/2)/np.pi-lev
        a,b=ts[i],ts[i+1]
        if f(a)*f(b)<0: roots.append(brentq(f,a,b,xtol=1e-13))
    return np.array(roots)
if __name__=='__main__':
    import sys
    X=float(sys.argv[1]); r=gonek_list(X,float(sys.argv[2])); np.save(f'gonekF_X{int(X)}.npy',r); print(len(r),
     r[:3],r[-1])
```

C.4. **horizonG.py.**

```
import numpy as np, json, sys
from scipy.optimize import least_squares
from core import *
g=np.load('gamma.npy'); tau=clock(60000)
q,s0,sp,M,INT,LAG=0.85,0.08,1.0,int(sys.argv[1]),float(sys.argv[2]),int(sys.argv[3])
r=np.load('gonekF_X8.npy'); kap=tau.copy(); kap[:len(r)]=r; tests=[]; level=[]; log=[]
for m in range(1,M+1):
    s=s0*q**(m-1); T=4.0/s; U=np.log(max(T,7)/(2*np.pi))+2.0
    new=[(float(c),s) for c in np.arange(9*s,U+1e-9,sp*s)]; tests+=new; level+=[m]*len(new)
    cs=np.array([t[0] for t in tests]); ss=np.array([t[1] for t in tests])
    ki=int(np.sum(tau<=INT/s))
    const=np.array([masspair(c,s_)-kpair(c,s_) for c,s_ in tests])
    tail=np.array([np.sum(Fphi(c,s_,kap[ki:])) for c,s_ in tests])  # edges & beyond: current values (clock)
    F=lambda y: np.sum(Fphi(cs[:,None],ss[:,None],y[None,:]),axis=1)+tail+const
    Jf=lambda y: dFphi(cs[:,None],ss[:,None],y[None,:])
    y0=kap[:ki].copy()
    sol=least_squares(F,y0,jac=Jf,method='trf',bounds=(tau[:ki]-3,tau[:ki]+3),xtol=1e-15,ftol=1e-15,gtol=1e-15,
     max_nfev=3000)
    kap=kap.copy(); kap[:ki]=np.sort(sol.x)
    Fall=F(kap[:ki]); lv=np.array(level)
    lag=np.max(np.abs(Fall[lv<=m-LAG])) if np.any(lv<=m-LAG) else float('nan')
    nk=min(ki,len(g)); dev=np.abs(kap[:nk]-g[:nk])
    front=lambda tol: (int(np.where(dev>tol)[0][0]) if np.any(dev>tol) else nk)
    rec=dict(m=m,s=s,T=T,interior=ki,tests=len(tests),res_all=float(np.max(np.abs(Fall))),res_lag=float(lag),
             dev5=float(dev[:5].max()),dev10=float(dev[:10].max()),dev20=float(dev[:min(20,nk)].max()),
             f6=front(1e-6),f4=front(1e-4),f2=front(1e-2),dev_tau=float(np.max(np.abs(kap[:ki]-tau[:ki]))))
    log.append(rec)
    print(f"m={m:2d} T={T:5.0f} interior={ki:3d} tests={len(tests):4d} res_all={rec['res_all']:.1e} res_lag{
     LAG}={rec['res_lag']:.1e} |k-g| k<=5:{rec['dev5']:.1e} k<=10:{rec['dev10']:.1e} k<=20:{rec['dev20']:.1e}
     front(1e-6/1e-4/1e-2)={rec['f6']}/{rec['f4']}/{rec['f2']} |k-tau|max={rec['dev_tau']:.2f}",flush=True)
json.dump(log,open(f'horizonG_M{M}_I{INT}_L{LAG}.json','w'),indent=1); np.save(f'horizonG_kap_M{M}_I{INT}.npy',
     kap)
```

C.5. **cert7.py.**

```
# Computer-assisted (gamma-free) existence certificate for finite square sections, at the
# receding-horizon iterate kappa^H (horizonG, Gonek X=8 initialization, M=16).
# Stage m: tests = levels <= m (spacing 1.0 s), unknowns = interior ordinates (tau_k <= 2.4/s_m),
# other ordinates frozen at kappa^H. Square subset of tests chosen by pivoted QR on J^T.
# F_j separable in kappa_k  =>  J(x)-J(x0) has columns f_j''(x0_k) delta_k + R_jk, |R_jk|<=A3_jk delta_k^2/2.
# x0 = kappa^H polished by 8 Newton steps on the square system (gamma-free).
# Banach for Psi(x)=x-Q F_sel(x) on {||x-x0||_inf<=r}:  k(r)=||QB||r + 0.5*|| |Q|A3 || r^2 < 1,  eta<=(1-k(r))
     r.
import numpy as np, json
from scipy.linalg import qr
from core import *
g=np.load('gamma.npy'); tau=clock(60000); kH=np.load('horizonG_kap_M16_I2.4.npy')
q,s0,sp,INT=0.85,0.08,1.0,2.4
```

```
def d2(c,s,k):
    a=s*s; e=s*SQ2PI*np.exp(-a*k*k/2)
    return e*(((a*k)**2-a-c*c)*np.cos(c*k)+2*a*k*c*np.sin(c*k))
def A3b(c,s,k,r):
    a=s*s; kl=np.maximum(np.abs(k)-r,0); ku=np.abs(k)+r; w=a*ku+c
    return s*SQ2PI*np.exp(-a*kl*kl/2)*(w**3+3*a*w)
tests=[]; out=[]
for m in range(1,17):
    s=s0*q**(m-1); T=4.0/s; U=np.log(max(T,7)/(2*np.pi))+2.0
    tests+=[(float(c),s) for c in np.arange(9*s,U+1e-9,sp*s)]
    n=int(np.sum(tau<=INT/s))
    cs=np.array([t[0] for t in tests]); ss=np.array([t[1] for t in tests])
    x0=kH[:n]
    J=dFphi(cs[:,None],ss[:,None],x0[None,:])
    _,_,piv=qr(J.T,pivoting=True); sel=np.sort(piv[:n])
    kk=kH.copy(); tsel=[tests[i] for i in sel]
    try:
        for it in range(8):                        # gamma-free Newton polishing on the square system
            Jc=dFphi(cs[sel][:,None],ss[sel][:,None],kk[None,:n]); kk[:n]=kk[:n]-np.linalg.solve(Jc,residual(
     tsel,kk))
    except np.linalg.LinAlgError:
        print(f"m={m:2d} Newton polishing failed (singular square Jacobian)"); break
    x0=kk[:n]; J=dFphi(cs[:,None],ss[:,None],x0[None,:])
    Q=np.linalg.inv(J[sel]); F=residual(tsel,kk)
    eta=np.max(np.abs(Q@F))
    QB=np.max(np.sum(np.abs(Q@d2(cs[sel][:,None],ss[sel][:,None],x0[None,:])),axis=1))
    best=None
    for r in np.geomspace(max(eta,1e-15)*1.0001,1.0,400):
        C3=np.max(np.sum(np.abs(Q)@A3b(cs[sel][:,None],ss[sel][:,None],x0[None,:],r),axis=1)) if best is None
     or r<10*best[0] else C3
        kr=QB*r+0.5*C3*r*r
        if kr<1 and eta<=(1-kr)*r: best=(r,kr); break
    margin=max((1-(QB*r+0.5*np.max(np.sum(np.abs(Q)@A3b(cs[sel][:,None],ss[sel][:,None],x0[None,:],r),axis=1))
     *r*r))*r for r in np.geomspace(1e-12,1e-1,60))
    rho_all=float(np.max(np.abs(residual(tests,kk))))
    Linf=np.max(np.sum(np.abs(Q),axis=1))
    nk=min(n,len(g)); devg=float(np.max(np.abs(x0[:nk]-g[:nk]))); devt=float(np.max(np.abs(x0-tau[:n])))
    rec=dict(m=m,T=T,n=n,Linf=float(Linf),eta=float(eta),QB=float(QB),certified=best is not None,margin=float(
     margin),rho_all_levels=rho_all,
             r=(float(best[0]) if best else None),k_r=(float(best[1]) if best else None),dev_gamma=devg,
     dev_tau=devt)
    out.append(rec)
    print(f"m={m:2d} T={T:5.0f} n={n:3d} Linf={Linf:.1e} eta={eta:.1e} ||QB||={QB:.1e} "
          + (f"CERTIFIED r={best[0]:.1e} k(r)={best[1]:.2f}" if best else "not certified")
          + f" margin={margin:.1e} rho(all tests<=m)={rho_all:.1e} | max|x0-tau|={devt:.2f} max|x0-gamma|={
     devg:.1e}",flush=True)
json.dump(out,open('cert7.json','w'),indent=1)
```

C.6. **entry.py.**

```
# Block-entry test (item (i) of the review): from the certified stage-m solution x^(m)
# (square section polished as in cert7), put the NEW band (tau_k in (2.4/s_m, 2.4/s_{m+1}]) and
# everything beyond at Gonek X=8 values, and ask whether a Banach certificate holds for
# (a) the band-only square system (new band unknowns, tests from level m+1),
# (b) the full square system of stage m+1, started there (no polishing).
import numpy as np, json
from scipy.linalg import qr
from core import *
exec(open('cert5.py').read().split("tests=[]; out=[]")[0].split("import numpy as np, json")[1])  # d2, A3b,
     data
rG=np.load('gonekF_X8.npy'); kG=tau.copy(); kG[:len(rG)]=rG
def certify(Q,F,QBn,A3fun):
    eta=np.max(np.abs(Q@F))
    for r in np.geomspace(max(eta,1e-15)*1.0001,1.0,300):
        kr=QBn*r+0.5*np.max(np.sum(np.abs(Q)@A3fun(r),axis=1))*r*r
```

```
        if kr<1 and eta<=(1-kr)*r: return eta,r
    return eta,None
def margin(QBn,Q,A3fun): return max((1-(QBn*r+0.5*np.max(np.sum(np.abs(Q)@A3fun(r),axis=1))*r*r))*r for r in
     np.geomspace(1e-9,1e-1,50))
tests=[]; lev=[]; out=[]; prev=None
for m in range(1,16):
    s=s0*q**(m-1); T=4.0/s; U=np.log(max(T,7)/(2*np.pi))+2.0
    new=[(float(c),s) for c in np.arange(9*s,U+1e-9,sp*s)]; tests+=new; lev+=[m]*len(new)
    n=int(np.sum(tau<=INT/s)); cs=np.array([t[0] for t in tests]); ss=np.array([t[1] for t in tests]); lv=np.
     array(lev)
    if prev is not None:
        x0=kG.copy(); x0[:prev[1]]=prev[0][:prev[1]]          # old interior: certified stage-(m-1) solution;
     rest Gonek
        band=np.arange(prev[1],n); idx=np.where(lv==m)[0]
        Jb=dFphi(cs[idx][:,None],ss[idx][:,None],x0[None,band]); _,_,pv=qr(Jb.T,pivoting=True); sb=idx[np.sort
     (pv[:len(band)])]
        Qb=np.linalg.inv(dFphi(cs[sb][:,None],ss[sb][:,None],x0[None,band])); Fb=residual([tests[i] for i in
     sb],x0)
        QBb=np.max(np.sum(np.abs(Qb@d2(cs[sb][:,None],ss[sb][:,None],x0[None,band])),axis=1))
        A3f=lambda r: A3b(cs[sb][:,None],ss[sb][:,None],x0[None,band],r)
        eb,rb=certify(Qb,Fb,QBb,A3f); mb=margin(QBb,Qb,A3f)
        J=dFphi(cs[:,None],ss[:,None],x0[None,:n]); _,_,pv=qr(J.T,pivoting=True); sf=np.sort(pv[:n])
        Qf=np.linalg.inv(J[sf]); Ff=residual([tests[i] for i in sf],x0)
        QBf=np.max(np.sum(np.abs(Qf@d2(cs[sf][:,None],ss[sf][:,None],x0[None,:n])),axis=1))
        A3g=lambda r: A3b(cs[sf][:,None],ss[sf][:,None],x0[None,:n],r)
        ef,rf=certify(Qf,Ff,QBf,A3g); mf=margin(QBf,Qf,A3g)
        bandG=float(np.max(np.abs(x0[band]-g[band]))) if band[-1]<len(g) else float('nan')
        rec=dict(m=m,T=T,band=len(band),eta_band=float(eb),margin_band=float(mb),cert_band=rb is not None,
                 eta_full=float(ef),margin_full=float(mf),cert_full=rf is not None,gonek_band_err=bandG)
        out.append(rec)
        print(f"m={m:2d} T={T:5.0f} band={len(band):2d} | band-only: eta={eb:.1e} margin={mb:.1e} {'CERT' if
     rb else 'no'} "
              f"| full: eta={ef:.1e} margin={mf:.1e} {'CERT' if rf else 'no'} | Gonek band err={bandG:.1e}",
     flush=True)
    kk=kG.copy() if prev is None else x0.copy(); tsel=None
    J=dFphi(cs[:,None],ss[:,None],kk[None,:n]); _,_,pv=qr(J.T,pivoting=True); sel=np.sort(pv[:n]); tsel=[tests
     [i] for i in sel]
    try:
        for it in range(10):
            Jc=dFphi(cs[sel][:,None],ss[sel][:,None],kk[None,:n]); kk[:n]-=np.linalg.solve(Jc,residual(tsel,kk
     ))
    except np.linalg.LinAlgError: print("singular at",m); break
    prev=(kk.copy(),n)
json.dump(out,open('entry.json','w'),indent=1)
```

C.7. **laplace_check.py.**

```
# Numerical check of the identity  (1/2) G'/G(s) = L[K](s)  with
# K = (1/2)e^{u/2} 1_+ - (1/2) g 1_+ - (1/4) fp(1_+/u) + c0 delta_0,
# g(u)=e^{-u/2}/(e^{2u}-1)-1/(2u),  c0=-(1/4)log(2 pi e^{gamma_E}),  z=s-1/2,
# L[fp(1_+/u)](z) = -log z - gamma_E.   Also checks k(u)=cosh(u/2)-e^{-u/2}/(2(1-e^{-2u})) = K|_(0,inf).
from mpmath import mp, mpf, exp, expm1, log, pi, euler, quad, digamma, cosh, inf, mpc
mp.dps=50
def g(u):
    if u<mpf('1e-12'): return -mpf(3)/4       # g(0+)=-3/4; avoids cancellation
    return exp(-u/2)/expm1(2*u)-1/(2*u)
def k_v18(u): return (exp(u/2)-exp(-u/2)/(exp(2*u)-1))/2
def k_mine(u): return cosh(u/2)-exp(-u/2)/(2*(1-exp(-2*u)))
def K_restr(u): return exp(u/2)/2-g(u)/2-1/(4*u)
for u in [mpf('0.1'),mpf(1),mpf('2.5')]:
    print('u=%s  k_v18=%s  k_mine=%s  K|=%s'%(u,k_v18(u),k_mine(u),K_restr(u)))
def halfGG(s): return (1/s+1/(s-1)-log(pi)/2+digamma(s/2)/2)/2
def LK(s):
    z=s-mpf(1)/2
    c0=-(euler+log(2)+log(pi))/4
```

```
    I1=1/(2*(z-mpf(1)/2))                    # L[(1/2)e^{u/2}]
    I2=-quad(lambda u: exp(-z*u)*g(u),[0,1,inf])/2
    I3=-(-log(z)-euler)/4
    return I1+I2+I3+c0
for s in [mpf(2),mpf('1.3'),mpc(2,5),mpc('1.5',-3)]:
    print('s=%s  (1/2)G\'/G=%s  L[K]=%s'%(s,halfGG(s),LK(s)))
```

C.8. **figs.py.**

```
# Figures for lemas_B_v4 / v19: (1) certificate scaling, (2) distance to gamma of clock and Gonek lists.
import numpy as np, json, matplotlib
matplotlib.use('Agg'); import matplotlib.pyplot as plt
from core import clock
g=np.load('gamma.npy'); K=400; tau=clock(2000)
def align(r,K):
    best=None
    for off in range(0,6):
        a=r[off:off+K]
        if len(a)<K: continue
        d=np.abs(a-g[:K]).mean()
        if best is None or d<best[0]: best=(d,off,a)
    return best[2]
sp=2*np.pi/np.log(g[:K]/(2*np.pi))
fig,ax=plt.subplots(figsize=(7.2,4.2))
ax.plot(np.arange(1,K+1),np.abs(tau[:K]-g[:K])/sp,'.',ms=3,color='0.3',label='midpoint clock $\\tau_k$')
rF=np.load('gonekF_X8.npy'); a=align(rF,K)
ax.plot(np.arange(1,K+1),np.abs(a-g[:K])/sp,'.',ms=3,color='C3',label='Gonek list, $X=8$')
K2=160; sp2=sp[:K2]
for X,c in [(3,'C0'),(20,'C2')]:
    r=np.load(f'gonek_X{X}.npy'); a=align(r,K2)
    ax.plot(np.arange(1,K2+1),np.abs(a-g[:K2])/sp2,'.',ms=3,color=c,label=f'Gonek list, $X={X}$')
ax.axhline(0.25,ls='--',color='k',lw=0.8); ax.text(5,0.26,'$0.25$ spacings',fontsize=8)
ax.set_xlabel('$k$'); ax.set_ylabel('$|\\kappa_k-\\gamma_k|/\\mathrm{spacing}$'); ax.set_ylim(0,0.75)
ax.legend(fontsize=8,loc='upper right'); ax.set_title('Distance to the zeta ordinates, in local mean spacings
    $2\\pi/\\log(\\gamma_k/2\\pi)$',fontsize=9)
fig.tight_layout(); fig.savefig('fig_gonek_vs_clock.png',dpi=160)
d=json.load(open('cert7.json'))
T=np.array([x['T'] for x in d]); L=np.array([x['Linf'] for x in d]); QB=np.array([x['QB'] for x in d]); mg=np.
    array([x['margin'] for x in d])
fig,ax=plt.subplots(figsize=(7.2,4.2))
for y,lab,c in [(L,'$L_m=\\|Q_m\\|_{\\infty\\to\\infty}$','C0'),(QB,'$\\|Q_mB_m\\|_{\\infty\\to\\infty}$','C1'
    ),(1/mg,'1 / (Banach margin)','C3')]:
    p=np.polyfit(np.log(T[3:]),np.log(y[3:]),1); ax.loglog(T,y,'o-',ms=4,color=c,label=lab+f'  (slope {p[0]:.2
    f})')
e=json.load(open('entry.json')); Te=np.array([x['T'] for x in e]); eb=np.array([x['eta_band'] for x in e]); mb
    =np.array([x['margin_band'] for x in e])
ax.loglog(Te,1/mb,'s--',ms=4,color='C2',label='1 / (band-only margin)')
ax.loglog(Te,1/eb,'^--',ms=4,color='C4',label='1 / (Gonek entry residual $\\eta$, band)')
ax.set_xlabel('height $T_m$'); ax.legend(fontsize=7.5); ax.set_title('Certificate constants of the square
    finite sections (floating point)',fontsize=9)
fig.tight_layout(); fig.savefig('fig_cert_scaling.png',dpi=160)
print('ok')
```

Appendix D. Console output of the programs of Section 17

D.1. **RESULTS.txt.**

```
RVM rung system - staged certificate numerics (2026-10-10)
Tests: Gaussians phi(u)=exp(-(u-c)^2/(2 s^2)), c>=9s; stage m: s_m=0.08*0.85^(m-1), T_m=4/s_m,
centres spaced 1.0*s_m in [9 s_m, log(max(T_m,7)/2pi)+2]. Interior unknowns: tau_k <= 2.4/s_m.
Masses a_n = Lambda(n)/(2 sqrt n); k(u)=cosh(u/2)-e^{-u/2}/(2(1-e^{-2u})).
Gonek list X=8: roots of (theta(t)+alpha_X(t)-pi/2)/pi in Z (gonek_fast.py).
gamma (400 zeros, mpmath) is used ONLY for diagnostics (columns 'gamma'), never in the constructions.

FILES
 core.py          residual, Jacobian, clock, k(u), mass pairing
```

```
 validate.py      R(gamma)~1e-16 normalisation check            -> out_validate.txt
 gonek*.py        Gonek lists, distance to gamma                 -> out_gonek.txt, gonek*.npy
 horizon.py       receding horizon from the clock (fails)        -> horizon_M16_I2.4_L4.json
 horizonG.py      receding horizon from Gonek X=8 (works)        -> horizonG_M16_I2.4_L4.json,
    horizonG_kap_M16_I2.4.npy
 certG.py         l-inf Kantorovich at Gonek list (fails)         -> out_certG.txt
 cert2/3/4.py     l2 / Gauss-Newton / diagonal l-inf constants at Gonek list -> out_cert2/3/4.txt
 cert5/6/7.py     Banach certificates of square finite sections (cert7 = final) -> out_cert5/6/7.txt, cert*.
    json
 entry.py         block-entry test (item (i))                    -> out_entry.txt, entry.json
 staged.py, lsq*.py, floor.py, basin*.py, reg.py, coupling.py: earlier exploratory runs (logs *.json)

horizon_M16_I2.4_L4.json
 m= 1 T=   50 n=  4 res_all=1.4e-02 res_lag4=nan max|k-g|(k<=20)=8.1e-02 front(1e-2)=3 max|k-tau|=0.77
 m= 2 T=   59 n=  5 res_all=1.4e-02 res_lag4=nan max|k-g|(k<=20)=4.6e-03 front(1e-2)=5 max|k-tau|=0.69
 m= 3 T=   69 n=  7 res_all=2.1e-02 res_lag4=nan max|k-g|(k<=20)=1.9e-02 front(1e-2)=5 max|k-tau|=0.69
 m= 4 T=   81 n=  9 res_all=1.9e-02 res_lag4=nan max|k-g|(k<=20)=1.2e-02 front(1e-2)=7 max|k-tau|=0.83
 m= 5 T=   96 n= 12 res_all=1.9e-02 res_lag4=2.9e-05 max|k-g|(k<=20)=3.2e-02 front(1e-2)=10 max|k-tau|=0.84
 m= 6 T=  113 n= 16 res_all=1.3e-02 res_lag4=1.7e-04 max|k-g|(k<=20)=5.0e-02 front(1e-2)=11 max|k-tau|=0.88
 m= 7 T=  133 n= 20 res_all=1.7e-02 res_lag4=6.9e-05 max|k-g|(k<=20)=1.1e-01 front(1e-2)=17 max|k-tau|=0.90
 m= 8 T=  156 n= 26 res_all=1.3e-02 res_lag4=8.6e-05 max|k-g|(k<=20)=3.5e-03 front(1e-2)=23 max|k-tau|=0.90
 m= 9 T=  183 n= 34 res_all=1.2e-02 res_lag4=1.8e-05 max|k-g|(k<=20)=3.6e-04 front(1e-2)=31 max|k-tau|=0.90
 m=10 T=  216 n= 43 res_all=2.6e-02 res_lag4=1.5e-04 max|k-g|(k<=20)=9.1e-04 front(1e-2)=30 max|k-tau|=0.90
 m=11 T=  254 n= 54 res_all=3.3e-02 res_lag4=7.2e-04 max|k-g|(k<=20)=1.4e-03 front(1e-2)=31 max|k-tau|=0.90
 m=12 T=  299 n= 68 res_all=4.2e-02 res_lag4=3.1e-03 max|k-g|(k<=20)=2.0e-03 front(1e-2)=30 max|k-tau|=0.90
 m=13 T=  352 n= 85 res_all=5.6e-02 res_lag4=9.0e-03 max|k-g|(k<=20)=1.2e-03 front(1e-2)=33 max|k-tau|=0.90
 m=14 T=  414 n=107 res_all=6.3e-02 res_lag4=1.7e-02 max|k-g|(k<=20)=1.3e-03 front(1e-2)=31 max|k-tau|=0.90
 m=15 T=  487 n=133 res_all=8.0e-02 res_lag4=3.0e-02 max|k-g|(k<=20)=1.4e-03 front(1e-2)=32 max|k-tau|=0.90
 m=16 T=  572 n=165 res_all=8.6e-02 res_lag4=3.7e-02 max|k-g|(k<=20)=1.3e-03 front(1e-2)=32 max|k-tau|=0.90
horizonG_M16_I2.4_L4.json
 m= 1 T=   50 n=  4 res_all=1.1e-04 res_lag4=nan max|k-g|(k<=20)=4.4e-04 front(1e-2)=4 max|k-tau|=0.69
 m= 2 T=   59 n=  5 res_all=1.7e-04 res_lag4=nan max|k-g|(k<=20)=3.1e-04 front(1e-2)=5 max|k-tau|=0.69
 m= 3 T=   69 n=  7 res_all=2.7e-04 res_lag4=nan max|k-g|(k<=20)=1.6e-03 front(1e-2)=7 max|k-tau|=0.69
 m= 4 T=   81 n=  9 res_all=3.8e-04 res_lag4=nan max|k-g|(k<=20)=3.1e-03 front(1e-2)=9 max|k-tau|=0.84
 m= 5 T=   96 n= 12 res_all=7.4e-04 res_lag4=3.0e-06 max|k-g|(k<=20)=2.1e-03 front(1e-2)=12 max|k-tau|=0.84
 m= 6 T=  113 n= 16 res_all=5.7e-04 res_lag4=1.2e-06 max|k-g|(k<=20)=5.9e-04 front(1e-2)=16 max|k-tau|=0.90
 m= 7 T=  133 n= 20 res_all=1.1e-03 res_lag4=2.8e-06 max|k-g|(k<=20)=1.8e-03 front(1e-2)=20 max|k-tau|=0.90
 m= 8 T=  156 n= 26 res_all=1.8e-03 res_lag4=2.4e-06 max|k-g|(k<=20)=2.5e-04 front(1e-2)=26 max|k-tau|=0.90
 m= 9 T=  183 n= 34 res_all=1.1e-03 res_lag4=4.2e-06 max|k-g|(k<=20)=1.4e-04 front(1e-2)=33 max|k-tau|=0.96
 m=10 T=  216 n= 43 res_all=2.0e-03 res_lag4=3.9e-06 max|k-g|(k<=20)=3.0e-05 front(1e-2)=43 max|k-tau|=0.99
 m=11 T=  254 n= 54 res_all=1.5e-03 res_lag4=4.5e-06 max|k-g|(k<=20)=2.3e-05 front(1e-2)=54 max|k-tau|=0.99
 m=12 T=  299 n= 68 res_all=2.3e-03 res_lag4=6.4e-06 max|k-g|(k<=20)=1.7e-05 front(1e-2)=68 max|k-tau|=0.99
 m=13 T=  352 n= 85 res_all=1.9e-03 res_lag4=3.4e-06 max|k-g|(k<=20)=6.4e-06 front(1e-2)=85 max|k-tau|=0.99
 m=14 T=  414 n=107 res_all=2.5e-03 res_lag4=2.1e-06 max|k-g|(k<=20)=2.8e-06 front(1e-2)=107 max|k-tau|=0.99
 m=15 T=  487 n=133 res_all=2.1e-03 res_lag4=3.8e-06 max|k-g|(k<=20)=4.1e-06 front(1e-2)=133 max|k-tau|=0.99
 m=16 T=  572 n=165 res_all=1.9e-03 res_lag4=8.7e-06 max|k-g|(k<=20)=4.7e-06 front(1e-2)=165 max|k-tau|=0.99

Scaling fits (cert7, m>=4): Linf ~ T^1.81, ||QB|| ~ T^2.10, Banach margin ~ T^-2.08
```

D.2. **out_cert7.txt.**

```
m= 1 T=   50 n=  4 Linf=4.1e+01 eta=1.7e-15 ||QB||=3.1e+01 CERTIFIED r=1.7e-15 k(r)=0.00 margin=7.0e-03 rho(
    all tests<=m)=7.7e-08 | max|x0-tau|=0.69 max|x0-gamma|=7.4e-07
m= 2 T=   59 n=  5 Linf=2.8e+01 eta=1.3e-15 ||QB||=2.2e+01 CERTIFIED r=1.3e-15 k(r)=0.00 margin=1.0e-02 rho(
    all tests<=m)=1.1e-07 | max|x0-tau|=0.69 max|x0-gamma|=2.2e-07
m= 3 T=   69 n=  7 Linf=3.8e+01 eta=1.8e-15 ||QB||=2.9e+01 CERTIFIED r=1.8e-15 k(r)=0.00 margin=7.1e-03 rho(
    all tests<=m)=1.6e-07 | max|x0-tau|=0.69 max|x0-gamma|=1.0e-06
m= 4 T=   81 n=  9 Linf=5.0e+01 eta=4.3e-15 ||QB||=7.3e+01 CERTIFIED r=4.3e-15 k(r)=0.00 margin=3.1e-03 rho(
    all tests<=m)=2.6e-07 | max|x0-tau|=0.84 max|x0-gamma|=1.1e-06
m= 5 T=   96 n= 12 Linf=6.7e+01 eta=2.0e-14 ||QB||=1.4e+02 CERTIFIED r=2.0e-14 k(r)=0.00 margin=1.7e-03 rho(
    all tests<=m)=9.4e-08 | max|x0-tau|=0.84 max|x0-gamma|=7.5e-07
m= 6 T=  113 n= 16 Linf=1.1e+02 eta=1.0e-14 ||QB||=1.4e+02 CERTIFIED r=1.0e-14 k(r)=0.00 margin=1.7e-03 rho(
    all tests<=m)=3.5e-07 | max|x0-tau|=0.90 max|x0-gamma|=2.6e-06
m= 7 T=  133 n= 20 Linf=1.8e+02 eta=3.4e-14 ||QB||=2.0e+02 CERTIFIED r=3.4e-14 k(r)=0.00 margin=1.2e-03 rho(
    all tests<=m)=1.0e-06 | max|x0-tau|=0.90 max|x0-gamma|=2.0e-05
```

```
m= 8 T=  156 n= 26 Linf=1.5e+02 eta=2.2e-14 ||QB||=3.2e+02 CERTIFIED r=2.2e-14 k(r)=0.00 margin=7.5e-04 rho(
     all tests<=m)=9.6e-07 | max|x0-tau|=0.90 max|x0-gamma|=1.3e-05
m= 9 T=  183 n= 34 Linf=1.6e+02 eta=2.5e-14 ||QB||=3.7e+02 CERTIFIED r=2.5e-14 k(r)=0.00 margin=6.6e-04 rho(
     all tests<=m)=1.1e-06 | max|x0-tau|=0.99 max|x0-gamma|=8.7e-06
m=10 T=  216 n= 43 Linf=3.4e+02 eta=4.0e-14 ||QB||=5.5e+02 CERTIFIED r=4.0e-14 k(r)=0.00 margin=4.5e-04 rho(
     all tests<=m)=2.3e-06 | max|x0-tau|=0.99 max|x0-gamma|=1.8e-05
m=11 T=  254 n= 54 Linf=4.9e+02 eta=6.8e-14 ||QB||=9.2e+02 CERTIFIED r=6.8e-14 k(r)=0.00 margin=2.6e-04 rho(
     all tests<=m)=3.2e-06 | max|x0-tau|=0.99 max|x0-gamma|=4.5e-05
m=12 T=  299 n= 68 Linf=4.6e+02 eta=1.8e-13 ||QB||=1.1e+03 CERTIFIED r=1.8e-13 k(r)=0.00 margin=2.2e-04 rho(
     all tests<=m)=5.5e-06 | max|x0-tau|=0.99 max|x0-gamma|=4.5e-05
m=13 T=  352 n= 85 Linf=6.8e+02 eta=2.2e-13 ||QB||=1.5e+03 CERTIFIED r=2.2e-13 k(r)=0.00 margin=1.6e-04 rho(
     all tests<=m)=3.6e-05 | max|x0-tau|=0.99 max|x0-gamma|=5.0e-04
m=14 T=  414 n=107 Linf=1.0e+03 eta=2.2e-13 ||QB||=2.4e+03 CERTIFIED r=2.2e-13 k(r)=0.00 margin=1.0e-04 rho(
     all tests<=m)=3.7e-04 | max|x0-tau|=0.99 max|x0-gamma|=5.4e-03
m=15 T=  487 n=133 Linf=1.6e+03 eta=5.3e-13 ||QB||=3.6e+03 CERTIFIED r=5.3e-13 k(r)=0.00 margin=6.7e-05 rho(
     all tests<=m)=3.3e-03 | max|x0-tau|=0.99 max|x0-gamma|=4.3e-02
m=16 Newton polishing failed (singular square Jacobian)
```

D.3. `out_entry.txt`.

```
m= 2 T=   59 band= 1 | band-only: eta=1.7e-03 margin=8.8e-02 CERT | full: eta=5.5e-03 margin=1.0e-02 CERT |
     Gonek band err=5.2e-03
m= 3 T=   69 band= 2 | band-only: eta=1.5e-02 margin=3.6e-02 CERT | full: eta=1.1e-02 margin=7.1e-03 no |
     Gonek band err=5.7e-03
m= 4 T=   81 band= 2 | band-only: eta=1.1e-02 margin=4.4e-02 CERT | full: eta=1.5e-02 margin=3.3e-03 no |
     Gonek band err=1.7e-02
m= 5 T=   96 band= 3 | band-only: eta=1.1e-02 margin=3.1e-02 CERT | full: eta=1.2e-02 margin=1.8e-03 no |
     Gonek band err=1.0e-02
m= 6 T=  113 band= 4 | band-only: eta=3.1e-02 margin=2.7e-02 no | full: eta=1.8e-02 margin=1.3e-03 no | Gonek
     band err=2.4e-02
m= 7 T=  133 band= 4 | band-only: eta=3.2e-02 margin=2.0e-02 no | full: eta=3.3e-02 margin=1.2e-03 no | Gonek
     band err=4.4e-02
m= 8 T=  156 band= 6 | band-only: eta=7.0e-02 margin=1.9e-02 no | full: eta=4.7e-02 margin=8.8e-04 no | Gonek
     band err=4.0e-02
m= 9 T=  183 band= 8 | band-only: eta=1.5e-01 margin=1.7e-02 no | full: eta=9.3e-02 margin=5.5e-04 no | Gonek
     band err=8.5e-02
m=10 T=  216 band= 9 | band-only: eta=2.6e-01 margin=8.0e-03 no | full: eta=1.1e-01 margin=3.8e-04 no | Gonek
     band err=5.1e-02
m=11 T=  254 band=11 | band-only: eta=1.6e-01 margin=3.2e-03 no | full: eta=2.9e-01 margin=2.9e-04 no | Gonek
     band err=6.7e-02
singular at 11
```

D.4. `out_certG.txt`.

```
m= 1 T=   50 interior=  4 L=4.16e+01 H=1.23e+01 eta_sel=1.31e-03 eta_all=1.31e-03 A=2L^2H eta=5.61e+01  2L*eta
     =1.09e-01
m= 2 T=   59 interior=  5 L=2.83e+01 H=1.42e+01 eta_sel=1.99e-03 eta_all=2.12e-03 A=2L^2H eta=4.52e+01  2L*eta
     =1.13e-01
m= 3 T=   69 interior=  7 L=4.01e+01 H=1.96e+01 eta_sel=2.48e-03 eta_all=2.53e-03 A=2L^2H eta=1.56e+02  2L*eta
     =1.99e-01
m= 4 T=   81 interior=  9 L=5.63e+01 H=2.26e+01 eta_sel=2.81e-03 eta_all=3.19e-03 A=2L^2H eta=4.03e+02  2L*eta
     =3.16e-01
m= 5 T=   96 interior= 12 L=6.77e+01 H=2.80e+01 eta_sel=4.47e-03 eta_all=4.47e-03 A=2L^2H eta=1.15e+03  2L*eta
     =6.05e-01
m= 6 T=  113 interior= 16 L=9.78e+01 H=3.32e+01 eta_sel=4.61e-03 eta_all=5.00e-03 A=2L^2H eta=2.93e+03  2L*eta
     =9.02e-01
m= 7 T=  133 interior= 20 L=1.41e+02 H=3.97e+01 eta_sel=4.56e-03 eta_all=6.93e-03 A=2L^2H eta=7.17e+03  2L*eta
     =1.28e+00
m= 8 T=  156 interior= 26 L=1.84e+02 H=5.16e+01 eta_sel=8.89e-03 eta_all=8.89e-03 A=2L^2H eta=3.09e+04  2L*eta
     =3.26e+00
m= 9 T=  183 interior= 34 L=2.60e+02 H=6.45e+01 eta_sel=9.13e-03 eta_all=1.01e-02 A=2L^2H eta=7.97e+04  2L*eta
     =4.75e+00
m=10 T=  216 interior= 43 L=3.59e+02 H=8.53e+01 eta_sel=1.45e-02 eta_all=1.45e-02 A=2L^2H eta=3.17e+05  2L*eta
     =1.04e+01
m=11 T=  254 interior= 54 L=4.69e+02 H=1.02e+02 eta_sel=1.51e-02 eta_all=1.83e-02 A=2L^2H eta=6.80e+05  2L*eta
     =1.41e+01
```

```
m=12 T=  299 interior= 68 L=7.53e+02 H=1.29e+02 eta_sel=1.66e-02 eta_all=2.11e-02 A=2L^2H eta=2.43e+06  2L*eta
    =2.50e+01
m=13 T=  352 interior= 85 L=6.45e+02 H=1.61e+02 eta_sel=2.36e-02 eta_all=2.61e-02 A=2L^2H eta=3.16e+06  2L*eta
    =3.04e+01
m=14 T=  414 interior=107 L=1.02e+03 H=2.03e+02 eta_sel=2.83e-02 eta_all=2.83e-02 A=2L^2H eta=1.20e+07  2L*eta
    =5.79e+01
```

D.5. **out_cert3.txt.**

```
m= 1 T=   50 n=  4 tests=  42 beta=  8.38 K=6.36e+00 delta=8.85e-03 h=4.72e-01 |kG-g|=8.75e-03 after1GN=4.77e
    -04
m= 2 T=   59 n=  5 tests=  96 beta=  5.19 K=9.92e+00 delta=1.00e-02 h=5.16e-01 |kG-g|=1.02e-02 after1GN=3.30e
    -04
m= 3 T=   69 n=  7 tests= 164 beta=  6.83 K=1.31e+01 delta=1.27e-02 h=1.13e+00 |kG-g|=1.18e-02 after1GN=1.65e
    -03
m= 4 T=   81 n=  9 tests= 248 beta=  6.99 K=1.59e+01 delta=2.03e-02 h=2.25e+00 |kG-g|=2.26e-02 after1GN=2.98e
    -03
m= 5 T=   96 n= 12 tests= 353 beta=  7.16 K=1.87e+01 delta=2.47e-02 h=3.31e+00 |kG-g|=2.50e-02 after1GN=2.10e
    -03
m= 6 T=  113 n= 16 tests= 482 beta=  8.00 K=2.12e+01 delta=4.45e-02 h=7.55e+00 |kG-g|=4.51e-02 after1GN=7.97e
    -04
m= 7 T=  133 n= 20 tests= 641 beta=  8.18 K=2.36e+01 delta=7.09e-02 h=1.37e+01 |kG-g|=7.14e-02 after1GN=9.39e
    -04
m= 8 T=  156 n= 26 tests= 836 beta=  8.06 K=2.59e+01 delta=9.47e-02 h=1.98e+01 |kG-g|=9.58e-02 after1GN=5.21e
    -03
m= 9 T=  183 n= 34 tests=1074 beta=  9.83 K=2.80e+01 delta=1.60e-01 h=4.42e+01 |kG-g|=1.53e-01 after1GN=4.14e
    -02
m=10 T=  216 n= 43 tests=1364 beta= 10.26 K=3.00e+01 delta=1.72e-01 h=5.28e+01 |kG-g|=1.80e-01 after1GN=1.60e
    -02
m=11 T=  254 n= 54 tests=1717 beta= 17.15 K=3.18e+01 delta=2.33e-01 h=1.27e+02 |kG-g|=2.43e-01 after1GN=2.26e
    -02
m=12 T=  299 n= 68 tests=2146 beta= 15.39 K=3.36e+01 delta=2.66e-01 h=1.37e+02 |kG-g|=2.79e-01 after1GN=2.68e
    -02
m=13 T=  352 n= 85 tests=2667 beta= 10.99 K=3.52e+01 delta=3.38e-01 h=1.31e+02 |kG-g|=3.55e-01 after1GN=3.56e
    -02
m=14 T=  414 n=107 tests=3298 beta= 15.67 K=3.67e+01 delta=4.05e-01 h=2.33e+02 |kG-g|=4.27e-01 after1GN=5.19e
    -02
```

D.6. **out_gonek.txt.**

```
clock : mean|.-g|/sp=0.187 max=0.506  #>0.25sp=50
Gonek X= 3 (offset -3): mean|.-g|/sp=0.048 max=0.184 #>0.25sp=0  k=34: 110.923 (g=111.030)
Gonek X= 5 (offset -3): mean|.-g|/sp=0.029 max=0.081 #>0.25sp=0  k=34: 111.034 (g=111.030)
Gonek X= 8 (offset -3): mean|.-g|/sp=0.019 max=0.053 #>0.25sp=0  k=34: 111.071 (g=111.030)
Gonek X=12 (offset -3): mean|.-g|/sp=0.014 max=0.047 #>0.25sp=0  k=34: 111.092 (g=111.030)
Gonek X=20 (offset -3): mean|.-g|/sp=0.008 max=0.035 #>0.25sp=0  k=34: 111.075 (g=111.030)
```

D.7. **out_laplace_check.txt.**

```
u=0.1  k_v18=-1.6225522892294519278916331593609088459137814857449  k_mine
    =-1.6225522892294519278916331593609088459137814857449  K
    |=-1.622552289229451927891633159360908845913781485744 9
u=1.0  k_v18=0.776894260793253581707337576838870172282688498067 97  k_mine
    =0.776894260793253581707337576838870172282688498067 97  K
    |=0.77689426079325358170733757683887017228268849806797
u=2.5  k_v18=1.744199703895470525246010376878366387012250050533 3  k_mine
    =1.744199703895470525246010376878366387012250050533 3  K
    |=1.74419970389547052524601037687836638701225005053 33
s=2.0  (1/2)G'/G=0.319513612312266741312515139641134714327636462786 19  L[K
    ]=0.31951361231226674131251526910712335613061628485609
s=1.3  (1/2)G'/G=1.422512214600398573948309601670381704130033469369 7  L[K
    ]=1.42251221460039857394830973113637034599573086493 48
s=(2.0 + 5.0j)  (1/2)G'/G=(-0.000004483925696515754762541200984984498851920143985085 5 +
    0.160338457354031730177741946245978073485991510733 49j)  L[K
    ]=(-0.000004483925696515754762411734996342695870171439857178 +
    0.160338457354031730177741946245978073038010423445 5j)
```

```
s=(1.5 - 3.0j)  (1/2)G'/G=(-0.092240558736457654753515338039303026644761224786081 -
    0.054136617330789788303900306539987383185752403110009j)  L[K
    ]=(-0.092240558736457654753515208573314384796983337328152 -
    0.054136617330789788303900306539987382916962760406523j)
```

Appendix E. Endpoint-only closure, interpolation tests, and autonomous residual questions

This addendum records the outcome of the subsequent interpolation and finite-section tests. It does not alter any theorem above. Its purpose is to separate the statements that are exact but equivalent to the final step from the one clock-only subproblem that remains autonomous.

E.1. **Three interpolations and their status.** There are three natural ways to interpolate between the midpoint clock and the Riemann zeta function.

The ordinate interpolation. The family ζ_λ of Sections 10–14 moves the ordinates while keeping every nontrivial zero on $\Re s = \frac{1}{2}$. It is an exact analytic family, but for every $0 < \lambda \le 1$ its continuous prime measure is not in the ordinary Dirichlet category. Thus the property that is needed at the endpoint is not preserved along the path; this is exactly the role of condition (D).

The exact-atomic interpolation. One may instead ask for real ordinates κ_k and masses a_n supported at $u = \log n$ throughout an interpolation. Finite sections of this problem were tested numerically after v16. The endpoints can be resolved, whereas the linear interpolation of the target distribution develops a persistent interior residual in the tested sections. This is evidence against that particular interpolation, not a proof that no infinite exact-atomic path exists. In any case the existence of such a path is not supplied by the results of this paper.

The additive interpolation of functions. There is an exact elementary interpolation

$$\mathcal{G}_\varepsilon(s) := (1-\varepsilon)\zeta_{\mathrm{RVM}}(s) + \varepsilon\zeta(s), \qquad 0 \le \varepsilon \le 1. \tag{E.1}$$

Both summands satisfy the same functional equation and reality symmetry, so the same is true of $\mathcal{G}_\varepsilon$. This family exists for every ε, but it does not create a new control quantity: it is a level-set representation of the already defined quotient $\mathcal{Q}_{\mathrm{RVM}} = \zeta/\zeta_{\mathrm{RVM}}$.

Proposition E.1 (The additive family is a level-set reformulation)**.** *Away from the zeros of* ζ_{RVM}*,*

$$\mathcal{G}_\varepsilon(s) = \zeta_{\mathrm{RVM}}(s)\big((1-\varepsilon) + \varepsilon\mathcal{Q}_{\mathrm{RVM}}(s)\big). \tag{E.2}$$

Hence for $0 < \varepsilon < 1$ *every zero of* $\mathcal{G}_\varepsilon$ *which is not a zero of* ζ_{RVM} *lies on the level set*

$$\mathcal{Q}_{\mathrm{RVM}}(s) = -\frac{1-\varepsilon}{\varepsilon}. \tag{E.3}$$

On the critical line $e^{i\vartheta(t)}\mathcal{G}_\varepsilon(\frac{1}{2}+it)$ *is real. At* $\varepsilon = 1$ *the statement that all its nontrivial zeros lie on the critical line is exactly RH.*

Proof. Equation (E.2) is the definition of $\mathcal{Q}_{\mathrm{RVM}}$. The functional equation and the reality statement follow by linearity from those of the two summands. The last assertion is $\mathcal{G}_1 = \zeta$. □

Thus the additive family is useful for visualizing the zero motion, but without an additional monotonicity or flow law it is circular as a route to RH: the terminal level is precisely the original zero problem.

E.2. **The clock as the first transition term.** The transition theorem gives the following $1+\chi$ form on every closed subband of $\frac{1}{2} < \Re s < 1$:

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1+\chi(s))\big(1+O(|t|^{-1})\big). \tag{E.4}$$

Independently, on $s = \frac{1}{2} + it$, where $\chi(s) = e^{-2i\vartheta(t)}$, the exact trigonometric identity

$$e^{i\vartheta(t)}(1+\chi(\tfrac{1}{2}+it)) = 2\cos\vartheta(t). \tag{E.5}$$

Consequently the midpoint clock is asymptotically the zero set selected by the $n = 1$ term of the Hardy–Littlewood/Riemann–Siegel approximate functional equation. This is a structural interpretation of the transition theorem, not an exact identification: the $O(1/t)$ term is visible already in the first ordinates.

For a finite Dirichlet polynomial

$$\mathsf{P}_{\varepsilon,M}(s) := 1 + \varepsilon \sum_{2\le n\le M} n^{-s}, \qquad \mathsf{R}_{\varepsilon,M}(s) := \mathsf{P}_{\varepsilon,M}(s) + \chi(s)\mathsf{P}_{\varepsilon,M}(1-s), \tag{E.6}$$

the functional equation is exact. On the critical line,

$$e^{i\vartheta(t)}\mathsf{R}_{\varepsilon,M}(\tfrac{1}{2}+it) = 2\Re\Big(e^{i\vartheta(t)}\mathsf{P}_{\varepsilon,M}(\tfrac{1}{2}+it)\Big). \tag{E.7}$$

At Gram points the $n = 1$ term fixes the sign whenever

$$\varepsilon \sum_{2\le n\le M} n^{-1/2} < 1, \tag{E.8}$$

so in every interval where ϑ is monotone there is a critical-line zero between consecutive Gram points. This dominance statement is only sufficient; it does not count all zeros of $\mathsf{R}_{\varepsilon,M}$ and is not used elsewhere in the paper.

E.3. **What remains genuinely live.** After these tests the status is narrower than the programme suggested by the interpolation pictures.

(L1) *The endpoint problem remains the endpoint problem.* Proving that the cell defect vanishes at $\lambda = 0$, equivalently proving (D), remains the only step that closes the main route. The additive family and the exact-atomic interpolation do not simplify this statement.

(L2) *The clock-only injectivity problem is autonomous.* The one subproblem that does not contain the genuine ordinates is the injectivity of the linearized rung system at the midpoint clock. In the meromorphic form of Remark 15.7, the target statement is: if a bounded real sequence $(\dot\kappa_k)$ and coefficients $\dot{\mathsf{a}}_n = O_\varepsilon(n^{-1/2+\varepsilon})$ give a Dirichlet series H satisfying

$$\mathsf{H}(z) = \nu_8 + \sum_{k\ge1} \dot\kappa_k \left(\frac{\tau_k}{z^2+\tau_k^2} - \frac{1}{\tau_k}\right) = \sum_{n\ge2} \frac{\dot{\mathsf{a}}_n}{\log n}\, n^{-z} \tag{E.9}$$

through a common right half-plane, under the growth hypotheses of the linearization, then $\mathsf{H} \equiv 0$. Theorem 15.6 proves the analogous conclusion at an actual solution, while Remark 15.7 explains why it is not automatic for a general list. Equation (E.9) is therefore a closed clock-only rigidity problem, naturally related to Fourier crystalline uniqueness, but by itself it does not prove (D).

(L3) *Bounded finite sections are a numerical existence programme, not yet an analytic route.* Direct finite sections of the endpoint rung system can be solved while constraining both the ordinate displacement and the free masses. If one could prove, for an exhausting nested family of tests, a uniform boundedness and residual-to-zero statement and then pass to a distributional limit, Theorem 15.5 would identify the limit. Such a theorem would already be an existence theorem for the final system and is therefore not a shortcut around (D).

In particular, the individual displacements $\gamma_k - \tau_k$ are not autonomous data. They are used below only for post-solution numerical diagnostics. None of the live statement (E.9), nor the definition of the bounded finite-section problem, requires them as input.

E.4. **A priori bounded displacement for exact rung solutions.** The following statement is independent of the existence of an interior exact-atomic interpolation. It says only that, if an exact real solution exists and has the stated local density, then its ordinates cannot drift arbitrarily far from the midpoint clock.

Proposition E.2 (A priori displacement bound for an exact rung solution)**.** *Fix $\theta \in [0,1]$. Let $\boldsymbol{\kappa} = (\kappa_k)_{k\ge 1}$ satisfy $0 < \kappa_1 \le \kappa_2 \le \cdots$ and $\kappa_k \to \infty$, and let $(\mathsf{a}_n)_{n\ge 2}$ be real masses satisfying, in $\mathcal{D}'(0,\infty)$,*

$$\sum_{k\ge 1} \cos(\kappa_k u) - (1-\theta) \sum_{k\ge 1} \cos(\tau_k u) - \theta\, \mathsf{k}(u) = -\sum_{n\ge 2} \mathsf{a}_n\, \delta(u - \log n). \tag{E.10}$$

Assume that for every $\epsilon_0 > 0$ the local count satisfies

$$N_{\boldsymbol{\kappa}}(T+1) - N_{\boldsymbol{\kappa}}(T) \ll_{\epsilon_0} (1+T)^{\epsilon_0}. \tag{E.11}$$

Pairing the exact rung identity with the smooth cell partition identifies a_n with the corresponding cell mass difference between the solution and the fixed clock/archimedean terms. The proof of Theorem 13.10, with Lemma 13.9 replaced by the stated local-density bound and with the powers 2^j treated by (13.21)*, then gives the derived estimate*

$$\sum_{X < n \le 2X} |\mathsf{a}_n|^2 \ll_{\epsilon_0} X^{1+\epsilon_0}. \tag{E.12}$$

There is therefore a constant C^*, uniform over every class with fixed constants in (E.11) *and depending only on those constants and the fixed normalization of the rung system, such that*

$$|\kappa_k - \tau_k| \le C^* \qquad (k \ge 1). \tag{E.13}$$

In particular the bound is uniform in θ.

Proof. The local-density hypothesis, taken with any fixed $\epsilon_0 < 1$, implies $N_{\boldsymbol{\kappa}}(T) = O(T^{1+\epsilon_0})$ and hence $\sum_k \kappa_k^{-2} < \infty$; thus the symmetric canonical product $P_{\boldsymbol{\kappa}}(s) = \prod_k (1 + q(s)/(\frac14 + \kappa_k^2))$ is well defined and entire. Pair (E.10) with $e^{-(s-1/2)u}$ first in a right half-plane $\Re s > \frac32$ where the mass term is absolutely convergent by (E.12). The identity

$$\int_0^\infty e^{-(s-1/2)u} \cos(\kappa u)\, du = \frac{s-\frac12}{(s-\frac12)^2 + \kappa^2} = \frac12\, \partial_s \log\!\left(1 + \frac{s(s-1)}{\frac14 + \kappa^2}\right)$$

gives, after integration in s from $+\infty$ and with the real normalization there,

$$P_{\boldsymbol{\kappa}}(s) = C_\theta \big(P^{\mathrm{crit}}_{\mathrm{RVM}}(s)\big)^{1-\theta}\, G(s)^\theta\, Y_\theta(s) = C_\theta G(s) \zeta_{\mathrm{RVM}}(s)^{1-\theta} Y_\theta(s), \tag{E.14}$$

where

$$Y_\theta(s) = e^{\ell_\theta} \exp\!\left(\sum_{n\ge 2} \frac{2\mathsf{a}_n}{\log n}\, n^{1/2-s} \right). \tag{E.15}$$

The Laplace transform of $2\mathsf{k}$ is G'/G by the digamma expansion used in (12.17); hence the fixed factor produced by the k-term is exactly G up to a nonzero normalization constant C_θ, absorbed into e^{ℓ_θ}. The fractional power is needed only in the zero-free right half-plane, where the logarithm of ζ_{RVM} is fixed by its positive real values; the zero count below is performed on the entire function $P_{\boldsymbol{\kappa}}$ itself.

By (E.12) and Cauchy–Schwarz on dyadic blocks, the Dirichlet logarithm in (E.15) converges absolutely and uniformly on every half-plane $\Re s \ge \sigma_0 > \frac32$. Indeed, for $X < n \le 2X$,

$$\sum_{X < n \le 2X} |\mathsf{a}_n| n^{1/2-\sigma_0} \ll_{\epsilon_0} X^{3/2-\sigma_0+\epsilon_0/2},$$

so choosing $\epsilon_0 < 2(\sigma_0 - \frac32)$ and summing dyadically gives

$$\sup_{\Re s \ge \sigma_0} |\log Y_\theta(s)| \le B_{\sigma_0} < \infty. \tag{E.16}$$

The constant is uniform in θ and in the particular exact solution once the constants in (E.12) are fixed. No displacement bound is used here.

The quotient $P_{\boldsymbol{\kappa}}/G$ has a simple pole at $s=1$, so the pole-cleared function

$$\mathcal{Z}_{\boldsymbol{\kappa}}(s) := (s-1)\frac{P_{\boldsymbol{\kappa}}(s)}{G(s)}$$

is entire of order one. This is the function to which Phragmén–Lindelöf is applied; using $\log|P/G|$ itself across $s=1$ would be illegitimate. Fix $\sigma_0 > \frac{3}{2}$. On the right boundary $\Re s = \sigma_0$, (E.14), Theorem 8.10, and (E.16) give

$$\log|\mathcal{Z}_{\boldsymbol{\kappa}}(\sigma_0+it)| \le B_{\sigma_0} + O(1) + \log(2+|t|),$$

uniformly in θ. The symmetry of $P_{\boldsymbol{\kappa}}$ gives the same functional equation as for every member of the canonical family; reflecting to $1-\sigma_0+it$ and using Stirling's formula adds only a fixed multiple of $\log(2+|t|)$. The order-one growth and the Phragmén–Lindelöf principle therefore give a logarithmic upper bound for $\log|\mathcal{Z}_{\boldsymbol{\kappa}}|$ throughout the strip.

Apply Backlund's form of Jensen's argument to $P_{\boldsymbol{\kappa}}$ (equivalently to the pole-cleared $\mathcal{Z}_{\boldsymbol{\kappa}}$ after the known archimedean factors are restored) in a disc centred at σ_0+iT with fixed radii $R > r = \sigma_0 - \frac{1}{2}$. The preceding upper bound, together with the lower bound at the centre supplied by (E.14) and the two-sided bound for ζ_{RVM}, gives

$$|N_{\boldsymbol{\kappa}}(T) - N_{\mathrm{RVM}}(T)| \le A^*\log(2+T) + A_1 B_{\sigma_0} + A_2, \tag{E.17}$$

with constants independent of θ and of the particular solution. The argument uses $P_{\boldsymbol{\kappa}}$, which is entire, so no branch of a fractional factor enters the zero count.

Finally $\mathcal{N}'_{\mathrm{RVM}}(T) = (2\pi)^{-1}\log(T/2\pi)$. The count estimate (E.17) implies, at the kth ordinate and with multiplicity included,

$$\mathcal{N}_{\mathrm{RVM}}(\kappa_k) = k + O(\log(2+\kappa_k)). \tag{E.18}$$

Indeed the jump of $N_{\boldsymbol{\kappa}}$ at a multiple ordinate is itself $O(\log(2+\kappa_k))$ by (E.17). Since $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k-\frac{1}{2}$, (E.18) first gives $\kappa_k \asymp \tau_k$: the error $O(\log\kappa_k)$ is $o(k)$, while $\mathcal{N}_{\mathrm{RVM}}(t) \asymp t\log t$. The mean-value theorem on the increasing branch then yields

$$|\kappa_k - \tau_k|\,\frac{1}{2\pi}\log\frac{\min(\kappa_k,\tau_k)}{2\pi} \ll \log(2+\kappa_k),$$

and the comparability just obtained makes the ratio of the logarithms bounded. Thus $|\kappa_k - \tau_k| \le C^*$ for all sufficiently large k, and the finitely many remaining indices are absorbed into the same constant. The constant depends only on the uniform local-density constants and the fixed normalization and is independent of θ. □

Remark E.3. Proposition E.2 is an a priori theorem, not an existence theorem. The exact-atomic interpolation has no established interior solution for $0<\theta<1$; the proposition only says that any exact solution satisfying its hypotheses lies in one uniform bounded-displacement class.

E.5. **Additional finite-section diagnostics.** The post-v16 calculations solve bounded finite sections directly from the midpoint clock. Here the normalized residual is the Euclidean residual divided by the square root of the number of test equations; a test bump is a fixed C_c^∞ weight in the u-variable; and a permanent block is a set of test equations that, once introduced, is retained in every later nested solve. The repository scripts and their console/CSV outputs are included with version 18. A representative run imposes

$$|\kappa_k - \tau_k| \le 1.2, \qquad |a_n| \le 0.5, \tag{E.19}$$

with the unresolved ordinate tail kept on the midpoint clock. The bounds in (E.19) are numerical regularizations, not hypotheses of any theorem. The genuine ordinates and the von Mangoldt masses are read only after the solve for diagnostics.

With free masses and the bounds (E.19), the sections $N = 20, 34, 50$ have normalized residuals of order 10^{-4}, and the low ordinates and low masses remain close to their diagnostic zeta/von-Mangoldt

values. Cross-validation with test bumps not used in the solve gives residuals of the same order. These endpoint calculations are distinct from the linearized clock problem: in the latter, it is the ordinate-direction coefficients h_k that become large while a residual of roughly 15–18% persists. Imposing bounds there does not remove an artifact; it exposes the lack of an exact solution to that tested linear problem.

A second test uses a nested family of permanent test blocks: once a test block is introduced it remains in all later solves. For the bounded sections $N = 50, 70, 100$, the RMS residuals on the old blocks stay small and in the $N = 100$ section are between $2.5 \cdot 10^{-8}$ and $6.1 \cdot 10^{-8}$ on the blocks $N = 13, 20, 34, 50, 70$, while the newest $N = 100$ block has RMS $2.2 \cdot 10^{-6}$. This is computational evidence for local stabilization only; no monotone convergence theorem is claimed.

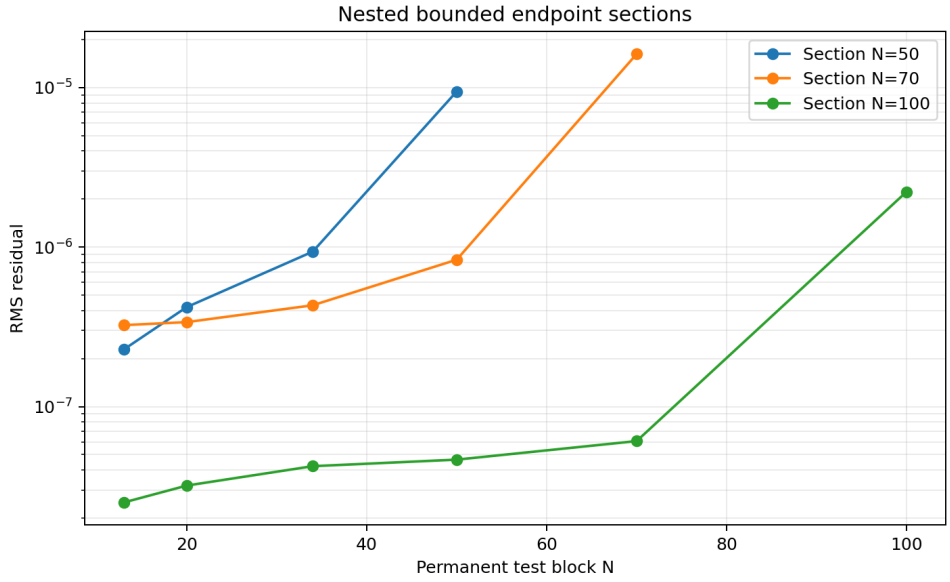


FIGURE 18. Nested bounded endpoint sections. Each curve shows the residual of one solved section on all permanent test blocks already introduced. The ordinate shifts and free masses are bounded during the solve. This is a finite computation and is not an existence proof for the infinite rung system.

Status. The additive interpolation is exact but is a level-set reformulation of $\mathcal{Q}_{\mathrm{RVM}}$; the exact-atomic interpolation has no established interior path; the bounded finite-section computations do not prove an infinite solution. The midpoint-clock injectivity problem (E.9) is a closed autonomous rigidity question, and Theorem 14.44 gives the accompanying multiscale linear information. Proposition E.2 gives an a priori displacement bound for any exact rung solution satisfying its hypotheses. None of these statements supplies existence; the main endpoint identity (D) remains open.

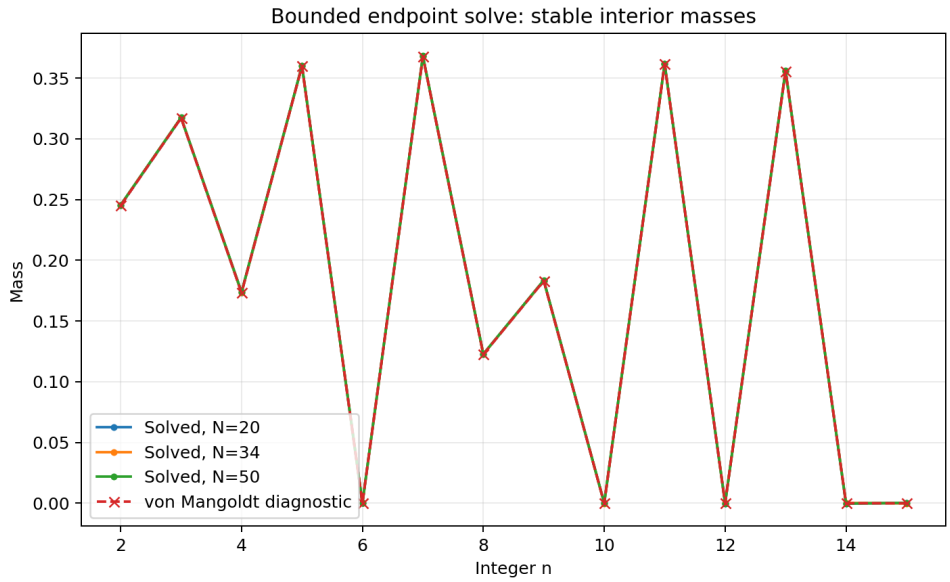


FIGURE 19. Recovered free masses in the interior $n \leq 15$ of bounded endpoint sections. The von Mangoldt values are shown only as a post-solution diagnostic and are not supplied to the solver; section-boundary masses are omitted from the plot.